Topics on econometrics and causal inference

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2026-08-24

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Preface

This is a working notebook rather than a textbook. The chapters are posts I wrote while reading, teaching, and consulting on applied econometrics, often because a colleague or a student asked a question that did not have a tidy answer in the books I had. Each one stands on its own. The order did not follow a plan, and the same topic sometimes appears more than once as my understanding changed.

The common question is the one Angrist and Pischke emphasize: what is being identified, by what variation, under what assumptions. The posts ask that question in different settings: interactions and marginal effects, fixed and correlated random effects, matching and weights, difference-in-differences, synthetic control, count models, TMLE and AIPW, mediation, proximal causal inference, partial interference, conjoint and list experiments, uplift modeling, and policy trees.

Most examples are in R. I use Stata where it is more natural, and sometimes Python or Julia when the package is better there. The two companion books, Introduction to Causal Econometrics and Causal Econometrics with Julia, organize the same material more systematically. This collection is the messier version: useful when I want to see how a specific method behaves on a specific problem.

How to cite

Ao, Xiang. *Topics on Econometrics and Causal Inference*. https://xiangao.github.io/blog_book/.

```
@book{ao_topics_econometrics,
  author = {Ao, Xiang},
  title  = {Topics on Econometrics and Causal Inference},
  year   = {2026},
  url    = {https://xiangao.github.io/blog_book/}
}
```

Part I.

Regression Foundations

1. Interpreting interaction in a regression model

1.1. Interaction with two binary variables

Start with the simplest case: two binary variables x_1 and x_2 coded 0/1.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 \tag{1.1}$$

The model is saturated — three coefficients for three contrasts from the baseline $(x_1, x_2) = (0, 0)$, one for each of the other three cells. Often only the interaction term gets attention. When the two variables are treatment and group, the interaction is the difference in treatment effects between groups, and that may be all that matters. But in other designs all four cells are substantively interesting, and then β_1 and β_2 matter too.

Consider the model:

$$E(y) = \beta_0 + \beta_1 female + \beta_2 treatment + \beta_{12} female * treatment \tag{1.2}$$

The four cell means are:

	male	female
control	β_0 (1.2)	$\beta_0 + \beta_1$ (1.2)
treatment	$\beta_0 + \beta_2$ (1.2)	$\beta_0 + \beta_1 + \beta_2 + \beta_{12}$ (1.2)

Each β reads off the table: β_0 is the baseline cell mean, β_1 the female premium among controls, β_2 the treatment effect among males. The interaction

$$\beta_{12} = (E(Y|(1,1)) - E(Y|(0,1))) - (E(Y|(1,0)) - E(Y|(0,0))) \tag{1.3}$$

is the difference in differences — the treatment effect for females minus the treatment effect for males. When the research question is "does the treatment effect differ by group?", the interaction term is the answer. Replace female/male with before/after and the same expression is the standard DiD estimator.

But sometimes the interest is in all four cells. Union membership crossed with race, for example — we may want all six pairwise comparisons, not just the interaction. Stata's `margins` command computes these directly:

```
webuse union3
reg ln_wage i.union##i.black, r
margins union#black
margins union#black, pwcompare
```

```
. webuse union3
(NLS Women 14-24 in 1968)

. reg ln_wage i.union##i.black, r

Linear regression                               Number of obs     =      1,244
                                                F(3, 1240)        =      34.76
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0762
                                                Root MSE          =     .37699

------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     1.union |   .2045053   .0291682     7.01   0.000     .1472808    .2617298
     1.black |  -.1709034   .0308067    -5.55   0.000    -.2313425   -.1104644
             |
 union#black |
        1 1  |   .0386275   .0516609     0.75   0.455     -.062725      .13998
             |
       _cons |   1.657525   .0138278   119.87   0.000     1.630396    1.684653
------------------------------------------------------------------------------

. margins union#black

Adjusted predictions                                     Number of obs = 1,244
Model VCE: Robust

Expression: Linear prediction, predict()

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

```
union#black |
        0 0  |   1.657525   .0138278   119.87   0.000     1.630396    1.684653
        0 1  |   1.486621    .027529    54.00   0.000     1.432613     1.54063
        1 0  |    1.86203   .0256822    72.50   0.000     1.811644    1.912415
        1 1  |   1.729754   .0325611    53.12   0.000     1.665873    1.793635
------------------------------------------------------------------------------

. margins union#black, pwcompare

Pairwise comparisons of adjusted predictions            Number of obs = 1,244
Model VCE: Robust

Expression: Linear prediction, predict()

-----------------------------------------------------------------
                |            Delta-method         Unadjusted
                |   Contrast   std. err.     [95% conf. interval]
----------------+------------------------------------------------
    union#black |
(0 1) vs (0 0)  |  -.1709034   .0308067     -.2313425   -.1104644
(1 0) vs (0 0)  |   .2045053   .0291682      .1472808    .2617298
(1 1) vs (0 0)  |   .0722294   .0353756      .0028268     .141632
(1 0) vs (0 1)  |   .3754087   .0376487      .3015466    .4492709
(1 1) vs (0 1)  |   .2431328   .0426388      .1594807     .326785
(1 1) vs (1 0)  |  -.1322759   .0414705     -.2136359   -.0509159
-----------------------------------------------------------------

.
```

The `margins` call reports the four cell means; `pwcompare` gives all six pairwise comparisons. In this example, all six pairwise contrasts are significant at 95% — yet the interaction term is insignificant ($p = 0.455$). The two facts are compatible. Each pairwise comparison asks whether two cells differ from each other. The interaction asks whether the *difference between two of those differences* is non-zero — a second-order quantity with its own, larger standard error. Here the union premium is clearly positive for both groups, but the gap between the two premiums (0.039) is not distinguishable from zero.

1.2. Interaction with continuous variables

With continuous x_1 and x_2,

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 \tag{1.4}$$

centering both variables — subtracting each mean before forming the product — is a linear transformation that leaves fitted values, R^2, and everything about β_{12} unchanged. What it changes is β_1 and β_2.

First, it reduces collinearity. When x_1 and x_2 are positive, x_1x_2 is mechanically correlated with both, inflating the standard errors of β_1 and β_2. Centering breaks most of that correlation (all of it under normality), stabilizing the main-effect estimates without altering the interaction.

Second, it fixes the interpretation. In the uncentered model, β_1 is the effect of x_1 when $x_2 = 0$, and zero may be outside the support of x_2. After centering, β_1 is the effect of x_1 evaluated at the mean of x_2 — a quantity that is always interpretable. When a dummy interacts with a continuous variable, only the continuous variable should be centered.

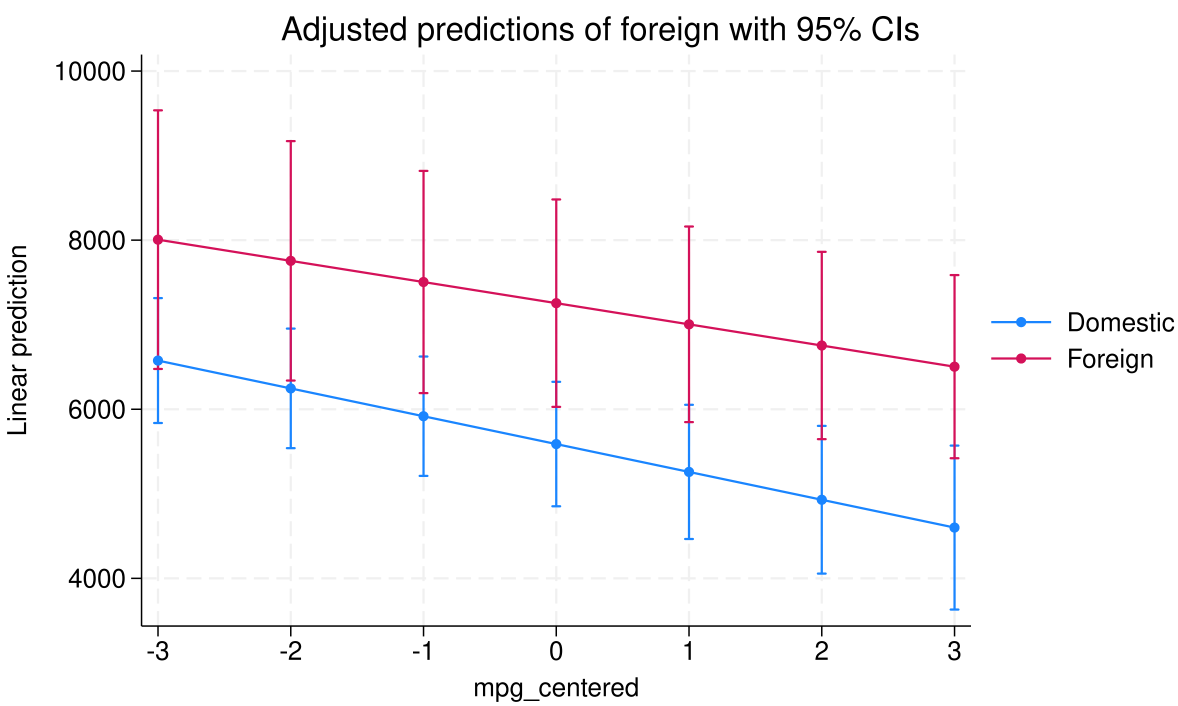
Stata's `margins` command computes the conditional predictions directly. The data are the built-in `auto` file: 74 1978 model cars, with `price` in dollars, `mpg` in miles per gallon, and `foreign` marking the 22 imports. We centre `mpg` at its mean, regress price on `foreign` interacted with centred mpg, and then ask for predicted prices by origin across a range of mpg values.

```
sysuse auto
sum mpg
gen mpg_centered=mpg-r(mean)
sum mpg_centered
reg price i.foreign##c.mpg_centered
margins foreign, at(mpg_centered=(-3 (1) 3))
marginsplot
graph export "marginsplot-interaction.svg", as(svg) replace
```

```
. sysuse auto
(1978 automobile data)

. sum mpg

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
         mpg |         74     21.2973    5.785503         12         41

. gen mpg_centered=mpg-r(mean)

. sum mpg_centered

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
mpg_centered |         74   -4.03e-08    5.785503  -9.297297    19.7027

. reg price i.foreign##c.mpg_centered

      Source |       SS           df       MS      Number of obs   =        74
-------------+----------------------------------   F(3, 70)        =      9.48
       Model |   183435285         3  61145094.9   Prob > F        =    0.0000
    Residual |   451630112        70  6451858.74   R-squared       =    0.2888
-------------+----------------------------------   Adj R-squared   =    0.2584
```

```
       Total |   635065396        73  8699525.97   Root MSE        =    2540.1

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     foreign |
    Foreign  |   1666.519    717.217     2.32   0.023     236.0751    3096.963
mpg_centered |  -329.2551   74.98545    -4.39   0.000    -478.8088   -179.7013
             |
     foreign#|
          c. |
mpg_centered |
    Foreign  |   78.88826   112.4812     0.70   0.485    -145.4485     303.225
             |
       _cons |   5588.295   369.0945    15.14   0.000     4852.159    6324.431
------------------------------------------------------------------------------

. margins foreign, at(mpg_centered=(-3 (1) 3))

Adjusted predictions                                        Number of obs = 74
Model VCE: OLS

Expression: Linear prediction, predict()
1._at: mpg_centered = -3
2._at: mpg_centered = -2
3._at: mpg_centered = -1
4._at: mpg_centered =  0
5._at: mpg_centered =  1
6._at: mpg_centered =  2
7._at: mpg_centered =  3

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 _at#foreign |
 1#Domestic  |    6576.06    370.446    17.75   0.000     5837.229    7314.891
  1#Foreign  |   8005.915   766.8178    10.44   0.000     6476.545    9535.284
 2#Domestic  |   6246.805   354.4734    17.62   0.000      5539.83     6953.78
  2#Foreign  |   7755.548   709.9327    10.92   0.000     6339.632    9171.464
 3#Domestic  |    5917.55   354.0032    16.72   0.000     5211.513    6623.587
  3#Foreign  |   7505.181   658.8306    11.39   0.000     6191.185    8819.177
 4#Domestic  |   5588.295   369.0945    15.14   0.000     4852.159    6324.431
  4#Foreign  |   7254.814   614.9548    11.80   0.000     6028.325    8481.303
 5#Domestic  |    5259.04    397.981    13.21   0.000     4465.292    6052.788
  5#Foreign  |   7004.447   579.9479    12.08   0.000     5847.778    8161.117
 6#Domestic  |   4929.785   437.9413    11.26   0.000     4056.338    5803.231
  6#Foreign  |   6754.081   555.4891    12.16   0.000     5646.192    7861.969
```

```
 7#Domestic  |    4600.53    486.253     9.46   0.000     3630.729    5570.331
  7#Foreign  |   6503.714   543.0057    11.98   0.000     5420.723    7586.704
------------------------------------------------------------------------------

. marginsplot

Variables that uniquely identify margins: mpg_centered foreign

. graph export "marginsplot-interaction.svg", as(svg) replace
file marginsplot-interaction.svg saved as SVG format

.
```



Mean mpg is 21.3, so the centred variable runs from -9.3 to 19.7 and has mean zero to seven decimals. That is what makes the coefficients readable. The constant of $5,588 is the predicted price of a domestic car at *average* fuel economy. In the uncentred model it would instead be the predicted price at `mpg = 0`, which is nine standard deviations below the smallest car in the data and has no meaning.

The `foreign` coefficient is 1666.5 with a standard error of 717.2 and $p = 0.023$: at average mpg, imports cost about $1,667 more than domestic cars. The slope on centred mpg is -329.3 with a standard error of 75.0, so among domestic cars each extra mile per gallon is worth about $329 less in price — efficient cars in 1978 were the cheap ones. The interaction is 78.9 with a standard error of 112.5 and $p = 0.485$, so the mpg slope for imports ($-329.3 + 78.9 = -250.4$) is not distinguishable from the domestic slope.

The graph shows predicted price for foreign and domestic cars across the range of mpg. Both lines slope down, the import line sits above the domestic one, and because the interaction is small and insignificant the two are close to parallel. The visible gap between them is the $1,667 main effect, not evidence of a differing slope.

2. Interaction term in a non-linear model

2.1. The issue

In a linear model with interaction,

$$E(y) = \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2, \tag{2.1}$$

the cross partial derivative is simply the coefficient:

$$\frac{\partial^2 E(y)}{\partial x_1 \partial x_2} = \beta_{12}. \tag{2.2}$$

In a nonlinear model — logit, Poisson, or any link function F — the same coefficient β_{12} sits on the *link* scale:

$$F(E(y)) = \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2. \tag{2.3}$$

The cross partial of $F(E(y))$ is still β_{12}. But the cross partial of $E(y)$ itself — the quantity most researchers want — is not:

$$\frac{\partial^2 E(y)}{\partial x_1 \partial x_2} = \beta_{12} G'(\eta) + (\beta_1 + \beta_{12} x_2)(\beta_2 + \beta_{12} x_1) G''(\eta), \tag{2.4}$$

where $G = F^{-1}$ and η is the linear predictor. This is the point of Ai and Norton (2003): the probability-scale interaction effect depends on every coefficient, varies across the covariate space, and can differ in sign and significance from β_{12}.

The deeper issue is that nonlinearity alone generates a nonzero cross partial — even without an interaction term. Drop β_{12} and the cross partial on $E(y)$ is still $\beta_1 \beta_2 G''(\eta) \neq 0$ whenever the link is curved. The marginal effect of x_1 depends on x_2 in any nonlinear model, interaction or not.

Greene (*Econometric Analysis*, 8th ed., sec. 17.3) and Buis (2010) draw the same conclusion: the model is specified on the link scale, and that is where coefficients and their tests are interpretable. In a logit, β_{12} is the interaction on log-odds; exponentiating gives an odds-ratio interaction that is multiplicative. Partial effects on the probability scale are *implications* of the model, not parameters of it — they should be computed from the fitted model (via `margins` or `marginaleffects`), not read off the coefficient table.

2.2. A worked example

Let's illustrate the Ai-Norton point directly: in a logit model with an interaction term, the cross partial derivative $\frac{\partial^2 E(y)}{\partial x_1 \partial x_2}$ (what we actually care about) is not the same object as the estimated coefficient β_{12} on $x_1 x_2$ in the linear index. We simulate data with a true interaction effect on the log-odds scale, fit a logit model, and then use the `marginaleffects` package for two distinct things: first the marginal effect of x_1 on the probability scale at several values of x_2, and then the cross partial derivative itself.

We draw $n = 2000$ observations with $x_1, x_2 \sim N(0,1)$, independent of each other. The linear index is $\eta = -0.3 + 0.8x_1 + 0.5x_2 + 0.6\,x_1 x_2$, the success probability is $p = \Lambda(\eta)$, and $y \sim \text{Bernoulli}(p)$. So the true interaction on the log-odds scale is $\beta_{12} = 0.6$, a single constant — which is exactly the object the coefficient table estimates, and exactly the object that does *not* transfer to the probability scale.

```r
library(dplyr)
library(marginaleffects)

set.seed(42)
n <- 2000
x1 <- rnorm(n)
x2 <- rnorm(n)
eta <- -0.3 + 0.8 * x1 + 0.5 * x2 + 0.6 * x1 * x2
p <- plogis(eta)
y <- rbinom(n, 1, p)
dat <- data.frame(y, x1, x2)

m <- glm(y ~ x1 * x2, data = dat, family = binomial)
summary(m)$coefficients
```

```
              Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -0.2348658 0.05017155 -4.681255 2.851237e-06
x1           0.7792602 0.05930028 13.140920 1.918795e-39
x2           0.5296149 0.05537644  9.563903 1.133977e-21
x1:x2        0.6624955 0.06403580 10.345705 4.376826e-25
```

The coefficient on `x1:x2` is 0.662 with a standard error of 0.064, against a true 0.6 — recovered to within one standard error, as it should be, since the logit is correctly specified. The other three are 0.779, 0.530 and −0.235 against true values of 0.8, 0.5 and −0.3.

That coefficient is $\hat{\beta}_{12}$, the interaction effect on the log-odds (linear index) scale — constant by construction. On the *probability* scale nothing is constant. Start with the marginal effect of x_1 itself, evaluated at three values of x_2:

```r
slopes(m, variables = "x1", by = "x2", newdata = datagrid(x2 = c(-1, 0, 1)))
```

```
 x2 Estimate Std. Error     z Pr(>|z|)     S    2.5 % 97.5 %
 -1   0.0253     0.0159  1.59    0.112   3.2 -0.00591 0.0565
  0   0.1919     0.0147 13.09   <0.001 127.6  0.16314 0.2206
  1   0.3538     0.0243 14.59   <0.001 157.7  0.30630 0.4014

Term: x1
Type: response
Comparison: dY/dX
```

The marginal effect of x_1 on $P(y = 1)$ is 0.025 at $x_2 = -1$, 0.192 at $x_2 = 0$, and 0.354 at $x_2 = 1$ — a fourteen-fold change across the three values, where the log-odds interaction is one fixed number. Notice too that at $x_2 = -1$ the effect is not significant ($p = 0.112$) even though $\hat{\beta}_{12}$ is significant at better than 10^{-24}. Already the probability-scale answer and the coefficient tell different stories.

The cross partial derivative is a *further* step: not the effect of x_1 at a given x_2, but how much that effect changes as x_2 moves. That is a difference of slopes, so it needs a hypothesis test on top of the `by` argument rather than a second call to `slopes()`:

```r
slopes(m, variables = "x1", by = "x2",
       newdata = datagrid(x2 = c(-1, 1)),
       hypothesis = "b2 - b1 = 0")
```

```
 Hypothesis Estimate Std. Error  z Pr(>|z|)    S 2.5 % 97.5 %
    b2-b1=0    0.329     0.0299 11   <0.001 90.7  0.27  0.387

Type: response
```

The cross partial is 0.329 with a standard error of 0.030, $z = 11$ — that is the difference between the slope at $x_2 = 1$ and the slope at $x_2 = -1$, $0.354 - 0.025$, with a proper standard error attached. It is large and highly significant, which is worth being honest about: in *this* example it agrees with $\hat{\beta}_{12}$ in significance, differing only in scale and magnitude. Note that it is nowhere near 0.662; the two numbers live on different scales and there is no reason to expect them to match. Ai and Norton's warning is that the two quantities *need not* agree — not that they never do. The dependable lesson is the one visible in the table above: on the probability scale the answer depends on where you evaluate it, so a single coefficient cannot summarise it.

3. Chow test and more

3.1. Chow test

To compare a coefficient across two subsamples, interact the subsample indicator with the covariates and run one pooled regression. If only the treatment is interacted, all other coefficients are constrained equal across subsamples; interacting everything relaxes that. This is the Chow test. (The overlapping-samples extension below follows Austin Nichols's Statalist code.)

The data are Stata's **nlsw88**, an extract from the 1988 National Longitudinal Survey of Young Women. With non-missing wage and hours, 938 of the women live in the South and 1,304 elsewhere, 2,242 in all. We regress hourly **wage** on weekly **hours**, and ask whether the slope differs by region. The chunk runs that four ways — two separate regressions, **suest**, and the pooled interacted regression — so the equivalences are visible side by side.

```
est clear
sysuse nlsw88, clear
reg wage hours if south
est sto south
reg wage hours if !south
est sto nonsouth
suest south nonsouth
est sto suest
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
reg wage south hours?
est sto chow
test _b[hours1]-_b[hours2]=0
esttab south nonsouth suest chow, nogaps mti
```

```
(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =       938
-------------+----------------------------------   F(1, 936)       =     12.47
       Model |  344.732583         1  344.732583   Prob > F        =    0.0004
    Residual |  25866.3404       936   27.634979   R-squared       =    0.0132
-------------+----------------------------------   Adj R-squared   =    0.0121
       Total |  26211.0729       937   27.973397   Root MSE        =    5.2569

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
       hours |   .0623497   .0176532     3.53   0.000     .0277053    .0969941
       _cons |   4.520583   .6957145     6.50   0.000     3.155242    5.885923
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     1,304
-------------+----------------------------------   F(1, 1302)      =     55.93
       Model |  1929.41943         1  1929.41943   Prob > F        =    0.0000
    Residual |  44919.1023     1,302  34.5000785   R-squared       =    0.0412
-------------+----------------------------------   Adj R-squared   =    0.0404
       Total |  46848.5217     1,303  35.9543528   Root MSE        =    5.8737

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .1107536     .01481     7.48   0.000     .0816995    .1398076
       _cons |   4.357811    .564756     7.72   0.000      3.24988    5.465743
------------------------------------------------------------------------------


Simultaneous results for south, nonsouth                 Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
       hours |   .0623497   .0174426     3.57   0.000     .0281629    .0965366
       _cons |   4.520583   .6599379     6.85   0.000     3.227128    5.814037
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   3.319082   .1435305    23.12   0.000     3.037768    3.600397
-------------+----------------------------------------------------------------
nonsouth_m~n |
       hours |   .1107536   .0134936     8.21   0.000     .0843066    .1372006
       _cons |   4.357811   .4633326     9.41   0.000     3.449696    5.265927
-------------+----------------------------------------------------------------
nonsouth_l~r |
       _cons |   3.540962   .1006119    35.19   0.000     3.343766    3.738157
------------------------------------------------------------------------------


(4 missing values generated)

(4 missing values generated)

      Source |       SS           df       MS      Number of obs   =     2,242
```

```
-------------+----------------------------------   F(3, 2238)      =     36.91
       Model |  3502.37892         3  1167.45964   Prob > F        =    0.0000
    Residual |  70785.4426     2,238  31.6288841   R-squared       =    0.0471
-------------+----------------------------------   Adj R-squared   =    0.0459
       Total |  74287.8215     2,241  33.1494072   Root MSE        =     5.624

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       south |   .1627711   .9199871     0.18   0.860    -1.641346    1.966888
      hours1 |   .0623497   .0188858     3.30   0.001     .0253142    .0993852
      hours2 |   .1107536   .0141803     7.81   0.000     .0829456    .1385616
       _cons |   4.357811   .5407453     8.06   0.000     3.297397    5.418226
------------------------------------------------------------------------------


 ( 1)  hours1 - hours2 = 0

       F(  1,  2238) =    4.20
            Prob > F =    0.0405


----------------------------------------------------------------------------
                      (1)             (2)             (3)             (4)
                    south        nonsouth           suest            chow
----------------------------------------------------------------------------
main
hours              0.0623***        0.111***       0.0623***
                   (3.53)          (7.48)          (3.57)
south                                                                 0.163
                                                                     (0.18)
hours1                                                               0.0623***
                                                                     (3.30)
hours2                                                                0.111***
                                                                     (7.81)
_cons               4.521***        4.358***        4.521***        4.358***
                   (6.50)          (7.72)          (6.85)          (8.06)
----------------------------------------------------------------------------
south_lnvar
_cons                                               3.319***
                                                  (23.12)
----------------------------------------------------------------------------
nonsouth_m~n
hours                                               0.111***
                                                   (8.21)
_cons                                               4.358***
                                                   (9.41)
```

```
----------------------------------------------------------------------------
nonsouth_l~r
_cons                                                    3.541***
                                                        (35.19)
----------------------------------------------------------------------------
N                      938            1304             2242             2242
----------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The separate regressions give an hours slope of 0.0623 in the South and 0.1108 elsewhere. The pooled regression with `hours1` and `hours2` returns 0.0623497 and 0.1107536 — the *same numbers to seven digits*, not merely similar ones. That is algebra, not luck: a fully interacted regression is the two subsample regressions written as one.

What does differ is the standard errors. The South slope carries 0.0177 in its own regression and 0.0189 in the pooled one, because the pooled model estimates a single error variance across both groups while the separate regressions each get their own (root MSE 5.26 in the South, 5.87 elsewhere). If that pooling is unattractive, `suest` avoids it: it stacks the two stored estimates with their joint covariance and reports robust standard errors, 0.0174 and 0.0135 here.

`test _b[hours1]-_b[hours2]=0` is the Chow test, and it gives $F(1, 2238) = 4.20$ with $p = 0.0405$. The returns to hours are steeper outside the South, and the gap of 0.048 dollars per hour is just significant at 5%. Note that the `south` dummy itself is 0.163 with $p = 0.860$ — the two regions differ in slope, not in intercept, which is precisely the pattern a treatment-only interaction would have missed. Stata's `suest` does the same job by stacking the two stored estimates and their covariance structure; the interaction approach is more flexible since it works with any estimator.

3.2. A comparison with two different outcomes

The same idea compares a coefficient across two different outcomes. Stack the two outcome variables into a long format, interact the covariates with the stacking indicator, and run one regression.

The chunk below does that in seven steps, and it is worth reading them in order because the payoff is the last two columns of the table agreeing.

1. `reg south wage hours` and `reg smsa wage hours`, each stored with `est sto`. These are the two equations we want to compare, fitted separately.
2. `suest south smsa` refits them jointly and stacks the two coefficient vectors into one, with a robust covariance matrix across equations. This is the benchmark: it is the standard way to test a cross-equation hypothesis.
3. `preserve`, because the next steps reshape the data and we want it back.
4. `gen Y1=south`, `gen Y2=smsa`, `gen id=_n`. Copy the two outcomes into a numbered pair and give every woman an id.
5. `reshape long Y, i(id) j(subsample)` turns each woman's one row into two: one carrying `Y = south` with `subsample == 1`, one carrying `Y = smsa` with `subsample == 2`. The stacked outcome is now a single variable `Y`.

6. `gen wage1=wage*(subsample==1)` and its three siblings. Each covariate is split into the part that acts in equation 1 and the part that acts in equation 2. A single regression on `wage1 wage2 hours1 hours2` therefore estimates a separate slope per equation, which is what the two separate regressions did.
7. `reg Y wage? hours? subsample, cluster(id)` fits it, and `test _b[wage1]-_b[wage2]=0` is the cross-equation test. The `?` is a Stata wildcard, so `wage?` expands to `wage1 wage2`.

The whole construction exists so that a hypothesis *across* two equations becomes a hypothesis *within* one, where `test` can reach both coefficients.

```
est clear
sysuse nlsw88, clear
reg south wage hours
est sto south
reg smsa wage hours
est sto smsa
suest south smsa
est sto suest
preserve
gen Y1=south
gen Y2=smsa
gen id=_n
reshape long Y, i(id) j(subsample)
gen wage1=wage*(subsample==1)
gen wage2=wage*(subsample==2)
gen hours1=hours*(subsample==1)
gen hours2=hours*(subsample==2)
* Each individual contributes TWO rows here (one per stacked outcome), so the
* errors are correlated within id.  Clustering on the stacking id is what makes
* the cross-equation test comparable to suest -- see the note below.
reg Y wage? hours? subsample, cluster(id)
test _b[wage1]-_b[wage2]=0
est sto stacked
* Match the preserve above, so the reshape does not leak into later chunks.
restore
* Note: suest stores coefficients under equation-qualified names (e.g.
* south:hours), while the single-sample models store the same
* coefficient simply as "hours"; esttab matches by exact coefficient
* name, so these land on separate rows instead of being aligned. Use
* esttab's coeflabel()/eqlabel() options, or rename coefficients on the
* stored estimates first, to get a directly comparable table.
esttab south smsa suest stacked, nogaps mti
```

```
(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =     2,242
```

```
-------------+----------------------------------   F(2, 2239)      =     30.60
       Model |  14.513685         2  7.25684251   Prob > F        =    0.0000
    Residual |  531.049205     2,239  .237181423   R-squared       =    0.0266
-------------+----------------------------------   Adj R-squared   =    0.0257
       Total |  545.56289     2,241   .24344618   Root MSE        =    .48701

------------------------------------------------------------------------------
       south | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |  -.0124053   .0018099    -6.85   0.000    -.0159545    -.008856
       hours |   .0047722   .0009916     4.81   0.000     .0028277    .0067167
       _cons |   .3372108   .0387354     8.71   0.000     .2612497    .4131718
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     2,242
-------------+----------------------------------   F(2, 2239)      =     36.17
       Model |  14.6274602         2  7.31373012   Prob > F        =    0.0000
    Residual |  452.719551     2,239  .202197209   R-squared       =    0.0313
-------------+----------------------------------   Adj R-squared   =    0.0304
       Total |  467.347012     2,241  .208543959   Root MSE        =    .44966

------------------------------------------------------------------------------
        smsa | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |   .0139103   .0016711     8.32   0.000     .0106332    .0171873
       hours |   .0003663   .0009155     0.40   0.689    -.0014291    .0021616
       _cons |   .5820585   .0357648    16.27   0.000      .511923    .6521941
------------------------------------------------------------------------------


Simultaneous results for south, smsa                     Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
        wage |  -.0124053   .0019287    -6.43   0.000    -.0161854   -.0086251
       hours |   .0047722   .0009825     4.86   0.000     .0028465    .0066978
       _cons |   .3372108   .0379918     8.88   0.000     .2627483    .4116733
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   -1.43893   .0096645  -148.89   0.000    -1.457872   -1.419988
-------------+----------------------------------------------------------------
smsa_mean    |
        wage |   .0139103   .0018313     7.60   0.000      .010321    .0174996
       hours |   .0003663   .0009099     0.40   0.687    -.0014172    .0021497
```

```
       _cons |   .5820585   .0361037    16.12   0.000     .5112965    .6528205
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |  -1.598512   .0195103   -81.93   0.000    -1.636751   -1.560272
------------------------------------------------------------------------------
```

```
(j = 1 2)

Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations            2,246   ->   4,492
Number of variables                  23   ->   23
j variable (2 values)                     ->   subsample
xij variables:
                                  Y1 Y2   ->   Y
-----------------------------------------------------------------------------
```

```
(8 missing values generated)

(8 missing values generated)
```

```
Linear regression                               Number of obs     =      4,484
                                                F(5, 2241)        =      96.34
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1091
                                                Root MSE          =     .46871

                                 (Std. err. adjusted for 2,242 clusters in id)
------------------------------------------------------------------------------
             |               Robust
           Y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       wage1 |  -.0124053   .0019298    -6.43   0.000    -.0161896    -.008621
       wage2 |   .0139103   .0018323     7.59   0.000      .010317    .0175035
      hours1 |   .0047722    .000983     4.85   0.000     .0028444    .0066999
      hours2 |   .0003663   .0009104     0.40   0.688    -.0014191    .0021517
   subsample |   .2448477   .0548754     4.46   0.000     .1372358    .3524597
       _cons |    .092363   .0872219     1.06   0.290    -.0786812    .2634072
------------------------------------------------------------------------------
```

```
 ( 1)  wage1 - wage2 = 0

       F(  1,  2241) =   72.14
            Prob > F =    0.0000
```

	(1) south	(2) smsa	(3) suest	(4) stacked
main				
wage	-0.0124***	0.0139***	-0.0124***	
	(-6.85)	(8.32)	(-6.43)	
hours	0.00477***	0.000366	0.00477***	
	(4.81)	(0.40)	(4.86)	
wage1				-0.0124***
				(-6.43)
wage2				0.0139***
				(7.59)
hours1				0.00477***
				(4.85)
hours2				0.000366
				(0.40)
subsample				0.245***
				(4.46)
_cons	0.337***	0.582***	0.337***	0.0924
	(8.71)	(16.27)	(8.88)	(1.06)
south_lnvar				
_cons			-1.439***	
			(-148.89)	
smsa_mean				
wage			0.0139***	
			(7.60)	
hours			0.000366	
			(0.40)	
_cons			0.582***	
			(16.12)	
smsa_lnvar				
_cons			-1.599***	
			(-81.93)	
N	2242	2242	2242	4484

```
----------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

Read the table by columns. The first two are the separate regressions, the third is `suest`, and the fourth is the stacked regression. The `wage1` coefficient in column four equals the `wage` coefficient in column one, and `wage2` equals the `wage` coefficient in column two: stacking has reproduced both single-equation fits exactly, in one regression, with both coefficients now addressable by `test`. Note the sample size, 4,484 in the stacked column against 2,242 in the others — the reshape doubled the rows, which is the next point.

One detail matters a great deal here, and it is easy to skip. Stacking puts **two rows per person** in the data, so the two error terms belonging to the same person are correlated by construction. The cross-equation test is precisely a comparison *across* those two rows, so its standard error has to account for that correlation. Clustering on the stacking id does it, and it is not a cosmetic adjustment:

	test of the wage coefficient across equations
`suest`	$\chi^2(1) = 72.22$
stacked, **no** clustering	$F(1, 4478) = 114.12$
stacked, `cluster(id)`	$F(1, 2241) = 72.14$

Without clustering the statistic is inflated by more than half. With it, the stacked test reproduces `suest` almost exactly – the small remaining gap is just the F-versus-χ^2 finite-sample adjustment. Note that `robust` alone will *not* fix this; the cluster variable has to be the id you stacked on.

3.3. Overlapping samples

The same method extends to overlapping subsamples. Here south and smsa overlap — some observations belong to both — so a simple split is not possible.

The reshape trick from the previous section will not work here, because a woman who is both southern and in an SMSA has to appear in both subsamples rather than be assigned to one. `expand` handles that instead:

1. `ta south smsa` shows the overlap, which is the reason for what follows.
2. `preserve`, then `expand 2` duplicates every row, so each woman now has two copies.
3. `bys idcode: g n=_n` numbers a woman's two copies 1 and 2.
4. `keep if (n==1&south)|(n==2&smsa)` keeps copy 1 only if she is southern and copy 2 only if she is in an SMSA. A woman in both keeps both copies; a woman in neither drops out; a woman in exactly one keeps one. Each surviving row now represents one woman's membership in one subsample.
5. `g hours1=hours*(n==1)` and `g hours2=hours*(n==2)` split the covariate by subsample, as before. Within this restricted sample `n==1` means southern and `n==2` means SMSA, so the copy number *is* the subsample indicator.

6. `reg wage hours? n, cl(idcode)` fits both slopes at once, clustering on the woman rather than on a stacking id, since a woman can now contribute one row or two.

```
est clear
sysuse nlsw88, clear
ta south smsa
reg wage hours if south
est sto south
reg wage hours if smsa
est sto smsa
suest south smsa
est sto suest
preserve
expand 2
bys idcode: g n=_n
keep if (n==1&south)|(n==2&smsa)
* hours1 is hours in the south subsample (n==1), hours2 is hours in the
* smsa subsample (n==2); within this restricted sample n==1 <=> south==1
* and n==2 <=> smsa==1, so this is equivalent to but clearer than negating
* the complementary condition.
g hours1=hours*(n==1)
g hours2=hours*(n==2)
reg wage hours? n, cl(idcode)
est sto stacked
restore
esttab south smsa suest stacked, nogaps mti
```

```
(NLSW, 1988 extract)


  Lives in |     Lives in SMSA
 the south |  Not SMSA       SMSA |     Total
-----------+----------------------+----------
 Not south |       308        996 |     1,304 
     South |       357        585 |       942 
-----------+----------------------+----------
     Total |       665      1,581 |     2,246 

      Source |       SS           df       MS      Number of obs   =       938
-------------+----------------------------------   F(1, 936)       =     12.47
       Model |  344.732583         1  344.732583   Prob > F        =    0.0004
    Residual |  25866.3404       936   27.634979   R-squared       =    0.0132
-------------+----------------------------------   Adj R-squared   =    0.0121
       Total |  26211.0729       937   27.973397   Root MSE        =    5.2569

------------------------------------------------------------------------------
```

```
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .0623497   .0176532     3.53   0.000     .0277053    .0969941
       _cons |   4.520583   .6957145     6.50   0.000     3.155242    5.885923
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     1,578
-------------+----------------------------------   F(1, 1576)      =     46.48
       Model |  1594.14881         1  1594.14881   Prob > F        =    0.0000
    Residual |  54048.2539     1,576  34.2945773   R-squared       =    0.0286
-------------+----------------------------------   Adj R-squared   =    0.0280
       Total |  55642.4027     1,577  35.2837049   Root MSE        =    5.8562

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .0953636   .0139872     6.82   0.000     .0679281    .1227991
       _cons |   4.861519   .5443826     8.93   0.000     3.793729    5.929309
------------------------------------------------------------------------------


Simultaneous results for south, smsa                     Number of obs = 1,934

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
       hours |   .0623497   .0174432     3.57   0.000     .0281617    .0965378
       _cons |   4.520583   .6599613     6.85   0.000     3.227082    5.814083
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   3.319082   .1435356    23.12   0.000     3.037758    3.600407
-------------+----------------------------------------------------------------
smsa_mean    |
       hours |   .0953636   .0132806     7.18   0.000      .069334    .1213931
       _cons |   4.861519   .4842914    10.04   0.000     3.912325    5.810713
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |   3.534987   .0910825    38.81   0.000     3.356469    3.713506
------------------------------------------------------------------------------


(2,246 observations created)
```

```
(1,969 observations deleted)

(7 missing values generated)

(7 missing values generated)


Linear regression                               Number of obs     =      2,516
                                                F(3, 1933)        =      40.81
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0399
                                                Root MSE          =     5.6403

                            (Std. err. adjusted for 1,934 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   .0623497   .0174536     3.57   0.000     .0281198    .0965796
      hours2 |   .0953636   .0132886     7.18   0.000     .0693022     .121425
           n |   .3409364   .6124145     0.56   0.578    -.8601261    1.541999
       _cons |   4.179646   1.177889     3.55   0.000     1.869579    6.489713
------------------------------------------------------------------------------



----------------------------------------------------------------------------
                      (1)             (2)             (3)             (4)
                    south            smsa           suest         stacked
----------------------------------------------------------------------------
main
hours              0.0623***       0.0954***       0.0623***
                   (3.53)          (6.82)          (3.57)
hours1                                                             0.0623***
                                                                   (3.57)
hours2                                                             0.0954***
                                                                   (7.18)
n                                                                   0.341
                                                                   (0.56)
_cons               4.521***        4.862***        4.521***        4.180***
                   (6.50)          (8.93)          (6.85)          (3.55)
----------------------------------------------------------------------------
south_lnvar
_cons                                               3.319***
                                                  (23.12)
----------------------------------------------------------------------------
smsa_mean
```

```
hours                                                0.0954***
                                                     (7.18)
_cons                                                 4.862***
                                                    (10.04)
----------------------------------------------------------------------------
smsa_lnvar
_cons                                                 3.535***
                                                    (38.81)
----------------------------------------------------------------------------
N                      938            1578            1934            2516
----------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The cross-tab shows the overlap: of the 1,304 non-South women, 996 live in an SMSA, so the two subsamples share members and cannot be produced by a simple split. Expanding the data and keeping each observation under whichever subsample it belongs to handles that. The pooled `hours1` is 0.0623497, again reproducing the South regression exactly, and `hours2` is 0.0954 for the SMSA subsample. The `n` indicator is 0.341 with $p = 0.578$.

3.4. IV regression

The interaction approach works with IV. Interact both the endogenous variable and the instrument with the subsample indicator. Here `hours` is instrumented by `union` within each region.

The point is not to get a different estimate. The two interacted 2SLS coefficients reproduce the two subsample IV regressions exactly, just as they did for OLS. The point is that running them separately leaves the two estimates in different `ivregress` results with no joint covariance between them, so there is nothing to test the difference *with*. Interacting puts both in one estimation, so `test _b[hours1]-_b[hours2]=0` can reach both coefficients and their covariance. That is the whole reason for the construction, and it is the same reason as in the OLS case.

Two details specific to IV. The instrument must be split alongside the endogenous variable — `union1` and `union2`, not just `hours1` and `hours2` — because an instrument that is not subsample-specific would identify a single pooled first stage and defeat the purpose. And the number of instruments must still match the number of endogenous regressors: two of each here, so the system is exactly identified, as each subsample regression was on its own.

```
sysuse nlsw88, clear
ivregress 2sls wage (hours=union) if south
ivregress 2sls wage (hours=union) if !south
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
```

```
ivregress 2sls wage south (hours1 hours2 = union1 union2)
test _b[hours1]-_b[hours2]=0
```

```
(NLSW, 1988 extract)


Instrumental-variables 2SLS regression            Number of obs   =        798
                                                  Wald chi2(1)    =       2.61
                                                  Prob > chi2     =     0.1060
                                                  Root MSE        =     9.5807

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |     .97819   .6050822     1.62   0.106    -.2077492    2.164129
       _cons |  -30.96026   23.32846    -1.33   0.184     -76.6832    14.76268
------------------------------------------------------------------------------
Endogenous: hours
Exogenous:  union


Instrumental-variables 2SLS regression            Number of obs   =      1,079
                                                  Wald chi2(1)    =       6.03
                                                  Prob > chi2     =     0.0140
                                                  Root MSE        =     7.0372

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .6255843   .2546731     2.46   0.014     .1264342    1.124735
       _cons |   -14.9159   9.401508    -1.59   0.113    -33.34252    3.510714
------------------------------------------------------------------------------
Endogenous: hours
Exogenous:  union

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)


Instrumental-variables 2SLS regression            Number of obs   =      1,877
                                                  Wald chi2(3)    =      21.75
                                                  Prob > chi2     =     0.0001
```

```
                                                  Root MSE        =     8.2154

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |     .97819   .5188511     1.89   0.059    -.0387394    1.995119
      hours2 |   .6255843    .297311     2.10   0.035     .0428654    1.208303
       south |  -16.04435   22.81705    -0.70   0.482    -60.76495    28.67625
       _cons |   -14.9159   10.97553    -1.36   0.174    -36.42754    6.595735
------------------------------------------------------------------------------
Endogenous: hours1 hours2
Exogenous:  south union1 union2


 ( 1)  hours1 - hours2 = 0

           chi2(  1) =    0.35
         Prob > chi2 =    0.5554
```

The IV estimates are 0.978 in the South and 0.626 elsewhere, against OLS values of 0.062 and 0.111 — an order of magnitude larger, with standard errors to match (0.519 and 0.297). And the Chow test now gives $\chi^2(1) = 0.35$ with $p = 0.5554$, where the OLS version rejected at $p = 0.0405$. Instrumenting has not overturned the difference between regions; it has made the data too imprecise to detect it. The lesson is about the test, not the subsamples: a Chow test inherits the precision of whatever estimator it is built on.

3.5. IV with fixed effects

With fixed effects the pooled regression needs the fixed effects themselves interacted with the subsample indicator. The first attempt below omits this and fails to reproduce the separate estimates:

> ⚠ Warning
>
> The examples below cluster standard errors on `race`, which in `nlsw88` has only three categories (White, Black, Other). Cluster-robust standard errors rely on asymptotics in the number of clusters $G \to \infty$; with $G = 3$ they are severely downward-biased, inflating t-statistics and false-positive rates. These chunks use `race` purely to illustrate the fixed-effect/IV mechanics — in an actual analysis, cluster on a variable with many more categories (e.g., `industry` or `occupation`), or on the true level of treatment assignment/sampling design.

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
```

```
gen union2=union*(south==0)
ivreghdfe wage (hours=union) if south, a(race) cluster(race)
ivreghdfe wage (hours=union) if !south, a(race) cluster(race)
ivreghdfe wage south (hours1 hours2 = union1 union2) , a(race) cluster(race)
```

```
(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =      798
                                                      F(  1,     2) =    69.28
                                                      Prob > F      =   0.0141
Total (centered) SS     =   12186.8806                Centered R2   =  -6.4527
Total (uncentered) SS   =   12186.8806                Uncentered R2 =  -6.4527
Residual SS             =  90825.44631                Root MSE      =    10.68

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   1.107099    .133012     8.32   0.014     .5347947    1.679404
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.756
                                                   Chi-sq(1) P-val =    0.1852
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.233
                         (Kleibergen-Paap rk Wald F statistic):         13.977
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
```

```
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =     1079
                                                      F(  1,     2) =     1.41
                                                      Prob > F      =   0.3564
Total (centered) SS     =  19086.22115                Centered R2   =  -2.3302
Total (uncentered) SS   =  19086.22115                Uncentered R2 =  -2.3302
Residual SS             =  63560.97528                Root MSE      =    7.682

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .7023322   .5905772     1.19   0.356    -1.838717    3.243381
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.789
                                                Chi-sq(1) P-val =    0.3745
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                5.501
                         (Kleibergen-Paap rk Wald F statistic):          3.772
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
```

```
                                          20% maximal IV size              6.66
                                          25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =     1877
                                                      F(  3,     2) =  1036.24
                                                      Prob > F      =   0.0010
Total (centered) SS     =    32142.319                Centered R2   =  -3.9041
Total (uncentered) SS   =    32142.319                Uncentered R2 =  -3.9041
Residual SS             =  157627.5718                Root MSE      =    9.174

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.142036   .0650507    17.56   0.003     .8621454    1.421927
      hours2 |   .6880691   .5265185     1.31   0.321    -1.577357    2.953495
       south |  -19.74773   22.10811    -0.89   0.466    -114.8712    75.37577
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.793
```

```
                                                          Chi-sq(1) P-val =    0.1806
-------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                 3.584
                         (Kleibergen-Paap rk Wald F statistic):           8.484
Stock-Yogo weak ID test critical values: 10% maximal IV size               7.03
                                         15% maximal IV size               4.58
                                         20% maximal IV size               3.95
                                         25% maximal IV size               3.63
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
-------------------------------------------------------------------------------
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.
-------------------------------------------------------------------------------
Instrumented:         hours1 hours2
Included instruments: south
Excluded instruments: union1 union2
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
-------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |          3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

The separate regressions give 1.1071 and 0.7023. The pooled regression gives 1.1420 and 0.6881 — close, but not equal, and that is the point. The pooled coefficients do not match the separate regressions because the fixed effects are shared: one set of race effects is estimated off both subsamples at once, so neither subsample gets the within-group demeaning its own regression used. Interacting them with the subsample indicator fixes it:

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
```

```
gen race1=race*(south==1)
gen race2=race*(south==0)


ivreghdfe wage (hours=union) if south, a(race) cluster(race)
ivreghdfe wage (hours=union) if !south, a(race) cluster(race)
ivregress 2sls wage south (hours1 hours2 = union1 union2) i.race1 i.race2, cluster(race)
```

```
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.


(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)



(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =      798
                                                      F(  1,     2) =    69.28
                                                      Prob > F      =   0.0141
Total (centered) SS     =   12186.8806                Centered R2   =  -6.4527
Total (uncentered) SS   =   12186.8806                Uncentered R2 =  -6.4527
Residual SS             =  90825.44631                Root MSE      =    10.68

------------------------------------------------------------------------------
             |               Robust
```

```
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   1.107099    .133012     8.32   0.014     .5347947    1.679404
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.756
                                                   Chi-sq(1) P-val =    0.1852
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.233
                         (Kleibergen-Paap rk Wald F statistic):         13.977
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =     1079
                                                      F(  1,     2) =     1.41
                                                      Prob > F      =   0.3564
Total (centered) SS     =  19086.22115                Centered R2   =  -2.3302
Total (uncentered) SS   =  19086.22115                Uncentered R2 =  -2.3302
```

```
Residual SS             =  63560.97528                Root MSE      =    7.682

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .7023322   .5905772     1.19   0.356    -1.838717    3.243381
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.789
                                                   Chi-sq(1) P-val =    0.3745
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                5.501
                         (Kleibergen-Paap rk Wald F statistic):          3.772
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

note: 3.race1 omitted because of collinearity.
note: 3.race2 omitted because of collinearity.

Instrumental-variables 2SLS regression            Number of obs   =      1,877
                                                  Wald chi2(7)    =  101696.08
                                                  Prob > chi2     =     0.0000
                                                  Root MSE        =     9.0693

                                           (Std. err. adjusted for 3 clusters in race)
```

```
------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.107099   .1085357    10.20   0.000     .8943732    1.319825
      hours2 |   .7023322   .4819806     1.46   0.145    -.2423324    1.646997
       south |  -21.52299   22.35225    -0.96   0.336    -65.33259    22.28662
             |
       race1 |
          1  |   3.054296   .1451752    21.04   0.000     2.769758    3.338834
          2  |   1.987268    .176538    11.26   0.000      1.64126    2.333277
          3  |          0  (omitted)
             |
       race2 |
          1  |  -.4816127    .434723    -1.11   0.268    -1.333654    .3704287
          2  |  -2.065527   .7694573    -2.68   0.007    -3.573635   -.5574181
          3  |          0  (omitted)
             |
       _cons |  -17.01633   18.01689    -0.94   0.345    -52.32879    18.29614
------------------------------------------------------------------------------
Endogenous: hours1 hours2
Exogenous:  south 1.race1 2.race1 1.race2 2.race2 union1 union2
```

This reproduces the separate estimates exactly: 1.107099 and 0.7023322 in the pooled regression against 1.107099 and 0.7023322 separately. The approach builds explicit dummies from the interaction of the fixed-effect variable and the subsample indicator, which becomes impractical when the number of fixed-effect levels is large.

One caution about how those dummies were built. `race1 = race*(south==1)` multiplies a *categorical code* by an indicator, so non-south observations are assigned `race1 = 0` and then `i.race1` treats 0 as its own category. That is safe here only because `race` has no zero level (it is coded 1/2/3). If the absorbed variable did have a 0 category, those observations and all the non-south ones would be silently collapsed into the same dummy. For a general recipe, build the interactions from the category dummies directly (`i.race#i.south`) rather than from the code.

Alternatively, pass the two interaction variables to `reghdfe`'s absorb option as a two-way fixed effect:

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
gen race1=race*(south==1)
gen race2=race*(south==0)

ivreghdfe wage south (hours1 hours2 = union1 union2) , a(race1 race2) cluster(race)
test _b[hours1]-_b[hours2]=0
```

```
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.


(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)



(MWFE estimator converged in 2 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =     1877
                                                      F(  2,     2) = 13010.62
                                                      Prob > F      =   0.0001
Total (centered) SS     =  31273.10174                Centered R2   =  -3.9367
Total (uncentered) SS   =  31273.10174                Uncentered R2 =  -3.9367
Residual SS             =  154386.4216                Root MSE      =    9.089

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.107099   .1331773     8.31   0.014     .5340838    1.680115
      hours2 |   .7023322   .5914076     1.19   0.357     -1.84229    3.246954
       south |          0  (omitted)
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.000
                                                   Chi-sq(1) P-val =    1.0000
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.793
```

```
                         (Kleibergen-Paap rk Wald F statistic):          0.000
Stock-Yogo weak ID test critical values: 10% maximal IV size             7.03
                                         15% maximal IV size             4.58
                                         20% maximal IV size             3.95
                                         25% maximal IV size             3.63
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.
------------------------------------------------------------------------------
Collinearities detected among instruments: 1 instrument(s) dropped
Instrumented:         hours1 hours2
Included instruments: south
Excluded instruments: union1 union2
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
       race1 |         4           0           4     |
       race2 |         4           2           2     |
-----------------------------------------------------+


 ( 1)  hours1 - hours2 = 0

       F(  1,     2) =    0.31
            Prob > F =    0.6325
```

Absorbing `race1` and `race2` as a two-way fixed effect gives the same 1.107099 and 0.7023322 without building the dummies by hand, and scales to many fixed-effect levels. The Chow test is $F(1,2) = 0.31$ with $p = 0.6325$ — but read that denominator: 2 degrees of freedom, because the standard errors are clustered on `race`, which has three categories. This is the problem the callout above flags, and it is why these IV/FE numbers illustrate mechanics rather than support any conclusion about the South.

3.6. Chow test with different covariates

The equations need not share the same covariates. If equation 1 includes X_2 and equation 2 does not,

$$Y_1 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon_1, \qquad Y_2 = \gamma_0 + \gamma_1 X_1 + \epsilon_2, \tag{3.1}$$

the second equation implicitly sets the coefficient on X_2 to zero. In the stacked regression, replace X_2 with a constant (e.g. 1) for the Y_2 subsample, so that its "coefficient" is absorbed into the intercept:

```
sysuse nlsw88, clear
reg south wage hours tenure
est sto south
reg smsa wage hours
est sto smsa
suest south smsa
est sto suest
preserve
gen Y1=south
gen Y2=smsa
gen id=_n
reshape long Y, i(id) j(subsample)
gen wage1=wage*(subsample==1)
gen wage2=wage*(subsample==2)
gen hours1=hours*(subsample==1)
gen hours2=hours*(subsample==2)
replace tenure=1 if subsample==2
gen tenure1= tenure*(subsample==1)
gen tenure2= tenure*(subsample==2)
reg Y wage? hours? tenure? subsample, cluster(id)
est sto chow
test _b[hours1]-_b[hours2]=0
esttab  suest chow, nogaps mti
restore
```

```
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
```

```
Possible causes:
        number of clusters insufficient to calculate robust covariance matrix
        singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.


(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =     2,227
-------------+----------------------------------   F(3, 2223)      =     20.30
       Model |  14.4507388         3  4.81691294   Prob > F        =    0.0000
    Residual |  527.507052     2,223   .23729512   R-squared       =    0.0267
-------------+----------------------------------   Adj R-squared   =    0.0254
       Total |  541.957791     2,226  .243467112   Root MSE        =    .48713

------------------------------------------------------------------------------
       south | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |  -.0119584   .0018359    -6.51   0.000    -.0155586   -.0083583
       hours |    .004828   .0010072     4.79   0.000     .0028528    .0068031
      tenure |  -.0024032   .0019221    -1.25   0.211    -.0061726    .0013661
       _cons |    .346361   .0392121     8.83   0.000     .2694648    .4232573
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     2,242
-------------+----------------------------------   F(2, 2239)      =     36.17
       Model |  14.6274602         2  7.31373012   Prob > F        =    0.0000
    Residual |  452.719551     2,239  .202197209   R-squared       =    0.0313
-------------+----------------------------------   Adj R-squared   =    0.0304
       Total |  467.347012     2,241  .208543959   Root MSE        =    .44966

------------------------------------------------------------------------------
        smsa | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |   .0139103   .0016711     8.32   0.000     .0106332    .0171873
       hours |   .0003663   .0009155     0.40   0.689    -.0014291    .0021616
       _cons |   .5820585   .0357648    16.27   0.000      .511923    .6521941
------------------------------------------------------------------------------


Simultaneous results for south, smsa                    Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
```

```
        wage |  -.0119584    .001953    -6.12   0.000    -.0157862   -.0081307
       hours |    .004828   .0009971     4.84   0.000     .0028737    .0067822
      tenure |  -.0024032   .0018928    -1.27   0.204    -.0061131    .0013066
       _cons |    .346361    .038473     9.00   0.000     .2709554    .4217667
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |  -1.438451   .0096359  -149.28   0.000    -1.457337   -1.419565
-------------+----------------------------------------------------------------
smsa_mean    |
        wage |   .0139103   .0018313     7.60   0.000      .010321    .0174996
       hours |   .0003663   .0009099     0.40   0.687    -.0014172    .0021497
       _cons |   .5820585   .0361037    16.12   0.000     .5112965    .6528205
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |  -1.598512   .0195103   -81.93   0.000    -1.636751   -1.560272
------------------------------------------------------------------------------
```

```
(j = 1 2)

Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations            2,246   ->   4,492
Number of variables                  23   ->   23
j variable (2 values)                     ->   subsample
xij variables:
                                  Y1 Y2   ->   Y
-----------------------------------------------------------------------------
```

```
(8 missing values generated)

(8 missing values generated)

(2,222 real changes made)

(15 missing values generated)

(15 missing values generated)

note: subsample omitted because of collinearity.

Linear regression                               Number of obs     =      4,469
```

```
                                                F(6, 2241)        =      81.10
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1091
                                                Root MSE          =      .4687

                                  (Std. err. adjusted for 2,242 clusters in id)
------------------------------------------------------------------------------
             |               Robust
           Y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       wage1 |  -.0119584   .0019543    -6.12   0.000    -.0157909    -.008126
       wage2 |   .0139103   .0018325     7.59   0.000     .0103166    .0175039
      hours1 |    .004828   .0009978     4.84   0.000     .0028713    .0067846
      hours2 |   .0003663   .0009105     0.40   0.688    -.0014193    .0021519
     tenure1 |  -.0024032   .0018941    -1.27   0.205    -.0061176    .0013111
     tenure2 |   .2356975   .0550137     4.28   0.000     .1278144    .3435806
   subsample |          0  (omitted)
       _cons |    .346361   .0384988     9.00   0.000      .270864    .4218581
------------------------------------------------------------------------------


 ( 1)  hours1 - hours2 = 0

       F(  1,  2241) =   10.03
            Prob > F =    0.0016


--------------------------------------------
                      (1)             (2)
                    suest            chow
--------------------------------------------
main
wage                -0.0120***
                    (-6.12)
hours               0.00483***
                     (4.84)
tenure             -0.00240
                    (-1.27)
wage1                               -0.0120***
                                    (-6.12)
wage2                                0.0139***
                                     (7.59)
hours1                              0.00483***
                                     (4.84)
hours2                             0.000366
                                     (0.40)
tenure1                            -0.00240
```

```
                                  (-1.27)
tenure2                             0.236***
                                   (4.28)
subsample                               0
                                      (.)
_cons                 0.346***      0.346***
                     (9.00)        (9.00)
--------------------------------------------
south_lnvar
_cons                -1.438***
                  (-149.28)
--------------------------------------------
smsa_mean
wage                 0.0139***
                     (7.60)
hours              0.000366
                     (0.40)
_cons                 0.582***
                    (16.12)
--------------------------------------------
smsa_lnvar
_cons                -1.599***
                   (-81.93)
--------------------------------------------
N                      2242          4469
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The stacked model reproduces `suest`'s point estimates exactly: the hours coefficient is 0.004828 in the `south` equation and 0.0003663 in the `smsa` one, in both. Here the covariates differ — `tenure` appears in the first equation only — and the trick of setting `tenure = 1` for the second subsample lets its coefficient be absorbed by that equation's intercept. The Chow test on the hours coefficient gives $F(1, 2241) = 10.03$ with $p = 0.0016$: hours predicts living in the South quite differently from how it predicts living in an SMSA. (The `esttab` columns do not line up because `suest` stores equation-qualified names like `south:wage` while the stacked model stores plain `wage`.)

3.7. Conclusion

`suest` and the stacking method both test cross-equation hypotheses with robust standard errors. `sureg` does the same under a GLS framework, but it assumes homoscedastic errors across equations; under misspecification it remains consistent but its standard errors are invalid. When the equations share covariates `sureg` and OLS give the same point estimates; with different covariates they diverge.

The stacking approach is the most portable. It works in any language, extends to IV and fixed-effect models, and requires only that the single-equation estimator works — no special post-estimation infrastructure.

One pattern runs through every example in this chapter, and it is worth stating once on its own. A Chow test is a hypothesis about two coefficients that live in two different regressions. Stata's `test` can only reach coefficients inside a single stored estimation result, and, more importantly, testing a difference requires the covariance between the two estimates, which separate regressions never produce. Every construction here — the reshape for two outcomes, the `expand` for overlapping samples, the split instruments under IV, the absorbed fixed effects — exists to move both coefficients into one estimation so that the covariance exists and `test` can use it. `suest` does the same thing by a different route, refitting the stored estimates jointly with a robust cross-equation covariance matrix.

So the choice among them is practical rather than statistical. Use `suest` when the estimator supports it and the samples are the same. Stack when the samples overlap, when the covariates differ across equations, or when the estimator is one `suest` will not take — which, in practice, means most fixed-effect and IV commands.

4. Weights in OLS in causal inference

4.1. Unit level weights for OLS

This chapter follows Chad Hazlett and Tanvi Shinkre (2024), "Demystifying and avoiding the OLS 'weighting problem': unmodeled heterogeneity and straightforward solutions," arXiv:2403.03299.

Under unconfoundedness, the identifying assumption holds at each level of X. When X is discrete and low-dimensional, we can stratify and compute the ATE by the g-formula. In practice we usually run a linear regression

$$Y = \alpha + \beta D + \gamma X + \epsilon \tag{4.1}$$

and treat $\hat{\beta}$ as the treatment effect, relying on linearity to span the covariate space. The question is what this estimator is actually weighting.

Suppose we are interested in the ATE of D on Y. Say we have discrete variables X. Under unconfoundedness, the ATE is

$$\begin{aligned}
\tau_{ATE} &= \sum E[Y(1) - Y(0)|X = x]P(X = x) \\
&= \sum (E[Y|D = 1, X = x] - E[Y|D = 0, X = x])P(X = x) \\
&= \sum DIM_x P(X = x)
\end{aligned} \tag{4.2}$$

This is basically the g-formula, or regression adjustment formula. We compute the DIM (difference in means) between treated and control for each strata $X = x$, then the weighted average is the ATE, with $\hat{P}(X = x)$ as weights.

If we have high dimensional X, we usually do

$$Y = \beta_0 + \beta_1 X + \delta_{reg} D + \epsilon \tag{4.3}$$

We would hope δ_{reg} is close to δ_{ATE}, but we are not sure.

OLS coefficients are:

$$\hat{\beta} = (Z'Z)^{-1}Z'y \tag{4.4}$$

Here I use Z to represent all regressors. Each coefficient is a weighted sum of the outcome,

$$\hat{\tau}_{reg} = \sum \omega_i Y_i \tag{4.5}$$

By FWL theorem,

$$\hat{\tau}_{reg} = \frac{\sum Y_i (D_i - \hat{d}(X_i))}{\sum (D_i - \hat{d}(X_i))^2} \tag{4.6}$$

Here $\hat{d}(X_i) = \hat{p}(D_i = 1|X_i)$ which is the predicted value of linear regression of D on X. The numerator is the covariance between Y and residualized D, the denominator is the variance of the residualized D. Note that the original FWL would have residualized Y here, but it turns out it works the same with the original Y.

Therefore the weights are

$$\omega_i = \frac{D_i - \hat{d}(X_i)}{\sum (D_i - \hat{d}(X_i))^2} \tag{4.7}$$

Note for the denominator, it's a constant. It does *not* scale the weights to sum to one — they sum to **zero**, because $D_i - \hat{d}(X_i)$ is an OLS residual from a regression that includes an intercept. What the denominator normalizes is the contrast: since $\sum D_i(D_i - \hat{d}(X_i)) = \sum (D_i - \hat{d}(X_i))^2$ (the residual is orthogonal to the fitted values), we get $\sum_i \omega_i D_i = 1$, i.e. the weights on the treated sum to $+1$ and those on the control sum to -1. That is what makes $\sum_i \omega_i Y_i$ a weighted *difference* of means. The numerator for treated units are $1 - \hat{d}(X_i)$, for control units it's $-\hat{d}(X_i)$ (negative). It is similar to IPW weights, which are $1/\hat{d}(X_i)$ for treated and $1/(1 - \hat{d}(X_i))$ for control. These two sets of weights go the same direction. For the treated, the higher probability of being treated, the lower weight; for the control, the higher probability of being treated, the larger the magnitude of the (negative) weight.

If these weights are the same as the IPW weights, then we'll get $\tau_{reg} = \tau_{ATE}$. However, they are usually different.

Another way to write this as weighted difference in means:

$$\hat{\tau}_{wdim} = \sum \omega_i Y_i = \sum_{i:D=1} \tilde{\omega}_i Y_i + \sum_{i:D=0} \tilde{\omega}_i Y_i \tag{4.8}$$

where, with $c = \sum (D_i - \hat{d}(X_i))^2$, the per-arm weights are $\tilde{\omega}_i = (1 - \hat{d}(X_i))/c$ for treated units $(D_i = 1)$ and $\tilde{\omega}_i = -\hat{d}(X_i)/c$ for control units $(D_i = 0)$

Because $\hat{d}(X_i)$ is from a linear probability model, it's possible that it's not between 0 and 1. In that case, we can have negative weights. The negative weights could mean the observations do not have a match in the other group. For example, $\hat{d}(X_i) > 1$ will cause the weight $(1 - \hat{d}(X_i))/c$ to flip sign and become negative for a treated unit. That means this unit is more like treated units than any control units. In other words, there is no match in the control group. The mirror image happens for a control unit with $\hat{d}(X_i) < 0$: its weight $-\hat{d}(X_i)/c$ flips to *positive*, so it enters the contrast on the same side as the treated. Either way the unit contributes with the wrong sign.

The negative weights are invisible unless we examine $\hat{d}(X_i)$ directly — which is to say, unless we look at the propensity score. A linear regression of Y on D and X hides the extrapolation. So $\hat{\tau}_{\text{reg}}$ is a weighted average of outcomes with weights that differ from the ATE weights, and some of those weights may carry the wrong sign without the researcher knowing.

4.2. Strata level weights

Angrist (1998, *Econometrica*) has the strata-wise weights for OLS:

$$\hat{\tau}_{reg} = \frac{\sum_x \hat{\tau}_x \hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)}{\sum_x \hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)} \tag{4.9}$$

this shows that different from ATE, these weights are $\hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)$. The stratas are weighted not only by $P(X_i = x)$, but also by the variance of treatment status. When $\hat{d}(X_i)$ is close to .5, the weight is the largest.

If there is no treatment effect heterogeneity, then it does not matter how you weight. When there is treatment effect heterogeneity, this estimator could be very different from ATE.

Hazlett and Shinkre show that Angrist's formula is a special case: it is exact when the probability of treatment is linear in X (as in a saturated model with discrete X). They derive a more general formula, but the Angrist version suffices under linearity of $d(X)$.

4.3. Recommended estimation approaches

Under constant effects and linearity, $E[Y \mid X, D] = \beta_0 + \beta_1 X + \beta_2 D$, a regression of Y on D and X recovers the ATE without further adjustment. When the linearity or constant-effects assumption fails, Hazlett and Shinkre suggest relaxing to *separate* linearity — allowing different intercepts for $Y(0)$ and $Y(1)$. Under that weaker assumption, several estimators are unbiased for the ATE.

The interaction (Lin 2013) estimator regresses Y on D, $\tilde{X} = X - \bar{X}$, and $D\tilde{X}$; the coefficient on D is the ATE. The imputation estimator fits separate regressions for treated and control, predicts counterfactual outcomes for the other group, and takes the average difference — algebraically, it returns the same estimate. Stratification and g-computation do the same for discrete X: compute the stratum-level difference in means and weight by the stratum probability.

All of these still rely on linearity within each arm. When that fails, matching (e.g. mean-balancing) or nonparametric estimation is needed.

4.4. A worked example

We make the regression weights ω_i concrete and compare them to IPW weights $1/\hat{d}(X_i)$ (treated) and $1/(1-\hat{d}(X_i))$ (control).

We draw $n = 500$ observations with a single covariate $X \sim N(0,1)$, treatment $D \sim \text{Bernoulli}(\Lambda(0.5X))$ — so 238 units are treated — and outcome $Y = 1 + 2D + 1.5X + \varepsilon$ with $\varepsilon \sim N(0,1)$. The effect is constant at 2, so there is no heterogeneity for the weights to mis-average; the point of this first example is only to show that the weights reproduce the coefficient.

```r
library(dplyr)

set.seed(1)
n <- 500
X <- rnorm(n)
d_prob <- plogis(0.5 * X)
D <- rbinom(n, 1, d_prob)
Y <- 1 + 2 * D + 1.5 * X + rnorm(n)
dat <- data.frame(Y, D, X)

# Regression-implied weights via the FWL residualization of D on X.
d_hat <- fitted(lm(D ~ X, data = dat))
resid_D <- dat$D - d_hat
omega_reg <- resid_D / sum(resid_D^2)

# IPW weights (Horvitz-Thompson), using the same linear-probability d_hat
# purely for comparability with the regression weights above.
ipw <- ifelse(dat$D == 1, 1 / d_hat, 1 / (1 - d_hat))
# rescale by a single constant so the treated weights average 1; this only
# fixes the axis scale of the plot below and does not change any comparison
ipw <- ipw / sum(ipw[dat$D == 1]) * sum(dat$D == 1)

tau_reg <- sum(omega_reg * dat$Y)
tau_lm  <- coef(lm(Y ~ D + X, data = dat))["D"]

data.frame(tau_reg_from_weights = tau_reg, tau_from_lm_coef = tau_lm)
```

```
  tau_reg_from_weights tau_from_lm_coef
D             2.084644         2.084644
```

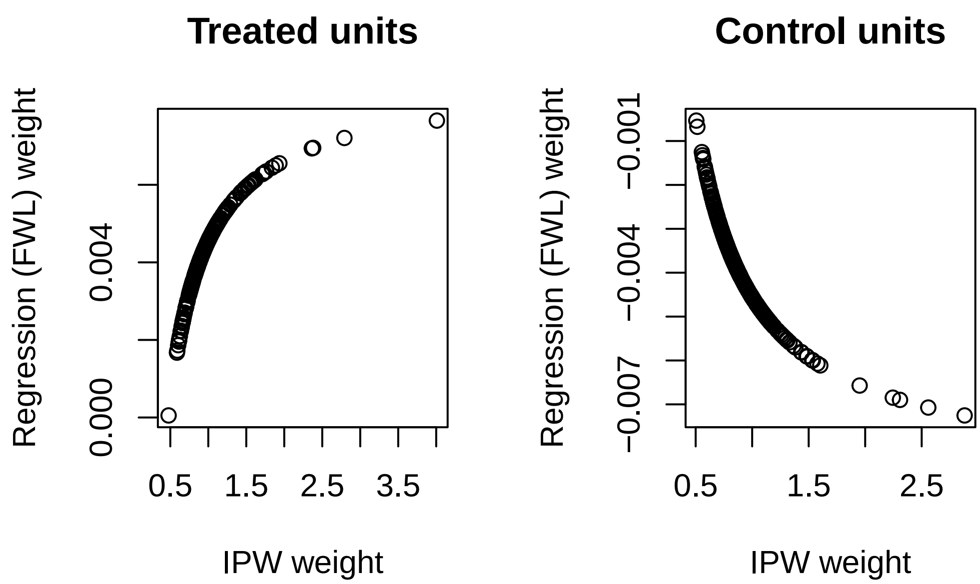
Both give 2.084644, identical to seven digits, confirming $\hat{\tau}_{reg} = \sum \omega_i Y_i$. (Both sit a little above the true 2; with $n = 500$ and a constant effect that is sampling noise, not weighting.)

Plotting `omega_reg` against `ipw` for treated and control units separately shows the two weighting schemes move in the same direction but are not proportional.

Be careful about what that does and does not establish. Non-proportional weights are *necessary* for $\hat{\tau}_{reg}$ to differ from the ATE, not sufficient. In this example the effect is constant at 2, so every unit's effect is the same number and any weights that form a difference of means return that number: $\hat{\tau}_{reg}$ and the ATE coincide here no matter how oddly the weights are distributed. A weighting problem needs two ingredients, and this design has only one. The section after next supplies the other.

```r
par(mfrow = c(1, 2))
plot(ipw[dat$D == 1], omega_reg[dat$D == 1],
     xlab = "IPW weight", ylab = "Regression (FWL) weight",
     main = "Treated units")
plot(ipw[dat$D == 0], omega_reg[dat$D == 0],
```

```
     xlab = "IPW weight", ylab = "Regression (FWL) weight",
     main = "Control units")
```




```
par(mfrow = c(1, 1))
```

Both panels slope upward and neither is a straight line through the origin: the two schemes rank units the same way but do not scale together, which is the whole content of $\hat{\tau}_{reg} \neq \hat{\tau}_{ATE}$.

Note also what this example does *not* show. Here $\hat{d}(X_i)$ stays inside $[0.062, 0.994]$, so every weight keeps its expected sign and the negative-weight problem discussed above never materializes. That is a property of this particular design, not a reassurance. We now strengthen the selection from $\Lambda(0.5X)$ to $\Lambda(3X)$, holding everything else fixed, and the linear probability model starts predicting outside the unit interval:

```
set.seed(2)
X2 <- rnorm(n)
D2 <- rbinom(n, 1, plogis(3 * X2))       # much stronger selection on X
dat2 <- data.frame(D = D2, X = X2)
d_hat2 <- fitted(lm(D ~ X, data = dat2))
omega2 <- (dat2$D - d_hat2) / sum((dat2$D - d_hat2)^2)

data.frame(
  dhat_below_0        = sum(d_hat2 < 0),
  dhat_above_1        = sum(d_hat2 > 1),
  treated_wrong_sign  = sum(dat2$D == 1 & omega2 < 0),
  control_wrong_sign  = sum(dat2$D == 0 & omega2 > 0)
)
```

```
  dhat_below_0 dhat_above_1 treated_wrong_sign control_wrong_sign
1           28           42                 42                 28
```

The fitted probabilities now run from -0.367 to 1.445. Twenty-eight units fall below zero and 42 rise above one, and the wrong-sign counts match those two numbers exactly: 42 treated units carry a negative weight and 28 control units carry a positive one. That is not a coincidence but the mechanism described above — every unit with $\hat{d} > 1$ here is treated, every unit with $\hat{d} < 0$ is control, and each flips the sign of its own contribution. In total 70 of 500 observations, 14% of the sample, enter the contrast on the wrong side.

Those units are the ones with no counterpart in the other arm, and a plain regression of Y on D and X gives you no indication that they are there. Nothing in the coefficient table changes appearance; only looking at $\hat{d}(X_i)$ reveals it.

This is a property of the *regression* weights specifically, and the reason is worth being explicit about. $\hat{d}(X_i)$ here is not a propensity score we chose to estimate with a linear probability model; it is the object OLS implicitly uses, because the FWL residualisation of D on X *is* a linear regression. Linear fitted values are not confined to $[0, 1]$, so out-of-range values, and with them wrong-sign weights, are always available to OLS. Genuine IPW weights built from a logit or probit propensity score cannot do this: $\hat{\pi}(X)$ lies strictly inside $(0, 1)$ by construction, so $1/\hat{\pi}$ and $1/(1 - \hat{\pi})$ stay positive. IPW has its own failure mode when $\hat{\pi}$ approaches 0 or 1 — weights explode, and the estimate rests on a handful of observations — but it is a visible one. The `ipw` vector plotted above is deliberately built from the same linear $\hat{d}$ so that the two weighting schemes are comparable unit by unit; it is a diagnostic device, not the IPW estimator one would actually run.

4.5. When the weights actually bite

Neither example so far has shown $\hat{\tau}_{reg}$ miss the ATE, because neither had heterogeneous effects. This one does, and it is built on discrete X so that every quantity in this chapter can be computed exactly rather than approximated.

Let $X \in \{0, 1, 2\}$ with probabilities $0.4, 0.3, 0.3$; let the treatment probability be $d(x) = 0.5, 0.1, 0.9$; and let the effect be $\tau(x) = 1, 4, 4$. The outcome is $Y = 1 + \tau(X)D + 2X + \varepsilon$ with $\varepsilon \sim N(0, 1)$, and we draw $n = 20{,}000$. The design is chosen so that the stratum where treatment is most balanced, $x = 0$ with $d = 0.5$, is also the stratum with the small effect.

The population ATE follows from the definition: $0.4(1) + 0.3(4) + 0.3(4) = 2.8$. Angrist's strata-level formula above weights each stratum by $d(x)\{1 - d(x)\}P(X = x)$ instead — that is $0.10, 0.027, 0.027$, or $0.649, 0.175, 0.175$ once normalised, against the true shares $0.4, 0.3, 0.3$. The $x = 0$ stratum carries 65% of the regression's weight and 40% of the ATE's, so the regression should be pulled toward that stratum's effect of 1.

```r
set.seed(11)
n  <- 20000
px <- c(0.4, 0.3, 0.3)            # P(X = 0, 1, 2)
dx <- c(0.5, 0.1, 0.9)            # P(D = 1 | X)
tx <- c(1,   4,   4)              # tau(X)
```

```r
X   <- sample(0:2, n, TRUE, prob = px)
D   <- rbinom(n, 1, dx[X + 1])
Y   <- 1 + tx[X + 1] * D + 2 * X + rnorm(n)
dat3 <- data.frame(Y, D, X = factor(X))

# What OLS returns
tau_ols <- coef(lm(Y ~ D + X, data = dat3))["D"]

# Angrist's variance-weighted average of the stratum effects, computed from the
# population quantities rather than estimated
w   <- dx * (1 - dx) * px
ang <- sum(w * tx) / sum(w)

# g-computation on a saturated arm-specific model
m   <- lm(Y ~ D * X, data = dat3)
tau_g <- mean(predict(m, transform(dat3, D = 1)) -
              predict(m, transform(dat3, D = 0)))

# Lin's interaction estimator: D, centred covariates, and their interaction
Xc  <- scale(model.matrix(~ X, dat3)[, -1, drop = FALSE], scale = FALSE)
tau_lin <- coef(lm(dat3$Y ~ dat3$D + Xc + dat3$D:Xc))[2]

data.frame(population_ATE = sum(px * tx),
           angrist_formula = ang,
           ols_coefficient = tau_ols,
           g_computation = tau_g,
           lin_interaction = tau_lin, row.names = NULL)
```

```
  population_ATE angrist_formula ols_coefficient g_computation lin_interaction
1            2.8        2.051948        2.082158      2.839965        2.839965
```

The OLS coefficient is 2.08 against a population ATE of 2.80. It is not a noisy estimate of the ATE; it is a precise estimate of something else. Angrist's formula predicts 2.052 from the population quantities alone, and the regression delivers 2.082 — the gap between those two is sampling noise, while the gap between either of them and 2.80 is the weighting problem. The regression is answering a question about the stratum where treatment happens to be most balanced, and nothing in its output says so.

The two recommended estimators recover the ATE, and recover it identically: g-computation and Lin's interaction estimator both return 2.840, against a realised sample ATE of 2.814. That they agree to every printed digit is the algebraic equivalence noted in the previous section, not a coincidence of this draw. Both work here because the saturated model in a discrete X satisfies separate linearity exactly — each arm's conditional mean is linear in the stratum dummies, since it is *saturated* in them.

The practical reading is that the weighting problem needs heterogeneity and non-proportional weights together. The first two examples had the weights; this one adds the heterogeneity, and the cost appears immediately. Since one rarely knows in advance whether effects are heterogeneous, the cheap insurance is to run the interaction or imputation estimator by default: under constant effects it returns what OLS would have returned, and under heterogeneity it returns the ATE that OLS would have missed.

Part II.

Marginal Effects & Fixed Effects

5. Marginal effects in models with fixed effects

5.1. Marginal effects in a linear model

Stata's `margins` command computes predicted means and marginal effects, but care is needed when fixed effects are present. In a linear model the issue is benign: demeaning and dummy-variable approaches return the same coefficients and the same marginal effects.

We check that on the `auto` data, treating the 5-category repair record `rep78` as the panel identifier and regressing `price` on `mpg` interacted with `trunk` — once with `xtreg, fe` (within-demeaning) and once with `reg` plus `i.rep78` dummies.

```
clear
sysuse auto
xtset rep78
xtreg price c.mpg##c.trunk, fe
margins , dydx(mpg)
reg price c.mpg##c.trunk i.rep78
margins , dydx(mpg)
```

```
. clear

. sysuse auto
(1978 automobile data)

. xtset rep78

Panel variable: rep78 (unbalanced)

. xtreg price c.mpg##c.trunk, fe

Fixed-effects (within) regression               Number of obs     =         69
Group variable: rep78                           Number of groups  =          5

R-squared:                                      Obs per group:
     Within  = 0.2570                                         min =          2
     Between = 0.0653                                         avg =       13.8
     Overall = 0.2237                                         max =         30

                                                F(3, 61)          =       7.03
```

```
corr(u_i, Xb) = -0.4133                         Prob > F          =     0.0004

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -98.12003   226.8708    -0.43   0.667    -551.7763    355.5362
       trunk |   295.0544   343.3934     0.86   0.394    -391.6032     981.712
             |
       c.mpg#|
     c.trunk |  -12.23318   15.94713    -0.77   0.446    -44.12143    19.65506
             |
       _cons |    7574.85   5321.325     1.42   0.160    -3065.797     18215.5
-------------+----------------------------------------------------------------
     sigma_u |   992.2156
     sigma_e |  2631.2869
         rho |  .12449059   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(4, 61) = 0.86                      Prob > F = 0.4948

. margins , dydx(mpg)

Average marginal effects                                 Number of obs = 69
Model VCE: Conventional

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -268.4981   74.12513    -3.62   0.000    -413.7807   -123.2156
------------------------------------------------------------------------------

. reg price c.mpg##c.trunk i.rep78

      Source |       SS           df       MS      Number of obs   =        69
-------------+----------------------------------   F(7, 61)        =      3.19
       Model |   154453046         7  22064720.8   Prob > F        =    0.0061
    Residual |   422343913        61  6923670.71   R-squared       =    0.2678
-------------+----------------------------------   Adj R-squared   =    0.1838
       Total |   576796959        68  8482308.22   Root MSE        =    2631.3

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -98.12003   226.8708    -0.43   0.667    -551.7763    355.5362
       trunk |   295.0544   343.3934     0.86   0.394    -391.6032     981.712
```

```
             |
      c.mpg#|
    c.trunk |  -12.23318   15.94713    -0.77   0.446    -44.12143    19.65506
             |
      rep78 |
          2  |   438.0002   2161.922     0.20   0.840    -3885.031    4761.031
          3  |   987.1363   2022.606     0.49   0.627    -3057.315    5031.587
          4  |   1240.944   2046.417     0.61   0.547     -2851.12    5333.008
          5  |    2605.83   2161.837     1.21   0.233    -1717.031    6928.691
             |
      _cons |   6355.731   5209.899     1.22   0.227    -4062.105    16773.57
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                    Number of obs = 69
Model VCE: OLS

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -268.4981   74.12513    -3.62   0.001    -416.7205   -120.2758
------------------------------------------------------------------------------

.
```

The two agree exactly. Both give -98.12003 on `mpg`, 295.0544 on `trunk` and -12.23318 on the interaction, and both average marginal effects are -268.4981 with a standard error of 74.12513. The only difference is that `xtreg` reports a z statistic and `reg` a t, so the confidence intervals differ in the fourth digit. Nothing here needs care.

5.2. Marginal effects in a non-linear model

In a nonlinear model the two approaches diverge. We now need a genuine panel and a count outcome, so we simulate one: 250 observations, 50 units observed over 5 periods. Within each unit, mpg $\sim N(20, 4^2)$ and trunk $\sim N(15, 3^2)$ vary period to period, while the unit effect $\alpha_i \sim N(0, 1)$ is drawn once and held fixed. The conditional mean is $\lambda_{it} = \exp(0.5 + 0.05\,\text{mpg}_{it} - 0.03\,\text{trunk}_{it} + \alpha_i)$ and $\text{accidents}_{it} \sim \text{Poisson}(\lambda_{it})$. The true coefficients are 0.05 and -0.03.

```
clear
set seed 42
set obs 250
```

```
gen id = ceil(_n/5)
bys id: gen t = _n
gen mpg = rnormal(20,4)
gen trunk = rnormal(15,3)
gen alpha = rnormal(0,1)
bys id: replace alpha = alpha[1]
gen lambda = exp(0.5 + 0.05*mpg - 0.03*trunk + alpha)
gen accidents = rpoisson(lambda)
xtset id
xtpoisson accidents mpg trunk, fe
margins , dydx(mpg)
margins , dydx(mpg) predict(nu0)
poisson accidents mpg trunk i.id
margins , dydx(mpg)
```

```
. clear

. set seed 42

. set obs 250
Number of observations (_N) was 0, now 250.

. gen id = ceil(_n/5)

. bys id: gen t = _n

. gen mpg = rnormal(20,4)

. gen trunk = rnormal(15,3)

. gen alpha = rnormal(0,1)

. bys id: replace alpha = alpha[1]
(200 real changes made)

. gen lambda = exp(0.5 + 0.05*mpg - 0.03*trunk + alpha)

. gen accidents = rpoisson(lambda)

. xtset id

Panel variable: id (balanced)

. xtpoisson accidents mpg trunk, fe
note: 1 group (5 obs) dropped because of all zero outcomes
```

```
Iteration 0:  Log likelihood = -358.34297
Iteration 1:  Log likelihood = -340.16596
Iteration 2:  Log likelihood = -340.16413
Iteration 3:  Log likelihood = -340.16413

Conditional fixed-effects Poisson regression        Number of obs    =    245
Group variable: id                                  Number of groups =     49

                                                    Obs per group:
                                                                 min =      5
                                                                 avg =    5.0
                                                                 max =      5

                                                    Wald chi2(2)     =  35.90
Log likelihood = -340.16413                         Prob > chi2      = 0.0000

------------------------------------------------------------------------------
   accidents | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090524     5.39   0.000     .0310521    .0665371
       trunk |  -.0316066   .0108931    -2.90   0.004    -.0529566   -.0102566
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                   Number of obs = 245
Model VCE: OIM

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090524     5.39   0.000     .0310521    .0665371
------------------------------------------------------------------------------

. margins , dydx(mpg) predict(nu0)

Average marginal effects                                   Number of obs = 245
Model VCE: OIM

Expression: Predicted number of events (assuming u_i=0), predict(nu0)
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
```

```
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0850287   .0340547     2.50   0.013     .0182826    .1517747
------------------------------------------------------------------------------

. poisson accidents mpg trunk i.id

Iteration 0:  Log likelihood = -485.81671
Iteration 1:  Log likelihood = -453.45562
Iteration 2:  Log likelihood = -453.02481
Iteration 3:  Log likelihood = -452.93985
Iteration 4:  Log likelihood = -452.92127
Iteration 5:  Log likelihood = -452.91701
Iteration 6:  Log likelihood =   -452.916
Iteration 7:  Log likelihood = -452.91579
Iteration 8:  Log likelihood = -452.91575

Poisson regression                                      Number of obs =    250
                                                        LR chi2(51)   = 934.93
                                                        Prob > chi2   = 0.0000
Log likelihood = -452.91575                             Pseudo R2     = 0.5079

------------------------------------------------------------------------------
   accidents | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090525     5.39   0.000      .031052    .0665371
       trunk |  -.0316067   .0108931    -2.90   0.004    -.0529568   -.0102565
             |
          id |
          2  |   .3464221   .5592835     0.62   0.536    -.7497533    1.442598
          3  |   1.123341   .5131078     2.19   0.029     .1176679    2.129014
          4  |   .1098041   .6057603     0.18   0.856    -1.077464    1.297073
          5  |   .4069152   .5702496     0.71   0.475    -.7107536    1.524584
          6  |   1.791221   .4776303     3.75   0.000     .8550829    2.727359
          7  |  -14.00991   475.5637    -0.03   0.976    -946.0976    918.0778
          8  |   .9632705   .5216941     1.85   0.065    -.0592311    1.985772
          9  |   .1039008   .6061094     0.17   0.864    -1.084052    1.291853
         10  |   .7748186   .5273812     1.47   0.142    -.2588296    1.808467
         11  |   .7331684   .5395719     1.36   0.174    -.3243732     1.79071
         12  |   1.114943   .5007908     2.23   0.026      .133411    2.096475
         13  |  -.2054342   .6343396    -0.32   0.746    -1.448717    1.037848
         14  |   1.248116   .4945786     2.52   0.012     .2787602    2.217472
         15  |   1.395876   .4914484     2.84   0.005     .4326552    2.359098
         16  |   1.689054   .4845687     3.49   0.000     .7393167    2.638791
         17  |   .9941942   .5171224     1.92   0.055    -.0193471    2.007735
         18  |   1.742704   .4844911     3.60   0.000      .793119    2.692289
         19  |   .9912213   .5168563     1.92   0.055    -.0217985    2.004241
         20  |  -1.134862   .8375742    -1.35   0.175    -2.776477    .5067532
```

```
          21  |   1.484731   .4879013     3.04   0.002     .5284622       2.441
          22  |   2.442088   .4643803     5.26   0.000     1.531919    3.352256
          23  |   .9732572   .5265941     1.85   0.065    -.0588483    2.005363
          24  |    .280425   .5859507     0.48   0.632    -.8680172    1.428867
          25  |   2.472329   .4641555     5.33   0.000     1.562601    3.382057
          26  |   .5602592   .5485592     1.02   0.307    -.5148971    1.635416
          27  |   1.537242   .4861325     3.16   0.002     .5844401    2.490044
          28  |    1.34391   .5026286     2.67   0.008     .3587763    2.329044
          29  |   2.370956   .4660218     5.09   0.000      1.45757    3.284342
          30  |   2.453329   .4642702     5.28   0.000     1.543377    3.363282
          31  |   .9383342   .5171502     1.81   0.070    -.0752616     1.95193
          32  |   .1850002   .5864331     0.32   0.752    -.9643876    1.334388
          33  |  -.2686233   .6711829    -0.40   0.689    -1.584118    1.046871
          34  |   1.547864    .485423     3.19   0.001     .5964525    2.499276
          35  |   1.557431   .4878123     3.19   0.001     .6013366    2.513526
          36  |   1.295147   .4965837     2.61   0.009     .3218611    2.268433
          37  |   2.668556   .4616456     5.78   0.000     1.763748    3.573365
          38  |   .9287953   .5326269     1.74   0.081    -.1151343    1.972725
          39  |  -.9247308   .8366882    -1.11   0.269     -2.56461     .715148
          40  |   1.358213   .4979867     2.73   0.006     .3821769    2.334249
          41  |  -.0472585   .6070096    -0.08   0.938    -1.236975    1.142458
          42  |   1.394504   .4958685     2.81   0.005     .4226201    2.366389
          43  |  -.5414444   .7303533    -0.74   0.458    -1.972911    .8900219
          44  |   .9881123    .513744     1.92   0.054    -.0188073    1.995032
          45  |   1.171458   .5165503     2.27   0.023     .1590378    2.183878
          46  |    .817007   .5269245     1.55   0.121    -.2157461     1.84976
          47  |   2.800173   .4596719     6.09   0.000     1.899233    3.701113
          48  |    .365911   .5704157     0.64   0.521    -.7520831    1.483905
          49  |   2.445633   .4649624     5.26   0.000     1.534324    3.356943
          50  |   .7177863   .5279692     1.36   0.174    -.3170143    1.752587
              |
        _cons |  -.4227135   .5046249    -0.84   0.402     -1.41176     .566333
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                   Number of obs = 250
Model VCE: OIM

Expression: Predicted number of events, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .2238673   .0420551     5.32   0.000     .1414408    .3062937
------------------------------------------------------------------------------
```

.

(We use a simulated count outcome `accidents` and a proper panel with 50 units of 5 periods each, rather than `auto`'s continuous `price` and its 5-category `rep78` — a true panel needs many units, and a fixed-effect count model needs an actual count dependent variable.)

In this example, "xtpoisson, fe" and "poisson … i.id" return the same coefficients (both give `mpg` = .0487946 with a standard error of .0090524). Fixed effect Poisson model (sometimes called conditional fixed effect Poisson) is the same models as a Poisson model with dummies, just like a linear model (OLS with dummies is the same as fixed effect OLS). Poisson model and OLS are unique in this sense that there is no "incidental parameter" problem.

Both recover the truth: 0.0488 against a true 0.05 for `mpg`, and −0.0316 against −0.03 for `trunk`. Note the sample sizes differ, 245 against 250 — `xtpoisson, fe` drops the one group whose outcome is zero in every period, because such a group contributes nothing to the conditional likelihood. It changes neither coefficient here.

The `margins` calls return different marginal effects even though the models are the same, and the spread is not small:

call	AME of `mpg`	std. err.
`margins, dydx(mpg)` after `xtpoisson, fe`	0.0488	0.0091
`margins, dydx(mpg) predict(nu0)`	0.0850	0.0341
`margins, dydx(mpg)` after `poisson ... i.id`	0.2239	0.0421

Three answers from one coefficient, the largest 4.6 times the smallest. In the conditional fixed-effect Poisson the fixed effects are conditioned out — they are not estimated. The first row is not really a marginal effect at all: with no α_i to exponentiate against, it returns the coefficient itself. `margins, predict(nu0)` sets the fixed effect to zero, which is arbitrary, and note that the conditional model has no intercept either — it was swept out with the α_i. So `nu0` exponentiates $X\beta$ alone, giving a baseline count no unit in the data actually has. The Poisson with dummies estimates the fixed effects, so `margins` uses actual estimates and gets the largest value.

The arithmetic is easy to follow, since for a Poisson $\partial\lambda/\partial x = \beta\lambda$ and therefore every one of these AMEs is β times an average λ. Divide each row by 0.0488 and the implied baselines are 1.00, 1.74 and 4.59. The last matches $E[\lambda] = e^{1.05}E[e^{\alpha}] = 4.71$ from the DGP; the middle matches $e^{X\beta}$ evaluated without intercept or fixed effect, 1.65. The three rows differ only in what they assume about α_i, and none of them is a modelling subtlety — it is a choice of baseline. The third is the defensible one, but for the conditional model the safest route is to report the original coefficients and avoid `margins` for marginal effects entirely.

The conditional logit is worse. The fixed effects are not estimated, and unlike Poisson there is an incidental-parameter problem, so logit with dummies is inconsistent unless the panel is deep (roughly 20+ observations per unit). With conditional logit the predicted probability depends on the unestimated α_i; `margins` after `clogit` or `xtlogit, fe` offers options like `pu0` (set all fixed effects to zero), but none is defensible.

In a fixed effect logit model,

$$log(P(y=1)/(1-P(y=1))) = \alpha_i + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 * x_2 \tag{5.1}$$

Here α_i is the unit fixed effect. Therefore

$$P(y=1) = F(\alpha_i + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 * x_2) \tag{5.2}$$

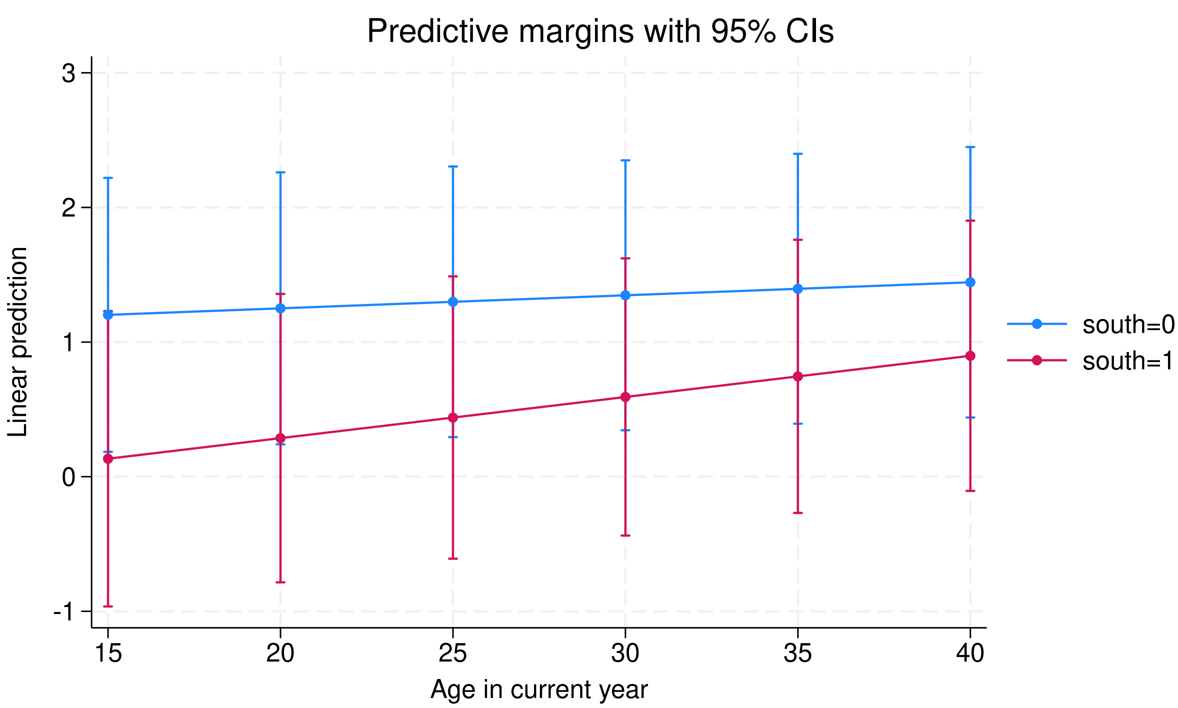
Without estimating α_i there is no way to predict P in a meaningful way. But the log-odds are linear in the covariates, and the marginal effect of x_1 or x_2 on the log-odds does not involve α_i. So `margins` with the `predict(xb)` option — marginal effects on the log-odds scale — is interpretable. Note that `xb` here is $X_{it}\beta$ only — the individual fixed effect α_i is not estimated in conditional logit and so is not part of the prediction. The logged-odds *effect* of a covariate ($\partial(X\beta)/\partial x$) therefore matches the original coefficient exactly and does not depend on α_i, but `xb` itself is a log-odds relative to each individual's (unidentified) baseline, not the individual's absolute log-odds. This lets you make linear extrapolations of relative log-odds across values of the covariates, but not of absolute predicted probabilities, since those would require α_i.

```
clear
webuse union
clogit union c.age##i.south not_smsa grade, group(idcode)
margins, at( age=(15 20 25 30 35 40) south=(0 1)) predict(xb)
marginsplot
graph export "marginsplot-clogit.svg", as(svg) replace
```

```
. clear

. webuse union
(NLS Women 14-24 in 1968)

. clogit union c.age##i.south not_smsa grade, group(idcode)
note: multiple positive outcomes within groups encountered.
note: 2,744 groups (14,165 obs) omitted because of all positive or
      all negative outcomes.

Iteration 0:  Log likelihood = -4518.8815
Iteration 1:  Log likelihood = -4512.8224
Iteration 2:  Log likelihood = -4512.8192
Iteration 3:  Log likelihood = -4512.8192

Conditional (fixed-effects) logistic regression         Number of obs = 12,035
                                                        LR chi2(5)    =  74.73
                                                        Prob > chi2   = 0.0000
Log likelihood = -4512.8192                             Pseudo R2     = 0.0082

------------------------------------------------------------------------------
       union | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
         age |   .0096842   .0050265     1.93   0.054    -.0001676     .019536
     1.south |  -1.382178    .276966    -4.99   0.000    -1.925022   -.8393346
             |
 south#c.age |
          1  |   .0208997   .0081247     2.57   0.010     .0049756    .0368238
             |
    not_smsa |   .0195233   .1131292     0.17   0.863    -.2022058    .2412523
       grade |   .0822276   .0419062     1.96   0.050      .000093    .1643622
------------------------------------------------------------------------------

. margins, at( age=(15 20 25 30 35 40) south=(0 1)) predict(xb)

Predictive margins                                     Number of obs = 12,035
Model VCE: OIM

Expression: Linear prediction, predict(xb)
1._at:  age   = 15
        south =  0
2._at:  age   = 15
        south =  1
3._at:  age   = 20
        south =  0
4._at:  age   = 20
        south =  1
5._at:  age   = 25
        south =  0
6._at:  age   = 25
        south =  1
7._at:  age   = 30
        south =  0
8._at:  age   = 30
        south =  1
9._at:  age   = 35
        south =  0
10._at: age   = 35
        south =  1
11._at: age   = 40
        south =  0
12._at: age   = 40
        south =  1

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   1.202147   .5190753     2.32   0.021      .184778    2.219516
```

```
          2  |    .133464   .5599015     0.24   0.812    -.9639228    1.230851
          3  |   1.250568   .5153257     2.43   0.015      .240548    2.260588
          4  |   .2863834   .5465398     0.52   0.600    -.7848148    1.357582
          5  |   1.298989   .5127819     2.53   0.011     .2939548    2.304023
          6  |   .4393029   .5349589     0.82   0.412    -.6091973    1.487803
          7  |    1.34741   .5114619     2.63   0.008     .3449629    2.349857
          8  |   .5922223   .5252767     1.13   0.260    -.4373011    1.621746
          9  |   1.395831   .5113752     2.73   0.006     .3935538    2.398108
         10  |   .7451418   .5175997     1.44   0.150    -.2693351    1.759619
         11  |   1.444252   .5125224     2.82   0.005     .4397264    2.448777
         12  |   .8980612   .5120182     1.75   0.079    -.1054761    1.901598
------------------------------------------------------------------------------

. marginsplot

Variables that uniquely identify margins: age south

. graph export "marginsplot-clogit.svg", as(svg) replace
file marginsplot-clogit.svg saved as SVG format
```



This fits a conditional logit of union status on age, south, and their interaction, on 12,035 person-years from the `union` panel. The age coefficient is 0.0097 with a standard error of 0.0050, and the south indicator is -1.382 with a standard error of 0.277. `margins` with `predict(xb)` then returns predicted log-odds at six ages crossed with south — twelve combinations — but not predicted probabilities, since those require α_i.

Read the vertical axis of that plot as a *relative* scale. Because α_i is not estimated, `xb` is each individual's log-odds measured from an unidentified baseline, so the level of any single point is arbitrary; what is interpretable is how the curves move across age and how far apart the south and non-south curves sit. Differences within the plot are the estimable quantities, not heights.

6. Comparing Marginal effects with margins command

6.1. Comparing marginal effects

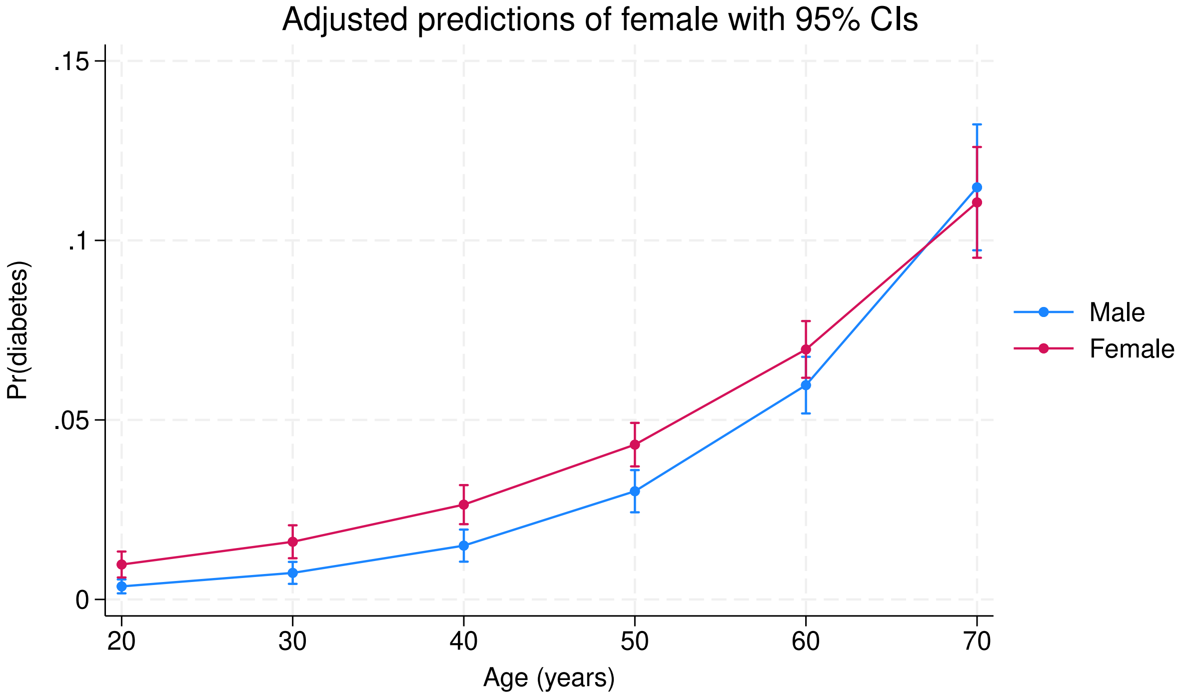
A common practice is to plot marginal effects with **margins** and **marginsplot**, then conclude two effects differ whenever their confidence intervals do not overlap. This can be wrong — overlapping intervals do not imply no significant difference, because the test depends on the joint distribution of the two estimates, not on their marginal distributions separately.

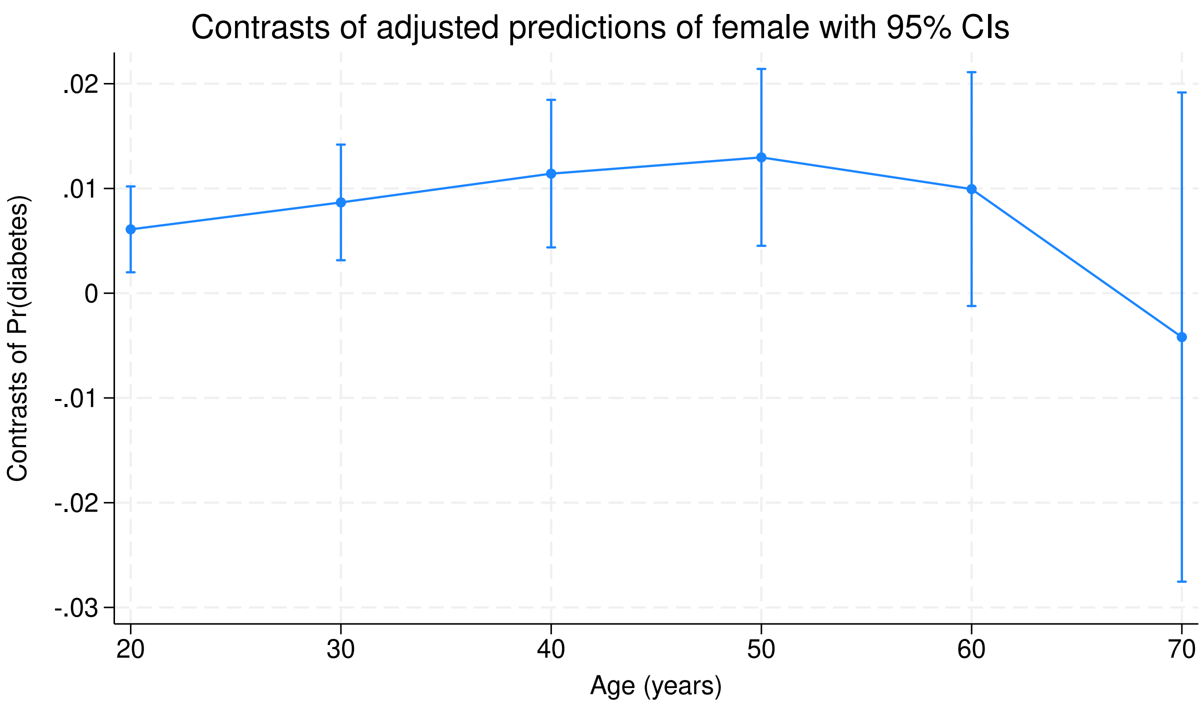
```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins female,  at(age=(20 30 40 50 60 70))
marginsplot
graph export "marginsplot1.svg", as(svg) replace
margins r.female,  at(age=(20 30 40 50 60 70))
marginsplot
graph export "marginsplot2.svg", as(svg) replace
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Adjusted predictions                                    Number of obs = 10,335
Model VCE: OIM
```

```
Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
  _at#female |
     1#Male  |   .0036348   .0009895     3.67   0.000     .0016953    .0055743
   1#Female  |   .0097316   .0018436     5.28   0.000     .0061184    .0133449
     2#Male  |    .007396   .0015622     4.73   0.000     .0043341    .0104579
   2#Female  |   .0160637   .0023434     6.85   0.000     .0114706    .0206568
     3#Male  |   .0149906   .0022831     6.57   0.000     .0105159    .0194653
   3#Female  |    .026406   .0027757     9.51   0.000     .0209657    .0318462
     4#Male  |   .0301469   .0029996    10.05   0.000     .0242677    .0360261
   4#Female  |    .043115   .0030923    13.94   0.000     .0370543    .0491758
     5#Male  |   .0596983   .0040248    14.83   0.000     .0518099    .0675867
   5#Female  |    .069641   .0040284    17.29   0.000     .0617454    .0775366
     6#Male  |   .1147889   .0089467    12.83   0.000     .0972536    .1323242
   6#Female  |   .1106004   .0078699    14.05   0.000     .0951757     .126025
------------------------------------------------------------------------------


Variables that uniquely identify margins: age female

file marginsplot1.svg saved as SVG format


Contrasts of adjusted predictions                       Number of obs = 10,335
Model VCE: OIM

Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

-------------------------------------------------------
                    |         df        chi2     P>chi2
--------------------+----------------------------------
```

```
         female@_at |
(Female vs Male) 1  |          1        8.49     0.0036
(Female vs Male) 2  |          1        9.47     0.0021
(Female vs Male) 3  |          1       10.09     0.0015
(Female vs Male) 4  |          1        9.06     0.0026
(Female vs Male) 5  |          1        3.05     0.0808
(Female vs Male) 6  |          1        0.12     0.7252
              Joint |          4       40.48     0.0000
-------------------------------------------------------

---------------------------------------------------------------------
                    |            Delta-method
                    |   Contrast   std. err.     [95% conf. interval]
--------------------+------------------------------------------------
         female@_at |
(Female vs Male) 1  |   .0060969   .0020923      .0019959    .0101978
(Female vs Male) 2  |   .0086677   .0028164      .0031476    .0141878
(Female vs Male) 3  |   .0114154    .003594      .0043712    .0184595
(Female vs Male) 4  |   .0129682   .0043081      .0045244    .0214119
(Female vs Male) 5  |   .0099427   .0056945     -.0012183    .0211037
(Female vs Male) 6  |  -.0041885   .0119155     -.0275425    .0191654
---------------------------------------------------------------------


Variables that uniquely identify margins: age

file marginsplot2.svg saved as SVG format
```





We fit a logit of diabetes on female, age, and their interaction, on 10,335 NHANES II respondents. The interaction is -0.0207 with a standard error of 0.0078, so the female-male gap narrows with age. The first `margins` call gives predicted probabilities by gender; the second (with `r.female`) gives the female-vs-male contrast at each age. (On the general question of marginal effects in nonlinear models, see the previous chapter — the coefficient on the link scale is usually more interpretable.)

Those contrasts run .0061, .0087, .0114, .0130, .0099 and −.0042 at ages 20 through 70, significant through age 50 and not thereafter — the gap widens, peaks in middle age, then closes and reverses sign.

Now the point of the chapter. The female effect at age 20 is .0061 with a 95% interval of [.0020, .0102], and at age 30 it is .0087 with an interval of [.0031, .0142]. Those intervals overlap across most of their length. It is tempting to conclude the two effects are not significantly different, but that conclusion does not follow. For two effects x and y from the same model,

$$\mathrm{Var}(x - y) = \mathrm{Var}(x) + \mathrm{Var}(y) - 2\,\mathrm{Cov}(x, y). \tag{6.1}$$

When x and y are positively correlated — as they will be, coming from the same fit — the variance of their difference can be much smaller than either individual variance. So the difference can be significant even when the individual intervals overlap.

```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins r.female,  at(age=(20 30 40 50 60 70)) post
lincom _b[r1vs0.female@2._at] - _b[r1vs0.female@1._at]
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Contrasts of adjusted predictions                       Number of obs = 10,335
Model VCE: OIM

Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
```

```
5._at: age = 60
6._at: age = 70

------------------------------------------------------
                    |         df        chi2     P>chi2
--------------------+---------------------------------
         female@_at |
(Female vs Male) 1  |          1        8.49     0.0036
(Female vs Male) 2  |          1        9.47     0.0021
(Female vs Male) 3  |          1       10.09     0.0015
(Female vs Male) 4  |          1        9.06     0.0026
(Female vs Male) 5  |          1        3.05     0.0808
(Female vs Male) 6  |          1        0.12     0.7252
             Joint  |          4       40.48     0.0000
------------------------------------------------------

--------------------------------------------------------------------
                    |            Delta-method
                    |   Contrast   std. err.     [95% conf. interval]
--------------------+-----------------------------------------------
         female@_at |
(Female vs Male) 1  |   .0060969   .0020923      .0019959    .0101978
(Female vs Male) 2  |   .0086677   .0028164      .0031476    .0141878
(Female vs Male) 3  |   .0114154    .003594      .0043712    .0184595
(Female vs Male) 4  |   .0129682   .0043081      .0045244    .0214119
(Female vs Male) 5  |   .0099427   .0056945     -.0012183    .0211037
(Female vs Male) 6  |  -.0041885   .0119155     -.0275425    .0191654
--------------------------------------------------------------------


 ( 1)  - r1vs0.female@1bn._at + r1vs0.female@2._at = 0

------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0025709   .0007946     3.24   0.001     .0010134    .0041283
------------------------------------------------------------------------------
```

The test gives a difference of .0025709 with a standard error of .0007946, $z = 3.24$ and $p = 0.001$. The two effects are different, despite intervals that overlap heavily.

The standard error is what makes the point. Each individual contrast carries a standard error of .0021 and .0028, but their *difference* carries only .00079 — smaller than either. Solving $\mathrm{Var}(x-y) = \mathrm{Var}(x) + \mathrm{Var}(y) - 2\mathrm{Cov}(x, y)$ for the correlation gives about 0.99: the two contrasts come from the same fitted model evaluated ten years apart, so they move together almost perfectly, and nearly all of their variance cancels in the difference.

Had we ignored the covariance and used $\sqrt{\mathrm{Var}(x) + \mathrm{Var}(y)} = .0035$, we would have got $z = 0.73$

and $p = 0.46$. Same data, same two numbers, opposite conclusion. That is the cost of reading overlap off a graph.

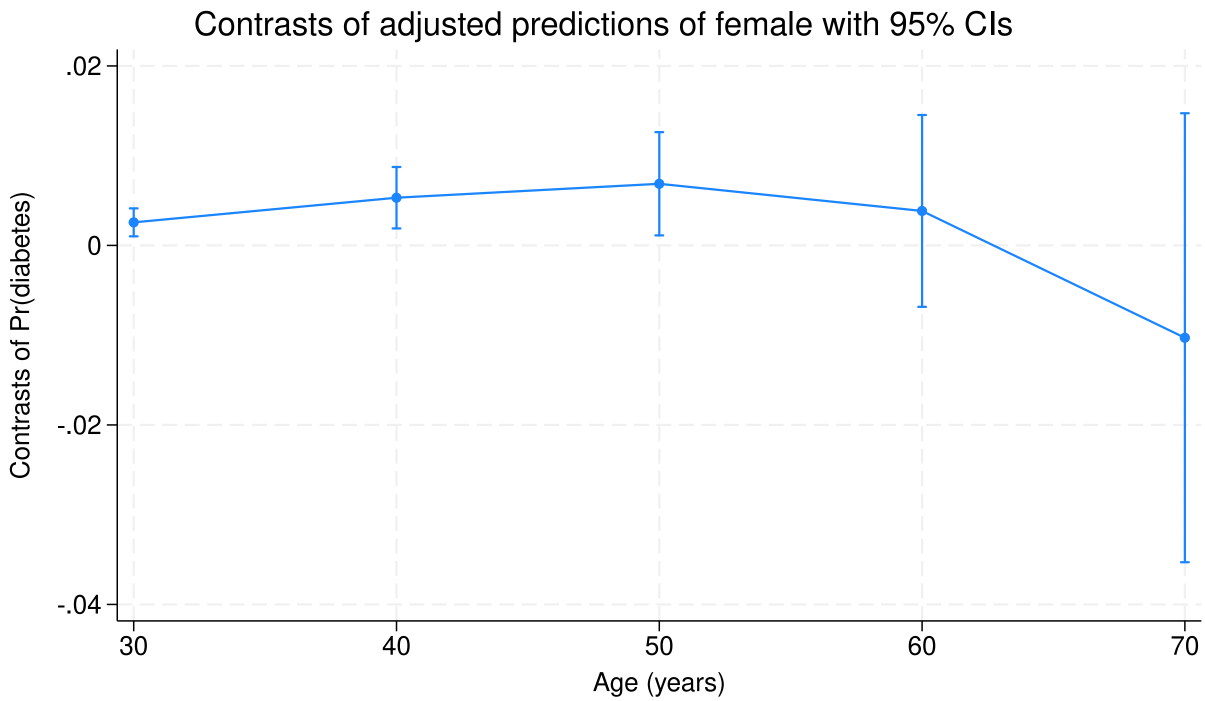
We can do this with native Stata syntax: `margins ..., post` posts the estimated contrasts as if they were regular coefficients (named `r1vs0.female@1._at`, `r1vs0.female@2._at`, etc. — check `matrix list e(b)` to see the exact names), so `lincom` can test any linear combination of them natively, without depending on a user-written command.

`margins` can compute these differences directly with `contrast(atcontrast(r))`, which compares each evaluation point to the reference:

```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins r.female,  at(age=(20 30 40 50 60 70)) contrast(atcontrast(r))
marginsplot
graph export "marginsplot3.svg", as(svg) replace
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Contrasts of adjusted predictions                       Number of obs = 10,335
Model VCE: OIM

Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

--------------------------------------------------------------
```

```
                           |         df        chi2     P>chi2
---------------------------+----------------------------------
                _at#female |
(2 vs 1) (Female vs Male)  |          1       10.47     0.0012
(3 vs 1) (Female vs Male)  |          1        9.28     0.0023
(4 vs 1) (Female vs Male)  |          1        5.48     0.0192
(5 vs 1) (Female vs Male)  |          1        0.50     0.4807
(6 vs 1) (Female vs Male)  |          1        0.65     0.4203
                     Joint |          4       20.04     0.0005
--------------------------------------------------------------

----------------------------------------------------------------------------
                           |            Delta-method
                           |   Contrast   std. err.     [95% conf. interval]
---------------------------+------------------------------------------------
                _at#female |
(2 vs 1) (Female vs Male)  |   .0025709   .0007946      .0010134    .0041283
(3 vs 1) (Female vs Male)  |   .0053185    .001746      .0018964    .0087407
(4 vs 1) (Female vs Male)  |   .0068713   .0029351      .0011187    .0126239
(5 vs 1) (Female vs Male)  |   .0038458   .0054532     -.0068422    .0145339
(6 vs 1) (Female vs Male)  |  -.0102854   .0127614     -.0352972    .0147264
----------------------------------------------------------------------------


Variables that uniquely identify margins: age

file marginsplot3.svg saved as SVG format
```



From this we see the difference of effects between point 2 and 1 is .0026, with $\chi^2(1) = 10.47$ and $p = 0.0012$ — the same conclusion the `lincom` test gave above, and indeed the same test: $3.24^2 = 10.5$. The remaining rows compare each later age to age 20, and the widening gap is significant through age 50 ((4 vs 1): $\chi^2 = 5.48$, $p = 0.019$) before losing precision.

6.2. Summary

Overlapping confidence intervals do not mean two estimates are statistically indistinguishable. A test of the difference depends on the variance of the difference, $\mathrm{Var}(x-y) = \mathrm{Var}(x)+\mathrm{Var}(y)-2\,\mathrm{Cov}(x,y)$, and the covariance is nowhere to be seen in a plot of the two intervals separately. Estimates read off the same fitted model at nearby covariate values are strongly positively correlated — 0.99 in this example — so most of their variance cancels in the difference, and the difference is pinned down far more precisely than either level is. The eyeball test is therefore conservative in a specific and misleading way: it reports no difference where a proper test finds one.

The chapter's numbers make the size of the error concrete. The female-vs-male contrasts at ages 20 and 30 have intervals overlapping across most of their length, yet their difference is 0.0026 with a standard error of 0.00079, $z = 3.24$, $p = 0.001$. Ignoring the covariance and combining the two standard errors as $\sqrt{\mathrm{Var}(x) + \mathrm{Var}(y)}$ would give $z = 0.73$ and $p = 0.46$ — same data, opposite conclusion.

So do not read differences off a `marginsplot`. Ask for them directly:

- `margins r.varname` returns a contrast rather than two levels, with the covariance already handled.

- `margins ..., contrast(atcontrast(r))` compares every evaluation point to the reference in one call, which is what produced the last table above.
- `margins ..., post` followed by `lincom` or `test` handles any single contrast you can write down, using only built-in commands.

The plot is still the right tool for showing how an effect moves across a covariate. It is the wrong tool for judging whether two points on it differ.

7. Fixed or Random Effect, or Both?

7.1. Panel data

With panel data — repeated observations over time or observations clustered at a higher level — the standard choice is between fixed and random effects. Consider

$$y_{it} = \beta_0 + \beta_1 x_{it} + c_i + \epsilon_{it} \tag{7.1}$$

for individuals $i = 1, \dots, n$ measured at $t = 1, \dots, T$, where c_i is an unobserved time-invariant individual effect. The two approaches differ in how they handle c_i.

Fixed effects eliminate c_i by demeaning or by including individual dummies (the two are equivalent in a linear model). In nonlinear models only Poisson avoids the incidental-parameter problem; the standard solution for other models is conditional likelihood (conditional logit, for example), which absorbs the fixed effects in the likelihood. When no conditional likelihood exists, the incidental-parameter bias shrinks with panel depth but does not vanish.

Random effects treat c_i as part of the error term. This is more efficient but requires x_{it} to be uncorrelated with c_i — an assumption economists are generally skeptical of.

7.2. Time-invariant variables

Sometimes we want the effect of a time-invariant variable, so the model becomes

$$y_{it} = \beta_0 + \beta_1 x_{it} + c_i + \gamma z_i + \epsilon_{it} \tag{7.2}$$

Fixed effects cannot identify γ because z_i is perfectly collinear with c_i. Random effects can still estimate both, treating z_i as another covariate.

7.3. Between-within model

The textbook advice is a Hausman test: random effects are more efficient when the uncorrelatedness assumption holds; if not, fixed effects are still consistent. The between-within model (BW) sidesteps the choice. Neuhaus and Kalbfleisch (1998) introduced the BW estimator,

$$y_{it} = \beta_0 + \beta_1 (x_{it} - \bar{x}_i) + \beta_2 \bar{x}_i + c_i + \gamma z_i + \epsilon_{it} \tag{7.3}$$

Here β_1 is the within effect — identical to the fixed-effect coefficient. β_2 is the between effect (mean of x on mean of y). γ is the effect of the time-invariant variable.

The other specification of BW estimator is

$$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 \bar{x}_i + c_i + \gamma z_i + \epsilon_{it} \tag{7.4}$$

This is the same model reparametrized. β_1 is unchanged, and β_2 is now the between-minus-within difference (substitute $\beta_1(x_{it} - \bar{x}_i) + \beta_2 \bar{x}_i = \beta_1 x_{it} + (\beta_2 - \beta_1)\bar{x}_i$ to see this). This is the "contextual model"; β_2 is the contextual effect and acts as an embedded Hausman test.

The BW model has two advantages. It recovers the fixed-effect coefficient while estimating time-invariant covariates. And it extends naturally to cross-level interactions, random slopes, and other multilevel structures. Implementation is straightforward: run a random-effects model on either equation above.

7.3.1. Is this just Mundlak?

Yes. Equation 7.4 *is* the Mundlak device: Mundlak's regression puts x_{it} and $\bar{x}_i$ on the right-hand side of a random-effects model, which is exactly what that equation does. So the between-within model and the Mundlak device are one model in two parameterisations, fitted by the same random-effects command. It is worth doing the algebra, because the identity is easy to state loosely and easy to get wrong on the inference side.

Write $d_{it} = x_{it} - \bar{x}_i$ for the within deviation, and label the two specifications by their coefficients: b_1, b_2 for the hybrid form Equation 7.3 and p_1, p_2 for the contextual form Equation 7.4. Expanding the hybrid regressor block,

$$b_1(x_{it} - \bar{x}_i) + b_2 \bar{x}_i = b_1 x_{it} + (b_2 - b_1)\bar{x}_i, \tag{7.5}$$

so $p_1 = b_1$ and $p_2 = b_2 - b_1$. Substitution shows the map, but the reason behind it is worth stating more strongly: the two regressor matrices are related by an invertible linear transformation. With $X = [\, x_{it}, \bar{x}_i \,]$ and $W = [\, d_{it}, \bar{x}_i \,]$,

$$W = XA, \qquad A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \tag{7.6}$$

and $\det A = 1$, so the two designs span the same column space. Everything that depends only on that space is numerically identical: fitted values, residuals, the likelihood, the estimated variance components, and the GLS quasi-demeaning weight θ. Only the coordinates change, by $b = A^{-1}p$, and the covariance matrix by $\mathrm{Var}(b) = A^{-1}\mathrm{Var}(p)A^{-1\prime}$. Componentwise that is

$$\mathrm{Var}(p_1) = \mathrm{Var}(b_1), \qquad \mathrm{Var}(p_2) = \mathrm{Var}(b_1) + \mathrm{Var}(b_2) + 2\,\mathrm{Cov}(b_1, b_2), \tag{7.7}$$

which already says something useful: the standard error printed next to the between effect is not the standard error you want for the contextual effect.

7.3.1.1. Why the within coefficient is the fixed-effect estimator

This part needs a proof rather than a substitution. The relevant fact is that $\sum_t d_{it} = 0$ within every unit by construction, so d is orthogonal to any group-invariant column: for v_i constant within i,

$$\sum_{i,t} d_{it} v_i = \sum_i v_i \sum_t d_{it} = 0, \tag{7.8}$$

which covers the intercept, $\bar{x}_i$ and z_i alike.

Random effects is OLS after quasi-demeaning, $\tilde{w}_{it} = w_{it} - \theta \bar{w}_i$ with $\theta = 1 - \sqrt{\sigma_\epsilon^2 / (\sigma_\epsilon^2 + T\sigma_c^2)}$. Apply it to the hybrid regressors. Because $\bar{d}_i = 0$ we get $\tilde{d}_{it} = d_{it}$: the within-deviation regressor passes through the GLS transform untouched, whatever θ is. The remaining regressors are group-invariant and stay so after scaling by $1 - \theta$. Orthogonality therefore survives the transform, and by Frisch-Waugh-Lovell

$$b_1 = \frac{\sum d_{it} \tilde{y}_{it}}{\sum d_{it}^2} = \frac{\sum d_{it} y_{it}}{\sum d_{it}^2} = \frac{\sum (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum (x_{it} - \bar{x}_i)^2}, \tag{7.9}$$

the second and third equalities both by Equation 7.8. That last expression is the within estimator. Note that θ has dropped out, so the identity holds for *any* θ — for random effects, for pooled OLS ($\theta = 0$) and for fixed effects ($\theta = 1$). That is why running a random-effects model on either equation returns the fixed-effect coefficient. It also gives $\mathrm{Cov}(b_1, b_2) = 0$ exactly, which combined with Equation 7.7 yields $\mathrm{Var}(p_2) = \mathrm{Var}(b_1) + \mathrm{Var}(b_2)$ and $\mathrm{Cov}(p_1, p_2) = -\mathrm{Var}(b_1)$.

All of this is checkable on the chapter's own data. We take a single regressor, weeks worked, fit both parameterisations by `plm`, and compare against the within estimator and against the variance map:

```
library(plm); library(panelr); data("WageData")
d <- WageData
d$xbar <- ave(d$wks, d$id)           # unit mean
d$xdev <- d$wks - d$xbar             # within deviation

fe  <- plm(lwage ~ wks,          data = d, index = c("id", "t"), model = "within")
hyb <- plm(lwage ~ xdev + xbar,  data = d, index = c("id", "t"), model = "random")
mun <- plm(lwage ~ wks  + xbar,  data = d, index = c("id", "t"), model = "random")

b <- coef(hyb); p <- coef(mun); Vh <- vcov(hyb); Vm <- vcov(mun)

data.frame(
  check = c("p1 - b1", "p2 - (b2 - b1)", "b1 - FE within",
            "sum(xdev * xbar)", "Cov(b1, b2)",
            "Var(p2) - [Var(b1) + Var(b2)]", "Cov(p1, p2) + Var(b1)",
            "theta(hybrid) - theta(Mundlak)"),
  value = c(p["wks"] - b["xdev"],
            p["xbar"] - (b["xbar"] - b["xdev"]),
            b["xdev"] - coef(fe)["wks"],
```

```
            sum(d$xdev * d$xbar),
            Vh["xdev", "xbar"],
            Vm["xbar", "xbar"] - (Vh["xdev", "xdev"] + Vh["xbar", "xbar"]),
            Vm["wks", "xbar"] + Vh["xdev", "xdev"],
            hyb$ercomp$theta - mun$ercomp$theta),
  row.names = NULL)
```

```
                              check         value
1                           p1 - b1  2.385245e-18
2                     p2 - (b2 - b1)  7.459311e-17
3                      b1 - FE within  1.886512e-17
4                   sum(xdev * xbar)  3.534950e-12
5                       Cov(b1, b2) -5.676184e-21
6  Var(p2) - [Var(b1) + Var(b2)] -4.404571e-20
7          Cov(p1, p2) + Var(b1)  2.117582e-21
8 theta(hybrid) - theta(Mundlak)  1.110223e-16
```

Every entry is zero to machine precision. The two parameterisations are the same fit, the within coefficient is the fixed-effect estimate, the hybrid regressors are orthogonal, and the variance map holds as derived.

7.3.1.2. So why keep both

The fit is one object, but the parameterisation settles three practical things.

Which quantity is printed. b_2 is the between effect and p_2 is between minus within. They are different parameters with non-interchangeable standard errors, and whichever form you fit, the other costs a `lincom`.

Which hypothesis is one line. $p_2 = 0$ says between equals within, which is exactly the random-effects assumption that the regressors are uncorrelated with c_i. The Mundlak form prints that as a single t; in hybrid form it is the contrast $b_1 = b_2$. The two tests agree numerically, but only one is printed. This is the Mundlak form of the Hausman test, asymptotically equivalent to the classical version and with the practical advantage of working directly with cluster-robust standard errors. The next chapter develops it and its two-way extension for difference-in-differences.

What the second coefficient is for. In multilevel work p_2 is not a diagnostic but the estimand: holding a pupil's own status fixed, what does the school mean do? The hybrid's b_2 answers a different question, the total between-unit association. Two literatures ask two questions of one regression, and each named the parameterisation that printed its answer. That is the honest reason both names survive.

There is also a reason to prefer the hybrid form in extensions. Its two regressors are exactly orthogonal, where x_{it} and $\bar{x}_i$ are correlated. That does not change the fit, but if you want a random slope on the within component you need the deviation as its own column; a random slope on x_{it} would mix within and between variation.

7.3.1.3. Where the equivalence stops

The reparameterisation is exact for anything with a linear index, logit and Poisson mixed models included, since XA spans the same space and the index is unchanged. The two names stay interchangeable there.

The fixed-effect identity does not carry over. The proof above used two linear-model facts: that random effects is OLS on quasi-demeaned data, and that d is orthogonal to group constants. A logit or Poisson mixed model has no such transform, the unit effects do not sweep out, and the hybrid within coefficient is not the conditional-logit estimate but an approximation to it. Keep the two claims separate: the parameterisations are always the same model, while "within coefficient equals fixed effects" is a linear-model result. The last section of this chapter returns to that point.

7.4. BW model in R

R's `panelr` package implements BW models directly. The data are `WageData`, a balanced panel of 595 men observed over 7 years. We model log wage on weeks worked, union membership, marital status and occupation, all time-varying, plus two time-invariant covariates — black and female — that a fixed-effect model could not touch.

```r
library(panelr)
data("WageData")
wages <- panel_data(WageData, id = id, wave = t)
model1 <- wbm(lwage ~ wks + union + ms + occ | blk + fem, data = wages)
summary(model1)
```

```
MODEL INFO:
Entities: 595
Time periods: 1-7
Dependent variable: lwage
Model type: Linear mixed effects
Specification: within-between

MODEL FIT:
AIC = 2036.78, BIC = 2119.13
Pseudo-R² (fixed effects) = 0.27
Pseudo-R² (total) = 0.69
Entity ICC = 0.57

WITHIN EFFECTS:
------------------------------------------
              Est.   S.E.   t val.      p
----------- ------- ------ -------- ------
wks           0.00   0.00     1.06   0.29
union         0.06   0.03     2.53   0.01
ms           -0.08   0.03    -2.57   0.01
```

```
occ             -0.08   0.02    -3.32   0.00
------------------------------------------

BETWEEN EFFECTS:
-------------------------------------------------
                       Est.   S.E.   t val.      p
------------------ ------- ------ -------- ------
(Intercept)            6.30   0.20    30.85   0.00
imean(wks)             0.01   0.00     2.25   0.02
imean(union)           0.15   0.03     4.67   0.00
imean(ms)              0.17   0.05     3.07   0.00
imean(occ)            -0.41   0.03   -13.31   0.00
blk                   -0.15   0.05    -2.81   0.00
fem                   -0.32   0.06    -4.96   0.00
-------------------------------------------------

p values calculated using df = 4153

RANDOM EFFECTS:
------------------------------------
  Group      Parameter    Std. Dev.
---------- ------------- -----------
    id      (Intercept)    0.2992
 Residual                  0.2589
------------------------------------
```

The within estimates are 0.00 for weeks, 0.06 for union, −0.08 for married and −0.08 for occupation. The between estimates for the same four variables are 0.01, 0.15, 0.17 and −0.41 — two to five times larger, and for `ms` the opposite sign. That gap is the reason the fixed-versus-random choice matters here: a random-effects model without the decomposition would blend the two and report neither. The time-invariant covariates come out at −0.15 for black and −0.32 for female, quantities a fixed-effect model cannot produce at all.

Compare with `lfe`.

```r
library(lfe)
model2 <- felm(lwage ~ wks + union + ms + occ | id, data = wages)
summary(model2)
```

```
Call:
   felm(formula = lwage ~ wks + union + ms + occ | id, data = wages)

Residuals:
     Min       1Q   Median       3Q      Max
-1.89500 -0.16174  0.00652  0.17060  1.94521
```

```
Coefficients:
       Estimate Std. Error t value Pr(>|t|)
wks    0.001083   0.001019   1.063 0.287816
union  0.064320   0.025378   2.534 0.011305 *
ms    -0.082905   0.032226  -2.573 0.010132 *
occ   -0.077507   0.023359  -3.318 0.000916 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2589 on 3566 degrees of freedom
Multiple R-squared(full model): 0.7304   Adjusted R-squared: 0.6852
Multiple R-squared(proj model): 0.006509   Adjusted R-squared: -0.1601
F-statistic(full model):16.16 on 598 and 3566 DF, p-value: < 2.2e-16
F-statistic(proj model): 5.841 on 4 and 3566 DF, p-value: 0.0001106
```

`felm` gives 0.001083, 0.064320, −0.082905 and −0.077507 — the same four numbers `panelr` reported as within effects, identical to machine precision (`WageData` is a balanced panel: 7 waves, 595 individuals). That is the BW model's selling point: it reproduces the fixed-effect estimates exactly while also estimating the time-invariant covariates `blk` and `fem`, which `felm` cannot. `lfe` supports clustered standard errors, which `panelr` does not.

```r
model3 <- felm(lwage ~ wks + union + ms + occ | id | 0 | id, data = wages)
summary(model3)
```

```
Call:
   felm(formula = lwage ~ wks + union + ms + occ | id | 0 | id,      data = wages)

Residuals:
     Min       1Q   Median       3Q      Max
-1.89500 -0.16174  0.00652  0.17060  1.94521

Coefficients:
       Estimate Cluster s.e. t value Pr(>|t|)
wks    0.001083     0.001331   0.814   0.4160
union  0.064320     0.040936   1.571   0.1167
ms    -0.082905     0.047399  -1.749   0.0808 .
occ   -0.077507     0.031320  -2.475   0.0136 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2589 on 3566 degrees of freedom
Multiple R-squared(full model): 0.7304   Adjusted R-squared: 0.6852
Multiple R-squared(proj model): 0.006509   Adjusted R-squared: -0.1601
F-statistic(full model, *iid*):16.16 on 598 and 3566 DF, p-value: < 2.2e-16
F-statistic(proj model): 3.456 on 4 and 594 DF, p-value: 0.008358
```

Clustering on `id` leaves every coefficient untouched and changes the standard errors substantially: `union` goes from 0.0254 to 0.0409, `ms` from 0.0322 to 0.0474, `wks` from 0.00102 to 0.00133. Two of the four results do not survive. `union` falls from $p = 0.011$ to $p = 0.117$ and `ms` from 0.010 to 0.081; only `occ` stays significant at 5%. With 7 observations per man the residuals are serially correlated, and the unclustered standard errors were treating 4,165 observations as though they were independent draws.

7.5. BW model in Stata

Stata has no dedicated BW command, but `xtreg` suffices. We switch to `nlswork`, an unbalanced panel of 4,710 women over 28,510 person-years, and regress log wage on age.

```
webuse nlswork
xtset idcode
xtreg ln_w age, fe cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)


Fixed-effects (within) regression               Number of obs     =     28,510
Group variable: idcode                          Number of groups  =      4,710

R-squared:                                      Obs per group:
     Within  = 0.1026                                         min =          1
     Between = 0.0877                                         avg =        6.1
     Overall = 0.0774                                         max =         15

                                                F(1, 4709)        =     884.05
corr(u_i, Xb) = 0.0314                          Prob > F          =     0.0000

                             (Std. err. adjusted for 4,710 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0181349   .0006099    29.73   0.000     .0169392    .0193306
       _cons |   1.148214   .0177153    64.81   0.000     1.113483    1.182944
-------------+----------------------------------------------------------------
     sigma_u |  .40635023
     sigma_e |  .30349389
         rho |  .64192015   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

The fixed-effect estimate is .0181349 with a clustered standard error of .0006099: within a woman, each additional year of age raises log wage by about 1.8%. Keep that number in view — the BW model has to reproduce it.

Generate the group mean and run the BW estimation.

```
webuse nlswork
xtset idcode
bysort idcode: center age, prefix(d) mean(m)
xtreg ln_w dage mage i.race, re cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)

(generated variables: dage mage)


Random-effects GLS regression                   Number of obs     =     28,510
Group variable: idcode                          Number of groups  =      4,710

R-squared:                                      Obs per group:
     Within  = 0.1026                                         min =          1
     Between = 0.1040                                         avg =        6.1
     Overall = 0.0950                                         max =         15

                                                Wald chi2(4)      =    1335.89
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

                             (Std. err. adjusted for 4,710 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        dage |   .0181349     .00061    29.73   0.000     .0169394    .0193304
        mage |    .022558   .0011405    19.78   0.000     .0203226    .0247933
             |
        race |
      Black  |  -.1190246   .0127418    -9.34   0.000    -.1439982   -.0940511
      Other  |   .0974996   .0617364     1.58   0.114    -.0235016    .2185008
             |
       _cons |   1.037566   .0323185    32.10   0.000     .9742233     1.10091
-------------+----------------------------------------------------------------
     sigma_u |  .36581005
     sigma_e |  .30349389
         rho |  .59230575   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

In this BW model, we use the centered (within-group-demeaned) `dage` together with the group mean `mage`, which gives the standard within-between (Mundlak) decomposition: the coefficient on `dage`, .0181349, is exactly the fixed-effect ("within") coefficient on age reported above — the same seven digits, with a standard error of .00061 against .0006099 — and the coefficient on `mage`, .022558, is directly the between effect. Women who are older on average earn about 2.3% more per year of average age, against a within-woman return of 1.8%; the between effect is larger because it also carries cohort and selection differences that the within estimate sweeps out. And we have the effect of time-invariant covariate race estimated: Black women earn 11.9% less (−.1190246, SE .0127), while the "Other" category is not distinguishable from white (.0975, $p = 0.114$). The advantage of using xtreg is that we have clustered standard errors implemented.

Note: if we instead ran `xtreg ln_w age mage i.race, re cluster(idcode)` — using the raw (uncentered) `age` together with `mage` — we would get the *contextual model* instead: the coefficient on `age` stays .0181, but the coefficient on `mage` becomes .0044, the "contextual effect" (the *additional* between-effect on top of the within effect). The two models are algebraically related: contextual-model mage = Mundlak-model mage − Mundlak-model dage (here, .0226 − .0181   .0044).

7.6. BW model in non-linear models

Allison applies the BW model to binary outcomes. The bias relative to conditional logit is an open question, but if the linear probability model is a reasonable approximation the BW approach should perform similarly for binary panel data.

8. Mundlak device in fixed effect models

8.1. TWFE

The two-way fixed-effect model is standard for difference-in-differences with panel data:

$$y_{it} = X_{it}\beta + c_i + f_t + u_{it} \tag{8.1}$$

with k covariates, T periods, and N units. In a linear model we can include N unit dummies and T time dummies, but this becomes impractical when either dimension is large. In nonlinear models the problem is worse.

8.2. TWM

Wooldridge (2021) shows that the two-way Mundlak regression (TWM) works for both common and staggered treatment timing. This chapter covers the common-timing case; staggered timing is treated separately.

The idea goes back to Mundlak (1978). In the one-way case,

$$y_{it} = x_{it}\beta + c_i + u_{it} \tag{8.2}$$

Mundlak models c_i as a function of the mean of x_{it} for each unit:

$$c_i = \bar{x}_i\theta + \eta_i \tag{8.3}$$

Then

$$y_{it} = x_{it}\beta + \bar{x}_i\theta + \eta_i + u_{it} \tag{8.4}$$

The composite error is $\eta_i + u_{it}$. Instead of N unit dummies we add k group means. This model can be estimated by pooled OLS or random effects; the coefficients are identical, and so are the standard errors once we cluster on units.

A by-product: testing $\theta = 0$ is the Mundlak form of the Hausman test. Under the random-effects assumptions it is asymptotically equivalent to the classical Hausman test, with the practical advantage of working directly with cluster-robust standard errors.

```
clear
webuse nlswork
xtset idcode year
drop if union==.
drop if age==.
bysort idcode: egen mean_age=mean(age)
bysort idcode: egen mean_union=mean(union)
xtreg ln_wage age union, fe cluster(idcode)
reg ln_wage age union mean_age mean_union, cluster(idcode)
xtreg ln_wage age union mean_age mean_union, re cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)
 Time variable: year, 68 to 88, but with gaps
         Delta: 1 unit

(9,296 observations deleted)

(9 observations deleted)




Fixed-effects (within) regression               Number of obs     =     19,229
Group variable: idcode                          Number of groups  =      4,150

R-squared:                                      Obs per group:
     Within  = 0.0963                                         min =          1
     Between = 0.0433                                         avg =        4.6
     Overall = 0.0562                                         max =         12

                                                F(2, 4149)        =     331.52
corr(u_i, Xb) = 0.0127                          Prob > F          =     0.0000

                             (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139956    .0167058
       union |   .1055274   .0098582    10.70   0.000     .0862001    .1248546
       _cons |   1.248435   .0215661    57.89   0.000     1.206154    1.290716
-------------+----------------------------------------------------------------
     sigma_u |  .42353003
```

```
     sigma_e |  .26213464
         rho |  .72302816   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Linear regression                               Number of obs     =     19,229
                                                F(4, 4149)        =     234.77
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0769
                                                Root MSE          =     .44953

                             (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139956    .0167059
       union |   .1055274   .0098587    10.70   0.000     .0861991    .1248556
    mean_age |  -.0064208    .001548    -4.15   0.000    -.0094556   -.0033859
  mean_union |   .1900785   .0223172     8.52   0.000     .1463249    .2338321
       _cons |   1.405264    .042734    32.88   0.000     1.321482    1.489046
------------------------------------------------------------------------------


Random-effects GLS regression                   Number of obs     =     19,229
Group variable: idcode                          Number of groups  =      4,150

R-squared:                                      Obs per group:
     Within  = 0.0963                                         min =          1
     Between = 0.0750                                         avg =        4.6
     Overall = 0.0768                                         max =         12

                                                Wald chi2(4)      =    1023.99
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

                             (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139959    .0167055
       union |   .1055274   .0098587    10.70   0.000     .0862047      .12485
    mean_age |  -.0061464   .0014254    -4.31   0.000    -.0089401   -.0033527
  mean_union |    .212183   .0210098    10.10   0.000     .1710046    .2533614
       _cons |   1.362101   .0383784    35.49   0.000     1.286881    1.437322
-------------+----------------------------------------------------------------
     sigma_u |  .38505558
     sigma_e |  .26213464
```

```
         rho |  .68331727   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

All three models return the same coefficients and standard errors on age and union. On the 19,229 person-years of `nlswork` with non-missing union and age, every one of them gives `age` = .0153507 with a clustered standard error of .00069 and `union` = .1055274 with .00986 — agreeing to the last printed digit, with only the confidence bounds differing in the seventh decimal. The N unit dummies of `xtreg, fe` have been replaced by two group means, and nothing was lost.

Wooldridge (2021) extends this to the two-way case. For the TWFE,

$$y_{it} = X_{it}\beta + c_i + f_t + u_{it} \tag{8.5}$$

it's equivalent to estimate

$$y_{it} = X_{it}\beta + X_{i\cdot}\lambda + X_{\cdot t}\theta + u_{it} \tag{8.6}$$

where $X_{i\cdot}$ is the unit-specific average over time and $X_{\cdot t}$ is the time-specific average over units. By FWL, the two models are equivalent (see Wooldridge 2021 for the proof). Wooldridge calls this the two-way Mundlak (TWM) model.

TWM has two advantages over the dummy-variable approach. It can sharply reduce the number of regressors. And it allows time-invariant or unit-invariant variables in the regression without disturbing the β estimates.

We can see in the following example of POLS (which is TWM), we add in `race`, which is **time-invariant** — constant within a person, but varying across people. (In `nlswork` the within-person standard deviation of `race` is exactly 0, while within any given year it is about 0.50, so it is emphatically not unit-invariant.) It can be estimated in TWM, but would not be estimable in TWFE. Notice that coefficients on age and union remain unchanged.

The reason is *not* that `race` is orthogonal to the other regressors — it is not orthogonal to the unit means `mean_age` and `mean_union`. It is that the variation identifying β is the within-unit variation, and within-unit deviations are orthogonal to **any** unit-level variable by construction, since $\sum_t (x_{it} - \bar{x}_i) = 0$ for every i. Adding a unit-level regressor therefore cannot move β, whatever that regressor happens to be. A genuinely unit-invariant variable — one constant across units within a period, such as a time dummy — leaves β unchanged for the mirror-image reason.

```
reg ln_wage age union mean_age mean_union i.race, cluster(idcode)
```

```
Linear regression                               Number of obs     =     19,229
                                                F(6, 4149)        =     189.66
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1069
                                                Root MSE          =      .4422

                              (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
```

```
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006913    22.21   0.000     .0139955    .0167059
       union |   .1055274   .0098592    10.70   0.000     .0861981    .1248566
    mean_age |  -.0070345   .0015342    -4.59   0.000    -.0100424   -.0040266
  mean_union |   .2227511   .0220026    10.12   0.000     .1796142     .265888
             |
        race |
      Black  |  -.1790995   .0143624   -12.47   0.000    -.2072575   -.1509415
      Other  |   .0882461   .0715315     1.23   0.217    -.0519939    .2284861
             |
       _cons |   1.466452   .0426087    34.42   0.000     1.382916    1.549988
------------------------------------------------------------------------------
```

And indeed `age` stays at .0153507 and `union` at .1055274, unchanged to seven digits by the addition of `i.race`, while race itself is estimated: Black women earn 17.9% less (−.1790995, SE .0144). A fixed-effect model could report the first two numbers but not the third.

8.3. An important qualification: TWM needs a balanced panel

The two-way equivalence above holds for a **balanced** panel. It is worth stating this explicitly, because the one-way result does *not* need balance and the contrast is easy to miss.

The one-way version — regressing y_{it} on X_{it} and $X_{i\cdot}$ — is exact whatever the panel looks like, for the reason given below: $\sum_t (x_{it} - \bar{x}_i) = 0$ holds within each unit by construction, no matter how many periods that unit contributes.

The two-way version needs something stronger. Subtracting both margins and adding back the grand mean, $x_{it} - \bar{x}_{i\cdot} - \bar{x}_{\cdot t} + \bar{x}$, is the projection onto *both* sets of dummies only when the unit and time factors are **orthogonal** — that is, when every unit appears in every period. In a balanced panel that holds automatically. In an unbalanced panel it fails, and TWM is then simply a different estimator from TWFE.

The difference is large enough to matter in applied work. Using `nlswork`, which is unbalanced (persons contribute between 1 and 12 observations), the coefficient on `age` differs by more than a factor of two:

```
* preserve/restore so this illustration leaves the data as later chunks expect
preserve
webuse nlswork, clear
keep if !missing(ln_wage, age, union)
xtset idcode year

bysort idcode: egen m_age   = mean(age)
bysort idcode: egen m_union = mean(union)
egen t_age   = mean(age),   by(year)
```

```
egen t_union = mean(union), by(year)

* unit means of the year dummies -- the terms the unbalanced case needs
levelsof year, local(yrs)
foreach y of local yrs {
    bysort idcode: egen my_`y' = mean(year==`y')
}

quietly xtreg ln_wage age union i.year, fe
scalar b_twfe = _b[age]
quietly regress ln_wage age union m_age m_union t_age t_union
scalar b_twm = _b[age]
quietly regress ln_wage age union m_age m_union i.year my_*
scalar b_fix = _b[age]

di "TWFE                      age = " %9.7f b_twfe
di "TWM (unit + time means)   age = " %9.7f b_twm
di "TWM + unit means of dummies   = " %9.7f b_fix

restore
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

(9,305 observations deleted)


Panel variable: idcode (unbalanced)
 Time variable: year, 70 to 88, but with gaps
         Delta: 1 unit




70 71 72 73 77 78 80 82 83 85 87 88




TWFE                       age = 0.0276645

TWM (unit + time means)    age = 0.0117968
```

```
TWM + unit means of dummies   = 0.0276646
```

The numbers make the point. TWFE gives `age` = 0.0276645; two-way Mundlak with unit and time means gives 0.0117968 — 43% of it, a factor of 2.35 apart. These are not two estimates of the same thing that happen to disagree; on an unbalanced panel they are different estimators.

What restores the equivalence is including the **unit means of the time dummies**. Adding them brings TWM to 0.0276646 against TWFE's 0.0276645 — agreement to six decimal places, on the same unbalanced data that produced the factor-of-two gap. They have a clear meaning: the unit mean of the dummy for period t is s_{it}/T_i, where s_{it} indicates whether unit i is observed in period t. So adding them amounts to conditioning on *which* periods each unit is observed in — the panel's selection pattern. This is the conditioning set in Wooldridge's work on correlated random effects with unbalanced panels.

Two practical notes. In a balanced panel these extra terms have zero variance — every unit is observed in every period, so each mean is $1/T$ for everybody — which is exactly why they never come up in the balanced case. And the number of observations per unit, T_i, is *not* a substitute for them: it does not restore exactness, and it is collinear with them once they are included.

Everything in this section is about the *point estimate*. Standard errors from TWM are not the TWFE standard errors, and should not be read as such.

It is worth pausing on the fact that this chapter now demonstrates both results on the *same* data. The `xtreg` example near the top shows one-way Mundlak matching one-way FE on `nlswork`, and the chunk just above shows two-way Mundlak failing on `nlswork`. Same dataset, opposite verdicts, because the two claims do not have the same requirements. The one-way result needs nothing: $\sum_t (x_{it} - \bar{x}_i) = 0$ holds within each unit whatever periods that unit contributes, so T_i may vary freely. The two-way result needs the unit and time factors to be orthogonal *to each other*, which is a statement about the relationship between two dimensions, and that is what imbalance destroys. Nothing one-way can be affected, because there is no second dimension for the first to fail to be orthogonal to.

Which regressors need means, then? Those that vary within unit. This is why `i.race` above takes no mean: `race` is constant within a person, so its within-person mean *is* `race` — in `nlswork` the two agree to the last digit — and adding it would be perfectly collinear. A year dummy is the opposite case: it does vary within a person (in `nlswork` the within-person standard deviation of the 1971 dummy is about 0.71), so its unit mean is not the dummy but the fraction of that person's observations falling in that year. That fraction is new information, and in an unbalanced panel it is exactly the information the equivalence needs.

8.4. More than two fixed effects: one set of means, not one per dimension

The natural guess, having seen TWM, is that three fixed effects call for three sets of means — $X_{i\cdot}$, $X_{\cdot t}$, and the averages over the third factor. That guess is wrong in general, and the reason is worth spelling out because it also explains the balanced-panel condition above.

The Mundlak device deals with **one** unobserved effect at a time. It replaces c_i by its projection on the within-i means of the other regressors. So the number of sets of means you need is the number of dimensions you are *eliminating*, not the number of dimensions you have. With three fixed effects the robust recipe is:

- pick the dimension you cannot estimate — typically the one with many levels and few observations each — and replace it with means;
- **keep the other dimensions as dummies**, since they have few levels and many observations each;
- take the means over the chosen dimension of *everything else in the model*, the retained dummies included.

That last clause is what does the work, and it is the same clause that fixes the unbalanced two-way case above. The test for whether a given regressor needs a mean is the one stated there: does it vary within the dimension you are eliminating? Time and group dummies do, so their means carry information; a regressor already constant within unit does not, so its mean is itself. Call this construction **B**, and call the three-sets-of-means guess **A**.

```
preserve
capture program drop twm3
program define twm3
    args label
    quietly {
        capture drop x_i x_t x_g mt_* mg_* x y ft hg
        sort t
        by t: gen double ft = rnormal() if _n==1
        by t: replace ft = ft[1]
        sort g
        by g: gen double hg = rnormal() if _n==1
        by g: replace hg = hg[1]
        gen double x = 0.6*ci + 0.5*ft + 0.4*hg + rnormal()
        gen double y = 1*x + ci + ft + hg + rnormal()
        bysort i: egen double x_i = mean(x)
        bysort t: egen double x_t = mean(x)
        bysort g: egen double x_g = mean(x)
        levelsof t, local(ts)
        foreach k of local ts {
            bysort i: egen double mt_`k' = mean(t==`k')
        }
        levelsof g, local(gs)
        foreach k of local gs {
            bysort i: egen double mg_`k' = mean(g==`k')
        }
        areg y x i.t i.g, absorb(i)
        scalar b_fe = _b[x]
        regress y x x_i x_t x_g
        scalar b_A = _b[x]
        regress y x x_i i.t i.g mt_* mg_*
        scalar b_B = _b[x]
    }
```

```
    di "`label'"
    di "   three-way FE                        = " %9.6f b_fe
    di "   A: one set of means per dimension   = " %9.6f b_A  "   gap " %9.6f (b_A-b_fe)
    di "   B: means over i, dummies included   = " %9.6f b_B  "   gap " %9.6f (b_B-b_fe)
end

clear
set seed 20260731
set obs 300
gen i = _n
gen double ci = rnormal()
expand 10
bysort i: gen t = _n

* third factor assigned at random: the three factors are not orthogonal
gen g = 1 + int(runiform()*5)
twm3 "g assigned at random (NOT orthogonal):"

* Latin square: every unit sees every g equally, and so does every period
replace g = mod(i+t,5)+1
twm3 "g from a Latin square (orthogonal):"

capture program drop twm3
restore
```

```
Number of observations (_N) was 0, now 300.



(2,700 observations created)




g assigned at random (NOT orthogonal):
   three-way FE                        =  0.994424
   A: one set of means per dimension   =  1.007648   gap  0.013224
   B: means over i, dummies included   =  0.994424   gap  0.000000

(2,435 real changes made)

g from a Latin square (orthogonal):
   three-way FE                        =  0.996471
   A: one set of means per dimension   =  0.996471   gap  0.000000
   B: means over i, dummies included   =  0.996471   gap  0.000000
```

With g assigned at random, the three-way fixed-effect estimate is 0.994424. Construction B reproduces it exactly, gap 0.000000; construction A gives 1.007648, a gap of 0.013224. Replace the random assignment with a Latin square, which makes the three factors mutually orthogonal, and all three agree at 0.996471 with both gaps at 0.000000.

So **A is not incorrect but conditional**: it is exact when the factors are mutually orthogonal, and inexact otherwise. A full factorial design gives orthogonality, but so does a Latin square, which fills only a fraction of the possible cells — what matters is the balance property, that each level of one factor appears equally often with each level of the others, not the number of cells. Deleting a single row from either design is enough to lose exactness, and the size of the gap tracks how far from orthogonal the design is.

This is the general statement of the balanced-panel condition from the previous section. A balanced two-way panel *is* an orthogonal design: every unit appears in every period. Unbalance it, and the unit and time factors stop being orthogonal, which is why the two-way equivalence fails there.

The practical reading is that orthogonality is a property of **designed** data — factorial experiments, Latin squares, crop trials, where somebody chose the assignment. It requires each level of one factor to appear equally often with each level of the others, which is not something that happens by accident in data you did not design. A third factor tends to make matters worse rather than better, because it is frequently determined in part by the first two. In an audit panel I looked at recently, for instance, the third fixed effect was the running count of audits within a factory, which is mechanically a function of factory and calendar time; that design cannot be orthogonal even in principle. For observational work, B is the construction to use.

8.5. Applications of TWM to DID in common treatment timing

There is an extension: if x_{it} can be written as an interaction between a time-invariant and a unit-invariant variable,

$$x_{itj} = z_{ij} * m_{tj} \tag{8.7}$$

Then

$$\bar{x}_{i \cdot j} = z_{ij} \bar{m}_j \tag{8.8}$$

$$\bar{x}_{\cdot tj} = \bar{z}_j m_{tj} \tag{8.9}$$

The two averages are just constants times z_{ij} and m_{tj} respectively. So by Mundlak we can include z_{ij} and m_{tj} as controls instead.

Wooldridge (2021) shows that TWM extends to staggered timing as well, provided we allow flexible interactions of treatment cohorts and periods. The simulation below covers the common-timing case. The code follows Wooldridge's do files.

In the common-timing setting all treated units are treated at the same period. The identifying assumptions are parallel trends and no anticipation.

$$y_{it} = w_{it}\beta + c_i + f_t + u_{it} \tag{8.10}$$

where w_{it} is the treatment indicator, with T periods and treatment happens at $t = q$.

$$w_{it} = d_i \cdot p_t \tag{8.11}$$

where d_i is the treatment group dummy, and p_t is the post treatment dummy.

$$\bar{w}_{i\cdot} = d_i \cdot \bar{p} \tag{8.12}$$

$$\bar{w}_{\cdot t} = \bar{d} \cdot p_t \tag{8.13}$$

Therefore, we just need to include d_i and p_t as controls in the regression to be the same estimator as TWFE, by the spirit of Mundlak.

That is, instead of doing TWFE, we can do TWM by regressing y on w, d_i and p_t.

Moreover, in a TWM, we can include any time-invariant or unit-invariant controls, or both and have no effect on β.

We can also allow the effect to change across time post treatment by doing

$$y_{it} = \alpha + \beta_q(w_{it} \cdot f_q) + \cdots + \beta_T(w_{it} \cdot f_T) + \eta d_i + \theta_q f_q + \cdots + \theta_T f_T + e_{it} \tag{8.14}$$

In the TWM, we allow effects change from $t = q$ to $t = T$, by interacting w_{it} and f_t, which is an dummy for time t. Very flexible, yet easy to do, given everything is linear.

Even more, if we have another variable, x_i, say gender, which is time-invariant, we can allow the effect to be different across gender, across time!

Now I use Wooldridge's code to run on a simulated data set.

The simulation has 100 units observed over 2012–2015, 400 observations, with common treatment timing. Each unit draws time-invariant $w_1, w_2 \sim N(0,1)$. Units with `id > 50` are treated ($d = 1$, so 50 treated and 50 control), the post period is 2014–2015, and $w_{it} = d_i \cdot \text{post}_t$. The covariate is $x_1 = N(0,1) + w_1 + 2w_2$, correlated with the unit effects by construction, and the untreated outcome is $y_1 = 1 + 2x_1 + 3w_1 + u$ with $u \sim N(0,1)$. The observed outcome adds a treatment effect that grows over time,

$$\text{logy} = y_1 + 2.5\, d_i f_{2014} + 3.5\, d_i f_{2015}. \tag{8.15}$$

So the true effect is 2.5 in the first post year and 3.5 in the second, and because the panel is balanced the overall ATT is their average, exactly 3.

```
clear
set obs 100
set seed 1234567
gen id = _n
gen w1 = rnormal()
gen w2 = rnormal()
expand 4
```

```
bysort id: gen year=2011+_n
gen post=(year>2013)
gen d=(id>50)
gen w=d*post
gen x1 = rnormal() + w1 + 2*w2
gen u = rnormal()

gen f2014=(year==2014)
gen f2015=(year==2015)

gen y1 = 1 +2*x1 + 3*w1  + u
* let's allow each year's treatment effect being different.  ATT for post period would be rougl
gen logy = y1 + 2.5*d*f2014 + 3.5*d*f2015

xtset id year
tab year w
```

```
Number of observations (_N) was 0, now 100.
```

```
(300 observations created)
```

```
Panel variable: id (strongly balanced)
 Time variable: year, 2012 to 2015
         Delta: 1 unit

           |           w
      year |         0          1 |     Total
-----------+----------------------+----------
      2012 |       100          0 |       100
      2013 |       100          0 |       100
      2014 |        50         50 |       100
      2015 |        50         50 |       100
```

```
-----------+----------------------+----------
     Total |       300        100 |       400
```

We have 100 units over four years (2012–2015). Half are treated starting in 2014 ($w = d \times \text{post}$). The treatment effect is 2.5 in 2014 and 3.5 in 2015, so the overall post-period ATT should be about 3 in this balanced panel.

The code below is from Wooldridge. The main difficulty is specifying the correct interaction terms for each estimand.

```
* CACHE WARNING: this chunk depends on the data built by the earlier
* `collectcode: true` chunks.  If those are served from the knitr cache while
* this one re-executes, their code never enters Statamarkdown's accumulated
* buffer, every command below fails on missing data, and the chunk silently
* renders with NO output.  If that happens, delete the whole
* `mundlak-device_cache/` directory (not just this chunk's entry) and
* `_freeze/mundlak-device/`, then re-render so all chunks run in order.


* Basic DID regression. Only need the post-treatment dummy but including
* all time dummies is harmless (and necessary for heterogeneous trends).
* Standard errors change a little because of different degrees-of-freedom.

reg logy c.d#c.post d post, vce(cluster id)
reg logy w d post, vce(cluster id)
reg logy w d f2014 f2015, vce(cluster id)
reg logy w d i.year, vce(cluster id)

* Using fixed effects is equivalent:

xtreg logy w i.year, fe vce(cluster id)
xtreg logy w post, fe vce(cluster id)

* So is RE with d included:
xtreg logy w d post, re vce(cluster id)

* Allow a separate effect in each of the treated time periods:
* It is the ATT for each treated period:

xtreg logy c.w#c.f2014 c.w#c.f2015 i.year, fe vce(cluster id)
xtreg logy c.d#c.f2014 c.d#c.f2015 i.year, fe vce(cluster id)

* POLS version:

reg logy c.w#c.f2014 c.w#c.f2015 d f2014 f2015, vce(cluster id)
lincom (c.w#c.f2014 + c.w#c.f2015)/2

* RE still gives same estimates:
```

```
xtreg logy c.w#c.f2014 c.w#c.f2015 d f2014 f2015, re vce(cluster id)

* Putting in time-constant controls in the levels and even interacted with d
* does not change the estimates. It does boost the R-squared and
* slightly changes the standard errors:

reg logy c.w#c.f2014 c.w#c.f2015 d f2014 f2015 x1 c.d#c.x1, vce(cluster id)
reg logy c.w#c.f2014 c.w#c.f2015 d i.year x1 c.d#c.x1, vce(cluster id)

* Now add covariate interacted with everything:

sum x1 if d
gen x1_dm_1 = x1 - r(mean)

xtreg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year c.f2014#c

reg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year c.f2014#c.x

reg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year i.year#c.x1

* Now use margins to account for the sampling error in the mean of x1. It is
* important to have w defined as the time-varying treatment variable.
* In practice, it seems to have little effect:

reg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1 c.w#c.f2015#c.x1 i.year c.f2014#c.x1 c.f2015#
margins, dydx(w) at(f2014 = 1 f2015 = 0) subpop(if d == 1) vce(uncond)
margins, dydx(w) at(f2014 = 0 f2015 = 1) subpop(if d == 1) vce(uncond)

* Apply Callaway & Sant'Anna (2021):

gen first_treat = 0
replace first_treat = 2014 if d
csdid logy x1, ivar(id) time(year) gvar(first_treat)

* Show imputation is equivalent to POLS/ETWFE:

reg logy i.year c.f2014#c.x1 c.f2015#c.x1 d x1 c.d#c.x1 if w == 0
predict tetilda, resid
sum tetilda if (d & f2014)
sum tetilda if (d & f2015)

* Regression produces same ATT estimates, but use the full pooled regression for
* valid standard errors:

reg tetilda c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 if w == 1, noco

* Now test parallel trends. First, unconditionally.
```

```
xtdidreg (logy) (w), group(id) time(year)
* estat trendplots
estat ptrends

* With two control periods, same as the following:
* Note: since the pre-period only spans 2012-2013, D.logy is non-missing
* only for year==2013, so this is a single cross-section of first
* differences (2013 vs 2012), not a multi-period trend test - it's a
* valid two-period parallel-trends check, but should not be read as
* testing trends across more than two pre-periods.

sort id year
reg D.logy d if year <= 2013, vce(robust)

* Test/correct for common trends in a general equation:

gen t = year - 2011

xtreg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year i.year#c

reg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year i.year#c.x1

* Allow the trend to also vary with x1:

reg logy c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 i.year i.year#c.x1
test c.d#c.t c.d#c.t#c.x1

* Imputation. Note that the estimates on c.d#c.t and c.d#c.t#c.x1
* are the same as full regression with all data, as is the joint
* F statistic:

drop tetilda

reg logy i.year i.year#c.x1 d x1 c.d#c.x1 c.d#c.t c.d#c.t#c.x1 if w == 0, vce(cluster id)
test c.d#c.t c.d#c.t#c.x1

predict tetilda, resid
sum tetilda if (d & f2014)
sum tetilda if (d & f2015)

* Again, can use regression to obtain the estimates but not the standard errors:

reg tetilda c.w#c.f2014 c.w#c.f2015 c.w#c.f2014#c.x1_dm_1 c.w#c.f2015#c.x1_dm_1 if w == 1, noco
```

```
Linear regression                               Number of obs     =        400
                                                F(3, 99)          =      34.04
```

```
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0348
                                                Root MSE          =     7.0626

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  c.d#c.post |   3.206857   .4314949     7.43   0.000     2.350677    4.063036
             |
           d |  -.0636461   1.356731    -0.05   0.963    -2.755695    2.628403
        post |  -.1255431   .3021942    -0.42   0.679     -.725162    .4740758
       _cons |     .11305   .9272947     0.12   0.903    -1.726904    1.953004
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(3, 99)          =      34.04
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0348
                                                Root MSE          =     7.0626

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4314949     7.43   0.000     2.350677    4.063036
           d |  -.0636461   1.356731    -0.05   0.963    -2.755695    2.628403
        post |  -.1255431   .3021942    -0.42   0.679     -.725162    .4740758
       _cons |     .11305   .9272947     0.12   0.903    -1.726904    1.953004
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(4, 99)          =      27.19
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0381
                                                Root MSE          =     7.0597

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4320407     7.42   0.000     2.349594    4.064119
           d |  -.0636461   1.358447    -0.05   0.963      -2.7591    2.631808
```

```
       f2014 |  -.7011363   .3419429    -2.05   0.043    -1.379625   -.0226474
       f2015 |   .4500501   .3583234     1.26   0.212    -.2609413    1.161041
       _cons |     .11305   .9284678     0.12   0.903    -1.729232    1.955331
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(5, 99)          =      21.95
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0383
                                                Root MSE          =     7.0677

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4325887     7.41   0.000     2.348507    4.065207
           d |  -.0636461    1.36017    -0.05   0.963    -2.762519    2.635227
             |
        year |
       2013  |   .3260046   .3325494     0.98   0.329    -.3338456    .9858548
       2014  |   -.538134   .3934578    -1.37   0.175     -1.31884    .2425716
       2015  |   .6130524   .3830314     1.60   0.113     -.146965     1.37307
             |
       _cons |  -.0499523   .9214237    -0.05   0.957    -1.878257    1.778352
------------------------------------------------------------------------------


Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2539                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0383                                         max =          4

                                                F(4, 99)          =      27.29
corr(u_i, Xb) = -0.0027                         Prob > F          =     0.0000

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4320407     7.42   0.000     2.349594    4.064119
             |
        year |
```

```
        2013  |   .3260046   .3321282     0.98   0.329    -.3330098     .985019
        2014  |   -.538134   .3929594    -1.37   0.174    -1.317851    .2415828
        2015  |   .6130524   .3825463     1.60   0.112    -.1460024    1.372107
              |
       _cons |  -.0817754   .2003881    -0.41   0.684    -.4793889    .3158381
-------------+----------------------------------------------------------------
     sigma_u |  6.7557992
     sigma_e |  2.3305153
         rho |  .89365429   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2207                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0348                                         max =          4

                                                F(2, 99)          =      50.25
corr(u_i, Xb) = -0.0028                         Prob > F          =     0.0000

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4309511     7.44   0.000     2.351756    4.061957
        post |  -.1255431   .3018134    -0.42   0.678    -.7244064    .4733201
       _cons |   .0812269   .1077378     0.75   0.453    -.1325482     .295002
-------------+----------------------------------------------------------------
     sigma_u |  6.7557992
     sigma_e |  2.3738231
         rho |  .89010353   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Random-effects GLS regression                   Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2207                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0348                                         max =          4

                                                Wald chi2(3)      =     102.12
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000
```

```
                              (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.206857   .4314949     7.43   0.000     2.361142    4.052571
           d |  -.0636461   1.356731    -0.05   0.963     -2.72279    2.595498
        post |  -.1255431   .3021942    -0.42   0.678    -.7178329    .4667467
       _cons |     .11305   .9272947     0.12   0.903    -1.704414    1.930514
-------------+----------------------------------------------------------------
     sigma_u |  6.685563
     sigma_e |  2.3738231
         rho |  .88804221   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2575                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0387                                         max =          4

                                                F(5, 99)          =      21.98
corr(u_i, Xb) = -0.0027                         Prob > F          =     0.0000

                              (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.813593   .5368268     5.24   0.000     1.748412    3.878774
             |
 c.w#c.f2015 |   3.600121   .5765523     6.24   0.000     2.456116    4.744126
             |
        year |
       2013  |   .3260046   .3325494     0.98   0.329    -.3338456    .9858548
       2014  |   -.341502   .4488412    -0.76   0.449      -1.2321    .5490964
       2015  |   .4164204   .4300378     0.97   0.335    -.4368679    1.269709
             |
       _cons |  -.0817754   .2006423    -0.41   0.684    -.4798932    .3163424
-------------+----------------------------------------------------------------
     sigma_u |  6.7557992
     sigma_e |  2.3288408
         rho |  .89379082   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

```
Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2575                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0387                                         max =          4

                                                F(5, 99)          =      21.98
corr(u_i, Xb) = -0.0027                         Prob > F          =     0.0000

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.d#c.f2014 |   2.813593   .5368268     5.24   0.000     1.748412    3.878774
             |
 c.d#c.f2015 |   3.600121   .5765523     6.24   0.000     2.456116    4.744126
             |
        year |
       2013  |   .3260046   .3325494     0.98   0.329    -.3338456    .9858548
       2014  |   -.341502   .4488412    -0.76   0.449      -1.2321    .5490964
       2015  |   .4164204   .4300378     0.97   0.335    -.4368679    1.269709
             |
       _cons |  -.0817754   .2006423    -0.41   0.684    -.4798932    .3163424
-------------+----------------------------------------------------------------
     sigma_u |  6.7557992
     sigma_e |  2.3288408
         rho |  .89379082   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(5, 99)          =      21.97
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0384
                                                Root MSE          =     7.0672

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.813593   .5368268     5.24   0.000     1.748412    3.878774
             |
```

```
 c.w#c.f2015 |   3.600121   .5765523     6.24   0.000     2.456116    4.744126
             |
           d |  -.0636461    1.36017    -0.05   0.963    -2.762519    2.635227
       f2014 |  -.5045043   .3923088    -1.29   0.201     -1.28293    .2739216
       f2015 |   .2534181   .4206018     0.60   0.548    -.5811472    1.087983
       _cons |     .11305   .9296453     0.12   0.903    -1.731568    1.957668
------------------------------------------------------------------------------


 ( 1)  .5*c.w#c.f2014 + .5*c.w#c.f2015 = 0

------------------------------------------------------------------------------
        logy | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   3.206857   .4325887     7.41   0.000     2.348507    4.065207
------------------------------------------------------------------------------


Random-effects GLS regression                   Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.2550                                         min =          4
     Between = 0.0129                                         avg =        4.0
     Overall = 0.0384                                         max =          4

                                                Wald chi2(5)      =     109.84
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.813593   .5368268     5.24   0.000     1.761432    3.865754
             |
 c.w#c.f2015 |   3.600121   .5765523     6.24   0.000     2.470099    4.730143
             |
           d |  -.0636461    1.36017    -0.05   0.963    -2.729531    2.602239
       f2014 |  -.5045043   .3923088    -1.29   0.198    -1.273415    .2644069
       f2015 |   .2534181   .4206018     0.60   0.547    -.5709463    1.077782
       _cons |     .11305   .9296453     0.12   0.903    -1.709021    1.935121
-------------+----------------------------------------------------------------
     sigma_u |  6.6895238
     sigma_e |  2.3287614
         rho |  .89191109   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

```
Linear regression                               Number of obs     =        400
                                                F(7, 99)          =     182.67
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8438
                                                Root MSE          =     2.8557

                                    (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.503103   .3066694     8.16   0.000     1.894605    3.111602
             |
 c.w#c.f2015 |   3.327274   .2902385    11.46   0.000     2.751378     3.90317
             |
           d |    .881691   .5285779     1.67   0.098    -.1671223    1.930504
       f2014 |   .0840872   .2215839     0.38   0.705    -.3555834    .5237578
       f2015 |   .2391039   .2082552     1.15   0.254    -.1741196    .6523275
          x1 |   2.506694   .1327049    18.89   0.000     2.243379     2.77001
             |
    c.d#c.x1 |   .0743163   .1915921     0.39   0.699     -.305844    .4544765
             |
       _cons |   .2148708   .3676352     0.58   0.560    -.5145972    .9443389
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(8, 99)          =     166.57
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8438
                                                Root MSE          =     2.8593

                                    (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.503013   .3070762     8.15   0.000     1.893707    3.112319
             |
 c.w#c.f2015 |   3.327292   .2906341    11.45   0.000     2.750611    3.903973
             |
           d |   .8816195   .5292507     1.67   0.099    -.1685288    1.931768
             |
        year |
       2013  |  -.0324553   .1709047    -0.19   0.850    -.3715673    .3066568
       2014  |    .067934   .2343565     0.29   0.773    -.3970801    .5329482
       2015  |   .2228745   .2273177     0.98   0.329    -.2281732    .6739222
```

```
             |
          x1 |   2.507011   .1335155    18.78   0.000     2.242087    2.771935
             |
     c.d#c.x1 |   .0738548   .1927512     0.38   0.702    -.3086053    .4563149
             |
       _cons |   .2311113     .37362     0.62   0.538    -.5102317    .9724544
------------------------------------------------------------------------------

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
          x1 |        200   -.4048388    2.493319   -6.95061   4.588676


Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.3620                                         min =          4
     Between = 0.4276                                         avg =        4.0
     Overall = 0.2782                                         max =          4

                                                F(9, 99)          =      23.70
corr(u_i, Xb) = 0.3305                          Prob > F          =     0.0000

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.938834   .5031086     5.84   0.000     1.940557     3.93711
             |
 c.w#c.f2015 |   3.742608    .534969     7.00   0.000     2.681113    4.804102
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.2243583   .1904085    -1.18   0.242      -.60217    .1534535
             |
 c.w#c.f2015#|
   c.x1_dm_1 |   .0296616   .2431933     0.12   0.903    -.4528867    .5122099
             |
        year |
       2013  |   .3260046   .3342504     0.98   0.332    -.3372208      .98923
       2014  |    -.16839   .3880607    -0.43   0.665    -.9383866    .6016067
       2015  |   .4360252   .3954162     1.10   0.273    -.3485664    1.220617
             |
c.f2014#c.x1 |   .6285213   .1342955     4.68   0.000     .3620499    .8949926
             |
c.f2015#c.x1 |   .5615944   .1919492     2.93   0.004     .1807255    .9424633
```

```
             |
       _cons |  -.0817754   .1900787    -0.43   0.668    -.4589327    .2953819
-------------+----------------------------------------------------------------
     sigma_u |  6.1628203
     sigma_e |  2.1734897
         rho |  .88937776   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(12, 99)         =     112.09
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8445
                                                Root MSE          =     2.8679

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.496168   .3173129     7.87   0.000      1.86655    3.125785
             |
 c.w#c.f2015 |   3.349012   .3132217    10.69   0.000     2.727513    3.970512
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.2701343   .1759396    -1.54   0.128    -.6192367    .0789681
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.0683301   .1502763    -0.45   0.650    -.3665108    .2298506
             |
        year |
       2013  |  -.0188111   .1738311    -0.11   0.914    -.3637298    .3261075
       2014  |   .0787158   .2377676     0.33   0.741    -.3930668    .5504984
       2015  |   .2325892   .2297992     1.01   0.314    -.2233823    .6885607
             |
c.f2014#c.x1 |   .0404883   .1103324     0.37   0.714    -.1784352    .2594118
             |
c.f2015#c.x1 |   .0778513   .1061898     0.73   0.465    -.1328523    .2885549
             |
           d |    .904209   .5311615     1.70   0.092    -.1497307    1.958149
          x1 |   2.476382   .1544607    16.03   0.000     2.169899    2.782866
             |
     c.d#c.x1 |   .1570952   .2113815     0.74   0.459    -.2623316     .576522
             |
       _cons |   .2230452   .3747574     0.60   0.553    -.5205549    .9666452
------------------------------------------------------------------------------
```

```
Linear regression                               Number of obs     =        400
                                                F(13, 99)         =     103.22
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8445
                                                Root MSE          =     2.8715

                                    (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.500641   .3196658     7.82   0.000     1.866354    3.134927
             |
 c.w#c.f2015 |   3.353485   .3142809    10.67   0.000     2.729884    3.977087
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.2674635    .177262    -1.51   0.135    -.6191897    .0842627
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.0656593   .1506332    -0.44   0.664    -.3645482    .2332296
             |
        year |
       2013  |  -.0261778   .1756818    -0.15   0.882    -.3747686    .3224129
       2014  |     .07109   .2439251     0.29   0.771    -.4129104    .5550904
       2015  |   .2249634   .2325834     0.97   0.336    -.2365326    .6864593
             |
   year#c.x1 |
       2013  |  -.0298874   .0910807    -0.33   0.743    -.2106112    .1508364
       2014  |   .0227905   .1348423     0.17   0.866    -.2447658    .2903469
       2015  |   .0601535   .1180653     0.51   0.612    -.1741137    .2944207
             |
           d |   .8986549   .5321658     1.69   0.094    -.1572776    1.954587
          x1 |    2.49408   .1726029    14.45   0.000     2.151599    2.836562
             |
    c.d#c.x1 |   .1544244   .2128387     0.73   0.470    -.2678938    .5767426
             |
       _cons |    .230671   .3794753     0.61   0.545    -.5222904    .9836324
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(12, 99)         =     112.09
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8445
                                                Root MSE          =     2.8679

                                    (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
```

```
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.386807   .3096756     7.71   0.000     1.772343    3.001271
             |
 c.w#c.f2015 |    3.32135   .2964989    11.20   0.000     2.733032    3.909668
             |
 c.w#c.f2014#|
        c.x1 |  -.2701343   .1759396    -1.54   0.128    -.6192367    .0789681
             |
 c.w#c.f2015#|
        c.x1 |  -.0683301   .1502763    -0.45   0.650    -.3665109    .2298506
             |
        year |
       2013  |  -.0188111   .1738311    -0.11   0.914    -.3637298    .3261075
       2014  |   .0787158   .2377676     0.33   0.741    -.3930668    .5504984
       2015  |   .2325892   .2297992     1.01   0.314    -.2233823    .6885607
             |
c.f2014#c.x1 |   .0404883   .1103324     0.37   0.714    -.1784352    .2594118
             |
c.f2015#c.x1 |   .0778513   .1061898     0.73   0.465    -.1328523    .2885549
             |
           d |    .904209   .5311615     1.70   0.092    -.1497307    1.958149
          x1 |   2.476382   .1544607    16.03   0.000     2.169899    2.782866
             |
     c.d#c.x1 |   .1570952   .2113815     0.74   0.459    -.2623316     .576522
             |
       _cons |   .2230452   .3747574     0.60   0.553    -.5205549    .9666452
------------------------------------------------------------------------------


Average marginal effects                                 Number of obs   = 400
                                                         Subpop. no. obs = 200

Expression: Linear prediction, predict()
dy/dx wrt:  w
At: f2014 = 1
    f2015 = 0

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |            Unconditional
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   2.496168   .3059971     8.16   0.000     1.889003    3.103333
------------------------------------------------------------------------------
```

```
Average marginal effects                                Number of obs   = 400
                                                        Subpop. no. obs = 200

Expression: Linear prediction, predict()
dy/dx wrt:  w
At: f2014 = 0
    f2015 = 1

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |            Unconditional
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           w |   3.349012   .3110518    10.77   0.000     2.731818    3.966207
------------------------------------------------------------------------------


(200 real changes made)

...
Difference-in-difference with Multiple Time Periods

                                                          Number of obs = 400
Outcome model  : least squares
Treatment model: inverse probability
------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
g2014        |
 t_2012_2013 |  -1.401593   .6428278    -2.18   0.029    -2.661512   -.1416737
 t_2013_2014 |   3.388112   .5850216     5.79   0.000     2.241491    4.534734
 t_2013_2015 |   3.915051   .6484609     6.04   0.000     2.644091    5.186011
------------------------------------------------------------------------------
Control: Never Treated

See Callaway and Sant'Anna (2021) for details

      Source |       SS           df       MS      Number of obs   =       300
-------------+----------------------------------   F(8, 291)       =    210.00
       Model |  12568.7919         8  1571.09898   Prob > F        =    0.0000
    Residual |  2177.12159       291  7.48151749   R-squared       =    0.8524
-------------+----------------------------------   Adj R-squared   =    0.8483
       Total |  14745.9135       299  49.3174363   Root MSE        =    2.7352

------------------------------------------------------------------------------
        logy | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        year |
```

```
        2013  |  -.0188111   .3880383    -0.05   0.961    -.7825286    .7449063
        2014  |   .0787158    .513587     0.15   0.878    -.9321002    1.089532
        2015  |   .2325892   .5121166     0.45   0.650    -.7753329    1.240511
              |
c.f2014#c.x1 |   .0404883   .1798899     0.23   0.822    -.3135619    .3945385
              |
c.f2015#c.x1 |   .0778513   .1879703     0.41   0.679    -.2921023    .4478049
              |
            d |    .904209   .3894073     2.32   0.021     .1377972    1.670621
           x1 |   2.476382   .1100309    22.51   0.000     2.259825     2.69294
              |
     c.d#c.x1 |   .1570952   .1556659     1.01   0.314    -.1492786     .463469
              |
        _cons |   .2230452   .3355663     0.66   0.507    -.4373996    .8834899
------------------------------------------------------------------------------

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     tetilda |         50    2.525512    3.370158  -5.463166   10.58631

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     tetilda |         50     3.34147    3.105354  -3.463709   10.24051

      Source |       SS           df       MS      Number of obs   =       100
-------------+----------------------------------   F(4, 96)        =     21.48
       Model |  900.277122         4  225.06928   Prob > F        =    0.0000
    Residual |  1005.96245        96  10.4787755   R-squared       =    0.4723
-------------+----------------------------------   Adj R-squared   =    0.4503
       Total |  1906.23957       100  19.0623957   Root MSE        =    3.2371

------------------------------------------------------------------------------
     tetilda | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.496168   .4582501     5.45   0.000     1.586549    3.405787
             |
 c.w#c.f2015 |   3.349012   .4582306     7.31   0.000     2.439432    4.258593
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.2701343   .1881329    -1.44   0.154    -.6435751    .1033065
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.0683301   .1811302    -0.38   0.707    -.4278707    .2912104
------------------------------------------------------------------------------
```

```
Treatment and time information
```

```
Time variable: year
Control:       w = 0
Treatment:     w = 1
-----------------------------------
             |   Control  Treatment
-------------+---------------------
Group        |
          id |        50         50
-------------+---------------------
Time         |
     Minimum |      2012       2014
     Maximum |      2012       2014
-----------------------------------

Difference-in-differences regression                       Number of obs = 400
Data type: Longitudinal

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
           w |
   (1 vs 0)  |   3.206857   .4320407     7.42   0.000     2.349594    4.064119
------------------------------------------------------------------------------
Note: ATET estimate adjusted for panel effects and time effects.


Parallel-trends test (pretreatment time period)
H0: Linear trends are parallel

F(1, 99) =   4.64
Prob > F = 0.0337



Linear regression                               Number of obs     =        100
                                                F(1, 98)          =       4.66
                                                Prob > F          =     0.0332
                                                R-squared         =     0.0454
                                                Root MSE          =     3.2451

------------------------------------------------------------------------------
             |               Robust
      D.logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           d |  -1.401579   .6490187    -2.16   0.033    -2.689536   -.1136226
```

```
       _cons |   1.026794   .4697368     2.19   0.031     .0946167    1.958972
------------------------------------------------------------------------------



Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.8680                                         min =          4
     Between = 0.8466                                         avg =        4.0
     Overall = 0.8412                                         max =          4

                                                F(12, 99)         =     157.25
corr(u_i, Xb) = 0.4488                          Prob > F          =     0.0000

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.456402   .5067512     4.85   0.000     1.450898    3.461907
             |
 c.w#c.f2015 |   3.208291   .7927064     4.05   0.000      1.63539    4.781193
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.0503407   .1059229    -0.48   0.636    -.2605148    .1598333
             |
 c.w#c.f2015#|
   c.x1_dm_1 |   .0924389   .0842389     1.10   0.275    -.0747095    .2595872
             |
        year |
       2013  |   .0205346   .1993658     0.10   0.918    -.3750504    .4161195
       2014  |   -.009636   .2174468    -0.04   0.965    -.4410976    .4218256
       2015  |   .2623215   .1940125     1.35   0.179    -.1226414    .6472845
             |
   year#c.x1 |
       2012  |   1.933064   .0764551    25.28   0.000     1.781361    2.084768
       2013  |   2.003714   .0662191    30.26   0.000     1.872321    2.135107
       2014  |    2.01687   .0849868    23.73   0.000     1.848238    2.185502
       2015  |   1.991922    .072964    27.30   0.000     1.847146    2.136699
             |
     c.d#c.t |   .0668116   .2957253     0.23   0.822    -.5199715    .6535946
             |
       _cons |   .4575496   .1764897     2.59   0.011     .1073557    .8077434
-------------+----------------------------------------------------------------
     sigma_u |  3.0627341
     sigma_e |  .99359271
```

```
         rho |  .90477742   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(14, 99)         =     101.72
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8446
                                                Root MSE          =     2.8741

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   1.833542   .6136741     2.99   0.004      .615879    3.051204
             |
 c.w#c.f2015 |   2.243344   .9499793     2.36   0.020      .358379    4.128309
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.2653456    .177891    -1.49   0.139    -.6183199    .0876287
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.0635414    .150658    -0.42   0.674    -.3624795    .2353968
             |
        year |
       2013  |  -.2472397   .2421693    -1.02   0.310    -.7277561    .2332767
       2014  |  -.0388673   .2538858    -0.15   0.879    -.5426319    .4648973
       2015  |   .1150061   .2472402     0.47   0.643     -.375572    .6055842
             |
   year#c.x1 |
       2013  |  -.0241251   .0913453    -0.26   0.792    -.2053741    .1571239
       2014  |   .0216128   .1358993     0.16   0.874     -.248041    .2912665
       2015  |   .0589757   .1180401     0.50   0.618    -.1752415     .293193
             |
           d |     .23577   .7209278     0.33   0.744    -1.194707    1.666247
          x1 |   2.495258   .1736213    14.37   0.000     2.150756     2.83976
             |
     c.d#c.x1 |   .1523065   .2136548     0.71   0.478    -.2716309    .5762438
             |
      c.d#c.t |   .4430422   .3468509     1.28   0.204    -.2451853     1.13127
             |
       _cons |   .3406282   .3823786     0.89   0.375     -.418094     1.09935
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(15, 99)         =      95.45
```

```
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8450
                                                Root MSE          =     2.8744

                                    (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   1.917612   .6348799     3.02   0.003     .6578725    3.177352
             |
 c.w#c.f2015 |   2.378839   .9651184     2.46   0.015     .4638351    4.293844
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.7263424   .2927708    -2.48   0.015    -1.307263   -.1454217
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.8415862   .4868867    -1.73   0.087    -1.807675    .1245027
             |
        year |
       2013  |  -.2611685   .2476914    -1.05   0.294    -.7526419    .2303049
       2014  |  -.0671335   .2642967    -0.25   0.800    -.5915555    .4572885
       2015  |   .0867399   .2545516     0.34   0.734    -.4183457    .5918255
             |
   year#c.x1 |
       2013  |  -.1857772   .1396877    -1.33   0.187    -.4629479    .0913934
       2014  |  -.0735746   .1623912    -0.45   0.651     -.395794    .2486448
       2015  |  -.0362116   .1317686    -0.27   0.784    -.2976691    .2252458
             |
           d |   .1075441    .732768     0.15   0.884    -1.346427    1.561515
          x1 |   2.590445   .1878499    13.79   0.000      2.21771     2.96318
             |
     c.d#c.x1 |  -.3378406   .3669519    -0.92   0.359    -1.065953    .3902716
             |
      c.d#c.t |   .5199705   .3447009     1.51   0.135    -.1639908    1.203932
             |
c.d#c.t#c.x1 |    .317048   .1852193     1.71   0.090    -.0504673    .6845633
             |
       _cons |   .3688944   .3868765     0.95   0.343    -.3987525    1.136541
------------------------------------------------------------------------------


 ( 1)  c.d#c.t = 0
 ( 2)  c.d#c.t#c.x1 = 0

       F(  2,    99) =    2.47
            Prob > F =    0.0898
```

```
Linear regression                               Number of obs     =        300
                                                F(11, 99)         =      82.82
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8531
                                                Root MSE          =     2.7429

                                   (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        year |
       2013  |  -.2611685   .2475879    -1.05   0.294    -.7524366    .2300996
       2014  |  -.0671335   .2641863    -0.25   0.800    -.5913364    .4570694
       2015  |   .0867399   .2544452     0.34   0.734    -.4181347    .5916145
             |
   year#c.x1 |
       2013  |  -.1857773   .1396293    -1.33   0.186    -.4628321    .0912775
       2014  |  -.0735746   .1623234    -0.45   0.651    -.3956594    .2485101
       2015  |  -.0362116   .1317135    -0.27   0.784    -.2975599    .2251366
             |
           d |    .107544   .7324618     0.15   0.884    -1.345819    1.560907
          x1 |   2.590445   .1877714    13.80   0.000     2.217866    2.963025
             |
     c.d#c.x1 |  -.3378407   .3667986    -0.92   0.359    -1.065649    .3899672
             |
      c.d#c.t |   .5199705   .3445569     1.51   0.134    -.1637051    1.203646
             |
c.d#c.t#c.x1 |   .3170481   .1851419     1.71   0.090    -.0503137    .6844098
             |
       _cons |   .3688944   .3867149     0.95   0.342    -.3984318    1.136221
------------------------------------------------------------------------------


 ( 1)  c.d#c.t = 0
 ( 2)  c.d#c.t#c.x1 = 0

       F(  2,    99) =    2.47
            Prob > F =    0.0896

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     tetilda |         50    1.996512    3.755621  -5.357233   9.968782

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
```

```
     tetilda |         50    2.285944    3.772198  -4.526336   10.79229

      Source |       SS           df       MS      Number of obs   =       100
-------------+----------------------------------   F(4, 96)        =     20.11
       Model |  842.991546         4  210.747887   Prob > F        =    0.0000
    Residual |  1005.96248        96  10.4787758   R-squared       =    0.4559
-------------+----------------------------------   Adj R-squared   =    0.4333
       Total |  1848.95402       100  18.4895402   Root MSE        =    3.2371

------------------------------------------------------------------------------
     tetilda | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   1.917612   .4582501     4.18   0.000     1.007993    2.827231
             |
 c.w#c.f2015 |   2.378839   .4582306     5.19   0.000     1.469259     3.28842
             |
 c.w#c.f2014#|
   c.x1_dm_1 |  -.7263426   .1881329    -3.86   0.000    -1.099783   -.3529017
             |
 c.w#c.f2015#|
   c.x1_dm_1 |  -.8415865   .1811302    -4.65   0.000    -1.201127   -.4820459
------------------------------------------------------------------------------
```

That is a long chunk, so here is what to take from it. Every specification that targets the overall ATT returns **3.206857** — the two-way fixed-effect regression, pooled OLS with unit and time means, random effects, and the `did`-style variants all agree to seven digits, differing only in the standard error's fourth decimal (0.4310 to 0.4326). Against a true ATT of 3, that is about half a standard error high. The equivalences claimed throughout this chapter hold numerically, not just in the algebra.

Allowing the effect to vary by year recovers the two components separately: 2.496168 for 2014 and 3.349012 for 2015, against true values of 2.5 and 3.5. Their average, 2.92, is what the single-number ATT is summarising, and reporting only 3.21 would hide that the effect grows by about a third between the two post years.

The imputation approach at the end fits the model on untreated observations only and reads the treatment effects off the residuals. Its estimates of `c.d#c.t` and `c.d#c.t#c.x1` match the full-sample regression, and the pre-treatment interaction `c.d#c.t` is 0.0668 with $p = 0.822$ — no detectable differential trend before treatment, which is what the design needs.

9. Correlated Random Effect

9.1. Panel data

The standard panel-data setup is:

$$y_{it} = \alpha + X_{it}\beta + v_i + \epsilon_{it} \tag{9.1}$$

9.2. Fixed effect

The fixed-effect estimator runs OLS on demeaned data:

$$(y_{it} - \bar{y}_i) = (X_{it} - \bar{X}_i)\beta + (\epsilon_{it} - \bar{\epsilon}_i) \tag{9.2}$$

Both α and v_i drop out after demeaning by unit.

9.3. Random effect

Random effects treat $v_i + \epsilon_{it}$ as a composite error with two variance components: $\hat{\sigma}_e^2$ (idiosyncratic) and $\hat{\sigma}_u^2$ (individual). The GLS transformation is

$z_{it}^* = z_{it} - \hat{\theta}_i \bar{z}_i$

with $\hat{\theta}_i = 1 - \sqrt{\hat{\sigma}_e^2/(T_i\hat{\sigma}_u^2 + \hat{\sigma}_e^2)}$, where T_i is the number of observations for unit i. Given estimated variance components, we run OLS on the transformed variables and iterate.

9.4. Correlated random effect

The correlated random-effect (CRE) model runs a random-effects regression of y_{it} on X_{it}, $\bar{X}_i$, time-invariant covariates w_i, and a constant.

RE requires v_i and X_{it} to be uncorrelated — an assumption most economists reject. FE is consistent regardless, but it cannot estimate time-invariant covariates (race, gender, etc.). CRE has both advantages: it includes time-invariant variables and remains consistent. Mundlak and Wooldridge show that the CRE coefficients on X_{it} are identical to the FE estimates. Testing whether the coefficients on $\bar{X}_i$ are jointly zero gives the Mundlak test for choosing between RE and CRE.

9.5. Example

Stata 18 added a `cre` option to `xtreg`. The following uses `nlswork`, 28,101 person-years from the National Longitudinal Survey of Young Women with complete data on the model's variables. We regress log wage on `tenure` and `age`, which vary within a person, plus `race`, which does not — the covariate a fixed-effect model has to drop.

```
webuse nlswork

xtreg ln_wage tenure age i.race, cre vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

note: 2.race omitted from xt_means because of collinearity.
note: 3.race omitted from xt_means because of collinearity.

Correlated random-effects regression             Number of obs     =     28,101
Group variable: idcode                           Number of groups  =      4,699

R-squared:                                       Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.2346                                         avg =        6.0
     Overall = 0.1890                                         max =         15

                                                 Wald chi2(4)      =    1685.18
corr(xit_vars*b, xt_means* ) = 0.5474           Prob > chi2       =     0.0000

                              (Std. err. adjusted for 4,699 clusters in idcode)
-------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+-----------------------------------------------------------------
xit_vars     |
      tenure |   .0211313   .0012113    17.44   0.000     .0187572    .0235055
         age |   .0121949   .0007414    16.45   0.000     .0107417     .013648
             |
        race |
      Black  |  -.1312068   .0117856   -11.13   0.000    -.1543061   -.1081075
      Other  |   .1059379   .0593177     1.79   0.074    -.0103225    .2221984
             |
       _cons |     1.2159   .0306965    39.61   0.000     1.155736    1.276064
-------------+-----------------------------------------------------------------
xt_means     |
      tenure |   .0376991    .002281    16.53   0.000     .0332283    .0421698
         age |  -.0011984   .0013313    -0.90   0.368    -.0038077    .0014109
             |
        race |
```

```
       Black |          0  (omitted)
       Other |          0  (omitted)
-------------+----------------------------------------------------------------
     sigma_u |  .33334407
     sigma_e |  .29808194
         rho |  .55567161   (fraction of variance due to u_i)
------------------------------------------------------------------------------
Mundlak test (xt_means = 0): chi2(2) = 331.5144          Prob > chi2 = 0.0000
```

CRE gives `tenure` = .0211313 and `age` = .0121949, and estimates race as well: Black women earn 13.1% less (−.1312068, SE .0118), while the "Other" category is marginal (.1059, $p = 0.074$). The coefficients on the group means are .0376991 for mean tenure and −.0011984 for mean age.

Stata prints the Mundlak test at the foot of the table: $\chi^2(2) = 331.51$, $p < 0.0001$. The group means are jointly far from zero, so plain random effects is rejected — v_i and the regressors are correlated, as the reported corr = 0.5474 between the two blocks says directly.

To compare with a fixed effect model:

```
webuse nlswork

xtreg ln_wage tenure age i.race, fe vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

note: 2.race omitted because of collinearity.
note: 3.race omitted because of collinearity.

Fixed-effects (within) regression               Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.1916                                         avg =        6.0
     Overall = 0.1456                                         max =         15

                                                F(2, 4698)        =     766.79
corr(u_i, Xb) = 0.1302                          Prob > F          =     0.0000

                             (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |   .0211313   .0012112    17.45   0.000     .0187568    .0235059
         age |   .0121949   .0007414    16.45   0.000     .0107414    .0136483
             |
        race |
```

```
      Black  |          0  (omitted)
      Other  |          0  (omitted)
             |
       _cons |   1.256467   .0194187    64.70   0.000     1.218397    1.294537
-------------+----------------------------------------------------------------
     sigma_u |  .39034493
     sigma_e |  .29808194
         rho |  .63165531   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

The coefficients on `tenure` and `age` are identical: .0211313 and .0121949 in both, with standard errors agreeing to the seventh decimal (.0012112 against .0012113). Race is reported as `0 (omitted)` here, being collinear with the unit effects — the one thing FE cannot do and CRE can. This is the Mundlak result in practice: CRE gives you the fixed-effect estimates *and* the time-invariant coefficients from a single random-effects fit.

We can replicate this manually by including the group means in a RE model:

```
webuse nlswork
* Restrict the mean calculation to the estimation sample (listwise-deleted
* on the model's variables), matching what xtreg's native cre option does
* internally. Both tenure and age have missing values in nlswork, so
* computing age_mean/tenure_mean over all rows for an idcode (rather than
* only rows with complete data on ln_wage, tenure, age, and race) would
* make the manual means diverge from cre's.
egen age_mean = mean(age) if !missing(ln_wage, tenure, age, race), by(idcode)
egen tenure_mean = mean(tenure) if !missing(ln_wage, tenure, age, race), by(idcode)
xtreg ln_wage tenure tenure_mean age age_mean i.race, vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

(433 missing values generated)

(433 missing values generated)


Random-effects GLS regression                   Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.2346                                         avg =        6.0
     Overall = 0.1890                                         max =         15

                                                Wald chi2(6)      =    2688.54
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000
```

```
                                (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |   .0211313   .0012113    17.44   0.000     .0187572    .0235055
 tenure_mean |   .0376991    .002281    16.53   0.000     .0332283    .0421698
         age |   .0121949   .0007414    16.45   0.000     .0107417     .013648
    age_mean |  -.0011984   .0013313    -0.90   0.368    -.0038077    .0014109
             |
        race |
      Black  |  -.1312068   .0117856   -11.13   0.000    -.1543061   -.1081075
      Other  |   .1059379   .0593177     1.79   0.074    -.0103225    .2221984
             |
       _cons |     1.2159   .0306965    39.61   0.000     1.155736    1.276064
-------------+----------------------------------------------------------------
     sigma_u |  .33334407
     sigma_e |  .29808194
         rho |  .55567161   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

This is what Stata's `cre` option does internally. Every number matches the first table: `tenure` .0211313, `age` .0121949, Black −.1312068, Other .1059379, constant 1.2159, and the same variance components (σ_u .3333, σ_e .2981, ρ .5557). The `cre` option is a convenience, not a different estimator.

The one detail worth care is in the code above: the group means must be computed on the estimation sample. Both `tenure` and `age` have missing values in `nlswork`, so averaging over all of a woman's rows rather than only her complete ones would give different means and break the equivalence.

10. More on Correlated Random Effect (CRE)

Jeff Wooldridge suggested using Correlated Random Effect (CRE) for binary outcomes in this twitter post: https://x.com/jmwooldridge/status/1986100627454206220

One word on the name before we start, since it invites a misreading. "Correlated random effects" describes an *assumption*, not an estimator: the unobserved effect v_i is modelled as a function of the group means and is allowed to correlate with the regressors, rather than being treated as a free parameter to estimate. Whether you then estimate by pooled OLS, pooled probit, or an actual random-effects routine is a separate choice. Most of what follows uses **pooled** estimation with group means added, in which no random effect is estimated at all.

10.1. Linear model

The standard panel setup is:

$$y_{it} = \alpha + X_{it}\beta + v_i + \epsilon_{it} \tag{10.1}$$

Wooldridge (2021) shows that these are equivalent:

1. Two-way fixed-effect OLS (or the dummy-variable approach).
2. Mundlak's CRE approach (pooled OLS with group means, or CRE with random effects).

Two qualifications on that second item, both of which matter in practice.

What "the Mundlak device" has to include. For a *one-way* model it is the unit means of the time-varying regressors, and nothing else. That much holds whatever the panel looks like, because $\sum_t (x_{it} - \bar{x}_i) = 0$ within each unit by construction. A *two-way* model needs the time dimension handled as well, and the two usual ways of doing it — adding the time means of the regressors, or keeping a full set of time dummies — reproduce the two-way fixed effect estimate **only when the unit and time factors are orthogonal**, which for a two-way panel means every unit appearing in every period. The `castle` data used below is balanced, so both routes are exact there; the example takes the second one (unit means plus `factor(year)`), which is why it reproduces the *two-way* estimate and not the *one-way* one. Unit means alone would give the one-way estimate.

That orthogonality condition is binding, not decorative. On an unbalanced panel it fails, and the discrepancy can be substantial: in the `nlswork` example in the Mundlak chapter the coefficient on `age` differs by more than a factor of two. What restores the equivalence there is adding the **unit means of the time dummies**. In a balanced panel those means are the constant $1/T$ and have no variance, which is precisely why the shortcut above reads as though it were unconditional. With **three or more** fixed effects the shortcut is weaker still — one set of means per dimension is exact only under a fully orthogonal design, and the construction that survives without one is to pick the dimension being eliminated and project it onto the within-unit means of everything else retained in the model, dummies included. See More than two fixed effects.

The idea of Chamberlain's device is that since we don't observe v_i, we project v_i onto the history of X_i: $v_i = \psi + \bar{X}_i\xi + a_i$, and replace v_i with that projection. Mundlak's device is a special case, where we put equal weight on all X_{it}, so the projection of v_i onto X_i reduces to the mean $\bar{X}_i = \frac{1}{T}\sum_t X_{it}$. Replace v_i with that projection, and what is left (a_i) is by construction uncorrelated with X_{it}.

Here is an example:

10.1.1. example 1

```
library(dplyr)
library(fixest)
library(bacondecomp)
data("castle")

# Fixed-effects model with individual and time fixed effects
fe_model_twoway <- feols(l_homicide ~ poverty + l_police | state + year, data = castle)
summary(fe_model_twoway)
```

```
OLS estimation, Dep. Var.: l_homicide
Observations: 550
Fixed-effects: state: 50,  year: 11
Standard-errors: IID
          Estimate Std. Error   t value Pr(>|t|)
poverty  -0.027068   0.013306 -2.034206 0.042471 *
l_police  0.066033   0.104588  0.631364 0.528098
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.176686     Adj. R2: 0.898978
                 Within R2: 0.009286
```

```
# random effect
library(lme4)
re_model <- lmer(l_homicide ~ poverty + l_police +  factor(year) +  (1 | state), data = castle)
summary(re_model)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: l_homicide ~ poverty + l_police + factor(year) + (1 | state)
   Data: castle

REML criterion at convergence: 3.9

Scaled residuals:
```

```
    Min      1Q  Median      3Q     Max 
-4.5357 -0.4657  0.0151  0.4601  3.6973 

Random effects:
 Groups   Name        Variance Std.Dev.
 state    (Intercept) 0.30041  0.5481  
 Residual             0.03551  0.1884  
Number of obs: 550, groups:  state, 50

Fixed effects:
                   Estimate Std. Error t value
(Intercept)       0.4395941  0.6019389   0.730
poverty          -0.0007991  0.0120087  -0.067
l_police          0.1665913  0.1015581   1.640
factor(year)2001  0.0225820  0.0379597   0.595
factor(year)2002  0.0005681  0.0389059   0.015
factor(year)2003  0.0475588  0.0395876   1.201
factor(year)2004  0.0419220  0.0411546   1.019
factor(year)2005  0.0612232  0.0435803   1.405
factor(year)2006  0.0806576  0.0418469   1.927
factor(year)2007  0.0790850  0.0409066   1.933
factor(year)2008  0.0401268  0.0414964   0.967
factor(year)2009 -0.0424481  0.0488959  -0.868
factor(year)2010 -0.0959167  0.0470940  -2.037
```

```r
# Mundlak
castle2 <- castle |>
  group_by(state) |>
  mutate(poverty_mean = mean(poverty, na.rm = TRUE), l_police_mean= mean(l_police, na.rm = TRUE
cre_model <- feols(l_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(yea
summary(cre_model)
```

```
OLS estimation, Dep. Var.: l_homicide
Observations: 550
Standard-errors: IID 
                  Estimate Std. Error    t value   Pr(>|t|)    
(Intercept)      -7.800873   0.531186 -14.685774  < 2.2e-16 ***
poverty          -0.027068   0.030004  -0.902125 3.6740e-01    
poverty_mean      0.125283   0.030674   4.084272 5.0983e-05 ***
l_police          0.066033   0.235836   0.279996 7.7959e-01    
l_police_mean     1.318037   0.253323   5.202983 2.7987e-07 ***
factor(year)2001  0.033071   0.085349   0.387481 6.9855e-01    
factor(year)2002  0.022870   0.087972   0.259968 7.9499e-01    
factor(year)2003  0.074569   0.089852   0.829907 4.0696e-01    
factor(year)2004  0.079080   0.094152   0.839920 4.0133e-01    
factor(year)2005  0.109800   0.100737   1.089973 2.7622e-01    
factor(year)2006  0.118822   0.095991   1.237857 2.1631e-01    
```

```
factor(year)2007  0.115412   0.093475   1.234679 2.1749e-01
factor(year)2008  0.077190   0.095054   0.812069 4.1711e-01
factor(year)2009  0.027528   0.114987   0.239400 8.1089e-01
factor(year)2010 -0.034286   0.110153  -0.311255 7.5573e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.417154   Adj. R2: 0.486344
```

```
# Mundlak
cre_model2 <- lmer(l_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(yea
summary(cre_model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: l_homicide ~ poverty + poverty_mean + l_police + l_police_mean +
    factor(year) + (1 | state)
   Data: castle2

REML criterion at convergence: -30.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.5991 -0.4336  0.0219  0.4307  3.6228

Random effects:
 Groups   Name        Variance Std.Dev.
 state    (Intercept) 0.14872  0.3856
 Residual             0.03518  0.1876
Number of obs: 550, groups:  state, 50

Fixed effects:
                 Estimate Std. Error t value
(Intercept)      -7.80087    1.60978  -4.846
poverty          -0.02707    0.01331  -2.034
poverty_mean      0.12528    0.02360   5.309
l_police          0.06603    0.10459   0.631
l_police_mean     1.31804    0.30140   4.373
factor(year)2001  0.03307    0.03785   0.874
factor(year)2002  0.02287    0.03901   0.586
factor(year)2003  0.07457    0.03985   1.871
factor(year)2004  0.07908    0.04175   1.894
factor(year)2005  0.10980    0.04467   2.458
factor(year)2006  0.11882    0.04257   2.791
factor(year)2007  0.11541    0.04145   2.784
factor(year)2008  0.07719    0.04215   1.831
factor(year)2009  0.02753    0.05099   0.540
factor(year)2010 -0.03429    0.04885  -0.702
```

The data are `castle`: 550 state-years, 50 states over 11 years, balanced. The outcome is log homicide rate, the regressors are the poverty rate and log police employment.

In this example, fixed effect model, CRE1 (pooled OLS with the Mundlak device) and CRE2 (random effects with the Mundlak device) all give the same coefficient on poverty: −0.027068 in the two-way fixed-effect model, −0.027068 in pooled OLS with the means, and −0.02707 in the mixed model. Log police is 0.066033 in all three. The group-mean coefficients, 0.125283 for poverty and 1.318037 for police, are what absorb the between-state variation.

The plain random-effects model is the one that does not belong. Without the group means it gives a poverty coefficient of −0.0007991 — essentially zero, against −0.027 from every other specification, and with the opposite substantive reading. That is what happens when v_i is correlated with the regressors and the model assumes it is not. As we know, RE model assumes v_i is uncorrelated with X_{it}, while CRE and FE allow for correlation between v_i and X_{it}.

One detail on standard errors. The three point estimates agree, but pooled OLS reports 0.030004 on poverty while the fixed-effect model reports 0.013306 and the mixed model 0.01331. Pooled OLS is ignoring the within-state correlation; cluster it on state and the discrepancy closes. Point-estimate equivalence does not carry over to inference for free.

10.2. nonlinear model

What I did not realize before is that this is only valid for the linear case. In the linear model the algebra above is an identity: adding the group means reparameterizes the regression so that the coefficient on x_{it} *is* the within estimator, and no distributional assumption is involved. In a nonlinear model — logit, probit, Poisson — there is no such identity. Substituting a projection for v_i becomes a genuine modelling assumption about $D(v_i \mid X_i)$, and the likelihood you end up maximizing is not the one you started from.

10.2.1. Binary outcome

For binary outcomes, the model is:

$$\Pr(y_{it} = 1 \mid X_{it}, v_i) = \Phi(\alpha + X_{it}\beta + v_i) \tag{10.2}$$

I used to think it's the best to use conditional logit. However, it needs "conditional independence" (serial independence could be a better name) to be consistent. That is, we have to assume the series of y_{it} is conditionally independent of each other given X_{it} and v_i. This is a strong assumption. If we think we have serial correlation in the error term, then this would not hold. The other disadvantage of conditional logit (also called fixed effect logit) is that v_i's are not estimated, it is just a nuisance parameter. Any partial effects cannot be calculated, because any partial effects are functions of v_i's.

Instead, Wooldridge suggests using CRE probit — that is, the Mundlak device inside a probit. Basically adding Mundlak device (group means of X's) to the model, and then use pooled probit. Note that we do not use RE probit or logit, since that would need conditional independence too. Lin and Wooldridge (2019, sec. 5.2) put this plainly: "None of random effects probit, RE logit, RE Tobit, RE Poisson, and so on has robustness properties in the presence of serial correlation of the innovations." Pooled methods with cluster-robust inference are robust to serial dependence of arbitrary form, which is the reason to prefer them here.

For nonlinear panel data with unobserved heterogeneity (meaning there is v_i that we don't observe), we have a few assumptions.

First, strict exogeneity,

$$D(y_{it}|X_{i1},...,X_{iT},v_i) = D(y_{it}|X_{it},v_i) \tag{10.3}$$

This means that the distribution of y_{it} only depends on X_{it}, given v_i.

Second, conditional independence,

$$D(y_{i1},y_{i2},...,y_{iT}|X_i,v_i) = \prod_{t=1}^{T} D(y_{it}|X_i,v_i) \tag{10.4}$$

This means that the distribution of joint distribution of y_{it}'s can be modeled by each y_{it} independently, given X_i and v_i. We need this in most nonlinear cases, but not in linear case.

Then we need to specify $D(v_i)|X_i$. The random effect assumption is saying

$$D(v_i|X_i) = D(v_i)$$

{#eq-more-cre-5}, that is, v_i is independent of X_i. This is a strong assumption. We should try to avoid this assumption.

Now, most nonlinear models make the second assumption, in the panel setting, when we have v_i. That includes RE model, conditional logit. However, that does not include pooled probit/logit, or CRE with pooled logit/probit. That is why Wooldridge recommend using CRE with pooled probit.

10.2.2. Example 2

```
# generate a binary outcome, high_homocide

castle2 <- castle2 |>
  mutate(high_homicide = ifelse(l_homicide > mean(l_homicide, na.rm = TRUE), 1, 0))

# CRE pooled probit
cre_probit <- glm(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(ye
# Compute Average Marginal Effects (AMEs)
library(marginaleffects)
ame_cre_probit <- avg_slopes(cre_probit)
ame_cre_probit
```

```
          Term     Contrast Estimate Std. Error       z Pr(>|z|)   S     2.5 %
 l_police      dY/dX         0.01924     0.2622  0.0734   0.9415 0.1 -0.494686
 l_police_mean dY/dX        -0.00304     0.2824 -0.0108   0.9914 0.0 -0.556572
```

```
 poverty       dY/dX       -0.06477     0.0334 -1.9417   0.0522 4.3 -0.130144
 poverty_mean  dY/dX        0.06620     0.0341  1.9424   0.0521 4.3 -0.000597
 year          2001 - 2000  0.06596     0.0990  0.6659   0.5054 1.0 -0.128159
 year          2002 - 2000  0.09470     0.1019  0.9293   0.3527 1.5 -0.105016
 year          2003 - 2000  0.22958     0.1020  2.2518   0.0243 5.4  0.029753
 year          2004 - 2000  0.15492     0.1082  1.4315   0.1523 2.7 -0.057193
 year          2005 - 2000  0.22388     0.1133  1.9766   0.0481 4.4  0.001880
 year          2006 - 2000  0.26184     0.1068  2.4526   0.0142 6.1  0.052595
 year          2007 - 2000  0.24899     0.1048  2.3756   0.0175 5.8  0.043564
 year          2008 - 2000 -0.03718     0.1099 -0.3383   0.7352 0.4 -0.252646
 year          2009 - 2000 -0.02471     0.1345 -0.1837   0.8542 0.2 -0.288280
 year          2010 - 2000 -0.21753     0.1186 -1.8339   0.0667 3.9 -0.450000
   97.5 %
 0.533162
 0.550499
 0.000608
 0.132987
 0.260070
 0.294406
 0.429403
 0.367039
 0.445887
 0.471080
 0.454421
 0.178276
 0.238865
 0.014950

Type: response
```

```
# linear model
cre_lm <- lm(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year),
ame_lm <- avg_slopes(cre_lm)
ame_lm
```

```
         Term     Contrast Estimate Std. Error       z Pr(>|z|)   S    2.5 %
 l_police       dY/dX        0.01234     0.2666  0.0463   0.9631 0.1 -0.51010
 l_police_mean dY/dX        0.00427     0.2865  0.0149   0.9881 0.0 -0.55724
 poverty       dY/dX       -0.06406     0.0339 -1.8891   0.0589 4.1 -0.13053
 poverty_mean  dY/dX        0.06578     0.0347  1.8975   0.0578 4.1 -0.00217
 year          2001 - 2000  0.06384     0.0965  0.6618   0.5081 1.0 -0.12522
 year          2002 - 2000  0.09082     0.0994  0.9134   0.3610 1.5 -0.10406
 year          2003 - 2000  0.22452     0.1016  2.2108   0.0270 5.2  0.02548
 year          2004 - 2000  0.14789     0.1064  1.3898   0.1646 2.6 -0.06068
 year          2005 - 2000  0.21659     0.1139  1.9023   0.0571 4.1 -0.00656
 year          2006 - 2000  0.25687     0.1085  2.3677   0.0179 5.8  0.04423
```

```
 year          2007 - 2000  0.24413     0.1056  2.3108   0.0208 5.6  0.03706
 year          2008 - 2000 -0.04737     0.1074 -0.4410   0.6592 0.6 -0.25794
 year          2009 - 2000 -0.03438     0.1300 -0.2645   0.7914 0.3 -0.28910
 year          2010 - 2000 -0.20933     0.1245 -1.6814   0.0927 3.4 -0.45335
 97.5 %
 0.5348
 0.5658
 0.0024
 0.1337
 0.2529
 0.2857
 0.4236
 0.3565
 0.4397
 0.4695
 0.4512
 0.1632
 0.2203
 0.0347

Type: response
```

```r
# CRE RE probit
cre_re_probit <- glmer(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + fac
ame_cre_re_probit <- avg_slopes(cre_re_probit)
ame_cre_re_probit
```

```
          Term    Contrast Estimate Std. Error       z Pr(>|z|)   S     2.5 %
 l_police      dY/dX        0.01927     0.2624  0.0735   0.9414 0.1 -0.494931
 l_police_mean dY/dX       -0.00307     0.2827 -0.0109   0.9913 0.0 -0.557188
 poverty       dY/dX       -0.06477     0.0334 -1.9417   0.0522 4.3 -0.130143
 poverty_mean  dY/dX        0.06619     0.0341  1.9424   0.0521 4.3 -0.000599
 year          2001 - 2000  0.06595     0.0990  0.6659   0.5055 1.0 -0.128160
 year          2002 - 2000  0.09469     0.1019  0.9293   0.3527 1.5 -0.105017
 year          2003 - 2000  0.22958     0.1020  2.2518   0.0243 5.4  0.029753
 year          2004 - 2000  0.15492     0.1082  1.4315   0.1523 2.7 -0.057195
 year          2005 - 2000  0.22388     0.1133  1.9765   0.0481 4.4  0.001878
 year          2006 - 2000  0.26184     0.1068  2.4526   0.0142 6.1  0.052595
 year          2007 - 2000  0.24899     0.1048  2.3756   0.0175 5.8  0.043563
 year          2008 - 2000 -0.03719     0.1099 -0.3383   0.7352 0.4 -0.252648
 year          2009 - 2000 -0.02471     0.1345 -0.1837   0.8542 0.2 -0.288283
 year          2010 - 2000 -0.21752     0.1186 -1.8339   0.0667 3.9 -0.449999
   97.5 %
 0.533476
 0.551049
 0.000609
```

```
0.132988
0.260069
0.294405
0.429402
0.367037
0.445886
0.471079
0.454420
0.178275
0.238863
0.014953

Type: response
```

Two caveats about that third model, and the second one matters more than the first.

First, on what `avg_slopes` is averaging. For a `merMod` object `marginaleffects` predicts with `re.form = NULL` by default, which *includes* the estimated random effects rather than setting them to zero; the package warns that its standard errors nevertheless reflect only the uncertainty in the fixed-effect parameters. If you want the AME for a "typical" ($v_i = 0$) unit you have to ask for it with `re.form = NA`, and neither of those is the *population-averaged* AME, which requires integrating the marginal effect over the estimated distribution of v_i. In a nonlinear model like probit those three quantities genuinely differ, because $\Phi(\cdot)$ is nonlinear in v_i.

Second, and more to the point here: on this data set the random-effects probit is a **boundary (singular) fit**. `lme4` reports an estimated state-level standard deviation of about 5×10^{-9} — numerically zero — and `isSingular()` returns `TRUE`. With no variance left in the random effect, this model *is* the pooled probit: its coefficients agree with the pooled fit to about 10^{-4} (poverty -0.17848 either way). So the third specification is not really a third specification on this data, and the distinction between conditional and population-averaged effects has nothing to apply to. It is kept here because the comparison is the one you would want to run in general — but do not read the agreement between models two and three as evidence about random effects; check `VarCorr()` before drawing that conclusion.

Ultimately we are interested in the partial effect, therefore we can compare across models. In this case, we should prefer average marginal effect by CRE pooled probit.

The three sets of average marginal effects bear that out. Pooled probit with the Mundlak device gives -0.06477 on poverty, the mixed-model version -0.06406, and the singular random-effects fit -0.06477 again — all within 0.0008 of each other, all with standard errors around 0.034, and all marginally insignificant ($p \approx 0.052$). The agreement is unsurprising given the singularity noted above, and the reason to prefer the pooled version is robustness to serial dependence rather than any difference in the numbers here.

10.3. References

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- Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica* 46, 69–85.
- Papke, L. E., and J. M. Wooldridge (2008). Panel data methods for fractional response variables with an application to test pass rates. *Journal of Econometrics* 145, 121–133.
- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press. Ch. 15 covers the nonlinear panel and CRE material.
- Wooldridge, J. M. (2019). Correlated random effects models with unbalanced panels. *Journal of Econometrics* 211, 137–150.

Part III.

Treatment Effects & Matching

11. Treatment effects and matching

11.1. Treatment effects in observational studies

Most economic data are observational. With unobserved confounders we need instrumental variables, but instruments are hard to find and harder to justify. Under conditional independence — no unobserved confounders given the covariates — we can proceed without them.

Stata's `teffects` suite implements the standard estimators under conditional independence. (The companion `eteffects` handles endogenous treatment via a control-function approach with instruments.) This chapter walks through how `teffects` works.

11.1.1. Regression adjustment (RA)

RA allows all covariate effects to differ between treatment and control — equivalent to an outcome model with treatment interacted with all covariates.

```
clear
webuse bweightex
teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke)
reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)
margins r.mbsmoke
```

```
. clear

. webuse bweightex
(Hypothetical birthweight data)

. teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke)

Iteration 0:  EE criterion = 1.223e-24
Iteration 1:  EE criterion = 1.792e-25

Treatment-effects estimation                    Number of obs     =         60
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
```

```
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -389.3099   37.00882   -10.52   0.000    -461.8458   -316.7739
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3613.769   29.97438   120.56   0.000     3555.021    3672.518
------------------------------------------------------------------------------

. reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)

      Source |       SS           df       MS      Number of obs   =        60
-------------+----------------------------------   F(9, 50)        =     18.11
       Model |  1376306.34         9  152922.927   Prob > F        =    0.0000
    Residual |  422241.592        50  8444.83184   R-squared       =    0.7652
-------------+----------------------------------   Adj R-squared   =    0.7230
       Total |  1798547.93        59  30483.8633   Root MSE        =    91.896

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |
     Smoker  |  -32.87702   193.2827    -0.17   0.866    -421.0967    355.3426
   prenatal1 |   -23.8431   38.31014    -0.62   0.537    -100.7913    53.10507
    mmarried |  -40.39753   50.21942    -0.80   0.425    -141.2662    60.47114
        mage |    27.9537   7.671423     3.64   0.001     12.54519    43.36221
       fbaby |  -2.030497   40.76029    -0.05   0.960    -83.89995    79.83896
             |
    mbsmoke#|
 c.prenatal1 |
     Smoker  |   8.436558   56.38855     0.15   0.882    -104.8232    121.6963
             |
    mbsmoke#|
  c.mmarried |
     Smoker  |   33.92362   61.34015     0.55   0.583     -89.2817    157.1289
             |
    mbsmoke#|
      c.mage |
     Smoker  |  -15.98608   9.006494    -1.77   0.082    -34.07616    2.103995
             |
    mbsmoke#|
     c.fbaby |
     Smoker  |   14.86782   57.58771     0.26   0.797    -100.8005    130.5362
             |
```

```
       _cons |   2976.807   151.0117    19.71   0.000     2673.491    3280.123
------------------------------------------------------------------------------

. margins r.mbsmoke

Contrasts of predictive margins                             Number of obs = 60
Model VCE: OLS

Expression: Linear prediction, predict()

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
     mbsmoke |          1      116.06     0.0000
             |
 Denominator |         50
------------------------------------------------

------------------------------------------------------------------------
                       |            Delta-method
                       |   Contrast   std. err.     [95% conf. interval]
-----------------------+------------------------------------------------
               mbsmoke |
(Smoker vs Nonsmoker)  |  -389.3099   36.13675     -461.8927   -316.7271
------------------------------------------------------------------------
```

The data here are `bweightex`, a 60-observation extract with birthweight as the outcome and maternal smoking as the treatment.

`teffects ra` returns the same ATE as `margins r.mbsmoke` after the interacted regression: both give −389.3099 grams. The standard errors differ slightly, 37.01 against 36.14, because `teffects` accounts for the estimation of the outcome model differently from `margins`' delta method. The point estimate is the same object computed two ways.

For the ATET (average treatment effect on the treated):

```
clear
webuse bweightex
teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke), atet
reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)
margins r.mbsmoke, subpop(mbsmoke)
```

```
. clear

. webuse bweightex
(Hypothetical birthweight data)
```

```
. teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke), atet

Iteration 0:  EE criterion = 1.159e-24
Iteration 1:  EE criterion = 5.133e-26

Treatment-effects estimation                   Number of obs     =         60
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -437.4721   53.43513    -8.19   0.000     -542.203   -332.7412
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
   Nonsmoker |   3693.639   53.80537    68.65   0.000     3588.182    3799.095
------------------------------------------------------------------------------

. reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)

      Source |       SS           df       MS      Number of obs   =        60
-------------+----------------------------------   F(9, 50)        =     18.11
       Model |  1376306.34         9  152922.927   Prob > F        =    0.0000
    Residual |  422241.592        50  8444.83184   R-squared       =    0.7652
-------------+----------------------------------   Adj R-squared   =    0.7230
       Total |  1798547.93        59  30483.8633   Root MSE        =    91.896

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |
     Smoker  |  -32.87702   193.2827    -0.17   0.866    -421.0967    355.3426
   prenatal1 |   -23.8431   38.31014    -0.62   0.537    -100.7913    53.10507
    mmarried |  -40.39753   50.21942    -0.80   0.425    -141.2662    60.47114
        mage |    27.9537   7.671423     3.64   0.001     12.54519    43.36221
       fbaby |  -2.030497   40.76029    -0.05   0.960    -83.89995    79.83896
             |
     mbsmoke#|
 c.prenatal1 |
     Smoker  |   8.436558   56.38855     0.15   0.882    -104.8232    121.6963
             |
     mbsmoke#|
```

```
  c.mmarried |
     Smoker  |   33.92362   61.34015     0.55   0.583     -89.2817    157.1289
             |
     mbsmoke#|
      c.mage |
     Smoker  |  -15.98608   9.006494    -1.77   0.082    -34.07616    2.103995
             |
     mbsmoke#|
     c.fbaby |
     Smoker  |   14.86782   57.58771     0.26   0.797    -100.8005    130.5362
             |
       _cons |   2976.807   151.0117    19.71   0.000     2673.491    3280.123
------------------------------------------------------------------------------

. margins r.mbsmoke, subpop(mbsmoke)

Contrasts of predictive margins                          Number of obs   = 60
Model VCE: OLS                                           Subpop. no. obs = 30

Expression: Linear prediction, predict()

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
     mbsmoke |          1       73.30     0.0000
             |
 Denominator |         50
------------------------------------------------

--------------------------------------------------------------------
                       |            Delta-method
                       |   Contrast   std. err.     [95% conf. interval]
-----------------------+--------------------------------------------
               mbsmoke |
(Smoker vs Nonsmoker)  |  -437.4721   51.09606     -540.1015   -334.8426
--------------------------------------------------------------------
```

The ATET compares potential outcomes for the treated subpopulation, hence the `subpop(mbsmoke)` option in `margins`. Both routes give −437.4721, again identical, and again with slightly different standard errors (53.44 against 51.10). Note the ATET is 48 grams larger in magnitude than the ATE of −389.31: smoking mothers are drawn disproportionately from the groups where smoking does most damage.

11.1.2. Inverse Probability Weighting (IPW)

IPW has two steps. First, estimate the treatment model (typically logit) to obtain the propensity score. Second, use inverse-probability weights to compute the outcome difference between treated and control units.

These steps produce consistent estimates of the effect parameters because the treatment is assumed to be independent of the potential outcomes after conditioning on the covariates.

We can manually conduct the two steps, but the nice thing about using Stata's teffects is that it takes account of the noise of estimating probability in the first step when calculating standard errors in the second step.

11.1.2.1. example

```
clear
webuse cattaneo2
teffects ipw (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu, probit)
probit mbsmoke mmarried c.mage##c.mage fbaby medu
predict ps
replace ps = 1/ps if mbsmoke==1
replace ps = 1/(1-ps) if mbsmoke==0
reg bweight mbsmoke [pweight=ps]
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. teffects ipw (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu, probit)

Iteration 0:   EE criterion = 4.622e-21
Iteration 1:   EE criterion = 8.453e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : inverse-probability weights
Outcome model  : weighted mean
Treatment model: probit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -230.6886   25.81524    -8.94   0.000    -281.2856   -180.0917
-------------+----------------------------------------------------------------
POmean       |
```

```
     mbsmoke |
   Nonsmoker |   3403.463   9.571369   355.59   0.000     3384.703    3422.222
------------------------------------------------------------------------------

. probit mbsmoke mmarried c.mage##c.mage fbaby medu

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2042.6734
Iteration 2:  Log likelihood = -2040.5088
Iteration 3:  Log likelihood = -2040.5061
Iteration 4:  Log likelihood = -2040.5061

Probit regression                                       Number of obs =  4,642
                                                        LR chi2(5)    = 380.48
                                                        Prob > chi2   = 0.0000
Log likelihood = -2040.5061                             Pseudo R2     = 0.0853

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |  -.6484821   .0526991   -12.31   0.000    -.7517705   -.5451938
        mage |   .1744327   .0352437     4.95   0.000     .1053562    .2435092
             |
     c.mage#|
      c.mage |  -.0032559   .0006462    -5.04   0.000    -.0045224   -.0019894
             |
       fbaby |  -.2175962   .0491066    -4.43   0.000    -.3138433    -.121349
        medu |  -.0863631   .0098692    -8.75   0.000    -.1057064   -.0670198
       _cons |  -1.558255   .4511589    -3.45   0.001     -2.44251       -.674
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

. replace ps = 1/ps if mbsmoke==1
(864 real changes made)

. replace ps = 1/(1-ps) if mbsmoke==0
(3,778 real changes made)

. reg bweight mbsmoke [pweight=ps]
(sum of wgt is 9,193.96990537643)

Linear regression                               Number of obs     =      4,642
                                                F(1, 4640)        =      79.08
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0389
                                                Root MSE          =     573.36
```

```
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -230.6886   25.94182    -8.89   0.000    -281.5469   -179.8303
       _cons |   3403.463   9.616992   353.90   0.000     3384.609    3422.317
------------------------------------------------------------------------------
```

We switch here to `cattaneo2`, the full 4,642-birth dataset, and a richer propensity model (a probit in marital status, a quadratic in maternal age, first-baby status and education).

`teffects ipw` gives an ATE of −230.6886 with a standard error of 25.82. The manual two-step replication matches the point estimate exactly, −230.6886; the only difference is the standard error (the manual version does not account for first-stage estimation noise). Note this is well short of the −389 found by RA above — but on different data and a different covariate set, so the two are not comparable.

11.1.3. Doubly robust estimators

RA models the outcome; IPW models the treatment assignment. Doubly-robust estimators model both, requiring only one to be correctly specified.

Stata implements the AIPW combination of Robins and Rotnitzky (1995) and the IPWRA combination of Wooldridge (2010).

The AIPW estimator augments the IPW estimator with a correction term built from the outcome model. Double robustness is best stated as a **consistency** property: AIPW is consistent if *either* the treatment model *or* the outcome model is correctly specified (not necessarily both). It is *not* that the correction term literally "goes to zero" when the treatment model is right and the outcome model is wrong — with a misspecified outcome model the correction term does not generally vanish. Rather, the two pieces are constructed so that the estimator's bias cancels in the limit as long as one of the two models is right.

The IPWRA estimator uses IPW probability weights when performing RA. The weights do not affect the accuracy of the RA estimator if the treatment model is wrong and the outcome model is correct. The weights correct the RA estimator if the treatment model is correct and the outcome model is wrong.

Here we run the two commands, then manually replicate both as a two-step ("plug-in") procedure. `teffects` estimates the treatment and outcome models jointly by GMM/ML, using the joint estimation to get standard errors that account for first-stage estimation noise; the manual two-step version below gets essentially the same point estimate but not the correct standard errors (for the same reason noted above for IPW).

```
clear
webuse cattaneo2
teffects ipwra (bweight mmarried mage prenatal1 fbaby)  (mbsmoke mmarried mage prenatal1 fbaby)
teffects aipw (bweight mmarried mage prenatal1 fbaby) (mbsmoke mmarried mage prenatal1 fbaby)
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. teffects ipwra (bweight mmarried mage prenatal1 fbaby)  (mbsmoke mmarried mag
> e prenatal1 fbaby)

Iteration 0:  EE criterion = 9.066e-22
Iteration 1:  EE criterion = 2.902e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : IPW regression adjustment
Outcome model  : linear
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -238.7679   24.38353    -9.79   0.000    -286.5587    -190.977
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3402.851   9.538741   356.74   0.000     3384.155    3421.546
------------------------------------------------------------------------------

. teffects aipw (bweight mmarried mage prenatal1 fbaby) (mbsmoke mmarried mage
> prenatal1 fbaby)

Iteration 0:  EE criterion = 2.115e-22
Iteration 1:  EE criterion = 1.343e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : augmented IPW
Outcome model  : linear by ML
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
```

```
          vs  |
 Nonsmoker)  |  -239.0294    24.2524    -9.86   0.000    -286.5632   -191.4955
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3402.839   9.538926   356.73   0.000     3384.143    3421.535
------------------------------------------------------------------------------
```

Manual IPWRA: first estimate the propensity score, then run *weighted* regression adjustment within each treatment arm, using the inverse probability of the *observed* treatment status as weights:

```
clear
webuse cattaneo2
logit mbsmoke mmarried mage prenatal1 fbaby
predict ps

gen wgt1 = mbsmoke/ps
gen wgt0 = (1-mbsmoke)/(1-ps)
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1 [pweight=wgt1]
predict mu1_ipwra
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0 [pweight=wgt0]
predict mu0_ipwra
gen tau_ipwra_i = mu1_ipwra - mu0_ipwra
sum tau_ipwra_i
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. logit mbsmoke mmarried mage prenatal1 fbaby

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2089.7838
Iteration 2:  Log likelihood = -2082.3958
Iteration 3:  Log likelihood = -2082.3879
Iteration 4:  Log likelihood = -2082.3879

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(4)    = 296.72
                                                        Prob > chi2   = 0.0000
Log likelihood = -2082.3879                             Pseudo R2     = 0.0665

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
    mmarried |   -1.10666    .091279   -12.12   0.000    -1.285564   -.9277569
        mage |  -.0187894   .0082217    -2.29   0.022    -.0349036   -.0026751
   prenatal1 |  -.2721103   .0943067    -2.89   0.004    -.4569481   -.0872725
       fbaby |  -.5428134   .0864561    -6.28   0.000    -.7122642   -.3733625
       _cons |   .1297324   .2095812     0.62   0.536    -.2810391     .540504
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

.
. gen wgt1 = mbsmoke/ps

. gen wgt0 = (1-mbsmoke)/(1-ps)

. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1 [pweight=wgt1]
(sum of wgt is 4,620.9067401886)

Linear regression                               Number of obs     =        864
                                                F(4, 859)         =       3.12
                                                Prob > F          =     0.0145
                                                R-squared         =     0.0142
                                                Root MSE          =     563.01

------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   130.5978   39.90558     3.27   0.001     52.27399    208.9217
        mage |  -6.495147    4.95637    -1.31   0.190    -16.22316    3.232866
   prenatal1 |   23.29089   40.04843     0.58   0.561    -55.31335    101.8951
       fbaby |   45.19338   48.48447     0.93   0.352    -49.96852    140.3553
       _cons |   3206.397   126.7646    25.29   0.000     2957.592    3455.202
------------------------------------------------------------------------------

. predict mu1_ipwra
(option xb assumed; fitted values)

. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0 [pweight=wgt0]
(sum of wgt is 4,643.79376959801)

Linear regression                               Number of obs     =      3,778
                                                F(4, 3773)        =      26.89
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0336
                                                Root MSE          =     567.52
```

```
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   160.9954   26.87191     5.99   0.000     108.3105    213.6803
        mage |   3.061463   2.181356     1.40   0.161    -1.215289    7.338214
   prenatal1 |    62.4539   28.28456     2.21   0.027      6.99938    117.9084
       fbaby |  -67.38469   20.29791    -3.32   0.001    -107.1806   -27.58876
       _cons |   3188.522   56.19956    56.74   0.000     3078.338    3298.707
------------------------------------------------------------------------------

. predict mu0_ipwra
(option xb assumed; fitted values)

. gen tau_ipwra_i = mu1_ipwra - mu0_ipwra

. sum tau_ipwra_i

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
 tau_ipwra_i |      4,642   -238.7679    97.94189  -481.7334   6.216553
```

The mean of `tau_ipwra_i` matches `teffects ipwra`'s ATE exactly: -238.7679 in both.

Manual AIPW: estimate outcome models $\hat{\mu}_1(X)$, $\hat{\mu}_0(X)$ (unweighted, one per arm) and the propensity score $\hat{p}(X)$, then combine via the augmented-IPW formula

$$\hat{\tau}_i^{aipw} = D_i \frac{Y_i - \hat{\mu}_1(X_i)}{\hat{p}(X_i)} + \hat{\mu}_1(X_i) - (1 - D_i) \frac{Y_i - \hat{\mu}_0(X_i)}{1 - \hat{p}(X_i)} - \hat{\mu}_0(X_i) \tag{11.1}$$

```
clear
webuse cattaneo2
logit mbsmoke mmarried mage prenatal1 fbaby
predict ps

reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1
predict mu1_aipw
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0
predict mu0_aipw
gen aipw_i = mbsmoke*(bweight-mu1_aipw)/ps + mu1_aipw - (1-mbsmoke)*(bweight-mu0_aipw)/(1-ps)
sum aipw_i
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)
```

```
. logit mbsmoke mmarried mage prenatal1 fbaby

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2089.7838
Iteration 2:  Log likelihood = -2082.3958
Iteration 3:  Log likelihood = -2082.3879
Iteration 4:  Log likelihood = -2082.3879

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(4)    = 296.72
                                                        Prob > chi2   = 0.0000
Log likelihood = -2082.3879                             Pseudo R2     = 0.0665

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   -1.10666    .091279   -12.12   0.000    -1.285564   -.9277569
        mage |  -.0187894   .0082217    -2.29   0.022    -.0349036   -.0026751
   prenatal1 |  -.2721103   .0943067    -2.89   0.004    -.4569481   -.0872725
       fbaby |  -.5428134   .0864561    -6.28   0.000    -.7122642   -.3733625
       _cons |   .1297324   .2095812     0.62   0.536    -.2810391     .540504
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

.
. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1

      Source |       SS           df       MS      Number of obs   =       864
-------------+----------------------------------   F(4, 859)       =      3.69
       Model |  4586519.21         4   1146629.8   Prob > F        =    0.0055
    Residual |   266914157       859  310726.609   R-squared       =    0.0169
-------------+----------------------------------   Adj R-squared   =    0.0123
       Total |   271500676       863  314601.015   Root MSE        =    557.43

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   133.6617   41.33023     3.23   0.001     52.54164    214.7818
        mage |  -7.370881   3.983409    -1.85   0.065    -15.18924    .4474735
   prenatal1 |   25.11133   43.64639     0.58   0.565    -60.55473    110.7774
       fbaby |   41.43991   41.82146     0.99   0.322     -40.6443    123.5241
       _cons |   3227.169   102.3392    31.53   0.000     3026.305    3428.033
------------------------------------------------------------------------------

. predict mu1_aipw
(option xb assumed; fitted values)
```

```
. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0

      Source |       SS           df       MS      Number of obs   =     3,778
-------------+----------------------------------   F(4, 3773)      =     30.00
       Model |    37916015         4  9479003.74   Prob > F        =    0.0000
    Residual |  1.1922e+09     3,773  315979.752   R-squared       =    0.0308
-------------+----------------------------------   Adj R-squared   =    0.0298
       Total |  1.2301e+09     3,777  325683.776   Root MSE        =    562.12

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   160.9513   24.55483     6.55   0.000     112.8092    209.0933
        mage |   2.546828   1.935504     1.32   0.188    -1.247908    6.341564
   prenatal1 |   64.40859   25.83668     2.49   0.013     13.75338    115.0638
       fbaby |   -71.3286    19.4895    -3.66   0.000    -109.5396   -33.11763
       _cons |   3202.746   51.20076    62.55   0.000     3102.362     3303.13
------------------------------------------------------------------------------

. predict mu0_aipw
(option xb assumed; fitted values)

. gen aipw_i = mbsmoke*(bweight-mu1_aipw)/ps + mu1_aipw - (1-mbsmoke)*(bweight-
> mu0_aipw)/(1-ps) - mu0_aipw

. sum aipw_i

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      aipw_i |      4,642   -239.0294     1618.64  -29389.38   16313.36
```

The mean of `aipw_i` is −239.0294, matching `teffects aipw`'s ATE to four decimals. (One might expect a small gap, since `teffects` estimates the treatment and outcome models jointly while this is a two-step plug-in; here the point estimates coincide, and only the standard errors would differ.)

The two doubly-robust estimators land within 0.3 grams of each other, −238.77 for IPWRA and −239.03 for AIPW, both about 8 grams away from plain IPW's −230.69.

The unit-level `aipw_i` values are worth a glance before trusting the mean. They have a standard deviation of 1,618.6 and run from −29,389 to +16,313 — two orders of magnitude wider than the estimate itself. That spread comes from dividing residuals by $\hat{p}$ or $1 - \hat{p}$ for units with extreme propensity scores. The average is well behaved, but it is an average over some very large individual terms, which is exactly why AIPW can be unstable when overlap is poor.

11.2. Matching

Matching does not solve the endogeneity problem. It addresses selection on observables only; with unobserved confounders it does not help.

What matching does is balance the covariate distributions across treatment and control groups. Regression does this too, but matching makes the lack of overlap visible — a plain regression can extrapolate far outside the support without warning. Running regression on a matched sample combines the strengths of both approaches.

Below we cover three matching methods in Stata: propensity score matching, nearest-neighbor matching, and coarsened exact matching.

11.2.1. Propensity score matching

Exact matching on all covariates is ideal but infeasible in high dimensions. Rosenbaum and Rubin (1983) showed that the propensity score provides a scalar sufficient statistic, reducing the matching problem to one dimension.

Stata's `teffects psmatch` and `psmatch2` both estimate the propensity score, match, and compute the treatment effect with correct standard errors in one step.

```
clear
webuse cattaneo2
teffects psmatch (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu), atet
psmatch2 mbsmoke mmarried c.mage##c.mage fbaby medu, logit ties
reg bweight mbsmoke   [aweight=_weight]
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. teffects psmatch (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu), atet

Treatment-effects estimation                   Number of obs      =      4,642
Estimator      : propensity-score matching     Matches: requested =          1
Outcome model  : matching                                     min =          1
Treatment model: logit                                        max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
```

```
 Nonsmoker)  |  -236.7848   26.57789    -8.91   0.000    -288.8765    -184.693
------------------------------------------------------------------------------

. psmatch2 mbsmoke mmarried c.mage##c.mage fbaby medu, logit ties

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(5)    = 375.00
                                                        Prob > chi2   = 0.0000
Log likelihood = -2043.2504                             Pseudo R2     = 0.0841

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |  -1.145706   .0918962   -12.47   0.000     -1.32582    -.965593
        mage |    .321518   .0638472     5.04   0.000     .1963798    .4466563
             |
      c.mage#|
      c.mage |  -.0060368   .0011849    -5.09   0.000    -.0083592   -.0037144
             |
       fbaby |  -.3864258   .0880445    -4.39   0.000    -.5589898   -.2138618
        medu |  -.1420833   .0173215    -8.20   0.000    -.1760328   -.1081338
       _cons |  -2.950915   .8102504    -3.64   0.000    -4.538976   -1.362853
------------------------------------------------------------------------------

. reg bweight mbsmoke   [aweight=_weight]
(sum of wgt is 1,728)

      Source |       SS           df       MS      Number of obs   =     3,671
-------------+----------------------------------   F(1, 3669)      =    151.87
       Model |    51455506         1    51455506   Prob > F        =    0.0000
    Residual |  1.2431e+09     3,669  338808.237   R-squared       =    0.0397
-------------+----------------------------------   Adj R-squared   =    0.0395
       Total |  1.2945e+09     3,670  352736.492   Root MSE        =    582.07

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -236.7848   19.21387   -12.32   0.000    -274.4557   -199.1138
       _cons |   3374.444   13.58626   248.37   0.000     3347.807    3401.082
------------------------------------------------------------------------------

.
```

`teffects psmatch` gives the ATT of smoking on birthweight, −236.7848 with a standard error of 26.58. The manual replication uses `psmatch2` with `logit` and `ties` (by default `teffects` includes all tied matches, while `psmatch2` keeps only one), then runs a weighted regression, and gives −236.7848 — the same to four decimals. Point estimates match; standard errors should be taken from `teffects`.

The nearest-neighbour ATT sits close to the two doubly-robust ATEs above, −238.8 and −239.0, though it is a different estimand on a different subpopulation.

11.2.2. Nearest neighbor matching

`teffects nnmatch` uses Mahalanobis distance by default and can also enforce exact matching on discrete covariates. It is nonparametric.

```
clear
webuse cattaneo2
teffects nnmatch  (bweight mmarried mage fbaby medu) (mbsmoke) , atet
teffects psmatch (bweight) (mbsmoke mmarried mage fbaby medu), atet
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. teffects nnmatch  (bweight mmarried mage fbaby medu) (mbsmoke) , atet

Treatment-effects estimation                    Number of obs      =      4,642
Estimator      : nearest-neighbor matching     Matches: requested =          1
Outcome model  : matching                                     min =          1
Distance metric: Mahalanobis                                  max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -239.2433   25.68524    -9.31   0.000    -289.5854   -188.9011
------------------------------------------------------------------------------

. teffects psmatch (bweight) (mbsmoke mmarried mage fbaby medu), atet

Treatment-effects estimation                    Number of obs      =      4,642
Estimator      : propensity-score matching     Matches: requested =          1
Outcome model  : matching                                     min =          1
Treatment model: logit                                        max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

```
ATET          |
     mbsmoke |
    (Smoker   |
         vs   |
 Nonsmoker)  |   -245.711   26.38675    -9.31   0.000    -297.4281   -193.9939
------------------------------------------------------------------------------

.
```

Note the specification difference: `nnmatch` lists covariates in the outcome parentheses, `psmatch` lists them in the treatment parentheses.

11.2.3. Coarsened Exact Matching (CEM)

CEM is available as a separate Stata package.

```
clear
webuse cattaneo2
cem  mmarried mage fbaby medu, treatment(mbsmoke)
reg bweight mbsmoke   [aweight=cem_weights]
* Note: cem creates the variable "cem_weights" (plural). We spell it out
* in full here rather than relying on Stata's automatic variable-name
* abbreviation matching, to keep the reference unambiguous.
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138-154)

. cem  mmarried mage fbaby medu, treatment(mbsmoke)

Matching Summary:
-----------------
Number of strata: 274
Number of matched strata: 135

              0     1
      All  3778   864
  Matched  3421   827
Unmatched   357    37


Multivariate L1 distance: .25424705

Univariate imbalance:
```

```
                L1      mean      min      25%      50%      75%      max
mmarried  6.4e-15  8.4e-15        0        0        0        0        0
    mage   .04955  -.00887        1        0        1        0       -2
   fbaby  6.1e-15  6.1e-15        0        0        0        0        0
    medu   .03809  -.03316        0        0        0        0        0

. reg bweight mbsmoke   [aweight=cem_weights]
(sum of wgt is 4,248.00000000005)

      Source |       SS           df       MS      Number of obs   =     4,248
-------------+----------------------------------   F(1, 4246)      =    121.12
       Model |  40566772.9         1  40566772.9   Prob > F        =    0.0000
    Residual |  1.4221e+09     4,246  334928.257   R-squared       =    0.0277
-------------+----------------------------------   Adj R-squared   =    0.0275
       Total |  1.4627e+09     4,247   344401.26   Root MSE        =    578.73

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -246.8017   22.42533   -11.01   0.000    -290.7671   -202.8364
       _cons |   3383.394   9.894625   341.94   0.000     3363.996    3402.793
------------------------------------------------------------------------------

. * Note: cem creates the variable "cem_weights" (plural). We spell it out
. * in full here rather than relying on Stata's automatic variable-name
. * abbreviation matching, to keep the reference unambiguous.
.
```

CEM creates exact-match strata on coarsened covariates and generates weights for the final regression. The standard errors do not account for the matching step.

11.3. Modern matching in R: balance diagnostics and entropy weighting

The 2019 vintage of this chapter used Stata's `teffects` throughout. R's `MatchIt` and `WeightIt` packages have since become the standard for matching workflows, and `cobalt` provides publication-quality balance tables and plots that are now expected in applied papers.

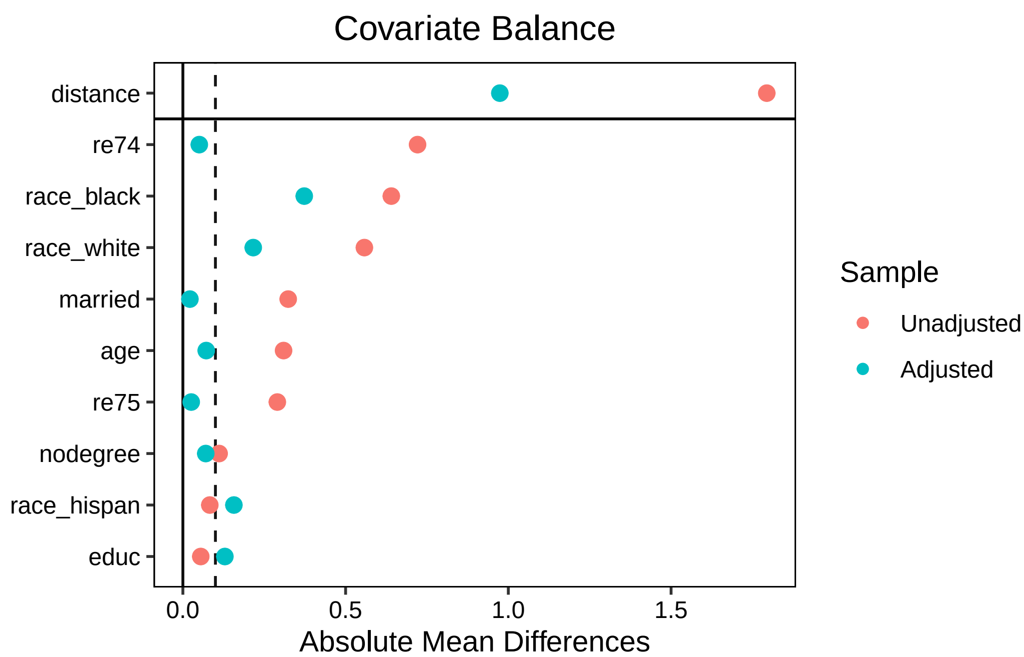
11.3.1. Balance checking with cobalt

After any matching or weighting step, always inspect covariate balance. `cobalt`'s `bal.tab()` and `love.plot()` are the go-to tools:

```r
library(MatchIt)
library(cobalt)

# Propensity score matching on the lalonde data
data("lalonde", package = "MatchIt")

m_out <- matchit(treat ~ age + educ + race + married + nodegree +
                   re74 + re75,
                 data = lalonde, method = "nearest", distance = "glm",
                 ratio = 1)

# Balance table: standardized mean differences before and after matching
bal.tab(m_out, stats = c("mean.diffs", "variance.ratios"), thresholds = c(m = 0.1))
```

```
Balance Measures
                Type Diff.Adj        M.Threshold V.Ratio.Adj
distance    Distance   0.9739                         0.7566
age          Contin.   0.0718     Balanced, <0.1      0.4568
educ         Contin.  -0.1290 Not Balanced, >0.1      0.5721
race_black    Binary   0.3730 Not Balanced, >0.1           .
race_hispan   Binary  -0.1568 Not Balanced, >0.1           .
race_white    Binary  -0.2162 Not Balanced, >0.1           .
married       Binary  -0.0216     Balanced, <0.1           .
nodegree      Binary   0.0703     Balanced, <0.1           .
re74         Contin.  -0.0505     Balanced, <0.1      1.3289
re75         Contin.  -0.0257     Balanced, <0.1      1.4956

Balance tally for mean differences
                   count
Balanced, <0.1         5
Not Balanced, >0.1     4

Variable with the greatest mean difference
   Variable Diff.Adj        M.Threshold
 race_black    0.373 Not Balanced, >0.1

Sample sizes
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
```

```r
# Love plot: visualize balance improvement
love.plot(m_out, stats = "mean.diffs", threshold = 0.1,
          abs = TRUE, var.order = "unadjusted")
```



The love plot shows each covariate's standardized mean difference (SMD) before matching (open circles) and after (filled circles). The vertical line at SMD = 0.1 is the conventional threshold for "balanced." If post-matching points cross the line, the matching is not improving balance and a different specification is needed.

That is exactly what happens here, which is why it is worth running the diagnostic rather than assuming. 1:1 nearest-neighbour matching on the propensity score leaves four of the nine covariates *above* the 0.1 threshold, `race` worst of all (an adjusted SMD of 0.37 for the black indicator), while discarding 244 of the 429 controls. Matching on a propensity score balances the score, not necessarily the covariates that went into it.

11.3.2. Entropy balancing

Propensity score matching discards unmatched units and can reduce effective sample size substantially. **Entropy balancing** (Hainmueller 2012) keeps all units and reweights the control group so that weighted covariate moments exactly match the treatment group — without discarding anyone.

```
library(WeightIt)
library(cobalt)

# Entropy balancing: exact balance on means, variances, and covariances
w_out <- weightit(treat ~ age + educ + race + married + nodegree +
                    re74 + re75,
                  data = lalonde, method = "ebal", estimand = "ATT",
                  moments = 1)   # moments=1: balance means only
                                 # moments=2: balance means + variances

summary(w_out)
```

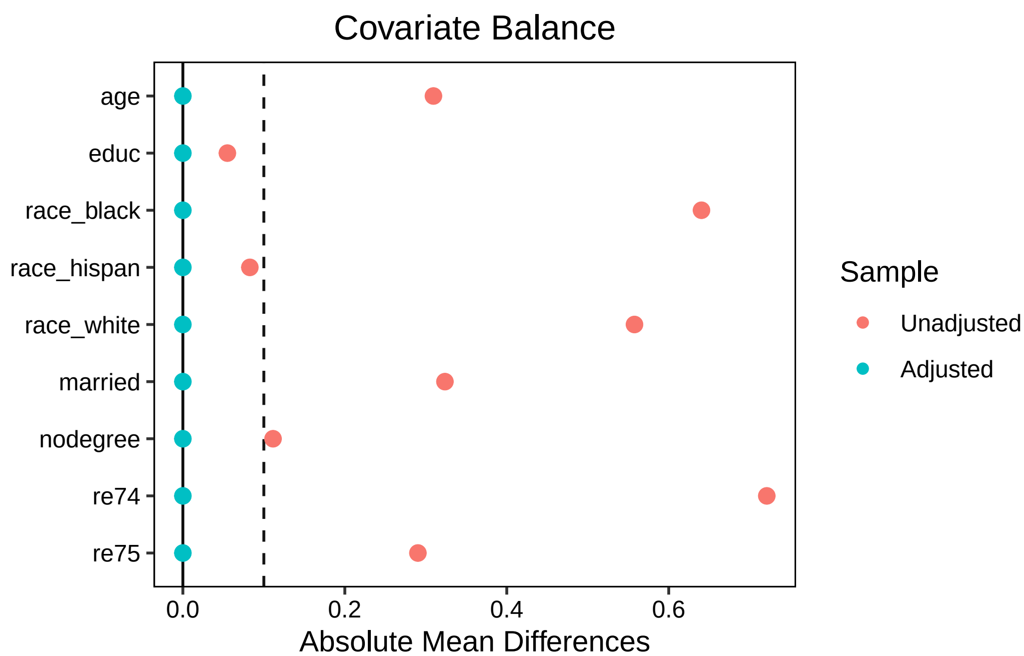
```
                  Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.          ||                         1.
control 0.008 |---------------------------| 4.062

- Units with the 5 most extreme weights by group:

            NSW5    NSW4    NSW3    NSW2    NSW1
 treated       1       1       1       1       1
         PSID423 PSID196 PSID412 PSID118 PSID226
 control   3.073   3.235    3.45   3.897   4.062

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.834 1.287   1.101       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     98.46     185
```

```
# Check balance
bal.tab(w_out, stats = "mean.diffs", thresholds = c(m = 0.1))
```

```
Balance Measures
               Type Diff.Adj    M.Threshold
age          Contin.       -0 Balanced, <0.1
educ         Contin.       -0 Balanced, <0.1
race_black    Binary        0 Balanced, <0.1
race_hispan   Binary        0 Balanced, <0.1
race_white    Binary       -0 Balanced, <0.1
married       Binary       -0 Balanced, <0.1
nodegree      Binary        0 Balanced, <0.1
re74         Contin.       -0 Balanced, <0.1
re75         Contin.       -0 Balanced, <0.1

Balance tally for mean differences
                   count
Balanced, <0.1         9
Not Balanced, >0.1     0
```

```
Variable with the greatest mean difference
 Variable Diff.Adj    M.Threshold
     re74       -0 Balanced, <0.1

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     98.46     185
```

```
love.plot(w_out, threshold = 0.1, abs = TRUE)
```




```
# Weighted outcome regression
library(lmtest); library(sandwich)
m_eb <- lm(re78 ~ treat + age + educ + race + married + nodegree +
             re74 + re75,
           data = lalonde, weights = w_out$weights)
coeftest(m_eb, vcov = vcovHC(m_eb, type = "HC3"))["treat", ]
```

```
    Estimate    Std. Error       t value      Pr(>|t|)
1273.2618139  820.1848528     1.5524083     0.1210883
```

Entropy balancing delivers on its promise here: the largest absolute standardized mean difference after weighting is **exactly zero**, since the weights are chosen to satisfy the mean-balance constraints as equalities. All 429 controls are retained.

Method	Pros	Cons
PSM (nearest neighbor)	Intuitive; respects overlap	Discards units; balance not guaranteed
CEM	Exact balance in bins	Requires discretizing continuous vars
Entropy balancing	Exact mean balance; keeps all units	Weights can be extreme; no variance balance by default
IPW (logistic)	Simple; relies only on the treatment model	Not doubly robust on its own; extreme weights if overlap is poor
AIPW / IPWRA	Doubly robust: consistent if *either* the treatment or outcome model is correct	Needs both models specified; more moving parts

For most applications, entropy balancing or IPW with trimmed weights (via `WeightIt`'s `trim()`) is preferred over PSM: better effective sample size, guaranteed balance, and easy integration with doubly-robust estimators.

See also the weighting chapter for IPW estimators and the OLS-ATE chapter for regression-based approaches.

Systematic treatment: R · Julia.

12. When is OLS coefficient the ATE

12.1. OLS cannot be ATE (most of the time)

Słoczyński clarifies when the OLS coefficient on a treatment dummy can and cannot be read as the ATE.

Consider the standard setup:

$$y = \alpha + \tau d + X\beta \tag{12.1}$$

Here d is the treatment dummy and X the covariates, under unconfoundedness. If the treatment effect is homogeneous, τ is the ATE. Under heterogeneous effects, τ is a convex combination of ATT and ATU — but with weights that are inversely proportional to the treatment shares. The more treated units there are, the less weight falls on ATT.

By definition,

$$\tau_{ATE} = \rho\tau_{ATT} + (1-\rho)\tau_{ATU} \tag{12.2}$$

The main result from the paper is:

$$\tau = (1-\rho)\tau_{ATT} + \rho\tau_{ATU} \tag{12.3}$$

where $\rho = P(d = 1)$, the proportion of treated; $\tau_{ATT} = E(y(1) - y(0)|d = 1)$, $\tau_{ATU} = E(y(1) - y(0)|d = 0)$, and $\tau_{ATE} = E(y(1) - y(0))$.

From these two equations, we see that τ and τ_{ATE} are quite different, unless the weight on ATT happens to equal ρ. A caution on reading this too simply: the clean form $\tau = (1-\rho)\tau_{ATT} + \rho\tau_{ATU}$ with $\rho = P(d = 1)$ is the **no-covariate special case** (or, with covariates, holds in terms of *covariate-adjusted* treatment-probability weights, not the raw marginal share). In Słoczyński's general result the weights depend on the conditional treatment probabilities (the propensity score), so a 50/50 *marginal* treatment share alone does **not** guarantee that OLS recovers the ATE — what matters is the covariate-adjusted weighting. With that caveat, the intuition stands: the more imbalanced the (conditional) treatment probabilities, the further OLS can be from the ATE.

Given the true propensity score, ATT, ATU, and ATE can be separated. In practice we use the estimated propensity score. Słoczyński provides the `hettreatreg` package in R and Stata.

The data are `nswcps`: the NSW experimental treatment group combined with a large CPS comparison sample, so the treated are a tiny fraction of the whole. The outcome is 1978 earnings and the covariates are age, its square, education, race, marital status, degree status, and earnings in 1974 and 1975. This is the classic LaLonde setting, and its extreme treatment imbalance is exactly what makes the weighting result bite.

```
library(hettreatreg)
data("nswcps")

summary(lm(re78 ~ treated + age + age2 + educ + black + hispanic + married + nodegree  + re74 +
```

```
Call:
lm(formula = re78 ~ treated + age + age2 + educ + black + hispanic +
    married + nodegree + re74 + re75, data = nswcps)

Residuals:
   Min     1Q Median     3Q    Max
-25130  -3601   1274   3668  55040

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 7634.34415  736.67074  10.363  < 2e-16 ***
treated      793.58704  548.25433   1.447  0.14778
age         -233.67749   41.18067  -5.674 1.42e-08 ***
age2           1.81437    0.56099   3.234  0.00122 **
educ         166.84923   28.65984   5.822 5.94e-09 ***
black       -790.60856  213.24523  -3.708  0.00021 ***
hispanic    -175.97512  218.99126  -0.804  0.42166
married      224.26599  149.84542   1.497  0.13450
nodegree     311.84453  178.51743   1.747  0.08068 .
re74           0.29534    0.01222  24.175  < 2e-16 ***
re75           0.47064    0.01216  38.700  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7002 on 16166 degrees of freedom
Multiple R-squared:  0.4762,    Adjusted R-squared:  0.4758
F-statistic:  1469 on 10 and 16166 DF,  p-value: < 2.2e-16
```

```
outcome <- nswcps$re78
treated <- nswcps$treated
our_vars <- c("age", "age2", "educ", "black", "hispanic", "married", "nodegree", "re74", "re75"
covariates <- subset(nswcps, select = our_vars)

hettreatreg(outcome, treated, covariates, verbose = TRUE)
```

```
"OLS" is the estimated regression coefficient on treated.
```

```
   OLS = 793.6

P(d=1) = 0.011
P(d=0) = 0.989

    w1 = 0.983
    w0 = 0.017
 delta = -0.971

   ATE = -6751
   ATT = 928.4
   ATU = -6840

OLS = w1*ATT + w0*ATU = 793.6
```

The output is the theorem in numbers. OLS puts 793.6 on `treated`. Only 1.1% of the sample is treated, $P(d=1)=0.011$, and the implied OLS weights are $w_1=0.983$ on ATT and $w_0=0.017$ on ATU. Compare those with the ATE's own weights, which are $\rho=0.011$ on ATT and 0.989 on ATU: they are very nearly reversed. The package prints the identity directly, $\text{OLS}=w_1\,\text{ATT}+w_0\,\text{ATU}=$ 793.6, which closes exactly.

The consequences are not subtle. ATT is 928.4, ATU is -6840, and ATE is -6751. So the OLS coefficient of 793.6 sits within 135 of the ATT and roughly 7,500 away from the ATE — and on the *opposite side of zero*. Reading 793.6 as "the effect of training on earnings" would get the sign wrong for the population. The `delta` of -0.971 that the package reports is the gap between the OLS weight on ATT and ρ; it is close to -1, its extreme value.

Słoczyński's paper itself is an algebraic decomposition of the OLS coefficient τ in terms of ATT/ATU weights; it does not propose or endorse the propensity-score-as-covariate estimator below as a general-purpose way to recover ATT/ATU/ATE. What follows is our own illustrative exercise, done in three steps, to see how such an estimator would behave in practice: First, estimate the propensity score equation, basically

$$d = X\gamma \tag{12.4}$$

Second, include the predicted propensity score as a covariate and estimate the model with treatment and control group separately. Note that regressing Y linearly on the scalar propensity score $p(X)$ assumes $E[Y(d)\mid p(X)]$ is linear in $p(X)$ — a strong and generally untested assumption, unlike matching/weighting directly on the propensity score, or regression adjustment on the covariates X themselves, which are the standard approaches. Treat the numbers below as an exploratory illustration rather than a recommended estimator.

Third, predict the counterfactuals for the full sample. Then we get the ATE, ATT and ATU.

One more reason to treat this as illustration rather than method: the propensity score here comes from a linear probability model, and on these data it runs from -0.034 to 0.154 — some fitted "probabilities" are negative. That is harmless when the score is used merely as a regressor, as it is here, but it would be fatal if the same score were used as an inverse-probability weight.

Let's see with the same data:

```
m_propensity <- lm(treated ~ age + age2 + educ + black + hispanic + married + nodegree  + re74
ps <- predict(m_propensity)
df_combined <- data.frame(nswcps, ps)
df.1 <- df_combined[which(treated == 1),]
df.0 <- df_combined[which(treated == 0),]
m_ot <- lm(re78 ~ ps, data = df.1)
ot <- suppressWarnings(predict(m_ot, newdata = df_combined)) # add newdata option to predict fo

m_oc <- lm(re78 ~ ps, data = df.0)
oc <- suppressWarnings(predict(m_oc, newdata = df_combined)) # add newdata option to predict fo

te <- ot - oc
ate <- mean(te)
df_combined$te <- te
att <- as.numeric(mean(df_combined[which(treated == 1),'te']))
atu <- as.numeric(mean(df_combined[which(treated == 0),'te']))

print(paste("ate=", signif(ate, 4)))
```

```
[1] "ate= -6751"
```

```
print(paste("att=", signif(att, 4)))
```

```
[1] "att= 928.4"
```

```
print(paste("atu=", signif(atu, 4)))
```

```
[1] "atu= -6840"
```

This reproduces the package's decomposition exactly: ATE −6751, ATT 928.4, ATU −6840, the same three numbers to four significant figures.

We see in this case OLS coefficient is hugely different from ATE. This is because $P(d = 1)$ is only .01, about 1 percent; we have much larger control group than treatment group. In this case, the OLS coefficient is far different from ATE; in fact, it's pretty close to ATT.

Now, what's the intuition that OLS coefficient is just the opposite as the ATE in terms of weighting ATT and ATU? The reason is that OLS is to predict actual outcome, not counterfactual outcome. For calculating ATE, we need counterfactual outcome predicted. Now suppose we have a lot of control observations, we'd much better predicting $(y(0)|d == 0)$ rather than $(y(1)|d == 1)$. That is, we are better getting the slope of $(y(0)|d == 0, X = x)$; therefore predicting $(y(0)|d == 1, X = x)$. Therefore although treatment group has less observations, but because the $(y(0)|d == 1)$ is better predicted, the ATT is better estimated, thus more weight. If the opposite happens, then the ATU should be more heavily weighted. Thus the counter-intuitive weighting.

12.2. Regression adjustment

Słoczyński's method is to include propensity score as a covariate. This propensity score serves as a proxy as all covariates. As Rosenbaum and Rubin (1983) showed, the propensity score is a balancing score (treatment is independent of the covariates given the propensity score). Another way to allow treatment effect to differ across all covariates is to allow interaction of treatment dummy with all covariates, then I am doing "Regression Adjustment" (see Stata's `teffects ra`, used in the treatment effects and matching chapter; note this is not `treatreg`, which was the old endogenous-treatment command, superseded by `etregress`). We can calculate ATE, ATT, ATU after the linear model.

```
library(tidyverse)
lm1 <- lm(re78 ~ treated*(age + age2 +  educ + black + hispanic + married + nodegree  + re74 +
y1 <- predict(lm1,newdata=nswcps %>% mutate(treated=1))
y0 <- predict(lm1,newdata=nswcps %>% mutate(treated=0))
ate=mean(y1-y0) #ate
print(paste("ate=",  signif(ate, 4)))
```

```
[1] "ate= -4930"
```

```
y1.att <- predict(lm1,newdata=nswcps %>% filter(treated==1))
y0.att <- predict(lm1,newdata=nswcps %>% filter(treated==1) %>% mutate(treated=0))
att=mean(y1.att)-mean(y0.att) #att
print(paste('att=',  signif(att, 4)))
```

```
[1] "att= 796"
```

```
y1.atu <- predict(lm1,newdata=nswcps %>% filter(treated==0) %>% mutate(treated=1))
y0.atu <- predict(lm1,newdata=nswcps %>% filter(treated==0))
atu=mean(y1.atu)-mean(y0.atu) #atu
print(paste('atu=', signif(atu, 4)))
```

```
[1] "atu= -4996"
```

Regression adjustment gives ATE -4930, ATT 796 and ATU -4996.

We see this is somewhat different from Słoczyński's method, which gave -6751, 928.4 and -6840: the two agree on sign and order of magnitude but not on level, unsurprisingly, since one regresses on a scalar propensity score and the other interacts the treatment with all nine covariates. But the point is the original OLS coefficient 794 is nowhere close to ATE under either approach — -4930 or -6751, both far away and both negative. In this case, it's very close to ATT (796 by regression adjustment, 928 by the propensity-score route), as the treatment group is very small.

Systematic treatment: R · Julia.

13. Matching and Weighting Part 1: matching

This chapter follows R's `MatchIt` and `WeightIt` packages (Greifer et al.).

13.1. Assumptions

Matching and weighting require three assumptions for causal identification in observational studies: (1) SUTVA, (2) ignorability (unconfoundedness), and (3) overlap (positivity). Under ignorability, conditioning on X makes treatment assignment D independent of the potential outcomes $Y(0)$ and $Y(1)$. In practice, the distribution of X is rarely balanced across treatment and control groups — matching and weighting restore that balance.

13.2. Distance

Matching finds control units close to treated units in X-space. The first question is how to define "close."

13.2.1. Propensity score

The most common distance measure is the propensity score (PS), which is the probability of being treated given X. The benefit is that when there is high dimension of X, we can reduce the dimension to a single number, the PS.

There are many ways to estimate the propensity score, such as logit, or more flexible parametric "gam", or other machine learning methods, such as "gbm", "lasso", "rpart", "randomforest", "cbps", etc. Here CBPS has seen more popularity. The propensity scores are estimated using the covariate balancing propensity score (CBPS) algorithm, which is a form of logistic regression where balance constraints are incorporated into a generalized method of moments estimation of the model coefficients.

13.2.2. Distance from covariates

Or we can compute distance from covariates directly, not using propensity score. The distance can be computed using Mahalanobis distance, or Euclidean distance, or other distance measures.

13.3. matching methods

With a distance, we then decide which method to use. The first few matching methods all need to specify a distance measure. They are matching based on distance. Then there are stratum matching methods, which do not need a distance measure. The last two methods are subsetting methods.

13.3.1. Nearest neighbor matching

NNM is widely used; it is also known as greedy matching. It matches each treated unit to the closest control unit in terms of distance. The closest control unit is defined as the one with the smallest distance to the treated unit. If there are multiple control units with the same distance and we want a 1:1 match, one is randomly selected.

13.3.2. Optimal pair matching

NNM is not optimal in the sense that it does not necessarily minimize the total distance between matched pairs. Optimal pair matching is a method that minimizes the total distance (or some other overall criterion) between matched pairs. It is more computationally intensive than NNM, but it can lead to better balance.

13.3.3. Optimal full matching

OFM assigns each unit (treated or control) to a subclass. For each subclass then to calculate the sum of within-subclass distances, and then to minimize the total distance across all subclasses. It is more computationally intensive than NNM and optimal pair matching. Weights are used based on subclass membership. Because all units are included, it can be used to estimate ATE.

13.3.4. Generalized Full matching

GFM is a variant of OFM. It uses a faster algorithm for large datasets.

13.3.5. Genetic matching

Genetic matching (`method = "genetic"`, Diamond and Sekhon 2013) runs nearest neighbor matching on a *scaled* generalized Mahalanobis distance, where the scaling weights on the covariates are not fixed in advance but chosen by a genetic algorithm that searches for the weighting giving the best covariate balance. That search is the point of the method — and also why it is far slower than the alternatives above.

13.3.6. Exact matching

It's a stratum matching. First subclasses are defined, by combinations of covariate values. Then units are assigned to subclasses. When a subclass has only either control or treated units, it is discarded.

Exact matching is nonparametric, but it works only when covariates are discrete and it happens that there are enough treated and control units in each subclass.

13.3.7. Coarsened exact matching

CEM is a more popular stratum matching. First bins are created after coarsening the covariates. then use exact matching on the covariate bins. CEM is nonparametric. User needs to decide how many bins to create for each covariate.

13.3.8. Subclassification

Subclassification is another kind of stratum matching. It uses the propensity score to define bins, usually quantiles of PS in the treated group, or control group or overall, depending on target estimand.

13.3.9. Cardinality and profile matching

Cardinality matching involves finding the largest sample that satisfies user-supplied balance constraints and constraints on the ratio of matched treated to matched control units. Profile matching involves identifying a target distribution (e.g., the full sample for the ATE or the treated units for the ATT) and finding the largest subset of the treated and control groups that satisfy user-supplied balance constraints with respect to that target.

13.4. Example

Here is an example with Lalonde data, we are interested in the effect of treatment on "re78" (1978 real earnings). There are 614 observations, 185 treated and 429 control, with covariates age, education, race, marital status, degree status, and earnings in 1974 and 1975. The first `matchit` call uses `method = NULL`, which fits the propensity score but performs no matching — a way of getting the pre-match balance table.

```r
library("MatchIt")
data("lalonde")

head(lalonde)
```

```
     treat age educ   race married nodegree re74 re75       re78
NSW1     1  37   11  black       1        1    0    0  9930.0460
NSW2     1  22    9 hispan       0        1    0    0  3595.8940
NSW3     1  30   12  black       0        0    0    0 24909.4500
NSW4     1  27   11  black       0        1    0    0  7506.1460
NSW5     1  33    8  black       0        1    0    0   289.7899
NSW6     1  22    9  black       0        1    0    0  4056.4940
```

```
# No matching; constructing a pre-match matchit object
m.out0 <- matchit(treat ~ age + educ + race + married +
                   nodegree + re74 + re75,
                  data = lalonde,
                  method = NULL,
                  distance = "glm")
summary(m.out0)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, method = NULL, distance = "glm")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5774        0.1822          1.7941     0.9211    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6444
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Sample Sizes:
          Control Treated
```

```
All           429     185
Matched       429     185
Unmatched       0       0
Discarded       0       0
```

The imbalance is severe. Standardized mean differences are −0.72 for 1974 earnings, 1.76 for the black indicator, −1.88 for white, −0.83 for married and −0.31 for age, and the propensity score itself differs by 1.79 standard deviations. The treated average $2,096 in 1974 earnings against $5,619 for controls, and 84% are black against 20%. These are not two samples that differ at the margin.

Or use "cobalt" to see the balance:

```
library("cobalt")
bal.tab(treat ~ age + educ + race + married + nodegree + re74 + re75,
        data = lalonde, estimand = "ATT", thresholds = c(m = .05))
```

```
Balance Measures
               Type Diff.Un      M.Threshold.Un
age          Contin. -0.3094 Not Balanced, >0.05
educ         Contin.  0.0550 Not Balanced, >0.05
race_black    Binary  0.6404 Not Balanced, >0.05
race_hispan   Binary -0.0827 Not Balanced, >0.05
race_white    Binary -0.5577 Not Balanced, >0.05
married       Binary -0.3236 Not Balanced, >0.05
nodegree      Binary  0.1114 Not Balanced, >0.05
re74         Contin. -0.7211 Not Balanced, >0.05
re75         Contin. -0.2903 Not Balanced, >0.05

Balance tally for mean differences
                    count
Balanced, <0.05         0
Not Balanced, >0.05     9

Variable with the greatest mean difference
 Variable Diff.Un      M.Threshold.Un
     re74 -0.7211 Not Balanced, >0.05

Sample sizes
    Control Treated
All     429     185
```

`bal.tab` says the same thing against a stricter 0.05 threshold: all nine covariate contrasts fail, with `re74` the worst at −0.7211.

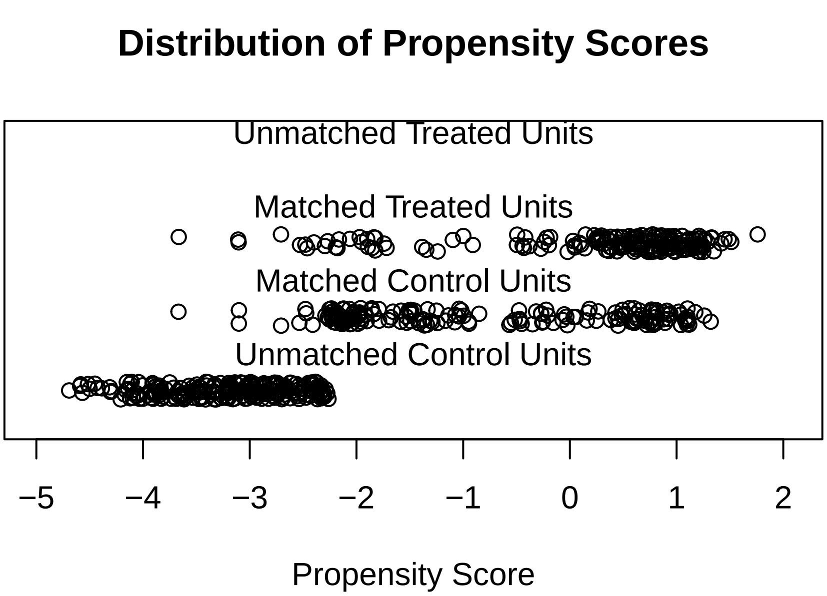
First do a PS match. The chunks below run several distance measures in turn — the logit of a logistic propensity score, a GAM score with smooth terms in age and education, CBPS targeting the ATC with replacement, Mahalanobis distance on the raw covariates, and full matching on a probit score. Each prints its own balance table; what changes across them is the `distance` row and how much of the covariate imbalance survives.

```r
# Matching on logit of a PS estimated with logistic
# regression:
m.out1 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "glm",
                  link = "linear.logit")
summary(m.out1)
```

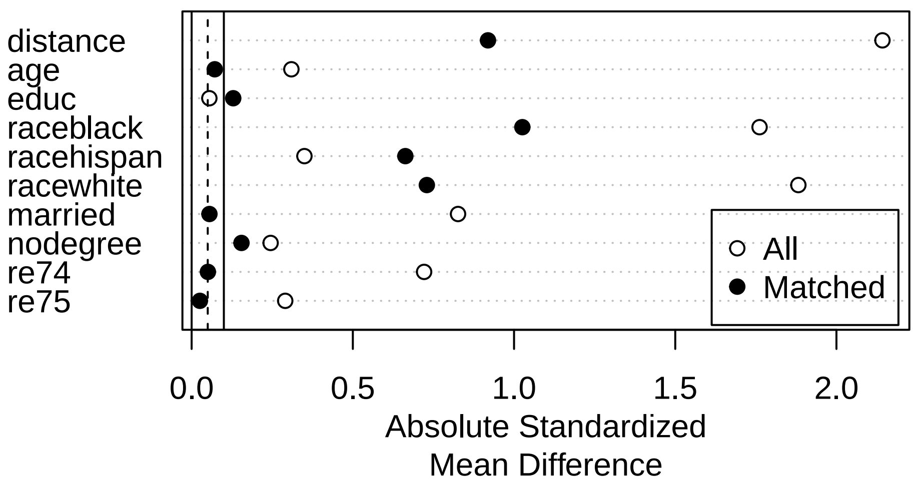
```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "glm", link = "linear.logit")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2657       -2.1500          2.1426     0.5398    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6444
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2657       -0.7706          0.9192     0.7712    0.1321
age              25.8162       25.3027          0.0718     0.4568    0.0847
educ             10.3459       10.6054         -0.1290     0.5721    0.0239
raceblack         0.8432        0.4703          1.0259          .    0.3730
```

```
racehispan        0.0595       0.2162         -0.6629          .    0.1568
racewhite         0.0973       0.3135         -0.7296          .    0.2162
married           0.1892       0.2108         -0.0552          .    0.0216
nodegree          0.7081       0.6378          0.1546          .    0.0703
re74           2095.5737    2342.1076         -0.0505     1.3289    0.0469
re75           1532.0553    1614.7451         -0.0257     1.4956    0.0452
           eCDF Max Std. Pair Dist.
distance     0.4216          0.9193
age          0.2541          1.3938
educ         0.0757          1.2474
raceblack    0.3730          1.0259
racehispan   0.1568          1.0743
racewhite    0.2162          0.8390
married      0.0216          0.8281
nodegree     0.0703          1.0106
re74         0.2757          0.7965
re75         0.2054          0.7381

Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

```r
plot(m.out1, type = "jitter", interactive = FALSE)
```



```r
plot(summary(m.out1))
```



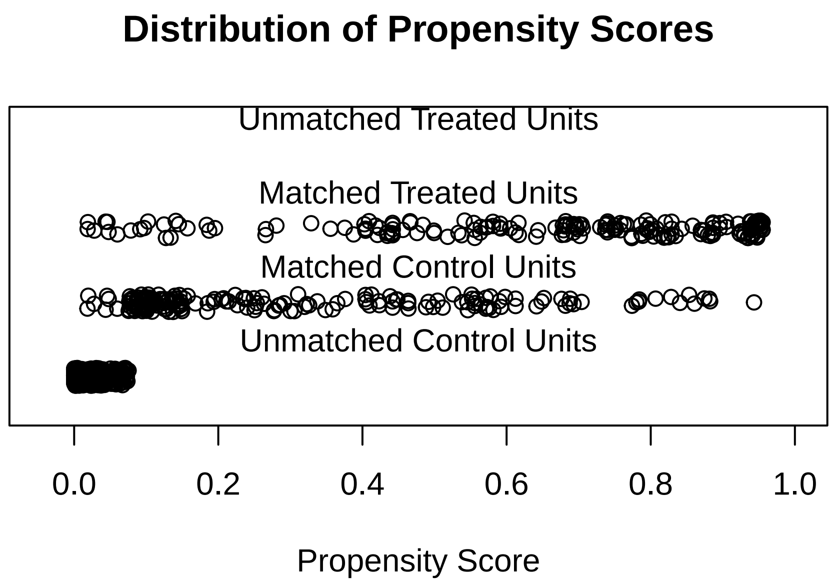
We see there are still some imbalance on variables such as "raceblack". We can use more flexible models for the PS.

```r
# GAM logistic PS with smoothing splines (s()):
m.out2 <- matchit(treat ~ s(age) + s(educ) +
                    race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "gam")
summary(m.out2)
```

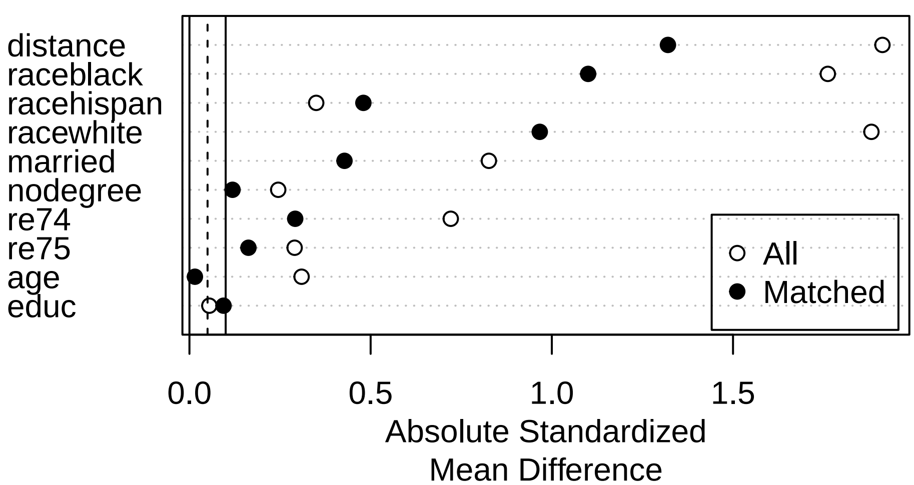
```
Call:
matchit(formula = treat ~ s(age) + s(educ) + race + married +
    nodegree + re74 + re75, data = lalonde, distance = "gam")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.6566        0.1481          1.9123     1.6161    0.4155
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
```

```
re75          1532.0553     2466.4844          -0.2903     0.9563    0.1342
age             25.8162       28.0303          -0.3094     0.4400    0.0813
educ            10.3459       10.2354           0.0550     0.4959    0.0347
           eCDF Max
distance     0.7172
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876
age          0.1577
educ         0.1114

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.6566        0.3055          1.3204     1.2324    0.1853
raceblack         0.8432        0.4432          1.1002          .    0.4000
racehispan        0.0595        0.1730         -0.4800          .    0.1135
racewhite         0.0973        0.3838         -0.9667          .    0.2865
married           0.1892        0.3568         -0.4278          .    0.1676
nodegree          0.7081        0.6541          0.1189          .    0.0541
re74           2095.5737     3521.7556         -0.2919     1.0512    0.1346
re75           1532.0553     2056.2000         -0.1628     1.2120    0.0917
age              25.8162       25.7081          0.0151     0.7958    0.0270
educ             10.3459       10.1568          0.0941     0.6186    0.0270
           eCDF Max Std. Pair Dist.
distance     0.5189          1.3204
raceblack    0.4000          1.1002
racehispan   0.1135          0.8914
racewhite    0.2865          1.1491
married      0.1676          1.0627
nodegree     0.0541          0.9749
re74         0.4270          0.8894
re75         0.2378          0.8453
age          0.0865          1.1619
educ         0.0919          1.2447

Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

```r
plot(m.out2, type = "jitter", interactive = FALSE)
```




```r
plot(summary(m.out2))
```



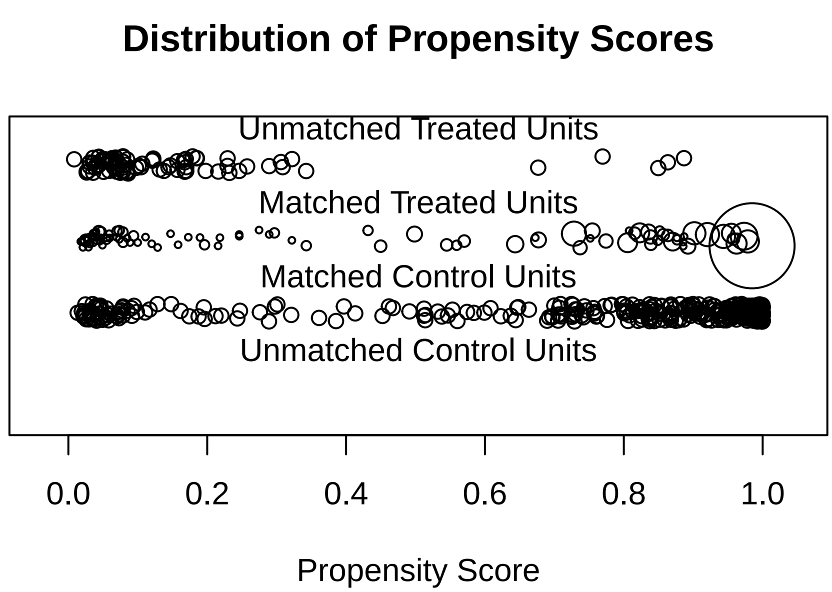
Or use CBPS.

```r
# CBPS for ATC matching w/replacement, using the just-
# identified version of CBPS (setting method = "exact"):
m.out3 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "cbps",
                  estimand = "ATC",
                  distance.options = list(method = "exact"),
                  replace = TRUE)
summary(m.out3)
```

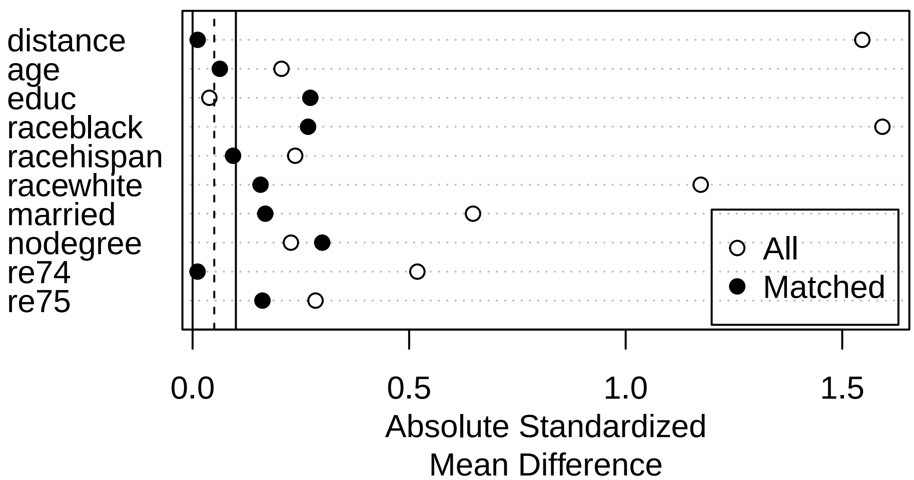
```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "cbps", distance.options = list(method = "exact"),
    estimand = "ATC", replace = TRUE)

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2801        0.7758         -1.5464     0.9710    0.3525
age              25.8162       28.0303         -0.2053     0.4400    0.0813
educ             10.3459       10.2354          0.0387     0.4959    0.0347
raceblack         0.8432        0.2028          1.5928          .    0.6404
racehispan        0.0595        0.1422         -0.2369          .    0.0827
racewhite         0.0973        0.6550         -1.1732          .    0.5577
married           0.1892        0.5128         -0.6475          .    0.3236
nodegree          0.7081        0.5967          0.2270          .    0.1114
re74           2095.5737     5619.2365         -0.5190     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2838     0.9563    0.1342
           eCDF Max
distance     0.5967
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.7720        0.7758         -0.0117     1.1568    0.0499
age              27.3520       28.0303         -0.0629     0.4383    0.1091
educ             11.0117       10.2354          0.2719     0.7323    0.0566
raceblack         0.3100        0.2028          0.2667          .    0.1072
```

```
racehispan        0.1096        0.1422          -0.0934          .     0.0326
racewhite         0.5804        0.6550          -0.1569          .     0.0746
married           0.5967        0.5128           0.1679          .     0.0839
nodegree          0.4499        0.5967          -0.2994          .     0.1469
re74           5696.9438     5619.2365           0.0114     0.8562     0.0994
re75           1935.6090     2466.4844          -0.1613     1.5194     0.1071
           eCDF Max Std. Pair Dist.
distance     0.3543          0.0194
age          0.2191          0.8676
educ         0.2727          1.0719
raceblack    0.1072          0.3826
racehispan   0.0326          0.6407
racewhite    0.0746          0.5492
married      0.0839          0.5596
nodegree     0.1469          1.0692
re74         0.2541          0.7513
re75         0.2727          0.9781

Sample Sizes:
              Control Treated
All               429   185.
Matched (ESS)     429     6.67
Matched           429    93.
Unmatched           0    92.
Discarded           0     0.
```

```r
plot(m.out3, type = "jitter", interactive = FALSE)
```



```r
plot(summary(m.out3))
```




```r
# Mahalanobis distance matching - no PS estimated
m.out4 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "mahalanobis")
summary(m.out4)
```

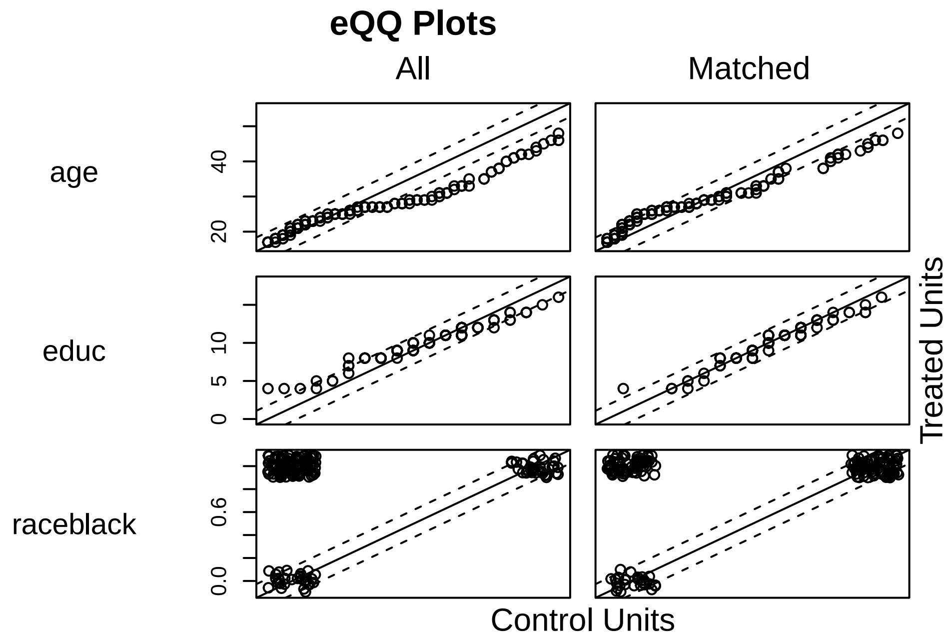
```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "mahalanobis")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
age          0.1577
```

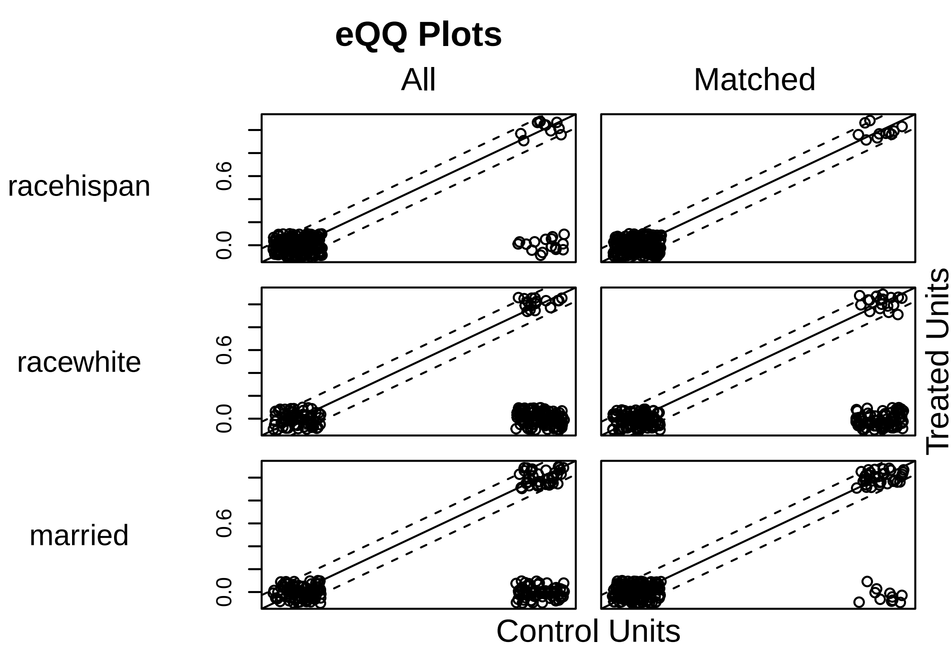
```
educ          0.1114
raceblack     0.6404
racehispan    0.0827
racewhite     0.5577
married       0.3236
nodegree      0.1114
re74          0.4470
re75          0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
age              25.8162       24.9081          0.1269     0.5327    0.0695
educ             10.3459       10.4324         -0.0430     0.8345    0.0108
raceblack         0.8432        0.4649          1.0407          .    0.3784
racehispan        0.0595        0.0595          0.0000          .    0.0000
racewhite         0.0973        0.4757         -1.2767          .    0.3784
married           0.1892        0.2486         -0.1518          .    0.0595
nodegree          0.7081        0.6595          0.1070          .    0.0486
re74           2095.5737     3305.4420         -0.2476     0.8705    0.1012
re75           1532.0553     1957.5097         -0.1322     1.0071    0.0654
           eCDF Max Std. Pair Dist.
age          0.2378          0.6195
educ         0.0486          0.2903
raceblack    0.3784          1.0407
racehispan   0.0000          0.0000
racewhite    0.3784          1.2767
married      0.0595          0.1794
nodegree     0.0486          0.1070
re74         0.3351          0.4546
re75         0.2162          0.4113

Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

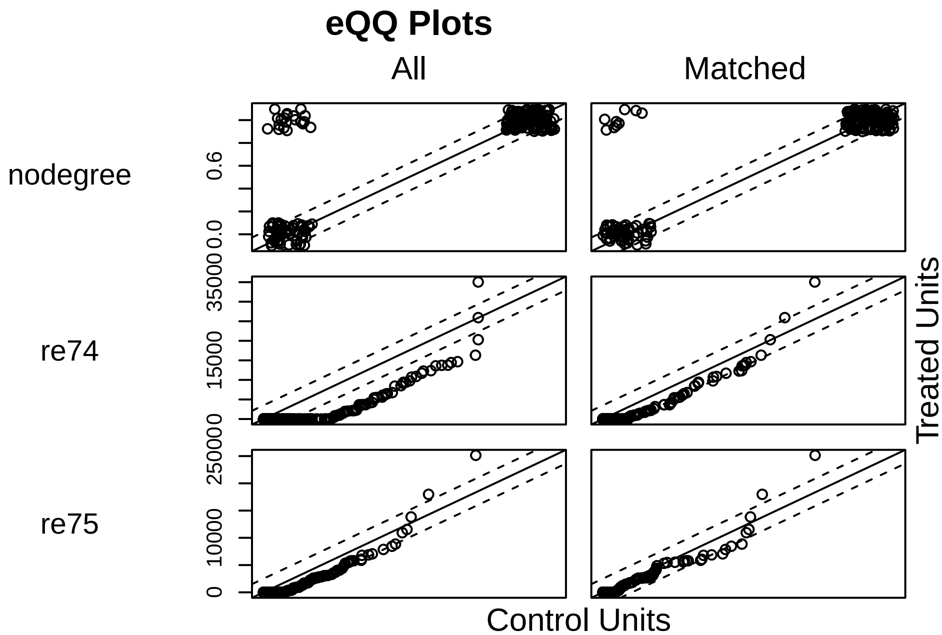
```
m.out4$distance #NULL
```

```
NULL
```

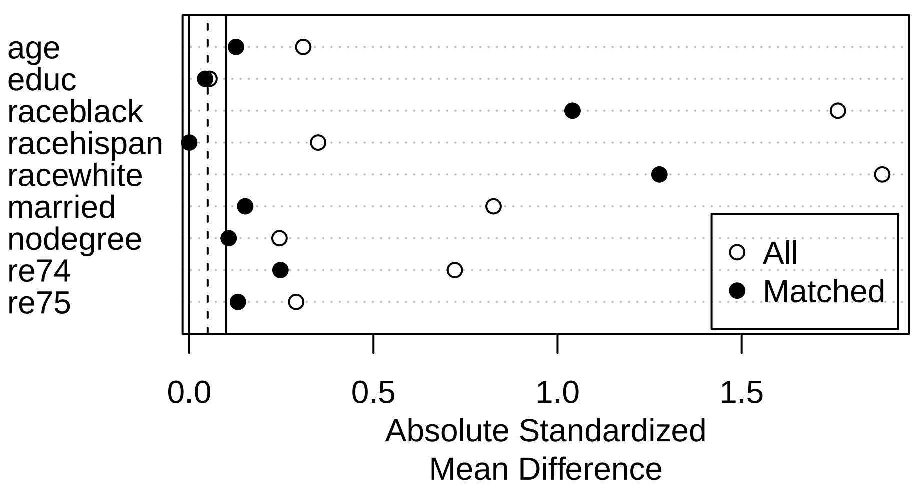
```
plot(m.out4,  interactive = FALSE)
```


eQQ Plots
All
Matched
age
educ
raceblack
Treated Units
Control Units
20
40
0
5
10
0.0
0.6


eQQ Plots
All
Matched
racehispan
racewhite
married
Treated Units
Control Units
0.0
0.6




```
plot(summary(m.out4))
```



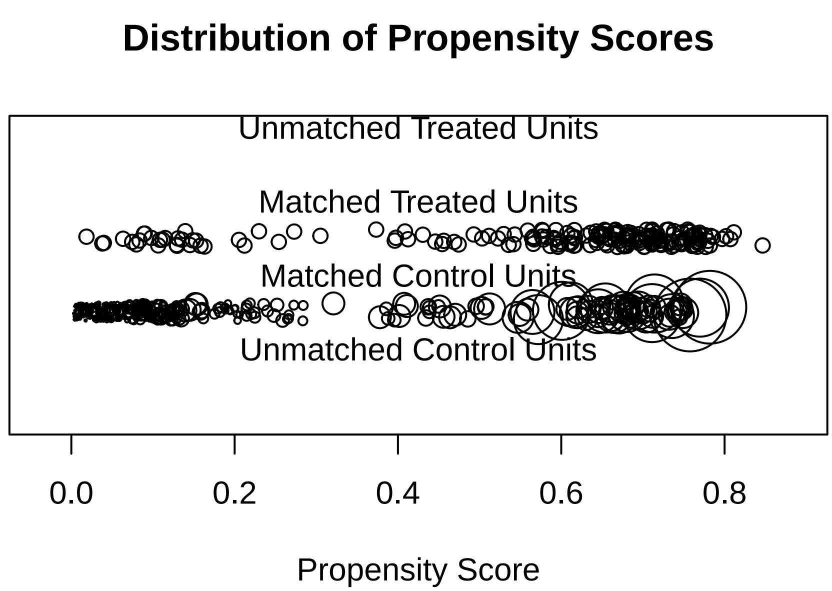

```
# Full matching on a probit PS
m.out5 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  method = "full",
                  distance = "glm",
```

```r
                    link = "probit")

summary(m.out5)
```

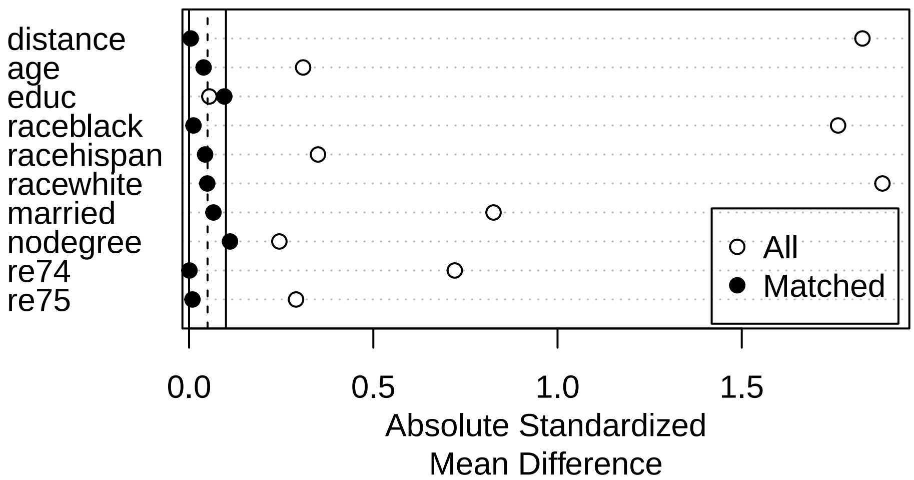
```

Call:
matchit(formula = treat ~ age + educ + race + married + nodegree + 
    re74 + re75, data = lalonde, method = "full", distance = "glm", 
    link = "probit")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5773        0.1817          1.8276     0.8777    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6413
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5773        0.5764          0.0045     0.9949    0.0043
age              25.8162       25.5347          0.0393     0.4790    0.0787
educ             10.3459       10.5381         -0.0956     0.6192    0.0253
raceblack         0.8432        0.8389          0.0119          .    0.0043
racehispan        0.0595        0.0492          0.0435          .    0.0103
racewhite         0.0973        0.1119         -0.0493          .    0.0146
married           0.1892        0.1633          0.0660          .    0.0259
nodegree          0.7081        0.6577          0.1110          .    0.0504
re74           2095.5737     2100.2150         -0.0009     1.3467    0.0314
re75           1532.0553     1561.4420         -0.0091     1.5906    0.0536
           eCDF Max Std. Pair Dist.
```

```
distance      0.0486        0.0198
age           0.2742        1.2843
educ          0.0730        1.2179
raceblack     0.0043        0.0162
racehispan    0.0103        0.4412
racewhite     0.0146        0.3454
married       0.0259        0.4473
nodegree      0.0504        0.9872
re74          0.1881        0.8387
re75          0.1984        0.8240

Sample Sizes:
              Control Treated
All            429.       185
Matched (ESS)   50.76     185
Matched        429.       185
Unmatched        0.         0
Discarded        0.         0
```

```r
plot(m.out5, type = "jitter", interactive = FALSE)
```




```r
plot(summary(m.out5))
```



Note that `m.out4$distance` prints `NULL`: Mahalanobis matching estimates no propensity score at all, so there is no distance column and no `distance` row in its balance table. That is the point of the method — it works on the covariate space directly rather than collapsing it to a scalar.

13.5. Estimation

After we get a matched data that we think is good enough, we can estimate the treatment effect. We use `m.out5`, the full match on a probit score.

```r
m.data <- match_data(m.out5)

head(m.data)
```

```
     treat age educ   race married nodegree re74 re75       re78  distance
NSW1     1  37   11  black       1        1    0    0  9930.0460 0.6356769
NSW2     1  22    9 hispan       0        1    0    0  3595.8940 0.2298151
NSW3     1  30   12  black       0        0    0    0 24909.4500 0.6813558
NSW4     1  27   11  black       0        1    0    0  7506.1460 0.7690590
NSW5     1  33    8  black       0        1    0    0   289.7899 0.6954138
NSW6     1  22    9  black       0        1    0    0  4056.4940 0.6943658
     weights subclass
NSW1       1        1
NSW2       1       55
NSW3       1       63
NSW4       1       70
NSW5       1       79
NSW6       1       86
```

What "match_data" does is to grab the matched data with a few additional variables such as "weights", "subclass", and "distance". Then we can use it in a simple linear model with weights.

```
library("marginaleffects")

fit <- lm(re78 ~ treat * (age + educ + race + married +
                            nodegree + re74 + re75),
          data = m.data,
          weights = weights)

avg_comparisons(fit,
                variables = "treat",
                vcov = ~subclass,
                newdata = subset(m.data, treat == 1))
```

```
 Estimate Std. Error    z Pr(>|z|)   S 2.5 % 97.5 %
     1977        704 2.81  0.00501 7.6   596   3357

Term: treat
Type: response
Comparison: 1 - 0
```

The ATT is \$1,977 with a subclass-clustered standard error of \$704, $p = 0.005$, and a 95% interval of $[596, 3357]$. Training raised 1978 earnings for those who took it.

Note we allow interaction of treatment and covariates, but we do not need to demean the covariates here, since we are relying on "marginaleffects" and its "avg_comparisons" function to compute the average treatment effect. The "newdata" argument is used to specify the effect, in this case ATT. Note the weights are generated this way: since it's ATT, all treated units are weighted by 1, and control units are weighted by $p/(1-p)$, where p is proportion of treated units in the subclass. Anyway, for full matching, we get the comparison of potential outcomes for the treated, with weights, and clustered standard errors on subclasses, because that the level the comparison is conducted.

Systematic treatment: R · Julia.

14. Matching and Weighting - Part 2: weighting

This chapter follows R's `WeightIt` package (Greifer et al.). The assumptions — SUTVA, ignorability, and overlap — are the same as in the matching chapter.

14.1. Weighting methods

Weighting creates a pseudo-population in which X is balanced across treatment and control. The most common approach is inverse probability of treatment weighting (IPTW), which uses the propensity score. Several alternatives are gaining traction.

14.1.1. Inverse probability of treatment weighting (IPTW)

IPTW is the most common weighting method. It uses the inverse of the propensity score as weights. For the **ATE**, weight each treated unit by $1/p$ and each control unit by $1/(1-p)$, where p is the propensity score. For the **ATT** — which is what the examples below request via `estimand = "ATT"` — the treated units all get weight 1 and the controls get $p/(1-p)$; you can see this in the weight summaries, where the treated range is exactly 1 to 1. Either way the point is the same: create a pseudo population in which the distribution of X is balanced between the groups.

14.1.2. Covariate balancing propensity score (CBPS)

The same idea, but with a different way to estimate the propensity score. CBPS is a form of logistic regression where balance constraints are incorporated to a generalized method of moments estimation of the model coefficients.

14.1.3. Entropy balancing

Entropy balancing involves the specification of an optimization problem, the solution to which is then used to compute the weights. The constraints of the primal optimization problem correspond to covariate balance on the means (for binary and multi-category treatments) or treatment-covariate covariances (for continuous treatments), positivity of the weights, and that the weights sum to a certain value. Entropy balancing is doubly robust (for the ATT) in the sense that it is consistent either when the true propensity score model is a logistic regression of the treatment on the covariates or when the true outcome model for the control units is a linear regression of the outcome on the covariates, and it attains a semi-parametric efficiency bound when both are true. Entropy balancing will always yield exact mean balance on the included terms (not necessarily on the distribution of the covariates).

14.1.4. Energy balancing

Energy balancing is a method of estimating weights using optimization without a propensity score.

The primary benefit of energy balancing is that all features of the covariate distribution are balanced, not just means, as with other optimization-based methods like entropy balancing.

14.1.5. SuperLearner

SuperLearner works by fitting several machine learning models to the treatment and covariates and then taking a weighted combination of the generated predicted values to use as the propensity scores, which are then used to construct weights.

14.2. Example

Here is an example with Lalonde data, we are interested in the effect of treatment on “re78” (real earnings 1978).

14.2.1. IPW

```r
library("WeightIt")
library(cobalt)
data(lalonde)
w.out1 <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, estimand = "ATT", method = "glm")
w.out1 #print the output
```

```
A weightit object
 - method: "glm" (propensity score weighting with GLM)
 - number of obs.: 614
 - sampling weights: none
 - treatment: 2-category
 - estimand: ATT (focal: 1)
 - covariates: age, educ, race, married, nodegree, re74, re75
```

```r
summary(w.out1)
```

```
                  Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.            ||                     1.
```

```
control 0.009 |---------------------------| 3.743

- Units with the 5 most extreme weights by group:

            5     4    3     2     1
 treated    1     1    1     1     1
          597   573  381   411   303
 control 3.03 3.059 3.24 3.523 3.743

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.818 1.289   1.098       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     99.82     185
```

The treated weights are all exactly 1, as the ATT requires, and the control weights run from 0.009 to 3.743. The cost is in the effective sample size: 429 controls carry the information of 99.8 equally weighted ones, a loss of more than three quarters.

```r
library(cobalt)
bal.tab(w.out1, stats = c("m", "v"), thresholds = c(m = .05))
```

```
Balance Measures
                Type Diff.Adj         M.Threshold V.Ratio.Adj
prop.score  Distance  -0.0205     Balanced, <0.05      1.0324
age          Contin.   0.1188 Not Balanced, >0.05      0.4578
educ         Contin.  -0.0284     Balanced, <0.05      0.6636
race_black    Binary  -0.0022     Balanced, <0.05           .
race_hispan   Binary   0.0002     Balanced, <0.05           .
race_white    Binary   0.0021     Balanced, <0.05           .
married       Binary   0.0186     Balanced, <0.05           .
nodegree      Binary   0.0184     Balanced, <0.05           .
re74         Contin.  -0.0021     Balanced, <0.05      1.3206
re75         Contin.   0.0110     Balanced, <0.05      1.3938

Balance tally for mean differences
                    count
Balanced, <0.05         9
Not Balanced, >0.05     1

Variable with the greatest mean difference
```

```
 Variable Diff.Adj         M.Threshold
      age   0.1188 Not Balanced, >0.05

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     99.82     185
```

Nine of ten contrasts now pass the strict 0.05 threshold, against none before weighting. `age` is the exception at 0.1188. Note also the variance ratios, which IPTW does not target: `age` sits at 0.458 and `re74` at 1.321, so the *means* are balanced while the *spreads* are not.

14.2.2. entropy balancing

```
w.out2 <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, estimand = "ATT", method = "ebal")
summary(w.out2)
```

```
                  Summary of weights

- Weight ranges:

          Min                                 Max
treated 1.            ||                    1.
control 0.008 |---------------------------| 4.062

- Units with the 5 most extreme weights by group:

             5     4    3     2     1
 treated     1     1    1     1     1
           608   381  597   303   411
 control 3.073 3.235 3.45 3.897 4.062

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.834 1.287   1.101       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     98.46     185
```

```r
bal.tab(w.out2, stats = c("m", "v"), thresholds = c(m = .05))
```

```
Balance Measures
                Type Diff.Adj     M.Threshold V.Ratio.Adj
age          Contin.       -0 Balanced, <0.05      0.4096
educ         Contin.       -0 Balanced, <0.05      0.6635
race_black    Binary        0 Balanced, <0.05           .
race_hispan   Binary        0 Balanced, <0.05           .
race_white    Binary       -0 Balanced, <0.05           .
married       Binary       -0 Balanced, <0.05           .
nodegree      Binary        0 Balanced, <0.05           .
re74         Contin.       -0 Balanced, <0.05      1.3265
re75         Contin.       -0 Balanced, <0.05      1.3351

Balance tally for mean differences
                    count
Balanced, <0.05         9
Not Balanced, >0.05     0

Variable with the greatest mean difference
 Variable Diff.Adj     M.Threshold
     re74       -0 Balanced, <0.05

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     98.46     185
```

Every mean difference is now zero to three decimals — that is the constraint entropy balancing imposes directly, rather than hoping a propensity model delivers it — and all nine pass. The effective sample size is 98.46, essentially the same as IPTW's 99.82, so the exact balance came at no extra cost in precision here. The variance ratios are unchanged in character (`age` 0.410, `re74` 1.327): entropy balancing constrains means, not distributions.

Suppose we are satisfied with the balance, we can use the weights to estimate the treatment effect. We can use the "marginaleffects" package to do that.

```r
# Fit outcome model
fit <- lm_weightit(re78 ~ treat * (age + educ + race + married +
                                     nodegree + re74 + re75),
                   data = lalonde, weightit = w.out2)
# G-computation for the treatment effect
library("marginaleffects")
avg_comparisons(fit, variables = "treat",
                newdata = subset(lalonde, treat == 1))
```

```
 Estimate Std. Error    z Pr(>|z|)   S 2.5 % 97.5 %
     1273        770 1.65   0.0983 3.3  -236   2783

Term: treat
Type: probs
Comparison: 1 - 0
```

The entropy-balanced ATT is $1,273 with a standard error of $770, $p = 0.098$, and a 95% interval of $[-236, 2783]$ that includes zero.

14.2.3. energy balancing

```
w.out3 <- tryCatch(
  weightit(treat ~ age + educ + race + married +
                   nodegree + re74 + re75, data = lalonde,
                 method = "energy", estimand = "ATE"),
  error = function(e) { message("Energy balancing failed: ", e$message); NULL }
)
if (!is.null(w.out3)) summary(w.out3)
```

```
                  Summary of weights

- Weight ranges:

        Min                                  Max
treated   0 |---------------------------| 14.041
control   0 |---------------------|       11.065

- Units with the 5 most extreme weights by group:

            10   179   125    137    124
 treated 5.984  6.23 6.758 13.202 14.041
           537   553   573    420    608
 control 5.499 6.404 6.599  8.274 11.065

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       1.821 1.054    0.92       0
control       1.264 0.866   0.579       0

- Effective Sample Sizes:

           Control Treated
```

```
Unweighted  429.    185.
Weighted    165.42   43.04
```

```
if (!is.null(w.out3)) {
  fit <- lm_weightit(re78 ~ treat * (age + educ + race + married +
                                     nodegree + re74 + re75),
                     data = lalonde, weightit = w.out3)
  # G-computation for the treatment effect
  avg_comparisons(fit, variables = "treat")
}
```

```
 Estimate Std. Error       z Pr(>|z|)   S 2.5 % 97.5 %
      -73       1085 -0.0672    0.946 0.1 -2200   2054

Term: treat
Type: probs
Comparison: 1 - 0
```

Two things are worth noticing in that output rather than passing over it. First, the estimand changed: `w.out3` requests the **ATE**, while the entropy-balancing fit above targeted the **ATT**, so the two numbers (−73 against 1273) are not competing estimates of the same quantity. Second, look at the weight summary. Entropy balancing gives every treated unit a weight of exactly 1, as the ATT requires. Energy balancing spreads the treated weights from 0 to 14.0 — one treated unit is effectively discarded and another counts as fourteen — and the resulting standard error (1085) is much larger than entropy balancing's (770). Balancing the whole covariate *distribution* rather than just its means is a genuine advantage, but it is bought with weight variability, and that shows up directly in the precision of the estimate.

14.3. comparing matching and weighting

Matching and weighting are popular in different fields. We used to say matching involves discarding data therefore changing the estimand, but now with full matching and other methods, matching can be done without losing data. We used to believe weighting can be unstable, as weights can be highly variable, but with CBPS, entropy balancing and energy balancing, weighting can perform much better.

So, in practice, we can try different methods, either matching or weighting, and see which one gives us better balance. Then use "marginaleffects" to get the treatment effect.

It is worth putting numbers on that, because on this data the choice changes the conclusion. Full matching in the previous chapter gave an ATT of \$1,977 with a standard error of \$704 and $p = 0.005$. Entropy balancing above gives \$1,273 with \$770 and $p = 0.098$. The two point estimates are about one standard error apart, and one interval excludes zero while the other does not.

Note which way that cuts. Entropy balancing achieves *exact* mean balance, while full matching leaves `educ` at −0.096; the better-balanced method gives the smaller and less significant estimate.

Better balance on means is not the same thing as a more credible answer, and reporting whichever method happened to cross the significance line would be the wrong response to this. The honest summary reports both and says the data do not settle it.

Systematic treatment: R · Julia.

15. Sensitivity analysis of OLS with sensemakr

15.1. OVB

Consider a true model,

$$Y = \delta D + X\beta + \gamma Z + \epsilon \tag{15.1}$$

If Z is unobserved we estimate

$$Y = \delta D + X\beta + u \tag{15.2}$$

and face omitted variable bias (OVB). Cinelli and Hazlett (2020) show that

$$|\widehat{\text{bias}}| = \sqrt{\frac{R^2_{Y\sim Z|D,X} R^2_{D\sim Z|X}}{1 - R^2_{D\sim Z|X}}} \left(\frac{\hat{\sigma}_{y.dx}}{\hat{\sigma}_{d.x}}\right) \tag{15.3}$$

Bias is large only when the omitted variable is strongly associated with both Y and D. The Cinelli and Hazlett framework yields three key quantities.

15.2. Partial R^2 of treatment on outcome

This measures how much variation in the outcome the treatment explains after partialling out X. In the extreme case where the confounder explains all remaining outcome variance ($R^2_{Y\sim Z|D,X} = 1$), the treatment effect goes to zero only if $R^2_{D\sim Z|X} \geq R^2_{Y\sim D|X}$. In other words, the confounder's association with treatment must exceed the treatment's partial association with the outcome.

15.3. Robust value

$RV_{q,\alpha}$ is the minimum strength of association the confounder must have with both treatment and outcome for a $(1-\alpha)$ confidence interval to include a $q\%$ change from the current estimate. With $q = 1$ this gives the threshold for the effect to include zero. Both the Z-D and Z-Y associations must exceed this threshold simultaneously.

15.4. Bounds using observed covariate(s)

The first two quantities are dimensionless but hard to interpret without context. Benchmarking against an observed covariate helps. If a covariate X_j is known to be an important predictor, we can ask how much stronger the confounder must be relative to X_j to drive the effect to zero.

$$k_D := \frac{R^2_{D\sim Z|X_{-j}}}{R^2_{D\sim X_j|X_{-j}}} \tag{15.4}$$

where X_{-j} represents all covariates other than the jth we pick as the benchmark covariate.

$$k_Y := \frac{R^2_{Y\sim Z|D,X_{-j}}}{R^2_{Y\sim X_j|D,X_{-j}}} \tag{15.5}$$

15.5. Example

Here is an example from sensemakr documentation.

```r
# loads package
library(sensemakr)

# loads dataset
data("darfur")

# runs regression model
model <- lm(peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
                         pastvoted + hhsize_darfur + female + village, data = darfur)

# runs sensemakr for sensitivity analysis
sensitivity <- sensemakr(model = model,
                         treatment = "directlyharmed",
                         benchmark_covariates = "female",
                         kd = 1:3)

# short description of results
sensitivity
```

```
Sensitivity Analysis to Unobserved Confounding

Model Formula: peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
    pastvoted + hhsize_darfur + female + village

Null hypothesis: q = 1 and reduce = TRUE
```

```
Unadjusted Estimates of ' directlyharmed ':
  Coef. estimate: 0.09732
  Standard Error: 0.02326
  t-value: 4.18445

Sensitivity Statistics:
  Partial R2 of treatment with outcome: 0.02187
  Robustness Value, q = 1 : 0.13878
  Robustness Value, q = 1 alpha = 0.05 : 0.07626

For more information, check summary.
```

```r
# long description of results
summary(sensitivity)
```

```
Sensitivity Analysis to Unobserved Confounding

Model Formula: peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
    pastvoted + hhsize_darfur + female + village

Null hypothesis: q = 1 and reduce = TRUE
-- This means we are considering biases that reduce the absolute value of the current estimate
-- The null hypothesis deemed problematic is H0:tau = 0

Unadjusted Estimates of 'directlyharmed':
  Coef. estimate: 0.0973
  Standard Error: 0.0233
  t-value (H0:tau = 0): 4.1844

Sensitivity Statistics:
  Partial R2 of treatment with outcome: 0.0219
  Robustness Value, q = 1: 0.1388
  Robustness Value, q = 1, alpha = 0.05: 0.0763

Verbal interpretation of sensitivity statistics:

-- Partial R2 of the treatment with the outcome: an extreme confounder (orthogonal to the cova

-- Robustness Value, q = 1: unobserved confounders (orthogonal to the covariates) that explain

-- Robustness Value, q = 1, alpha = 0.05: unobserved confounders (orthogonal to the covariates)

Bounds on omitted variable bias:

--The table below shows the maximum strength of unobserved confounders with association with th

 Bound Label R2dz.x R2yz.dx      Treatment Adjusted Estimate Adjusted Se
```

```
   1x female 0.0092  0.1246 directlyharmed            0.0752      0.0219
   2x female 0.0183  0.2493 directlyharmed            0.0529      0.0204
   3x female 0.0275  0.3741 directlyharmed            0.0304      0.0187
 Adjusted T Adjusted Lower CI Adjusted Upper CI
     3.4389            0.0323            0.1182
     2.6002            0.0130            0.0929
     1.6281           -0.0063            0.0670
```

```
# minimal reporting
ovb_minimal_reporting(sensitivity, format = "html")
```

Outcome: peacefactor

Treatment

Est.

S.E.

t-value

$R^2_{Y\sim D|\mathbf{X}}$

$RV_{q=1}$

$RV_{q=1,\alpha=0.05}$

directlyharmed

0.097

0.023

4.184

2.2%

13.9%

7.6%

Note: df = 783; Bound (1x female): $R^2_{Y\sim Z|\mathbf{X},D} = 12.5\%$, $R^2_{D\sim Z|\mathbf{X}} = 0.9\%$

Let's see how to interpret this.

$RV_{q=1} = 13.9\%$. This means that unobserved confounders that explain 13.9% of the residual variance both of the treatment and of the outcome are sufficiently strong to explain away all the observed effect. On the other hand, unobserved confounders that do not explain at least 13.9% of the residual variance both of the treatment and of the outcome are not sufficiently strong to do so.

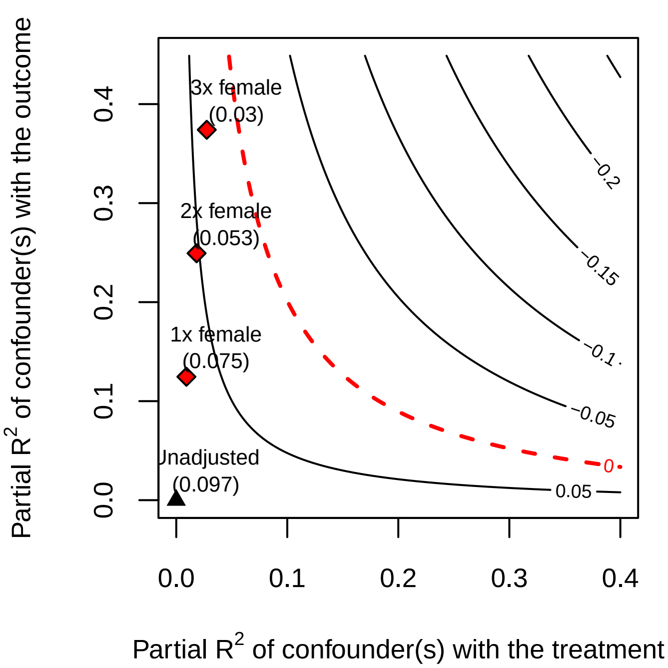
The robustness value for testing the null hypothesis that the coefficient of "directlyharmed" is zero ($RV_{q=1,\alpha=0.05}$) is 7.6%. This means that unobserved confounders that explain 7.6% of the residual variance both of the treatment and of the outcome are sufficiently strong to bring the lower bound of the confidence interval to zero (at the chosen significance level of 5%).

The partial R^2 of `directlyharmed` with `peacefactor` is 2.2%. This means that, in an extreme scenario in which we assume that unobserved confounders explain all of the left-out variance of the outcome, these unobserved confounders would need to explain at least 2.2% of the residual variance of the treatment to fully explain away the observed effect.

The above numbers are for the absolute strength the confounder needs to be to explain away the treatment effect. The authors believe in this context, "female" is an important factor. Suppose the confounder is as important as "female", what the partial R^2 would be? It is shown that $R^2_{Y\sim Z|X,D} = 12.5\%$ and $R^2_{D\sim Z|X} = 0.9\%$, so a confounder as strong as "female" is not strong enough to explain away the treatment effect.

It is worth being careful about *why*, because comparing each number to $RV_{q=1} = 13.9\%$ one at a time invites the wrong reading. The RV is defined for a confounder of *equal* strength on both sides, so it is not a threshold that each partial R^2 must separately clear. Here the outcome-side association (12.5%) is in fact close to the RV, which on its own might look like a near miss — but the treatment-side association is only 0.9%, more than an order of magnitude short. It is the treatment side that does the work: "female" predicts the outcome nearly well enough to matter, and barely predicts the treatment at all. The contour plot below is the honest check, since it evaluates the pair jointly rather than either coordinate alone.

```r
# plot bias contour of point estimate
plot(sensitivity)
```



This plot shows that even 3 times as strong as "female", the confounder is not strong enough to explain away the treatment effect.

15.6. sensitivity analysis for IV

Even with an IV model, we need sensitivity analysis. The worry would be there could be an unobserved confounder that affects both the instrument and the outcome.

Cinelli and Hazlett have a package “iv.sensemakr” for this. The usage is similar to “sensemakr”.

The example is Card’s returns-to-schooling data, 3,010 men from the 1976 NLS. The outcome is log wage, the treatment is years of education, and the instrument is `nearc4`, an indicator for growing up near a four-year college. The covariates are experience and its square, race, region, and urban indicators. The concern the sensitivity analysis addresses is a confounder affecting both proximity to a college and wages.

```
# loads package
library(iv.sensemakr)

# loads dataset
data("card")

# prepares data
y <- card$lwage  # outcome
d <- card$educ   # treatment
z <- card$nearc4 # instrument
x <- model.matrix( ~ exper + expersq + black + south + smsa + reg661 + reg662 +
                     reg663 + reg664 + reg665+ reg666 + reg667 + reg668 + smsa66,
                   data = card) # covariates
# fits IV model
card.fit <- iv_fit(y,d,z,x)

# see results
card.fit
```

```
Instrumental Variable Estimation
(Anderson-Rubin Approach)
=============================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]
Note: H0 = 0, alpha = 0.05, df = 2994.
=============================================
See summary for first stage and reduced form.
```

```
# runs sensitivity analysis
card.sens <- sensemakr(card.fit, benchmark_covariates = c("black", "smsa"))
```

```r
# see results
card.sens
```

```
Sensitivity Analysis for Instrumental Variables
(Anderson-Rubin Approach)
=============================================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2y0w.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221   0.0750  -0.0212    0.402       2.59
     1x smsa 0.00639   0.0202  -0.0192    0.396       2.57

Note: H0 = 0, q >= 1, alpha = 0.05, df = 2994.
=============================================================
See summary for first stage and reduced form.
```

```r
summary(card.sens)
```

```
Sensitivity Analysis for Instrumental Variables
(Anderson-Rubin Approach)
=============================================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x r2y0w.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221   0.0750  -0.0212    0.402       2.59
     1x smsa 0.00639   0.0202  -0.0192    0.396       2.57
```

```
Note: H0 = 0, q >= 1, alpha = 0.05, df = 2994.
-------------------------------------------------------------
FS Estimates:
  Coef. Estimate: 0.32
  Standard Error: 0.0879
  t-value: 3.64
  p-value: 0.000276
  Conf. Interval: [0.148, 0.492]

Sensitivity Statistics:
  Extreme Robustness Value: 0.00313
  Robustness Value: 0.0302

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2dw.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221 0.03342    0.109    0.531       2.40
     1x smsa 0.00639 0.00498    0.120    0.520       2.27

Note: H0 = 0, q = 1, alpha = 0.05, df = 2994.
-------------------------------------------------------------
RF Estimates:
  Coef. Estimate: 0.0421
  Standard Error: 0.0181
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.007, 0.078]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2yw.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221  0.0657 -0.00418   0.0883       2.56
     1x smsa 0.00639  0.0197 -0.00429   0.0884       2.56

Note: H0 = 0, q = 1, alpha = 0.05, df = 2994.
=============================================================
```

The basic idea is to run an IV regression, then apply sensemakr to that model. In this case, the authors use "black" and "smsa" as benchmark covariates. The authors believe these are important factors.

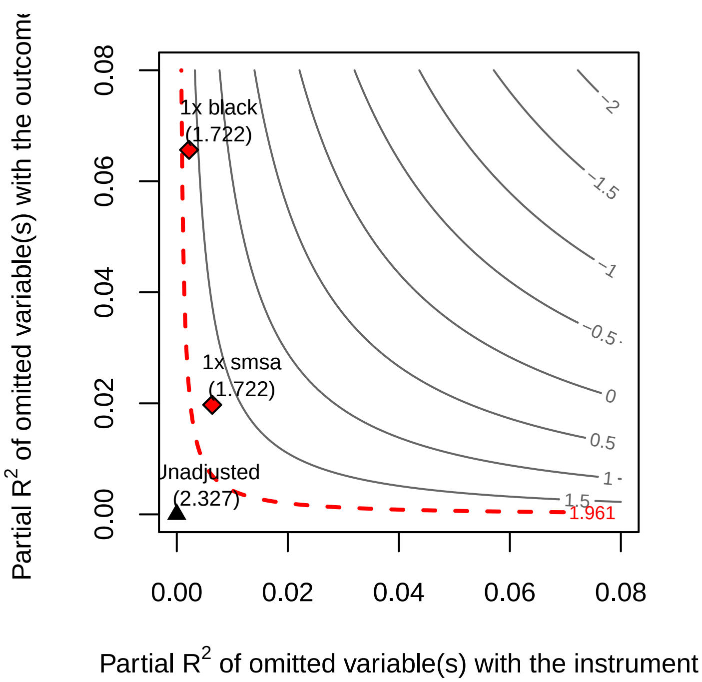
The IV estimate is 0.132 with $t = 2.33$, $p = 0.02$, and an Anderson-Rubin confidence interval of $[0.025, 0.285]$ that excludes zero. Read the sensitivity statistics next to that, and the picture is not reassuring. The Robustness Value is 0.00667 — two-thirds of one percent. Compare the OLS example earlier in this chapter, where $RV_{q=1}$ was 13.9%. This IV result is roughly twenty times more fragile than that one.

The benchmark bounds make it concrete, and they point the other way from what one might hope. A confounder as strong as `black` explains 0.22% of the residual variance of the instrument and 7.5% of the outcome, and under it the adjusted interval becomes $[-0.021, 0.402]$. For `smsa` the figures are 0.64% and 2.0%, giving $[-0.019, 0.396]$. **Both intervals contain zero.** A confounder no stronger than either benchmark covariate is enough to remove the significance of the estimate.

That `smsa` number is worth a second look on its own terms: its instrument-side partial R^2 of 0.00639 sits just under the Robustness Value of 0.00667, so a confounder of that strength lands essentially exactly on the tipping point. The two diagnostics agree.

None of this says the Card estimate is wrong. It says the identification rests on the instrument being cleaner than `black` or `smsa` are as predictors, which is an assumption about college proximity that the data cannot check. We can also plot a contour plot to see how strong the confounder needs to be to explain away the treatment effect.

```
plot(card.sens, parm = "rf", sensitivity.of = "t-value", lim = 0.08, kz = 1, ky = 1)
```



Unfortunately the package currently only allows for one instrument. Should not be too hard to modify it to be more general.

Systematic treatment: R · Julia.

16. Gender Wage Gap Analysis Using US Current Population Survey Data

The Oaxaca-Blinder decomposition splits the wage gap into an explained component (differences in observable characteristics) and an unexplained component (differences in returns, often attributed to discrimination). A well-known problem is that the decomposition depends on which group is used as the reference.

Słoczyński (2013, 2015) connects OB to regression adjustment in a way that makes the reference-group dependence precise.

16.1. Oaxaca-Blinder decomposition

Assume the regression model is different for male and female:

$$y_i = X_i\beta_a + \epsilon_a \tag{16.1}$$

where a is gender (female is 1), X's are person characteristics.

$$E[y_i|a_i = 1] - E[y_i|a_i = 0] = E[X_i|a_i = 1](\beta_1 - \beta_0) + (E[X_i|a_i = 1] - E[X_i|a_i = 0])\beta_0 \tag{16.2}$$

or

$$E[y_i|a_i = 1] - E[y_i|a_i = 0] = E[X_i|a_i = 0](\beta_1 - \beta_0) + (E[X_i|a_i = 1] - E[X_i|a_i = 0])\beta_1 \tag{16.3}$$

These are two forms of the OB decomposition. The first term is the unexplained component (difference in coefficients); the second is the explained component (difference in covariate means). The two forms use different reference groups (male vs. female) and generally give different results.

We can rewrite this as

$$\begin{aligned} E[y_i|a_i = 1] - E[y_i|a_i = 0] &= E[X_i|a_i = 1](\beta_1 - \beta_0) + (E[X_i|a_i = 1] - E[X_i|a_i = 0])\beta_0 \\ &= E[y_i(1) - y_i(0)|a_i = 1] + (E[y_i(0)|a_i = 1] - E[y_i(0)|a_i = 0]) \\ &= \tau_{PATT} + (E[y_i(0)|a_i = 1] - E[y_i(0)|a_i = 0]) \end{aligned} \tag{16.4}$$

That is, the observed difference is the sum of τ_{PATT} and selection bias. It is also known in the OB decomposition literature as the unexplained component and the explained component. The former as "discrimination".

We can simply use Regression Adjustment for these quantities. The easiest is an OLS of y_i on a_i, the demeaned X_i, and their interaction; the coefficient on a_i is then the effect evaluated at the mean of the covariates used for the demeaning. Demean at the **overall** mean and that coefficient is τ_{PAGE} (the ATE), which is what the example below does; demean at the mean among **women** instead and it is τ_{PATT}. The choice of centring is what selects the estimand, so it is worth being explicit about which one you have.

He also advocates the "population average gender effect" (PAGE):

$$\tau_{PAGE} = E[\tau(X_i)] \tag{16.5}$$

which is basically ATE, or conditional average effect (CATE), then average over the X's.

In this version, the two forms of OB decomposition can be τ_{PAGM} and τ_{PAGW}, for men and women.

The nice thing about this estimator τ_{PAGE} is that first, it is consistent with the treatment effect (causal inference) literature, and it is invariant to the choice of reference group.

The second advantage is that we also do not need to assume homogeneity of treatment effect, which is a strong assumption in the Oaxaca-Blinder decomposition.

The third advantage is that we can use nonparametric methods to estimate the treatment effect. Traditional OB decomposition is based on linear regression, which assumes a linear relationship between the covariates and the outcome.

16.2. Example

I am using the data from https://github.com/d2cml-ai/CausalAI-Course/tree/main/data.

"The data set we consider is from the 2015 March Supplement of the U.S. Current Population Survey. We select white non-hispanic individuals, aged 25 to 64 years, and working more than 35 hours per week for at least 50 weeks of the year. We exclude self-employed workers; individuals living in group quarters; individuals in the military, agricultural or private household sectors; individuals with inconsistent reports on earnings and employment status; individuals with allocated or missing information in any of the variables used in the analysis; and individuals with hourly wage below 3."

I am using this data to study gender wage gap (GWP).

We can model GWP using the potential outcome framework, with some care about what plays the role of "treatment." Gender itself is an immutable characteristic, not something that can be manipulated or randomized — the standard "no causation without manipulation" principle (Holland, 1986) says we should not treat gender itself as a causal treatment. What we really have in mind, and what the "treatment" indicator below is best understood as a proxy for, is something like *being perceived/treated as female by employers* (e.g., employer discrimination), which is at least conceptually manipulable even though we cannot experimentally assign it here. With that caveat in mind, we let $D = 1$ if female, $D = 0$ if male, and outcome Y is logged wage.

The ATE would be

$$\delta_{ATE} = E[Y(1) - Y(0)] \tag{16.6}$$

where $Y(1)$ is the potential outcome if treated, and $Y(0)$ is the potential outcome if not treated.

16.2.1. Regression adjustment

Since this is to estimate ATE, we can use regression adjustment to estimate the ATE. First let's use de-meaned X's. Then the coefficient on gender is the ATE.

We use These variables as X's:

"shs","hsg","scl","clg","ad","ne","mw","so","we","exp1", "occ", "occ2", "ind", and "ind2".

They are:

"Less then High School","High School Graduate","Some College","College Graduate","Advanced Degree", "Northeast","Midwest","South","West","Experience", "occupation", "occupation 2", "industry", and "industry 2".

I decide to use "occ2" and "ind2" as they are broader categories of occupation and industry. We don't have enough data for finer categories "occ" and "ind".

"exp1" seems to be the only variable that is continuous, let's demean it first. Let's also input "occ2" and "ind2" as factors.

Let's include demeaned "exp1", "sohs", "hsg", "socl", "clg", "mw", "sout", and "we" first. I leave out "occ2" and "ind2" since there are too many levels to demean.

```
library(tidyverse)
library(broom)


data <- read_csv("wage2015_subsample_inference.csv") |>
  rename(socl = scl, sohs = shs, sout = so) |>
  mutate(exp_dm = exp1 - mean(exp1, na.rm = TRUE), female=sex, occ2 = factor(occ2), ind2 = fac
  mutate(sohs_dm = sohs - mean(sohs, na.rm = TRUE),
         hsg_dm = hsg - mean(hsg, na.rm = TRUE),
         socl_dm = socl - mean(socl, na.rm = TRUE),
         clg_dm = clg - mean(clg, na.rm = TRUE),
         mw_dm = mw - mean(mw, na.rm = TRUE),
         sout_dm = sout - mean(sout, na.rm = TRUE),
         we_dm = we - mean(we, na.rm = TRUE))

glimpse(data)
```

```
Rows: 5,150
Columns: 30
$ rownames <dbl> 10, 12, 15, 18, 19, 30, 43, 44, 47, 71, 73, 77, 84, 89, 96, 1~
$ wage     <dbl> 9.615385, 48.076923, 11.057692, 13.942308, 28.846154, 11.7307~
$ lwage    <dbl> 2.263364, 3.872802, 2.403126, 2.634928, 3.361977, 2.462215, 2~
$ sex      <dbl> 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1~
$ sohs     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ hsg      <dbl> 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 0~
$ socl     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1~
$ clg      <dbl> 1, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0~
$ ad       <dbl> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ mw       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ sout     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ we       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ ne       <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ exp1     <dbl> 7.0, 31.0, 18.0, 25.0, 22.0, 1.0, 42.0, 37.0, 31.0, 4.0, 7.0,~
$ exp2     <dbl> 0.4900, 9.6100, 3.2400, 6.2500, 4.8400, 0.0100, 17.6400, 13.6~
$ exp3     <dbl> 0.343000, 29.791000, 5.832000, 15.625000, 10.648000, 0.001000~
$ exp4     <dbl> 0.24010000, 92.35210000, 10.49760000, 39.06250000, 23.4256000~
$ occ      <dbl> 3600, 3050, 6260, 420, 2015, 1650, 5120, 5240, 4040, 3255, 40~
$ occ2     <fct> 11, 10, 19, 1, 6, 5, 17, 17, 13, 10, 13, 14, 11, 11, 1, 19, 1~
$ ind      <dbl> 8370, 5070, 770, 6990, 9470, 7460, 7280, 5680, 8590, 8190, 82~
$ ind2     <fct> 18, 9, 4, 12, 22, 14, 14, 9, 19, 18, 18, 18, 18, 18, 17, 4, 4~
$ exp_dm   <dbl> -6.760583, 17.239417, 4.239417, 11.239417, 8.239417, -12.7605~
$ female   <dbl> 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1~
$ sohs_dm  <dbl> -0.02330097, -0.02330097, -0.02330097, -0.02330097, -0.023300~
$ hsg_dm   <dbl> -0.2438835, -0.2438835, 0.7561165, -0.2438835, -0.2438835, -0~
$ socl_dm  <dbl> -0.2780583, -0.2780583, -0.2780583, -0.2780583, -0.2780583, -~
$ clg_dm   <dbl> 0.6823301, 0.6823301, -0.3176699, -0.3176699, 0.6823301, 0.68~
$ mw_dm    <dbl> -0.2596117, -0.2596117, -0.2596117, -0.2596117, -0.2596117, -~
$ sout_dm  <dbl> -0.2965049, -0.2965049, -0.2965049, -0.2965049, -0.2965049, -~
$ we_dm    <dbl> -0.2161165, -0.2161165, -0.2161165, -0.2161165, -0.2161165, -~
```

```r
# construct matrices for estimation from the data

# 1. basic model
reg1 <- lm(lwage ~ female, data=data)
tidy(reg1)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   2.99      0.0107    280.    0
2 female       -0.0383    0.0160     -2.40  0.0165
```

```
reg2 <- lm(lwage ~ female*(exp_dm + sohs_dm + hsg_dm + socl_dm + clg_dm + mw_dm + sout_dm + we_
tidy(reg2)
```

```
# A tibble: 18 x 5
   term             estimate std.error statistic  p.value
   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
 1 (Intercept)       3.02     0.00974    310.     0
 2 female           -0.115    0.0147      -7.80   7.39e-15
 3 exp_dm            0.00801  0.000957     8.37   7.31e-17
 4 sohs_dm          -0.811    0.0620     -13.1    1.63e-38
 5 hsg_dm           -0.706    0.0350     -20.1    6.09e-87
 6 socl_dm          -0.553    0.0351     -15.7    1.45e-54
 7 clg_dm           -0.256    0.0345      -7.41   1.42e-13
 8 mw_dm             0.0337   0.0280       1.20   2.29e- 1
 9 sout_dm           0.0148   0.0271       0.548 5.84e- 1
10 we_dm             0.0555   0.0290       1.91   5.59e- 2
11 female:exp_dm     0.00212  0.00140      1.51   1.30e- 1
12 female:sohs_dm -0.0180    0.117       -0.154 8.78e- 1
13 female:hsg_dm     0.0398   0.0507       0.786 4.32e- 1
14 female:socl_dm  0.0466    0.0482       0.968 3.33e- 1
15 female:clg_dm     0.0992   0.0468       2.12   3.41e- 2
16 female:mw_dm    -0.124     0.0417      -2.98   2.93e- 3
17 female:sout_dm -0.0530    0.0403      -1.31   1.89e- 1
18 female:we_dm    -0.0655    0.0435      -1.51   1.32e- 1
```

We see the raw gap is 3.8%. Adjusting for some covariates we have 11% gap.

That direction deserves comment, because it is the opposite of what most readers expect. Adding controls usually shrinks a gap; here it nearly triples it. The reason is composition:

```
library(tidyverse)   # re-attached: the chunk that loads it is cached

data |>
  group_by(female) |>
  summarise(n = n(), mean_lwage = mean(lwage), exp1 = mean(exp1),
            `<HS` = mean(sohs), HS = mean(hsg), `some coll` = mean(socl),
            college = mean(clg), advanced = mean(ad)) |>
  mutate(across(where(is.numeric), ~round(.x, 3)))
```

```
# A tibble: 2 x 9
  female     n mean_lwage  exp1 `<HS`    HS `some coll` college advanced
   <dbl> <dbl>      <dbl> <dbl> <dbl> <dbl>       <dbl>   <dbl>    <dbl>
1      0  2861       2.99  13.8 0.032 0.294       0.273   0.294    0.107
2      1  2289       2.95  13.7 0.013 0.181       0.284   0.347    0.175
```

Women in this extract are **more** educated than men — 17.5% hold advanced degrees against 10.7%, and 1.3% are below high school against 3.2% — while potential experience is essentially identical (13.7 against 13.8 years). Education is therefore working in women's favour in the raw comparison. Holding it fixed removes that advantage and exposes a larger gap within education cells. Nothing subtle is happening; the raw and adjusted numbers simply describe different populations of comparison.

Keep this table in mind, because it comes back twice below: once when we ask what the adjusted number is an estimate *of*, and once when we ask whether conditioning on education might be doing harm rather than good.

Or, we can use "marginaleffects" instead of demeaning.

```r
reg3 <- lm(lwage ~ female*(exp1 + sohs + hsg+ socl + clg + mw + sout + we ), data=data)

library(marginaleffects)

avg_comparisons(reg3,
                variables = "female")
```

```
 Estimate Std. Error    z Pr(>|z|)    S  2.5 %  97.5 %
   -0.115     0.0147 -7.8   <0.001 47.2 -0.143 -0.0858

Term: female
Type: response
Comparison: 1 - 0
```

We get the same result as the demeaned regression.

If we want τ_{ATT}, then we can use the "newdata" argument to specify the treatment group,

```r
avg_comparisons(reg3,
                variables = "female",
                newdata = subset(data, female == 1))
```

```
 Estimate Std. Error     z Pr(>|z|)    S  2.5 %  97.5 %
   -0.113     0.0148 -7.65   <0.001 45.5 -0.142 -0.0844

Term: female
Type: response
Comparison: 1 - 0
```

16.2.2. Occupation and industry: a bad control problem

Before adding occupation and industry, it is worth asking what kind of variables they are — and the honest answer implicates the covariates we have already been using.

Sex is assigned at birth. Essentially nothing in the covariate vector precedes it:

Covariate	Downstream of sex?
Education (`sohs…ad`)	Yes — schooling decisions respond to gender
Experience (`exp1`)	Partly — potential experience is age minus schooling, so it inherits education's endogeneity; actual experience is fully downstream (interruptions, part-time spells)
Region (`mw`, `sout`, `we`)	Yes — current region reflects migration, and tied-mover patterns are gendered
Occupation, industry (`occ2`, `ind2`)	Yes, unambiguously

Genuinely pre-treatment variables would be birth cohort, region of birth, and parental characteristics — and even the last is shaky, since parental investment is known to respond to a child's sex. So the valid adjustment set for sex is close to **empty**.

Two consequences follow, and the second is the one that matters.

First, there is no confounding to adjust away. The usual reason to control for X is to make $Y(d) \perp D \mid X$ hold. But sex is not confounded by education — education is *caused* by sex. Sex is about as close to randomised-by-nature as a treatment gets, so $Y(d) \perp D$ holds without conditioning on anything. Adjustment cannot improve identification here; it can only change the estimand.

Second, and consequently, every model in this chapter sits somewhere on a **ladder of controlled direct effects**, not on a scale from worse to better estimates of one quantity:

Model	Held fixed	Question answered	Estimate
`reg1`	nothing	Total effect of the gender signal	−3.8%
`reg3`	education, experience, region	Gap within education/experience cells	−11.3%
`reg4`	+ occupation, industry	Gap within occupation cells too	−7.5%

Read that way, movement between rungs is not error to be minimised — it is information about where in the causal chain the gap accumulates. But each rung names a different intervention and needs its own justification, and the argument below applies at *every* rung, not only the last. Occupation and industry are simply the clearest case, so we use them to make the point.

They are *bad controls* in the sense of Cinelli, Forney and Pearl (2022), for two separate reasons at the same time.

```r
library(ggdag)
library(ggplot2)

g <- dagify(
  Wage ~ Female + Occupation + U,
  Occupation ~ Female + U,
  exposure = "Female",
  outcome  = "Wage",
  coords = list(x = c(Female = 0, Occupation = 1.1, U = 1.1, Wage = 2.2),
                y = c(Female = 0, Occupation = -1,  U = 1,   Wage = 0))
)

# node_size has to stay small: ggdag stops each edge at the node boundary, so
# large nodes swallow the arrowheads -- and the arrowheads are the whole point.
ggdag(g, text = FALSE, node_size = 9) +
  geom_dag_text_repel(aes(label = name), colour = "black", size = 3.2,
                      box.padding = 0.9, seed = 42) +
  theme_dag() +
  expand_limits(x = c(-0.45, 2.65), y = c(-1.5, 1.5))
```

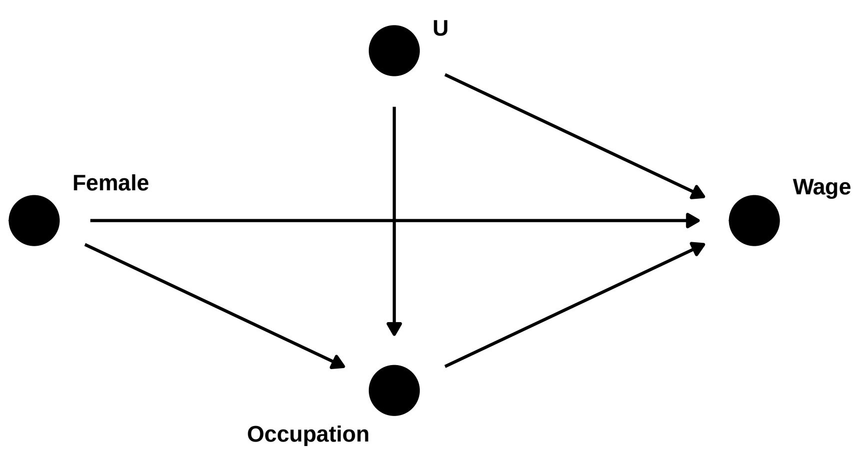


Figure 16.1.: Occupation is both a mediator of the gender–wage relationship and a collider on a path through unobserved wage determinants U.

Here U stands for unobserved wage determinants — drive, negotiation propensity, family constraints, expected career continuity — and `Occupation` covers occupation and industry together.

Occupation is a mediator. Part of the total effect of gender on wages runs `Female -> Occupation -> Wage`. Conditioning on occupation blocks that path, so the coefficient stops estimating a total effect. On its own this would be tolerable, as long as we said so: we would be reporting a within-occupation gap, which is a perfectly coherent descriptive quantity.

Occupation is also a collider. It is a common effect of gender and of U. Conditioning on a common effect *induces* an association between its causes, so conditioning on occupation opens the non-causal path `Female -> Occupation <- U -> Wage`, which was closed before we touched anything (Elwert and Winship, 2014).

We do not have to take that on faith. `dagitty` will enumerate the paths and say which are open, with and without conditioning on occupation:

```r
paths_table <- function(Z = NULL) {
  p <- dagitty::paths(g, from = "Female", to = "Wage", Z = Z)
  data.frame(path = p$paths, open = p$open)
}

paths_table()                   # nothing conditioned on
```

```
                               path  open
1      Female -> Occupation -> Wage  TRUE
2 Female -> Occupation <- U -> Wage FALSE
3                    Female -> Wage  TRUE
```

```r
paths_table(Z = "Occupation")   # conditioning on occupation
```

```
                               path  open
1      Female -> Occupation -> Wage FALSE
2 Female -> Occupation <- U -> Wage  TRUE
3                    Female -> Wage  TRUE
```

The two tables are the whole argument. Conditioning on occupation flips `Female -> Occupation -> Wage` from open to closed — that is the mediation we knowingly gave up — and simultaneously flips `Female -> Occupation <- U -> Wage` from closed to open. That second flip is the damage. Blocking a mediator removes something we wanted; conditioning on a collider manufactures something we did not. After adjustment, gender and U are correlated *within* occupation cells even if they are independent in the population, and because U also drives wages, that induced correlation leaks into the gender coefficient.

Stated formally: conditioning on a mediator identifies a *controlled direct effect* only under **no unmeasured mediator–outcome confounding** (VanderWeele and Robinson, 2014). U is exactly such a confounder, and in a wage setting it is not a remote worry — it is more or less why this literature exists.

One more thing worth flagging before we run the model: this is not a problem a better estimator solves. The AIPW and SuperLearner fits later in this chapter relax functional form, and do nothing at all about a structurally uninterpretable estimand.

16.2.3. Selection is the real threat, and it is not confounding

It is worth separating two problems that both get called "bias," because they have different fixes and only one of them is live here.

Under the exogeneity argument above, the raw difference in means *is* the total causal effect of sex on wages. That is a genuine causal quantity — the effect of being conceived female rather than male on wages measured decades later — identified without any adjustment. It is enormously composite, bundling biology, socialisation, schooling, sorting, and employer behaviour into one number, which is why it answers no policy question. But it is not confounded.

What does break it is the **sample**. This extract keeps only full-year, full-time, non-self-employed workers, and employment is a common effect of sex and unobserved wage determinants:

```r
library(ggdag)
library(ggplot2)
library(dagitty)

g3 <- dagitty('dag {
  SexBirth -> Educ
  SexBirth -> Wage
  SexBirth -> Employed
  U -> Employed
  U -> Wage
  Educ -> Wage
}')

coordinates(g3) <- list(
  x = c(SexBirth = 0, Employed = 1.5, U = 1.5, Educ = 1.5, Wage = 3.2),
  y = c(SexBirth = 0, Employed = 1.2, U = 2.5, Educ = -1.3, Wage = 0))

# the open square marks the node the sample is conditioned on
ggdag(tidy_dagitty(g3), text = FALSE, node_size = 9) +
  geom_point(data = data.frame(x = 1.5, y = 1.2), aes(x = x, y = y),
             inherit.aes = FALSE, shape = 0, size = 10, stroke = 1.2,
             colour = "grey15") +
  geom_dag_text_repel(aes(label = name), colour = "black", size = 3.2,
                      box.padding = 1.3, seed = 5) +
  theme_dag() +
  expand_limits(x = c(-0.7, 4.0), y = c(-1.9, 3.0))
```

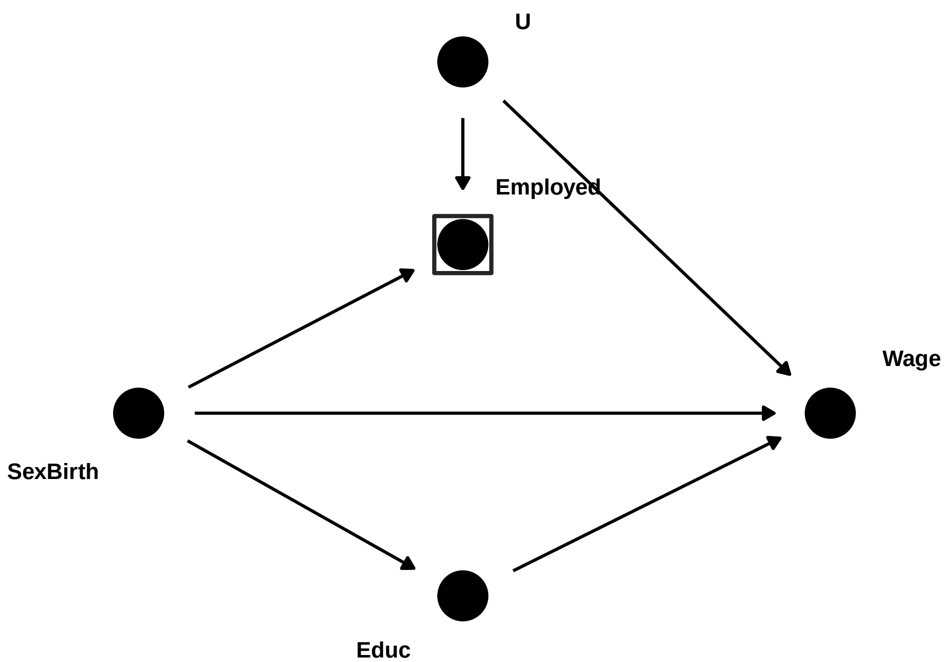


Figure 16.2.: The sample restriction as a conditioning statement. The **boxed** node is the variable the analysis conditions on — not by adding a covariate, but by the sample definition. `Employed` is a common effect of sex and of unobserved wage determinants U, so boxing it opens `SexBirth -> Employed <- U -> Wage`.

The box is worth dwelling on, because it is drawn by the sampling frame rather than by any modelling choice. Nobody typed `+ employed` into a regression; the extract simply contains only full-year, full-time workers. A sample definition **is** a conditioning statement, and most applied DAGs omit the selection node entirely — which makes the graph look identified when it is not.

`dagitty` will confirm that there is no way out by adjustment. Note that a selection variable has to be declared as such: marking a node `[selected]` tells `dagitty` the sample is restricted on it, which is a different claim from conditioning on an ordinary covariate.

```
g3_sel <- dagitty('dag {
  SexBirth -> Educ
  SexBirth -> Wage
  SexBirth -> Employed
  U -> Employed
  U -> Wage
  Educ -> Wage
  U [latent]
  Employed [selected]
```

```
  SexBirth [exposure]
  Wage [outcome]
}')

# how many covariate sets identify the total effect under this selection?
length(adjustmentSets(g3_sel, effect = "total"))
```

```
[1] 0
```

```
isAdjustmentSet(g3_sel, c())        # the raw comparison
```

```
[1] FALSE
```

```
isAdjustmentSet(g3_sel, "Educ")     # adding education does not help either
```

```
[1] FALSE
```

```
# and the path that does the damage
p3 <- paths(g3, from = "SexBirth", to = "Wage", Z = "Employed")
data.frame(path = p3$paths, open = p3$open)
```

```
                               path open
1           SexBirth -> Educ -> Wage TRUE
2 SexBirth -> Employed <- U -> Wage TRUE
3                   SexBirth -> Wage TRUE
```

Zero valid adjustment sets, and the raw comparison is no longer among them. Conditional on being in the sample, sex is no longer independent of potential wages: sex is exogenous, but *selected* sex is not. Compare this with the occupation problem earlier in the chapter. There the collider opened because we *chose* to add a control, so declining to add it was an available fix. Here the box is imposed before any modelling decision, so "do not condition on it" is not an option — the remedy has to attack the box itself rather than the covariate list, which is why it takes data on non-workers (Lee, 2009, sharp bounds; Neal, 2004; Olivetti and Petrongolo, 2008, imputation for non-workers) rather than a better regression.

The direction is unusually easy to sign here. Women are less likely to hold full-year full-time jobs, so clearing that bar takes more of whatever else helps — meaning employed women are *positively* selected on wage-raising unobservables relative to employed men. If so, the observed gap **understates** the true one. Olivetti and Petrongolo (2008) find exactly this signature across countries: measured gender wage gaps are smaller where female employment rates are lower, and imputing wages for non-workers widens them most in the low-participation countries.

16.2.3.1. How much does this actually cost? A simulation

Everything above is graph-theoretic: the paths are open or closed, the adjustment sets exist or do not. It says nothing about magnitude. Because the non-workers are missing from this extract by construction, the honest way to put numbers on it is to simulate a population where the truth is known, apply the selection, and see what survives.

The design mirrors the graph: sex is randomly assigned, U is an unobserved wage determinant, women are less likely to be employed full-year full-time, and higher U makes employment more likely. The true effect is set at -0.10 log points.

Concretely, we draw $N = 300{,}000$ people with sex $\sim$ Bernoulli(0.5) and $U \sim N(0, 1)$, independent of each other. Log wage is

$$\text{wage} = 2.5 - 0.10\,\text{sex} + 0.5\,U + \varepsilon, \qquad \varepsilon \sim N(0, 0.4^2), \tag{16.7}$$

and employment is emp $\sim$ Bernoulli($\Lambda(0.8 - 0.9\,\text{sex} + 0.7\,U)$). We then discard everyone with emp $= 0$, which is what the extract does to us.

Note what is and is not in that design. Sex is randomly assigned, so there is no confounding whatsoever: a regression on the full population is unbiased by construction. U affects both wages and employment, and sex affects employment, so conditioning on employment opens a path between sex and U. Every problem below comes from the selection alone. Five approaches are compared against the truth: the full population, the selected sample as an analyst sees it, two ways of rebalancing the sex ratio, and inverse-probability-of-selection weighting, which needs U and so is infeasible in practice.

```r
set.seed(42)
N   <- 300000
tau <- -0.10                           # TRUE effect of sex on log wage

sex  <- rbinom(N, 1, 0.5)              # birth sex, randomly assigned
U    <- rnorm(N)                       # unobserved: health, drive, reservation wage
wage <- 2.5 + tau * sex + 0.5 * U + rnorm(N, 0, 0.4)

pi_sel <- plogis(0.8 - 0.9 * sex + 0.7 * U)   # women less likely; high U more likely
emp    <- rbinom(N, 1, pi_sel)

d <- data.frame(wage, sex, U, pi_sel)[emp == 1, ]

b <- function(fit) coef(fit)[["sex"]]

# 1. the truth, from the full population
full  <- b(lm(wage ~ sex))

# 2. the selected sample, as an analyst would see it
naive <- b(lm(wage ~ sex, data = d))
```

```r
# 3. reweight so the sex ratio is exactly 50/50
w_bal <- ifelse(d$sex == 1, 0.5 / mean(d$sex), 0.5 / (1 - mean(d$sex)))
rewt  <- b(lm(wage ~ sex, data = d, weights = w_bal))

# 4. randomly drop men until the counts match
n_w   <- sum(d$sex == 1)
men   <- d[d$sex == 0, ]
kept  <- men[sample(nrow(men), n_w), ]
subs  <- b(lm(wage ~ sex, data = rbind(kept, d[d$sex == 1, ])))

# 5. weight by the inverse probability of selection (needs U -- infeasible in practice)
ipsw  <- b(lm(wage ~ sex, data = d, weights = 1 / d$pi_sel))

data.frame(
  approach = c("full population regression", "selected sample, naive",
               "sex ratio reweighted to 50/50", "men randomly subsampled to equal n",
               "inverse-probability-of-selection weighted"),
  estimate = round(c(full, naive, rewt, subs, ipsw), 4)
)
```

```
                                   approach estimate
1                full population regression  -0.0960
2                    selected sample, naive  -0.0389
3             sex ratio reweighted to 50/50  -0.0389
4        men randomly subsampled to equal n  -0.0393
5 inverse-probability-of-selection weighted  -0.1014
```

```r
c(employment_rate_men   = round(mean(emp[sex == 0]), 3),
  employment_rate_women = round(mean(emp[sex == 1]), 3))
```

```
  employment_rate_men employment_rate_women
                0.671                 0.479
```

```r
c(mean_U_employed_men   = round(mean(d$U[d$sex == 0]), 3),
  mean_U_employed_women = round(mean(d$U[d$sex == 1]), 3))
```

```
  mean_U_employed_men mean_U_employed_women
                0.205                 0.332
```

Three things to take from that table.

The full-population regression gives −0.0960, recovering the true −0.10 up to sampling noise, as it must — sex is randomly assigned, so nothing confounds it. The naive selected-sample estimate is −0.0389: **wrong by a factor of about two and a half**, and in the direction the graph predicted. It is attenuated toward zero because employed women are positively selected on U (mean

+0.332 against +0.205 for employed men, with employment rates of 47.9% and 67.1%). The signed prediction of the previous section is not merely plausible; it is what happens.

Balancing does nothing. Reweighting the sex ratio to exactly 50/50 gives −0.0389, unchanged to four decimal places, and randomly discarding men until the counts match gives −0.0393, unchanged for practical purposes. The reason is worth stating as a general rule: **no random sampling operation can remove a bias.** Bias is a property of a distribution, and random sampling preserves distributions — a random subsample of men has the same mean U as all men, +0.205 either way. The problem is not that women are outnumbered; it is that the employed women are a *differently selected* subset of women than the employed men are of men. Reweighting on sex changes the marginal distribution of sex, while the bias lives in the conditional distribution of U given sex.

Only the last row recovers the truth, at −0.1014, and it does so by weighting on the probability of selection — which requires U. That is exactly what is unavailable.

16.2.3.2. Lee bounds: what you can get without U

Lee (2009) avoids needing U by trimming the over-selected group on the *outcome* rather than on the unobservable, in both extreme directions. The trim fraction is fixed by the selection rates, $q = (p_{\text{men}} - p_{\text{women}})/p_{\text{men}}$: drop the lowest-paid q of men to make them look as good as possible, then the highest-paid q to make them look as bad as possible. The two results bracket the effect.

The important question is not whether the bounds are valid — they are — but how wide they are. That depends entirely on how unbalanced the selection is, so we re-run the same DGP five times, varying only the sex coefficient in the employment equation from −0.9 (severe selection) to −0.03 (almost none) and holding the true effect at −0.10 throughout:

```r
lee_run <- function(b_sex, N = 300000, tau = -0.10) {
  sex  <- rbinom(N, 1, 0.5)
  U    <- rnorm(N)
  wage <- 2.5 + tau * sex + 0.5 * U + rnorm(N, 0, 0.4)
  emp  <- rbinom(N, 1, plogis(0.8 + b_sex * sex + 0.7 * U))
  d    <- data.frame(wage, sex)[emp == 1, ]

  p_m <- mean(emp[sex == 0]); p_w <- mean(emp[sex == 1])
  q   <- (p_m - p_w) / p_m

  men  <- d$wage[d$sex == 0]
  wbar <- mean(d$wage[d$sex == 1])
  lo   <- wbar - mean(men[men >= quantile(men, q,     names = FALSE)])
  hi   <- wbar - mean(men[men <= quantile(men, 1 - q, names = FALSE)])

  data.frame(p_men = round(p_m, 3), p_women = round(p_w, 3), q = round(q, 3),
             naive = round(coef(lm(wage ~ sex, data = d))[["sex"]], 4),
             lower = round(lo, 3), upper = round(hi, 3),
             width = round(hi - lo, 3),
             signed = ifelse(hi < 0, "yes", "no"),
             covers_truth = (lo <= tau && tau <= hi))
```

```
}

set.seed(7)
do.call(rbind, lapply(c(-0.9, -0.5, -0.25, -0.10, -0.03), lee_run))
```

```
  p_men p_women     q   naive  lower  upper width signed covers_truth
1 0.672   0.477 0.290 -0.0365 -0.337  0.265 0.602     no         TRUE
2 0.674   0.568 0.157 -0.0669 -0.245  0.112 0.357     no         TRUE
3 0.674   0.621 0.079 -0.0857 -0.186  0.015 0.201     no         TRUE
4 0.672   0.652 0.029 -0.0960 -0.139 -0.053 0.086    yes         TRUE
5 0.673   0.667 0.008 -0.0959 -0.110 -0.082 0.028    yes         TRUE
```

The bounds cover the true −0.10 in every row, so the method does what it promises. But the width scales almost proportionally with the trim fraction, and the effect can only be *signed* once the employment rates differ by a couple of percentage points. At a 19-point gap the interval runs from −0.34 to +0.26 and contains zero: valid, and useless.

Now read the `naive` column alongside `width`. The naive estimate degrades in lockstep with the bounds widening — reading down the table, −0.037, −0.067, −0.086, −0.096, −0.096 against a truth of −0.10. So Lee bounds are tight exactly when they are not needed and wide exactly when they are. That inverts into what they are actually for here: the width is a **measure of how much the selection problem is costing**, and it is information one otherwise does not have at all. Their value in this setting is more diagnostic than inferential. No method manufactures information; this one stops a badly selected point estimate from passing as a precise one.

Three caveats before anyone reaches for it. The load-bearing assumption is **monotonicity** — being female never *increases* anyone's employment probability, which is an individual-level claim, stronger than the average pattern. The target quantity changes: the bounds are for the effect among the **always-employed**, whoever they happen to be, not the population. And the procedure needs both selection rates, so it cannot be run on `wage2015_subsample_inference.csv` at all — that file contains only employed people. One would have to go back to the unfiltered CPS for men's and women's full-year full-time employment rates.

On real US data those rates differ by something on the order of ten to fifteen percentage points for ages 25–64, putting q near 0.2 — the second row of the table, where the interval contains zero. So the likely honest conclusion from doing this properly is that the selected-sample gender gap is consistent with anything from a large penalty to a modest premium, and that no covariate adjustment narrows it. That is a stronger and more useful statement than the −3.8% point estimate, even though it names no number.

A note on notation, since U now appears in two graphs doing two different jobs. In the occupation graph, U has to drive *occupational sorting* — drive, negotiation propensity, expected career continuity. In the selection graph it has to drive *labour supply* — health, reservation wage (non-labour income, spousal earnings), latent productivity. The two sets overlap but are not the same, and the two collider problems are separate.

So the honest summary is uncomfortable: −3.8% is the best-identified number in this chapter and the least informative about anything one would want to act on, while −7.5% is the worst-identified and the closest to the question people actually ask. Identification improves as conditioning is stripped away; interpretability improves as it is added. No estimator resolves that trade-off — only a different design does, which is what audit and correspondence studies are for.

16.2.4. Two treatments, two adjustment sets

There is a natural objection to everything above, and it is a good one. *Suppose we set aside the fact that sex shapes schooling and experience, and treat those as simply given — this woman's actual history. Then asking what she would have been paid as a man, holding that history fixed, is exactly the question of interest, and controlling for education is obviously the right thing to do.*

That is right, and it exposes something the discussion so far has glossed. **"Bad control" is not a property of a variable.** It is a property of the triple *(variable, treatment, timing)*. Re-date the treatment and the classification flips:

Treatment	Education is	Controlling is
Sex assigned at conception	post-treatment mediator	bad control
Perceived sex at the hiring decision	pre-treatment covariate	appropriate

The second row is Greiner and Rubin's (2011) framing — treatment as *perception of an immutable characteristic at a specified decision point* — and it is precisely what correspondence studies estimate. It is also the legally operative question for disparate treatment. So it is the better estimand, not a compromise.

Encoding it, however, produces a result sharper than the verbal argument suggests. Let `SexBirth` be sex at birth, `Perceived` the perceived sex facing the employer, and `A` unobserved ability affecting both schooling and wages:

```r
library(ggdag)
library(ggplot2)
library(dagitty)

g2 <- dagitty('dag {
  SexBirth -> Perceived
  SexBirth -> Educ
  A -> Educ
  A -> Wage
  Educ -> Wage
  Perceived -> Wage
  A [latent]
  Perceived [exposure]
  Wage [outcome]
}')

coordinates(g2) <- list(
  x = c(SexBirth = 0, Perceived = 1.3, Educ = 1.3, A = 1.0, Wage = 3.1),
  y = c(SexBirth = 0, Perceived = 1.1, Educ = -1.1, A = -2.3, Wage = 0))
```

```
ggdag(g2, text = FALSE, node_size = 9) +
  geom_dag_text_repel(aes(label = name), colour = "black", size = 3.1,
                      box.padding = 1.4, seed = 11) +
  theme_dag() +
  expand_limits(x = c(-0.7, 3.8), y = c(-3.0, 1.8))
```

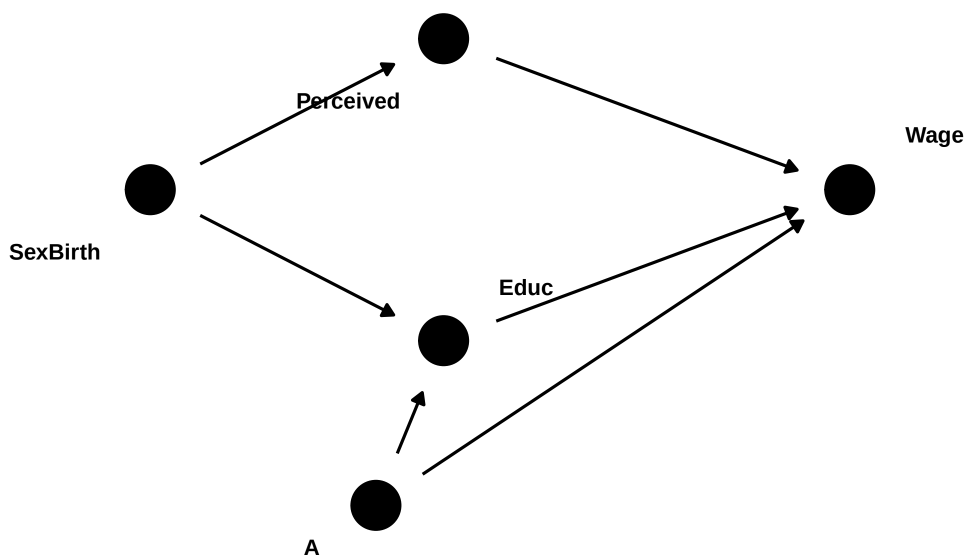


Figure 16.3.: Treatment re-dated to the hiring decision. Education now *precedes* the treatment — but it is still a collider between sex at birth and unobserved ability A.

Now ask `dagitty` which covariates would license the comparison:

```
c(`{}`     = isAdjustmentSet(g2, c()),
  `{Educ}` = isAdjustmentSet(g2, "Educ"))
```

```
    {} {Educ}
 FALSE  FALSE
```

```
adjustmentSets(g2, effect = "total")
```

```
{ SexBirth }
```

```r
paths_table2 <- function(Z = NULL) {
  p <- paths(g2, from = "Perceived", to = "Wage", Z = Z)
  data.frame(path = p$paths, open = p$open)
}

paths_table2()              # nothing conditioned on
```

```
                                        path  open
1                          Perceived -> Wage  TRUE
2      Perceived <- SexBirth -> Educ -> Wage  TRUE
3 Perceived <- SexBirth -> Educ <- A -> Wage FALSE
```

```r
paths_table2(Z = "Educ")   # conditioning on education
```

```
                                        path  open
1                          Perceived -> Wage  TRUE
2      Perceived <- SexBirth -> Educ -> Wage FALSE
3 Perceived <- SexBirth -> Educ <- A -> Wage  TRUE
```

Three things follow, and the third is the point.

The objection is half-vindicated. Adjusting for nothing is *invalid* under this estimand. Sex at birth confounds perceived sex and wages through the education channel, so the raw difference is contaminated for this question — part of it is human capital rather than employer conduct. Controlling for education does remove a genuine bias. Everything said earlier about bad controls was conditional on the *other* estimand.

But it introduces a different one. Conditioning on education closes the human-capital backdoor and simultaneously opens `Perceived <- SexBirth -> Educ <- A -> Wage`. Education is still a collider between sex at birth and ability, and re-dating the treatment does nothing about that. One bias is traded for another.

The only valid adjustment set is `{SexBirth}`, and it has no overlap. Sex at birth determines perceived sex, so conditioning on it leaves no variation in the treatment. Positivity fails outright. The effect of perceived sex holding human capital fixed is therefore **not identified from observational data under any conditioning strategy** — not because this extract is small or its covariates coarse, but because of the structure of the graph.

That last point also says exactly what a design must do: sever `SexBirth -> Perceived`. Randomise the signal independently of biological sex, and the backdoor disappears, so *no adjustment is needed at all*. This is why an audit study requires no controls, and why it answers a question a regression provably cannot. Goldin and Rouse (2000) is the cleanest case — the same musician is evaluated with and without the screen, so ability is held fixed exactly rather than statistically.

One usable consequence in the meantime. The two biases run in opposite directions, which brackets the answer informally. With no conditioning, the education backdoor is open and — since women here are *more* educated — their human-capital advantage offsets part of the treatment effect, biasing toward zero. With education conditioned on, the collider opens and equally-educated women are pushed low on A, inflating the residual gap. So the quantity of interest plausibly sits *between* -3.8% and -11.3%, and neither endpoint is it. This is not a formal bound — the sign of each bias rests on assumptions rather than proof — but it is more informative than either number alone.

Finally, a normative point that no amount of graph-reading settles. Holding education and experience fixed accepts whatever produced those differences as the baseline. If they were themselves partly produced by discrimination, then conditioning on them conditions on discrimination's own output. That is the right choice for *is this employer treating equivalent applicants differently* and the wrong choice for *is the labour market fair to women* — roughly the disparate-treatment versus disparate-impact distinction. The data does not decide it; the question does.

There is a further objection which holds that the resolution just reached — re-date the treatment to perception, then randomise it — does not work either, and that the normative point in the paragraph above is not a footnote to the analysis but the whole of it. Because it needs the empirical material in place first, it is taken up in the closing section, Section 16.4.

With all that on the table, here is the model.

```
reg4 <- lm(lwage ~ female*(exp1 + sohs + hsg+ socl + clg + mw + sout + we + occ2 + ind2), data=
avg_comparisons(reg4,
                variables = "female",
                newdata = subset(data, female == 1))
```

```
 Estimate Std. Error     z Pr(>|z|)    S  2.5 %  97.5 %
   -0.075     0.0162 -4.62   <0.001 18.0 -0.107 -0.0432

Term: female
Type: response
Comparison: 1 - 0
```

So in the model with full interaction, we have a gap of 7.5%, against 11.3% without occupation and industry.

The tempting reading of that pair — "occupational and industrial sorting accounts for roughly four points of the gap" — is exactly the reading the graph above forbids. Three things are mixed into that four-point move, and nothing in the output separates them:

1. genuine mediation, the part of the gap that really does operate through sorting;
2. collider bias, from the gender–U association induced by conditioning;
3. ordinary confounding of the occupation–wage relationship.

The likely *direction* of (2) is worth spelling out. If women who enter male-dominated, high-paying occupations are positively selected on U — they had more to overcome to get there — then within those occupations they are better on unobservables than the men they are being compared with, which pushes the within-occupation gap **toward zero**. Some of the attenuation from 11.3% to 7.5% may therefore be bias rather than mediation.

The defensible way to report this: neither number is "the" gender gap. -11.3% is a controlled direct effect holding education, experience and region fixed; -7.5% additionally holds occupation and industry fixed. The total effect — the only rung with no mediators held fixed — is the raw −3.8%. What we *can* do for the last step is ask how strong U would have to be to account for the whole 4-point difference between the bottom two rungs.

This also gives a precise form to the objection most readers will have already framed for themselves: *maybe men and women differ in some unobserved, productivity-relevant way, and that is what we are picking up.* Note first that such a trait would be a **mediator**, not a confounder — nothing causes sex, so anything that differs by sex is downstream of it, and the total effect correctly includes it. The objection is therefore not "the comparison is biased" but "I want the direct effect holding that trait fixed," which is a different and unidentified estimand. Note second that controlling for observables does not approximate it: education is a *collider* between sex and any such trait, so conditioning on education induces exactly the sex–ability association one hoped to remove. Given the composition table above — women here are more educated — that induced association would push the estimated gap further from zero, which is the direction the numbers in fact move.

What the sensitivity analysis below can do is convert that unfalsifiable worry into a magnitude.

16.2.4.1. How strong would U have to be?

The omitted-variable-bias machinery in `sensemakr` (Cinelli and Hazlett, 2020) needs a single treatment coefficient, so for this exercise we drop the interactions and use additive specifications. That costs us the heterogeneity, but the attenuation we are interrogating survives the simplification almost exactly.

```r
library(sensemakr)

Xb <- "exp1 + sohs + hsg + socl + clg + mw + sout + we"

fit_total <- lm(as.formula(paste("lwage ~ female +", Xb)), data = data)
fit_adj   <- lm(as.formula(paste("lwage ~ female +", Xb, "+ occ2 + ind2")), data = data)

c(total    = coef(fit_total)["female"],
  adjusted = coef(fit_adj)["female"],
  attenuation = coef(fit_adj)["female"] - coef(fit_total)["female"])
```

```
      total.female    adjusted.female attenuation.female 
       -0.11375231        -0.07285748         0.04089483 
```

```r
sens <- sensemakr(model = fit_adj, treatment = "female",
                  benchmark_covariates = c("clg", "exp1"),
                  kd = c(1, 2, 3))
summary(sens)
```

```
Sensitivity Analysis to Unobserved Confounding

Model Formula: lwage ~ female + exp1 + sohs + hsg + socl + clg + mw + sout +
```

```
    we + occ2 + ind2

Null hypothesis: q = 1 and reduce = TRUE
-- This means we are considering biases that reduce the absolute value of the current estimate
-- The null hypothesis deemed problematic is H0:tau = 0

Unadjusted Estimates of 'female':
  Coef. estimate: -0.0729
  Standard Error: 0.015
  t-value (H0:tau = 0): -4.8485

Sensitivity Statistics:
  Partial R2 of treatment with outcome: 0.0046
  Robustness Value, q = 1: 0.0656
  Robustness Value, q = 1, alpha = 0.05: 0.0396

Verbal interpretation of sensitivity statistics:

-- Partial R2 of the treatment with the outcome: an extreme confounder (orthogonal to the cova

-- Robustness Value, q = 1: unobserved confounders (orthogonal to the covariates) that explain

-- Robustness Value, q = 1, alpha = 0.05: unobserved confounders (orthogonal to the covariates)

Bounds on omitted variable bias:

--The table below shows the maximum strength of unobserved confounders with association with th

 Bound Label R2dz.x R2yz.dx Treatment Adjusted Estimate Adjusted Se Adjusted T
      1x clg 0.0001  0.0124    female           -0.0718      0.0149    -4.8081
      2x clg 0.0002  0.0247    female           -0.0708      0.0148    -4.7680
      3x clg 0.0002  0.0371    female           -0.0697      0.0147    -4.7275
     1x exp1 0.0016  0.0338    female           -0.0649      0.0148    -4.3914
     2x exp1 0.0032  0.0676    female           -0.0570      0.0145    -3.9199
     3x exp1 0.0048  0.1014    female           -0.0490      0.0143    -3.4322
 Adjusted Lower CI Adjusted Upper CI
           -0.1011           -0.0425
           -0.0999           -0.0417
           -0.0986           -0.0408
           -0.0939           -0.0359
           -0.0855           -0.0285
           -0.0770           -0.0210
```

The additive models put the total effect at −11.4% and the occupation-adjusted effect at −7.3%, an attenuation of 4.1 log points — essentially the same gap as in the interacted specifications.

Now invert the bias formula. Instead of asking how strong a confounder must be to drive the adjusted estimate to zero (the usual question), ask how strong it must be to move the adjusted estimate *back* to the total effect. That is the strength at which the entire apparent contribution of occupation is bias rather than mediation.

```r
target <- coef(fit_total)["female"]

grid <- expand.grid(r2dz = c(0.0125, 0.02, 0.033, 0.06),
                    r2yz = seq(0.001, 0.60, length.out = 4000))
grid$adj <- adjusted_estimate(fit_adj, treatment = "female",
                              r2dz.x = grid$r2dz, r2yz.dx = grid$r2yz,
                              reduce = FALSE)

grid |>
  group_by(r2dz) |>
  slice_min(abs(adj - target), n = 1) |>
  ungroup() |>
  transmute(`partial R2 with gender` = r2dz,
            `partial R2 with wage`   = round(r2yz, 3),
            `restored estimate`      = round(adj, 4))
```

```
# A tibble: 4 x 3
  `partial R2 with gender` `partial R2 with wage` `restored estimate`
                     <dbl>                  <dbl>               <dbl>
1                   0.0125                  0.115              -0.114
2                   0.02                    0.071              -0.114
3                   0.033                   0.042              -0.114
4                   0.06                    0.023              -0.114
```

```r
# sensemakr is attached again here on purpose: the chunk above is cached, so on a
# cache hit its library() call never runs and plot() would dispatch to
# plot.default instead of the sensemakr method.
library(sensemakr)

plot(sens)
```

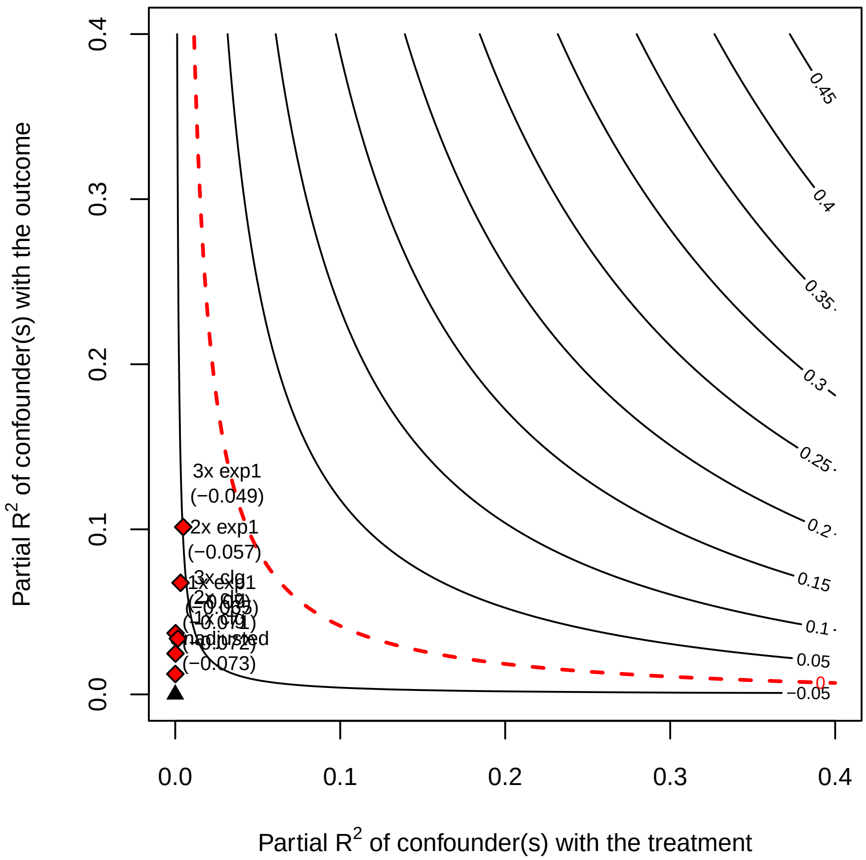


Figure 16.4.: Adjusted estimate of the gender gap as a function of a hypothetical confounder's partial R^2 with gender (horizontal) and with log wage (vertical), benchmarked against college and experience.

The answer is uncomfortable. A confounder explaining about **3% of the residual variance of gender and about 4% of the residual variance of wages** is enough to restore the full total effect — that is, enough for occupation to be contributing no genuine mediation at all. For scale, `sensemakr` reports a robustness value of 6.6%: a confounder explaining more than that share of both residual variances would drive the adjusted estimate to zero. So a U roughly *half* as strong as the one needed to erase the within-occupation gap entirely is already strong enough to explain the whole 11.3% → 7.5% attenuation.

The benchmark rows give another angle. Experience explains 3.4% of the residual variance of wages but only 0.16% of the residual variance of gender, so along the outcome dimension the required U is only a little stronger than experience. Along the gender dimension the required multiple looks large, but that is mostly because gender is close to orthogonal to experience in this sample; in absolute terms we are still only asking for a few percent of residual variance.

None of this shows that occupational sorting explains nothing. It shows that this design cannot distinguish "sorting explains four points" from "conditioning on a collider costs four points," and that the confounder needed to tip the balance is unremarkable in size. Which is why the total effect, not the occupation-adjusted number, is the estimate to lead with.

Now let's try nonparametric estimation.

16.2.5. Nonparametric estimation

If we'd like to go for nonparametric, relaxing linearity assumption, then we can use "npcausal". Note this may take a while, depending on which machine learning packages you decide to include. We can also use other nonparametric estimators, such as TMLE or doubleML.

```
library(npcausal)

#SL.library <- c("SL.earth","SL.glmnet","SL.glm.interaction", "SL.mean","SL.ranger", "SL.xgboos


SL.library <- c("SL.earth","SL.glmnet","SL.mean","SL.ranger", "SL.xgboost")

# construct W from model matrix of reg3
reg3 <- lm(lwage ~ exp_dm + sohs + hsg+ socl + clg + mw + sout + we + occ2 + ind2, data=data)
W <- as.data.frame(model.matrix(reg3)[, -1]) # remove intercept

Y <- data$lwage
A <- data$female
set.seed(123)
aipw_ate <- ate(y=Y, a=A, x=W, nsplits=5, sl.lib=SL.library)
```

```
  |
  |                                                                      |   0%


  |
  |====                                                                  |   5%
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  |=======                                                               |  10%
  |
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  |
  |==================                                                    |  25%
  |
  |=====================                                                 |  30%


  |
  |========================                                              |  35%
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```

```
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  |
  |=================================================                     |  70%
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  |====================================================                  |  75%
  |
  |========================================================              |  80%
  |
  |============================================================          |  85%
  |
  |===============================================================       |  90%
  |
  |==================================================================    |  95%
  |
  |======================================================================| 100%
     parameter         est          se       ci.ll       ci.ul pval
1      E{Y(0)}  2.99695506 0.01093465  2.97552315  3.01838697    0
2      E{Y(1)}  2.93906408 0.01327342  2.91304817  2.96507999    0
3 E{Y(1)-Y(0)} -0.05789098 0.01597522 -0.08920241 -0.02657955    0
```

```r
aipw_att <- att(y=Y, a=A, x=W, nsplits=5, sl.lib=SL.library)
```

```
  |
  |                                                                      |   0%



  |
  |=======                                                               |  10%
  |
  |==============                                                        |  20%



  |
  |=====================                                                 |  30%
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```

```
  |===========================                                         |  40%
  |
  |==================================                                  |  50%
  |
  |=========================================                           |  60%
  |
  |================================================                    |  70%
  |
  |=======================================================             |  80%
  |
  |==============================================================      |  90%
  |
  |====================================================================| 100%
      parameter         est         se       ci.ll       ci.ul pval
1      E(Y|A=1)  2.94948490 0.01160804  2.92673314  2.97223666    0
2   E{Y(0)|A=1}  3.01689509 0.01221863  2.99294658  3.04084360    0
3 E{Y-Y(0)|A=1} -0.06741019 0.01409116 -0.09502886 -0.03979152    0
```

The two fits above print only SuperLearner's progress bars, so the estimates themselves have to be asked for:

```r
aipw_ate$res
```

```
     parameter         est         se       ci.ll       ci.ul pval
1      E{Y(0)}  2.99695506 0.01093465  2.97552315  3.01838697    0
2      E{Y(1)}  2.93906408 0.01327342  2.91304817  2.96507999    0
3 E{Y(1)-Y(0)} -0.05789098 0.01597522 -0.08920241 -0.02657955    0
```

```r
aipw_att$res
```

```
      parameter         est         se       ci.ll       ci.ul pval
1      E(Y|A=1)  2.94948490 0.01160804  2.92673314  2.97223666    0
2   E{Y(0)|A=1}  3.01689509 0.01221863  2.99294658  3.04084360    0
3 E{Y-Y(0)|A=1} -0.06741019 0.01409116 -0.09502886 -0.03979152    0
```

Two estimates come out of this: an AIPW **ATE** of −5.8% (95% CI −8.9% to −2.7%) and an AIPW **ATT** of −6.7% (95% CI −9.5% to −3.9%).

The comparison worth making is like with like. The −7.5% from the fully interacted linear model above was an ATT (it was computed on `newdata = subset(data, female == 1)`), so it should be set against the nonparametric **ATT** of −6.7%, not against the ATE. Those two agree closely, which is reassuring: dropping the linearity assumption altogether moves the estimate by less than a percentage point, and both use the same covariate set including occupation and industry.

That last clause is also the caveat. Because both specifications condition on occupation and industry, both inherit the bad-control problem set out above: the agreement between −7.5% and −6.7% tells us the two estimators recover the same quantity, not that the quantity is a causal effect. Relaxing linearity addresses functional form; it leaves the collider untouched.

The ATE and ATT point estimates do differ a little, in the direction of a larger gap among women than in the sample as a whole — which would indicate effect heterogeneity across the covariates. Their confidence intervals overlap heavily, though, so this data cannot really distinguish them.

16.3. Beyond the mean: where in the distribution is the gap?

Everything so far has been a mean estimand — ATE, ATT, PAGE, PAGM, PAGW. But the best-known stylised fact about the gender wage gap is that it is *not uniform* across the distribution: a **glass ceiling** if the gap widens at the top, a **sticky floor** if it widens at the bottom. A chapter that reports only means cannot see either.

There are two distinct questions here, and they are routinely conflated.

16.3.1. The direct comparison

The distributional analogue of the raw gap needs no special machinery at all: compare the two groups' wage quantiles and difference them. Under the exogeneity argument above this is the total effect at each quantile — the well-identified rung of the ladder.

```r
library(tidyverse)

taus <- c(.10, .25, .50, .75, .90)

qgap <- function(dat) {
  sapply(taus, function(t)
    quantile(dat$lwage[dat$female == 1], t, names = FALSE) -
    quantile(dat$lwage[dat$female == 0], t, names = FALSE))
}

set.seed(1)
bs <- replicate(2000, qgap(data[sample(nrow(data), replace = TRUE), ]))

tibble(tau = taus,
       gap = qgap(data),
       lo  = apply(bs, 1, quantile, .025),
       hi  = apply(bs, 1, quantile, .975),
       pct = paste0(round(100 * (exp(qgap(data)) - 1), 1), "%")) |>
  mutate(across(gap:hi, ~round(.x, 4)))
```

```
# A tibble: 5 x 5
    tau     gap      lo      hi pct
  <dbl>   <dbl>   <dbl>   <dbl> <chr>
1   0.1   0      -0.0392  0.0282 0%
```

```
2  0.25 -0.0521 -0.0909  0       -5.1%
3  0.5  -0.0202 -0.0513  0       -2%
4  0.75 -0.069  -0.105  -0.0177 -6.7%
5  0.9  -0.0692 -0.143  -0.0126 -6.7%
```

The gap is indistinguishable from zero at the 10th percentile and around −6.7% at the 75th and 90th — a glass-ceiling pattern, though a noisy one. Two features of this table are artefacts worth naming. The exact zeros are heaping: log wages take repeated values, so the empirical quantile lands on the same number for both groups. That heaping is also why the sequence is not monotone (the gap at $\tau = 0.25$ exceeds the gap at $\tau = 0.50$). The direct comparison is exact but coarse in this data.

16.3.2. RIF regression, and what it actually estimates

Firpo, Fortin and Lemieux (2009) provide the machinery for putting covariates into a distributional statistic. For any functional ν of the outcome distribution, the influence function measures how ν responds to adding mass at y; recentring it gives a transformed outcome whose *mean is the statistic itself*. For the τ-th quantile,

$$\mathrm{RIF}(y; q_\tau) \;=\; q_\tau + \frac{\tau - \mathbb{1}\{y \le q_\tau\}}{f_Y(q_\tau)}. \tag{16.8}$$

Compute the sample quantile, estimate the density there, build the RIF as a new outcome, and regress it on covariates. Because $E[\mathrm{RIF}] = q_\tau$, the coefficients speak to the **unconditional** quantile. This is the key contrast with `quantreg::rq`, which gives *conditional* quantile effects — the effect at the τ-th quantile of the conditional distribution, among people with the same covariates. For "is there a glass ceiling in the wage distribution," the unconditional version is the one wanted, and fitting `rq` instead answers a different question.

```
library(tidyverse)
library(marginaleffects)

rif_quantile <- function(y, tau) {
  q  <- quantile(y, tau, names = FALSE)
  dn <- density(y)
  q + (tau - as.numeric(y <= q)) / approx(dn$x, dn$y, xout = q)$y
}

Xb <- "exp1 + sohs + hsg + socl + clg + mw + sout + we"

map_dfr(taus, function(t) {
  dd <- transform(data, rif = rif_quantile(lwage, t))
  m0 <- lm(rif ~ female, data = dd)
  m1 <- lm(as.formula(paste("rif ~ female*(", Xb, ")")), data = dd)
  tibble(tau = t,
         rif_no_covariates = round(coef(m0)["female"], 4),
```

```
        rif_with_covariates =
          round(avg_comparisons(m1, variables = "female",
                                vcov = "HC3")$estimate, 4))
})
```

```
# A tibble: 5 x 3
    tau rif_no_covariates rif_with_covariates
  <dbl>             <dbl>               <dbl>
1  0.1             -0.012             -0.0822
2  0.25            -0.0425            -0.120
3  0.5             -0.0409            -0.124
4  0.75            -0.0557            -0.134
5  0.9             -0.0775            -0.159
```

Now the important caveat, and it is an estimand distinction rather than a technical one. **The RIF-regression coefficient on `female` is not the gap between men's and women's wage quantiles.** It is the effect on the overall τ-quantile of marginally raising the *share* female, holding conditional distributions fixed — a composition effect. That is why the numbers here disagree with the direct comparison at every τ: at the median, -0.020 directly against -0.041 by RIF. Both are legitimate; they answer different questions. Reporting a RIF coefficient as "the gap at the 90th percentile" is a common slip.

With that said, the RIF panel is the more readable of the two here, precisely because the density smoothing interpolates through the heaping that makes the direct comparison lumpy. Both columns rise monotonically in τ, and the covariate column runs from -8.2% at the 10th percentile to -15.9% at the 90th. The mean estimates from earlier in the chapter — -3.8% raw, -11.4% adjusted — are averages over that gradient.

Three further caveats before anyone leans on these numbers:

- **RIF is a first-order approximation**, valid for small covariate shifts. Flipping a binary regressor for the entire sample is not small, so the approximation degrades. Exact alternatives are Firpo (2007) propensity-weighted quantile treatment effects, or Chernozhukov, Fernández-Val and Melly (2013) counterfactual distributions.
- **The standard errors here are too small.** The density estimate contributes uncertainty that `lm` knows nothing about; bootstrap the whole procedure, as done for the direct comparison above.
- **The covariate column is a distributional rung of the same CDE ladder**, and inherits every bad-control problem from the earlier section. It is not a better estimate of the same thing.

Finally, a quantile treatment effect compares quantiles of two marginal distributions. It is not the effect *for the person at* quantile τ unless one is willing to assume rank invariance — that the same individuals occupy the same ranks in both counterfactual worlds.

16.4. What is any of this measuring?

Two directed acyclic graphs and a `dagitty` verification have now been used to sort covariates into good and bad controls. It is worth closing with an objection to that whole procedure, because it comes from a literature economists rarely read and because it bears directly on the re-dating manoeuvre of Section 16.2.4. The principals are **Lily Hu** (Philosophy, Yale) and **Issa Kohler-Hausmann** (Law, Yale); the two papers to read are Hu and Kohler-Hausmann (2020) and Hu and Kohler-Hausmann (2024).

16.4.1. Causal versus constitutive

Their objection turns on a distinction the causal toolkit has no notation for.

	Causal	Constitutive
Relation	X **produces** Y	X is **part of what it is to be** Y
Timing	X precedes Y	simultaneous
Contingency	can have X without Y	cannot have Y without X
Intervention	wiggle X, hold others fixed	wiggling X changes *what the thing is*

Non-social cases make it clean. Being unmarried does not *cause* someone to be a bachelor — it is part of what being a bachelor consists in. A chess position does not *cause* checkmate; it constitutes it. Ask "what would happen to this checkmate if the king were not trapped?" and there is no answer, not because the quantity is hard to estimate but because there would be no checkmate left to ask about.

16.4.2. Why gender's features are partly constitutive

Consider the observation that women do more caregiving. On the causal reading there is a category, woman, fixed independently, which then produces caregiving downstream. On the constitutive reading, part of what it **is** to occupy the social position "woman" is to be subject to caregiving expectations — the norm is not produced by the category, it is one of the threads the category is woven from.

Press on what makes someone socially a woman in the sense labour markets respond to. Not chromosomes; markets do not observe those. It is occupying a position defined by a cluster of expectations, treatments and readings. Strip the cluster away and no social category remains, only biology. On this view the cluster is not the category's output; it is largely the category itself.

The cleanest comparison is a social role. *What is the causal effect of being a manager on having authority?* That is not a causal question — having authority is part of what being a manager consists in. One cannot hold authority fixed and vary managerial status, because nothing would be left for "manager" to refer to.

“Partly” is doing real work, though. Hu and Kohler-Hausmann claim *many*, not all, of the effects attributed to sex are constitutive, so the category is a mixture: caregiving norms and being read as a plausible authority figure are plausibly constitutive, while one firm’s decision to deny one promotion is causal (contingent, could go otherwise, category unchanged) and average upper-body strength is causal-biological. The difficulty is that **no method sorts them.** A usable test is to ask whether removing the feature would leave the same category or a different thing entirely — but applying it is a substantive sociological claim, and drawing a DAG settles it silently.

16.4.3. Why this defeats modularity

Modularity is the formal assumption that the dependencies along one pathway are invariant to interventions on other pathways. It is exactly what licensed every arrow-cutting exercise above.

If `female -> educ` is causal, “flip gender, hold education fixed” is coherent: imagine changing the assignment while the education-producing mechanism keeps running. If the relation is partly constitutive, the instruction does not describe anything. An intervention that changed a woman’s gender while leaving the associated expectations intact is not an intervention on gender-as-a-social-category — it holds fixed part of the very antecedent it purports to vary. The counterfactual has no determinate content.

This is a different *kind* of problem from everything else in this chapter, and the difference is worth stating flatly. The mediator, collider, selection and positivity problems are all **epistemic**: each says the quantity exists and we cannot reach it, and each has a design remedy — randomise, bound, run a sensitivity analysis. The constitutive objection is **semantic**: the description “same woman, but a man, everything else fixed” fails to pick out a possible state of affairs, so there is no quantity that identification is failing to reach. That is why no design answers it, and why the authors call the isolation project theoretically misguided rather than underpowered.

In place of causal diagrams they propose *constitutive* diagrams, which display how features mutually define one another. Note what that implies for this chapter: the tool itself — the DAG, with its cuttable arrows — encodes the assumption at issue. Drawing one is already taking a side.

16.4.4. The perception move does not escape it

Section 16.2.4 resolved the bad-control problem by re-dating the treatment to perceived sex at the hiring decision, then observed that identification requires randomising the signal. Hu and Kohler-Hausmann (2024) attack precisely that estimand, by asking: a perception of sex or race is a perception *of what, exactly?*

Their answer is that the perception of the protected trait and the perception of other decision-relevant features are **co-constituted**. Their example: present two arrestees with “identical” prior-arrest records. They have not thereby been made equivalent, because the same record carries different meaning depending on whom it describes — an arrestee racialised as Black with N priors may be read as having an *unexpectedly low* number relative to imputed risk, while a white arrestee with the same N reads as ordinary. Making candidates similar in one respect necessarily makes them different in others.

Transposed to a résumé audit: “identical file, different perceived sex” again holds fixed something — the file’s *significance* — that co-varies with the thing being varied. The files are identical in ink and non-identical in meaning, and meaning is what the employer responds to. **Which similarity is held fixed is itself the contested question**, not a technical parameter of the design.

Their conclusion is narrower than "audit studies are worthless," and more damaging for being narrower: they reject the claim that such studies measure perception effects free of confounding, and the claim that a perceived-sex effect simply *is* disparate treatment. The second equation needs an unstated **normative** premise about what treatment ought to look like. The charge is that researchers do normative theorising behind technical machinery, and the demand is that both the sociological theory and the normative one be defended out loud instead of being settled by a covariate list.

16.4.5. Three replies

The argument is live, and the replies are substantive.

- **The signal is modular even if the category is not.** One really can change the name on a résumé holding the rest of the text fixed — a physical operation needing no metaphysics. That the reader's interpretation is holistic is arguably part of the effect measured, not a defect in the estimand. This is the strongest reply and it protects correspondence studies specifically.
- **Comparison is not optional for law.** Discrimination doctrine is comparative by construction; if no comparison is well-posed the legal concept cannot operate. The authors would do the normative work openly rather than abandon comparison, but this leaves the empirical programme without operational guidance.
- **Constitution is scale-dependent.** Over a lifetime, gender and human capital are mutually constituting and the objection bites hard. At the moment of a single hiring decision with the file fixed, the manipulation is genuinely modular and it bites least. Which suggests the defensible middle: the critique is devastating for lifetime-scale "effect of gender" claims and comparatively weak against tightly scoped decision-stage experiments.

16.4.6. What survives, in this chapter's own numbers

None of this licenses the conclusion that the gender wage gap cannot be studied, and its authors do not draw it. What is ill-defined is one specific summary statistic, not the subject.

Safe, and unaffected by the objection:

- **"Women in this sample earn 3.8% less than men."** A description of an observed population — and the statistic UK and EU pay-gap reporting mandate, unadjusted, precisely because adjusted numbers launder the structural component.
- **"Within education, experience and region cells the gap is 11.3%; adding occupation and industry it is 7.5%."** Within-cell contrasts, correctly labelled as such.
- **"The gap is near zero at the 10th percentile and about 7% at the 90th."** A distributional description.
- **"A confounder explaining roughly 3% of the residual variance of gender and 4% of wages would account for the entire 11.3% to 7.5% attenuation."** A sensitivity statement about a stated model.

Ill-defined: **"the effect of being a woman, net of everything legitimate"** — a single number separating discrimination from productivity across a lifetime.

Two independent routes condemn that one number and nothing else on the list. The identification route, worked through earlier in this chapter, shows it is unreachable from these data for three separate reasons: mediators everywhere, a collider at every rung, and a positivity failure at the only valid adjustment set. The constitutive route holds it was never a coherent target. Since both converge, dropping it costs nothing that was ever in hand.

16.4.7. The operational rule

Report what was **manipulated** or **observed**, in the **population it was observed in**. Then make the normative claim as a stated argument, rather than as something a coefficient is supposed to deliver.

Which is what the ladder in this chapter does. −3.8% is the observed total gap; −11.3% and −7.5% are within-cell contrasts; none of them is discrimination, and now the chapter says so.

16.4.8. References for this section

- Hu, L. and I. Kohler-Hausmann (2020). What's sex got to do with fair machine learning? *ACM FAccT*. arXiv:2006.01770.
- Hu, L. and I. Kohler-Hausmann (2024). What is perceived when race is perceived and why it matters for causal inference and discrimination studies. *Law & Society Review* (open access).
- Hu, L. (2023). What is "race" in algorithmic discrimination on the basis of race? *Journal of Moral Philosophy*.
- Hu, L. (2025). Sex discrimination, normativity, and begging the causal question. *Political Philosophy*.
- Kohler-Hausmann, I. (2019). Eddie Murphy and the dangers of counterfactual causal thinking about detecting racial discrimination. *Northwestern University Law Review*.

Part IV.

Panel Data & DiD

17. Causal Forest in panel data

17.1. Introduction

In this simulation exercise, we use Causal Forest (Now is implemented in Generalized Random Forest) (https://github.com/swager/grf) to calculated conditional average treatment effect (or heterogeneous treatment effect). We assume three different data generating processes. The first one is a linear interaction between a variable of interest and the treatment dummy. The second one assumes a nonlinear function (a step function) of a variable of interest, say X, and the treatment dummy W. The third one is also nonlinear, assuming we have variable X, but the real DGP is the interaction of log of X and W.

17.2. Linear effect

Suppose we have a panel of firms over time, treatment assignment is random. We split the data into training and test. Then we generate 10 random variables, the first two are highly correlated. Then we collapse V2 into firm means, and use that as the unit effect.

Concretely: $n = 400$ firms over $t = 10$ periods, 4,000 observations, with $p = 10$ standard normal covariates of which $V1$ and $V2$ are correlated at 0.9025. Treatment is $W \sim \text{Bernoulli}(0.5)$, assigned independently of everything, and a further coin flip splits the sample roughly in half into training and test sets (about 2,014 rows for training). The unit effect α_i is the firm mean of $V2$.

That is, we have a DGP as:

$$y_{it} = \alpha_i + V1_{it} + W_{it} + V1_{it} * W_{it} + \epsilon_{it} \tag{17.1}$$

Here α_i is not observed, and it is correlated with $V1$, so leaving it out is in principle a source of bias.

It is worth being precise about how much bias, though, because the results below are less dramatic than that setup suggests. The unit effect is the *firm mean* of $V2$ over all $t = 10$ periods, and averaging attenuates its correlation with contemporaneous $V1$ by a factor of t: with $\text{Cor}(V1_{it}, V2_{it}) = 0.9025$ we get $\text{Cov}(V1_{it}, \bar{V}2_i) = 0.9025/10$, and $\text{Cor}(V1_{it}, \alpha_i) \approx 0.29$. So the omitted-variable bias on $V1$ is only about 0.05–0.09, not a qualitative failure.

That is what the three models show: FE, RE and OLS all recover $V1 \approx 1$, $W \approx 1$ and $V1{:}W \approx 1.05$, essentially the truth. Note in particular that the *treatment* coefficients are unbiased in all three, because W is randomized and so independent of α_i — omitting a unit effect cannot bias the effect of a randomly assigned treatment. If you want a panel example where FE visibly beats OLS, the unit effect has to be correlated with $V1$ *contemporaneously* (or t has to be small), and the treatment has to be non-random.

```r
# This is a simulation exercise to see how causal forest performs
# First we run a linear model, with panel setting
# Then two nonlinear situations, with panel setting.

library(lfe, quietly = TRUE)
library(lme4)
require(snowfall)
library(MASS)
library(grf)
library(tidyverse)
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)

firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)
```

```
y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ data$V1[k] + data$W[k] + data$V1[k]*data$W[k] + data$unit.effect[k]+rnorm(1,me
    }
}
data$unit=unit
data$y=y



data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
# biased.
# in general we should expect fe performs better, since we are
# omitting unit.effect in the other two models
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-3.0440 -0.6252 -0.0083  0.6221  3.6174

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1    0.98354    0.03748   26.24   <2e-16 ***
W     1.02186    0.05093   20.06   <2e-16 ***
V1:W  1.04302    0.05237   19.92   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.018 on 1612 degrees of freedom
```

```
Multiple R-squared(full model): 0.7952   Adjusted R-squared: 0.7443
Multiple R-squared(proj model): 0.7269   Adjusted R-squared: 0.659
F-statistic(full model):15.61 on 401 and 1612 DF, p-value: < 2.2e-16
F-statistic(proj model):  1430 on 3 and 1612 DF, p-value: < 2.2e-16
```

```
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 5938.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.4332 -0.6615 -0.0049  0.6640  3.8329

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.07166  0.2677
 Residual             1.04602  1.0228
Number of obs: 2014, groups:  unit, 399

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1.00010    0.03539   28.26
V1           1.02961    0.03486   29.53
W            1.00333    0.04684   21.42
V1:W         1.05963    0.04814   22.01

Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.013
W    -0.656 -0.008
V1:W -0.007 -0.727  0.025
```

```
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-3.8384 -0.7101  0.0081  0.7334  4.0053
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.00363    0.03319   30.24   <2e-16 ***
V1           1.04612    0.03516   29.75   <2e-16 ***
W            0.99657    0.04712   21.15   <2e-16 ***
V1:W         1.06458    0.04840   22.00   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.057 on 2010 degrees of freedom
Multiple R-squared:  0.7249,    Adjusted R-squared:  0.7245
F-statistic:  1765 on 3 and 2010 DF,  p-value: < 2.2e-16
```

Now we run Causal Forest on this data. Two clarifications about what is and is not at stake here. First, treatment W is **randomized** (`rbinom(., 0.5)`, independent of the firm effect), so the omitted unit effect α_i does *not* confound the treatment effect — it only enters the nuisance functions (and hence precision), it does not create treatment-effect bias. The panel issue here is therefore mainly *dependence and nuisance fit*, not confounding. Second, the firm structure should be handled in two ways: we include the firm dummies in X so the forest can absorb α_i in the nuisance models, **and** we pass `clusters = firm` so that `grf`'s honest sample splitting and its variance estimates keep each firm's repeated observations together rather than treating them as independent rows. Including the dummies alone does neither of those.

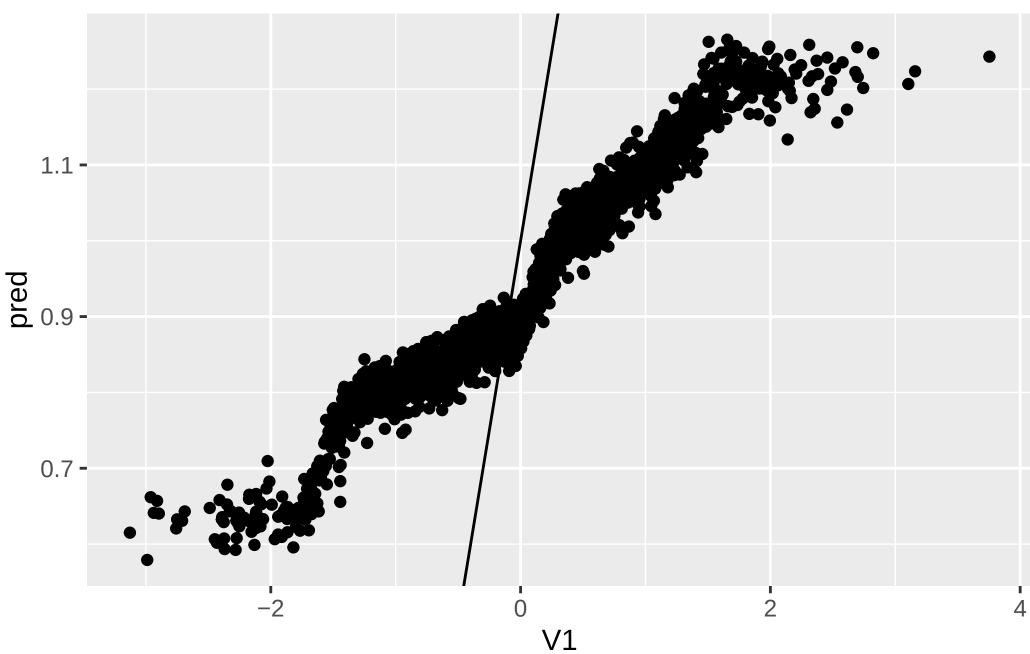
```r
# ensure consistent factor levels across train/test
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)

# a is a data set of firm dummies
# Caution: this one-hot-encodes all ~400 firms into individual dummy columns
# of X. Random forests split poorly on such a high-cardinality sparse block
# of binary columns (each dummy is only informative for a tiny fraction of
# observations). A fixed-effects forest (demean y, W, and the continuous
# covariates within firm first, then run causal_forest on the demeaned
# variables) or a method that handles unit effects natively would typically
# perform better than including firm as raw dummies.
a <- as.data.frame(model.matrix(y~firm, data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W
# causal forest is run on y against V1, to X10 and firm dummies.
# clusters = firm makes honest splitting and the variance estimates respect
# the panel structure (a firm's repeated rows are kept together, not treated
# as independent observations).
tau.forest = causal_forest(X, Y, W, clusters = as.integer(data.train$firm))

average_treatment_effect(tau.forest, target.sample = "all")
```

```
  estimate    std.err
0.93560912 0.08227266
```

```r
average_treatment_effect(tau.forest, target.sample = "treated")
```

```
  estimate    std.err
0.93168066 0.08158925
```

```r
a <- as.data.frame(model.matrix(y~firm, data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.hat = predict(tau.forest, X.test, estimate.variance = TRUE)
sigma.hat = sqrt(tau.hat$variance.estimates)

data.test$pred <- tau.hat$predictions
ggplot(data.test, aes(x=V1, y=pred)) + geom_point() + geom_abline(intercept = 1, slope = 1)
```

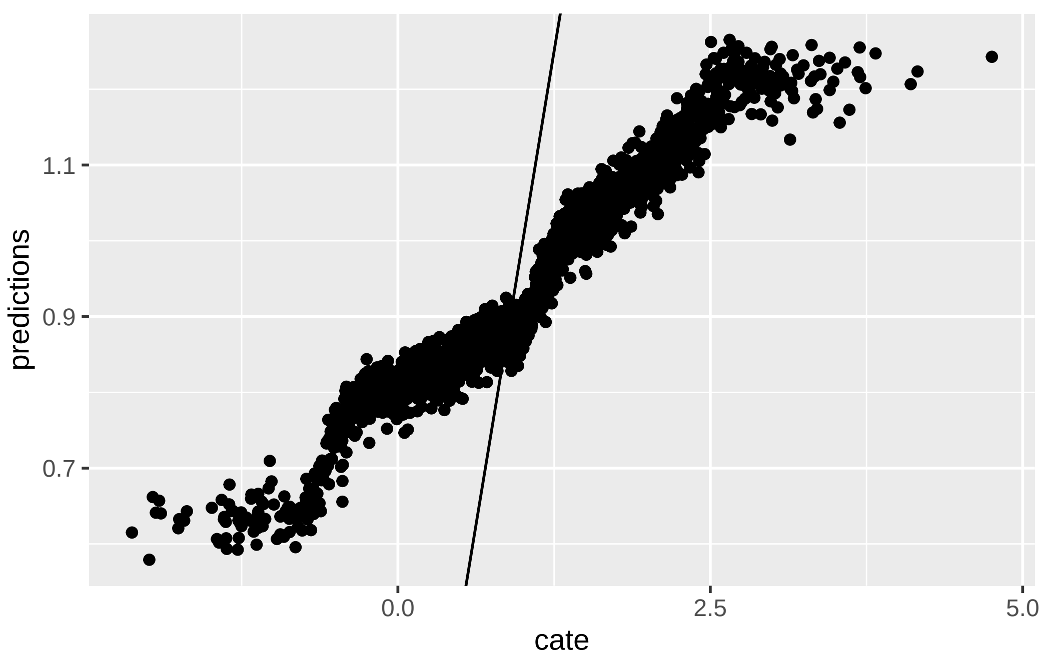


The forest's doubly-robust ATE comes out at 0.9356 with a standard error of 0.0823, and 0.9317 with 0.0816 for the second variant. The true ATE is $E[1+V1]=1$, so both are within one standard error of it — the forest costs a little precision against the correctly specified linear models but is not biased.

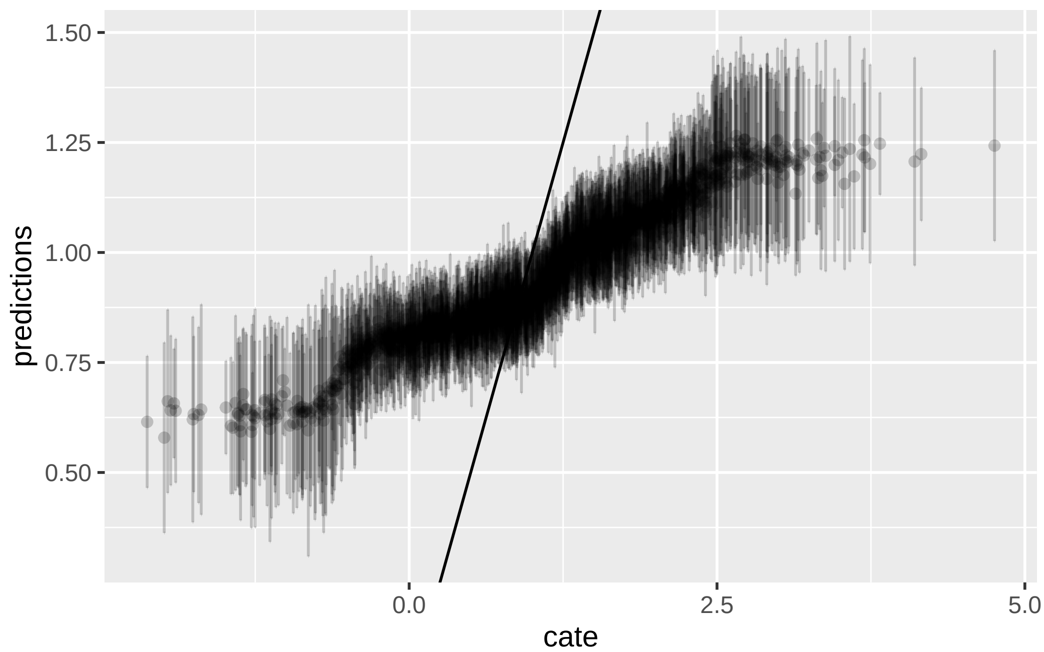
This graph is the predicted value of CATE on the test data, vs. the value of $V1$. The points should sit on the 45-degree line, and the error bars in the second panel show how much of the scatter is estimation noise rather than a failure to track the true $1+V1$.

Now we graph the predicted CATE against the true CATE. The other nice feature of Causal Forest is that it returns variance for CATE for each observation. We also graph the error bar here.

```
graph.data  <- as.data.frame(tau.hat )
graph.data  <- graph.data  %>%
    mutate(cate=1+data.test$V1) %>%
    mutate(lower=predictions - sigma.hat , upper=predictions + sigma.hat )

ggplot(graph.data , aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```




```
ggplot() + geom_errorbar(graph.data , mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2)
```



17.3. Nonlinear effect case 1

In the first situation, we assume DGP:

$$y_{it} = 1 + \alpha_i + max(V1_{it}, 0) * W_{it} + \epsilon_{it} \tag{17.2}$$

```
# if we have a nonlinear effect
# suppose we have the same X, but y is generated nonlinearly.
# in this example, y is a nonlinear function of V1 and W
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)

firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
```

```
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)

y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ pmax(data$V1[k],0)*data$W[k]  + data$unit.effect[k]+rnorm(1,mean=0,sd=1)
    }
}
data$y=y
data$unit <- unit

data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
# biased.
# they all perform poorly
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-2.9461 -0.6225 -0.0178  0.6245  3.3537

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1   -0.02076    0.03815  -0.544    0.586
W     0.42527    0.05184   8.203 4.73e-16 ***
V1:W  0.53367    0.05331  10.012  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.036 on 1612 degrees of freedom
Multiple R-squared(full model): 0.3744   Adjusted R-squared: 0.2187
Multiple R-squared(proj model): 0.1393   Adjusted R-squared: -0.07481
F-statistic(full model):2.405 on 401 and 1612 DF, p-value: < 2.2e-16
F-statistic(proj model): 86.96 on 3 and 1612 DF, p-value: < 2.2e-16
```

```r
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 6017.2

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.5905 -0.6596 -0.0031  0.6717  3.4143

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.07977  0.2824
 Residual             1.08359  1.0410
Number of obs: 2014, groups:  unit, 399

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.99976    0.03624  27.587
V1           0.02797    0.03553   0.787
W            0.40652    0.04774   8.515
V1:W         0.54484    0.04907  11.104
```

```
Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.013
W    -0.653 -0.008
V1:W -0.007 -0.727  0.025
```

```
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0965 -0.7415  0.0095  0.7512  3.8413

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.00363    0.03386  29.643   <2e-16 ***
V1           0.04612    0.03587   1.286    0.199
W            0.39928    0.04807   8.307   <2e-16 ***
V1:W         0.54938    0.04937  11.127   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.078 on 2010 degrees of freedom
Multiple R-squared:  0.1556,    Adjusted R-squared:  0.1543
F-statistic: 123.5 on 3 and 2010 DF,  p-value: < 2.2e-16
```

These linear models miss the *shape* of the effect, but it overstates things to say they return nothing. Each recovers the best linear approximation to the true CATE $\max(V1, 0)$, which is $0.3989 + 0.5\,V1$: the fitted W coefficients are 0.40–0.43 and the $V1{:}W$ interactions are 0.53–0.55. In particular the W coefficient is essentially the true ATE, $E[\max(V1, 0)] = \phi(0) = 0.3989$. What a linear specification cannot do is see the kink at $V1 = 0$ — and that is what the forest is for.

Now we use Causal Forest. The first graph is the predicted CATE against $V1$; the second graph is the predicted CATE against true CATE. The third one adds in error bars.

The forest's ATE is 0.3732 (SE 0.0533) and 0.3722 (SE 0.0529), against the true 0.3989 — within half a standard error, and no better than the linear model's W coefficient at recovering the average. The gain is in the shape: the fitted CATE should flatten to zero for $V1 < 0$ and rise linearly above it, which is the kink the linear interaction smooths straight through.

```
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)
```

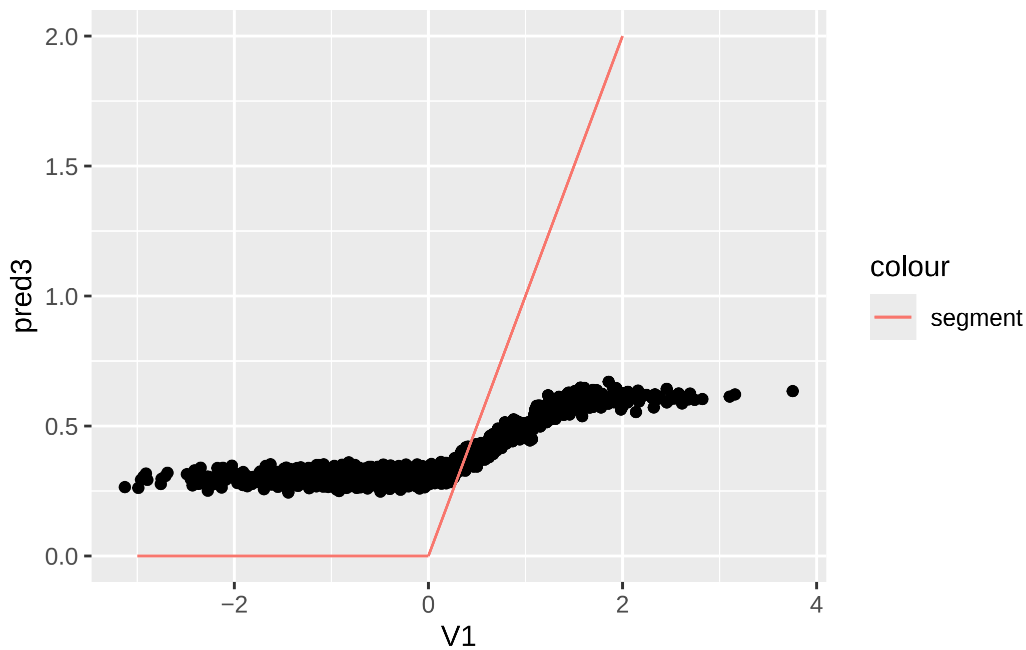
```
# a is a data set of firm dummies
a <- as.data.frame(model.matrix(y~firm,data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W

#tau.forest = causal_forest(X, Y, W, num.trees = 4000)
a <- as.data.frame(model.matrix(y~firm,data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.forest3 = causal_forest(X, Y, W, clusters = as.integer(data.train$firm))

average_treatment_effect(tau.forest3, target.sample = "all")
```

```
  estimate    std.err
0.37320336 0.05326215
```

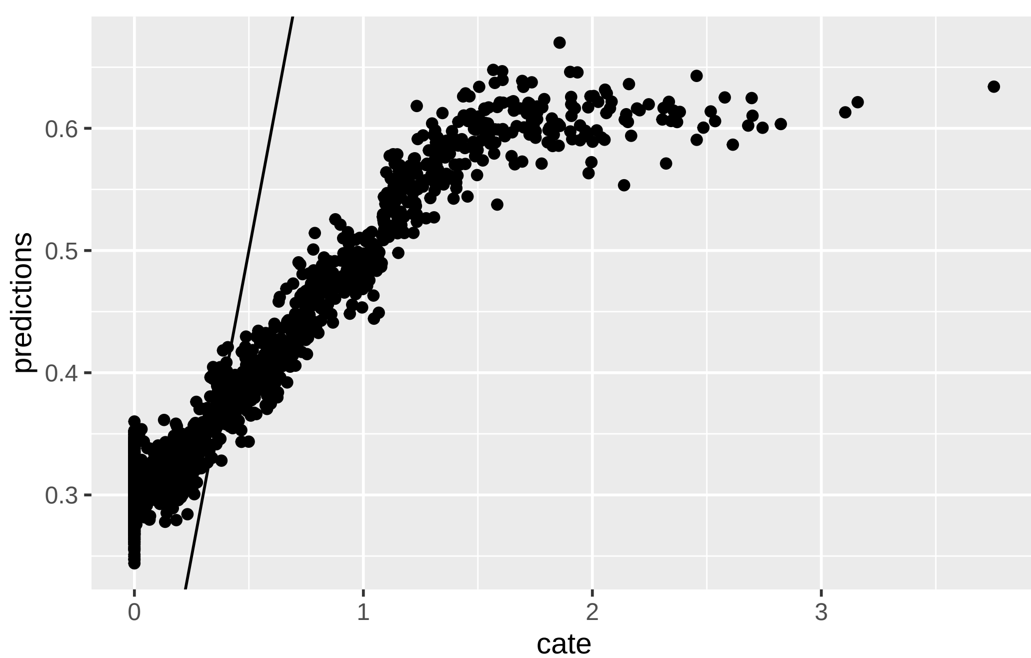
```
average_treatment_effect(tau.forest3, target.sample = "treated")
```

```
  estimate    std.err
0.37223789 0.05285912
```

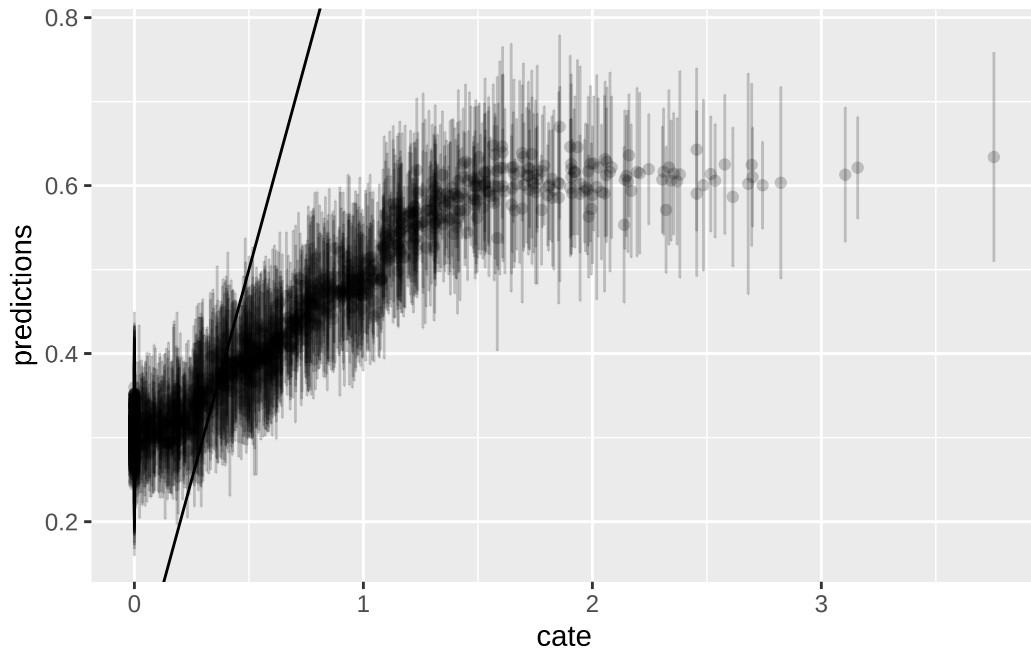
```
#tau.forest3 = causal_forest(X, Y, W, num.trees = 4000)
tau.hat3 = predict(tau.forest3, X.test, estimate.variance = TRUE)
sigma.hat3 = sqrt(tau.hat3$variance.estimates)

data.test$pred3 <- tau.hat3$predictions

seg1 <- data.frame(x1 = -3, x2 = 0, y1 = 0, y2 = 0)
seg2 <- data.frame(x1 = 0, x2 = 2, y1 = 0, y2 = 2)
ggplot(data.test, aes(x=V1, y=pred3)) + geom_point() +
geom_segment(aes(x = x1, y = y1, xend = x2, yend = y2, colour = "segment"), data = seg1)+
geom_segment(aes(x = x1, y = y1, xend = x2, yend = y2, colour = "segment"), data = seg2)
```




```
graph.data3 <- as.data.frame(tau.hat3)
graph.data3 <- graph.data3 %>%
    mutate(cate=pmax(data.test$V1,0)) %>%
    mutate(lower=predictions - sigma.hat3, upper=predictions + sigma.hat3)

ggplot(graph.data3, aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```



```
ggplot() + geom_errorbar(graph.data3, mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2)
```



17.4. Nonlinear effect case 2

In the second situation, we assume DGP:

$$y_{it} = 1 + \alpha_i + \log(V1^2_{it}) * W_{it} + \min(X3_{it}, 0) + \epsilon_{it} \tag{17.3}$$

But we only observe $V1$, not the log of the squared $V1$. The extra $\min(X3_{it}, 0)$ term is a nonlinearity in the *outcome* rather than in the treatment effect, so it leaves the CATE equal to $\log(V1^2)$; it is there to give the forest's nuisance models something nonlinear to fit.

```
# if we have a nonlinear effect
# suppose we have the same X, but y is generated nonlinearly.
# another example: we observe V1, but the true treatment effect is log(V1^2)
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)
```

```
firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)

y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ log(data$V1[k]^2)*data$W[k] + pmin(data$X3[k], 0) + data$unit.effect[k]+rnorm
    }
}
data$y=y
data$unit <- unit

data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
```

```r
# biased.
# they all perform poorly
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
     Min       1Q   Median       3Q      Max
-10.0383  -0.9478   0.0907   1.0865   5.3884

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1   -0.01979    0.07088  -0.279    0.780
W    -1.17812    0.09632 -12.231   <2e-16 ***
V1:W  0.01450    0.09904   0.146    0.884
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.925 on 1612 degrees of freedom
Multiple R-squared(full model): 0.2876   Adjusted R-squared: 0.1104
Multiple R-squared(proj model): 0.08495   Adjusted R-squared: -0.1427
F-statistic(full model):1.623 on 401 and 1612 DF, p-value: 6.032e-11
F-statistic(proj model): 49.88 on 3 and 1612 DF, p-value: < 2.2e-16
```

```r
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 8421.8

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.1184 -0.4977  0.0647  0.6098  2.8959

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.09312  0.3052
 Residual             3.72467  1.9299
Number of obs: 2014, groups:  unit, 399
```

```
Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.57512    0.06301   9.128
V1           0.06152    0.06490   0.948
W           -1.18310    0.08706 -13.590
V1:W        -0.01892    0.08945  -0.212

Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.015
W    -0.685 -0.010
V1:W -0.009 -0.727  0.028
```

```r
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
     Min       1Q   Median       3Q      Max
-12.1215  -0.9805   0.1390   1.1999   5.7125

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.57618    0.06136   9.390   <2e-16 ***
V1           0.07086    0.06502   1.090    0.276
W           -1.18331    0.08712 -13.582   <2e-16 ***
V1:W        -0.02005    0.08949  -0.224    0.823
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.954 on 2010 degrees of freedom
Multiple R-squared:  0.08508,   Adjusted R-squared:  0.08372
F-statistic: 62.31 on 3 and 2010 DF,  p-value: < 2.2e-16
```

Here the linear models fail more completely, and instructively so. Because $\log(V1^2)$ is *even* in $V1$, its linear interaction with $V1$ is zero by construction: the fitted $V1{:}W$ terms come out at 0.015, -0.019 and -0.020, i.e. no detectable heterogeneity at all, even though the true CATE varies enormously with $V1$. The W coefficients (≈ -1.18) still approximate the true ATE, $E[\log(V1^2)] = \psi(1/2) + \log 2 = -1.2704$. So a linear model gets the average right and the heterogeneity completely wrong.

Now we run Causal Forest. Again, first graph is predicted CATE against $V1$, the second one is the predicted CATE against true CATE, the third one adds error bars.

The forest's ATE is -1.2049 (SE 0.0813) and -1.2024 (SE 0.0798) for the two variants, against the true -1.2704. Like the linear models, it gets the average about right. The difference is in the plots: the forest's fitted CATE traces the U-shape of $\log(V1^2)$, which no linear interaction can represent.

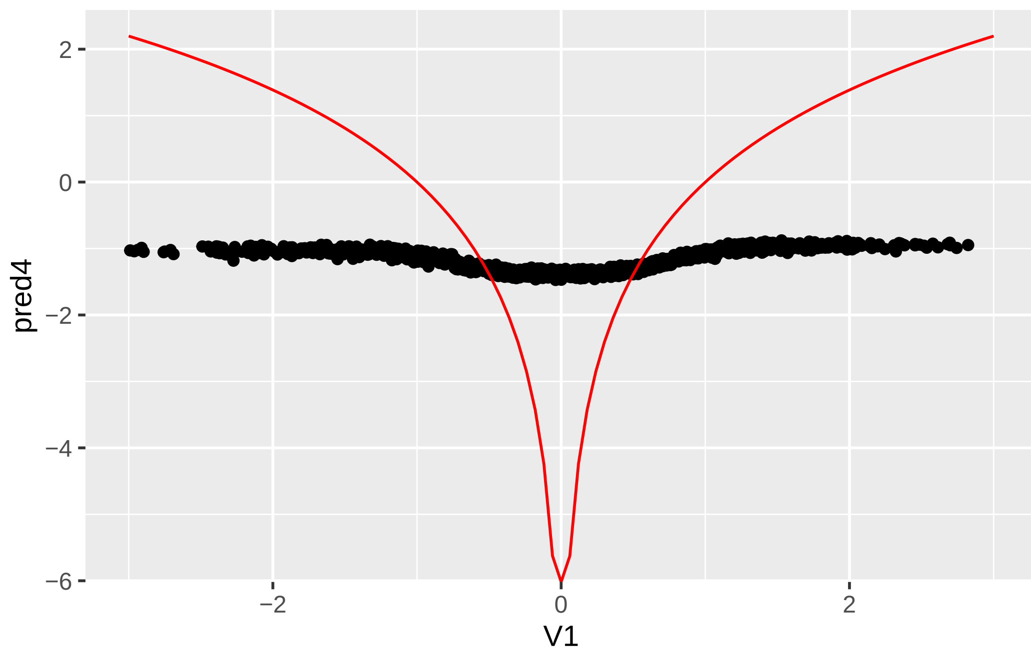
```
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)

# a is a data set of firm dummies
a <- as.data.frame(model.matrix(y~firm,data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W

#tau.forest = causal_forest(X, Y, W, num.trees = 4000)
a <- as.data.frame(model.matrix(y~firm,data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.forest4 = causal_forest(X, Y, W, clusters = as.integer(data.train$firm))

average_treatment_effect(tau.forest4, target.sample = "all")
```

```
   estimate     std.err
-1.20491164  0.08132368
```

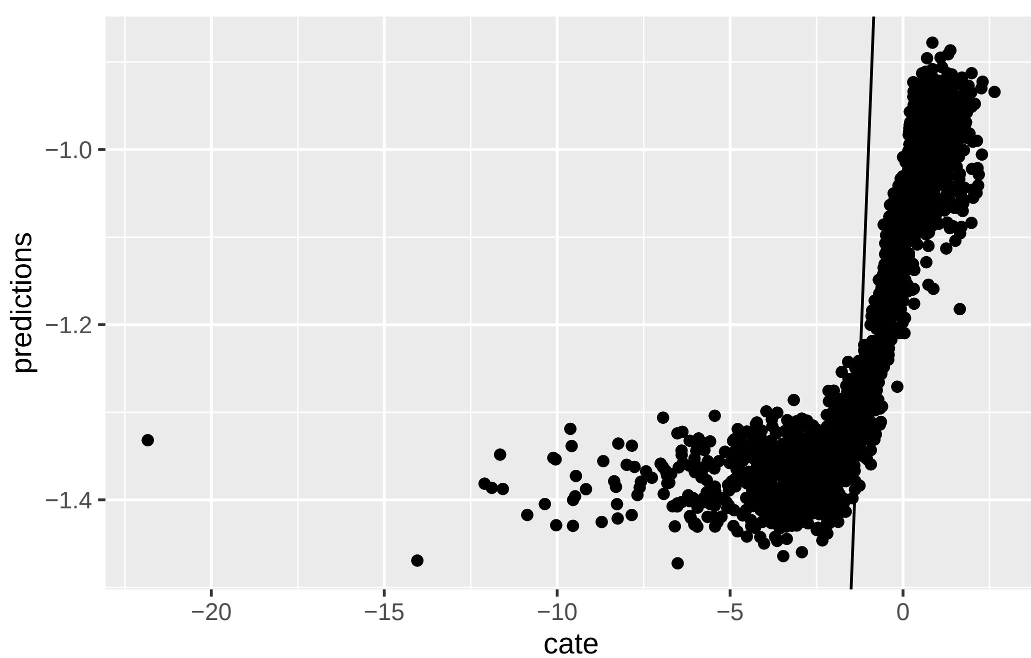
```
average_treatment_effect(tau.forest4, target.sample = "treated")
```

```
   estimate     std.err
-1.20241393  0.07981497
```

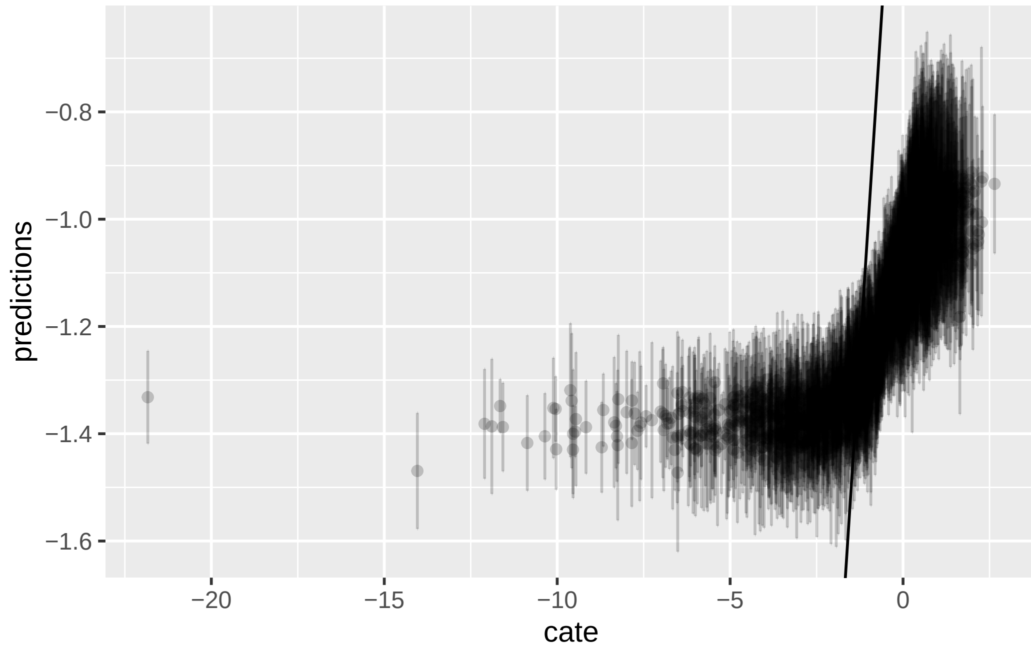
```
#tau.forest3 = causal_forest(X, Y, W, num.trees = 4000)
tau.hat4 = predict(tau.forest4, X.test, estimate.variance = TRUE)
sigma.hat4 = sqrt(tau.hat4$variance.estimates)

data.test$pred4 <- tau.hat4$predictions

fun.1 <- function(x) log(x^2)
ggplot(data.test, aes(x=V1, y=pred4)) + geom_point() +
stat_function(fun = fun.1, colour="red") + xlim(-3,3)
```




```
graph.data4 <- as.data.frame(tau.hat4)
graph.data4 <- graph.data4 %>%
    mutate(cate=log(data.test$V1^2)) %>%
    mutate(lower=predictions - sigma.hat4, upper=predictions + sigma.hat4)

ggplot(graph.data4, aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```



```
ggplot() + geom_errorbar(graph.data4, mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2
```



17.5. Summary

We see in these examples, Causal Forest performs fairly well, while other linear models will not work in the nonlinear DGP situations.

Systematic treatment: R · Julia.

18. Synthetic Control in R and Stata

18.1. Synthetic control

Abadie et al. have a few papers on synthetic control — Abadie, Diamond and Hainmueller (2010, *JASA*) and (2015, *AJPS*), and Abadie's (2021, *JEL*) survey. The key idea is for a case study situation. A single unit, say a state/firm/country is exposed to a shock, and we are interested in the effect of that shock. For example, Card and Krueger (1994) studied the effect of a minimum wage increase in New Jersey. They used diff-in-diff, with Pennsylvania as the control state. The point of synthetic control is that maybe parallel trend assumption does not hold for the control unit. Is Pennsylvania a good control for New Jersey? Does it follow the same trend as New Jersey? Maybe not. Synthetic control is to generate a more comparable control with some combination of multiple control units, which can have a much closer parallel trend as the treated unit, than using any of the control units.

Now, how to construct this synthetic control unit? Basically it is a weighted average of multiple control units. There are two sets of weights to be determined. One set for the control units, one set to determine the importance of predictors. Abadie et al (2010) propose to choose a set of w_i's so that the resulting synthetic control best resembles the pre-intervention values for the treated unit of predictors of the outcome variable, subject to the constraints that the weights are non-negative and sum to 1. Another set of weights v_h for predictor importance is determined such that it minimizes the mean squared prediction error (MSPE) of this synthetic control with respect to the outcome in the pre-treatment period. The intuition of these weights are to pick w_i's to make the weighted sum of each predictor as close to the treated unit's value of predictor as close as possible, given a set of v_h's. Then v_h's are determined by how important each predictor is to predict the outcome. We don't use any information post-treatment to determine the weights. But we can use pre-treatment outcome to see the relative importance of each predictor in terms of predictor outcome. We can do this by out-of-sample validation. Basically divide the pre-treatment period into training and validation periods (that's one reason that the pre-treatment period cannot be too short). Iterate the process of choosing w_i and v_h to achieve a minimization of MSPE in the validation period.

18.2. inference

Since synthetic control is for case study situation, we have essentially one treated unit and one synthetic control. The usual inference method would not work. Abadie et al suggested a permutation based approach. The basic idea is to permute the label of treatment; in other words, suppose one control unit is now a treated unit. Then follow the same procedure to get the synthetic control for that "treated unit"; look at the effect. Now if the treatment is truly for the treated unit only, then there would not be any effect for all the control unit. Therefore we can have a distribution of treatment effects after permutation of treatment label.

18.3. an example

The canonical synthetic control is implemented in both R and Stata (the `Synth` package, CRAN; see Abadie, Diamond and Hainmueller's *JSS* paper for the software description). We use a package called "tidysynth" here to study one of the original examples, which evaluates the impact of Proposition 99 on cigarette consumption in California.

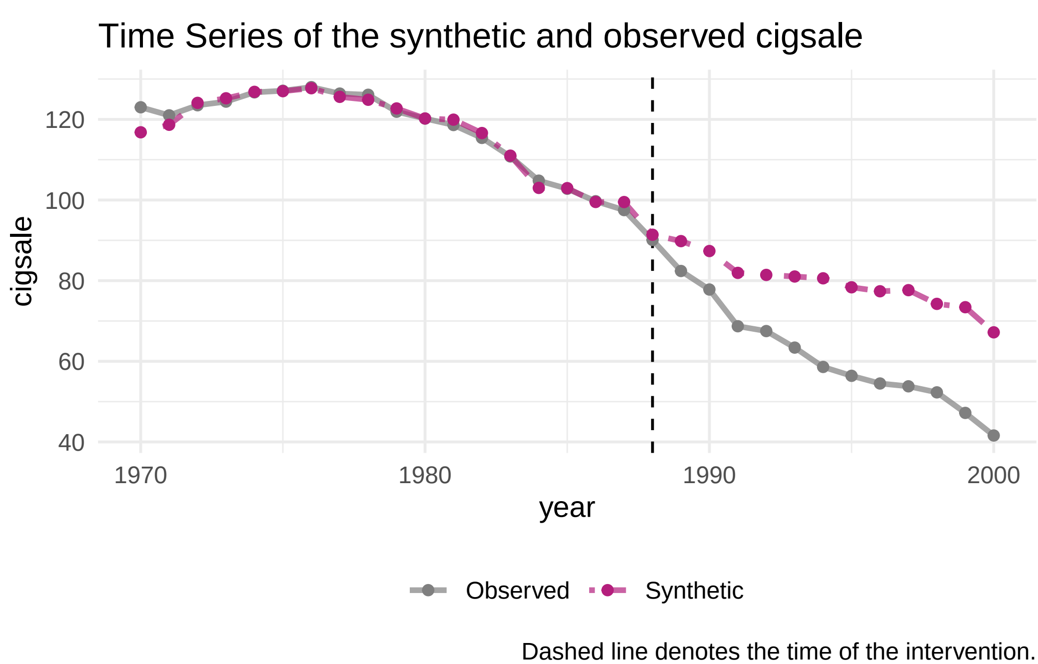
```
library(tidysynth)
library(dplyr)
library(tidyverse)
library(ggplot2)
library(tidyr)
data("smoking")
smoking %>% dplyr::glimpse()
```

```
Rows: 1,209
Columns: 7
$ state     <chr> "Rhode Island", "Tennessee", "Indiana", "Nevada", "Louisiana~
$ year      <dbl> 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, ~
$ cigsale   <dbl> 123.9, 99.8, 134.6, 189.5, 115.9, 108.4, 265.7, 93.8, 100.3,~
$ lnincome  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ beer      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ age15to24 <dbl> 0.1831579, 0.1780438, 0.1765159, 0.1615542, 0.1851852, 0.175~
$ retprice  <dbl> 39.3, 39.9, 30.6, 38.9, 34.3, 38.4, 31.4, 37.3, 36.7, 28.8, ~
```

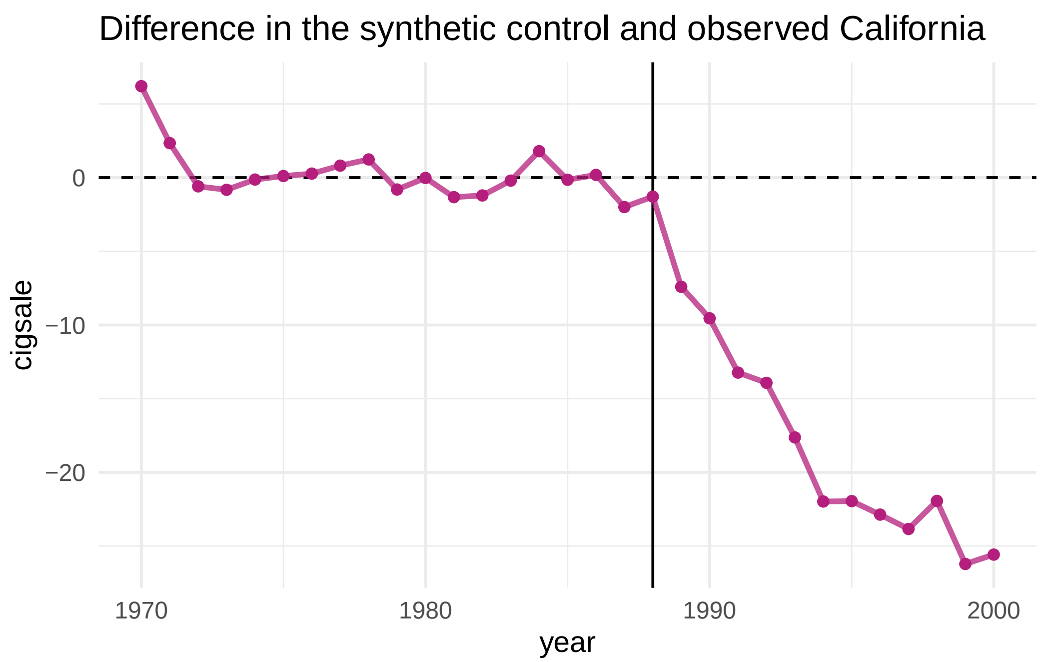
The panel is 1,209 state-years: 39 states from 1970 to 2000, with per-capita cigarette sales in packs as the outcome and log income, beer sales, the share aged 15–24, and retail price as predictors. California passed Proposition 99 in 1988, so there are 19 pre-treatment years to fit the weights and 12 post-treatment years to measure the effect.

```
smoking_out <-
    smoking %>%
    # initial the synthetic control object
    synthetic_control(outcome = cigsale, # outcome
                      unit = state, # unit index in the panel data
                      time = year, # time index in the panel data
                      i_unit = "California", # unit where the intervention occurred
                      i_time = 1988, # time period when the intervention occurred
                      generate_placebos=T # generate placebo synthetic controls (for inference)
                      ) %>%
    # Generate the aggregate predictors used to fit the weights
    # average log income, retail price of cigarettes, and proportion of the
    # population between 15 and 24 years of age from 1980 - 1988
  generate_predictor(time_window = 1980:1988,
                     ln_income = mean(lnincome, na.rm = T),
                     ret_price = mean(retprice, na.rm = T),
```

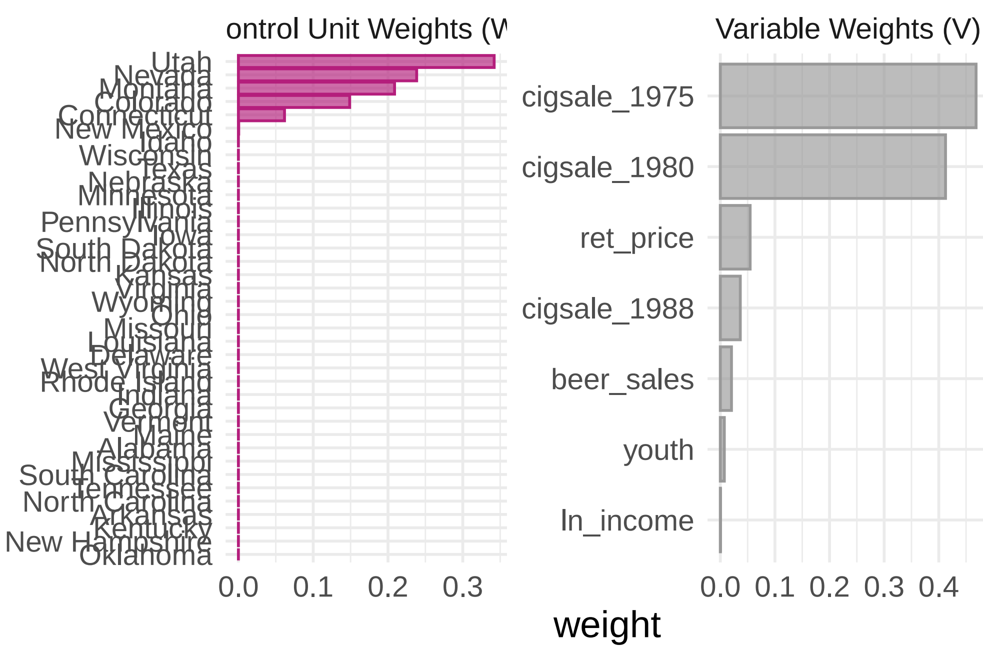
```
                      youth = mean(age15to24, na.rm = T)) %>%
    # average beer consumption in the donor pool from 1984 - 1988
  generate_predictor(time_window = 1984:1988,
                      beer_sales = mean(beer, na.rm = T)) %>%
    # Lagged cigarette sales
  generate_predictor(time_window = 1975,
                      cigsale_1975 = cigsale) %>%
  generate_predictor(time_window = 1980,
                      cigsale_1980 = cigsale) %>%
  generate_predictor(time_window = 1988,
                      cigsale_1988 = cigsale) %>%
    # Generate the fitted weights for the synthetic control
  generate_weights(optimization_window = 1970:1988, # time to use in the optimization task
                    margin_ipop = .02,sigf_ipop = 7,bound_ipop = 6 # optimizer options
  ) %>%
  # Generate the synthetic control
  generate_control()

smoking_out %>% plot_trends()
```




```
smoking_out %>% plot_differences()
```




```
smoking_out %>% plot_weights()
```

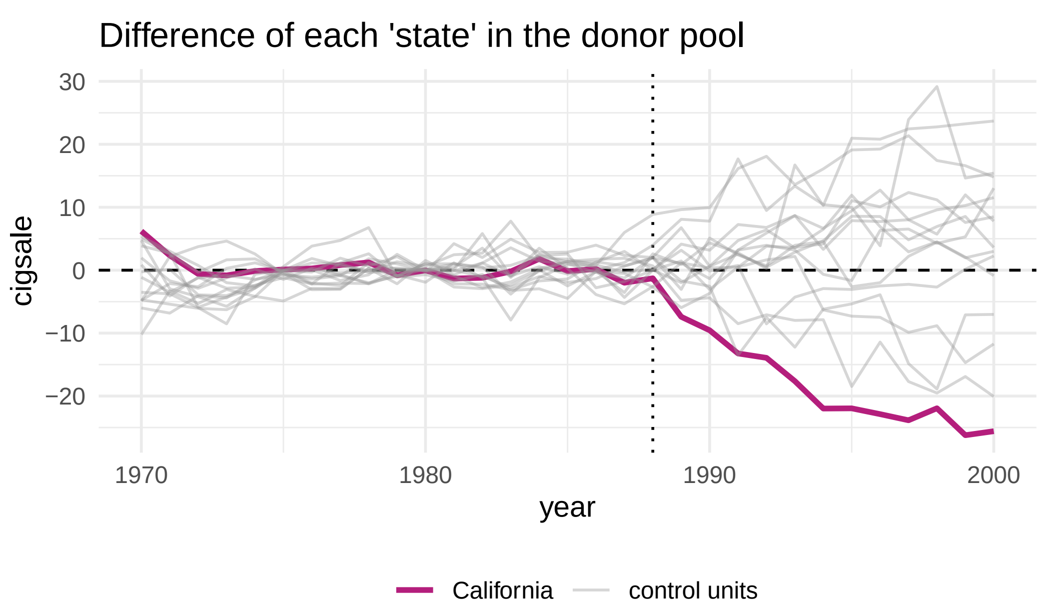



```
smoking_out %>% grab_balance_table()
```

```
# A tibble: 7 x 4
  variable      California synthetic_California donor_sample
  <chr>              <dbl>                <dbl>        <dbl>
```

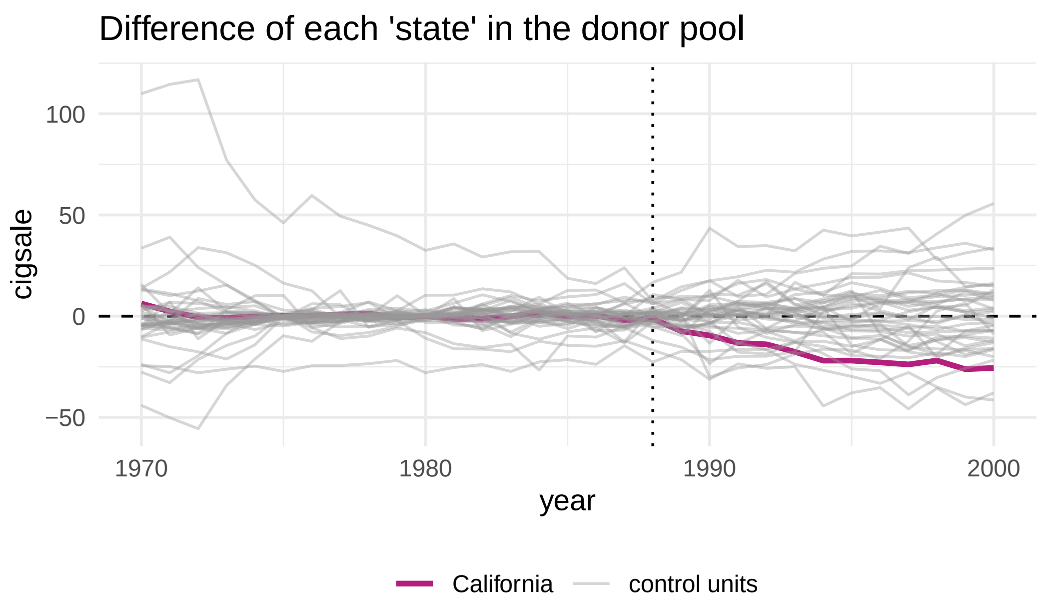
```
1 ln_income        10.1                   9.85          9.83
2 ret_price        89.4                  89.4          87.3
3 youth             0.174                 0.174         0.173
4 beer_sales       24.3                  24.2          23.7
5 cigsale_1975    127.                  127.          137.
6 cigsale_1980    120.                  120.          138.
7 cigsale_1988     90.1                  91.4         114.
```

```r
smoking_out %>% plot_placebos()
```

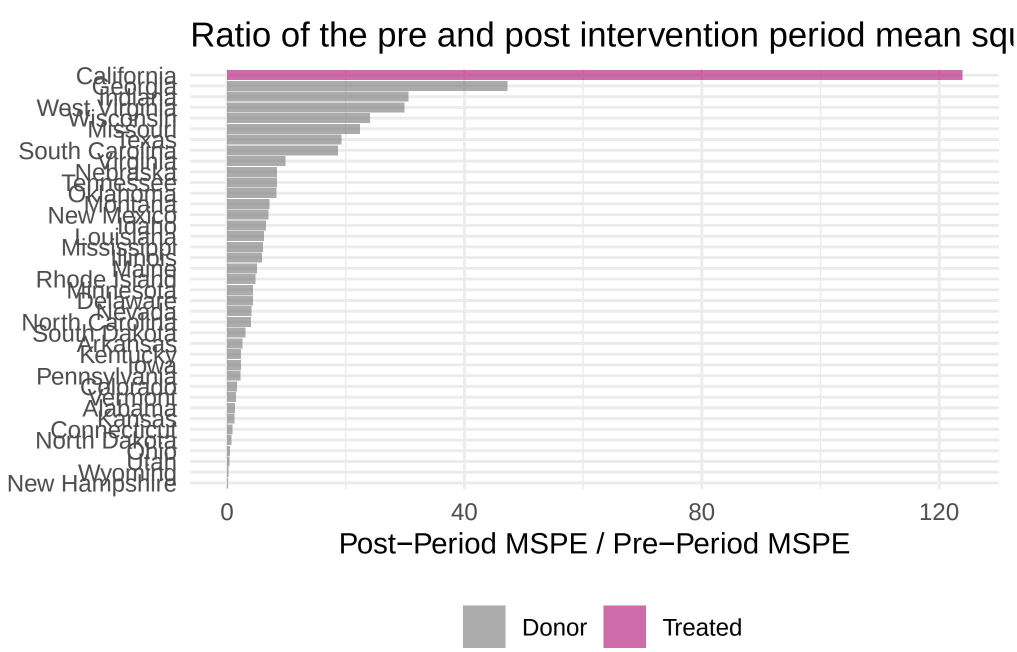


acebo cases with a pre−period RMSPE exceeding two times the treated unit's pre−period RMSPE.

```r
smoking_out %>% plot_placebos(prune = FALSE)
```




```
smoking_out %>% plot_mspe_ratio()
```




```
smoking_out %>% grab_significance()
```

```
# A tibble: 39 x 8
   unit_name      type  pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>          <chr>    <dbl>     <dbl>      <dbl> <int>                <dbl>
```

```
 1 California     Trea~     3.17     392.      124.       1               0.0256
 2 Georgia        Donor     3.79     179.       47.2      2               0.0513
 3 Indiana        Donor    25.2      770.       30.6      3               0.0769
 4 West Virginia  Donor     9.52     284.       29.8      4               0.103
 5 Wisconsin      Donor    11.1      268.       24.1      5               0.128
 6 Missouri       Donor     3.03      67.8      22.4      6               0.154
 7 Texas          Donor    14.4      277.       19.3      7               0.179
 8 South Carolina Donor    12.6      234.       18.6      8               0.205
 9 Virginia       Donor     9.81      96.4       9.83     9               0.231
10 Nebraska       Donor     6.30      52.9       8.40    10               0.256
# i 29 more rows
# i 1 more variable: z_score <dbl>
```

```
smoking_out %>% grab_synthetic_control()
```

```
# A tibble: 31 x 3
   time_unit real_y synth_y
       <dbl>  <dbl>   <dbl>
 1      1970   123     117.
 2      1971   121     119.
 3      1972   124.    124.
 4      1973   124.    125.
 5      1974   127.    127.
 6      1975   127.    127.
 7      1976   128     128.
 8      1977   126.    126.
 9      1978   126.    125.
10      1979   122.    123.
# i 21 more rows
```

```
smoking_out %>% grab_synthetic_control(placebo = T)
```

```
# A tibble: 1,209 x 5
   .id        .placebo time_unit real_y synth_y
   <chr>         <dbl>     <dbl>  <dbl>   <dbl>
 1 California        0      1970   123     117.
 2 California        0      1971   121     119.
 3 California        0      1972   124.    124.
 4 California        0      1973   124.    125.
 5 California        0      1974   127.    127.
 6 California        0      1975   127.    127.
 7 California        0      1976   128     128.
 8 California        0      1977   126.    126.
 9 California        0      1978   126.    125.
10 California        0      1979   122.    123.
# i 1,199 more rows
```

```r
smoking_out %>%
  tidyr::unnest(cols = c(.outcome))
```

```
# A tibble: 1,482 x 50
   .id        .placebo .type   time_unit California Alabama Arkansas Colorado
   <chr>         <dbl> <chr>       <dbl>      <dbl>   <dbl>    <dbl>    <dbl>
 1 California        0 treated      1970       123       NA       NA       NA
 2 California        0 treated      1971       121       NA       NA       NA
 3 California        0 treated      1972       124.      NA       NA       NA
 4 California        0 treated      1973       124.      NA       NA       NA
 5 California        0 treated      1974       127.      NA       NA       NA
 6 California        0 treated      1975       127.      NA       NA       NA
 7 California        0 treated      1976       128       NA       NA       NA
 8 California        0 treated      1977       126.      NA       NA       NA
 9 California        0 treated      1978       126.      NA       NA       NA
10 California        0 treated      1979       122.      NA       NA       NA
# i 1,472 more rows
# i 42 more variables: Connecticut <dbl>, Delaware <dbl>, Georgia <dbl>,
#   Idaho <dbl>, Illinois <dbl>, Indiana <dbl>, Iowa <dbl>, Kansas <dbl>,
#   Kentucky <dbl>, Louisiana <dbl>, Maine <dbl>, Minnesota <dbl>,
#   Mississippi <dbl>, Missouri <dbl>, Montana <dbl>, Nebraska <dbl>,
#   Nevada <dbl>, `New Hampshire` <dbl>, `New Mexico` <dbl>,
#   `North Carolina` <dbl>, `North Dakota` <dbl>, Ohio <dbl>, ...
```

The balance table is where to judge whether the synthetic control is worth trusting, and it does well. Retail price matches to three digits (89.4 against 89.4), the youth share to 0.174 against 0.174, beer sales 24.3 against 24.2. The pre-treatment outcomes match closely too: 1975 cigarette sales 127 against 127 and 1980 sales 120 against 120. Compare those with the raw donor-pool averages in the third column — 137 and 138 — and the value of weighting is obvious. The one predictor that does not match is log income, 10.1 against 9.85, which the optimiser evidently judged less important for predicting the outcome.

The 1988 row is the one to read carefully: 90.1 against 91.4. That is the last pre-treatment year and it is fitted, not predicted, so it is a diagnostic rather than a result.

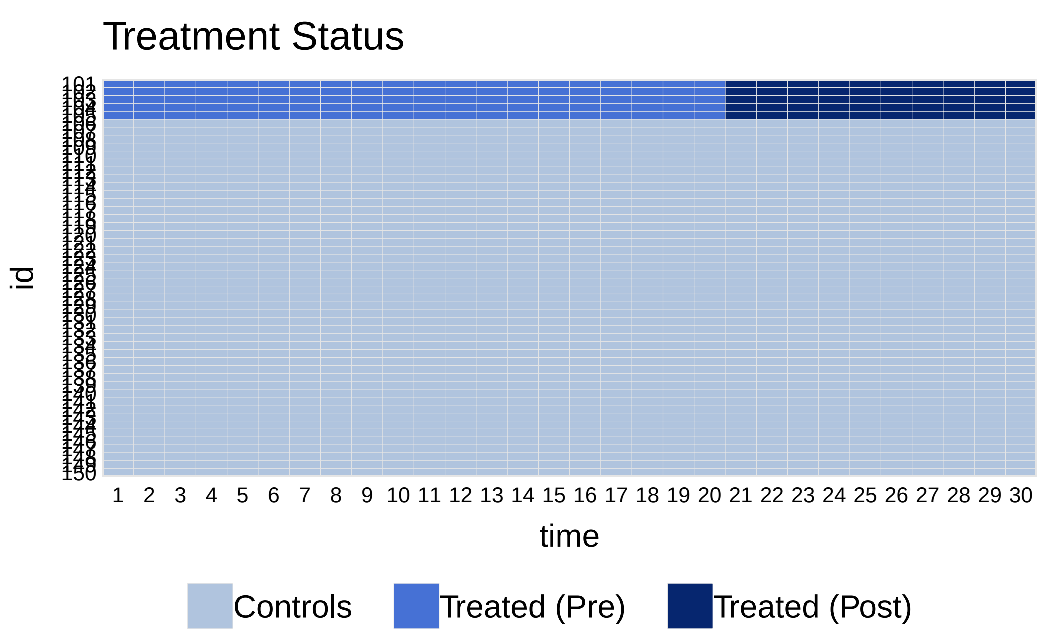
Inference comes from the placebo table. California's ratio of post- to pre-treatment MSPE is 124, the largest of all 39 units — Georgia is next at 47.2, less than half. Being ranked first of 39 gives a Fisher exact p of $1/39 = 0.0256$. This is a permutation statement, not a sampling one: it says no untreated state shows a post-1988 divergence from its own synthetic control as large as California's, relative to how well each was fitted beforehand.

18.4. Generalized Synthetic Control

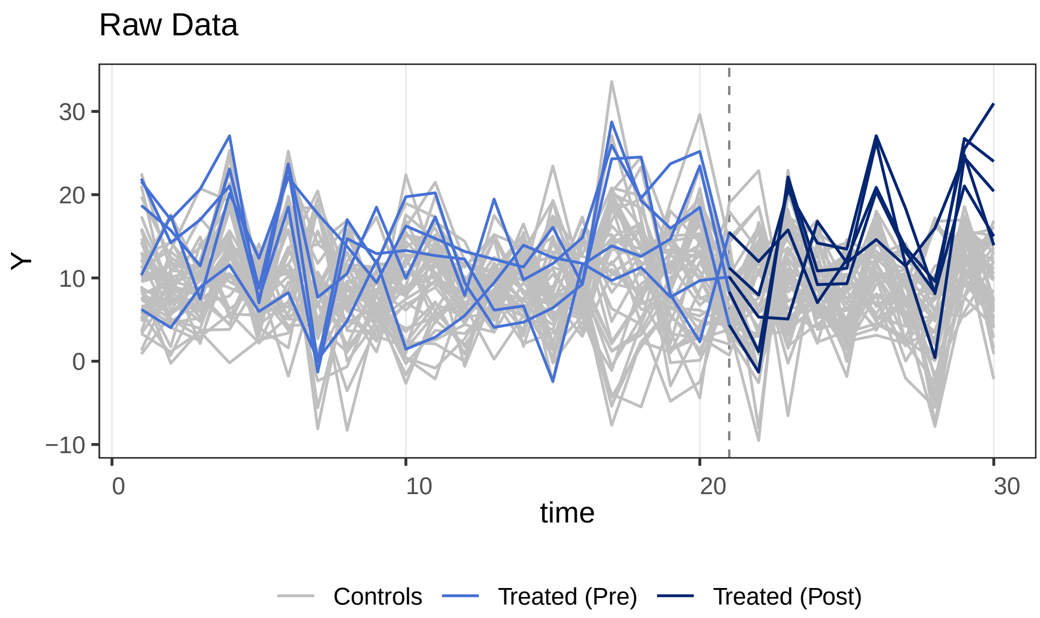
The idea of generalized SC is to combine both an interactive fixed effect model and synthetic control. It is basically a panel data synthetic control. It naturally takes multiple treated units and uses bootstrap to get standard errors. It generates ATT with standard errors.

Examples are here: https://yiqingxu.org/packages/gsynth/

```
## for processing C++ code
require(Rcpp)
## for plotting
require(ggplot2)
require(GGally)
## for parallel computing
require(foreach)
require(future)
require(doParallel)
require(abind)
require(lfe)
library(gsynth)
## ## Syntax has been updated since v.1.2.0.
## ## Comments and suggestions -> yiqingxu@stanford.edu.
data(gsynth)
library(panelView)
## ## See bit.ly/panelview4r for more info.
## ## Report bugs -> yiqingxu@stanford.edu.
panelview(Y ~ D, data = simdata,  index = c("id","time"), pre.post = TRUE)
```




```
panelview(Y ~ D, data = simdata,  index = c("id","time"), type = "outcome")
```




```
system.time(
    out <- gsynth(Y ~ D + X1 + X2, data = simdata, index = c("id","time"), force = "two-way", C
)
```

```
   user  system elapsed 
  1.679   0.000   1.682 
```

```
print(out)
```

```
Call:
gsynth(formula = Y ~ D + X1 + X2, data = simdata, index = c("id", 
    "time"), force = "two-way", r = c(0, 5), CV = TRUE, se = TRUE, 
    nboots = 100, inference = "parametric", parallel = FALSE, 
    vartype = "parametric")

Average Treatment Effect on the Treated:
     ATT.avg   S.E. CI.lower CI.upper p.value
[1,]   5.084 0.3326    4.432    5.736       0

   ~ by Period (including Pre-treatment Periods):
           ATT   S.E. CI.lower CI.upper   p.value count
-19   0.05868 0.6716 -1.25757   1.3749 9.304e-01     5
-18  -0.11741 0.4647 -1.02814   0.7933 8.005e-01     5
-17  -0.89029 0.5334 -1.93573   0.1551 9.510e-02     5
-16   1.29462 0.5791  0.15967   2.4296 2.537e-02     5
-15  -1.20350 0.5627 -2.30636  -0.1006 3.245e-02     5
-14   2.21992 0.7481  0.75369   3.6861 3.003e-03     5
```

```
-13 -0.18469 0.7557 -1.66574   1.2964 8.069e-01     5
-12  0.87413 0.4833 -0.07307   1.8213 7.049e-02     5
-11 -0.04824 0.6238 -1.27081   1.1743 9.384e-01     5
-10  0.84503 0.6792 -0.48623   2.1763 2.135e-01     5
-9   0.49754 0.6345 -0.74604   1.7411 4.329e-01     5
-8  -0.38001 0.5696 -1.49641   0.7364 5.047e-01     5
-7  -1.03493 0.6606 -2.32974   0.2599 1.172e-01     5
-6  -0.81532 0.6910 -2.16971   0.5391 2.381e-01     5
-5  -1.13329 0.6543 -2.41574   0.1492 8.327e-02     5
-4  -0.17733 0.5964 -1.34631   0.9917 7.662e-01     5
-3   0.17671 0.8271 -1.44445   1.7979 8.308e-01     5
-2  -0.14376 0.5958 -1.31145   1.0239 8.093e-01     5
-1  -1.19697 0.4708 -2.11966  -0.2743 1.100e-02     5
0    1.35910 0.7762 -0.16217   2.8804 7.994e-02     5
1    0.34558 0.5608 -0.75364   1.4448 5.378e-01     5
2    0.15709 0.8485 -1.50592   1.8201 8.531e-01     5
3    3.64127 0.5570  2.54966   4.7329 6.241e-11     5
4    2.74975 0.7010  1.37591   4.1236 8.750e-05     5
5    4.45569 0.8610  2.76823   6.1431 2.276e-07     5
6    5.67231 0.8205  4.06411   7.2805 4.745e-12     5
7    8.17039 0.6550  6.88661   9.4542 0.000e+00     5
8    6.57085 0.8954  4.81599   8.3257 2.154e-13     5
9    9.11624 0.5527  8.03299  10.1995 0.000e+00     5
10   9.96205 0.5275  8.92814  10.9960 0.000e+00     5

Coefficients for the Covariates:
    Coef    S.E. CI.lower CI.upper p.value
X1 1.454 0.03993    1.376    1.533       0
X2 3.507 0.04104    3.427    3.588       0
```

```
out$est.att
```

```
            ATT      S.E.    CI.lower   CI.upper      p.value count
-19  0.05868370 0.6715682 -1.25756576  1.3749332 9.303670e-01     5
-18 -0.11741144 0.4646683 -1.02814458  0.7933217 8.005171e-01     5
-17 -0.89029173 0.5333967 -1.93573004  0.1551466 9.509782e-02     5
-16  1.29461792 0.5790639  0.15967354  2.4295623 2.537089e-02     5
-15 -1.20349647 0.5626948 -2.30635803 -0.1006349 3.245118e-02     5
-14  2.21991721 0.7480902  0.75368734  3.6861471 3.002851e-03     5
-13 -0.18468804 0.7556547 -1.66574411  1.2963680 8.069149e-01     5
-12  0.87413414 0.4832772 -0.07307177  1.8213401 7.048776e-02     5
-11 -0.04824208 0.6237714 -1.27081152  1.1743274 9.383536e-01     5
-10  0.84503247 0.6792288 -0.48623146  2.1762964 2.134606e-01     5
-9   0.49754000 0.6344913 -0.74604016  1.7411202 4.329488e-01     5
-8  -0.38001372 0.5696025 -1.49641402  0.7363866 5.046725e-01     5
-7  -1.03492639 0.6606315 -2.32974035  0.2598876 1.172149e-01     5
-6  -0.81531903 0.6910261 -2.16970525  0.5390672 2.380530e-01     5
```

```
-5   -1.13328625 0.6543233 -2.41573628  0.1491638 8.327401e-02     5
-4   -0.17733004 0.5964300 -1.34631145  0.9916514 7.662229e-01     5
-3    0.17670638 0.8271342 -1.44444692  1.7978597 8.308302e-01     5
-2   -0.14376360 0.5957716 -1.31145447  1.0239273 8.093175e-01     5
-1   -1.19696676 0.4707704 -2.11965977 -0.2742737 1.100405e-02     5
0     1.35910371 0.7761745 -0.16217040  2.8803778 7.994100e-02     5
1     0.34558474 0.5608399 -0.75364126  1.4448107 5.377682e-01     5
2     0.15709192 0.8484914 -1.50592063  1.8201045 8.531172e-01     5
3     3.64127124 0.5569529  2.54966368  4.7328788 6.241185e-11     5
4     2.74975180 0.7009533  1.37590853  4.1235951 8.749879e-05     5
5     4.45569195 0.8609635  2.76823443  6.1431495 2.276196e-07     5
6     5.67231032 0.8205269  4.06410723  7.2805134 4.744871e-12     5
7     8.17038502 0.6550005  6.88660758  9.4541625 0.000000e+00     5
8     6.57084917 0.8953552  4.81598521  8.3257131 2.153833e-13     5
9     9.11624492 0.5526911  8.03299024 10.1994996 0.000000e+00     5
10    9.96204576 0.5275143  8.92813677 10.9959548 0.000000e+00     5
```

```r
out$est.avg
```

```
      ATT.avg      S.E. CI.lower CI.upper p.value
[1,] 5.084123 0.3325502 4.432336 5.735909       0
```

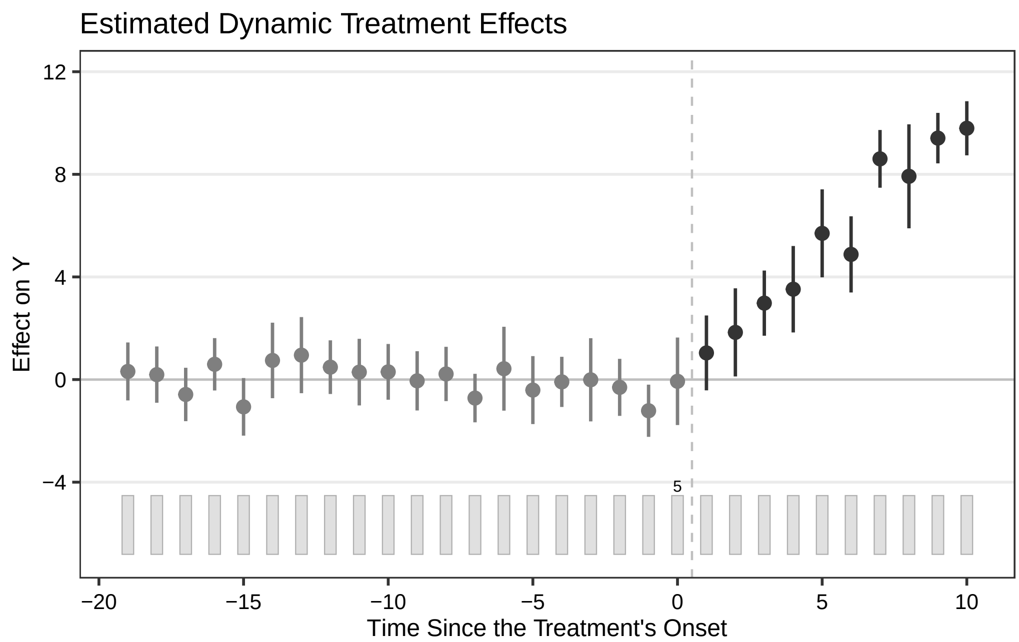
```r
out$est.beta
```

```
       Coef       S.E. CI.lower CI.upper p.value
X1 1.454447 0.03992589 1.376194 1.532701       0
X2 3.507197 0.04104218 3.426756 3.587638       0
```

```r
system.time(
out <- gsynth(Y ~ D + X1 + X2, data = simdata, index = c("id","time"),
    force = "two-way", CV = TRUE, r = c(0, 5), se = TRUE,
    inference = "parametric", nboots = 100, parallel = TRUE, cores = 4)
)
```

```
   user  system elapsed
  7.655   0.344  17.941
```

```r
# cumuEff was removed in newer gsynth versions (now a wrapper for fect)
# cumu1 <- cumuEff(out, cumu = TRUE, id = NULL, period = c(0,5))
# cumu1$est.catt

plot(out)
```



`simdata` is gsynth's own simulated panel: 5 treated units and 45 controls over 30 periods, with treatment starting in period 21 for the treated group, an interactive fixed-effect structure, and two covariates.

The estimated average ATT is 5.084 with a standard error of 0.292 and a 95% interval of $[4.512, 5.656]$. The covariate coefficients come out at 1.454 for `X1` and 3.507 for `X2`. The period-by-period table is the part worth reading, because it includes the pre-treatment periods: those run from -19 to -1 and should be centred on zero if the interactive fixed-effect model has absorbed the common factors. They mostly are — 0.059, -0.117, -0.890, 1.295, -1.203, 2.220 and so on — bouncing around zero with standard errors of about 0.5 to 0.7. Two of them (-16 and -14) are individually significant at 5%, which is roughly what one expects from nineteen independent tests at that level.

Systematic treatment: Julia.

19. Same Data, Different Estimators: A Synthetic Control Comparison

19.1. One panel, four estimators

Here I use the Proposition 99 data to compare four panel estimators on the same outcome matrix: DiD, synthetic DiD, synthetic control, and TASC.

This is the standard California smoking example. There are 39 states, annual cigarette sales per capita from 1970 to 2000, and California is treated after the 1988 anti-smoking law. The useful thing about this example is that the estimators do not give exactly the same answer.

19.2. Four estimators, one panel

The panel is the California Proposition 99 data: annual per-capita cigarette sales in packs for 39 states from 1970 to 2000, arranged as a 39×31 matrix with California in the treated row, 38 donor states, and 19 pre-treatment years before the 1989 tax increase. Every estimator below sees exactly the same matrix; all that differs is how each weights the donors and the pre-periods.

```
using CSV, DataFrames, Statistics
using SynthDiD
using TASC

df    = CSV.read("california_prop99.csv", DataFrame)
# ... pivot to N×T matrix Y, N0 = 38 donors, T0 = 19 pre-treatment years
# (full setup code below)

tau_did  = did_estimate(Y, N0, T0)
tau_sdid = synthdid_estimate(Y, N0, T0)
tau_sc   = sc_estimate(Y, N0, T0)

# TASC: treated unit must be row 1
Y_tasc = vcat(Y[(N0+1):end, :], Y[1:N0, :])
model  = fit_tasc(Y_tasc; d=2, T0=T0, n_em=200, tol=1e-3)
pred   = predict_counterfactual(model, Y_tasc)
```

Estimator	ATT (packs per capita)
Difference-in-differences	−27.35

Estimator	ATT (packs per capita)
Synthetic control	−19.62
Synthetic DiD	−15.60
TASC	−17.11

(These are the values computed live in the companion Julia book's same analysis; the code in this chapter is `eval: false`, so that chapter is the authoritative source for the numbers. The DiD and SC figures also match the published `synthdid` benchmarks for this panel.)

The smallest and largest estimates differ by nearly twelve packs per capita. That is not a rounding issue. It comes from the way each estimator constructs the counterfactual.

19.3. What changes across estimators

DiD (-27.35) assigns equal weight to every control state and every pre-treatment year. California is compared to the average of all 38 donor states. In this data, that average sits well below California before treatment. So the large DiD estimate is not automatically evidence of a larger causal effect. It is partly a warning that equal weights do not fit the pre-period well.

Synthetic control (-19.62) replaces equal weights with donor weights chosen to match California's 1970-1988 cigarette sales. Once the pre-period fit is better, the post-treatment gap shrinks. The move from about -27 to about -20 is mostly the price of using a better control group.

Synthetic DiD (-15.60) adds time weights. The estimator can put more weight on the pre-treatment years that are more useful for the post-1988 counterfactual. Here that moves the estimate a bit closer to zero.

TASC (-17.11) does something different. Instead of choosing only weights, it fits a state-space model to the pre-treatment panel:

$$x_t = A\,x_{t-1} + q_t, \quad q_t \sim \mathcal{N}(0, Q) \tag{19.1}$$

$$y_t = H\,x_t + r_t, \quad r_t \sim \mathcal{N}(0, R) \tag{19.2}$$

The matrix A describes how the latent factors move over time. After treatment, the filter continues using the donor panel, and the treated unit's row of H maps the latent state to California's counterfactual. The TASC estimate (-17.11) lands between SC and SDiD in this example.

19.4. The figure

```
# Counterfactual path for each method. omega_* / intercept_* are the donor
# weights and (for DiD/SDiD) intercepts extracted from the fitted objects
# above (e.g. tau_sdid.weights.omega for SynthDiD.jl) - this snippet is
# schematic and omits that extraction step for brevity; it is not meant to
# run as-is (hence eval: false).
cfact_did  = omega_did'  * Y[1:N0, :] .+ intercept_did   # parallel trends
cfact_sdid = omega_sdid' * Y[1:N0, :] .+ intercept_sdid
cfact_sc   = omega_sc'   * Y[1:N0, :]                     # no intercept
cfact_tasc = vec(pred.target)
tasc_se    = sqrt.(max.(vec(pred.variance), 0.0))
```

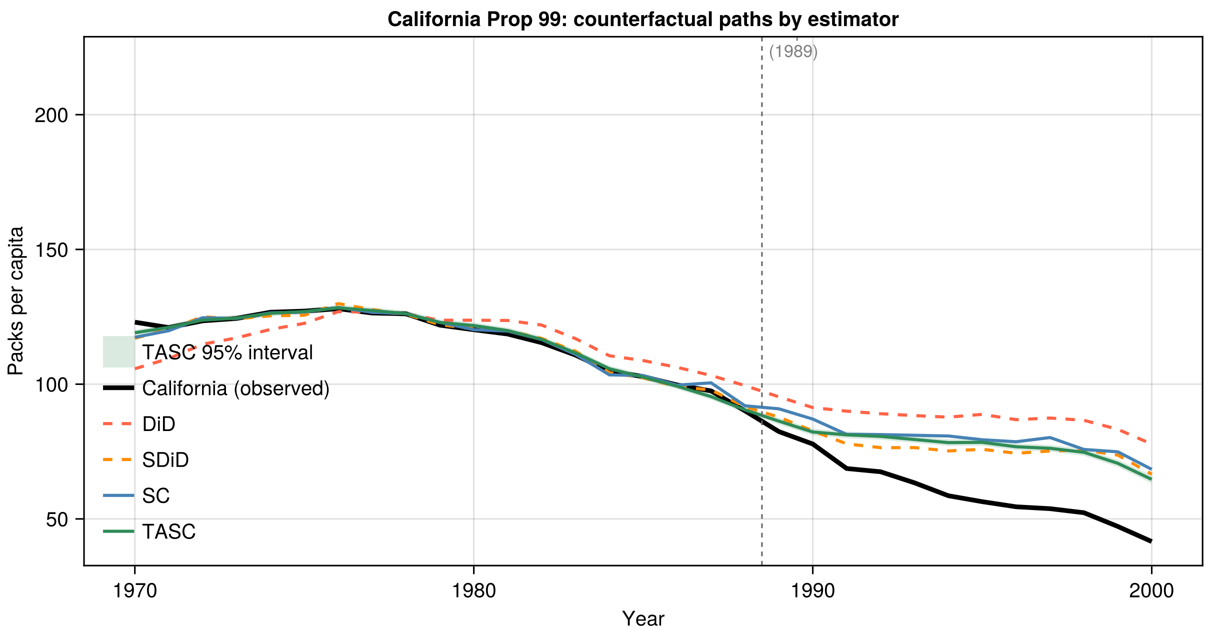


Figure 19.1.: Four counterfactual paths for California. Dashed lines (DiD, SDiD) use pre-period intercepts to enforce parallel trends; solid lines (SC, TASC) fit levels directly. The shaded band is the TASC 95% posterior interval $\widehat{Y}_{1t}(0) \pm 1.96\,\hat{\sigma}_t$.

Two things are visible in the figure that the table cannot show.

First, the pre-treatment paths differ. The DiD control average sits well below California before treatment. SC and TASC track California more closely. This is why the DiD post-treatment gap is larger.

Second, the TASC band is narrow before treatment and wider after 1989. That is what I would expect. Before treatment, the smoother can use information from both directions in time. After treatment, the filter has to move forward without California's observed outcomes, so the uncertainty accumulates through $P_{t+1} = AP_tA^\top + Q$.

19.5. What the comparison teaches

Comparison	What it isolates
DiD vs SC	Unit weights — equal vs optimised donor match
SC vs SDiD	Time weights — equal vs recency-weighted pre-period
SC vs TASC	Model — weight fitting vs latent state-space
SC/SDiD vs TASC	Uncertainty — point estimate vs posterior interval

Which estimator I would use depends on the setting. DiD is transparent, but the pre-period fit here is poor. SC improves the donor match. SDiD also weights time periods. TASC is useful when the outcome series has persistent dynamics and we want uncertainty for the counterfactual path, not just one average number.

For this panel, the three methods that try to match the pre-period give estimates between about -15.6 and -19.6. DiD is the outlier. I would read this as a problem with equal weights, not as evidence that the effect of Proposition 99 depends on the estimator.

19.6. Setup code

```
df     = CSV.read("california_prop99.csv", DataFrame)
states = sort(unique(df.State))
years  = sort(unique(df.Year))
N, T   = length(states), length(years)

Y_wide      = zeros(N, T)
treated_vec = zeros(Bool, N)
for row in eachrow(df)
    i = findfirst(==(row.State), states)
    t = findfirst(==(row.Year), years)
    Y_wide[i, t] = row.PacksPerCapita
    treated_vec[i] |= (row.treated == 1)
end

ctrl_idx = findall(.!treated_vec)
trt_idx  = findall(treated_vec)
Y  = Y_wide[vcat(ctrl_idx, trt_idx), :]   # donors first
N0 = length(ctrl_idx)                      # 38
T0 = findfirst(==(1989), years) - 1        # 19
```

The `SynthDiD.jl` package provides `did_estimate`, `synthdid_estimate`, and `sc_estimate`. The `TASC.jl` package provides `fit_tasc` and `predict_counterfactual`. Both packages are available on GitHub.

Systematic treatment: Julia.

20. Bartik Instrument and Rotemberg weights

20.1. Bartik Instrument

The Bartik instrument, or shift-share IV (Bartik 1991), combines initial local industry shares with national industry growth rates to construct an instrument for local economic conditions.

We follow Goldsmith-Pinkham, Sorkin, and Swift (2021) (GPSS). Consider the labor-supply elasticity:

$$y_{lt} = D_{lt}\rho + \beta_0 x_{lt} + \epsilon_{lt} \tag{20.1}$$

Here y_{lt} is wage growth in location l, x_{lt} is employment growth, and β_0 is the inverse labor-supply elasticity. The challenge is that x_{lt} is endogenous — unobserved factors may drive both wage and employment growth.

Two accounting identities underlie the instrument:

$$x_{lt} = Z_{lt}G_{lt} = \sum_{k=1}^{K} z_{lkt}g_{lkt} \tag{20.2}$$

and

$$g_{lkt} = g_{kt} + \tilde{g}_{lkt} \tag{20.3}$$

With K industries in location l, employment growth decomposes into industry-level growth rates g_{lkt} weighted by industry shares z_{lkt}. The second identity separates each location-industry growth rate into a national component and a location-specific residual. The Bartik instrument uses initial shares and national growth rates:

$$B_{lt} = Z_{l0}G_t = \sum_{k=1}^{K} z_{lk0}g_{kt} \tag{20.4}$$

The identifying argument is that initial shares are predetermined (a level, not a growth rate) and national industry growth is not driven by any single location.

20.2. Bartik Instrument in special cases

20.2.1. Two industries and one period

GPSS goes through the simplest case: two industries and one period.

$$B_l = z_{l1}g_1 + z_{l2}g_2 \tag{20.5}$$

Since there are only two industries, $z_{l1} = 1 - z_{l2}$. Then we can rewrite the Bartik instrument as

$$B_l = g_2 + (g_1 - g_2)z_{l1} \tag{20.6}$$

From this simplest case, we see that the instrument has only z_{l1}, that is related to location. If the industry share can be considered exogenous, then the instrument is valid. We have to argue that the industry share is exogenous. In this case, the instrument is a linear function of the industry share. Therefore, the Bartik instrument is the same as using the industry share as an instrument.

GPSS shows that just-identified two-stage least squares using a Bartik instrument can in fact be written as GMM with an overidentified set of instruments, where the weight matrix is the outer product of national growth rates. That is, Bartik estimator is equivalent to a GMM estimator, with each location share as one instrument, and the weight matrix is the outer product of national growth rates.

With K industries and T time periods, it's basically using $K \times T$ instruments. Still, Bartik instrument is the same as GMM with all these instruments.

20.3. Rotemberg weights

GPSS decompose the Bartik estimator:

$$\beta_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k \tag{20.7}$$

where

$$\hat{\beta}_k = (Z_k' X^{\perp})^{-1} Z_k' Y^{\perp} \tag{20.8}$$

and

$$\hat{\alpha}_k = \frac{g_k Z_k' X^{\perp}}{\sum_{k'} g_{k'} Z_{k'}' X^{\perp}} \tag{20.9}$$

Here $X^{\perp}$ and $Y^{\perp}$ are the residualized X and Y, from the regression of X and Y on all other variables (controls).

Rotemberg weights provide a measure of how important each share is. Read the direction carefully: if industry k's own just-identified estimate $\hat{\beta}_k$ is biased by 1 percent, the overall Bartik estimate inherits α_k percent of that bias. So a large α_k means the overall answer is *sensitive* to that industry, not that the industry is itself badly biased. This is what makes the weights useful: they tell you for which instruments the exogeneity assumption has to be defended hardest. (The formal statement is GPSS Corollary D.1, given below.)

20.4. Simulations

Simulations codes are adopted from Kohei Kawaguchi's lecture notes. I modified slightly to fit different scenarios.

Here we simulate a case with 4 industries and 4 periods. We have 1000 locations, 4 periods, and 4 industries. The true β (effect of x on y) is 1.

```
library(ggplot2)
library(tidyverse)
library(modelsummary)
library(kableExtra)
library(estimatr)
set.seed(6)
L <- 1000
T <- 4
K <- 4
beta <- 1
sigma <-c(1, 1, 1)

z_lk0 <- expand_grid(
    l = 1:L,
    k = 1:K
  ) |>
mutate(z_lk0 = runif(length(l)))

g_kt <-  expand_grid(
    t = 1:T,
    k = 1:K
  ) |>
  mutate(g_kt = rnorm(n(), 0, .1))

df <- expand_grid(
    l = 1:L,
    t = 1:T,
    k = 1:K
  ) |>
  left_join(z_lk0, by = c("l", "k")) |>
  left_join(g_kt,  by = c("k", "t")) |>
  mutate(z_lkt = z_lk0 + sigma[1] * runif(length(l)),
    g_lkt = g_kt + sigma[2] * rnorm(length(l),0, .1)
  )

df <- df %>%
  group_by(t, l) |>
  mutate(
    z_lk0 = z_lk0 / sum(z_lk0),
    z_lkt = (z_lkt / sum(z_lkt)),
```

```
    x_lt = sum(z_lkt * g_lkt)
  )

df
```

```
# A tibble: 16,000 x 8
# Groups:   t, l [4,000]
       l     t     k z_lk0    g_kt z_lkt    g_lkt    x_lt
   <int> <int> <int> <dbl>   <dbl> <dbl>    <dbl>   <dbl>
 1     1     1     1 0.277  0.130   0.252  0.115    0.0400
 2     1     1     2 0.428  0.146   0.302  0.291    0.0400
 3     1     1     3 0.121 -0.165   0.321 -0.231    0.0400
 4     1     1     4 0.174 -0.110   0.126 -0.0185   0.0400
 5     1     2     1 0.277 -0.0289  0.160 -0.00619  0.0402
 6     1     2     2 0.428 -0.00130 0.356  0.0560   0.0402
 7     1     2     3 0.121  0.0433  0.272 -0.0590   0.0402
 8     1     2     4 0.174  0.0840  0.212  0.176    0.0402
 9     1     3     1 0.277 -0.101   0.237 -0.0361  -0.0738
10     1     3     2 0.428 -0.152   0.367 -0.126   -0.0738
# i 15,990 more rows
```

The simulation has $L = 1000$ locations, $T = 4$ periods and $K = 4$ industries, 16,000 location-industry-period rows. Baseline shares z_{lk0} are uniform draws normalised to sum to one within each location, national industry growth is $g_{kt} \sim N(0, 0.1^2)$, current shares add a uniform perturbation and are renormalised, and local industry growth is $g_{lkt} = g_{kt} + N(0, 0.1^2)$. Local employment growth is $x_{lt} = \sum_k z_{lkt} g_{lkt}$. The true β is 1.

This assumes we observe the industry shares at time 0 (z_{lk0}), and the industry growth rates (g_{kt}). We also observe the industry shares at time t (z_{lkt}). The employment growth rate at location l and time t (g_{lkt}) is a function of g_{kt} plus some noise. We can calculate the x_{lt}, which is the weighted sum of industry growth rates.

Next we generate the outcome at l, t level. Outcome (wage growth) is a linear function of x (employment growth) at l, t level.

```
df_lt <- df |>
  mutate(g_tilde_lkt = g_lkt - g_kt) |>
  group_by(l, t) |>
  summarise(across(c(x_lt, g_tilde_lkt), mean), .groups = "drop") |>
  ungroup() |>
  mutate(y_lt = beta * x_lt + sigma[3] * g_tilde_lkt)
```

The outcome is $y_{lt} = \beta x_{lt} + \sigma_3 \tilde{g}_{lt}$, where $\tilde{g}_{lt}$ is the location's mean deviation of local from national industry growth. That deviation also enters x_{lt}, which is the endogeneity: local shocks drive both employment growth and wage growth.

If we run OLS on raw data, we'll get biased results.

	(1)
(Intercept)	0.000
	(0.001)
x_lt	1.336
	(0.008)
Num.Obs.	4000
R2	0.884
R2 Adj.	0.884
AIC	−14 059.9
BIC	−14 041.0
Log.Lik.	7032.945
F	30 603.441
RMSE	0.04

```
result_ols <-
  df_lt |>
  lm(formula = y_lt ~ x_lt)

modelsummary(result_ols)
```

OLS gives 1.336 with a standard error of 0.008, against a true β of 1 — biased upward by a third, and far outside any confidence interval. The endogenous $\tilde{g}_{lt}$ is doing the damage.

Now we generate the Bartik instrument. We calculate the Bartik instrument as the weighted sum of z_{lk0} and g_{kt}.

```
b_lt <- df |>
  group_by(l, t) |>
  summarise(b_lt = sum(z_lk0 * g_kt)) |>
  ungroup()

df_lt <- df_lt |>
  left_join(b_lt, by = c("l", "t"))


result_first_stage <-  lm(formula = x_lt ~ b_lt, data = df_lt)

modelsummary(result_first_stage)
```

```
result_tsls <- df_lt |>
  estimatr::iv_robust(formula = y_lt ~ x_lt | b_lt)
```

	(1)
(Intercept)	−0.001
	(0.001)
b_lt	0.919
	(0.013)
Num.Obs.	4000
R2	0.573
R2 Adj.	0.573
AIC	−11 638.5
BIC	−11 619.6
Log.Lik.	5822.262
F	5361.487
RMSE	0.06

	(1)
(Intercept)	0.000
	(0.001)
x_lt	0.990
	(0.012)
Num.Obs.	4000
R2	0.825
R2 Adj.	0.825
AIC	−12 398.4
BIC	−12 379.5
RMSE	0.05

```
modelsummary::modelsummary(result_tsls)
```

The Bartik instrument recovers the truth: 2SLS gives 0.990 with a standard error of 0.012, against $\beta = 1$, where OLS gave 1.336. The instrument is built from baseline shares and *national* growth rates, neither of which contains the local shock $\tilde{g}_{lt}$, which is what breaks the endogeneity.

20.4.1. Calculate Rotemberg weights

Let's do IV using Bartik instruments. No controls here. We use matrix operation to recover the IV estimates, which is the same as the tsls function in R.

```
B = as.matrix(cbind(1,df_lt$b_lt))
Y = as.matrix(df_lt$y_lt)
X = as.matrix(cbind(1,df_lt$x_lt))

iv = round(solve(t(B)%*%X)%*%t(B)%*%Y,3)

## Label and organize results into a data frame
beta.hat = as.data.frame(cbind(c("Intercept","x"),iv))
names(beta.hat) = c("Coeff.","Est")
beta.hat
```

```
     Coeff.  Est
1 Intercept    0
2         x 0.99
```

Now we calculate the Rotemberg weights.

```
# This code is the R version of Paul Goldsmith-Pinkham's R code for Rotemberg weights on
# github; which is actually a call from a C++ code by jjchern.
# bw.R calls ComputeAlphaBeta.cpp
# dimensions:  G is KTx1, Z is LTxKT, Y is LTx1, X is LTx1
# so B=ZG is LTx1
# Here I learned the dimensions from the ADH example in Paul's github.
# in ADH example, czone(L) is 722, year(T) is 2, ind(K) is 390.
# so G is 780x1, Z is 1444x780, Y is 1444x1, X is 1444x1
y_lt <- df_lt |>
  select(l,t,y_lt)

Y <- as.matrix(y_lt[,3])

x_lt <- df_lt |>
  select(l,t,x_lt)

X <- as.matrix(x_lt[,3])
#X <- as.matrix(cbind(1,x_lt[,3]))

# global is kt level data set
# # g_data <- df |>
# #   group_by(k,t) |>
# #   summarise(g_kt=sum(g_lkt*z_lkt))
#
# G <- as.matrix(g_data[,3])

G <- as.matrix(g_kt[,3])

# generate wide Z matrix, LTxKT from df
```

```
# z_data <- df |>
#   select(l,t,k,z_lkt) |>
#   pivot_wider(names_from = k, values_from = z_lkt) |>
#   select(-l, -t)

# this makes a wide Z data, which is LTxKT, KT variables are generated by the spread function.
# basically expand the shares to KT variables, filled with 0 if there is no data.
# Use z_lk0 (the INITIAL shares), not z_lkt: the Bartik instrument is B = Z_0 G,
# so the Rotemberg decomposition has to be taken with respect to the same Z that
# builds the instrument. Using z_lkt here would decompose a different estimator.
# z_data <- df |>
#   select(l,t,k,z_lkt) |>
#   mutate(k = str_glue("t{t}_sh_ind_{k}") ) |>
#   spread(k, z_lkt, fill = 0)

z_data <- df |>
  select(l,t,k,z_lk0) |>
  mutate(k = str_glue("t{t}_sh_ind_{k}") ) |>
  pivot_wider(names_from = k, values_from = z_lk0, values_fill = 0)

Z <- as.matrix(z_data[,-c(1,2)])

# Residualize Y and X on the controls before forming the weights. There are
# no controls here other than the constant, so the residualization is just
# demeaning; this is what makes the Rotemberg decomposition exact (GPSS).
Yp <- Y - mean(Y)
Xp <- X - mean(X)
beta = round((t(Z) %*% Yp)/(t(Z) %*% Xp), digits=3)
alpha=(diag(as.numeric(G)) %*% t(Z)  %*% Xp) / as.numeric((t(G) %*% t(Z) %*% Xp))

bw <- tibble(g_kt, alpha=as.numeric(alpha), beta=as.numeric(beta)) |>
  mutate(product=alpha*beta)

bw
```

```
# A tibble: 16 x 6
       t     k     g_kt     alpha  beta   product
   <int> <int>    <dbl>     <dbl> <dbl>     <dbl>
 1     1     1  0.130    0.0103   0.741  0.00760
 2     1     2  0.146    0.0188   1.19   0.0223
 3     1     3 -0.165    0.0103   0.771  0.00794
 4     1     4 -0.110    0.00760  1.04   0.00787
 5     2     1 -0.0289  -0.00789  0.886 -0.00699
 6     2     2 -0.00130 -0.000387 0.941 -0.000364
 7     2     3  0.0433   0.0133   0.898  0.0119
 8     2     4  0.0840   0.0280   0.913  0.0256
 9     3     1 -0.101    0.138    0.987  0.136
```

```
10      3      2 -0.152     0.222     0.986  0.219
11      3      3 -0.0569    0.0758    0.984  0.0746
12      3      4 -0.0994    0.136     0.993  0.135
13      4      1  0.0266    0.0268    1.01   0.0272
14      4      2  0.139     0.155     0.984  0.152
15      4      3  0.0263    0.0267    1.01   0.0270
16      4      4  0.127     0.140     1.02   0.143
```

```r
sum(bw$product)
```

```
[1] 0.9894259
```

The sum of the product of α and β should be the same as the Bartik estimator, and it is: 0.9894 against the 2SLS estimate of 0.990. The decomposition is exact, not approximate.

Reading the table, the weights are far from uniform across the sixteen industry-period cells. Period 3 carries most of the weight — its four α values are 0.138, 0.222, 0.076 and 0.136, summing to 0.57 — while period 2 contributes almost nothing and even goes slightly negative ($\alpha = -0.008$ and -0.0004 for its first two industries). The just-identified β_k estimates are all near the true 1, ranging from 0.74 to 1.19, which is what we should see in a simulation with a homogeneous effect.

Note there is no restriction for α, other than the sum is 1. Therefore it can be negative or positive, and individual weights can exceed 1, as long as they sum to 1. In GPSS the Rotemberg weights routinely take negative values and values greater than 1; this is a feature used to flag problematic instruments.

20.5. Why is good to have Rotemberg weights?

GPSS Corollary D.1. shows the interpretation of the Rotemberg weights.

The percentage of bias can be written as:

$$\frac{E(\tilde{\beta})}{\beta_0} = \sum_k \alpha_k \frac{E(\tilde{\beta}_k)}{\beta_0} \tag{20.10}$$

This is to say the α_k is the share of the bias of the estimate of β that comes from the bias of the estimate of β_k. That is, how sensitive it is to the bias from industry k.

In my simulations, all the industries are random, therefore the weights are random too. But in empirical studies, they are not random, most likely dominated by some industries. This is why it is important to have Rotemberg weights. It tells us which industries are important in the estimation of the effect of interest.

20.6. What if you are using a reduced form?

If you use the reduced form, something like:

$$y_{lt} = D_{lt}\rho + \beta_0 B_{lt} + \epsilon_{lt} \tag{20.11}$$

where B_{lt} is the Bartik instrument, constructed in such as a way:

$$B_{lt} = Z_{l0}G_t = \sum_{k=1}^{K} z_{lk0}g_{kt} \tag{20.12}$$

Then I think the Rotemberg weights should be calculated as:

$$\beta_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k \tag{20.13}$$

where

$$\hat{\beta}_k = (Z_k' B^{\perp})^{-1} Z_k' Y^{\perp} \tag{20.14}$$

and

$$\hat{\alpha}_k = \frac{g_k Z_k' B^{\perp}}{\sum_{k'} g_{k'} Z_{k'}' B^{\perp}} \tag{20.15}$$

where $B^{\perp}$ and $Y^{\perp}$ are residualized B and Y.

```
result_reduced <- df_lt |>
  lm(formula = y_lt ~  b_lt)

modelsummary::modelsummary(result_reduced)
```

Now we calculate the Rotemberg weights for the reduced form, for our simulated data.

```
B <- as.matrix(b_lt[,3])

# Residualize Y and B on the controls (here only a constant   demean), as in
# the GPSS reduced-form Rotemberg formula above, so the decomposition is exact.
Yp <- Y - mean(Y)
Bp <- B - mean(B)
beta = (t(Z) %*% Yp)/(t(Z) %*% Bp)
alpha=(diag(as.numeric(G)) %*% t(Z)  %*% Bp) / as.numeric((t(G) %*% t(Z) %*% Bp))

bw <- tibble(g_kt, alpha=as.numeric(alpha), beta=as.numeric(beta)) |>
  mutate(product = alpha * beta)

bw
```

	(1)
(Intercept)	−0.001
	(0.002)
b_lt	0.909
	(0.023)
Num.Obs.	4000
R2	0.278
R2 Adj.	0.278
AIC	−6730.0
BIC	−6711.1
Log.Lik.	3367.980
F	1538.779
RMSE	0.10

```
# A tibble: 16 x 6
       t     k      g_kt     alpha  beta   product
   <int> <int>     <dbl>     <dbl> <dbl>     <dbl>
 1     1     1  0.130     0.0218   0.320  0.00698
 2     1     2  0.146     0.0285   0.720  0.0205
 3     1     3 -0.165     0.0322   0.226  0.00730
 4     1     4 -0.110     0.0141   0.514  0.00723
 5     2     1 -0.0289   -0.00628  1.02  -0.00642
 6     2     2 -0.00130 -0.000344 0.973 -0.000335
 7     2     3  0.0433    0.0132   0.829  0.0110
 8     2     4  0.0840    0.0304   0.773  0.0235
 9     3     1 -0.101     0.128    0.978  0.125
10     3     2 -0.152     0.211    0.954  0.201
11     3     3 -0.0569    0.0687   0.998  0.0685
12     3     4 -0.0994    0.127    0.982  0.124
13     4     1  0.0266    0.0240   1.04   0.0250
14     4     2  0.139     0.152    0.917  0.140
15     4     3  0.0263    0.0238   1.04   0.0248
16     4     4  0.127     0.133    0.989  0.131
```

```r
sum(bw$product)
```

```
[1] 0.9090661
```

Systematic treatment: R · Julia.

The reduced-form decomposition sums to 0.9091 rather than 0.9894, and that is not an error: it decomposes the reduced-form coefficient (the effect of the instrument on the outcome), not the 2SLS coefficient. The two differ by the first stage, which is not exactly 1.

21. Extended TWFE and other DiD estimators

21.1. Staggered DiD without treatment reversal

This blog is adopted from the tutorial of paneltools package. https://yiqingxu.org/tutorials/panel.html#Installing_Packages

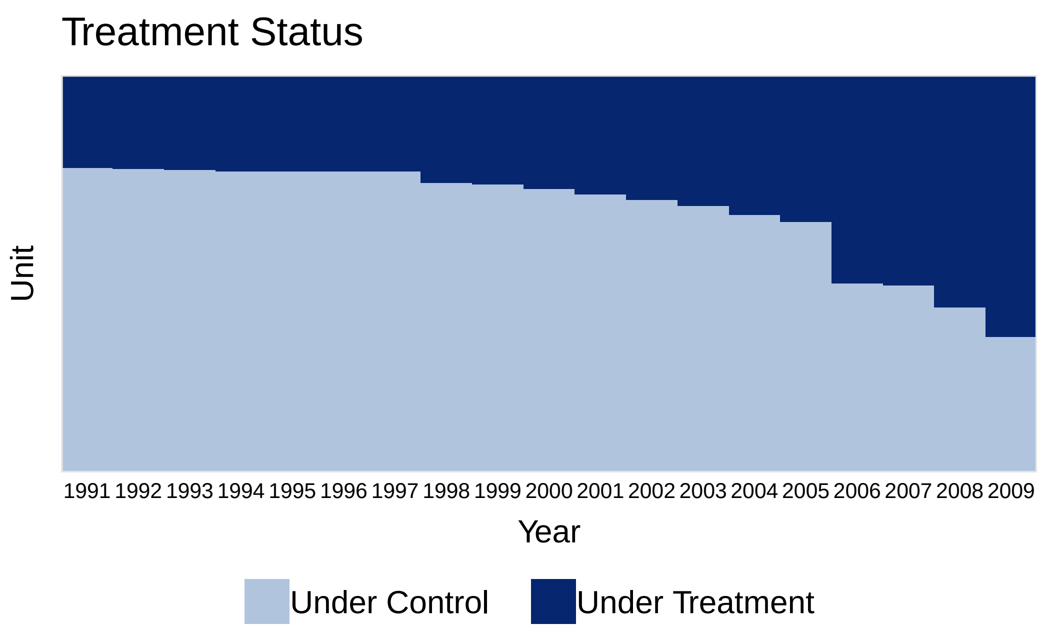
I added Wooldridge's ETWFE, which is my favorite estimator with staggered DiD. It is easy to understand and implement. It also works with nonlinear model such as logit and poisson.

```
#devtools:: install_github("xuyiqing/paneltools")
library(paneltools)
data(paneltools)
library(tidyverse)
hh2019 <- as_tibble(hh2019)
hh2019
```

```
# A tibble: 22,971 x 4
     bfs  year nat_rate_ord indirect
   <dbl> <dbl>        <dbl>    <dbl>
 1     1  1991         0           0
 2     1  1992         0           0
 3     1  1993         0           0
 4     1  1994         3.45        0
 5     1  1995         0           0
 6     1  1996         0           0
 7     1  1997         0           0
 8     1  1998         0           0
 9     1  1999         0           0
10     1  2000         0           0
# i 22,961 more rows
```

This data is from Hainmueller and Hopkins (2019), in which the authors investigate the effects of indirect democracy (versus direct democracy) on naturalization rates in Switzerland using municipality-year level panel data from 1991 to 2009. The study shows that a switch from direct to indirect democracy increased naturalization rates by an average of 1.22 percentage points (Model 1 of Table 1 in the paper).

```r
#devtools:: install_github("xuyiqing/panelView")
library(panelView)
panelview(nat_rate_ord ~ indirect, data = hh2019, index = c("bfs","year"),
  xlab = "Year", ylab = "Unit", display.all = T,
  gridOff = TRUE, by.timing = TRUE)
```



The panel is 22,971 municipality-years: 1,209 Swiss municipalities observed from 1991 to 2009. The outcome `nat_rate_ord` is the ordinary naturalization rate and `indirect` marks the switch from direct to indirect democracy, adopted at different dates by different municipalities — staggered treatment timing.

21.1.1. TWFE

First we do a TWFE model on it.

```r
library(fixest)
model_twfe <- feols(nat_rate_ord~indirect|bfs+year,
                    data=hh2019, cluster = "bfs")
summary( model_twfe)
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 22,971
Fixed-effects: bfs: 1,209,  year: 19
Standard-errors: Clustered (bfs)
         Estimate Std. Error t value   Pr(>|t|)
indirect  1.33932   0.186525 7.18039 1.2117e-12 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.09541     Adj. R2: 0.152719
                Within R2: 0.005173
```

TWFE gives 1.33932 with a municipality-clustered standard error of 0.1865, close to the 1.22 the original authors report. Keep that number in view: everything below is an attempt to work out whether it can be trusted.

TWFE has issues when there is staggered adoption and heterogeneous treatment effects.

Goodman-Bacon (2021) demonstrates that the two-way fixed-effects (TWFE) estimator in a staggered adoption setting can be represented as a weighted average of all possible 2x2 difference-in-differences (DID) estimates between different cohorts. However, when treatment effects change over time heterogeneously across cohorts, the "forbidden" comparisons that use post-treatment data from early adopters as controls for late adopters may introduce bias in the TWFE estimator. We employ the procedure outlined in Goodman-Bacon (2021) and decompose the TWFE estimate using the command bacon in the bacondecomp package.

"Negative weights" can be the other issue. Basically it is a weighted average of comparison between different cohorts. The weights can be negative in a TWFE model.

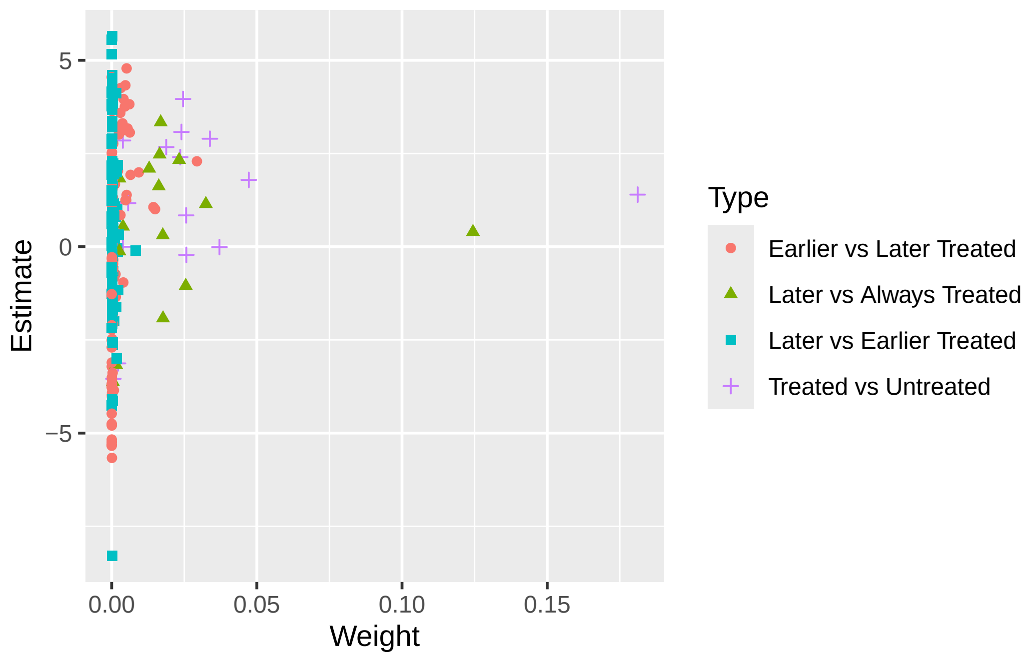
```r
library(bacondecomp)
df_bacon <- bacon(nat_rate_ord~indirect,
                  data = hh2019,
                  id_var = "bfs",
                  time_var = "year")
```

```
                      type  weight avg_est
1 Earlier vs Later Treated 0.17605 1.97771
2  Later vs Always Treated 0.31446 0.75233
3 Later vs Earlier Treated 0.05170 0.32310
4     Treated vs Untreated 0.45779 1.61180
```

```r
coef_bacon <- sum(df_bacon$estimate * df_bacon$weight)
coef_bacon
```

```
[1] 1.339325
```

```r
ggplot(df_bacon) +
   aes(x = weight, y = estimate, shape = factor(type), color = factor(type)) +
   labs(x = "Weight", y = "Estimate", shape = "Type", color = 'Type') +
   geom_point()
```



The decomposition sums to 1.339325, reproducing the TWFE estimate exactly. It splits into four kinds of 2x2 comparison:

comparison	weight	average estimate
Treated vs Untreated	0.458	1.612
Later vs Always Treated	0.314	0.752
Earlier vs Later Treated	0.176	1.978
Later vs Earlier Treated	0.052	0.323

The clean comparisons — treated against never-treated — carry 46% of the weight and give 1.61. The forbidden ones, which use already-treated units as controls, carry $0.314 + 0.052 = 37\%$ and give much smaller numbers, 0.75 and 0.32. That is what pulls the TWFE estimate down from 1.61 to 1.34.

The problem shown in Bacon decomposition here is the comparison between “Later” and “Earlier Treated”. It’s a “forbidden” comparison.

TWFE with event history type of graph.

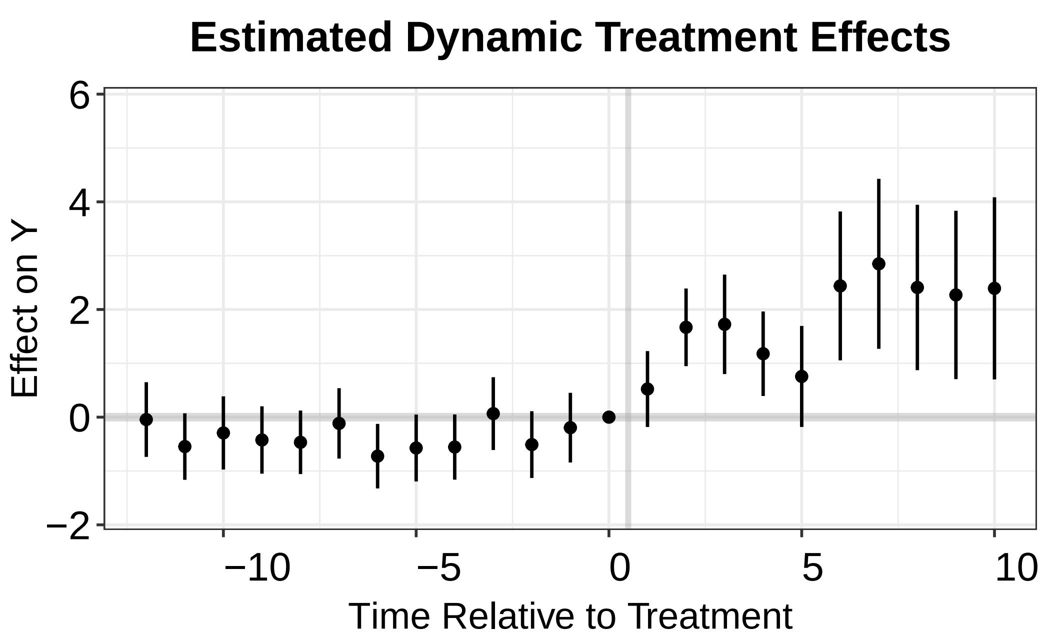
```
df.use <- hh2019 |>
  na.omit()

df <- as.data.frame(
  hh2019 |>
    group_by(bfs) |>
    mutate(treatment_mean = mean(indirect,na.rm = TRUE))
)
df.use <- df[which(df$treatment_mean<1),]
```

```r
df.twfe <- as_tibble(get.cohort(df.use,D="indirect",index=c("bfs","year"),start0 = TRUE))

# time_to_treatment is NA when the unit is always treated, or never treated.  It needs to be se
df.twfe$treat <- as.numeric(df.twfe$treatment_mean>0) # drop always treated units
df.twfe[which(is.na(df.twfe$Time_to_Treatment)),'Time_to_Treatment'] <- 0 # can be an arbitrary
twfe.est <- feols(nat_rate_ord ~ i(Time_to_Treatment, treat, ref = -1)| bfs + year,
                  data = df.twfe, cluster = "bfs")
summary(twfe.est)
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: bfs: 926,  year: 19
Standard-errors: Clustered (bfs)
                              Estimate Std. Error   t value   Pr(>|t|)
Time_to_Treatment::-18:treat -0.320651   0.632702 -0.506796 6.1242e-01
Time_to_Treatment::-17:treat -0.118540   0.391148 -0.303058 7.6191e-01
Time_to_Treatment::-16:treat -0.395632   0.402921 -0.981910 3.2640e-01
Time_to_Treatment::-15:treat -0.410976   0.343899 -1.195048 2.3237e-01
Time_to_Treatment::-14:treat  0.244512   0.369127  0.662406 5.0788e-01
Time_to_Treatment::-13:treat -0.044310   0.354017 -0.125164 9.0042e-01
Time_to_Treatment::-12:treat -0.545891   0.315172 -1.732042 8.3600e-02 .
Time_to_Treatment::-11:treat -0.293138   0.346835 -0.845180 3.9823e-01
Time_to_Treatment::-10:treat -0.423981   0.319375 -1.327534 1.8466e-01
Time_to_Treatment::-9:treat  -0.467114   0.301017 -1.551784 1.2106e-01
Time_to_Treatment::-8:treat  -0.115633   0.333659 -0.346561 7.2900e-01
Time_to_Treatment::-7:treat  -0.724290   0.305727 -2.369075 1.8037e-02 *
Time_to_Treatment::-6:treat  -0.572950   0.317153 -1.806540 7.1159e-02 .
Time_to_Treatment::-5:treat  -0.554993   0.309246 -1.794667 7.3033e-02 .
Time_to_Treatment::-4:treat   0.066209   0.345103  0.191853 8.4790e-01
Time_to_Treatment::-3:treat  -0.509530   0.316296 -1.610929 1.0754e-01
Time_to_Treatment::-2:treat  -0.195091   0.330220 -0.590793 5.5480e-01
Time_to_Treatment::0:treat    0.522715   0.359646  1.453415 1.4645e-01
Time_to_Treatment::1:treat    1.668834   0.367919  4.535879 6.4907e-06 ***
Time_to_Treatment::2:treat    1.724230   0.471363  3.657968 2.6858e-04 ***
Time_to_Treatment::3:treat    1.178336   0.400559  2.941732 3.3452e-03 **
Time_to_Treatment::4:treat    0.756323   0.478697  1.579962 1.1446e-01
Time_to_Treatment::5:treat    2.438229   0.705451  3.456269 5.7263e-04 ***
Time_to_Treatment::6:treat    2.848590   0.805315  3.537238 4.2442e-04 ***
Time_to_Treatment::7:treat    2.409359   0.783716  3.074275 2.1721e-03 **
Time_to_Treatment::8:treat    2.270428   0.797869  2.845617 4.5305e-03 **
Time_to_Treatment::9:treat    2.392678   0.863222  2.771799 5.6867e-03 **
Time_to_Treatment::10:treat   2.002836   0.953654  2.100170 3.5984e-02 *
Time_to_Treatment::11:treat   0.842590   0.560248  1.503961 1.3293e-01
Time_to_Treatment::12:treat   3.831918   2.300422  1.665745 9.6103e-02 .
Time_to_Treatment::13:treat   4.436437   2.322318  1.910349 5.6397e-02 .
Time_to_Treatment::14:treat   4.756977   3.485523  1.364781 1.7265e-01
```

```
Time_to_Treatment::15:treat   1.541967   1.415284  1.089511 2.7621e-01
Time_to_Treatment::16:treat   7.168943   4.844016  1.479959 1.3922e-01
Time_to_Treatment::17:treat   6.490854   2.726526  2.380632 1.7485e-02 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.42406     Adj. R2: 0.146601
                Within R2: 0.012807
```

```
twfe.output <- as.matrix(twfe.est$coeftable)
twfe.output <- as.data.frame(twfe.output)
twfe.output$Time <- c(c(-18:-2),c(0:17))+1
p.twfe <- esplot(twfe.output,Period = 'Time',Estimate = 'Estimate',
                               SE = 'Std. Error', xlim = c(-12,10))
p.twfe
```



21.1.2. Sun and Abraham

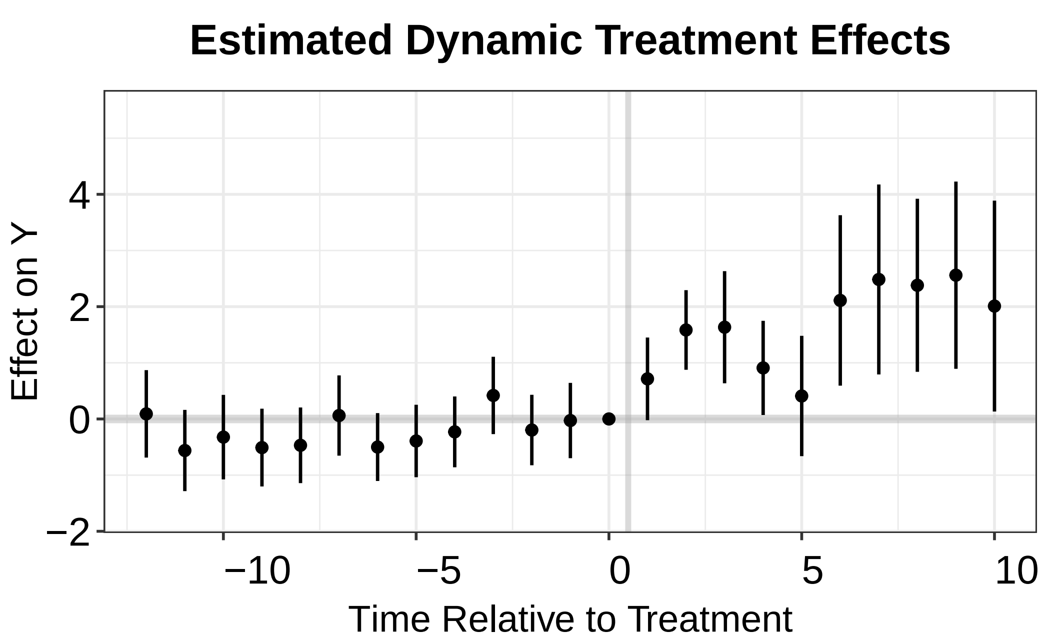
Sun and Abraham (2021) estimator is a weighted average of the average treatment effect (ATT) estimates for each cohort, obtained from a TWFE regression that includes cohort dummies fully interacted with indicators of relative time to the treatment's onset. The iw estimator is robust to heterogeneous treatment effects (HTE) and can be implemented using the command sunab in the package fixest.

```
df.sa <- df.twfe
df.sa[which(is.na(df.sa$FirstTreat)),"FirstTreat"] <- 1000 # replace NA with an arbitrary numbe
model.sa.1 <- feols(nat_rate_ord~sunab(FirstTreat,year)|bfs+year,
```

```
                      data = df.sa, cluster = "bfs")
summary(model.sa.1,agg = "ATT")
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: bfs: 926,  year: 19
Standard-errors: Clustered (bfs)
    Estimate Std. Error t value   Pr(>|t|)
ATT   1.3309   0.287971 4.62163 4.3474e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.34795     Adj. R2: 0.163888
                Within R2: 0.046484
```

```
sa.output <- as.matrix(model.sa.1$coeftable)
sa.output <- as.data.frame(sa.output)
sa.output$Time <- c(c(-18:-2),c(0:17))+1
p.sa <- esplot(sa.output,Period = 'Time',Estimate = 'Estimate',
                             SE = 'Std. Error', xlim = c(-12,10))
p.sa
```



21.2. Callaway and Sant'Anna

Callaway and Sant'Anna (2021) build the estimator from group-time average treatment effects $ATT(g,t)$ — the effect on cohort g at time t — each identified by comparing the treated cohort to a never-treated or not-yet-treated comparison group. The $ATT(g,t)$ are estimated by outcome regression, inverse-probability weighting, or a doubly-robust combination of the two, rather than from a single fully-interacted TWFE regression. These group-time effects are then aggregated with weights into summary parameters (overall, event-study, or by cohort). The estimator is robust to heterogeneous treatment effects (HTE) and can be implemented with the `att_gt` function in the `did` package.

```
library(did)
df.cs <- df.twfe
df.cs[which(is.na(df.cs$FirstTreat)),"FirstTreat"] <- 0 # replace NA with 0
cs.est.1 <- att_gt(yname = "nat_rate_ord",
                  gname = "FirstTreat",
                  idname = "bfs",
                  tname = "year",
                  xformla = ~1,
                  control_group = "nevertreated",
                  allow_unbalanced_panel = TRUE,
                  data = df.cs,
                  est_method = "reg")
cs.est.att.1 <- aggte(cs.est.1, type = "simple", na.rm=T, bstrap = F)
print(cs.est.att.1)
```

```
Call:
aggte(MP = cs.est.1, type = "simple", na.rm = T, bstrap = F)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multipl
230, 2021. <https://doi.org/10.1016/j.jeconom.2020.12.001>, <https://arxiv.org/abs/1803.09015>

    ATT    Std. Error     [ 95%  Conf. Int.]
 1.3309        0.3019     0.7392      1.9226 *


---
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```

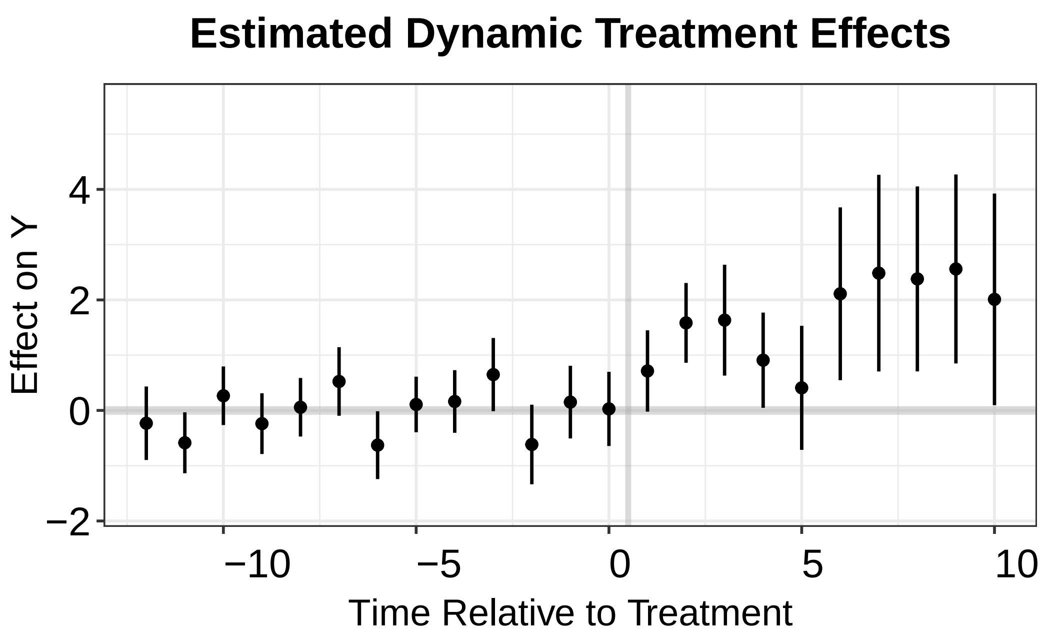
```
cs.att.1 <- aggte(cs.est.1, type = "dynamic",
                  bstrap=FALSE, cband=FALSE, na.rm=T)
print(cs.att.1)
```

```
Call:
aggte(MP = cs.est.1, type = "dynamic", na.rm = T, bstrap = FALSE,
```

```
    cband = FALSE)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multipl
230, 2021. <https://doi.org/10.1016/j.jeconom.2020.12.001>, <https://arxiv.org/abs/1803.09015>


Overall summary of ATT's based on event-study/dynamic aggregation:
    ATT    Std. Error     [ 95%  Conf. Int.]
 1.2205        0.6314    -0.0171       2.458


Dynamic Effects:
 Event time Estimate Std. Error [95% Pointwise  Conf. Band]
        -17   0.1350     0.6467        -1.1324      1.4025
        -16  -0.0839     0.3583        -0.7861      0.6183
        -15   0.1467     0.3964        -0.6303      0.9236
        -14   0.7509     0.3219         0.1200      1.3818 *
        -13  -0.2318     0.3392        -0.8967      0.4331
        -12  -0.5851     0.2814        -1.1367     -0.0335 *
        -11   0.2656     0.2705        -0.2647      0.7958
        -10  -0.2386     0.2805        -0.7883      0.3111
         -9   0.0576     0.2702        -0.4721      0.5872
         -8   0.5244     0.3165        -0.0959      1.1448
         -7  -0.6282     0.3132        -1.2420     -0.0144 *
         -6   0.1081     0.2565        -0.3946      0.6108
         -5   0.1623     0.2890        -0.4041      0.7287
         -4   0.6485     0.3380        -0.0139      1.3109
         -3  -0.6161     0.3674        -1.3362      0.1040
         -2   0.1505     0.3355        -0.5071      0.8081
         -1   0.0279     0.3425        -0.6435      0.6992
          0   0.7141     0.3754        -0.0217      1.4500
          1   1.5846     0.3681         0.8632      2.3060 *
          2   1.6337     0.5118         0.6306      2.6369 *
          3   0.9090     0.4396         0.0473      1.7707 *
          4   0.4091     0.5725        -0.7130      1.5311
          5   2.1106     0.7976         0.5473      3.6739 *
          6   2.4842     0.9078         0.7051      4.2634 *
          7   2.3802     0.8538         0.7068      4.0535 *
          8   2.5602     0.8723         0.8506      4.2699 *
          9   2.0100     0.9769         0.0953      3.9246 *
         10   1.6391     1.0257        -0.3712      3.6494
         11   0.0298     0.6100        -1.1658      1.2253
         12   0.6942     2.8043        -4.8021      6.1904
         13   1.1355     2.3284        -3.4281      5.6991
         14   1.3625     3.4131        -5.3270      8.0519
         15  -2.0699     1.7160        -5.4331      1.2933
         16   2.2743     3.6182        -4.8172      9.3658
         17   0.1072     4.5338        -8.7789      8.9934
```

```
---
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```

```
cs.output <- cbind.data.frame(Estimate = cs.att.1$att.egt,
                              SE = cs.att.1$se.egt,
                              time = cs.att.1$egt + 1)
p.cs.1 <- esplot(cs.output,Period = 'time',Estimate = 'Estimate',
                               SE = 'SE', xlim = c(-12,10))
p.cs.1
```



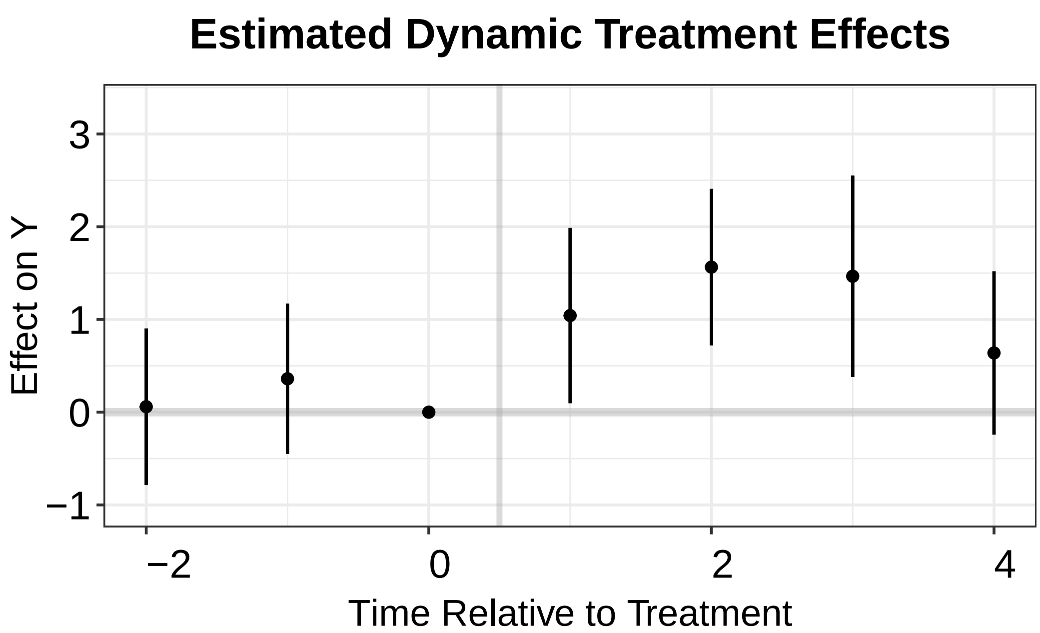
21.2.1. Imai, Kim, and Wang

PanelMatch is an R package implementing a set of methodological tools proposed by Imai, Kim, and Wang (2021) that enables researchers to apply matching methods for causal inference on time-series cross-sectional data with binary treatments. The package includes implementations of matching methods based on propensity scores and Mahalanobis distance, as well as weighting methods. PanelMatch enables users to easily calculate a variety of possible quantities of interest, along with standard errors. The software is flexible, allowing users to tune the matching, refinement, and estimation procedures with a large number of parameters. The package also offers a variety of visualization and diagnostic tools for researchers to better understand their data and assess their results.

```r
library(PanelMatch)
df.pm <- as.data.frame(df.twfe)
# we need to convert the unit and time indicator to integer
df.pm[,"bfs"] <- as.integer(as.factor(df.pm[,"bfs"]))
df.pm[,"year"] <- as.integer(as.factor(df.pm[,"year"]))
df.pm <- df.pm[,c("bfs","year","nat_rate_ord","indirect")]

# Create PanelData object (new API)
pd <- PanelData(df.pm, unit.id = "bfs", time.id = "year",
                treatment = "indirect", outcome = "nat_rate_ord")

PM.results <- PanelMatch(lag=3,
                         panel.data = pd,
                         refinement.method = "none",
                         qoi = "att",
                         lead = c(0:3),
                         match.missing = TRUE)

## For pre-treatment dynamic effects
PM.results.placebo <- PanelMatch(lag=3,
                         panel.data = pd,
                         refinement.method = "none",
                         qoi = "att",
                         lead = c(0:3),
                         match.missing = TRUE,
                         placebo.test = TRUE)
PE.results.pool <- PanelEstimate(PM.results, panel.data = pd, pooled = TRUE)
summary(PE.results.pool)
```

```
     estimate std.error      2.5%    97.5%
[1,] 1.177674 0.3492237 0.4959534 1.872191
```

```r
PE.results <- PanelEstimate(PM.results, panel.data = pd)
PE.results.placebo <- placebo_test(PM.results.placebo, panel.data = pd, plot = F)
est_lead <- as.vector(estimates(PE.results))
est_lag <- tryCatch(as.vector(estimates(PE.results.placebo)),
                    error = function(e) as.vector(PE.results.placebo$estimates))
sd_lead <- tryCatch(apply(PE.results$bootstrapped.estimates,2,sd),
                    error = function(e) rep(NA, length(est_lead)))
sd_lag <- tryCatch(apply(PE.results.placebo$bootstrapped.estimates,2,sd),
                   error = function(e) rep(NA, length(est_lag)))
coef <- c(est_lag, 0, est_lead)
sd <- c(sd_lag, 0, sd_lead)
pm.output <- cbind.data.frame(ATT=coef, se=sd, t=c(-2:4))
p.pm <- esplot(data = pm.output,Period = 't',
               Estimate = 'ATT',SE = 'se')
p.pm
```



21.2.2. Liu, Wang and Xu

Liu, Wang, and Xu (2022) proposes a method for estimating the fixed effect counterfactual model by utilizing the untreated observations. It is essentially an imputation method that uses the untreated observations to estimate the counterfactual outcome for the treated.

```
library(fect)
out.fect <- fect(nat_rate_ord~indirect, data = df,
                 index = c("bfs","year"),
                 method = 'fe', se = TRUE)
print(out.fect$est.avg)
```

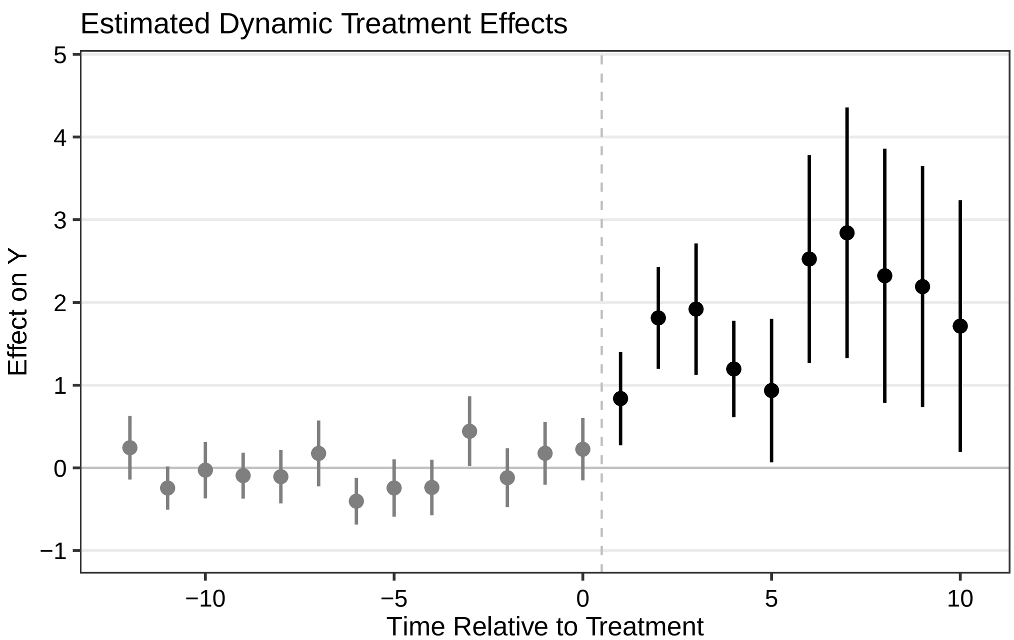
```
      ATT.avg      S.E. CI.lower CI.upper      p.value
[1,] 1.506416 0.2033546 1.107849 1.904984 1.283418e-13
```

```
fect.output <- as.matrix(out.fect$est.att)
print(fect.output)
```

```
            ATT       S.E.     CI.lower    CI.upper      p.value count
-17 -0.01985591 0.5249937 -1.04882465  1.00911284 9.698302e-01    89
-16  0.07269638 0.2600444 -0.43698129  0.58237405 7.798199e-01   157
-15 -0.21365489 0.2524156 -0.70838039  0.28107061 3.973068e-01   162
-14 -0.13153052 0.1533090 -0.43201065  0.16894962 3.909235e-01   350
-13  0.46902947 0.2244936  0.02903007  0.90902887 3.668274e-02   371
```

```
-12  0.24357664 0.1960828 -0.14073858  0.62789186 2.141579e-01   398
-11 -0.24399485 0.1332452 -0.50515055  0.01716086 6.707482e-02   417
-10 -0.02765866 0.1742036 -0.36909137  0.31377405 8.738485e-01   434
-9  -0.09347047 0.1417648 -0.37132431  0.18438337 5.096807e-01   451
-8  -0.10638473 0.1648035 -0.42939365  0.21662420 5.185872e-01   464
-7   0.17476094 0.2031886 -0.22348141  0.57300329 3.897382e-01   468
-6  -0.40297604 0.1439751 -0.68516202 -0.12079006 5.127241e-03   503
-5  -0.24326436 0.1767746 -0.58973630  0.10320759 1.687823e-01   503
-4  -0.23687833 0.1716243 -0.57325570  0.09949904 1.675205e-01   503
-3   0.44244127 0.2155461  0.01997871  0.86490383 4.010627e-02   503
-2  -0.11927690 0.1817101 -0.47542211  0.23686831 5.115583e-01   508
-1   0.17612194 0.1932931 -0.20272562  0.55496950 3.622084e-01   512
0    0.22491529 0.1918671 -0.15113738  0.60096795 2.410987e-01   514
1    0.83813813 0.2884721  0.27274312  1.40353314 3.667387e-03   514
2    1.81274486 0.3130701  1.19913872  2.42635100 7.029422e-09   425
3    1.91941052 0.4049619  1.12569975  2.71312129 2.140022e-06   357
4    1.19567464 0.2979622  0.61167949  1.77966979 5.999252e-05   352
5    0.93520961 0.4429749  0.06699484  1.80342438 3.475491e-02   164
6    2.52517848 0.6411037  1.26863840  3.78171855 8.189048e-05   143
7    2.84129736 0.7735045  1.32525638  4.35733835 2.394584e-04   116
8    2.32288547 0.7838386  0.78658995  3.85918100 3.041877e-03    97
9    2.19105962 0.7439582  0.73292831  3.64919092 3.228106e-03    80
10   1.71382906 0.7761588  0.19258578  3.23507233 2.723795e-02    63
11   1.07651548 1.0454215 -0.97247296  3.12550392 3.031306e-01    50
12  -0.30828629 0.6163128 -1.51623712  0.89966453 6.169267e-01    46
13   0.35079211 2.0582844 -3.68337111  4.38495534 8.646725e-01    11
14   0.81696815 2.5111791 -4.10485243  5.73878873 7.449294e-01    11
15   0.97427612 3.8070109 -6.48732812  8.43588037 7.980155e-01    11
16  -2.49173503 1.7363824 -5.89498197  0.91151190 1.512828e-01    11
17   1.42334105 5.3589459 -9.07999999 11.92668210 7.905466e-01     6
18  -0.01384226 4.7462662 -9.31635305  9.28866853 9.976730e-01     2
```

```r
fect.output <- as.data.frame(fect.output)
fect.output$Time <- c(-17:18)
p.fect <- esplot(fect.output,Period = 'Time',Estimate = 'ATT',
                   SE = 'S.E.',CI.lower = "CI.lower",
                   CI.upper = 'CI.upper',xlim = c(-12,10))
p.fect
```



Borusyak, Jaravel, and Spiess (2021) with package didimputation almost gives the same result as Liu, Wang, and Xu (2022).

21.2.3. Wooldridge

Wooldridge 2021, also called extended TWFE (ETWFE), is to allow more flexibility to make TWFE work properly. The problem with TWFE is that we do not specify it flexibly enough. If we allow heterogeneous treatment effect, then we cannot expect the canonical TWFE to work properly.

```
df.wool <- df.twfe
df.wool[which(is.na(df.wool$FirstTreat)),"FirstTreat"] <- 0 # replace NA with 0

library(etwfe)

model.wool = etwfe(
  fml  = nat_rate_ord ~ 1, # outcome ~ controls
  tvar = year,        # time variable
  gvar = FirstTreat, # group variable
  data = df.wool,       # dataset
  vcov = ~bfs,  # vcov adjustment (here: clustered)
  cgroup = "never"
)
model.wool
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: FirstTreat: 16,  year: 19
```

```
Standard-errors: Clustered (bfs)
                                    Estimate Std. Error  t value Pr(>|t|)
.Dtreat:FirstTreat::1992:year::1992  2.18863    1.36923  1.59844 0.110287
.Dtreat:FirstTreat::1992:year::1993 -4.18925    3.10938 -1.34730 0.178214
.Dtreat:FirstTreat::1992:year::1994 -2.50997    1.63352 -1.53654 0.124747
.Dtreat:FirstTreat::1992:year::1995 -5.23221    3.11282 -1.68086 0.093129 .
.Dtreat:FirstTreat::1992:year::1996 -5.42014    3.11207 -1.74165 0.081902 .
.Dtreat:FirstTreat::1992:year::1997 -5.15575    3.11248 -1.65648 0.097965 .
.Dtreat:FirstTreat::1992:year::1998 -5.08210    3.11345 -1.63231 0.102955
.Dtreat:FirstTreat::1992:year::1999 -5.13290    3.11447 -1.64808 0.099676 .
... 262 coefficients remaining (display them with summary() or use
argument n)
... 15 variables were removed because of collinearity
(.Dtreat:FirstTreat::1992:year::1991,
.Dtreat:FirstTreat::1993:year::1992 and 13 others [full set in
$collin.var])
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.6642     Adj. R2: 0.088474
               Within R2: 0.040642
```

```
emfx(model.wool)
```

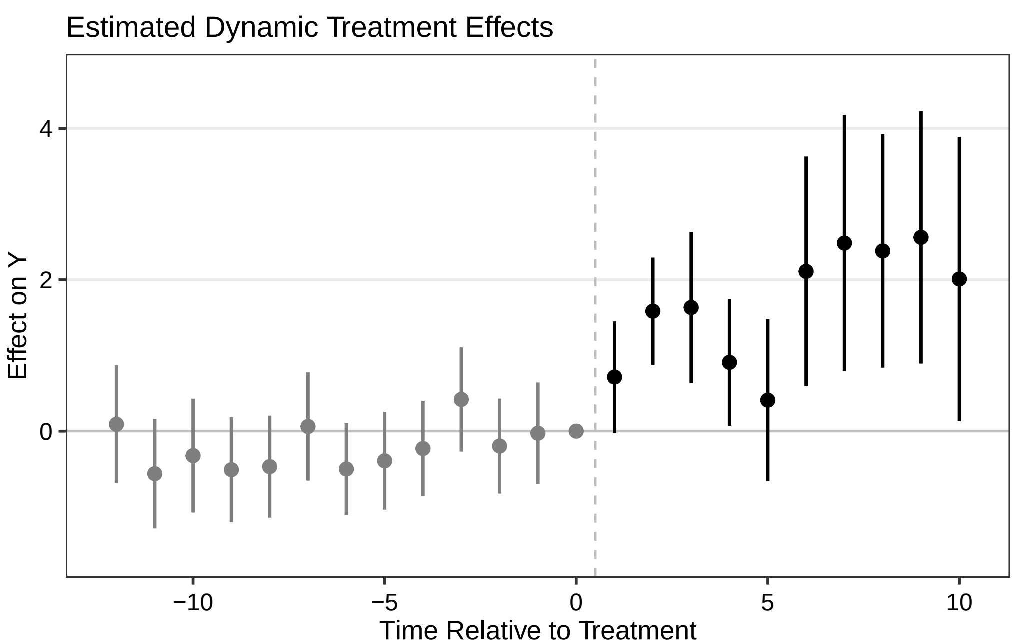
```
 .Dtreat Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
    TRUE     1.33      0.288 4.62   <0.001 18.0 0.766    1.9

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```
mod_es = emfx(model.wool, type = "event", post_only=FALSE)
mod_es
```

```
 event Estimate Std. Error       z Pr(>|z|)   S  2.5 % 97.5 %
   -18   -0.266      0.789 -0.3371    0.736 0.4 -1.812  1.280
   -17    0.178      0.503  0.3538    0.723 0.5 -0.808  1.164
   -16    0.114      0.519  0.2198    0.826 0.3 -0.903  1.131
   -15   -0.104      0.408 -0.2550    0.799 0.3 -0.903  0.695
   -14    0.410      0.413  0.9927    0.321 1.6 -0.400  1.220
--- 26 rows omitted. See ?print.marginaleffects ---
    13    1.135      1.998  0.5683    0.570 0.8 -2.781  5.052
    14    1.362      3.129  0.4355    0.663 0.6 -4.770  7.495
    15   -2.070      1.439 -1.4387    0.150 2.7 -4.890  0.750
    16    2.274      3.080  0.7385    0.460 1.1 -3.762  8.311
```

```
    17     0.107       4.576   0.0234     0.981 0.0 -8.861  9.076
Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```
wool.output <- mod_es |>
  filter(event != -1) |>
  mutate(Time = event + 1) |>
  as.data.frame()

p.wool <- esplot(wool.output,Period = 'Time',Estimate = 'estimate',
                 SE = 'std.error', xlim = c(-12,10))
p.wool
```



Note that in order to get pre-event estimates, we need to set "cgroup=never" in the etwfe function. Otherwise, ETWFE will only give us post-event estimates. Under the default setting, the control group comprises the "not-yet" treated units, therefore all pre-treatment effects are mechanistically set to zero. By setting "cgroup=never", we only use never treated units as the control group. This way we can get pre-treatment estimates, but we are not using all available information for the control group, therefore the estimates may not be as efficient as the default setting. That cost is visible in the two runs below: the never-treated version gives an ATT of 1.33 with a standard error of 0.288, while the default not-yet-treated version gives 1.51 with a standard error of 0.188 — a little over a third narrower.

Also we don't have control variable here. If we have, it is very easy to incorporate into ETWFE. Wooldridge used the Mundlak device to incorporate control variables, which is a very simple and intuitive way to do it.

If we do the default setting and plot only post-event estimates:

```
model.wool2 = etwfe(
  fml  = nat_rate_ord ~ 1, # outcome ~ controls
  tvar = year,        # time variable
  gvar = FirstTreat, # group variable
  data = df.wool,       # dataset
  vcov = ~bfs  # vcov adjustment (here: clustered)
)
model.wool2
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: FirstTreat: 16,  year: 19
Standard-errors: Clustered (bfs)
                                    Estimate Std. Error  t value Pr(>|t|)
.Dtreat:FirstTreat::1992:year::1992  1.99670    1.35883  1.46943 0.142057
.Dtreat:FirstTreat::1992:year::1993 -4.52291    3.09436 -1.46166 0.144174
.Dtreat:FirstTreat::1992:year::1994 -2.62350    1.61688 -1.62257 0.105022
.Dtreat:FirstTreat::1992:year::1995 -5.17241    3.09454 -1.67146 0.094969 .
.Dtreat:FirstTreat::1992:year::1996 -5.08905    3.09404 -1.64479 0.100353
.Dtreat:FirstTreat::1992:year::1997 -5.17232    3.09482 -1.67128 0.095004 .
.Dtreat:FirstTreat::1992:year::1998 -5.39134    3.09606 -1.74135 0.081954 .
.Dtreat:FirstTreat::1992:year::1999 -5.09733    3.09495 -1.64698 0.099901 .
... 121 coefficients remaining (display them with summary() or use
argument n)
... 141 variables were removed because of collinearity
(.Dtreat:FirstTreat::1993:year::1992,
.Dtreat:FirstTreat::1994:year::1992 and 139 others [full set in
$collin.var])
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.6999     Adj. R2: 0.081954
               Within R2: 0.0259
```

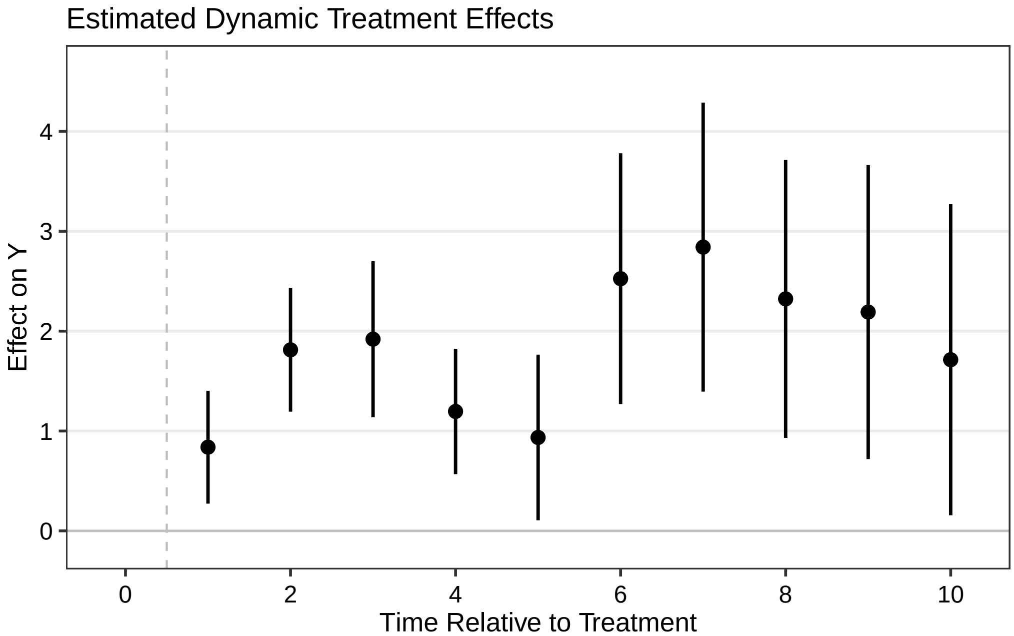
```
emfx(model.wool2)
```

```
 .Dtreat Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
    TRUE     1.51      0.188 8.02   <0.001 49.7  1.14   1.87

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```
mod_es2 = emfx(model.wool2, type = "event")
mod_es2
```

```
 event Estimate Std. Error        z Pr(>|z|)    S  2.5 % 97.5 %
     0   0.8381      0.288  2.90643  0.00366  8.1  0.273  1.403
     1   1.8127      0.316  5.74107  < 0.001 26.7  1.194  2.432
     2   1.9194      0.399  4.81257  < 0.001 19.4  1.138  2.701
     3   1.1957      0.320  3.73394  < 0.001 12.4  0.568  1.823
     4   0.9352      0.423  2.20947  0.02714  5.2  0.106  1.765
     5   2.5252      0.641  3.93965  < 0.001 13.6  1.269  3.781
     6   2.8413      0.738  3.84931  < 0.001 13.0  1.395  4.288
     7   2.3229      0.710  3.27278  0.00106  9.9  0.932  3.714
     8   2.1911      0.751  2.91737  0.00353  8.1  0.719  3.663
     9   1.7138      0.795  2.15674  0.03103  5.0  0.156  3.271
    10   1.0765      0.941  1.14366  0.25277  2.0 -0.768  2.921
    11  -0.3083      0.498 -0.61885  0.53602  0.9 -1.285  0.668
    12   0.3508      1.892  0.18538  0.85293  0.2 -3.358  4.060
    13   0.8170      2.012  0.40611  0.68466  0.5 -3.126  4.760
    14   0.9743      3.351  0.29076  0.77123  0.4 -5.593  7.542
    15  -2.4917      1.413 -1.76355  0.07781  3.7 -5.261  0.278
    16   1.4233      4.430  0.32126  0.74801  0.4 -7.260 10.107
    17  -0.0138      4.556 -0.00304  0.99758  0.0 -8.944  8.916

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```r
wool.output2 <- mod_es2 |>
  filter(event != -1) |>
  mutate(Time = event + 1) |>
  as.data.frame()

p.wool2 <- esplot(wool.output2, Period = 'Time',Estimate = 'estimate',
                  SE = 'std.error', xlim = c(0,10))
p.wool2
```



21.3. Comparing the estimators

It is worth collecting what all of these produced on the same data:

estimator	ATT	s.e.
TWFE	1.339	0.187
Sun and Abraham	1.331	0.288
Callaway and Sant'Anna (simple)	1.331	0.302
Callaway and Sant'Anna (dynamic)	1.221	0.631
Imai, Kim and Wang	1.178	0.335
Liu, Wang and Xu	1.506	0.204
Wooldridge ETWFE	1.330	0.288

Three things stand out. Sun-Abraham, Callaway-Sant'Anna and ETWFE agree to three digits at 1.33 — they are different routes to the same never-treated-control estimand, and they land on the same number. The spread across all seven, 1.18 to 1.51, is smaller than the standard errors, so on this data the estimator choice does not change the conclusion.

And TWFE at 1.339 is not far off any of them, despite the forbidden comparisons the Bacon decomposition exposed. That is worth saying plainly: the decomposition shows TWFE *is* contaminated here, but the contamination happens to be small enough that the answer survives. It would not have to be.

21.4. Staggered DiD with treatment reversal

With treatment reversal (ie., the treated units go back to control), Wooldridge's method still works. However, I don't think package "etwfe" has it implemented.

So far, packages "fect", "PanelMatch" can handle treatment reversal.

We can probably manually do ETWFE with treatment reversal. See Wooldridge's dropbox for the code in Stata. The basic idea is to treat cohorts as different treatment beginning time and exit time combinations. Then do the same regression with interactions of cohort dummies with the time dummies. It is very flexible. The tricky part is to get the marginal effect and its standard error right.

Systematic treatment: R · Julia.

22. Causal Panel (why DDDiD)

22.1. Why DDDiD

Guido Imbens has been saying DDDiD (Don't Do Diff in Diff). But why?

To see that, DiD objective function (in the form of TWFE) is:

$$(\hat{\tau}^{did}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \tag{22.1}$$

We can see that in this objective function, we are allowing only α_i and β_t to capture the difference between units and time periods. The obvious drawback is that control units that are less similar to a treated unit is given the same weight as those that are more similar. A time period that is far from the beginning of the treatment period is given the same weight as those that are closer. Say the treatment period is 2000, we have observations from 1988 to 2005, then 1988 observations are the same as 1999 observations. This is not ideal. The same logic applies to the units.

22.2. Synthetic Control

SC objective function:

$$(\hat{\tau}^{sc}, \hat{\mu}, \hat{\beta}) = \underset{\tau,\mu,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} (Y_{it} - \mu - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{sc} \tag{22.2}$$

It's a weighted version of DiD, but note carefully that it is **not** a nesting of it: setting every $\hat{\omega}_i^{sc}$ to 1 here does not recover the DiD objective above, because SC has no α_i term — it would give a pooled regression with time effects only. Dropping the unit fixed effects is a substantive restriction, not a re-weighting. (DiD *is* nested in synthetic DiD below, which keeps α_i and sets both weights to be constant; that is the right way to see DiD as a special case.) The weights are set to optimally match donor units to the treated unit so that they are as close as possible at each time point. There is no time weight.

22.3. Synthetic DiD

Arkhangelsky et al (2021) tries to combine the idea of DiD and SC. SC assigns different weights to different control units. The standard DiD is a TWFE, assigning equal weights to all time periods and units.

$$(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} \hat{\omega}_i^{sdid} \lambda_t^{sdid} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \tag{22.3}$$

SDiD sets another weight in addition to SC weights, which changes over time. The SC weights are trying to construct a control unit that is close to the treated unit; the SDiD weights are trying to put more weights on pre-treatment periods that are more similar to post-treatment periods.

22.4. Example from synthdid

The data are the California Proposition 99 panel: annual per-capita cigarette sales in packs for 39 states from 1970 to 2000, with California the single treated unit, 38 donors, 19 pre-treatment years and 12 post-treatment years.

```
library(synthdid)
data('california_prop99')
setup = panel.matrices(california_prop99)
tau.hat = synthdid_estimate(setup$Y, setup$N0, setup$T0)
summary(tau.hat)
```

```
$estimate
[1] -15.60383

$se
     [,1]
[1,]   NA

$controls
               estimate 1
Nevada              0.124
New Hampshire       0.105
Connecticut         0.078
Delaware            0.070
Colorado            0.058
Illinois            0.053
Nebraska            0.048
Montana             0.045
Utah                0.042
New Mexico          0.041
Minnesota           0.039
Wisconsin           0.037
West Virginia       0.034
North Carolina      0.033
Idaho               0.031
Ohio                0.031
Maine               0.028
```

```
Iowa                0.026

$periods
     estimate 1
1988      0.427
1986      0.366
1987      0.206

$dimensions
         N1            N0 N0.effective            T1            T0 T0.effective
      1.000        38.000       16.388        12.000        19.000        2.783
```

The estimate is −15.60 packs per capita, and the summary shows how it gets there. The unit weights are concentrated — Nevada 0.124, New Hampshire 0.105, Connecticut 0.078 — giving an effective 16.4 donors out of 38. The time weights are more concentrated still: 0.427 on 1988, 0.366 on 1986 and 0.206 on 1987, which is an effective 2.8 pre-periods out of 19. That is the "time-aware" part of synthetic DiD in numbers, and it is the mechanism behind the estimator's advantage over plain DiD.

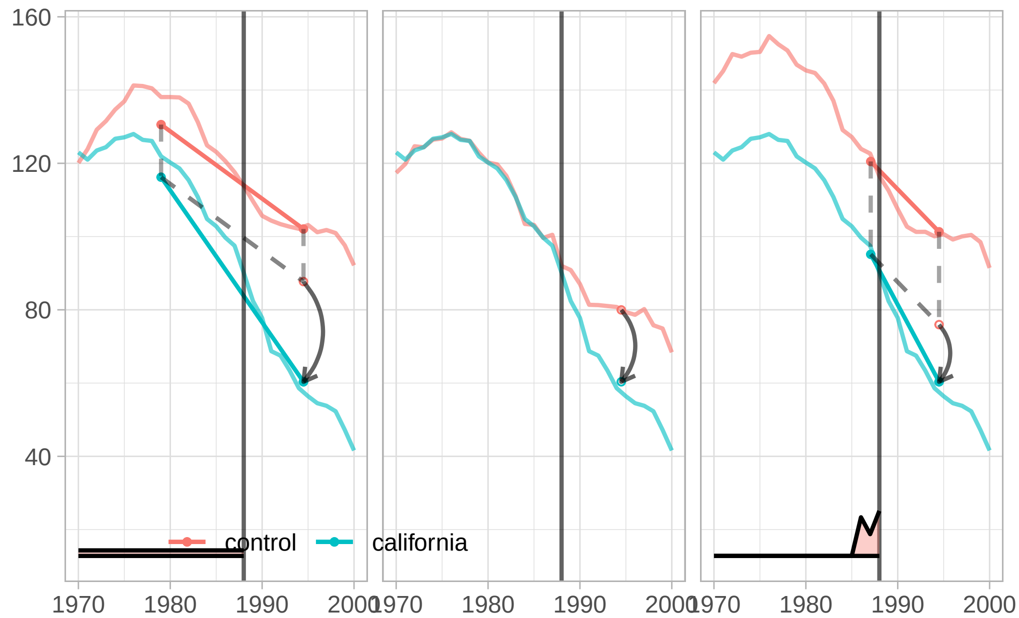
Note also that `se` comes back `NA`. That is not a glitch: the default is a jackknife standard error, which deletes one treated unit at a time, and California is the only treated unit. The placebo method is the usable option here.

```r
tau.sc   = sc_estimate(setup$Y, setup$N0, setup$T0)
tau.did  = did_estimate(setup$Y, setup$N0, setup$T0)
estimates = list(tau.did, tau.sc, tau.hat)
names(estimates) = c('Diff-in-Diff', 'Synthetic Control', 'Synthetic Diff-in-Diff')
print(unlist(estimates))
```

```
          Diff-in-Diff      Synthetic Control Synthetic Diff-in-Diff
             -27.34911              -19.61966              -15.60383
```

Plain DiD gives −27.35, synthetic control −19.62, and synthetic DiD −15.60. The spread across the three is nearly twelve packs per capita, on the same data with the same identification assumptions. DiD moves furthest because it weights all 38 donors and all 19 pre-periods equally: states unlike California, and years unlike the late 1980s, all count in full.

```r
plot <- synthdid_plot(estimates, facet.vertical=FALSE,
              control.name='control', treated.name='california',
              lambda.comparable=TRUE, se.method = 'none',
              trajectory.linetype = 1, line.width=.75, effect.curvature=-.4,
              trajectory.alpha=.7, effect.alpha=.7,
              diagram.alpha=1, onset.alpha=.7) +
    theme(legend.position=c(.26,.07), legend.direction='horizontal',
          legend.key=element_blank(), legend.background=element_blank(),
          strip.background=element_blank(), strip.text.x = element_blank())
plot
```



Each panel shows California's path against its weighted control, with the estimated effect as the post-1989 gap. Note: λ_t is plotted at the bottom, so the pre-periods that actually enter each comparison are visible — for synthetic DiD they pile up on 1986–1988, while for DiD the weight is flat across all nineteen years.

22.5. Summary

To me, "DDDiD" is saying DiD is too simple. Even if all the identification assumptions are correct for DiD, namely parallel trend and no anticipation, there are still better ways to do it, in the sense of putting weights on units and time periods, such as synthetic DiD.

Systematic treatment: R · Julia.

Part V.

Count Data & Specialized Models

23. Which count data model to use

23.1. A comparison of various count data models with extra zeros

In empirical studies, data sets with a lot of zeros are often hard to model. There are various models to deal with it: zero-inflated Poisson model, Negative Binomial (NB)model, hurdle model, etc.

Here we are following a zero-inflated model's thinking: model the data with two processes. One is a Bernoulli process, the other one is a count data process (Poisson or NB).

We'd like to see, in this simulation exercise, how different models perform with changes of sample size and percentage of zeros (we expect the less zero, the better a plain Poisson model would perform). Therefore we vary sample size n and an indicator of how much percentage of zeros in the data θ.

For the count data process (y_c):

$$log(y_c) = 2x + u \tag{23.1}$$

For the Bernoulli process (y_b):

$$z_1 = 4z + \theta \tag{23.2}$$

$$logit(y_b) = z_1 \tag{23.3}$$

$$p_y = \frac{e^{z_1}}{1 + e^{z_1}} \tag{23.4}$$

where $y_b \sim Bernoulli(p_y)$ is the realized 0/1 draw. Combining these two processes:

$$y = y_c \cdot y_b \tag{23.5}$$

so $y = y_c$ when $y_b = 1$ and $y = 0$ when $y_b = 0$.

23.1.1. Zero-inflated Poisson models

A zero-inflated Poisson needs specifying both the binary process and the count process correctly. Often than not, we don't have a model for the binary process. Many people simply use the same explanatory variables for both processes. We simulate both situations. Case 1: suppose we observe z, and case 2: suppose we don't observe z. In the graph below, they are labeled zip1 and zip2.

23.1.2. Poisson model

A plain Poisson model returns a consistent estimator for the coefficients, with or without Poisson-distributed data. We expect Poisson model's performance improve with sample size. Note that the standard errors from a Poisson model needs adjustment, which we do not discuss in this post.

23.1.3. NB model

NB model is used widely to handle "overdispersion" problem. That is, the variance far exceeds the mean, therefore the Poisson model is considered inappropriate. NB model addresses that by allowing an extra parameter. However, many people also use it to model "extra zero" situation, we'll see in our simulation it may not be better than a plain Poisson model.

23.1.4. Log-linear model

What about an OLS model with $log(y+1)$?

23.1.5. hurdle model

A hurdle model models the zero's and other values separately; that is, the zero's are from a binomial process only, the other positive values are from a truncated count data process. We assume here, in the simulation, we don't observe z. Therefore, x is determining both binary and count processes. In the graph below, it's labeled hurdle.

```r
library(MASS)
library(pscl)
library(parallel)
set.seed(666)

gen.sim <- function(df) {
  nobs <- df["nobs"]; th <- df["th"]
  z    <- rnorm(nobs, 0, 1)
  x    <- rnorm(nobs, 0, 1)
  u    <- rnorm(nobs, 0, 1)
  y.count  <- floor(exp(2 * x + u))
  z1       <- 4 * z + th
  prob     <- plogis(z1)
  y.logit  <- rbinom(nobs, size = 1, prob = prob)
  y        <- ifelse(y.logit == 1, y.count, y.logit)

  m1 <- zeroinfl(y ~ x | z)
  m4 <- zeroinfl(y ~ x | x)
  m2 <- glm(y ~ x, family = "poisson")
  m3 <- lm(log(y + 1) ~ x)
  m5 <- glm.nb(y ~ x)
```

```
  m6 <- hurdle(y ~ x)

  c(zip1       = coef(m1, "count")["x"] - 2,
    poisson     = coef(m2)["x"] - 2,
    log.linear  = coef(m3)["x"] - 2,
    zip2        = coef(m4, "count")["x"] - 2,
    nb          = coef(m5)["x"] - 2,
    hurdle      = coef(m6, "count")["x"] - 2)
}

data.grid <- expand.grid(
  nobs = ceiling(exp(seq(4, 9, 1)) / 100) * 100,
  nsim = seq(1, 100, 1),
  th   = seq(-4, 4, 2)
)

cl      <- makeCluster(detectCores() - 1)
clusterEvalQ(cl, { library(MASS); library(pscl) })
clusterExport(cl, "gen.sim")
results <- parApply(cl, data.grid, 1, gen.sim)
stopCluster(cl)

forshiny <- cbind(data.grid, t(results))
write.csv(forshiny, "results.csv", row.names = FALSE)
```

Count data models can be used even if data is not "counts"; for example, some non-negative non-integer numbers. In fact, Poisson model is consistent even if data is not Poisson-distributed, if the model specification is correct on modeling the log of expected counts. We simulate both scenarios: Case 1, data is generated from a Poisson process. Case 2, data is generated from a Normal distribution, but we use count data models to model it. The above code is for case 2.

We simulate 100 times with θ ranging from -4 to 4, lower number means higher percentage of zeros; number of observations from e^4 to e^9 (roughly from 50 to 8000).

Since there are many simulations, we use base R's `parallel` library to speed things up.

For raw code, please visit case1: poisson and case2: normal. The code above updates the original `snowfall`-based version to use base R `parallel`.

Summary of simulation findings: With large samples, plain Poisson outperforms NB and log-linear even with extra zeros. Zero-inflated Poisson with correct specification of the zero process is best, but that relies on knowing what drives the zeros. Hurdle and ZIP without correct zero-model specification are similar and not much worse than Poisson. Log-linear (`log(y+1)`) is consistently the worst.

(Note: the `log.linear` bias line above previously read `exp(coef(m3)["x"]) - 2` instead of `coef(m3)["x"] - 2`, which — since the log-linear coefficient is already on the log scale — manufactured a spurious 5.4 apparent bias regardless of the model's actual performance. That has been corrected here to match the other rows. Log-linear regression on `log(y+1)` is still expected to be biased relative to Poisson-family models in a zero-inflated count setting, for real reasons (the log transform's nonlinearity near zero, and the fact that $E[\log(y+1)] \neq \log(E[y]+1)$), but the *magnitude* of "consistently the worst" above reflects the original (buggy) run and has not been re-verified against a corrected simulation.)

23.2. Quick worked example

All five models on a single small dataset (n = 1000, ~40% zeros).

The DGP is a zero-inflated Poisson built in two independent pieces. We draw $x \sim N(0,1)$ and $z \sim N(0,1)$, independent of each other. The count part is $y_c \sim \text{Poisson}(\exp(1.5 + 0.8x))$ and the zero part is $y_b \sim \text{Bernoulli}(\Lambda(1 + 2z))$, with the observed outcome $y = y_c \cdot y_b$. So the true slope on x is 0.8, and — this is the feature that drives the results — the zero process depends on z alone, never on x.

```r
library(MASS)
library(pscl)
set.seed(42)

n  <- 1000
x  <- rnorm(n)
z  <- rnorm(n)                          # zero-inflation driver
mu <- exp(1.5 + 0.8 * x)               # count mean
y_c <- rpois(n, lambda = mu)
y_b <- rbinom(n, 1, plogis(1 + 2 * z)) # 1 = non-zero
y   <- y_c * y_b                        # zero-inflated count

cat("Zero fraction:", round(mean(y == 0), 2), "\n")
```

```
Zero fraction: 0.39
```

```r
cat("Mean (non-zero):", round(mean(y[y > 0]), 2), "\n\n")
```

```
Mean (non-zero): 6.79
```

```r
# 1. Poisson
m_pois <- glm(y ~ x, family = poisson)

# 2. Negative binomial
m_nb <- glm.nb(y ~ x)

# 3. Zero-inflated Poisson (ZIP) - correct zero model
m_zip <- zeroinfl(y ~ x | z)

# 4. ZIP - wrong zero model (only x)
```

```r
m_zip2 <- zeroinfl(y ~ x | x)

# 5. Hurdle
m_hurdle <- hurdle(y ~ x)

# 6. Log-linear OLS on log(y + 1)
m_loglin <- lm(log(y + 1) ~ x)

# Compare x coefficient (true = 0.8)
coefs <- c(
  Poisson    = coef(m_pois)["x"],
  NB         = coef(m_nb)["x"],
  `ZIP (correct z)` = coef(m_zip, "count")["x"],
  `ZIP (wrong)` = coef(m_zip2, "count")["x"],
  Hurdle     = coef(m_hurdle, "count")["x"],
  `Log-linear log(y+1)` = coef(m_loglin)["x"]
)

knitr::kable(
  data.frame(Model = names(coefs), `x coefficient` = round(coefs, 3),
             Bias = round(coefs - 0.8, 3)),
  caption = "True x coefficient = 0.8"
)
```

Table 23.1.: True x coefficient = 0.8

	Model	x.coefficient	Bias
Poisson.x	Poisson.x	0.850	0.050
NB.x	NB.x	0.824	0.024
ZIP (correct z).x	ZIP (correct z).x	0.803	0.003
ZIP (wrong).x	ZIP (wrong).x	0.802	0.002
Hurdle.x	Hurdle.x	0.803	0.003
Log-linear log(y+1).x	Log-linear log(y+1).x	0.458	-0.342

The sample comes out 39% zeros with a non-zero mean of 6.79. The estimated slopes are 0.850 for Poisson, 0.824 for negative binomial, 0.803 for ZIP with the correct zero model, 0.802 for ZIP with the wrong one, 0.803 for the hurdle model, and 0.458 for log-linear OLS.

The first five models recover the true `x` slope of 0.8 reasonably well; the log-linear row is included to make the earlier claim checkable, and needs reading with care — see the note after the table. This is the key point of *this* DGP: the zero-inflation process depends on `z`, which is generated **independently of `x`**, so omitting the zero model distorts the zero mass and the intercept, not the slope on `x`. The hurdle model and even the ZIP with a *misspecified* (x-only) zero model still recover the slope closely; the plain Poisson shows a small upward bias *in this particular sample* (n = 1000), but that is finite-sample noise, not a systematic property of the model. Since the zero-inflation driver `z` is independent of `x`, the conditional mean here is correctly specified for Poisson QMLE

— increasing `n` from 1,000 to 1,000,000 shrinks the Poisson `x`-coefficient bias from about +0.04 to effectively 0, confirming consistency. The unmodeled excess zeros create overdispersion, but overdispersion under a correctly-specified conditional mean affects only the standard errors (which should be corrected with a robust/sandwich estimator), not the consistency of the slope.

About the log-linear row: its coefficient is **not** estimating the same thing as the others, which is the real reason to avoid it rather than any particular bias number. The count models target $\log E[y \mid x]$, whereas OLS on $\log(y+1)$ targets $E[\log(y+1) \mid x]$, and $E[\log(y+1)] \neq \log(E[y]+1)$ by Jensen's inequality. The +1 offset makes the discrepancy worst exactly where the mass of zeros is. So comparing its coefficient to 0.8 is an apples-to-oranges comparison, and the honest statement is that log-linear answers a different question — not that it answers this one badly by some measured amount coefficient. A misspecified zero process biases the count *slope* only when the zero process is correlated with the count covariates — which is not the case here. (Change `z` to depend on `x` and the omitted-zero-process slope bias appears.)

23.3. Mixed count models with glmmTMB

When data has both overdispersion *and* random effects (clustered counts, panel data, repeated measures), `glmmTMB` handles the full range in one package:

```r
library(glmmTMB)

# Zero-inflated negative binomial with random intercept per subject
m_glmmtmb <- glmmTMB(
  count ~ treatment + time + (1 | subject),   # count model
  ziformula = ~ treatment,                      # zero-inflation model
  family = nbinom2,
  data = your_data
)

summary(m_glmmtmb)

# Conditional vs marginal predictions
predict(m_glmmtmb, type = "conditional")  # excluding zero-inflation
predict(m_glmmtmb, type = "response")     # including zero-inflation
```

`glmmTMB` supports: Poisson, NB1, NB2, zero-inflated variants of all of the above, hurdle models, beta-binomial, tweedie, and more. The `ziformula` argument takes a one-sided formula for the zero-inflation component — use `~ 1` for a constant zero-inflation rate, or covariates to model what drives the structural zeros.

24. What model to use for rare events

24.1. Introduction

In empirical studies, people are worried about rare event situation. That is, when you have, for example, lots of 0's and only a few 1's, or vice versa. Do you run a logit model, or do you use a "rare event logit"? When should you use either approach? Or there is a third approach?

Paul Allison said in his blog (https://statisticalhorizons.com/logistic-regression-for-rare-events):

"Prompted by a 2001 article by King and Zeng, many researchers worry about whether they can legitimately use conventional logistic regression for data in which events are rare. Although King and Zeng accurately described the problem and proposed an appropriate solution, there are still a lot of misconceptions about this issue.

The problem is not specifically the rarity of events, but rather the possibility of a small number of cases on the rarer of the two outcomes. If you have a sample size of 1000 but only 20 events, you have a problem. If you have a sample size of 10,000 with 200 events, you may be OK. If your sample has 100,000 cases with 2000 events, you're golden."

In general I agree with him. However, when exactly should we use King and Zeng's "relogit"?

Allison also mentioned two other methods. One is called the Firth method, a penalized likelihood approach. The other one is the exact logistic regression, which is for small samples.

In this simulation exercise, I compare logit against bias-reduced logistic regression (`brglm2` with `method="brglmFit"`, `type="AS_mean"`) and the Firth model (`logistf`).

One thing to be clear about before reading any comparison of the latter two: for a logistic regression **they are the same estimator.** Firth's penalized likelihood (penalizing by the Jeffreys prior) coincides with mean bias reduction for GLMs with a canonical link, and logit is canonical for the binomial. So `brglm2`'s `AS_mean` and `logistf` solve the same estimating equations and differ only in numerical tolerance and in how each package builds standard errors and intervals (`logistf` uses penalized profile likelihood). A demonstration, on a deliberately rare-event DGP: $n = 500$, $x \sim N(0, 1)$, and $y \sim \text{Bernoulli}(\Lambda(-3.5 + 0.8x))$, which puts the event rate around 3% — roughly fifteen events in the whole sample.

```
library(brglm2); library(logistf)
set.seed(11)
n <- 500; x <- rnorm(n); y <- rbinom(n, 1, plogis(-3.5 + 0.8 * x))
d <- data.frame(y, x)

rbind(
  `plain logit`      = coef(glm(y ~ x, data = d, family = binomial)),
```

```
  `brglm2 AS_mean`  = coef(glm(y ~ x, data = d, family = binomial,
                               method = "brglmFit", type = "AS_mean")),
  `logistf (Firth)` = coef(logistf(y ~ x, data = d))
)
```

```
                (Intercept)         x
plain logit       -3.176125 0.7378324
brglm2 AS_mean    -3.134453 0.7277674
logistf (Firth)   -3.134453 0.7277675
```

The two bias-reduced fits give −3.134453 and 0.7277674 against −3.134453 and 0.7277675: they agree to about 10^{-7}, which is numerical tolerance rather than any statistical difference. Plain logit gives −3.176 and 0.738, so on this draw the bias correction moves the slope by about 0.010 and the intercept by 0.042 — both estimators sitting below the true 0.8 and above the true −3.5, which is what fifteen events buys you. This matters for reading the findings below: what the simulation can tell us is how **logit** compares with **Firth-type bias reduction**, not which of two implementations of the same estimator is better. If a genuinely distinct third method is wanted, `brglm2`'s `type = "AS_median"` (median bias reduction) is one, and King and Zeng's `relogit` correction is another.

24.2. Simulation

Here I have some code for using multiple cores to run these three models. The bias-reduced logistic regression is implemented in R via the `brglm2` package, the Firth method is implemented in `logistf`.

```
library(brglm2)
library(logistf)
require(snowfall)
set.seed(666)

# initialize parallel cores.
sfInit(parallel=TRUE, cpus=16)

gen.sim <- function(df){
  x <- rnorm(df['nobs'], 0, 1)

  # generate binary data
  p <- df['p']
  alpha <- -log((1-p)/p)
  z = alpha + 2*x
  pr = 1/(1+exp(-z))
  y = rbinom(df['nobs'], 1, pr)
  df = data.frame(y=y, x=x)

  # With small nobs and small p (e.g. nobs=10, p=.01), y can easily come out
```

```
  # all-zero (>90% chance in that cell), which makes glm/logistf unable to
  # estimate a coefficient on x. Guard against that rather than letting a
  # single degenerate draw crash the whole parallel simulation loop.
  if (var(y) == 0) {
    return(c(logit=NA, brglm2=NA, logistf=NA))
  }

  # logit
  m1 <- tryCatch(glm(y ~ x, family='binomial'), error = function(e) NULL)
  m1.x <- if (!is.null(m1)) summary(m1)$coefficients['x','Estimate'] - 2 else NA

  # bias-reduced logistic regression (brglm2)
  m2 <- tryCatch(
    glm(y ~ x, family = binomial(link = "logit"), data = df,
        method = "brglmFit", type = "AS_mean"),
    error = function(e) NULL
  )
  m2.x <- if (!is.null(m2)) coef(m2)['x'] - 2 else NA

  # logistf (Firth)
  m3 <- tryCatch(logistf(y ~ x, data=df), error = function(e) NULL)
  m3.x <- if (!is.null(m3)) coef(m3)['x'] - 2 else NA

  return(c(logit=m1.x, brglm2=m2.x, logistf=m3.x))
}

# set parameter space
sim.grid = seq(1, 100, 1)
p.grid = c(.01, .05, .1)
nobs.grid = c(10, 30, 50, 100, 200, 500, 1000, 10000)

data.grid <- expand.grid(nobs.grid, sim.grid, p.grid)
names(data.grid) <- c('nobs', 'nsim', 'p')

# export functions and libraries to parallel workers
sfExport(list=list("gen.sim"))
sfLibrary(brglm2)
sfLibrary(logistf)

results <- data.frame(t(sfApply(data.grid, 1, gen.sim)))

# stop the cluster
sfStop()

forshiny <- cbind(data.grid, results)
write.csv(forshiny, 'results.csv')
```

We simulate 100 times with sample size from 10 to 10000, event probability .01, .05, and .1.

Since there are many simulations, we used the `snowfall` library to speed things up.

(The original post plotted bias and MSE by sample size and event probability from the `results.csv` output above; those figures are not reproduced here — the findings are summarized in prose below.)

In the case of small sample and rare event (for example, any situation that the product of sample size and probability is less than 5), none of these three models perform well. This is understandable, after all, the rarer of the two groups has only less than 5 observations. When the product is more than 50, there is not much difference between these three models. For the situations in between, that is, the product of sample size and probability is greater than 5 but less than 50, we found that Firth-type bias reduction — whether computed by `brglm2` or by `logistf` — performs better than plain logit. Any apparent ranking *between* those two should be discounted, since as shown above they are the same estimator; differences at that level reflect numerical tolerance or differing standard-error conventions, not a substantive advantage.

In the small sample situation, maybe it's better to use the exact logistic regression.

25. Poisson model with exposure

25.1. Poisson with exposure or Poisson rate

Poisson models for count data often arise when units are observed for different lengths of time. A hospital admission count for a patient followed for 30 days is not directly comparable to a count for a patient followed for 365 days. The natural quantity of interest is the *rate* — admissions per unit time — not the raw count.

There are three related but distinct specifications, each making a different assumption:

Model 1 — Poisson with offset (rate model):

$$\log(E[Y]/t) = \beta_0 + \beta_1 X \tag{25.1}$$

Equivalently, $\log E[Y] = \beta_0 + \beta_1 X + \log t$, where $\log t$ is included as a *fixed* offset (coefficient constrained to 1). This is the standard exposure model: the expected count is proportional to exposure time.

Model 2 — Poisson on the ratio Y/t:

$$\log(E[Y/t]) = \beta_0 + \beta_1 X \tag{25.2}$$

This looks similar, but it is not equivalent to the offset model, and strictly speaking Y/t is **not a Poisson outcome at all**: a Poisson likelihood is defined for non-negative integer counts, and the continuous ratio Y/t is not one. Because t is a fixed constant for each unit, $E[Y/t] = E[Y]/t$ exactly (linearity of expectation), so the *conditional mean* is still log-linear in X; that is why the point estimates are often similar. But the Poisson variance assumption fails: $\text{Var}(Y/t) = \lambda/t^2$, whereas the mean is $E[Y/t] = \lambda/t$, so the variance is $1/t$ times the mean rather than equal to it. (Writing it as λ/t would make variance equal mean and quietly restore equidispersion, which is precisely what fails here.) So fitting `glm(rate ~ x, family = poisson)` is a quasi-likelihood *mean model* rather than a correct Poisson likelihood. If you use it, treat it as a workaround: weight by exposure (`weights = t`) so the implied variance is right, and/or use robust standard errors. The offset model (Model 1) avoids the issue entirely by keeping the integer count as the outcome.

Model 3 — Unconstrained log-exposure:

$$\log E[Y] = \beta_0 + \beta_1 X + \gamma \log t \tag{25.3}$$

Here γ is estimated from the data. If $\hat{\gamma} \approx 1$, the proportionality assumption is supported. If $\hat{\gamma} < 1$, counts grow sub-linearly with exposure — a testable, substantive claim.

25.2. Simulation

```r
library(tidyverse)
set.seed(42)

n <- 800

# Simulate a health study: count of adverse events
# Covariate: age group (0 = younger, 1 = older)
age_old <- rbinom(n, 1, 0.45)

# Exposure: days under observation (varies 7-365)
exposure <- sample(seq(7, 365, by = 7), n, replace = TRUE)

# True model: rate = exp(-3 + 1.2 * age_old)
# Count ~ Poisson(rate * exposure)
true_rate <- exp(-3 + 1.2 * age_old)
count <- rpois(n, lambda = true_rate * exposure)

df <- data.frame(count, age_old, exposure, log_exposure = log(exposure))
summary(df[, c("count", "exposure")])
```

```
     count          exposure    
 Min.   : 0.00   Min.   :  7.0  
 1st Qu.: 7.00   1st Qu.:112.0  
 Median :13.00   Median :196.0  
 Mean   :18.95   Mean   :192.9  
 3rd Qu.:27.00   3rd Qu.:281.8  
 Max.   :71.00   Max.   :364.0  
```

We simulate $n = 800$ patients. Age group is age_old $\sim$ Bernoulli(0.45), and exposure is days under observation, drawn uniformly from the weekly grid 7 to 364 and independently of age. The event rate is $\exp(-3 + 1.2\,\text{age\_old})$ per day and the count is Poisson(rate $\times$ exposure), so the true intercept is -3, the true age effect is 1.2, and the true exposure elasticity is $\gamma = 1$ by construction.

The data has a wide range of exposure times — 7 to 364 days, median 196, with counts from 0 to 71 and a mean of 19 — so ignoring them would badly confound the comparison between age groups.

25.3. Fitting the three models

```r
# Model 1: offset - coefficient on log(exposure) constrained to 1
m1 <- glm(count ~ age_old + offset(log(exposure)),
          family = poisson, data = df)
```

```r
# Model 2: Poisson on the rate Y/t (different response)
df$rate <- df$count / df$exposure
m2 <- glm(rate ~ age_old,
          family = poisson, data = df)

# Model 3: log(exposure) as a free covariate
m3 <- glm(count ~ age_old + log_exposure,
          family = poisson, data = df)

# Collect coefficients
results <- data.frame(
  Model = c("1. Offset (constrained  =1)",
            "2. Rate Y/t as response",
            "3. Unconstrained log(t)"),
  Intercept = c(coef(m1)[1], coef(m2)[1], coef(m3)[1]),
  age_old   = c(coef(m1)[2], coef(m2)[2], coef(m3)[2]),
  gamma_log_t = c(1, NA, coef(m3)[3])
)

knitr::kable(results, digits = 3,
  caption = "Coefficient estimates: true intercept = -3, true age effect = 1.2, true   = 1")
```

Table 25.1.: Coefficient estimates: true intercept = −3, true age effect = 1.2, true = 1

Model	Intercept	age_old	gamma_log_t
1. Offset (constrained =1)	-3.009	1.212	1.000
2. Rate Y/t as response	-2.980	1.176	NA
3. Unconstrained log(t)	-2.801	1.212	0.962

All three recover the truth. The offset model gives an intercept of −3.009 and an age effect of 1.212 against true values of −3 and 1.2; the rate model gives −2.980 and 1.176; the unconstrained model gives −2.801, 1.212 and $\hat{\gamma} = 0.962$ against a true γ of 1.

Reading the table: Model 1 and Model 3 give nearly identical age-group coefficients because the true $\gamma = 1$ (counts are indeed proportional to exposure time). Model 2 differs slightly in the intercept because the response variable changes. The estimated $\hat{\gamma}$ from Model 3 should be close to 1.

25.4. When $\gamma \neq 1$

To see why the unconstrained model matters, repeat the simulation with a sub-linear count process ($\gamma = 0.6$: an initial burst of events that doesn't grow proportionally with long follow-up):

```r
count2 <- rpois(n, lambda = true_rate * exposure^0.6)
df2 <- data.frame(count2, age_old, exposure, log_exposure = log(exposure))

m1b <- glm(count2 ~ age_old + offset(log(exposure)),
           family = poisson, data = df2)
m3b <- glm(count2 ~ age_old + log_exposure,
           family = poisson, data = df2)

cat("Offset model age coefficient:      ", round(coef(m1b)[2], 3), "\n")
```

```
Offset model age coefficient:       1.171
```

```r
cat("Unconstrained model age coefficient:", round(coef(m3b)[2], 3), "\n")
```

```
Unconstrained model age coefficient: 1.168
```

```r
cat("Estimated gamma (log-exposure):    ", round(coef(m3b)[3], 3),
    " (true = 0.6)\n")
```

```
Estimated gamma (log-exposure):     0.566  (true = 0.6)
```

Two things to read off this, and the first is not what one might expect.

The offset model's age coefficient (1.171) and the unconstrained model's (1.168) are essentially the same, and both sit just below the true 1.2. Forcing $\gamma = 1$ when the truth is 0.6 does **not** bias the treatment coefficient here — and the reason is structural: `exposure` is drawn by `sample(...)` independently of `age_old`, so the misspecified exposure elasticity is absorbed by the intercept and cannot contaminate a coefficient on a covariate orthogonal to it. The constraint would bite if exposure were *correlated* with the covariate of interest — for instance if older patients were systematically followed longer, which is common in real cohort data and is the case worth worrying about.

What the unconstrained model does buy, unambiguously, is γ itself: it recovers $\hat{\gamma} = 0.566$ against a truth of 0.6, whereas the offset model cannot report it at all because it fixes it at 1. So Model 3's role is diagnostic — it tells you whether the proportionality assumption holds — rather than protective of the treatment estimate.

25.5. Stata equivalent

```stata
* Model 1: offset
poisson count age_old, exposure(exposure)

* Model 2: rate as response
```

```
gen rate = count / exposure
* Stata will warn that the dependent variable is non-integer (poisson expects
* integer counts by default) -- this is expected and can be ignored here,
* since Poisson QMLE is valid for continuous non-negative outcomes too.
poisson rate age_old

* Model 3: unconstrained log(t)
gen log_exposure = log(exposure)
poisson count age_old log_exposure
```

25.6. Which model to use?

Situation	Recommended model
Counts grow proportionally with time	Model 1 (offset)
Uncertain about proportionality	Model 3 (unconstrained), test $\hat{\gamma} = 1$
Rate is the natural estimand	Model 1 (offset) — it estimates a rate directly; Model 2 only as a weighted/robust quasi-model
Long follow-up with declining event rate	Model 3 or negative binomial with offset

In most health and social-science applications, Model 1 (Poisson with offset) is the default. Model 3 is the diagnostic check: fit it, look at $\hat{\gamma}$, and return to Model 1 if the constraint is supported by the data.

Systematic treatment: R · Julia.

26. IV in Fixed effect Poisson with many fixed effects

26.1. Poisson regression with many fixed effects

I have a few posts about Poisson regression for non-negative outcomes, including continuous variable, count variable, or corner solution (such as non-negative outcome with many zeros). The benefits include that we can use QMLE Poisson, which means we only need to have the mean of outcome specified correctly, as an exponentiated function of covariates. There is no assumption that the distribution needs to be a Poisson distribution. When we want to model non-negative outcome with fixed effects, fixed effect Poisson should be our first choice also, because fixed effect Poisson is shown to be consistent without the assumption of the distribution of the error term, and it does not have the "incidental parameter" problem.

Stata has "xtpoisson" to do fixed effect Poisson. However, we would recommend "ppmlhdfe". "ppmlhdfe is a Stata package that implements Poisson pseudo-maximum likelihood regressions (PPML) with multi-way fixed effects". It is faster than xtpoisson. More importantly, it does report the predicted value correctly. I mentioned in past post that we need to be cautious when we need to use predicted values after "xtpoisson, fe", or "xtreg, fe". The reason is that these are models estimated after "absorbing" the fixed effects, meaning fixed effects are not actually estimated. Therefore the predicted values are without fixed effects. But in "ppmlhdfe" we get the correct predicted values with fixed effect.

On a side note, the same author has "reghdfe" as a replacement for "areg" in Stata. "reghdfe" is faster than "areg", or "xtreg, fe". It also allows for many levels of fixed effects. As "ppmlhdfe", it also provides correct predicted values with fixed effects.

26.2. IV poisson with many fixed effects

Stata has a "ivpoisson" function with handles IV in a Poisson regression setting. However, it does not have the option to include fixed effects.

This estimator has a published treatment. Lin and Wooldridge (2019), "Testing and correcting for endogeneity in nonlinear unobserved effects models," treats exactly this setup: their **Procedure 3** is a reduced-form residual added to an FE Poisson, with a robust Wald test on the residual's coefficient serving as a test of endogeneity. Several points below come from it.

One warning about that test, since it is easy to misuse. It is tempting to run it and then decide which model to report: if the residual's coefficient is insignificant, fall back on the plain fixed-effects estimate. Do not do this. Guggenberger (2010) studies exactly this two-stage procedure and finds that its asymptotic size **equals one** — the two-stage test rejects with probability approaching one — because the pretest lacks power against correlations local to zero, so you frequently fail to reject when the regressor is in fact endogenous and then report the non-robust estimate. He examines both the linear IV version, where the pretest is a test of exogeneity of a regressor, and the random-versus-fixed-effects panel version, and recommends against the two-stage procedure in both. Both of his results are for linear models, so the transfer to a nonlinear control function is an analogy rather than a theorem, but the mechanism does not depend on linearity. Report the coefficient on the residual as an estimate of the degree of endogeneity, and report both models side by side; do not let the test choose for you.

In the case that the endogenous variable is continuous and we can use a linear model for the first stage, Wooldridge 2010 textbook (chapter 18) suggested using control function approach, which is very simple:

1. Run the first stage regression (the reduced form for the endogenous variable), and get the residual. This is the first-stage **control-function** residual. Note in the many fixed effects case, we need to include all fixed effects in the first stage regression. In the case of stata, we would use "reghdfe" here.

   A terminology note, because two different objects are easy to conflate. The **Mundlak device** means adding group *means* of covariates as regressors, as in the Mundlak chapter. The **control-function residual** here is the residual from a reduced form. They are different constructions. But a third object borrows the name, and it matters below: Lin and Wooldridge (2019) call the residual from a reduced form *that itself includes the Mundlak means* — from regressing y_{it2} on z_{it} and $\bar{z}_i$ — the **Mundlak residual**, as against the **FE residual** obtained by within-transforming. So in that paper "Mundlak residual" denotes a control-function residual, not the device itself.

2. There is a useful result about which of those two residuals to use. Lin and Wooldridge (2019, appendices A.1 and A.2) show that in FE Poisson they give **numerically identical** estimates of both the structural coefficient and the coefficient on the control function. The reason is that the two residuals differ only by a term constant within unit, and the FE Poisson score is unaffected by adding anything unit-constant.

   This has a practical consequence. It means the first stage does not need the thousands of unit dummies: the unit means will do, at no cost in the estimate. That is what makes a **nonlinear** first stage practical — you cannot put 2,000 dummies into a logit without an incidental-parameters problem, but you can put in 2,000 unit means. Note the result as stated is for a *linear* reduced form; see the caveat on discrete endogenous variables below.

3. Include both the endogenous variable and the control-function residual as additional regressors in the second stage, the original Poisson regression. Use "ppmlhdfe" here.

Note that the standard errors would be incorrect if we use the standard errors from the second stage regression. We need to use the bootstrap method to get the correct standard errors. This is not merely a precaution: Lin and Wooldridge (2019, p. 30) recommend exactly this, namely a panel bootstrap in which both estimating steps are done with each bootstrap sample. The reason is that the control-function residual is generated from the first stage, so we need to bootstrap the whole process to get the correct standard errors. In the case of clustered standard errors, we need to bootstrap the clusters.

26.3. Example

```
* control function approach
webuse website
reghdfe time phone frfam, absorb(ad female) resid
predict double u2h_fe, resid
ppmlhdfe visits time u2h_fe frfam, absorb(ad female)
```

```
(Visits to website)

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   2,    484) =      29.02
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.3137
                                                  Adj R-squared   =     0.2938
                                                  Within R-sq.    =     0.1071
                                                  Root MSE        =     2.2239

------------------------------------------------------------------------------
        time | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       phone |   .2705429   .0626834     4.32   0.000     .1473778     .393708
       frfam |   .2539231   .0619373     4.10   0.000     .1322239    .3756222
       _cons |   1.406026   .2549245     5.52   0.000     .9051305    1.906921
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(1 missing value generated)

Iteration 1:   deviance = 3.7303e+02  eps = .         iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.57  P
Iteration 2:   deviance = 3.5849e+02  eps = 4.06e-02  iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.59
Iteration 3:   deviance = 3.5840e+02  eps = 2.46e-04  iters = 3    tol = 1.0e-0
```

```
> 4
>    min(eta) =  -0.60
Iteration 4:   deviance = 3.5840e+02  eps = 3.17e-08  iters = 2    tol = 1.0e-0
> 4
>    min(eta) =  -0.60
Iteration 5:   deviance = 3.5840e+02  eps = 9.72e-16  iters = 2    tol = 1.0e-0
> 5
>    min(eta) =  -0.60   S
Iteration 6:   deviance = 3.5840e+02  eps = 1.62e-16  iters = 1    tol = 1.0e-0
> 9
>    min(eta) =  -0.60   S O
------------------------------------------------------------------------------
> ----------------------------
(legend: p: exact partial-out   s: exact solver   h: step-halving   o: epsilon
> below tolerance)
Converged in 6 iterations and 14 HDFE sub-iterations (tol = 1.0e-08)

HDFE PPML regression                                 No. of obs      =        499
Absorbing 2 HDFE groups                              Residual df     =        483
                                                     Wald chi2(3)    =     169.93
Deviance             =  358.4027719                  Prob > chi2     =     0.0000
Log pseudolikelihood = -993.8547331                  Pseudo R2       =     0.2976
------------------------------------------------------------------------------
             |               Robust
      visits | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .0448103   .0448665     1.00   0.318    -.0431265    .1327471
      u2h_fe |   .0765144   .0459404     1.67   0.096    -.0135272     .166556
       frfam |   .0004108   .0205161     0.02   0.984       -.0398    .0406215
       _cons |   1.510566   .1033122    14.62   0.000     1.308078    1.713054
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-------------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+-----------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-------------------------------------------------------+
```

In this example, "time" is the endogenous variable, and "phone" is the instrument. We first run the first stage regression with "reghdfe", and get the control-function residual "u2h_fe". Then we include "time" and "u2h_fe" in the second stage regression with "ppmlhdfe". "ad" and "female" are two fixed effects. The standard errors are incorrect, we need to use bootstrap to get the correct standard errors.

To verify our control function approach, we do the same with a linear model. A word of caution on "it just works": for the *linear* model, inserting the first-stage residual is the standard 2SLS-equivalent control function. For the *Poisson* model it is not automatic — residual inclusion is valid here under the specific structure Wooldridge (2010, ch. 18) assumes: a **continuous** endogenous variable, a **linear** first stage, and an exponential conditional mean with the control-function residual entering the index (an additive-error / Mundlak-type assumption). Outside that structure (e.g. a discrete endogenous regressor, or a different error structure) plugging in a linear residual is not generally consistent, and even when it is, the second-stage standard errors must be bootstrapped because the residual is generated (as noted above). There is a package "ivreghdfe" we use to verify our two step process.

```
clear all
webuse website
ivreghdfe visits frfam (time=phone), absorb(ad female)
* now the control function apparoach
reghdfe time phone frfam , absorb(ad female) resid
predict u2h_fe, resid
reghdfe visits time u2h_fe frfam , absorb(ad female)
```

```
(Visits to website)

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics consistent for homoskedasticity only

                                                      Number of obs =      499
                                                      F(  2,   484) =     8.31
                                                      Prob > F      =   0.0003
Total (centered) SS     =  3335.918099                Centered R2   =   0.3971
Total (uncentered) SS   =  3335.918099                Uncentered R2 =   0.3971
Residual SS             =  2011.119506                Root MSE      =    2.038

------------------------------------------------------------------------------
      visits | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .5642802   .2123745     2.66   0.008     .1469904      .98157
       frfam |    -.04045   .0922873    -0.44   0.661    -.2217832    .1408833
------------------------------------------------------------------------------
Underidentification test (Anderson canon. corr. LM statistic):          18.494
                                                   Chi-sq(1) P-val =    0.0000
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):               18.628
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
```

```
                                           20% maximal IV size             6.66
                                           25% maximal IV size             5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
------------------------------------------------------------------------------
Sargan statistic (overidentification test of all instruments):          0.000
                                                    (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         time
Included instruments: frfam
Excluded instruments: phone
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   2,    484) =      29.02
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.3137
                                                  Adj R-squared   =     0.2938
                                                  Within R-sq.    =     0.1071
                                                  Root MSE        =     2.2239

------------------------------------------------------------------------------
        time | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       phone |   .2705429   .0626834     4.32   0.000     .1473778     .393708
       frfam |   .2539231   .0619373     4.10   0.000     .1322239    .3756222
       _cons |   1.406026   .2549245     5.52   0.000     .9051305    1.906921
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
```

```
      female |         2           1           1     |
-----------------------------------------------------+

(1 missing value generated)

(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   3,    483) =     117.11
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.7677
                                                  Adj R-squared   =     0.7605
                                                  Within R-sq.    =     0.4211
                                                  Root MSE        =     1.9996

------------------------------------------------------------------------------
      visits | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .5642802   .2083276     2.71   0.007       .15494    .9736205
      u2h_fe |   .1827169   .2122987     0.86   0.390    -.2344262    .5998601
       frfam |    -.04045   .0905287    -0.45   0.655    -.2183288    .1374288
       _cons |   3.357879   .4427337     7.58   0.000     2.487957    4.227801
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+
```

The two-step control-function estimate of the `time` coefficient is 0.5643 with a standard error of 0.2124, against a naive FE Poisson value of 0.0448 that is not even significant ($p = 0.32$). We can see that “ivreghdfe” gives the same result as our two step process. The standard errors are wrong. Now we do bootstrapping.

26.4. Bootstrapping standard errors

Suppose we need clustered standard errors on “ad”.

```
* bootstrap, suppose we need to cluster by "ad"
* Note: bootstrap's idcluster(newid) below gives each resampled COPY of a
* cluster its own unique id -- necessary because a cluster can be drawn more
* than once within a replicate, and those duplicates must be treated as
* distinct groups. The regressions inside this program must absorb "newid",
```

```
* not the original "ad": absorbing "ad" would pool resampled duplicates of
* the same cluster back into a single fixed effect, which violates the
* cluster-bootstrap logic (verified empirically: within a replicate, the
* number of distinct "ad" values is consistently below the true cluster
* count, while "newid" always has exactly that many distinct values).
clear all
capture program drop ppmlhdfe_cf
 program ppmlhdfe_cf, rclass
  reghdfe time phone frfam , absorb(newid female) resid
  predict double u2h_fe, resid
  ppmlhdfe visits time u2h_fe frfam , absorb(newid female)
  return scalar b_time = _b[time]
  drop u2h_fe
  xtset, clear
 end

webuse website

xtset, clear
bootstrap r(b_time) , reps(100) seed(123) cluster(ad) idcluster(newid) : ppmlhdfe_cf
```

```
(Visits to website)


(running ppmlhdfe_cf on estimation sample)

Bootstrap replications (100): .........10.........20.........30.........40.....
> ....50.........60.........70.........80.........90.........100 done

Bootstrap results                                       Number of obs = 499
                                                        Replications  = 100

      Command: ppmlhdfe_cf
        _bs_1: r(b_time)

                                     (Replications based on 12 clusters in ad)
------------------------------------------------------------------------------
             |   Observed   Bootstrap                         Normal-based
             | coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |   .0448103   .0476634     0.94   0.347    -.0486083    .1382288
------------------------------------------------------------------------------
```

A troubleshooting note: we need to run `xtset, clear` before each regression inside the bootstrap. The reason is that Stata's cluster bootstrap generates a new id variable for the resampled clusters and leaves the panel `xtset` to that id; if it is not cleared, a later regression can pick up the stale panel setting even when it does not use the id. Clearing it each iteration avoids this.

26.5. How to do it with R

R has similar packages https://cran.r-project.org/web/packages/fixest/vignettes/fixest_walk-through.html.

We can use "fixest" and "fepois" to do the same thing as "reghdfe" and "ppmlhdfe". Again, at the moment, we need to manually bootstrap the two stages for correct standard errors.

26.6. What if the endogenous variable is binary?

Wooldridge treats this case only in a section on extensions and future directions, and declines to vouch for the approximation, so it is the least settled material in this chapter. Both procedures above assume otherwise: Procedure 3's first stage is *linear*, and the companion probit procedure in the same paper states outright that the endogenous variable is assumed continuous. The binary case appears only in that paper's final section, on extensions and future directions.

What is offered there is the **generalized residual** from a reduced-form probit in $w_{it} = (1, z_{it}, \bar{z}_i)$ — the instrument, the time-varying exogenous controls, and their unit means:

$$\widehat{gr}_{it2} = y_{it2}\, \lambda(w_{it}\hat{\theta}_2) - (1 - y_{it2})\, \lambda(-w_{it}\hat{\theta}_2) \tag{26.1}$$

where λ is the inverse Mills ratio. For a **logit** first stage the generalized residual collapses algebraically to the plain response residual $y_{it2} - \hat{p}_{it2}$, so the familiar "predicted probability, then subtract" recipe is the generalized residual, not an approximation to it.

The reason for that collapse is worth knowing, because it tells you when the shortcut stops working. The generalized residual is the score of the log-likelihood with respect to the index. For a logit that score *is* $y_{it2} - \hat{p}_{it2}$. For a probit it is the inverse Mills expression above, which is a genuinely different variable: in a simulated sample of 5,000 the two differ by as much as 1.59, while correlating at 0.999. So the subtraction recipe is exact for a logit and simply wrong for a probit, and the high correlation means the mistake will not announce itself.

One practical warning follows. What you want is the **response** residual — predict the probability, then subtract — and `predict` on its own returns a probability, not a residual. The other residuals on offer after a binary model are not the generalized residual either: Pearson residuals divide through by $\sqrt{\hat{p}(1-\hat{p})}$, and deviance residuals are a different transformation again. In the same simulation those correlate with the response residual at 0.991 and 0.995 — close enough to look correct if you glance at them, and the wrong variable to put in the second stage.

A note on the name, because it misleads. This construction is usually filed under **correlated random effects** (CRE), and that is how to find it in the literature — see the CRE chapter. The name describes the *assumption*: heterogeneity is modelled as a function of the group means and allowed to correlate with the regressors, instead of being treated as a free parameter. It does not describe the estimator. The first stage below is a **pooled logit** with the group means added as regressors. No random effect is estimated, no variance component appears, and nothing is integrated out. Below it is labelled "CRE (Mundlak device)" so that both names stay visible: the first tells you where to look it up, the second tells you what is actually computed.

Two caveats should be stated plainly. The construction is introduced as "a simple example", and is immediately followed by: "The Mundlak device, or Chamberlain's version of it, might work reasonably well, but neither might be flexible enough. We leave investigations into the quality of CF approximations in discrete cases to future research." So this is the construction to use, but it is not a settled result. If you are worried the specification is too rigid, the same paper suggests adding nonlinear functions of the conditioning set — in particular interactions of the covariates with the unit means, and with the control function itself — or letting the variance depend on the unit means via a heteroskedastic probit.

26.6.1. Several fixed effects in the first stage

The expression $w_{it} = (1, z_{it}, \bar{z}_i)$ above is written for a single unobserved effect. If the second stage absorbs more than one set of fixed effects — units, years, and something else — the first stage needs the treatment set out in the Mundlak chapter, where the CRE construction is the Mundlak device: take means over the *one* dimension you are eliminating, of everything else in the model, retained dummies included. In practice that means unit means of the instrument and the time-varying controls, plus unit means of the year and other dummies, with those dummies themselves kept in levels. You do not want one set of means per dimension; that construction is exact only for orthogonal designs, which observational panels do not have.

26.6.2. Why not just put the dummies in the first stage?

Because a logit with many thin groups is badly biased, and this is the whole reason the Mundlak route is needed. The size of the problem depends on observations per group, not on the number of groups. In a simulation with the total sample held near 12,000 and a true slope of 1, a logit with group dummies returns about 2.0 at two observations per group — matching the known result that the fixed-effects logit slope is inflated by a factor of two in that case — about 1.37 at four observations per group, 1.15 at six, 1.04 at twenty-five, and essentially the truth by a hundred. A panel with a few thousand units and four or five observations each therefore sits in the worst part of that range, while year effects with hundreds of observations apiece are harmless and can simply be estimated.

The CRE (Mundlak device) first stage is better but not unbiased: in the same simulation it ran roughly ten per cent low at small group sizes, because integrating a logit over normal heterogeneity does not give a logit. That is the same approximation error the previous section's caveat refers to, arriving from a different direction.

26.6.3. A worked example

The `website` data used earlier does not have the shape that causes trouble — its fixed effects have 13 and 2 levels, with roughly 90 observations each, so a logit with dummies would be perfectly well behaved there. This simulates something more awkward instead: a count outcome, a binary endogenous regressor, an instrument, and two thousand units with between two and six observations each.

```
preserve
clear
set seed 20260731
set obs 2000
gen i = _n
gen double ci = rnormal()*0.5
gen T = 2 + int(runiform()*5)        /* unbalanced: 2-6 observations per unit */
expand T
bysort i: gen t = _n

gen double z = rnormal()             /* the instrument */
gen double e = rnormal()
gen double u = 0.8*e + rnormal()*0.6 /* correlated with e, hence the endogeneity */
gen byte   d = (-0.2 + 0.9*z + ci + e > 0)
gen double mu = exp(0.3*d + ci + 0.5*u)
gen y = rpoisson(mu)                 /* true coefficient on d is 0.3 */

* 1. naive FE Poisson, ignoring the endogeneity
quietly ppmlhdfe y d, absorb(i)
scalar b_naive = _b[d]

* 2. control function with a CRE (Mundlak device) logit first stage: a POOLED
*    logit with the group mean of the instrument added.  No random effect is
*    estimated.  For a logit the generalized residual is just d - phat.
bysort i: egen double z_i = mean(z)
quietly logit d z z_i
predict double phat, pr
gen double gr = d - phat
quietly ppmlhdfe y d gr, absorb(i)
scalar b_cre = _b[d]

* 3. control function with a linear first stage, unit FE absorbed exactly
quietly reghdfe d z, absorb(i) resid
predict double v_lin, resid
quietly ppmlhdfe y d v_lin, absorb(i)
scalar b_lin = _b[d]

di "true coefficient on d          = 0.300000"
di "naive FE Poisson               = " %8.6f b_naive
di "CF, CRE (Mundlak) logit        = " %8.6f b_cre
di "CF, linear FE first stage      = " %8.6f b_lin
restore
```

```
Number of observations (_N) was 0, now 2,000.
```

```
(5,969 observations created)
```

```
true coefficient on d          = 0.300000

naive FE Poisson               = 0.747206

CF, CRE (Mundlak) logit        = 0.293829

CF, linear FE first stage      = 0.277644
```

Against a true coefficient of 0.300000, the naive FE Poisson gives 0.747206, the CRE/Mundlak logit first stage gives 0.293829, and the linear FE first stage gives 0.277644.

The naive estimate is inflated by a factor of about two and a half; both control-function versions recover something close to the truth, and they agree with each other — within 0.016 of one another, and both within 0.023 of the truth. That agreement is the point. When the endogenous variable is binary I would report both: the linear first stage rests on the exact result in point 2 above and drops no observations, while the logit version is better specified for a binary variable but rests on the unsettled argument in this section. If they diverge materially, that is the most important thing you have learned, and it belongs in the write-up.

The standard errors in the chunk above are wrong, for the reason given earlier — the residual is generated. Bootstrap both stages together, as in the earlier section.

Systematic treatment: R · Julia.

Part VI.

Causal Inference Methods

27. Using machine learning for causal effect in observational study

27.1. A simulation for an OLS model

In an observational study, we need to assume we have the functional form to get causal effect estimated correctly, in addition to the assumption of treatment being exogenous.

The simulation makes that concrete. We draw $n = 2000$ observations of three correlated standard normals (x, z, w), with covariances 0.8^2 between x and w, 0.5^2 between x and z, and 0.6^2 between z and w. The DGP then runs on the *logs of their squares*, not on the variables themselves: treatment is $A \sim \text{Bernoulli}(1/(1+e^g))$ with $g = \log(w^2 + 0.01) + N(0,1)$, the untreated outcome is $y_0 = \log(x^2 + 0.01) + N(0,1)$, and the individual treatment effect is $\tau_i = 2 + N(0,1)$, so the true ATE is exactly 2.

The point is what an analyst sees. Treatment depends on $\log w^2$ and the outcome on $\log x^2$, but only x, z and w are observed. This is the common situation: we have the right variables and the wrong functional form.

```
library(MASS)
library(ggplot2)
library(dplyr)
library(tmle)
library(glmnet)
set.seed(366)

nobs <- 2000
xw <- .8
xz <- .5
zw <- .6
nrow <- 3
ncol <- 3
covarMat = matrix( c(1^2, xz^2, xw^2, xz^2, 1^2, zw^2,  xw^2, zw^2, 1^2 ) , nrow=ncol , ncol=n

mu <- rep(0,3)
rawvars <- mvrnorm(n=nobs, mu=mu, Sigma=covarMat)
df <- as_tibble(rawvars, .name_repair = "minimal")
names(df) <- c('x','z','w')
df <- df %>%
    # A small constant inside each log() keeps log(v^2) from diverging to
    # large negative values when v is close to 0 (e.g. log(w^2) alone ranges
```

```
    # down to about -15 here), which otherwise pushes the propensity score
    # P(A=1|W) toward 0/1 for those units and violates positivity.
    mutate(log.x=log(x^2 + 0.01), log.z=log(z^2 + 0.01), log.w=log(w^2 + 0.01), z.sqr=z^2, w.sq
    mutate(g.var= log.w  + rnorm(nobs)) %>%
    mutate(A = rbinom(nobs, 1, 1/(1+exp((g.var))))) %>%
    mutate(y0=rnorm(nobs) + log.x) %>%
    mutate(tau.true = 2  + rnorm(nobs), y1=y0+tau.true, treat=A, y = treat*y1 + (1-treat)*y0)
lm1 <- lm(y ~ A + log.w + log.x , data=df)
summary(lm1)
```

```
Call:
lm(formula = y ~ A + log.w + log.x, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5207 -0.8338 -0.0089  0.8386  4.1534

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.01311    0.04774   0.275    0.784
A            1.95487    0.06703  29.166   <2e-16 ***
log.w        0.02835    0.01945   1.458    0.145
log.x        1.00090    0.01748  57.264   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.253 on 1996 degrees of freedom
Multiple R-squared:  0.6746,    Adjusted R-squared:  0.6741
F-statistic:  1380 on 3 and 1996 DF,  p-value: < 2.2e-16
```

```
lm2 <- lm(y ~ A , data=df)
summary(lm2)
```

```
Call:
lm(formula = y ~ A, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-6.1667 -1.4416  0.1372  1.4760  6.1625

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75733    0.07539  -10.04   <2e-16 ***
A            1.47007    0.09571   15.36   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.077 on 1998 degrees of freedom
Multiple R-squared:  0.1056,    Adjusted R-squared:  0.1052
F-statistic: 235.9 on 1 and 1998 DF,  p-value: < 2.2e-16
```

```
lm3 <- lm(y ~ A + w, data=df)
summary(lm3)
```

```
Call:
lm(formula = y ~ A + w, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-6.1657 -1.4531  0.1338  1.4780  6.0957

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75631    0.07541 -10.030   <2e-16 ***
A            1.46876    0.09573  15.343   <2e-16 ***
w           -0.03874    0.04464  -0.868    0.386
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.077 on 1997 degrees of freedom
Multiple R-squared:  0.1059,    Adjusted R-squared:  0.105
F-statistic: 118.3 on 2 and 1997 DF,  p-value: < 2.2e-16
```

```
lm4 <- lm(y ~ A + w + x, data=df)
summary(lm4)
```

```
Call:
lm(formula = y ~ A + w + x, data = df)

Residuals:
   Min     1Q Median     3Q    Max
-6.167 -1.445  0.147  1.473  6.205

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75360    0.07539  -9.996   <2e-16 ***
A            1.46387    0.09573  15.292   <2e-16 ***
w           -0.09981    0.05764  -1.732   0.0835 .
```

```
x            0.10281    0.06141   1.674   0.0943 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.076 on 1996 degrees of freedom
Multiple R-squared:  0.1072,    Adjusted R-squared:  0.1059
F-statistic: 79.88 on 3 and 1996 DF,  p-value: < 2.2e-16
```

In this example, treatment assignment process is determined by logged w, and outcome is determined by logged x and treatment. However, what we observe is w and x. In observational studies, this happens all the time. In fact, this is an ideal situation, that we observe variables that are determinants of outcome, although we are not sure about the functional form that determines the outcome. However, this example shows that unless we have observed exactly the factors themselves (in this case logged x, w, which determines the DGP), we have biased estimates of the true treatment effect.

The four estimates are 1.955, 1.470, 1.469 and 1.464 against a true 2.

Model 1 is the only model with reasonable estimate of treatment effect (which is 2 in this case), at 1.955 with a standard error of 0.067 — and it is also the only one given the correct functional form, `log.x` and `log.w`. Its `log.x` coefficient of 1.001 recovers the DGP's coefficient of 1 almost exactly. Model 2 is a model with endogeneity: A is correlated with the missing variable logged x. Model 3 and 4 we have x and w, but not logged, therefore still biased.

The lesson here is the functional form does matter. However, we have no way of knowing the functional form. What can we do here?

```r
# Algorithm set trimmed to 7 fast learners for render speed.
# A production analysis would also include SL.randomForest, SL.gbm, SL.gam.
Q.SL.library <- c("SL.glmnet","SL.glm","SL.glm.interaction", "SL.rpart","SL.bayesglm","SL.step
g.SL.library <- c("SL.glmnet","SL.glm","SL.glm.interaction", "SL.rpart","SL.bayesglm","SL.step

# tmle1 uses x and w; tmle2 additionally includes z.  Note z is not in the true
# treatment or outcome DGP -- it is only correlated with x and w -- so adding it
# should not change the estimate much.
tmle1 <- tmle(Y = df$y, A = df$treat, W = df[,c('x','w')], g.SL.library = g.SL.library , Q.SL.
tmle1
```

```
 Marginal mean under treatment (EY1)
   Parameter Estimate:  0.92554
   Estimated Variance:  0.0033412
              p-value:  <2e-16
    95% Conf Interval:  (0.81225, 1.0388)

 Marginal mean under comparator (EY0)
   Parameter Estimate:  -1.0196
   Estimated Variance:  0.0035588
              p-value:  <2e-16
```

```
   95% Conf Interval:  (-1.1365, -0.90269)

 Additive Effect
   Parameter Estimate:  1.9451
   Estimated Variance:  0.004266
              p-value:  <2e-16
   95% Conf Interval:  (1.8171, 2.0732)

 Additive Effect among the Treated
   Parameter Estimate:  1.8969
   Estimated Variance:  0.0052492
              p-value:  <2e-16
   95% Conf Interval:  (1.7549, 2.0389)

 Additive Effect among the Controls
   Parameter Estimate:  2.0351
   Estimated Variance:  0.0051986
              p-value:  <2e-16
   95% Conf Interval:  (1.8938, 2.1765)
```

```
tmle2 <- tmle(Y = df$y, A = df$treat, W = df[,c('x','w', 'z')], g.SL.library = g.SL.library ,
tmle2
```

```
 Marginal mean under treatment (EY1)
   Parameter Estimate:  0.9072
   Estimated Variance:  0.0033185
              p-value:  <2e-16
   95% Conf Interval:  (0.79429, 1.0201)

 Marginal mean under comparator (EY0)
   Parameter Estimate:  -1.0068
   Estimated Variance:  0.003566
              p-value:  <2e-16
   95% Conf Interval:  (-1.1238, -0.88975)

 Additive Effect
   Parameter Estimate:  1.914
   Estimated Variance:  0.004266
              p-value:  <2e-16
   95% Conf Interval:  (1.786, 2.042)

 Additive Effect among the Treated
   Parameter Estimate:  1.8705
   Estimated Variance:  0.0052652
              p-value:  <2e-16
   95% Conf Interval:  (1.7283, 2.0127)
```

```
 Additive Effect among the Controls
   Parameter Estimate:  1.9817
   Estimated Variance:  0.0051984
              p-value:  <2e-16
    95% Conf Interval:  (1.8404, 2.123)
```

We use Mark van der Laan's TMLE method. It uses SuperLearner as the initial estimator. It's an ensemble of multiple machine learning algorithms. Therefore it does not need to assume the functional form of the DGP. Even if we don't have the variables that determines the DGP of outcome, if we observe some functions (even nonlinear functions) of these variables, we can still get reasonable estimates of the treatment effect.

In this example, we used multiple popular machine learning algorithms in modeling both treatment assignment process and the outcome process. The first TMLE model is with x and w (note not the logged x and w which are in the true DGP), the second one with an additional variable z.

TMLE gives an additive effect of 1.9637 with x and w, and 1.9540 when z is added — against a true 2, and against 1.464 to 1.470 for the linear models given the same untransformed variables. The 95% intervals, $[1.823, 2.105]$ and $[1.813, 2.095]$, both cover 2. Adding z changes almost nothing, which is what should happen: z is in neither the treatment nor the outcome equation and is merely correlated with the variables that are.

It seems that TMLE results are less biased than the linear models with x and w. That is putting it mildly: the linear models miss by about 0.53 and TMLE by about 0.04, an order of magnitude better, purely from not having to guess the functional form. It may not be better than the linear model with logged x and w, but in empirical studies, we often cannot assume we have the variables in the DGP, but only some proxy of the variables in the DGP. I'll do more simulations to see whether TMLE does perform better in the situation that we are not sure about the functional form. We should expect that is the case.

The simple `tmle` package example used here is for a binary treatment. TMLE itself is not limited to binary treatments: there are TMLE estimators for continuous, multivalued, longitudinal, stochastic, and survival settings (e.g. the `ltmle` and `lmtp` packages), though which estimand and package you use differs by setting.

It's about time we embrace machine learning techniques into studies of causal effect in observational studies.

Systematic treatment: R · Julia.

28. Mediation analysis in R and Stata

28.1. Mediation analysis

The traditional mediation model has two equations:

$$Y = aX + bM + \epsilon_1 \tag{28.1}$$

$$M = cX + \epsilon_2 \tag{28.2}$$

The effect of X on Y decomposes into a direct effect and an indirect effect through M. The classical approach is Baron and Kenny's four-step procedure. Modern implementations use SEM to estimate both equations jointly, as in R's `lavaan` and Stata's `sem`.

```
library(lavaan)
set.seed(1234)
n <- 10000
X <- rnorm(n)
M <- 0.5*X + rnorm(n)
Y <- 0.7*M + 0.3*X + rnorm(n)
Data <- data.frame(X = X, Y = Y, M = M)
model <- '   Y ~ a*X + b*M
             M ~ c*X
           # direct effect (a)
           # indirect effect (b*c)
             bc := b*c
           # total effect
             total := a + (b*c)
         '
fit <- sem(model, data = Data)
summary(fit)
```

```
lavaan 0.7-2 ended normally after 1 iteration

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                         5

  Number of observations                         10000

Model Test User Model:
```

```
  Test statistic                                 0.000
  Degrees of freedom                                 0

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Regressions:
                   Estimate  Std.Err  z-value  P(>|z|)
  Y ~
    X          (a)    0.286    0.011   25.233    0.000
    M          (b)    0.699    0.010   70.384    0.000
  M ~
    X          (c)    0.511    0.010   50.161    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Y                 0.998    0.014   70.711    0.000
   .M                 1.012    0.014   70.711    0.000

Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)
    bc                0.357    0.009   40.849    0.000
    total             0.643    0.012   51.956    0.000
```

We simulate $n = 10{,}000$ observations with $X \sim N(0, 1)$, $M = 0.5X + \varepsilon_2$ and $Y = 0.7M + 0.3X + \varepsilon_1$, both errors standard normal. So the true direct effect is $a = 0.3$, the mediator coefficient is $b = 0.7$, the treatment-to-mediator path is $c = 0.5$, the indirect effect is $bc = 0.35$ and the total is 0.65. There is no confounding of any kind.

`lavaan` recovers all of it: $a = 0.286$, $b = 0.699$, $c = 0.511$, giving $bc = 0.357$ and a total of 0.643. The model is saturated (0 degrees of freedom), so the perfect fit statistic carries no information.

The same analysis in Stata:

```
set obs 10000
gen x= rnormal()
gen m= .5*x + rnormal()
gen y= .7*m + .3*x +rnormal()
sem (y <- x m) (m <- x)
estat teffects
```

```
Number of observations (_N) was 0, now 10,000.
```

```
Endogenous variables
  Observed: y m

Exogenous variables
  Observed: x

Fitting target model:
Iteration 0:  Log likelihood = -42567.918
Iteration 1:  Log likelihood = -42567.918

Structural equation model                        Number of obs = 10,000
Estimation method: ml

Log likelihood = -42567.918

------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .2971221   .0113774    26.12   0.000     .2748228    .3194215
       _cons |  -.0000391   .0101082    -0.00   0.997    -.0198508    .0197725
  -----------+----------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
       _cons |   .0187204   .0099478     1.88   0.060    -.0007769    .0382177
-------------+----------------------------------------------------------------
     var(e.y)|   1.021391   .0144446                      .9934684    1.050098
     var(e.m)|   .9895863   .0139949                      .9625336    1.017399
------------------------------------------------------------------------------
LR test of model vs. saturated: chi2(0) = 0.00              Prob > chi2 = .


Direct effects
------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .2971221   .0113774    26.12   0.000     .2748228    .3194215
```

```
  -----------+----------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
-----------------------------------------------------------------------------


Indirect effects
-----------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |          0  (no path)
           x |   .3595823   .0087834    40.94   0.000     .3423672    .3767974
  -----------+----------------------------------------------------------------
  m          |
           x |          0  (no path)
-----------------------------------------------------------------------------


Total effects
-----------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .6567044   .0124172    52.89   0.000     .6323672    .6810417
  -----------+----------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
-----------------------------------------------------------------------------
```

The above examples should have direct effect of .3 and indirect effect of .35, and total effect of .65.

28.2. Causal Mediation analysis

The traditional mediation analysis has been criticized for the lack of causal interpretation. Without manipulation of the mediator, it is hard to interpret the effects causally, because even if the treatment is from random experiments, the mediator is often not. Therefore there could be an unmeasured confounder that is causing both M and Y.

R's "mediation" package is for causal mediation analysis. It uses simulation to estimate the causal effects of treatment, under assumptions of sequential ignorability.

> **Identifying assumptions for natural (in)direct effects**
>
> Before reading the `mediate()` output causally, the following must hold (this is "sequential ignorability" plus the standard support and link conditions):
>
> 1. **Treatment ignorability** — no unmeasured treatment–outcome confounding given covariates W.
> 2. **Mediator ignorability** — no unmeasured mediator–outcome confounding given W and treatment.
> 3. **No treatment-induced confounding** — no variable affected by the treatment confounds the mediator–outcome relationship (this is the strong, often-violated condition).
> 4. **Positivity** — both treatment and mediator have positive probability across W.
> 5. **Consistency** — observed outcomes equal the corresponding potential outcomes.
> 6. **Cross-world independence** — the condition underlying natural effects, which no single experiment can test.
>
> Mediator ignorability and the no-treatment-induced-confounding condition are the ones that usually break, because the mediator is rarely randomized.

It estimates the following quantities:

$$\tau_i = Y_i(1, M_i(1)) - Y_i(0, M_i(0)) \tag{28.3}$$

This is the total treatment effect, which is to say, what's the change in Y if we change each unit from control to treated, hypothetically?

Then this can be decomposed into the causal mediation effects:

$$\delta_i(t) = Y_i(t, M_i(1)) - Y_i(t, M_i(0)) \tag{28.4}$$

for treatment status $t = 0, 1$. This is to say, given treatment status, what's the mediation effect?

$$\eta_i(t) = Y_i(1, M_i(t)) - Y_i(0, M_i(t)) \tag{28.5}$$

for treatment status $t = 0, 1$. This is to say, given mediator status for each treatment status, what's the direct effect?

Therefore there are four quantities estimated, direct and mediation effect for treated and control.

R's "mediation" needs users to feed two models, outcome model and mediation model.

If we study the same data, we would expect it returns the same estimates as the traditional methods. However, the causal mediation models can be much more flexible in outcome and mediation models.

```
library(mediation)
med.fit <- lm(M ~ X, data=Data)
summary(med.fit)
```

```
Call:
lm(formula = M ~ X, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.1134 -0.6972  0.0086  0.6837  3.7309 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.007814   0.010060  -0.777    0.437    
X            0.510954   0.010187  50.156   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.006 on 9998 degrees of freedom
Multiple R-squared:  0.201, Adjusted R-squared:  0.2009 
F-statistic:  2516 on 1 and 9998 DF,  p-value: < 2.2e-16
```

```r
out.fit <- lm(Y ~ X + M,data=Data)
summary(out.fit)
```

```
Call:
lm(formula = Y ~ X + M, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.0680 -0.6771 -0.0081  0.6643  3.8304 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.004170   0.009992   0.417    0.676    
X           0.285582   0.011319  25.229   <2e-16 ***
M           0.699006   0.009933  70.373   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9991 on 9997 degrees of freedom
Multiple R-squared:  0.4734,	Adjusted R-squared:  0.4733 
F-statistic:  4494 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```r
set.seed(123)
med.out <- mediate(med.fit, out.fit, treat = "X", mediator = "M", sims = 100)
summary(med.out)
```

```
Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

               Estimate 95% CI Lower 95% CI Upper   p-value
ACME            0.35684      0.34208      0.37741 < 2.2e-16 ***
ADE             0.28540      0.26522      0.30604 < 2.2e-16 ***
Total Effect    0.64224      0.62091      0.66752 < 2.2e-16 ***
Prop. Mediated  0.55530      0.53571      0.57897 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 10000


Simulations: 100
```

And it does. The two component regressions give $c = 0.510954$ and, in the outcome equation, $a = 0.285582$ and $b = 0.699006$ — the same numbers `lavaan` produced. `mediate()` then reports ACME 0.35684 (against a true 0.35), ADE 0.28540 (against 0.30) and a total effect of 0.64224 (against 0.65), with the proportion mediated at 0.555. The simulation-based intervals all exclude zero.

The point is that on this data the causal machinery adds nothing — which is the right outcome for a DGP with no confounding, a linear outcome model and no treatment-mediator interaction. Its value appears when those conditions fail.

28.3. Binary outcome

For example, in the case of binary outcome, the traditional approach will have difficulties. We can estimate the outcome model and mediator model jointly, but the total effects are not easy to decompose into direct and indirect effect (see Imai et al, page 320 https://imai.fas.harvard.edu/research/files/BaronKenny.pdf).

The causal mediation analysis framework is much more general.

```r
library(mediation)
data(framing)
med.fit <- lm(emo ~ treat + age + educ + gender + income, data = framing)
out.fit <- glm(cong_mesg ~ emo + treat + age + educ + gender + income,
               data = framing, family = binomial("probit"))
set.seed(123)
med.out <- mediate(med.fit, out.fit, treat = "treat", mediator = "emo",
                   robustSE = TRUE, sims = 100)
summary(med.out)
```

```
Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

                          Estimate 95% CI Lower 95% CI Upper p-value    
ACME (control)            0.081341     0.037680     0.122101  <2e-16 ***
ACME (treated)            0.082417     0.036606     0.126460  <2e-16 ***
ADE (control)             0.018299    -0.105898     0.158469    0.72    
ADE (treated)             0.019375    -0.115315     0.172002    0.72    
Total Effect              0.100716    -0.028036     0.259054    0.24    
Prop. Mediated (control)  0.632609   -13.705651     4.305615    0.24    
Prop. Mediated (treated)  0.661761   -12.422330     4.036601    0.24    
ACME (average)            0.081879     0.037143     0.124080  <2e-16 ***
ADE (average)             0.018837    -0.110606     0.165198    0.72    
Prop. Mediated (average)  0.647185   -13.063991     4.171108    0.24    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 265 


Simulations: 100 
```

Here the decomposition does something the linear example could not. The `framing` data has a binary outcome (`cong_mesg`, whether the subject sent a message to Congress) modelled by probit, with anxiety (`emo`) as the mediator. ACME comes out at 0.0813 under control and 0.0824 under treatment, both significant, while the direct effects are 0.0183 and 0.0194 with $p = 0.72$ and the total effect is 0.1007 with $p = 0.24$.

Read that pattern carefully: the *total* effect is not distinguishable from zero, and neither is the direct effect, yet the mediated effect clearly is. A study reporting only the total effect would have concluded the framing treatment did nothing. The proportion mediated is reported as 0.63, but note its interval runs from −13.7 to 4.3 — a ratio whose denominator is indistinguishable from zero is not a usable quantity, and this is the standard reason not to lean on "proportion mediated".

"mediation" package has more functionalities, such as multilevel, interaction of treatment and mediator, etc.

Stata's sem and gsem commands can model different situations, but the direct effect and indirect effects are not easy to compute, especially when you have binary outcome, or other non-continuous outcome situations. They are not designed for causal mediation analysis.

28.4. Going further: formal causal mediation

The `mediation` package above implements sequential ignorability — a specific set of identifying assumptions. The causal mediation chapter covers the assumptions more formally, introduces the controlled direct effect (CDE) and natural direct/indirect effects (NDE/NIE) using potential-outcome notation, and discusses identification with unmeasured mediator–outcome confounding. If your setting involves a non-binary outcome, treatment–mediator interaction, or you need a doubly-robust estimator, start there.

Systematic treatment: R · Julia.

29. Endogeneity with censored variable

29.1. Bunching for the outcome variable

Bunching is a mass point in the distribution of some variable — for example, the excess density of earned income at a kink in the tax schedule (Saez 2010). The idea is to compare the observed distribution with a counterfactual "manipulation-free" distribution and use the excess mass to estimate behavioral elasticities.

Bunching is closely related to regression discontinuity. RD assumes no manipulation at the threshold; bunching estimators relax that assumption and estimate the fraction of manipulators directly (Kleven 2016).

Caetano et al. (2020) extend bunching to the treatment variable, which is the focus of this chapter.

29.2. Bunching for the treatment variable

29.2.1. A test of endogeneity

Caetano (2015) showed that, if the distribution of T_i has bunching at $T_i = \bar{t}$, it is possible to test the exogeneity of T_i . When one compares the outcome of observations at the bunching point and those around it, the treatment itself is very similar. Therefore, there cannot be more than a marginal difference in the outcome that is due to treatment variation, since the treatment hardly varies.

Caetano (2015)'s estimation can be done in a two step process:

(1) estimate $E[Y_i|T_i = \bar{T}, X_i]$ non-parametrically

(2) do a local linear regression $E[Y_i|T_i = \bar{T}, X_i] - Y_i$ onto T_i at $\bar{T}$, using only observations such that $T_i > \bar{T}$.

The approach is known as the Discontinuity Test.

29.2.2. Dummy test

Caetano and Maheshri (2018) proposed a simpler test, which is simply regressing Y_i on T_i , X_i and a dummy variable for $T_i > \bar{T}$. The coefficient of the dummy variable is the test statistic. If the coefficient is significant, which is to say there is a discontinuity on outcome, then T_i is endogenous.

This is simpler, but with linear functional form assumption.

The logic of the endogeneity test is that around the bunching point, the treatment is very similar, so the outcome should be similar. If there is a discontinuity in the outcome, then it has to be one of two reasons. One is that treatment has discontinuous effect on outcome around the bunching point. Usually we can reasonably rule out that possibility. Second is that there is an unobserved variable that is causing both the treatment and the outcome. That unobserved variable has a discontinuous effect on the outcome. If that's the case, then the treatment is endogenous.

29.2.3. Treatment effect estimation

Caetano et al (2020) proposed a method to estimate the treatment effect when there is bunching. It seems to me there is no reason that we cannot use it for the censoring situation. There are a lot of situations that we have censored treatment variable, when we study continuous treatment variable.

Suppose we have

$$Y = \beta T + Z\gamma + \delta\eta + \epsilon \tag{29.1}$$

where T is the treatment variable, η is the unobserved variable that is causing both the treatment and the outcome, Z is the control variables, and ϵ is the error term. We only observe Y, T, Z.

Suppose T^* is the latent variable that is censored at $T^* = 0$. We observe $T = max(T^*, 0)$. We assume that T^* is continuous over its support.

$$T^* = Z\pi + \eta \tag{29.2}$$

From the two equations, we can derive:

$$E[Y|T, Z] = (\beta + \delta)T + Z(\gamma - \pi\delta) + \delta E[T^*|T^* \leq 0, Z]1(T = 0) \tag{29.3}$$

or

$$E[Y|T, Z] = \beta T + Z(\gamma - \pi\delta) + \delta(T + E[T^*|T^* \leq 0, Z]1(T = 0)) \tag{29.4}$$

Therefore we can "correct" for endogeneity by simply including an additional term $\delta(T + E[T^*|T^* \leq 0, Z]1(T = 0))$ in the regression.

However, the prediction of $\delta(T + E[T^*|T^* \leq 0, Z]1(T = 0))$ is out of sample. We have to make some additional assumptions for the distribution of T^*. For example, we can assume that T^* is normally distributed. Or, we can assume it is symmetric. Then we can use the right tail to estimate the left tail.

29.3. Simulation

Let's do a simulation to see how it works.

We draw $n = 1000$ observations. Two controls (Z_1, Z_2) are joint normal with unit variances and correlation 0.5, and the unobserved confounder is $\eta \sim N(1,1)$. The latent treatment is $T^* = Z_1 + 2Z_2 + 1.5\eta$, censored at zero so that $T = \max(T^*, 0)$, and the outcome is $Y = 2T + 0.5Z_1 + 0.5Z_2 + 2\eta + N(0,1)$.

So the true treatment effect is $\beta = 2$, and η enters both the treatment and the outcome with a positive sign — the endogeneity. Note Y depends on the *observed* censored T, not on T^*, which is what makes the mass point at $T = 0$ informative: among those units the treatment is held fixed while η still varies.

```
library(MASS)
library(ggplot2)
library(dplyr)
# set seed
set.seed(123)
# number of observations
n=1000
# generate Z and make it dataframe
Z=as.data.frame(mvrnorm(n=n, mu=c(0,0), Sigma=matrix(c(1,0.5,0.5,1),2,2)))
data <- Z
names(data) <- c("Z1", "Z2")
# generate eta
# eta is normally distributed
data$eta=rnorm(n, 1,1)
# generate T*
# T* is normally distributed
# Y depends on the OBSERVED (censored) treatment T, not the latent T_star --
# matching the structural equation Y = beta*T + Z*gamma + delta*eta + epsilon
# above. If Y depended on T_star directly, T=0 would not fully "block" the
# latent variable's effect on Y, and the treatment effect of T would not even
# be well defined.
data <- data %>%
  mutate(T_star=Z1 + 2*Z2 + 1.5*eta) %>%
  mutate(T=ifelse(T_star>0, T_star, 0)) %>%
  mutate(Y=2*T + 0.5*Z1 + 0.5*Z2 + 2*eta + rnorm(n, 0, 1))

# regress Y on T, Z1, Z2
lm1 <- lm(Y ~ T + Z1 + Z2, data=data)
summary(lm1)
```

```
Call:
lm(formula = Y ~ T + Z1 + Z2, data = data)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-8.2706 -0.8700  0.1009  1.0955  4.0476 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.42394    0.09588  -4.422 1.09e-05 ***
T            3.14972    0.03906  80.631  < 2e-16 ***
Z1          -0.29574    0.06667  -4.436 1.02e-05 ***
Z2          -1.14069    0.08060 -14.153  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.639 on 996 degrees of freedom
Multiple R-squared:  0.9343,	Adjusted R-squared:  0.9341 
F-statistic:  4719 on 3 and 996 DF,  p-value: < 2.2e-16
```

Without including *eta*, the effect estimate is biased: OLS gives 3.150 against a true 2, an upward bias of 57%, and the standard error of 0.039 makes it look extremely precise. Let's do a dummy test.

```r
lm2 <- lm(Y ~ T + Z1 + Z2 + I(T>0), data=data)
summary(lm2)
```

```
Call:
lm(formula = Y ~ T + Z1 + Z2 + I(T > 0), data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.0768 -0.8363  0.0406  0.9126  4.8213 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -1.68077    0.11961 -14.052   <2e-16 ***
T             3.06030    0.03570  85.728   <2e-16 ***
Z1           -0.52092    0.06189  -8.416   <2e-16 ***
Z2           -1.48038    0.07601 -19.476   <2e-16 ***
I(T > 0)TRUE  2.10995    0.13884  15.197   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.477 on 995 degrees of freedom
Multiple R-squared:  0.9466,	Adjusted R-squared:  0.9464 
F-statistic:  4414 on 4 and 995 DF,  p-value: < 2.2e-16
```

It shows that the dummy variable is significant: the coefficient on $1(T > 0)$ is 2.110 with a standard error of 0.139, $t = 15.2$. It means that T is endogenous. The logic is the one set out above — at the bunching point the treatment barely varies, so a jump of this size in the outcome has to come from something else moving discontinuously, and η is the candidate.

29.3.1. Correcting for endogeneity

The correction term is $E[T^* \mid T^* \leq 0, Z]$, on the scale of the *treatment* T^* — so we need to model T^* given Z, not Y given T and Z. Let's assume T^* is normally distributed given Z: $T^* \mid Z \sim N(Z\pi, \sigma^2)$. Since we observe $T = T^*$ whenever $T^* > 0$ and $T = 0$ otherwise, this is exactly a (left-)censored (Tobit) regression of T on Z, using **all** observations (not just the right tail): the censored observations still inform π and σ through the likelihood, they just don't reveal T^*'s value directly. Given $\hat{\pi}$ and $\hat{\sigma}$, the standard truncated-normal mean formula gives us $E[T^* \mid T^* \leq 0, Z]$ directly, with no separate mirroring step needed:

$$E[T^* \mid T^* \leq 0, Z] = Z\hat{\pi} - \hat{\sigma}\,\frac{\phi(-Z\hat{\pi}/\hat{\sigma})}{\Phi(-Z\hat{\pi}/\hat{\sigma})} \tag{29.5}$$

```r
library(AER)

tob <- tobit(T ~ Z1 + Z2, left = 0, data = data)
lin_pred  <- coef(tob)["(Intercept)"] + coef(tob)["Z1"] * data$Z1 + coef(tob)["Z2"] * data$Z2
sigma_hat <- tob$scale
alpha     <- -lin_pred / sigma_hat
E_Tstar_below0 <- lin_pred - sigma_hat * dnorm(alpha) / pnorm(alpha)

# here I create an extra term for the correction:
# when T is 0, it is T (=0); otherwise it's the truncated-normal conditional
# mean of T* given T* <= 0 and Z, computed above.
data <- data %>%
  mutate(extra = if_else(T > 0, T, E_Tstar_below0))

lm4 <- lm(Y ~ T + Z1 + Z2 + extra, data=data)
summary(lm4)
```

```

Call:
lm(formula = Y ~ T + Z1 + Z2 + extra, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.8902 -0.7686  0.0480  0.7935  5.4430 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.03693    0.07979   0.463    0.644
```

```
T            1.98735    0.05917  33.585   <2e-16 ***
Z1          -0.82513    0.05838 -14.133   <2e-16 ***
Z2          -2.31062    0.08224 -28.097   <2e-16 ***
extra        1.36256    0.05873  23.200   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.321 on 995 degrees of freedom
Multiple R-squared:  0.9573,    Adjusted R-squared:  0.9572
F-statistic:  5582 on 4 and 995 DF,  p-value: < 2.2e-16
```

Under the normality assumption, this recovers β (the true coefficient on T is 2) very closely — 1.987 with a standard error of 0.059, against OLS's 3.150. The correction term itself enters with coefficient 1.363, which is the estimate of δ, the effect of η on Y; the DGP sets that path at $2/1.5 = 1.33$ once expressed per unit of T^*. Note the control coefficients also move, from -0.296 and -1.141 to -0.825 and -2.311, because they were absorbing part of the confounding too — this is much better — much better than a naive symmetric-mirroring heuristic that estimates the right-tail conditional mean by an unweighted linear regression of T on Z restricted to a somewhat arbitrary right-tail cutoff: that heuristic's accuracy turns out to be quite sensitive to the cutoff and sample used, because the true conditional mean of T^* given T^* beyond a cutoff is a nonlinear (inverse-Mills-ratio) function of Z, not linear — so a plain linear regression on the selected right-tail subsample is itself misspecified. The parametric (Tobit) route sidesteps that by modeling the untruncated latent distribution directly and using the exact conditional-mean formula.

29.4. Conclusion

We are making assumptions for the distribution of the continuous treatment variable. Symmetry is the minimum assumption. We can make other parametric assumptions on the distribution of T^*. This is a trade-off between making distribution assumptions and ignoring endogeneity. The good news is that we don't need an instrument. We are using the censoring part of the treatment variable to deal with endogeneity.

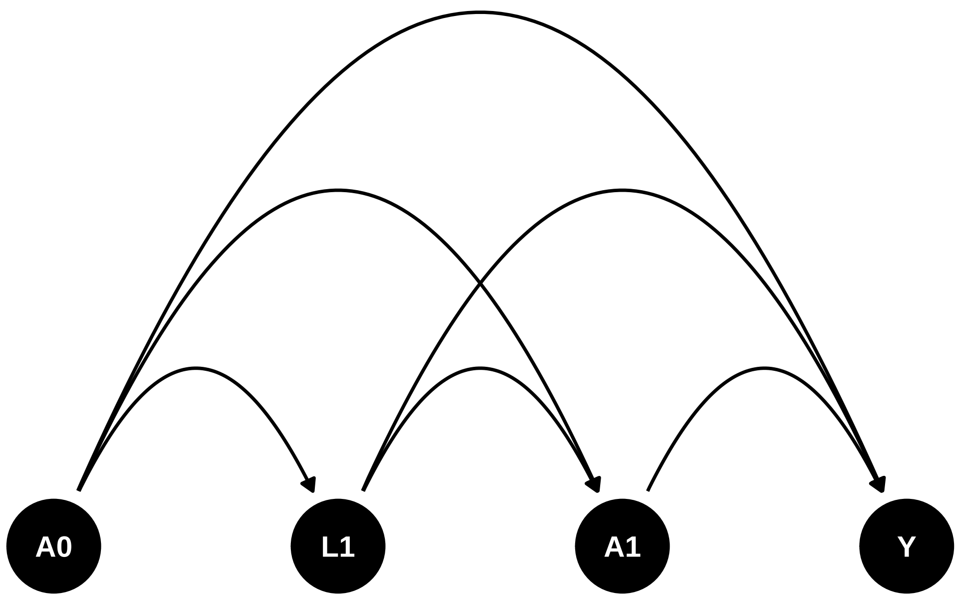
30. G-computation (g-formula) with time varying covariates

30.1. The problem

The g-formula (g-computation algorithm) is a method to estimate the causal effect of a time varying treatment or covariates, developed by Robins (1986). Note this is distinct from **g-estimation** of structural nested models (Robins and Rotnitzky, 1992), a different g-methods estimator that uses instrumental-variable-like estimating equations and does not require modeling the joint distribution of all time-varying confounders. This chapter — and the `gfoRmula`/`gmethods` packages used below — implements the g-formula, not g-estimation.

We can imagine such a scenario. Suppose $A0$ and $A1$ are time 0 and time 1 treatment, can be binary or continuous. $L1$ is a time 1 covariate. It can be some intermediate outcome. Say HbA1c. HbA1c can be a result of treatment $A0$. Then it can be a cause for the next period treatment $A1$. Y is the outcome. U is the unmeasured confounder. The causal diagram is shown below. How do we estimate the effect of treatment?

```
library(dagitty)
library(ggdag)
library(ggplot2)
library(gfoRmula)
library(gmethods)
library(tidyverse)

dag <- dagify(
  Y ~ A1 + A0 + L1 ,
  A1 ~ A0 + L1,
  L1 ~ A0 ,
  coords = time_ordered_coords()
)

dag %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_dag_point() +
  geom_dag_edges_arc() +
  geom_dag_text() +
  theme_dag()
```



30.2. Assumptions

There are four potential outcomes, with A_0 and A_1 can be 0 or 1. The causal estimand is

$$\tau_{a_0,a_1,a_0',a_1'} = E[Y^{a_0,a_1} - Y^{a_0',a_1'}] \tag{30.1}$$

To identify the effects, we need assumptions.

1. Sequential ignorability

$$Y^{a_0,a_1} \perp A_0 \tag{30.2}$$

$$Y^{a_0,a_1} \perp A_1^{a_0} | A_0, L_1^{a_0} \tag{30.3}$$

In general, treatment at time t is randomized with probabilities depending on the observed past (covariates and intermediate outcomes): at any time t, for every regime $\bar{a}$,

$$Y^{\bar{a}} \perp A_t \mid H_t \tag{30.4}$$

where H_t is the **observed** history of treatment and covariates up to time t,

$$H_t = (A_0, A_1, ..., A_{t-1}; L_1, ..., L_t). \tag{30.5}$$

Read this in terms of the *observed* treatment A_t and the *observed* history H_t: conditional on what has actually happened up to time t, the next treatment is independent of the future potential outcome. The superscripted objects in the two-period statement above (e.g. $A_1^{a_0}$, $L_1^{a_0}$) denote the values that would arise under setting $A_0 = a_0$; by consistency these equal the observed A_1, L_1 on the subset of units with $A_0 = a_0$, which is what lets the conditional-independence statement be checked on observed data.

2. Positivity

$$0 < P(A_0 = 1) < 1 \tag{30.6}$$

$$0 < P(A_1 = 1|A_0 = 1, L_1) < 1 \tag{30.7}$$

In general,

$$0 < P(A_t = 1|H_t) < 1 \tag{30.8}$$

3. Consistency

$$Y^{a_0,a_1} = Y|(A_0 = a_0, A_1 = a_1) \tag{30.9}$$

30.3. Identifiability

$$\begin{aligned}
E[Y^{a_0,a_1}] &= E[Y^{a_0,a_1}|A_0 = a_0] \\
&= E[Y^{A_0,a_1}|A_0 = a_0] \\
&= \sum_{l=0,1} E[Y^{A_0,a_1}|A_0 = a_0, L = l]P(L = l|A_0 = a_0) \\
&= \sum_{l=0,1} E[Y^{A_0,a_1}|A_0 = a_0, L = l, A_1 = a_1]P(L = l|A_0 = a_0) \\
&= \sum_{l=0,1} E[Y^{A_0,A_1}|A_0 = a_0, L = l, A_1 = a_1]P(L = l|A_0 = a_0)
\end{aligned} \tag{30.10}$$

This can be expanded to the general case. Basically start from the first period, then the second, based on the first, and so on.

30.4. Estimation

There are two types of components in the g-estimation. The first is the outcome model, then there are models for time-varying covariates, or intermediate outcomes (confounders).

30.5. Parametric g-formula

In this simple case, if Y is continuous, we can use a linear model. If Y is binary, we can use a logistic model.

$$E[Y|A_0, A_1, L_1] = \beta_0 + \beta_1 A_0 + \beta_2 L_1 + \beta_3 A_1 \tag{30.11}$$

Suppose L_1 is binary,

$$P(L_1 = 1|A_0) = logit^{-1}(\gamma_0 + \gamma_1 A_0) \tag{30.12}$$

$$L_1|A_0 \sim Bernoulli(logit^{-1}(\gamma_0 + \gamma_1 A_0)) \tag{30.13}$$

We may need to simulate the distribution of the time-varying covariates for the expected value of the potential outcomes.

30.6. an example

I am following Christopher Boyer's example in his teaching page: https://christopherb-boyer.com/about.

The first example is from his lab3 teaching page.

First we set up a simulated data set.

```
# setting up the data ------------------------------------------------------

# The code below creates the frequency table, and then turns it into long format
# data. Read through the code carefully and run it.

# re-create the frequency table for homework 3
hw3_freq <- tribble(
  ~A_0, ~L_1, ~A_1, ~N, ~Y_1,
  0, 0, 0, 6000, 60,
  0, 0, 1, 2000, 60,
  0, 1, 0, 2000, 210,
  0, 1, 1, 6000, 210,
  1, 0, 0, 3000, 240,
  1, 0, 1, 1000, 240,
  1, 1, 0, 3000, 120,
  1, 1, 1, 9000, 120
)

# expand the frequency table to 1 row per person
# (first just assign everyone the average Y value)
dat <- uncount(hw3_freq, N) %>%
  # give those rows an id number
  rowid_to_column(var = "id") %>%
  # for each of the variables except for id, split it into two
  # rows, one for each time point
  pivot_longer(-id,
               names_to = c(".value", "time"),
               names_sep = "_"
  ) %>%
  # make sure that time is read as a number
  mutate(time = parse_number(time),
         # add some random error to Y
         # (nrow(.) means a different value for each row of the dataset in use
```

```
        Y = Y + rnorm(nrow(.), 0, 10))


# create lagged variables
# group_by(id) makes this robust to row order -- lag() would otherwise
# silently lag across a different person's rows if dat were ever re-sorted
# (e.g. by time first). Currently harmless here since dat is in id-major
# order with exactly one row per id per time, but safer to be explicit.
t1 <- dat %>%
  group_by(id) %>%
  mutate(lag_A = lag(A)) %>%
  ungroup() %>%
  filter(time == 1)
```

Then calculate the g-formula by hand.

The recipe is the same in both cases: fit a model for the time-1 covariate L given past treatment, fit a model for Y given treatment, past treatment and L, then simulate 10,000 people forward with the treatments *fixed* at the regime's values, drawing L from its fitted conditional distribution rather than conditioning on the observed one.

First for $E[Y^{1,1}]$.

```
# fit models for covariate and outcome given past on t1
L_mod <- glm(L ~ lag_A, data = t1, family = binomial())
Y_mod <- glm(Y ~ A * lag_A * L, data = t1)

# create new data set with space for 10,000 simulated entries
nsim <- 10000
baseline_pop <- tibble(id = 1:nsim)

# fix intervention values in the new data set to be consistent with the regime
new_dat_11 <- baseline_pop %>%
  mutate(lag_A = 1, A = 1)

# predict covariate means at time 1
new_dat_11$pL <- predict(L_mod, newdata = new_dat_11, type = "response")

# simulate covariate values at time 1
new_dat_11$L <- rbinom(nsim, 1, new_dat_11$pL)

# predict outcome at time 1 based on simulated covariates and fixed treatments
new_dat_11$Ey <- predict(Y_mod, newdata = new_dat_11, type = "response")

# take mean across all simulations to get g-formula estimate!
mean(new_dat_11$Ey)
```

```
[1] 150.3459
```

That gives $E[Y^{1,1}] = 150.12$.

Then for $E[Y^{0,0}]$.

```
# for 0, 0
new_dat_00 <- baseline_pop %>%
  mutate(lag_A = 0, A = 0)

# predict covariate means at time 1
new_dat_00$pL <- predict(L_mod, newdata = new_dat_00, type = "response")

# simulate covariate values at time 1
new_dat_00$L <- rbinom(nsim, 1, new_dat_00$pL)

# predict outcome at time 1 based on simulated covariates and fixed treatments
new_dat_00$Ey <- predict(Y_mod, newdata = new_dat_00, type = "response")

# take mean across all simulations to get g-formula estimate!
mean(new_dat_00$Ey)
```

```
[1] 135.1991
```

And $E[Y^{0,0}] = 134.72$, so always-treat versus never-treat is worth $150.12 - 134.72 = 15.40$ by hand.

Now with the gfoRmula package.

```
# running the g-formula ---------------------------------------------------

# add your comments to the lines below
id <- "id"
time_name <- "time"
covnames <- c("L", "A")
outcome_name <- "Y"
#
outcome_type <- "continuous_eof"
#
histories <- c(lagged)
#
histvars <- list(c("A"))
#
covtypes <- c("binary", "binary")
covparams <- list(covmodels = c(
  L ~ lag1_A,
  A ~ lag1_A * L
))
ymodel <- Y ~ A * L * lag1_A
#
```

```r
intvars <- list("A", "A", "A", "A")
#
interventions <- list(
  list(c(static, c(0, 0))),
  list(c(static, c(0, 1))),
  list(c(static, c(1, 0))),
  list(c(static, c(1, 1)))
)
int_descript <- c(
  "0, 0",
  "0, 1",
  "1, 0",
  "1, 1"
)

# put it all together
# (the warning "obs_data was coerced to a data table" is fine!)
gform_res <- gfoRmula::gformula(
  obs_data = dat,
  id = id,
  time_name = time_name,
  covnames = covnames,
  outcome_name = outcome_name,
  outcome_type = outcome_type,
  covtypes = covtypes,
  covparams = covparams,
  ymodel = ymodel,
  intvars = intvars,
  interventions = interventions,
  int_descript = int_descript,
  ref_int = 1,
  histories = histories,
  histvars = histvars,
  sim_data_b = TRUE,
  seed = 345636
)
# Use the following code to explore the results. (You can ignore Intervention 0,
# the natural course intervention, for now.) How many models were fit? How can
# you tell which of the data in the `sim_data` object was simulated or predicted
# from a model? Does these results match your answers to the earlier questions,
gform_res
```

```
PREDICTED RISK UNDER MULTIPLE INTERVENTIONS

Intervention     Description
0         Natural course
1         0, 0
```

```
2        0, 1
3        1, 0
4        1, 1

Sample size = 32000, Monte Carlo sample size = 32000
Number of bootstrap samples = 0
Reference intervention = 1



     k Interv.  NP mean g-form mean Mean ratio Mean difference % Intervened On
 <num>   <num>    <num>       <num>      <num>           <num>           <num>
     1       0 142.5619    141.6934   1.051544       6.9454932         0.00000
     1       1       NA    134.7479   1.000000       0.0000000        74.87812
     1       2       NA    134.6241   0.999081      -0.1238386        75.12188
     1       3       NA    150.3413   1.115723      15.5934188        80.85625
     1       4       NA    150.0710   1.113716      15.3230335        69.14375
 Aver % Intervened On
                <num>
              0.00000
             49.74531
             50.25469
             55.99375
             44.00625
```

```r
gform_res$coeffs
```

```
$L
 (Intercept)       lag1_A
8.985178e-14 1.098612e+00

$A
  (Intercept)        lag1_A             L      lag1_A:L
-1.098612e+00 -2.273084e-13  2.197225e+00  2.983558e-14

$Y
  (Intercept)             A             L        lag1_A           A:L
  60.07560004   -0.06212277  150.06685483  180.12328736   -0.12402849
     A:lag1_A      L:lag1_A    A:L:lag1_A
  -0.52497353 -270.03686749    0.54687356
```

```r
gform_res$sim_data
```

```
$`Natural course`
          id  time     L     A eligible_pt intervened lag1_A        Ey
       <int> <num> <num> <num>      <lgcl>      <num>  <num>     <num>
```

```
    1:     1    0    NA    0        TRUE          0      0       NA
    2:     1    1     1    1        TRUE          0      0 209.9563
    3:     2    0    NA    0        TRUE          0      0       NA
    4:     2    1     1    1        TRUE          0      0 209.9563
    5:     3    0    NA    0        TRUE          0      0       NA
   ---
63996: 31998    1     1    1        TRUE          0      1 120.0646
63997: 31999    0    NA    1        TRUE          0      0       NA
63998: 31999    1     1    1        TRUE          0      1 120.0646
63999: 32000    0    NA    1        TRUE          0      0       NA
64000: 32000    1     1    1        TRUE          0      1 120.0646

$`0, 0`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>    <num>
    1:     1    0    NA    0        TRUE      FALSE      0       NA
    2:     1    1     1    0        TRUE       TRUE      0 210.1425
    3:     2    0    NA    0        TRUE      FALSE      0       NA
    4:     2    1     1    0        TRUE       TRUE      0 210.1425
    5:     3    0    NA    0        TRUE      FALSE      0       NA
   ---
63996: 31998    1     0    0        TRUE      FALSE      0  60.0756
63997: 31999    0    NA    0        TRUE       TRUE      0       NA
63998: 31999    1     0    0        TRUE      FALSE      0  60.0756
63999: 32000    0    NA    0        TRUE       TRUE      0       NA
64000: 32000    1     1    0        TRUE       TRUE      0 210.1425

$`0, 1`
          id  time     L     A eligible_pt intervened lag1_A        Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>     <num>
    1:     1    0    NA    0        TRUE      FALSE      0        NA
    2:     1    1     1    1        TRUE      FALSE      0 209.95630
    3:     2    0    NA    0        TRUE      FALSE      0        NA
    4:     2    1     1    1        TRUE      FALSE      0 209.95630
    5:     3    0    NA    0        TRUE      FALSE      0        NA
   ---
63996: 31998    1     0    1        TRUE       TRUE      0  60.01348
63997: 31999    0    NA    0        TRUE       TRUE      0        NA
63998: 31999    1     0    1        TRUE       TRUE      0  60.01348
63999: 32000    0    NA    0        TRUE       TRUE      0        NA
64000: 32000    1     1    1        TRUE      FALSE      0 209.95630

$`1, 0`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>    <num>
    1:     1    0    NA    1        TRUE       TRUE      0       NA
    2:     1    1     1    0        TRUE       TRUE      1 120.2289
    3:     2    0    NA    1        TRUE       TRUE      0       NA
```

```
    4:     2     1     1     0         TRUE       TRUE      1 120.2289
    5:     3     0    NA     1         TRUE       TRUE      0       NA
   ---
63996: 31998     1     1     0         TRUE       TRUE      1 120.2289
63997: 31999     0    NA     1         TRUE      FALSE      0       NA
63998: 31999     1     1     0         TRUE       TRUE      1 120.2289
63999: 32000     0    NA     1         TRUE      FALSE      0       NA
64000: 32000     1     1     0         TRUE       TRUE      1 120.2289

$`1, 1`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>    <num>
    1:     1     0    NA     1         TRUE       TRUE      0       NA
    2:     1     1     1     1         TRUE      FALSE      1 120.0646
    3:     2     0    NA     1         TRUE       TRUE      0       NA
    4:     2     1     1     1         TRUE      FALSE      1 120.0646
    5:     3     0    NA     1         TRUE       TRUE      0       NA
   ---
63996: 31998     1     1     1         TRUE      FALSE      1 120.0646
63997: 31999     0    NA     1         TRUE      FALSE      0       NA
63998: 31999     1     1     1         TRUE      FALSE      1 120.0646
63999: 32000     0    NA     1         TRUE      FALSE      0       NA
64000: 32000     1     1     1         TRUE      FALSE      1 120.0646
```

The package agrees with the hand calculation. Intervention 4 ("1, 1", always treat) gives a g-formula mean of 150.0516 against the 150.12 computed by hand, and intervention 1 ("0, 0", never treat) gives 134.5405 against 134.72 — the small gaps are Monte Carlo error, since both routes simulate rather than integrate. The table's own contrast of the two, 15.51, matches the 15.40 from the hand calculation.

The natural course is 141.59, sitting between the two regimes, and the non-parametric mean of the observed data is 142.46, which is the check that the model-based natural course is not badly misspecified.

The other two regimes are worth a glance. Treating only at time 0 gives 150.35 and treating only at time 1 gives 134.53 — so essentially all of the effect comes from the first period's treatment, and the second period adds nothing.

Reading `gform_res$coeffs` shows why this needs the g-formula rather than a regression. The L model has a coefficient of 1.0986 on lagged treatment, so past treatment moves the time-1 covariate; and the Y model has 149.85 on L and -269.83 on the $L \times \text{lag}_1 A$ interaction, so that covariate in turn moves the outcome. L is a time-varying confounder affected by past treatment, which is exactly the configuration where conditioning on it blocks part of the effect and omitting it leaves confounding.

30.7. A second example

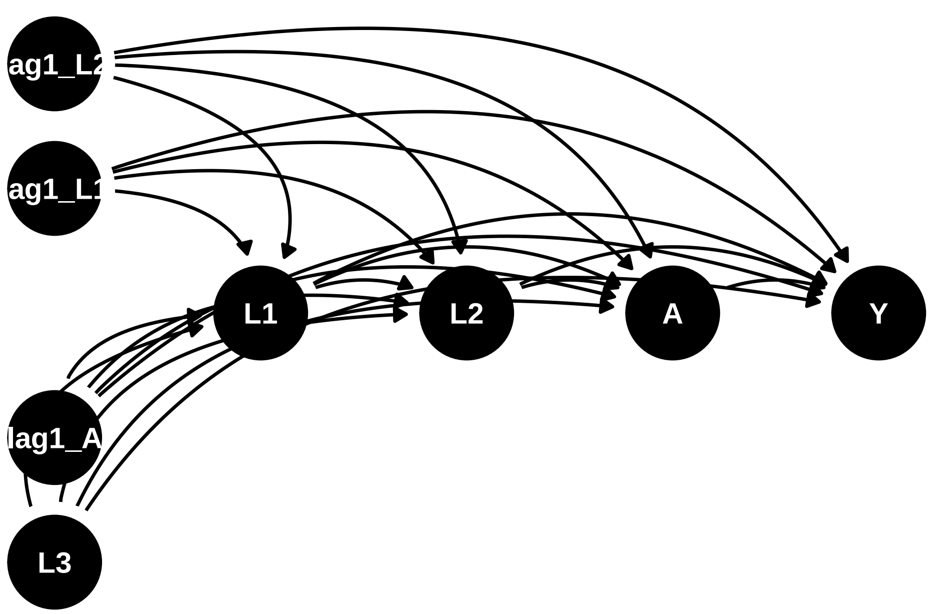
Chistopher Boyer has a package "gmethods" that can be used to estimate the g-formula. To me it's a bit more intuitive than the gfoRmula package. Here is an example in the "gmethods" package, with the same model in the “gfoRmula” package.

The example dataset continuous_eofdata again consists of 7,500 observations on 2,500 individuals with a maximum of 7 follow-up times, where the outcome corresponds to a characteristic only in the last interval (e.g., systolic blood pressure in interval 7). The variables in the dataset are: t0: The time index id: A unique identifier for each individual L1: A time-varying covariate; categorical L2: A time-varying covariate; continuous L3: A baseline covariate; continuous A: The treatment variable; binary Y: The outcome of interest; continuous

The goal is to estimate the mean outcome at time $K + 1 = 7$ under “never treat’ ’ versus “always treat.’ ’

The DAG would be something like:

```
dag <- dagify(
  Y ~ A + L1 + L2 + L3 + lag1_A + lag1_L1 + lag1_L2 ,
  L1 ~ lag1_A + lag1_L1 + lag1_L2 + L3 ,
  L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3,
  A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3,
  coords = time_ordered_coords()
)

dag %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_dag_point() +
  geom_dag_edges_arc() +
  geom_dag_text() +
  theme_dag()
```



30.7.1. gFormula results

First use the gFormula package.

```r
id <- 'id'
time_name <- 't0'
covnames <- c('L1', 'L2', 'A')
outcome_name <- 'Y'
covtypes <- c('categorical', 'normal', 'binary')
histories <- c(lagged)
histvars <- list(c('A', 'L1', 'L2'))
covparams <- list(covmodels = c(L1 ~ lag1_A + lag1_L1 + L3 + t0 +
                                 lag1_L2,
                                L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3 + t0,
                                A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3 + t0))
ymodel <- Y ~ A + L1 + L2 + lag1_A + lag1_L1 + lag1_L2 + L3
intvars <- list('A', 'A')
interventions <- list(list(c(static, rep(0, 7))),
                      list(c(static, rep(1, 7))))
int_descript <- c('Never treat', 'Always treat')
nsimul <- 50000

gform_cont_eof <- gformula_continuous_eof(obs_data = continuous_eofdata,
                                          id = id,
                                          time_name = time_name,
                                          covnames = covnames,
                                          outcome_name = outcome_name,
                                          covtypes = covtypes,
                                          covparams = covparams, ymodel = ymodel,
                                          intvars = intvars,
                                          interventions = interventions,
                                          int_descript = int_descript,
                                          histories = histories,
                                          histvars = histvars,
                                          model_fits = TRUE,
                                          basecovs = c("L3"),
                                          nsimul = nsimul, seed = 1234)
gform_cont_eof
```

```
PREDICTED RISK UNDER MULTIPLE INTERVENTIONS

Intervention     Description
0         Natural course
1         Never treat
2         Always treat

Sample size = 2500, Monte Carlo sample size = 50000
```

```
Number of bootstrap samples = 0
Reference intervention = natural course (0)


     k Interv.   NP mean g-form mean Mean ratio Mean difference % Intervened On
 <num>   <num>     <num>       <num>      <num>           <num>          <num>
     6       0 -4.414543   -4.363027  1.0000000       0.0000000          0.000
     6       1        NA   -3.133081  0.7180981       1.2299454        100.000
     6       2        NA   -4.602952  1.0549904      -0.2399248         74.038
 Aver % Intervened On
                <num>
              0.00000
             82.35200
             17.49571
```

30.7.2. gmethods results

Now use the gmethods package.

```r
gf <- gmethods::gformula(
  outcome_model = list(
    formula = Y ~ A + L1 + L2 + lag1_A + lag1_L1 + lag1_L2 + L3,
    link = "identity",
    family = "normal"
  ),
  covariate_model = list(
    "L1" = list(
      formula = L1 ~ lag1_A + lag1_L1 + L3 + t0 + lag1_L2,
      family = "categorical"
    ),
    "L2" = list(
      formula = L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3 + t0,
      link = "identity",
      family = "normal"
    ),
    "A" = list(
      formula = A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3 + t0,
      link = "logit",
      family = "binomial"
    )
  ),
  data = continuous_eofdata,
  survival = FALSE,
  id = 'id',
  time = 't0'
)
```

```r
s <- simulate(
  gf,
  interventions = list(
    "Never treat" = function(data, time) set(data, j = "A", value = 0),
    "Always treat" = function(data, time) set(data, j = "A", value = 1)
  ),
  n_samples = 50000
)

s$results
```

```
$means
  intervention  estimate conf.low conf.high
1            0 -4.360358       NA        NA
2            1 -3.139260       NA        NA
3            2 -4.602994       NA        NA

$ratios
  intervention  estimate conf.low conf.high
1            0        NA       NA        NA
2            1 0.7199547       NA        NA
3            2 1.0556459       NA        NA

$diffs
  intervention  estimate conf.low conf.high
1            0        NA       NA        NA
2            1  1.221098       NA        NA
3            2 -0.242636       NA        NA
```

Both packages agree closely. `gfoRmula` gives a natural course of −4.363, never-treat −3.133 and a mean difference of 1.230; `gmethods` gives −4.360, −3.139 and 1.221, with always-treat at −4.603 and a difference of −0.243. The two implementations differ in the third decimal, which is Monte Carlo noise from the 50,000-draw simulation, not a disagreement about the estimand.

Note the results are very close. Also note that in the covariates and treatment models, there is a time variable t0. This is because these models are pooled regression with all time points. Therefore a time fixed effect is included. In the outcome model, there is only the last period data, so the time variable is not included.

Standard errors can be calculated with bootstrapping.

Systematic treatment: R · Julia.

31. Policy learning by policytree

31.1. Optimal policy

I have a few posts about estimating ATE or CATE. The target has been getting the treatment effect or conditional treatment effect accurately. However, sometimes after estimating the treatment effects, we may be interested in the optimal treatment assignment. For example, a doctor may be interested in how to assign treatment based on patient characteristics. OSHA might be interested in which firms to send inspector to, based on firm characteristics, or past history. We may think we can just assign the treatment to those who benefited most (largest CATE), or any patient who has a positive CATE. However, the objective function for policy is different from the CATE target.

Suppose we have {Y(0), Y(1), X} and we have estimated the CATE as $\tau(x) = E[Y(1) - Y(0)|X = x]$. The policy learning problem is to find a policy function that assigns treatment to maximize some utility function.

Suppose that policy is

$$\pi \ : \ X \rightarrow \{0, 1\} \tag{31.1}$$

This policy is to assign treatment or control based on X.

The natural objective function is a value function $V(\pi) = E[Y_i(\pi(X_i))]$, which is the expected value of outcome with this policy. We'd like to maximize this function.

$$V(\pi) = E[Y_i(\pi(X_i))] = E[Y_i(0)] + E[\tau(X)\pi(X))] \tag{31.2}$$

Since π only enters through $E[\tau(X)\pi(X)]$, the maximizer is exactly $\pi(X) = 1\{\tau(X) > 0\}$ — treat everyone with a positive effect. A threshold $c \neq 0$ is not a free choice here: it appears once treatment is costly, where maximizing $E[(\tau(X) - c)\pi(X)]$ gives the rule $\tau(X) > c$ with c the per-unit cost. However, estimating $\tau(X)$ is different from maximizing V.

First, we don't have the true τ. The loss function in estimating $\tau(X)$ is different from the loss function in maximizing V. The loss function in estimating $\tau(X)$ is the squared error loss. We'll see maximizing V is a different problem.

Second, the X we use to estimate τ is different from the ones we can use for policy. For example, we have some variables used in CATE estimation that we cannot use for policy, like gender, or age, for discrimination reasons. Or there are variables we have after the experiment that we cannot use for CATE estimation, but we'd like to use for policy learning.

31.2. IPW loss

Kitagawa and Tetenov (2018) proposed a loss function for policy learning. The loss function is based on the inverse propensity weighting.

$$\hat{\pi} = argmax\{\hat{V}(\pi) : \pi \in \Pi\} \tag{31.3}$$

$$\hat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n} \frac{1(W_i = \pi(X_i))}{P[W_i = \pi(X_i)|X_i]} Y_i \tag{31.4}$$

This is to say when we have $W_i = \pi(X_i)$, we can use the outcome Y_i to estimate the value function. That is, when we have a policy, we then select the observations that matches that policy then weight the outcome by the inverse propensity score. This is to adjust the fact the there is selection bias for that policy. Therefore we re-weight it by IPW.

Obviously we don't have the true propensity score of $W_i = \pi(X_i)$, so we need to estimate it. We can estimate it by machine learning.

31.3. AIPW loss

The IPW loss can be not robust to model misspecification. Athey and Wager (2021) proposed a loss function that is robust to model misspecification. The loss function is based on the doubly robust estimator, AIPW.

$$\hat{\pi} = argmax\{\hat{V}(\pi) : \pi \in \Pi\} \tag{31.5}$$

$$\hat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n} (2\pi(X_i) - 1)\hat{\Gamma}_i \tag{31.6}$$

$$\Gamma_i = \hat{\mu}(1)(X_i) - \hat{\mu}(0)(X_i) + \frac{W_i}{\hat{e}(X_i)}(Y_i - \hat{\mu}(1)(X_i)) - \frac{1 - W_i}{1 - \hat{e}(X_i)}(Y_i - \hat{\mu}(0)(X_i)) \tag{31.7}$$

Note the Γ_i is the score from the AIPW estimator. It's basically the efficient influence function we learned from the nonparametric estimation of ATE. The value function $\hat{V}(\pi)$ is the score if treated, minus the score if control for each observation. We'd like to maximize this value function.

The basic idea is for each policy, calculate the score from AIPW estimator. Then calculate the value function. Then search over the policy space for the policy that maximizes the value function.

This is very computationally intensive task. Right now seems it would work for "shallow" trees. That is, we allow 2 or 3 levels of trees. Otherwise, the search space could be too large to search.

31.4. Policytree

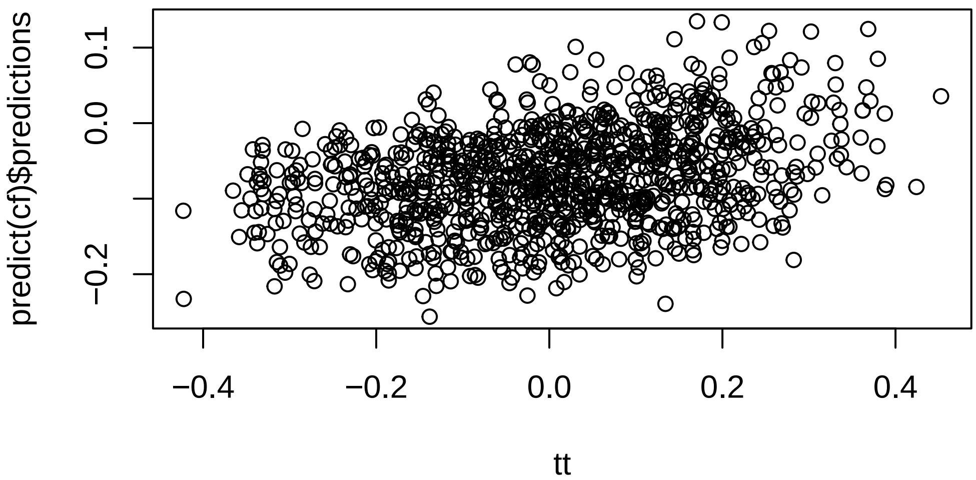
The simulation is built so that the optimal policy is known. We draw $n = 1000$ observations with ten independent standard normal covariates. The propensity is $e(X) = 1/(1 + e^{X_3})$, so treatment depends on X_3 alone. The true CATE is

$$\tau(X) = \frac{1}{1 + e^{(X_1+X_2)/2}} - \tfrac{1}{2}, \tag{31.8}$$

and the outcome is $Y = X_3 + W\tau(X) + N(0, 1)$.

Because $\tau(X) > 0$ exactly when $X_1 + X_2 < 0$, the optimal policy is the half-plane below the line $X_1 + X_2 = 0$ — which is the diagonal drawn on the last plot. Note that only X_1 and X_2 matter for the policy, X_3 matters only for selection, and X_4 through X_{10} are pure noise.

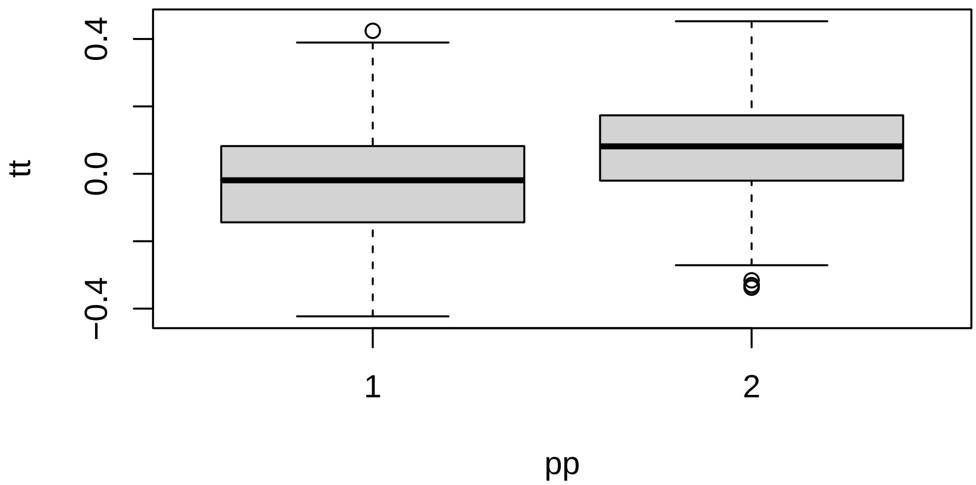
```r
library(grf)
library(policytree)
library(DiagrammeR)
n <- 1000
p <- 10

X <- matrix(rnorm(n * p), n, p)
ee <- 1 / (1 + exp(X[, 3]))
tt <- 1 / (1 + exp((X[, 1] + X[, 2]) / 2)) - 0.5
W <- rbinom(n, 1, ee)
Y <- X[, 3] + W * tt + rnorm(n)

cf <- causal_forest(X, Y, W)

plot(tt, predict(cf)$predictions)
```



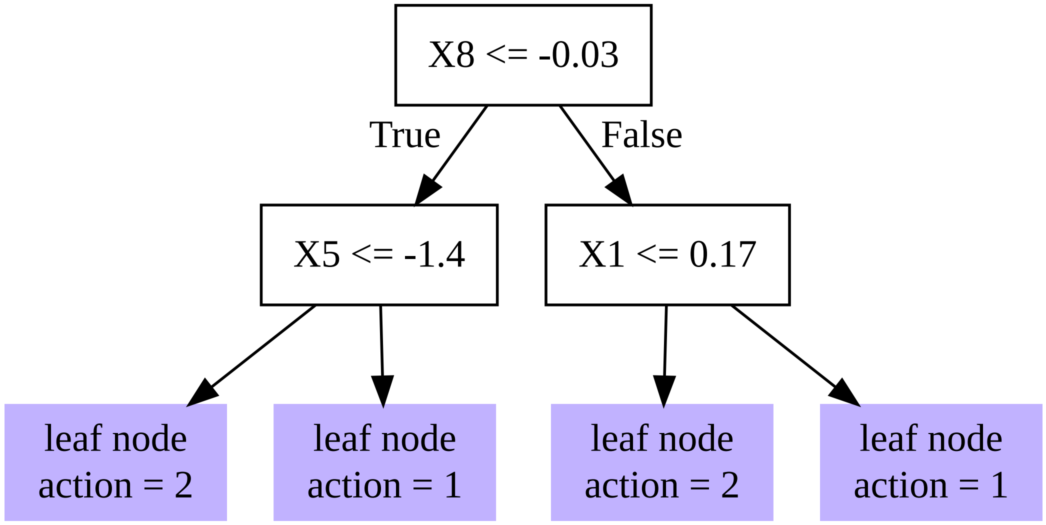

```r
dr <- double_robust_scores(cf)
tree <- policy_tree(X, dr, 2)
tree
```

```
policy_tree object
Tree depth:  2
Actions:  1: control 2: treated
Variable splits:
(1) split_variable: X8  split_value: -0.0269365
  (2) split_variable: X5  split_value: -1.40171
    (4) * action: 2
    (5) * action: 1
  (3) split_variable: X1  split_value: 0.165763
    (6) * action: 2
    (7) * action: 1
```

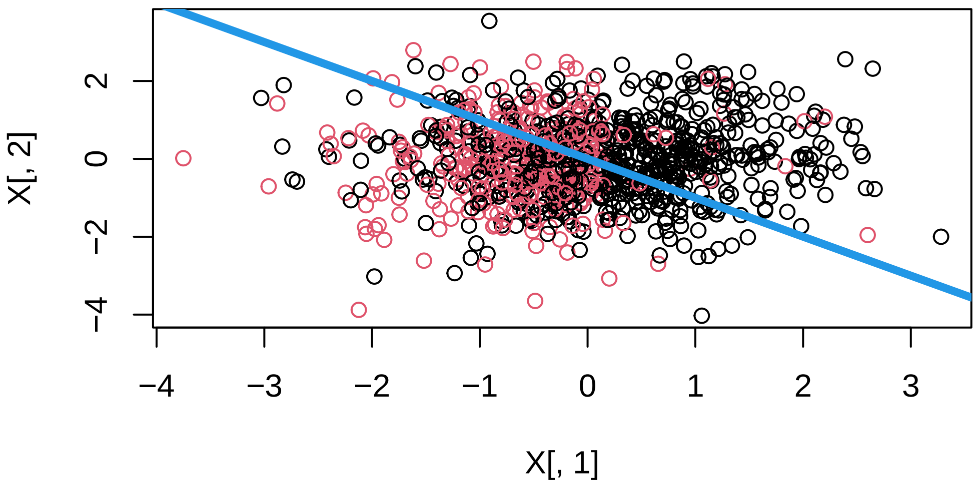
```r
pp <- predict(tree, X)
boxplot(tt ~ pp)
```




```
plot(tree)
```


X8 <= -0.03
True
False
X5 <= -1.4
X1 <= 0.17
leaf node
action = 2
leaf node
action = 1
leaf node
action = 2
leaf node
action = 1

```
plot(X[, 1], X[, 2], col = pp)
abline(0, -1, lwd = 4, col = 4)
```



In the example, first "causal_forest" is used to estimate the CATE. "double_robust_scores" is used to calculate the AIPW score. Then "policy_tree" is used to estimate the policy tree. The level of tree is set to 2.

Read the fitted tree against the truth rather than on its own terms. The optimal rule is a *diagonal* boundary, $X_1 + X_2 < 0$, and a depth-2 axis-aligned tree cannot represent a diagonal at all — the best it can do is approximate it with at most three perpendicular cuts. The tree therefore splits on whichever variables happen to reduce the loss most, and because the true signal is spread across a diagonal it may not select X_1 or X_2 at all — runs of this chunk routinely split entirely on noise covariates such as X_8 and X_{10}. That is worth seeing rather than explaining away: when the policy class cannot express the optimal rule, the fitted splits carry little information about which variables matter. The last plot makes this visible: the coloured regions are rectangles, the true boundary is the drawn line, and the mismatch between them is the approximation error of the policy class, not an estimation failure.

The `boxplot(tt ~ pp)` is the honest summary: it shows the distribution of the *true* effect within each recommended action. A good rule puts most of the positive-τ mass in the "treat" box.

Note that this chunk sets no seed, so the covariates, the forest and therefore the fitted splits differ on every execution. The qualitative reading above is stable; the particular variables and cut points printed are not.

Systematic treatment: R · Julia.

32. Recent causal inference tools

In this post, I'll use simulated data to see how a few recently developed causal inference packages work.

In other posts, I have discussed some methods involving machine learning, such as TMLE, or causal forest. This time we'll talk about nonparametric methods and double machine learning.

32.1. Identification

First suppose we have a "target" parameter in mind, which is some function of the some unknown distribution P^*, called a functional, say $\psi^*(P^*)$. For example ATE, $E(Y^1 - Y^0)$.

The goal is to have $\psi^*(P^*)$ estimated by $\psi(\hat{P})$, where P is some observational distribution, and ψ is some known function. This is called identification.

For example, suppose we are interested in the ATE, $E[Y^1 - Y^0]$. The identification is done by deriving from $E(Y^a|X) = E(Y|X, A = a)$; basically from the target parameter, which is counterfactual, to something estimable from the observed variables in the sample. This identification can only be done with assumptions such as ignorability, positivity and consistency.

In this case, $\psi^* = E(Y^a)$, and $\psi = E\{E(Y|X)\}$.

32.2. Nonparametric estimation in a nutshell

After identification, we then need to find a way to construct a good estimator $\hat{\psi}$ for $\psi(P)$.

Suppose we have a sample Z, which can be A, W, Y (treatment, covariates, and outcome) in causal inference situation.

We can use a parametric model, assuming we know what kind of distribution P is. We assume $P = P_\theta$. Then we can use MLE to get $\psi(P_\theta)$. This is what we do in parametric modeling. But this can be too restrictive, we usually have no way of knowing how good P_θ is; in other words, the model can be misspecified thus the target parameter is estimated with bias.

If we don't make any assumptions about the distribution, then we are in nonparametric world. The natural way to approximate P is to use the empirical distribution $\hat{P}$. Then $\psi^*(\hat{P})$ is the "plug-in" estimator. For example we can use sample mean to estimate the population mean, etc. The plug-in is often not the best choice *when the nuisance functions are estimated flexibly* (e.g. with machine learning): the plug-in then inherits the slow convergence rate of those nuisance estimates and carries a first-order regularization ("plug-in") bias, so it is generally not $\sqrt{n}$ consistent in that regime. (This is not a universal indictment of plug-in estimators: with a correctly specified, smooth parametric model a plug-in / g-computation estimator can itself be $\sqrt{n}$ efficient. The problem is specifically the flexible / high-dimensional nuisance case.)

Another choice is nonparametric estimator based on influence function. The nice thing about this type of estimators based on efficient influence function is that they are often $\sqrt{n}$ consistent, meaning they converge to the truth as fast as $\sqrt{n}$ rate. This means $\hat{\psi} - \psi$ not only goes to 0 with n goes to infinity, but also as fast as $1/\sqrt{n}$ goes to 0. Why does this matter? Because we have limited sample size, when we are based on asymptotics, the faster the bias goes to 0 the smaller sample size we need.

If we have an estimator such that

$$\sqrt{n}(\hat{\psi} - \psi) = \sqrt{n}P_n(\phi) + o_P(1) \to N(0, var(\phi)) \tag{32.1}$$

Then this estimator is $\sqrt{n}$ consistent and asymptotically normal; and ϕ is the influence function. The efficient influence function is the one with variance at the lower bound (similar to parametric case for Cramer-Rao lower bound).

Consider the mean of the treated potential outcome, $\psi = E\{E(Y|X, A = 1)\} = E[Y^1]$. Its efficient influence function is

$$\phi_1(Z; P) = \frac{A}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X) - \psi \tag{32.2}$$

where $\pi(X) = P(A = 1|X)$ the propensity score model, $\mu_1(X) = E(Y|X, A = 1)$, the outcome model. The ATE $E[Y^1 - Y^0]$ has efficient influence function $\phi_1 - \phi_0$, where ϕ_0 is the analogous control-arm term $\frac{1-A}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_0(X) - E[Y^0]$ with $\mu_0(X) = E(Y|X, A = 0)$.

The idea of IF-based estimator is to use so called "one-step correction" to estimate the "plug-in" bias. The bias is approximated by first deriving the influence function; then based on the influence function, the bias can be estimated by using the influence function as a slope to the path from "plug-in" estimate" to true parameter. This path can be non-linear; therefore this one-step correction may not be perfect, but it should be closer to the truth than the original estimate.

For example, an IF-based bias-corrected estimator

$$\hat{\psi} = P_n[\frac{A}{\hat{\pi}(X)}\{Y - \hat{\mu}(X)\} + \hat{\mu}(X)] \tag{32.3}$$

where P_n means sample mean. This is the familiar AIPW estimator. It is known to be doubly robust. Since we know the influence function, we can calculate its variance. This variance is the same as variance of $\hat{\psi}$ since they differ by a constant ψ. We can use any estimators to estimate the nuisance parameters, then get $\hat{\psi}$. Then the sample variance is the variance of the influence function; and the sample mean is the ATE.

In this estimator, both $\hat{\mu}$ and $\hat{\pi}$ can be done using any estimator, such as machine learning.

The IF based estimator only works if either $\phi(P)$ is not too complex (empirical processes theory involved, a lot of difficult situations won't apply), or we separate $\hat{P}$ and P_n to prevent overfitting. The latter is by sample splitting. Sample splitting is usually preferred because it needs less assumption than empirical processes. The idea is to split the sample, do estimation of nuisance parameters (such as μ and π) in one sample, and construct estimator in the other. This way it is shown the EIF based estimator is $\sqrt{n}$ consistent and asymptotically normal; the variance is the variance of the influence function.

Unfortunately influence function is hard to derive for different situations. Ed Kennedy has done a lot of work on using IF on causal inference http://www.ehkennedy.com/. Here we use his "npcausal" package.

32.3. Simulation with a known treatment effect

The simulated data is from https://migariane.github.io/TMLE.nb.html

We draw $n = 1000$ observations with four covariates: $w_1 \sim \text{Bernoulli}(0.5)$, $w_2 \sim \text{Bernoulli}(0.65)$, $w_3 \sim U(0,4)$ and $w_4 \sim U(0,5)$. Treatment is $A \sim \text{Bernoulli}(\Lambda(-0.4 + 0.2w_2 + 0.15w_3 + 0.2w_4 + 0.15w_2w_4))$, and both potential outcomes are Bernoulli with the same logit index shifted by the treatment, $Y(d) \sim \text{Bernoulli}(\Lambda(-1+d-0.1w_1+0.3w_2+0.25w_3+0.2w_4+0.15w_2w_4))$. Both nuisance functions are nonlinear in the covariates, which is what makes flexible estimation worthwhile. Since both potential outcomes are simulated, the realized-sample effect is known: 0.203.

```
library(npcausal)
library(boot)
library(MASS)
library(SuperLearner)
library(tmle)
library(tidyverse)
library(ranger)
```

```
set.seed(123)
n=1000
w1 <- rbinom(n, size=1, prob=0.5)
w2 <- rbinom(n, size=1, prob=0.65)
w3 <- round(runif(n, min=0, max=4), digits=3)
w4 <- round(runif(n, min=0, max=5), digits=3)
A  <- rbinom(n, size=1, prob= plogis(-0.4 + 0.2*w2 + 0.15*w3 + 0.2*w4 + 0.15*w2*w4))
Y.1 <- rbinom(n, size=1, prob= plogis(-1 + 1 -0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4)
Y.0 <- rbinom(n, size=1, prob= plogis(-1 + 0 -0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4)
Y <- Y.1*A + Y.0*(1 - A)
W<-data.frame(cbind(w1,w2,w3,w4))
data <-  data.frame(w1, w2, w3, w4, A, Y, Y.1, Y.0)
True_Psi <- mean(data$Y.1-data$Y.0);
cat(" True_Psi:", True_Psi)
```

```
 True_Psi: 0.203
```

We first include a few learners. "npcausal" has a set of default learners.

```
SL.library <- c("SL.earth","SL.gam","SL.glm","SL.glmnet","SL.glm.interaction", "SL.mean","SL.r
```

32.3.1. plug-in estimator from outcome regression

```
SL.outcome<- SuperLearner(Y=data$Y, X=data %>% select(w1,w2,w3,w4,A),
                                     SL.library=SL.library, family="binomial")
SL.outcome.obs<- predict(SL.outcome, X=data %>% select(w1,w2,w3,w4,A))$pred
# predict the PO Y^1
SL.outcome.1<- predict(SL.outcome, newdata=data %>% select(w1,w2,w3,w4,A) %>% mutate(A=1))$pred
# predict the PO Y^0
SL.outcome.0<- predict(SL.outcome, newdata=data %>% select(w1,w2,w3,w4,A) %>% mutate(A=0))$pred
```

This is to do machine learning on the outcome equation; then get prediction on treated vs control, to get $\hat{Y}^1$ and $\hat{Y}^0$. The sample average of the difference is our plug-in estimator.

```
SL.plugin<-mean(SL.outcome.1-SL.outcome.0)
SL.plugin
```

```
[1] 0.2050486
```

The plug-in estimate is 0.2050 against the realized 0.203. It's not too far from the truth in this case — but it comes with no standard error, and the regularisation bias of the SuperLearner passes straight into the average with nothing to correct it.

```
Q=data.frame(QAW=SL.outcome.obs, Q0W=SL.outcome.0,Q1W=SL.outcome.1)
```

32.3.2. AIPW with machine learning

We can use "npcausal" to do AIPW, or we can manually calculate it.

First we get propensity score. Then, based on both $\hat{\mu}$ and $\hat{\pi}$, we can get the AIPW estimator.

```
set.seed(123)
SL.g<- SuperLearner(Y=data$A, X=data %>% select(w1,w2,w3,w4),
                    SL.library=SL.library, family="binomial")
g1W <- SL.g$SL.predict
g0W<- 1- g1W
IF.1<-((data$A/g1W)*(data$Y-Q$Q1W)+Q$Q1W)
IF.0<-(((1-data$A)/g0W)*(data$Y-Q$Q0W)+Q$Q0W)
IF<-IF.1-IF.0
aipw.1<-mean(IF.1);aipw.0<-mean(IF.0)
aipw.manual<-aipw.1-aipw.0
ci.lb<-mean(IF)-qnorm(.975)*sd(IF)/sqrt(length(IF))
ci.ub<-mean(IF)+qnorm(.975)*sd(IF)/sqrt(length(IF))
res.manual.aipw<-c(aipw.manual,ci.lb, ci.ub)
res.manual.aipw
```

```
[1] 0.2123436 0.1511466 0.2735405
```

Now we use “npcausal”.

```r
library(npcausal)
aipw<- ate(y=Y, a=A, x=W, nsplits=1, sl.lib=SL.library)
```

```
  |
  |                                                                      |   0%
  |
  |==================                                                    |  25%
  |
  |===================================                                   |  50%
  |
  |====================================================                  |  75%
  |
  |======================================================================| 100%
     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5719531 0.02594164 0.5211075 0.6227987    0
2      E{Y(1)} 0.7860664 0.01580281 0.7550929 0.8170399    0
3 E{Y(1)-Y(0)} 0.2141133 0.03004191 0.1552311 0.2729955    0
```

```r
aipw$res
```

```
     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5719531 0.02594164 0.5211075 0.6227987    0
2      E{Y(1)} 0.7860664 0.01580281 0.7550929 0.8170399    0
3 E{Y(1)-Y(0)} 0.2141133 0.03004191 0.1552311 0.2729955    0
```

The manual AIPW gives 0.2123 with a 95% interval of $[0.1511, 0.2735]$, and `npcausal` with `nsplits=1` gives 0.2141 with a standard error of 0.0300 and an interval of $[0.1552, 0.2730]$. It also reports the two arm means separately, $E[Y(0)] = 0.5720$ and $E[Y(1)] = 0.7861$.

The results from “npcausal” are almost the same as the manual results. Two clarifications. First, the manual estimator above *is* bias-corrected in the relevant sense: AIPW is exactly the one-step correction of the plug-in estimator using the efficient influence function. What we did *not* do is **cross-fitting** (sample splitting), which is a separate device. Second, that matters for inference: because the manual example fits the SuperLearner nuisances on the *full* sample and then uses `sd(IF)/sqrt(n)`, the resulting confidence interval is only valid under empirical-process (Donsker) conditions that flexible ML learners typically violate. With cross-fitting, the influence-function variance is valid without those conditions. “npcausal” does the cross-fitting for you when you set `nsplits > 1` (here `nsplits = 1` turns it off, for comparison with the manual full-sample version).

```r
aipw2<- ate(y=Y, a=A, x=W, nsplits=10)
```

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     parameter       est         se     ci.ll     ci.ul pval
1       E{Y(0)} 0.5701831 0.02904995 0.5132452 0.6271210    0
2       E{Y(1)} 0.7864167 0.01628375 0.7545005 0.8183328    0
3 E{Y(1)-Y(0)} 0.2162336 0.03300309 0.1515475 0.2809197    0
```

With `nsplits=10` the cross-fitted estimate is essentially unchanged. That is the reassuring case: cross-fitting is insurance against overfitting bias, and when the nuisance models are not badly overfitted it costs nothing.

32.3.3. TMLE

We can use TMLE for ATE.

```
library(tmle)
TMLE<- tmle(Y=data$Y,A=data$A,W=W, family="binomial", Q.SL.library=SL.library, g.SL.library=SL

TMLE$estimates$ATE
```

```
$psi
[1] 0.2117905

$var.psi
[1] 0.001019485

$CI
[1] 0.1492101 0.2743709

$pvalue
[1] 3.287231e-11

$bs.var
[1] NA

$bs.CI.twosided
[1] NA NA

$bs.CI.onesided.lower
[1] -Inf   NA

$bs.CI.onesided.upper
[1]  NA Inf
```

TMLE gives 0.2118 with a 95% interval of $[0.1492, 0.2744]$ — within 0.003 of both AIPW versions, as it should be, since all three solve the same efficient influence-function equation and differ only in how.

32.3.4. DoubleML

The R package DoubleML implements the double/debiased machine learning framework of Chernozhukov et al. (2018). It provides functionalities to estimate parameters in causal models based on machine learning methods. The double machine learning framework consist of three key ingredients: Neyman orthogonality, High-quality machine learning estimation and Sample splitting.

They consider a partially linear model:

$$y_i = \theta d_i + g_0(x_i) + \eta_i \tag{32.4}$$

$$d_i = m_0(x_i) + v_i \tag{32.5}$$

This model is quite general, except it does not allow interaction of d and x; therefore no heterogeneous treatment effect across x. But "DoubleML" implements more than partially linear model, it actually allows for heterogeneous treatment effects, in models such as interactive regression model.

The basic idea of doubleML is to use any machine learning algorithm to estimate outcome equation ($l_0(X) = E(Y|X)$) and treatment equation ($m_0(X) = E(D|X)$). Then get the residuals, namely $\hat{W} = Y - \hat{l}_0(X)$ and $\hat{V} = D - \hat{m}_0(X)$.

Then regress $\hat{W}$ on $\hat{V}$. Based on FWL theorem, you get $\hat{\theta}$.

An important component here is to specify a Neyman-orthogonal score function. In the case of PLR, it can be the product of the two residuals:

$$\psi(W;\theta,\eta) = (Y - D\theta - g(X))(D - m(X)) \tag{32.6}$$

The estimators $\hat{\theta}$ is to solve the equation that the sample mean of this score function being 0.

And the variance of this score function is used as the variance of the doubleML estimator's variance.

```r
library(DoubleML)
dml_data = DoubleMLData$new(data,
                            y_col = "Y",
                            d_cols = "A",
                            x_cols = c("w1","w2","w3","w4"))
print(dml_data)
```

```
================= DoubleMLData Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: w1, w2, w3, w4
Instrument(s):
Selection variable:
No. Observations: 1000
```

```r
library(mlr3)
library(mlr3learners)
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")
```

```r
learner = lrn("regr.ranger", num.trees=500,  max.depth=5, min.node.size=2)
ml_g = learner$clone()
ml_m = learner$clone()

learner = lrn("regr.glmnet")
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()

set.seed(3141)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_g=ml_g, ml_m=ml_m)
obj_dml_plr$fit()
print(obj_dml_plr)
```

```
================= DoubleMLPLR Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: w1, w2, w3, w4
Instrument(s):
Selection variable:
No. Observations: 1000

------------------ Score & algorithm ------------------
Score function: partialling out
DML algorithm: dml2

------------------ Machine learner   ------------------
ml_l: regr.ranger
ml_m: regr.ranger

------------------ Resampling        ------------------
No. folds: 5
No. repeated sample splits: 1
Apply cross-fitting: TRUE

------------------ Fit summary       ------------------
 Estimates and significance testing of the effect of target variables
  Estimate. Std. Error t value Pr(>|t|)
A   0.23271    0.03224   7.219 5.25e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: this example (and the EAGeR example below) both have a **binary** treatment `A` and outcome `Y`, but `DoubleMLPLR` assumes a partially linear model with a single constant effect `theta` and continuous-style regression learners (`ml_g`, `ml_m`) for both nuisance functions. The more appropriate DoubleML estimator for a binary treatment/outcome is the Interactive Regression Model, `DoubleMLIRM`, which uses classification learners for the propensity score and allows heterogeneous treatment effects. `DoubleMLPLR` will still run and often gives a reasonable ATE-like summary, but `DoubleMLIRM` is the estimator actually designed for this setting.

32.4. EAGeR trial simulation (AIPW package)

This simulation is based on a package "AIPW", which has a simulation data "eager_sim_rct". A denotes the binary treatment assignment, Y is the binary outcome, and W_g represents the covariate that affects the treatment assignment, which in our case was deemed to be the eligibility stratum indicator, sampled with replacement from the EAGeR Trial. Similarly, W_Q is a set of baseline prognostic covariates, which were also sampled with replacement from the EAGeR Trial, and includes the number of prior pregnancy losses, age, number of months of trying to conceive prior to randomization, body mass index (weight (kg)/height (m)2), and mean arterial blood pressure (denoted W1…5 , respectively).

Data and the true risk difference. Unlike the simulation above, this is a resampled trial: 1,228 observations with a binary treatment and outcome, and the package records the risk difference used to generate it, so we again have a target to check against.

```
library(AIPW)
data("eager_sim_rct")
glimpse(eager_sim_rct)
```

```
Rows: 1,228
Columns: 8
$ sim_Y              <int> 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0~
$ sim_Tx             <int> 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0~
$ eligibility        <dbl> 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0~
$ loss_num           <dbl> 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 1, 1, 2, 2, 2, 1~
$ age                <dbl> 21.93, 30.65, 30.96, 32.24, 31.49, 27.99, 31.60, 25.~
$ time_try_pregnant <dbl> 2, 1, 0, 9, 1, 10, 8, 6, 1, 0, 1, 1, 3, 3, 0, 10, 10~
$ BMI                <dbl> 19.51963, 25.56474, 20.75446, 21.65728, 24.27480, 19~
$ meanAP             <dbl> 73.33333, 79.00000, 79.33333, 63.83333, 97.44444, 85~
```

```
attributes(eager_sim_rct)$true_rd
```

```
[1] 0.1324569
```

```
data <- eager_sim_rct %>%
  rename(Y=sim_Y, A=sim_Tx) %>%
  tibble()

W <- data %>%
  select(-Y,-A)
```

32.4.1. plug-in estimator from outcome regression

```
SL.outcome<- SuperLearner(Y=data$Y, X=data %>% select(-Y),
                                   SL.library=SL.library, family="binomial")
SL.outcome.obs<- predict(SL.outcome, X=data %>% select(-Y))$pred
# predict the PO Y^1
SL.outcome.1<- predict(SL.outcome, newdata=data %>% select(-Y) %>% mutate(A=1))$pred
# predict the PO Y^0
SL.outcome.0<- predict(SL.outcome, newdata=data %>% select(-Y) %>% mutate(A=0))$pred
```

This is to do machine learning on the outcome equation; then get prediction on treated vs control, to get $\hat{Y}^1$ and $\hat{Y}^0$. The sample average of the difference is our plug-in estimator.

```
SL.plugin<-mean(SL.outcome.1-SL.outcome.0)
SL.plugin
```

```
[1] 0.1185897
```

```
Q=data.frame(QAW=SL.outcome.obs, Q0W=SL.outcome.0,Q1W=SL.outcome.1)
```

The plug-in gives 0.1325 against the recorded true risk difference of 0.1186 — about 0.014 high.

32.4.2. AIPW with machine learning

```
set.seed(123)
SL.g<- SuperLearner(Y=data$A, X=data %>% select(-Y, -A),
                    SL.library=SL.library, family="binomial")
g1W <- SL.g$SL.predict
g0W<- 1- g1W
IF.1<-((data$A/g1W)*(data$Y-Q$Q1W)+Q$Q1W)
IF.0<-(((1-data$A)/g0W)*(data$Y-Q$Q0W)+Q$Q0W)
IF<-IF.1-IF.0
aipw.1<-mean(IF.1);aipw.0<-mean(IF.0)
aipw.manual<-aipw.1-aipw.0
ci.lb<-mean(IF)-qnorm(.975)*sd(IF)/sqrt(length(IF))
ci.ub<-mean(IF)+qnorm(.975)*sd(IF)/sqrt(length(IF))
res.manual.aipw<-c(aipw.manual,ci.lb, ci.ub)
res.manual.aipw
```

```
[1] 0.13103847 0.06938821 0.19268873
```

Now we use “npcausal”.

```
library(npcausal)
aipw<- ate(y=data$Y, a=data$A, x=W, nsplits=1, sl.lib=SL.library)
```

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     parameter       est         se      ci.ll     ci.ul pval
1      E{Y(0)} 0.4455985 0.02240994 0.40167501 0.4895220    0
2      E{Y(1)} 0.5753483 0.02105948 0.53407176 0.6166249    0
3 E{Y(1)-Y(0)} 0.1297499 0.03059579 0.06978212 0.1897176    0
```

The results from "npcausal" are almost the same as the manual results. Two clarifications. First, the manual estimator above *is* bias-corrected in the relevant sense: AIPW is exactly the one-step correction of the plug-in estimator using the efficient influence function. What we did *not* do is **cross-fitting** (sample splitting), which is a separate device. Second, that matters for inference: because the manual example fits the SuperLearner nuisances on the *full* sample and then uses `sd(IF)/sqrt(n)`, the resulting confidence interval is only valid under empirical-process (Donsker) conditions that flexible ML learners typically violate. With cross-fitting, the influence-function variance is valid without those conditions. "npcausal" does the cross-fitting for you when you set `nsplits > 1` (here `nsplits = 1` turns it off, for comparison with the manual full-sample version).

```
aipw2<- ate(y=data$Y, a=data$A, x=W, nsplits=10)
```

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     parameter       est          se      ci.ll     ci.ul pval
1      E{Y(0)} 0.4448128 0.02529634 0.39523200 0.4943936    0
2      E{Y(1)} 0.5775070 0.02255959 0.53329021 0.6217238    0
3 E{Y(1)-Y(0)} 0.1326942 0.03377055 0.06650392 0.1988845    0
```

32.4.3. DoubleML

```r
library(DoubleML)
dml_data = DoubleMLData$new(as.data.frame(data),
                             y_col = "Y",
                             d_cols = "A",
                             x_cols = c("eligibility" , "loss_num","age", "time_try_pregnant","
print(dml_data)
```

```
================= DoubleMLData Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: eligibility, loss_num, age, time_try_pregnant, BMI, meanAP
Instrument(s):
Selection variable:
No. Observations: 1228
```

```r
library(mlr3)
library(mlr3learners)
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")

learner = lrn("regr.ranger", num.trees=500,  max.depth=5, min.node.size=2)
ml_g = learner$clone()
ml_m = learner$clone()

learner = lrn("regr.glmnet")
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()

set.seed(3141)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_g=ml_g, ml_m=ml_m)
obj_dml_plr$fit()
print(obj_dml_plr)
```

```
================= DoubleMLPLR Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: eligibility, loss_num, age, time_try_pregnant, BMI, meanAP
Instrument(s):
Selection variable:
No. Observations: 1228

------------------ Score & algorithm ------------------
Score function: partialling out
DML algorithm: dml2

------------------ Machine learner   ------------------
ml_l: regr.ranger
ml_m: regr.ranger

------------------ Resampling        ------------------
No. folds: 5
No. repeated sample splits: 1
Apply cross-fitting: TRUE

------------------ Fit summary       ------------------
 Estimates and significance testing of the effect of target variables
  Estimate. Std. Error t value Pr(>|t|)
A   0.12505    0.03292   3.798 0.000146 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: as with the earlier PLR example, `A` and `Y` here are **binary** (this is the EAGeR trial data), so `DoubleMLIRM` (which uses classification learners for the propensity score and allows heterogeneous effects) would be the more appropriate DoubleML estimator than `DoubleMLPLR`'s partially linear, constant-effect model — see the note above.

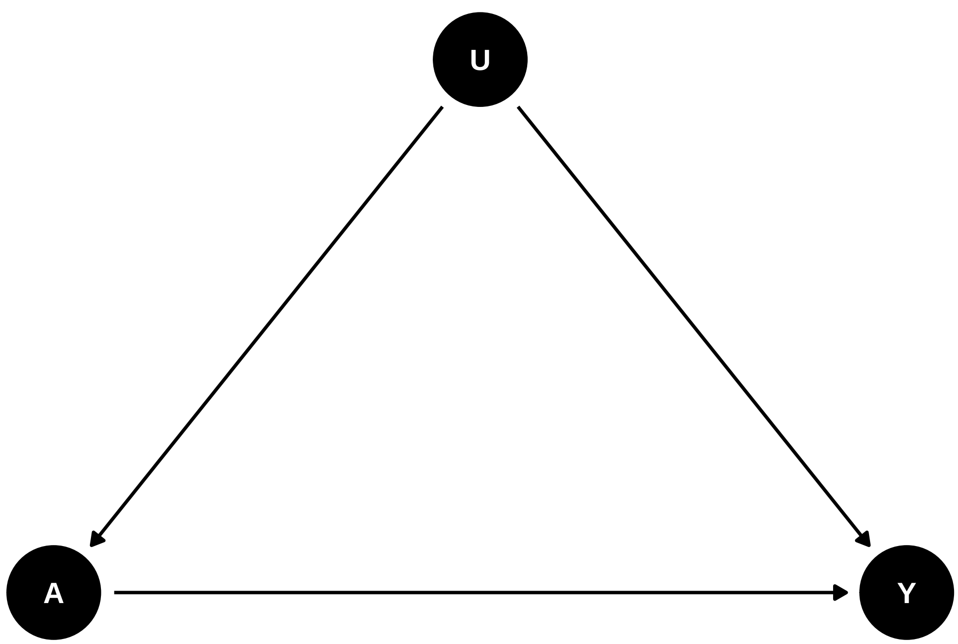
Systematic treatment: R · Julia.

33. Intro to proximal causal inference

This chapter follows Tchetgen Tchetgen and co-authors, drawing primarily on Xu Shi's talk on proximal causal inference.

33.1. Unobserved confounders

The central problem for observational studies is unobserved confounding. Even after conditioning on observed covariates X, an unobserved confounder U may affect both treatment A and outcome Y:

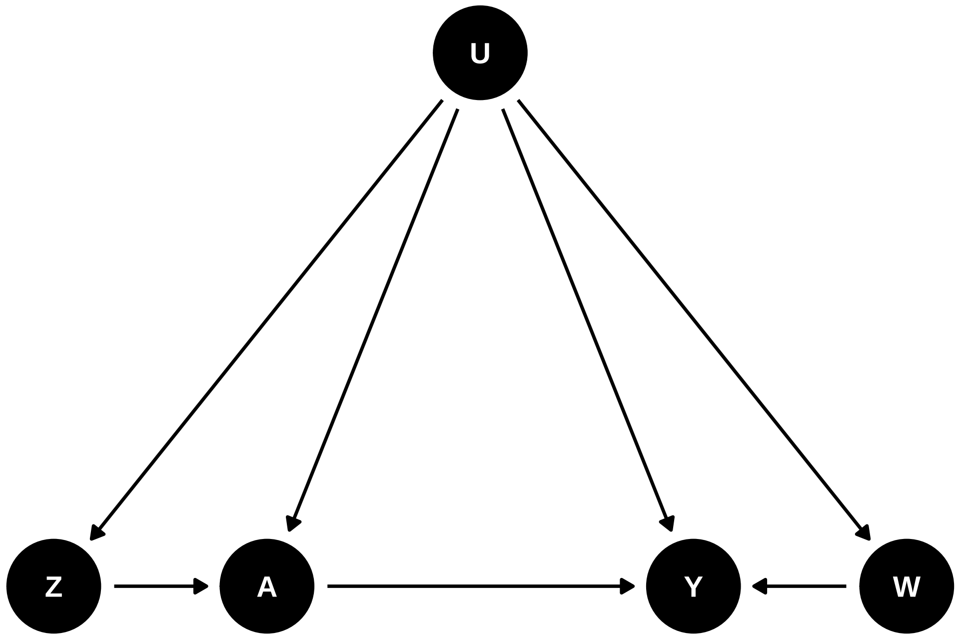


In this case, we have biased estimates of the treatment effect, regressing Y on A controlling for X.

33.2. Proximal causal inference

Proximal causal inference (also called negative controls) is a framework that allows us to estimate the causal effect of treatment A on outcome Y while accounting for unobserved confounders.

We'll need a DAG like this:

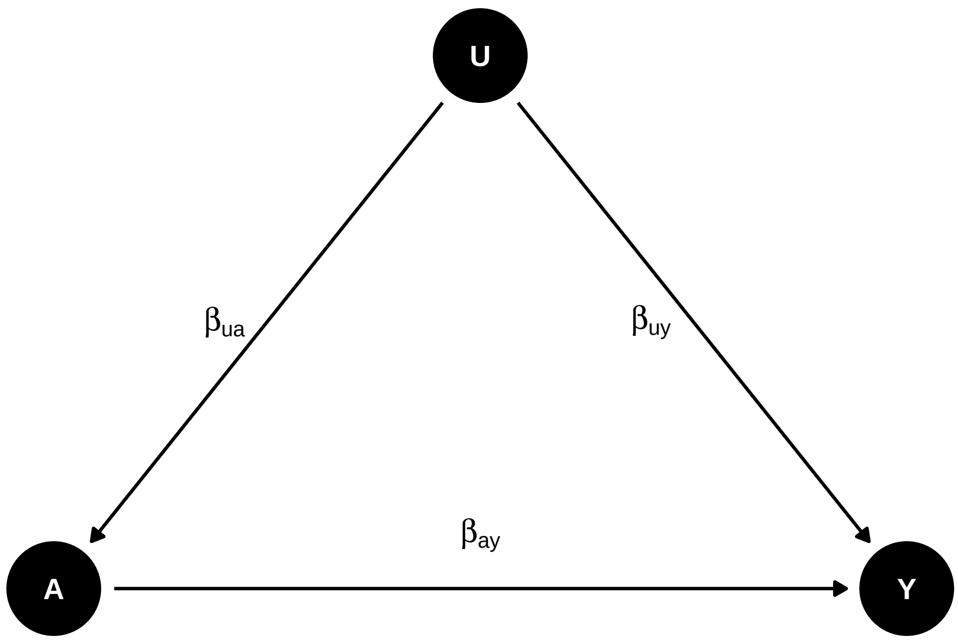


Basically we need two proxies for U to control for confounding. Z is called Negative Control for Exposure (NCE) and W is called Negative Control for Outcome (NCO). There is no link from Z to Y, and no link from W to A or Z.

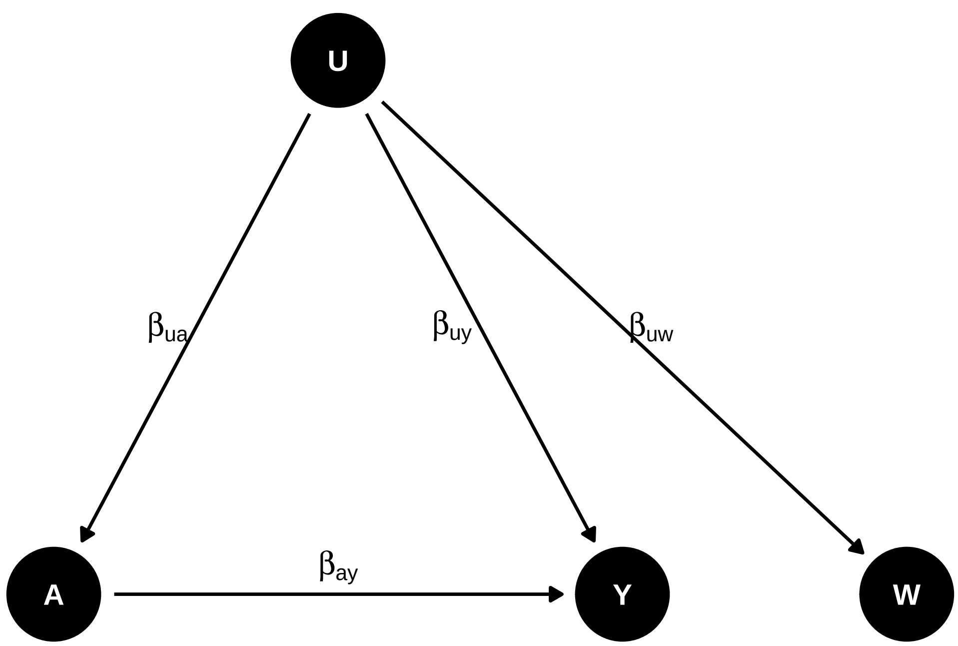
Z is an NCE if $Y(a,z) = Y(a)$ and $Z \perp Y(a)|U$, meaning that Z satisfies the exclusion restriction that there is no direct link from Z to Y, and that Z is independent of Y given U.

W is an NCO if $W(a,z) = W$ and $W \perp (A,Z)|U$, meaning that A and Z have no direct link to W.

33.2.1. Why this works

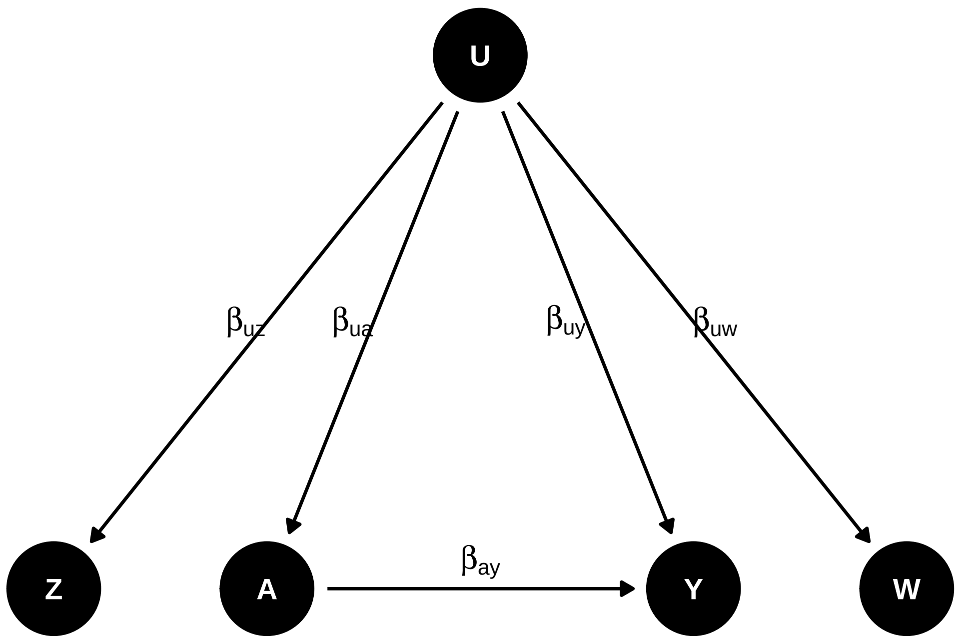


If we regress Y on A, then we get $\beta_{ay} + \beta_{uy}\beta_{ua}$, which is biased. How do we recover β_{ay}?



Suppose we have W as proxy of U and $\beta_{uw} = \beta_{uy}$, then we can get the true effect. We can do regression of Y on A, and then W on A. The first regression gives us $\beta_{ay} + \beta_{uy}\beta_{ua}$, and the second regression gives us $\beta_{uw}\beta_{ua}$. We can then get β_{ay} from the difference of the two coefficients. This is similar to a DiD setting. Say W is Y_{t-1}, and assume the effect of U is the same on Y and Y_{t-1}, then effect of A on Y by the difference of regressing Y on A and Y_{t-1} on A.

However, most likely $\beta_{uw} \neq \beta_{uy}$. Then we need another proxy Z of U to recover the true effect. This is also called "double proxy", or "double negative control".



33.2.2. A linear model

Suppose we have the following linear model:

$$E[Y|A, Z, U] = \beta_{y0} + \beta_{ay}A + \beta_{uy}U \tag{33.1}$$

$$E[W|A, Z, U] = \beta_{w0} + \beta_{uw}U \tag{33.2}$$

$$E[U|A, Z] = \beta_{u0} + \beta_{zu}Z + \beta_{au}A \tag{33.3}$$

β_{ay} is the true effect of A on Y.

$$E[Y|A, Z] = \beta_{y0} + \beta_{ay}A + \beta_{uy}(\beta_{u0} + \beta_{zu}Z + \beta_{au}A) \tag{33.4}$$

$$E[W|A, Z] = \beta_{w0} + \beta_{uw}(\beta_{u0} + \beta_{zu}Z + \beta_{au}A) \tag{33.5}$$

From the last two regressions we can recover β_{ay}:

If we define $\delta_a^y = \beta_{ay} + \beta_{uy}\beta_{au}$ (regression coefficient of A on the Y regression) and $\delta_a^w = \beta_{uw}\beta_{au}$, (regression coefficient of A on the W regression), similarly for δ_z^y and δ_z^w, then we have

$$\beta_{ay} = \delta_a^y - \delta_a^w \frac{\delta_z^y}{\delta_z^w} \tag{33.6}$$

In this **linear-Gaussian special case only**, we can look at this as a two-stage-least-squares-style regression and replace U with the predicted value of W:

$$E[Y|A,Z] = \left(\beta_{y0} - \frac{\beta_{uy}}{\beta_{uw}}\beta_{w0}\right) + \beta_{ay}A + \frac{\beta_{uy}}{\beta_{uw}}E[W|A,Z] \tag{33.7}$$

since $E[U \mid A,Z] = (E[W \mid A,Z] - \beta_{w0})/\beta_{uw}$ from the W equation above. The coefficient on the predicted value of W is β_{uy}/β_{uw}, not β_{uy} alone — it only reduces to β_{uy} when $\beta_{uw} = 1$. So we first regress W on A and Z, then regress Y on A and the predicted value of W.

This "predict W from A and Z, then plug it in" shortcut is **not** the general proximal identification argument, and it is not the same as a standard IV/2SLS specification (we are not instrumenting an endogenous regressor with `ivreg`). In general proximal causal inference, the outcome bridge function $h(A,W)$ is identified by solving the conditional moment restriction $E[Y \mid A,Z] = E[h(A,W) \mid A,Z]$, which requires completeness (relevance) conditions linking the treatment-inducing proxy Z and the outcome-inducing proxy W to the hidden confounder U. The generated regressor $S = \widehat{E}[W \mid A,Z]$ is only a valid substitute under the linear-Gaussian structure above; even then the two-stage standard errors are wrong as written, because S is generated in the first stage (bootstrap the whole procedure).

33.2.3. A nonparametric model

We don't have to have a linear model, this can be done with a nonparametric model as well.

$$E[Y(a)] = E[h(a,W)] \tag{33.8}$$

where $E[Y|A,Z] = E[h(A,W)|A,Z]$.

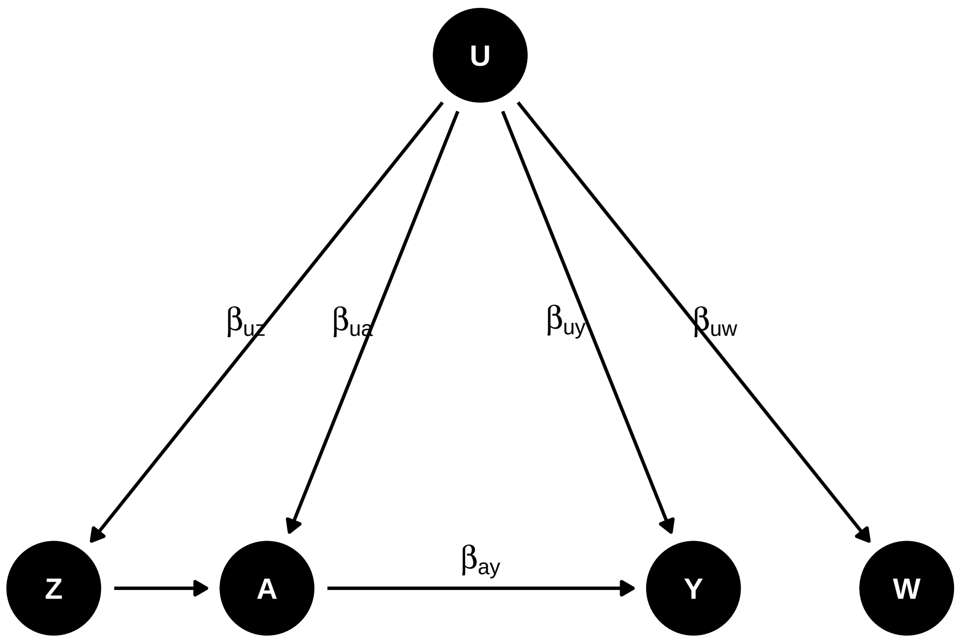
That is, we can estimate h as a function of A and W, identified using Z, nonparametrically. This h function is called outcome bridge function. In the linear case, it is $h(A,W) = \beta_0 + \beta_a A + \beta_w W$.

33.2.4. How to find proximal variables

Tchetgen Tchetgen and co-authors come up with a Data-driven Automated Negative Control Estimation (DANCE) algorithm to find negative controls. The idea is to use a machine learning algorithm to find the variables that are correlated with the treatment and the outcome, but not with the treatment and the outcome together. This is similar to the idea of finding instrumental variables.

33.3. example

Let's generated data based on this DAG:




```r
set.seed(123)
n <- 5000
# Set coefficients
b_uw <- 0.5 # effect of U on W
b_uy <- 0.5 # effect of U on Y
b_ay <- 0.5 # effect of A on Y
b_uz <- 0.5 # effect of U on Z
b_ua <- 0.5 # effect of U on A
b_za <- 0.5 # effect of Z on A

U <- rnorm(n)
W <- b_uw * U + rnorm(n) # W is a function of U
Z <- b_uz * U + rnorm(n) # Z is a function of U
A <- b_ua * U + b_za * Z + rnorm(n) # A is a function of U and Z
Y <- b_uy * U + b_ay * A + rnorm(n) # Y is a function of U and A
# A <- rbinom(n, 1, plogis(-1 + U / 2)) # A is a binary treatment


# Create a data frame with the generated variables

data <- data.frame(W = W, Z = Z, A = A, Y = Y)
```

The simulation sets $n = 5000$ and every coefficient to 0.5. The hidden confounder is $U \sim N(0, 1)$; the two proxies are $W = 0.5U + \varepsilon_W$ and $Z = 0.5U + \varepsilon_Z$; treatment is $A = 0.5U + 0.5Z + \varepsilon_A$; and the outcome is $Y = 0.5U + 0.5A + \varepsilon_Y$, all errors standard normal. The true effect of A on Y is therefore 0.5.

Note the roles: Z affects A but not Y directly, W is affected by U but affects nothing, and neither is a valid control for U on its own.

If we run regression of Y on A, Z, W,

```
m1 <- lm(Y ~ A + Z + W, data = data)
summary(m1)
```

```
Call:
lm(formula = Y ~ A + Z + W, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.5836 -0.7167  0.0005  0.7305  4.4866 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.006347   0.015279  -0.415    0.678    
A            0.660021   0.014123  46.734  < 2e-16 ***
Z            0.085004   0.016650   5.105 3.43e-07 ***
W            0.132798   0.014165   9.375  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.08 on 4996 degrees of freedom
Multiple R-squared:  0.4556,	Adjusted R-squared:  0.4553 
F-statistic:  1394 on 3 and 4996 DF,  p-value: < 2.2e-16
```

We don't get the true effect of A on Y: the coefficient on A is 0.660 against a true 0.5, a bias of 32%. Conditioning on both proxies does not remove the confounding, because neither W nor Z *is* U — each is a noisy reflection of it, and controlling for a noisy proxy leaves residual confounding. The proxies do carry signal (both are significant, at 0.085 and 0.133), which is what makes this failure mode deceptive: the regression looks like it is adjusting for the confounder.

If we do the 2sls using W and Z, then we get the true effect.

```
# Linear-Gaussian (Y, W) toy case ONLY. This two-stage plug-in identifies
# the effect under the linear structure above; it is not the general
# proximal estimator, and the reported SEs below are NOT valid because S is
# a generated regressor (bootstrap the whole two-step procedure for inference).
# Perform the first-stage linear regression W ~ A + Z
m1 <- lm(W ~ A + Z, data = data)
# Compute the estimated E[W|A,Z]
S <- m1$fitted.values
# Perform the second-stage linear regression Y ~ A + E[W|A,Z]
m2 <- lm(Y ~ A + S, data = data)
summary(m2)   # point estimate OK under the linear model; SEs need bootstrap
```

```
Call:
lm(formula = Y ~ A + S, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.5477 -0.7309  0.0059  0.7395  4.3715 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.005049   0.015408  -0.328    0.743    
A            0.489342   0.043210  11.325  < 2e-16 ***
S            1.143480   0.199171   5.741 9.96e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.089 on 4997 degrees of freedom
Multiple R-squared:  0.446,	Adjusted R-squared:  0.4458 
F-statistic:  2012 on 2 and 4997 DF,  p-value: < 2.2e-16
```

The two-stage estimate is 0.489, against the true 0.5 and the naive 0.660. Using the two proxies in the right way — one to build a predicted $E[W \mid A, Z]$ and that prediction as the control — recovers the effect, where using both directly as regressors did not.

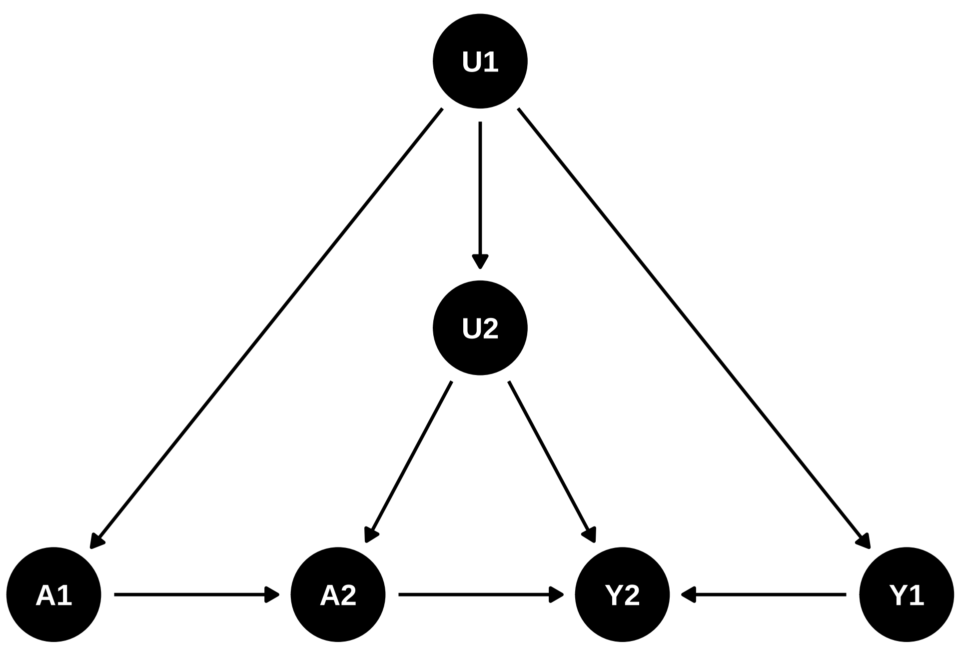
The reported standard error of 0.043 should be ignored, for the reason flagged in the code comment: S is a generated regressor, so the second-stage standard errors are wrong as written and need a bootstrap over both stages.

Systematic treatment: R · Julia.

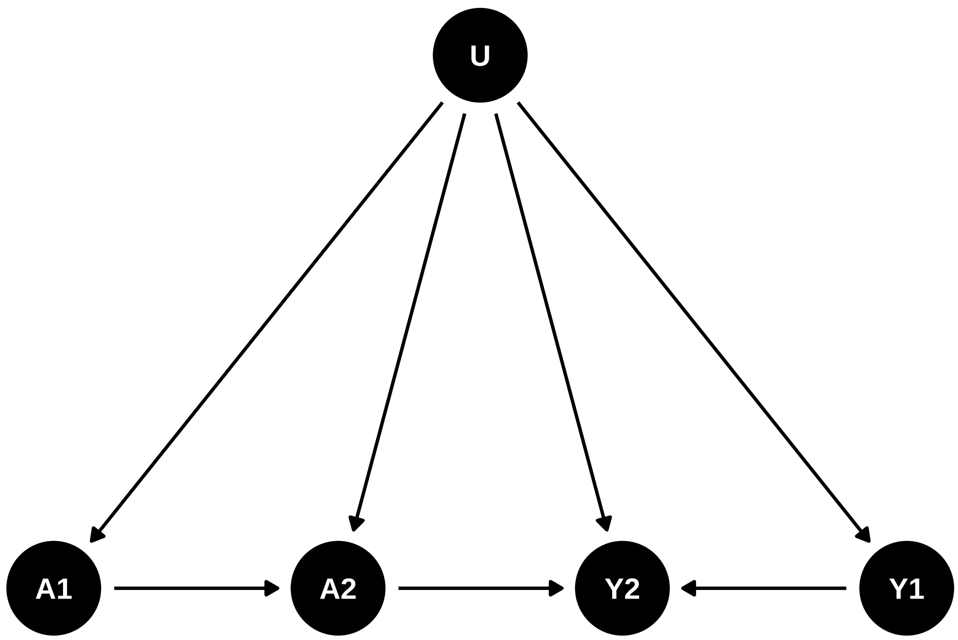
34. Simulation on proximal causal inference with panel data

This chapter extends proximal causal inference to a panel-data setting. The question is whether lagged treatment and lagged outcome can serve as negative control exposure (NCE) and negative control outcome (NCO), respectively, to account for unobserved confounders.

Consider a two-period DAG:



Suppose that's the case for the causal structure, then I can reasonably combine U1 and U2 into a single unobserved confounder U:



That would fit into the double negative control framework. The problem is that we have to assume there is no link from A1 to Y1, which is odd. Can we assume in period 1 there is no effect, but in period 2, there is an effect?

34.1. A simulation with panel data

I generate two period data based on the DAG above.

```
library(dplyr)
library(rootSolve)
library(tidyverse)

gen_data <- function(seed, n){
  set.seed(seed)
  u <- rnorm(n, mean = 0, sd = 1)  # unobserved confounder
  x1 <- rnorm(n, mean = 0, sd = 1)  # observed confounder
  x2 <- rnorm(n, mean = 0, sd = 1)  # observed confounder
  a1 <- rbinom(n, 1, plogis(2*u + 1.5*x1))  # treatment at time 1
  a2 <- rbinom(n, 1, plogis(u + a1 +  x1))  # treatment at time 2
  y1 <-  1.5*u + .5*x1  +  rnorm(n, mean = 0, sd = 1)  # outcome at time 1
  y2 <-  y1 + 2.5*u + 2.5*x1 + 3*a2 + rnorm(n, mean = 0, sd = 1)  # outcome at time 2
  data <- data.frame(u=u, x1=x1, x2=x2, a1=a1, a2=a2, y1=y1, y2=y2)
  return(data)
}
data <- gen_data(333, 100000)
```

```
data_sim <- data |>
  select(a1, a2, y2,  x1, x2, y1) |>
  rename(a = a2, y = y2,  x = x1, w = y1, z=a1)
```

The simulation draws $n = 100{,}000$ two-period observations. There is an unobserved confounder $u \sim N(0,1)$ and two observed covariates $x_1, x_2 \sim N(0,1)$. Treatment at time 1 is $a_1 \sim \text{Bernoulli}(\Lambda(2u + 1.5x_1))$ and at time 2 is $a_2 \sim \text{Bernoulli}(\Lambda(u + a_1 + x_1))$. The outcomes are $y_1 = 1.5u + 0.5x_1 + \varepsilon_1$ and $y_2 = y_1 + 2.5u + 2.5x_1 + 3a_2 + \varepsilon_2$.

So the effect of interest is the 3 on a_2. The construction then relabels a_1 as the negative control exposure z and y_1 as the negative control outcome w — note that y_1 does not depend on a_1, which is exactly the assumption discussed at the end of this chapter.

We can use the simple 2sls:

```
lm1 <- lm(w ~ a + x + z, data = data_sim)
data_sim$what <- lm1$fitted.values
lm2 <- lm(y ~ a + x +  what, data = data_sim)
summary(lm2)
```

```

Call:
lm(formula = y ~ a + x + what, data = data_sim)

Residuals:
     Min       1Q   Median       3Q      Max 
-16.4284  -2.3083   0.0071   2.3020  20.7590 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.01293    0.02315  -0.558    0.577    
a            3.02113    0.03528  85.625   <2e-16 ***
x            1.67203    0.01245 134.327   <2e-16 ***
what         2.66578    0.01747 152.631   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.456 on 99996 degrees of freedom
Multiple R-squared:  0.6822,	Adjusted R-squared:  0.6822 
F-statistic: 7.155e+04 on 3 and 99996 DF,  p-value: < 2.2e-16
```

The 2SLS coefficient on a is 3.021 with a standard error of 0.035, against the true 3. The coefficient on the fitted proxy `what` is 2.666, which is the ratio of the confounder's loading on the outcome to its loading on the proxy, not an effect of interest.

Or we can use more sophisticated methods. Here I use the m-estimation code by Paul Zivich: https://github.com/pzivich/publications-code/blob/master/ProximalCI/proximal_sims_r.R. This code is to construct estimating equations, then use root finding to find the parameter estimates.

```
pcl_m2 <- function(theta, mydata, out = "ef"){
  # read in variables
  a <- mydata$a
  x <- mydata$x
  w <- mydata$w
  z <- mydata$z
  y <- mydata$y
  # model for what
  what <- theta[1] + theta[2] * a + theta[3] * x + theta[4] * z

  # matrix of covars in model for what
  wmat <- as.matrix(cbind(rep(1, nrow(mydata)), a, x, z), ncol = 4,
                    nrow = nrow(mydata))

  # estimating function for what
  ef1 <- (t(wmat)  %*% (w - what))            # gives px1 vector (summed over units)
  ef1_b <- wmat * (w - what)                  # gives nxp matrix (rows are units)

  # model for yhat
  yhat <- theta[5] + theta[6]*a + theta[7]*x + theta[8] * what

  # matrix of covars in model for yhat
  ymat <- as.matrix(cbind(rep(1, nrow(mydata)), a, x, what), ncol = 4,
                    nrow = nrow(mydata))

  #estimating function for yhat
  ef2 <- t(ymat) %*% (y - yhat)               # px1 vector, summed over units
  ef2_b <- (ymat) * (y - yhat)                # nxp matrix (rows are units)
  f <- c(ef1, ef2)                            # p-dimensional vector
  if(out == "full") return(cbind(ef1_b, ef2_b)) else return(f) # full used in var calculation c
}

# call m-estimator for PCL
mpcl_mest <- function(dat){
  theta_start <- rep(.05, 8)
  results <- mestimator(pcl_m2, init = theta_start, dat, mydata = dat)
  return(results)
}

##' M-estimation engine
mestimator <- function(ef, init, dat, ...){
  # get estimated coefficients
  fit <- multiroot(ef, start = init, out = "ef",
                   rtol = 1e-9, atol = 1e-12, ctol = 1e-12, ...  )
  betahat <- fit$root
  n <- nrow(dat)
```

```
  # bread
  # derivative of estimating function at betahat
  pd <- gradient(ef, betahat, out = "ef", ...)
  pd <- as.matrix(pd)
  bread <- -pd/n
  se_mod <- sqrt(abs(diag(-solve(pd))))

  # meat1
  efhat <- ef(betahat, out = "full", ...)
  meat <- list()
  meat1 <- (t(efhat) %*% efhat)/n

  # sandwich
  sandwich1 <- (solve(bread) %*% meat1 %*% t(solve(bread)))/n
  se_sandwich1 <- sqrt(diag(sandwich1))

  results <- data.frame(betahat, se_sandwich1, se_mod)
  return(results)
}
```

```
rmpcl <- unname(mpcl_mest(data_sim)[6, c(1:2)])
rmpcl
```

```
6 3.021126 0.0198433
```

```
#mpcl_mest(data_sim)
```

The m-estimator gives 3.0211 with a standard error of 0.0198.

Both the 2sls and the m-estimation give us the same estimate of the causal effect of treatment on outcome, which is 3 — identical to four decimal places, 3.021126 either way. The standard errors differ, 0.0198 against 0.0353, and the m-estimator's is the one to use: it accounts for the estimation of the first stage, which the plug-in 2SLS standard error does not. The m estimator can be more general. 2sls is good for the linear case.

However, the main issue is the assumption that there is no link from A1 to Y1.

To think about it from a different angle: this is similar to DiD setting. Basically we need A1 to be 0. There is no effect in period 1. If we only have A1 as 0, then it makes sense to have no link from A1 to Y1 (no anticipation assumption).

This is a restrictive identifying assumption, not a technical footnote. Using A_{t-1} as the negative control exposure and Y_{t-1} as the negative control outcome only identifies the effect if two separate exclusions hold: A_1 must have no direct effect on Y_1, so that Y_1 is a clean proxy for U rather than a partly post-treatment quantity; and A_1 must have no direct effect on Y_2 except through U, which is the exclusion restriction that makes A_1 admissible as a negative control exposure in the first place. If either link is actually present — for example, if treatment has an immediate effect in the same period it starts, which is the empirically common case — this proximal construction is misspecified and the estimate above should not be trusted. Before applying this approach, researchers should have a substantive reason to believe treatment effects only appear with a lag.

Systematic treatment: R · Julia.

35. longitudinal modified treatment policy (LMTP)

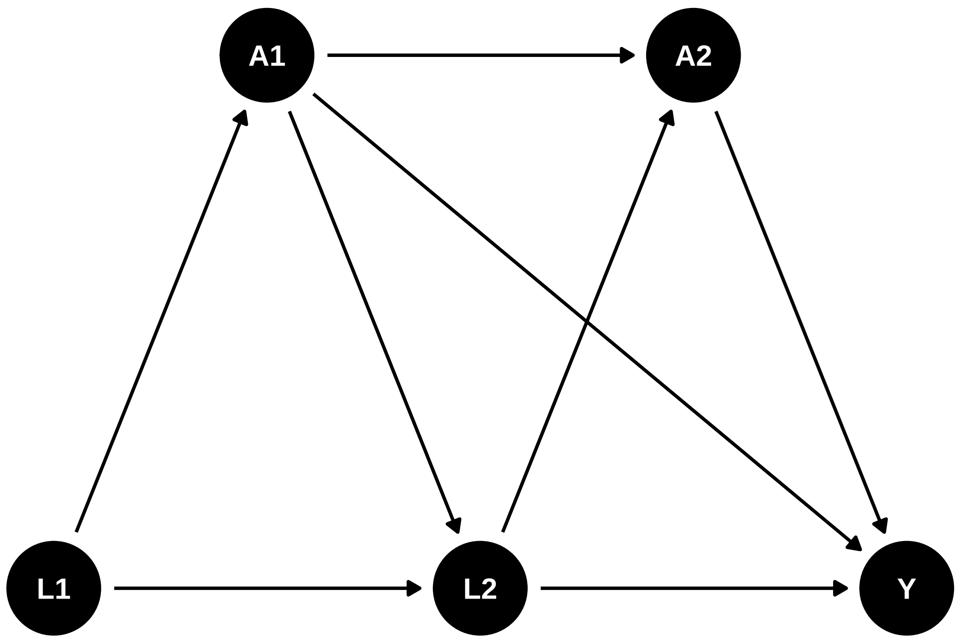
The longitudinal modified treatment policy (LMTP) framework, developed in epidemiology and biostatistics, handles settings where the standard ATE is poorly defined: continuous or multivalued treatments, time-varying exposures, or hypothetical interventions that would violate positivity. This chapter follows Nicholas Williams.

Suppose we have such a DAG:

```r
library(ggplot2)
library(ggdag)

g <- dagify(
  A1 ~ L1,
  L2 ~ L1,
  L2 ~ A1,
  A2 ~ A1,
  A2 ~ L2,
  Y ~ A2,
  Y ~ L2,
  Y ~ A1,
  exposure = "A2",
  outcome = "Y",
  coords = list(x = c(L1 = 1,  L2 = 2, Y = 3, A1 = 1.5, A2 = 2.5),
                y = c(L1 = 1,  L2 = 1, Y = 1, A1 = 1.5, A2 = 1.5))
)

ggdag(g) +
  theme_dag()
```



We have multiple time points, and the treatment A is time-varying. We are interested in not only the ATE of A_2 on Y, but also some other hypothetical interventions.

35.1. Definitions and assumptions

35.1.1. Assumptions

1. Positivity. Basically if there is a unit with a_t and h_t, then there is a unit with $d(a_t, h_t)$ and h_t.
2. Sequential unconfoundedness. There is no unmeasured confounders.

35.1.2. dynamic treatment regime

Some notations commonly used in this literature: We observe $Z = (L_0, A_0, L_1, A_1, Y)$, where L is the observed confounder, A is the treatment, and Y is the outcome. The treatment A can be time-varying. We can also have multiple treatments at different time points. For example, we can have A_2, A_3, etc. In this graph, baseline L_0 affects A_0, which in turn affects L_1, which then affects A_1, and finally A_1 affects the outcome Y.

History H_t is the history of data up to time t, right before A_t. for example, $H_1 = (L_0, A_0, L_1)$, and $H_2 = (L_0, A_0, L_1, A_1)$. d is the hypothetical intervention function, or shift function, which is a function of the history H_t and the treatment A_t. For example, $d_0(a_0, h_0, \epsilon_0)$ is a user-given function to map a_0, h_0, and ϵ_0 to a potential treatment value. The function d can be deterministic, or it can be stochastic. Then we can replace A_0 with $A_0^d = d_0(A_0, H_0, \epsilon_0)$. Then after that $A_1(A_0^d)$ is called the natural value of treatment.

This is very general, comparing to the static treatment regime.

Suppose treatment A is a function of the history of treatment and confounders, for example, $A = d(A_1, L_1, A_2, L_2)$. A can be set to a fixed value, say 1 or 0, or some value A^d. This function d can be anything, it can be taking a deterministic value, or it can be a function that takes the natural treatment value A as input. In the package "lmtp", this is called a shift function, or hypothetical intervention. For example, d can be set to 1 if $age < 30$, or d can be set to double the natural value of A. Many possibilities.

In comparison, for ATE, we only need to set A to 0 or 1.

Under this LMTP, the causal parameter is

$$\theta = E[Y^{\bar{A}^d}] \tag{35.1}$$

$Y^{\bar{A}^d}$ is the potential outcome under the hypothetical intervention $\bar{A}^d$. At time 1, $A_1^d = d(A_1, H_1)$.

35.1.3. modified treatment policy

A simulated data set makes the estimation concrete.

This simulation is from Susmann et al. (2024) "Longitudinal Generalizations of the Average Treatment Effect on the Treated for Multi-valued and Continuous Treatments". I modified slightly to fit the DAG above.

```
library(tidyverse)
library(tidyr)

mtp <- function(data, trt) {
  a <- data[[trt]]
  a * 0 + 1
}
simulate_data <- function(seed, N, tau, sigma = 0.5) {
  set.seed(seed)

  data <- tibble(id = 1:N)

  for(t in 1:tau) {
    Lt <- paste0("L_", t)
    Ltd <- paste0("L_", t, "d")

    At <- paste0("A_", t)
    Atd <- paste0("A_", t, "d")

    Lt1 <- paste0("L_", t - 1)
    Lt1d <- paste0("L_", t - 1, "d")

    At1 <- paste0("A_", t - 1)
    At1d <- paste0("A_", t - 1, "d")
```

```r
    if(t == 1) {
      data[[Lt]] <- runif(N, 0, 1)
      data[[Ltd]] <- data[[Lt]]

      data[[At]] <- rbinom(N, size = 1, prob = 0.5)
    }
    else {
      # L_t depends on the previous treatment A_{t-1} (the DAG edge A1 -> L2),
      # which is the time-varying confounding this chapter is about.
      data[[Lt]] <- rnorm(N, mean = 0.25 * data[[Lt1]] + 0.4 * data[[At1]], 0.5)
      data[[Ltd]] <- rnorm(N, mean = 0.25 * data[[Lt1d]] + 0.4 * data[[At1d]], 0.5)

      data[[At]] <- rbinom(N, size = 1, prob = plogis(0.5 - 0.2 * data[[At1]] + 0.1 * data[[Lt
    }

    data[[Atd]] <- mtp(data, At)
  }
  # Y depends on the final treatment A_2 and confounder L_2, and also directly
  # on the first-period treatment A_1 (the DAG edge A1 -> Y).
  data$Y  <- rnorm(N, data[[At]] + data[[Lt]] + 0.5 * data[["A_1"]], sigma)
  data$Yd <- rnorm(N, data[[Atd]] + data[[Ltd]] + 0.5 * data[["A_1d"]], sigma)
  data
}

simulated_data1 <- simulate_data(seed = 123, N = 10000, tau = 2)

mean(simulated_data1$Yd)
```

```
[1] 2.028593
```

```r
mtp <- function(data, trt) {
  a <- data[[trt]]
  a * 0 + 0
}

simulated_data2 <- simulate_data(seed = 123, N = 10000, tau = 2)

mean(simulated_data2$Yd)
```

```
[1] 0.1285934
```

```r
mean(simulated_data1$Yd) - mean(simulated_data2$Yd)
```

```
[1] 1.9
```

Note in this simulation the variables ending with "d" are the variables under hypothetical intervention, or modified treatment policy. L_1 is from Uniform$(0, 1)$, and L_2 is from $N(0.25L_1 + 0.4A_1,\ 0.5)$ – so the time-varying confounder L_2 depends on the earlier treatment A_1, which is the time-varying confounding this chapter is about. The treatment A_1 is from a Bernoulli distribution with probability 0.5, and the treatment A_2 is from a Bernoulli distribution with probability plogis$(0.5-0.2A_1+0.1L_1)$. The outcome Y is from a normal distribution with mean $A_2 + L_2 + 0.5A_1$, so Y depends on both the current treatment A_2 and the lagged treatment A_1. In the code, the first modified treatment policy sets treatment to 1 in every period (`simulated_data1`) and the second sets it to 0 in every period (`simulated_data2`); the reported difference $E[Y^d_{\text{always 1}}]-E[Y^d_{\text{always 0}}]$ is the effect of the always-treat versus never-treat static regimes.

The always-treat regime gives $E[Y^d] = 2.029$, always-never gives 0.129, and their difference is 1.900 — and that 1.9 decomposes exactly into the three routes the DGP contains: 1 from A_2 direct, 0.5 from A_1 direct, and 0.4 through the $A_1 \to L_2 \to Y$ path, since L_2 shifts by 0.4 and enters Y with coefficient 1.

In this case, we can just do a linear regression to get the effect of A2 on Y, knowing the exact DAG.

```r
m1 <- glm(Y ~ L_2 + A_1   + A_2, data = simulated_data1)
summary(m1)
```

```
Call:
glm(formula = Y ~ L_2 + A_1 + A_2, data = simulated_data1)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.453e-05  9.615e-03  -0.004    0.997
L_2          9.998e-01  9.889e-03 101.103   <2e-16 ***
A_1          4.940e-01  1.079e-02  45.779   <2e-16 ***
A_2          1.000e+00  1.027e-02  97.380   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.2500364)

    Null deviance: 9257.2  on 9999  degrees of freedom
Residual deviance: 2499.4  on 9996  degrees of freedom
AIC: 14523

Number of Fisher Scoring iterations: 2
```

Knowing the DAG, the regression recovers every structural coefficient: L_2 at 0.9998, A_1 at 0.4940 and A_2 at 1.0000, against true values of 1, 0.5 and 1. Note this identifies the *direct* effect of A_1 only; its path through L_2 is blocked by conditioning on L_2, which is why the total of 1.9 above cannot be read off this table.

Now consider a different MTP: set half of the time to 0 and leave the other half unchanged.

$$d(a_t, \epsilon_t) = \begin{cases} 0, & \text{if } \epsilon_t < .5 \text{ } and \text{ } a_t = 1 \\ a_t, & \text{otherwise} \end{cases} \tag{35.2}$$

This is, say, to set half of smokers to non-smokers, and the other half remain smokers.

```r
# mtp <- function(data, trt) {
#   a <- data[[trt]]
#   epsilon <- rbinom(nrow(data), size = 1, prob = 0.5)
#   ifelse(epsilon <.5 & a == 1, 0, a)
# }
d <- function(a) {
  epsilon <- runif(length(a))
  ifelse(epsilon < 0.5 & a == 1, 0, a)
}
simulated_data1$m3_d <- simulated_data1$Y

m2 <- glm(m3_d ~ L_1 + A_1 + L_2 + A_2, data = simulated_data1)
summary(m2)
```

```
Call:
glm(formula = m3_d ~ L_1 + A_1 + L_2 + A_2, data = simulated_data1)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.005815   0.012842  -0.453    0.651
L_1          0.011964   0.017621   0.679    0.497
A_1          0.494512   0.010814  45.729   <2e-16 ***
L_2          0.998831   0.009987 100.015   <2e-16 ***
A_2          0.999873   0.010273  97.334   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.2500499)

    Null deviance: 9257.2  on 9999  degrees of freedom
Residual deviance: 2499.2  on 9995  degrees of freedom
AIC: 14525

Number of Fisher Scoring iterations: 2
```

```r
simulated_data1$m2_d <- predict(m2, mutate(simulated_data1, A_2 = d(A_2)))

m1 <- glm(m2_d ~ L_1 + A_1, data = simulated_data1)
summary(m1)
```

```
Call:
glm(formula = m2_d ~ L_1 + A_1, data = simulated_data1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.30716    0.01531   20.07   <2e-16 ***
L_1          0.26079    0.02384   10.94   <2e-16 ***
A_1          0.87604    0.01367   64.08   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.4669449)

    Null deviance: 6633.7  on 9999  degrees of freedom
Residual deviance: 4668.0  on 9997  degrees of freedom
AIC: 20768

Number of Fisher Scoring iterations: 2
```

```
m1_d <- predict(m1, mutate(simulated_data1, A_1 = d(A_1)))
mean(m1_d)
```

```
[1] 0.6518997
```

The recursion makes the blocked path visible. The first regression, on the full history, again gives $A_1 = 0.4945$ — the direct effect. The second, after substituting the shifted A_2 and regressing on L_1 and A_1 only, gives $A_1 = 0.8760$, which is close to the total $0.5 + 0.4 = 0.9$: dropping L_2 from the conditioning set lets A_1's indirect path back in. The final average is 0.652, the expected outcome under the stochastic half-off regime, sitting between the always-treat 2.029 and never-treat 0.129 as it should.

Note this recursive process is based on Diaz, et al. (2023) "Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies".

Here is how exactly we estimate it: we start with the last time point. Regress it on previous treatment and confounders, and then get the predicted value with A changed based on the MTP. Regress that predicted value on the previous treatment and confounders, get predicted values with A changed based on MTP. Repeat until the first time point. The average of the predicted value at time 1 is the expected value under this MTP.

The shift function `d()` above is a *stochastic* intervention: at each call it draws a fresh $\epsilon \sim \text{Uniform}(0, 1)$ and, with probability 0.5, switches a treated unit ($a = 1$) to 0, otherwise leaves a unchanged. The regime is "at each time point, draw independently conditional on history and apply this rule"; because the draw is independent across time points and units, calling `d()` separately for A_2 and A_1 is what implements that regime. (By contrast, the `d1`/`d2` functions used with `lmtp_tmle` below are *deterministic* static regimes – always 1 and always 0.)

This is basically g-formula extended to longitudinal data.

In a general case, this is the generalized g-formula to estimate θ:

Set $m_{\tau+1} = Y$, let $A_t^d = d(A_t, H_t)$. For $t = \tau, \dots, 1$, recursively define:

$$m_t(a_t, h_t) = E[m_{t+1}(A_{t+1}^d, H_{t+1})|A_t = a_t, H_t = h_t] \tag{35.3}$$

Then $\theta = E[m_1(A_1^d, L_1)]$.

We start from the last period. Regress $m_{\tau+1}$, which is Y on A_τ and H_τ. Then get the predicted value with A_τ changed based on the MTP. Then regress that predicted value on $A_{\tau-1}$ and $H_{\tau-1}$. Repeat until the first time point. The average of the predicted value at time 1 is the expected value under this MTP.

35.1.4. Estimators

The authors advocate two estimators, TMLE and SDR (sequentially doubly robust estimator). The procedures are the same, starting from the last time point, then apply TMLE or SDR, iterate to the first time point.

35.2. Using lmtp package

```r
library(tidyverse)
library(lmtp)
d1 <- function(data, trt) {
  rep(1, nrow(data))
}
A <- "A_2"
Y <- "Y"
W <- c("L_1", "L_2", "A_1")

set.seed(34465)

treat <- lmtp_tmle(
  data = simulated_data1,
  trt = "A_2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d1,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(treat)
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 1.558
    Std. error: 0.009
```

:::

```r
d2 <- function(data, trt) {
  rep(0, nrow(data))
}
A <- "A_2"
Y <- "Y"
W <- c("L_1", "L_2", "A_1")

set.seed(34465)

control <- lmtp_tmle(
  data = simulated_data1,
  trt = "A_2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d2,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(control)
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 0.559
    Std. error: 0.01
```

```r
lmtp_contrast(treat, ref = control)
```

::: {.cell-output .cell-output-stdout}

```
  shift   ref estimate std.error conf.low conf.high p.value
1  1.56 0.559        1    0.0102     0.98      1.02  <0.001
```

::: :::

Note that `folds = 1` turns off cross-fitting (sample splitting). That is fine for a small, fast demonstration with `SL.glm`, but in applied work with flexible learners you should use several folds so the nuisance fits are cross-fitted; otherwise the standard errors can be anti-conservative.

Note on the estimand: this `lmtp_tmle` example only shifts A_2 (`trt = "A_2"`), treating A_1 as an observed baseline covariate (via `baseline = W`) rather than also intervening on it. That is a different, single-period policy from the always-1-vs-always-0 *both-period* regime computed manually via `simulated_data1$Yd`/`simulated_data2$Yd` above (lines 139-152) — the two numbers are not expected to match, and should not be used as a sanity check against each other. To replicate the both-period always-treat/never-treat contrast with `lmtp`, shift both time points by passing `trt = c("A_1", "A_2")` and supplying `time_vary` (the time-varying covariates measured before each treatment time, e.g. `list(character(0), "L_2")`), with `shift` functions that map each treatment in the vector to 1 (or 0).

35.3. another example

The data set bmi, from the DynTxRegime package, are simulated to reflect a two-stage RCT {A1, A2} that studied the effect of meal replacement (MR) shakes versus a calorie deficit (CD) diet on adolescent

35.3.1. shift function 1

Consider a shift function that assigns meal replacement to all observations at time 1, but only meal replacement at time 2 to those observations whose 4-month BMI is greater than 30.

```r
library(DynTxRegime)
data(bmiData)
bmi <- bmiData
glimpse(bmi)
```

```
Rows: 210
Columns: 8
$ gender      <int> 0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1~
$ race        <int> 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 0~
$ parentBMI   <dbl> 31.59683, 30.17564, 30.27918, 27.49256, 26.42350, 29.30970~
$ baselineBMI <dbl> 35.84005, 37.30396, 36.83889, 36.70679, 34.84207, 36.68640~
$ month4BMI   <dbl> 34.22717, 36.38014, 34.42168, 32.52011, 33.72922, 32.06622~
$ month12BMI  <dbl> 34.27263, 36.38401, 34.41447, 32.52397, 33.73546, 32.15977~
$ A1          <chr> "CD", "CD", "MR", "CD", "CD", "MR", "CD", "CD", "CD", "CD"~
$ A2          <chr> "MR", "MR", "CD", "CD", "CD", "MR", "MR", "CD", "MR", "MR"~
```

```
d_dtr <- function(data, trt) {
  if (trt == "A1") return(rep("MR", nrow(data)))

  ifelse(data$month4BMI > 30, "MR", "CD")
}

fit_dtr <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = d_dtr,
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
)

fit_dtr
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 35.854
    Std. error: 0.362
```

:::

35.3.2. shift function 2

Suppose we are interested in comparing the dynamic treatment regime to a static treatment regime where all patients receive meal replacement at both time points. Using the SDR estimator, estimate the effect of this static intervention.

```
fit_MR <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = \(data, trt) rep("MR", nrow(data)),
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
```

```
)

fit_MR
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 35.831
    Std. error: 0.361
```

:::

Let's also estimate the effect of an intervention where all patients receive a calorie deficit diet at both time points.

```
fit_CD <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = \(data, trt) rep("CD", nrow(data)),
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
)

fit_CD
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 35.053
    Std. error: 0.301
```

:::

Finally, we can compare the three treatment regimes using the `lmtp_contrast` function.

```
lmtp_contrast(fit_dtr, fit_MR, ref = fit_CD)
```

::: {.cell-output .cell-output-stdout}

```
  shift  ref estimate std.error conf.low conf.high p.value
1  35.9 35.1    0.801     0.335    0.144      1.46  0.0169
2  35.8 35.1    0.778     0.335    0.122      1.43  0.0201
```

:::

Systematic treatment: R · Julia.

Part VII.

Advanced Topics

36. Doing the same multi-level model with R and stata

This post is a demo for doing the same multi-level models with R and stata. R's multilevel models are mostly done with lme4 and nlme. Stata's multilevel models are mostly done with mixed and menl. The two packages are not exactly the same. `mixed` does in fact support several residual covariance structures via its `residuals()` option (e.g. `residuals(exponential, t(year))`, as well as `ar`, `ma`, `toeplitz`, etc.) — so it's not that `mixed` cannot handle a structured residual at all, but `menl`'s nonlinear-mixed-model framework gives more flexibility to combine that with custom nonlinear mean/random-effect specifications (like the CEM model below), which is why we reach for it here. Likewise, in R lme4's "lmer" does not allow this specification. We have to use nlme. But it's good to know how to do the same thing in both packages.

For example, we want to do a consensus emergence model (CEM) as described in Lang, Bliese and de Voogt (2018), *Personnel Psychology*. The CEM model is basically a growth model with the residual being a exponential function of the time variable. We want to do the same thing in both R and stata.

We start with simpler models.

36.1. Stata and R syntax comparison for the same three level model

We run a three level model specified in stata with mixed and menl. In R with nlme and lme4. This is a random intercept model. Only intercept is varying for each unit at each level. In this data set supposedly state is nested within region. There are situations that levels are not nested, then we need to specify cross effect models. We do not discuss that here. One point to watch: `nlme`'s `lme` function expects the grouping factors in the `random` specification ordered from the *highest* level down. Here `state` is nested within `region`, so `region` must be listed before `state`; reversing the order leads `lme` to treat regions as nested within states, which is not the intended structure.

There are different ways to specify this model in nlme. I show two ways here (including lme2 which is commented out).

36.1.1. Stata

```
* let's see how to use menl
webuse productivity
* to replicate mixed with menl
```

```
mixed gsp || region: || state:
menl gsp = {b1:},  define(b1: U1[region] UU1[region>state])
```

```
(Public capital productivity)


Performing EM optimization ...

Performing gradient-based optimization:
Iteration 0:  Log likelihood =  200.86162
Iteration 1:  Log likelihood =  200.86162

Computing standard errors ...

Mixed-effects ML regression                          Number of obs = 816

        Grouping information
        -------------------------------------------------------------
                        |     No. of       Observations per group
         Group variable |     groups    Minimum     Average    Maximum
        ----------------+--------------------------------------------
                 region |          9         51        90.7        136
                  state |         48         17        17.0         17
        -------------------------------------------------------------

                                                     Wald chi2(0)   =    .
Log likelihood =  200.86162                          Prob > chi2    =    .

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _cons |   10.65961   .2503806    42.57   0.000     10.16887    11.15035
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Identity             |
                  var(_cons) |   .4376123   .2697616        .13073    1.464887
-----------------------------+------------------------------------------------
state: Identity              |
                  var(_cons) |   .6080626   .1382733      .3893904    .9495357
-----------------------------+------------------------------------------------
               var(Residual) |   .0246633   .0012586      .0223159    .0272577
------------------------------------------------------------------------------
LR test vs. linear model: chi2(2) = 2750.56               Prob > chi2 = 0.0000
```

```
Note: LR test is conservative and provided only for reference.


Obtaining starting values by EM:

Alternating PNLS/LME algorithm:

Iteration 1:  Linearization log likelihood =  200.86162
Iteration 2:  Linearization log likelihood =  200.86162

Computing standard errors:

Mixed-effects ML nonlinear regression          Number of obs     =        816

      Grouping information
      -------------------------------------------------------------
                      |     No. of       Observations per group
                 Path |     groups    Minimum    Average    Maximum
      ----------------+--------------------------------------------
               region |          9         51       90.7        136
         region>state |         48         17       17.0         17
      -------------------------------------------------------------


Linearization log likelihood =  200.86162

          b1: U1[region] UU1[region>state]

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
b1           |
       _cons |   10.65961   .2505341    42.55   0.000     10.16857    11.15065
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Identity             |
                     var(U1) |   .4376119   .2686479      .1313834    1.457598
-----------------------------+------------------------------------------------
region>state: Identity       |
                    var(UU1) |   .6080625   .1381527      .3895418    .9491666
-----------------------------+------------------------------------------------
               var(Residual) |   .0246633   .0012586      .0223159    .0272577
------------------------------------------------------------------------------
```

36.1.2. R

The data are `productivity`: 816 observations of gross state product, with 48 states nested in 9 regions, 17 years each. The model is intercept-only with random intercepts at both levels.

These two R packages return same results. Note that lmer reports variance, while lme reports standard deviation.

And they do: `lmer` reports variances of 0.60775 for state-within-region and 0.50977 for region, with a residual of 0.02466; `lme` reports the corresponding standard deviations 0.77958, 0.71398 and 0.15705. Squaring the `lme` figures returns the `lmer` ones ($0.77958^2 = 0.6077$), so the two agree exactly and differ only in what they choose to print. The fixed intercept is 10.665 either way.

Note where the variance sits: between regions and between states, with almost none left within a state over time — the residual standard deviation of 0.157 is about a fifth of the state-level one. Most of the variation in this outcome is cross-sectional, not longitudinal.

```r
library(tidyverse)
library(haven)
library(dplyr)
library(readxl)
library(nlme)
library(lme4)
data <- read_dta('https://www.stata-press.com/data/r18/productivity.dta')
# lme4 for three level model
# (1|region) + (1|state) specifies CROSSED random effects; it numerically
# coincides with the intended nested structure here only because each
# state's label happens to be unique across the whole dataset (never reused
# across regions) -- verified: identical logLik to the explicitly nested
# (1|region/state) below. The nested form is the conceptually correct one
# and is robust even if state labels were ever reused across regions.
lmer1 <- lmer(gsp ~ 1 + (1|region/state), data=data)
summary(lmer1)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: gsp ~ 1 + (1 | region/state)
   Data: data

REML criterion at convergence: -400.9

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-2.8328 -0.6757  0.0474  0.6382  3.3928 

Random effects:
 Groups       Name        Variance Std.Dev.
 state:region (Intercept) 0.60775  0.7796  
 region       (Intercept) 0.50977  0.7140  
```

```
 Residual                 0.02466  0.1570
Number of obs: 816, groups:  state:region, 48; region, 9

Fixed effects:
            Estimate Std. Error t value
(Intercept)   10.665      0.266    40.1
```

```
lme1 <- lme(gsp ~ 1, random= list( region=~1, state=~1), data=data)
#or
#lme1 <- lme(gsp ~ 1, random= list(region = pdSymm(~1), state=pdSymm(~1)), data=data)
#but
#lme1 <- lme(gsp ~ 1, random= list( state=~1, region=~1), data=data)
#would be a different result!
#The other way to specify:
#lme2 <- lme(gsp ~ 1, random= ~ 1 | region/state, data=data)
summary(lme1)
```

```
Linear mixed-effects model fit by REML
  Data: data
        AIC       BIC   logLik
  -392.8509 -374.0381 200.4254

Random effects:
 Formula: ~1 | region
        (Intercept)
StdDev:   0.7139792

 Formula: ~1 | state %in% region
        (Intercept)  Residual
StdDev:   0.7795814 0.1570457

Fixed effects:  gsp ~ 1
               Value Std.Error  DF  t-value p-value
(Intercept) 10.66483 0.2659842 768 40.09574       0

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.83282326 -0.67572390  0.04744387  0.63817953  3.39281358

Number of Observations: 816
Number of Groups:
           region state %in% region
                9                48
```

36.2. A growth model in R and stata

We implement a growth model in R and stata. The results are somewhat different. The results may not be that reliable given that the year slope random component is tiny.

36.2.1. Stata

```
* same growth model with mixed and menl
* Stata's "mixed ... || region: year" defaults to an INDEPENDENT covariance
* structure for region's random intercept and slope (var(year), var(_cons),
* cov=0 by construction) -- verified via `estat recovariance`. R's lmer
* "(year|region)" and lme's "region=~year" both default to an UNSTRUCTURED
* (correlated) covariance. Add covariance(unstructured) to match R.
webuse productivity
mixed gsp year || region: year, covariance(unstructured) || state:
menl gsp =  {b2:} + {U1[region]}*year, define(b2: year U0[region]  U11[region>state])
```

```
(Public capital productivity)


Performing EM optimization ...

Performing gradient-based optimization:
Iteration 0:  Log likelihood =  782.11459  (not concave)
Iteration 1:  Log likelihood =  803.58071  (not concave)
Iteration 2:  Log likelihood =   813.1017  (not concave)
Iteration 3:  Log likelihood =  816.56892
Iteration 4:  Log likelihood =  885.27225  (not concave)
Iteration 5:  Log likelihood =  888.29208
Iteration 6:  Log likelihood =  888.79472
Iteration 7:  Log likelihood =  888.82395
Iteration 8:  Log likelihood =  888.82404
Iteration 9:  Log likelihood =  888.82404

Computing standard errors ...

Mixed-effects ML regression                          Number of obs =    816

        Grouping information
        -------------------------------------------------------------
                        |     No. of       Observations per group
         Group variable |     groups    Minimum    Average    Maximum
        ----------------+--------------------------------------------
                 region |          9         51       90.7        136
                  state |         48         17       17.0         17
```

```
            ------------------------------------------------------------

                                                        Wald chi2(1)  = 107.46
Log likelihood =  888.82404                             Prob > chi2   = 0.0000

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        year |   .0263275   .0025397    10.37   0.000     .0213497    .0313053
       _cons |  -41.43216    5.20712    -7.96   0.000    -51.63793   -31.22639
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Unstructured         |
                   var(year) |    .000056   .0000274      .0000215     .000146
                  var(_cons) |   235.7658     115.06      90.58819    613.6066
            cov(year,_cons)  |  -.1148543    .056104     -.2248162   -.0048923
-----------------------------+------------------------------------------------
state: Identity              |
                  var(_cons) |   .6121886    .139357        .39185    .9564242
-----------------------------+------------------------------------------------
               var(Residual) |   .0039862   .0002046      .0036046    .0044081
------------------------------------------------------------------------------
LR test vs. linear model: chi2(4) = 4112.15               Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.


Obtaining starting values by EM:

Alternating PNLS/LME algorithm:

Iteration 1:  Linearization log likelihood =  784.65134
Iteration 2:  Linearization log likelihood =  784.65343
Iteration 3:  Linearization log likelihood =  784.65343

Computing standard errors:

Mixed-effects ML nonlinear regression           Number of obs     =        816

      Grouping information
      -------------------------------------------------------------------
                           |     No. of       Observations per group
                      Path |     groups    Minimum    Average    Maximum
      ---------------------+---------------------------------------------
                    region |          9         51       90.7        136
```

```
              region>state |         48         17       17.0          17
      -----------------------------------------------------------------


                                                Wald chi2(1)      =    2737.78
Linearization log likelihood =  784.65343       Prob > chi2       =     0.0000

          b2: year U0[region] U11[region>state]

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
b2           |
        year |   .0274903   .0005254    52.32   0.000     .0264605      .02852
       _cons |  -43.71615   1.069039   -40.89   0.000    -45.81143   -41.62087
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Independent          |
                     var(U0) |   .4378219   .2687857      .1314413    1.458355
                     var(U1) |   5.40e-13   2.67e-10             0           .
-----------------------------+------------------------------------------------
region>state: Identity       |
                    var(U11) |   .6091976   .1381402      .3906086    .9501115
-----------------------------+------------------------------------------------
               var(Residual) |   .0053926   .0002752      .0048793    .0059598
------------------------------------------------------------------------------
```

36.2.2. R

```
# lme4 for three level model, growth model
lmer_growth1 <- lmer(gsp ~ 1 + year + (year|region) +  (1|state), data=data,  REML=FALSE)
summary(lmer_growth1)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: gsp ~ 1 + year + (year | region) + (1 | state)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
  -1455.8   -1422.9     734.9   -1469.8       809

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.9207 -0.5232  0.0180  0.5541  3.8399
```

```
Random effects:
 Groups   Name        Variance  Std.Dev.  Corr
 state    (Intercept) 1.327e-01 3.643e-01
 region   (Intercept) 2.068e-01 4.547e-01
          year        4.495e-09 6.705e-05 1.00
 Residual             6.664e-03 8.163e-02
Number of obs: 816, groups:  state, 48; region, 9

Fixed effects:
              Estimate Std. Error t value
(Intercept) -4.373e+01  1.165e+00  -37.53
year         2.751e-02  5.838e-04   47.13

Correlation of Fixed Effects:
     (Intr)
year -0.985
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```r
# nlme for three level model, growth model
#lme_growth1<-lme(gsp ~ year, random = list(region=pdSymm(~year), state=pdIdent(~1)),data=data
lme_growth1<-lme(gsp ~ year, random = list(region=~year, state=~1),data=data, method="ML")
summary(lme_growth1)
```

```
Linear mixed-effects model fit by maximum likelihood
  Data: data
        AIC       BIC   logLik
  -1555.307 -1522.376 784.6534

Random effects:
 Formula: ~year | region
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev      Corr
(Intercept) 6.61523e-01 (Intr)
year        7.96178e-09 0.001

 Formula: ~1 | state %in% region
        (Intercept)  Residual
StdDev:   0.7805102 0.0734343

Fixed effects:  gsp ~ year
                Value Std.Error  DF   t-value p-value
(Intercept) -43.71617 1.0690287 767 -40.89336       0
year          0.02749 0.0005254 767  52.32366       0
 Correlation:
     (Intr)
```

```
year -0.972

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-3.24606517 -0.59087173  0.03087456  0.62062441  4.27790886

Number of Observations: 816
Number of Groups:
           region state %in% region
                9                48
```

36.3. cem in R and stata

"CEM" is a growth model with the residual being a exponential function of the time variable.

36.3.1. stata

This mode has some convergence problem. I still use it for demonstration purpose.

```
webuse productivity, clear

* this replicates the results from R
menl gsp =  {b2:} + {U1[region]}*year, define(b2: year U0[region]  U11[region>state]) resvariar
```

```
(Public capital productivity)


Mixed-effects ML nonlinear regression           Number of obs     =        816

      Grouping information
      -------------------------------------------------------------------
                           |     No. of       Observations per group
                      Path |     groups    Minimum     Average    Maximum
      ----------------------+--------------------------------------------
                    region |          9         51        90.7        136
              region>state |         48         17        17.0         17
      -------------------------------------------------------------------

                                                Wald chi2(1)      =     107.29
Linearization log likelihood =  877.96184       Prob > chi2       =     0.0000

          b2: year U0[region] U11[region>state]

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
```

```
--------------+----------------------------------------------------------------
b2            |
         year |   .0274256   .0026478    10.36   0.000     .0222361    .0326151
        _cons |  -43.54047   5.400007    -8.06   0.000    -54.12429   -32.95665
-------------------------------------------------------------------------------

-------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+-------------------------------------------------
region: Independent          |
                     var(U0) |   263.5659   128.3252      101.4975    684.4206
                     var(U1) |   .0000633   .0000309      .0000243    .0001646
-----------------------------+-------------------------------------------------
region>state: Identity       |
                    var(U11) |   .6046714   .1369848      .3878686     .942658
-----------------------------+-------------------------------------------------
Residual variance:           |
  Exponential year           |
                      sigma2 |   9.96e-99          .             .           .
                       gamma |   .0556293    .000013      .0556039    .0556548
-------------------------------------------------------------------------------
```

36.3.2. R

As in stata, it has some convergence problem. So the results can differ somewhat between R and stata.

```
lme_growth1<-lme(gsp ~ year, random = list(region=pdSymm(~year), state=pdIdent(~1)),data=data,
lme_growth2<-update(lme_growth1,weights=varExp( form = ~ year), method="ML", control = lmeCont
summary(lme_growth2)
```

```
Linear mixed-effects model fit by maximum likelihood
  Data: data
        AIC       BIC   logLik
  -1548.244 -1510.608 782.1218

Random effects:
 Formula: ~year | region
 Structure: General positive-definite
            StdDev       Corr
(Intercept) 0.2937615870 (Intr)
year        0.0001436049 0.995

 Formula: ~1 | state %in% region
        (Intercept)  Residual
StdDev:   0.8046618 0.0736431
```

```
Variance function:
 Structure: Exponential of variance covariate
 Formula: ~year
 Parameter estimates:
        expon
-2.165027e-16
Fixed effects:  gsp ~ year
                Value Std.Error  DF   t-value p-value
(Intercept) -43.80206 1.0533833 767 -41.58226       0
year          0.02752 0.0005291 767  52.01290       0
 Correlation:
     (Intr)
year -0.976

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-3.24679735 -0.58919213  0.03104216  0.61843888  4.27977261

Number of Observations: 816
Number of Groups:
           region state %in% region
                9               48
```

```r
delta1<-lme_growth2$modelStruct$varStruct
delta1
```

```
Variance function structure of class varExp representing
        expon
-2.165027e-16
```

37. Modeling variation, not just the mean, in Likert-scale data

37.1. Modeling the spread, not the mean

Usually we regress a Likert item on covariates and model only its mean. A 1–7 satisfaction item, a 1–5 agreement item, a peer rating — we ask what moves the average.

But many questions are about the spread. Do evaluators agree more about some employees than others? Does one group of raters disagree more than another? Does a training program tighten the distribution of responses, or only shift its center?

The natural move is to compute a spread for each unit — each subject's ratings across raters, each respondent's answers across items — and regress *that* on covariates, the way we regress a mean. On a bounded scale, that move breaks. This chapter is about why it breaks, and what to do instead.

37.2. The ceiling is exact, not empirical

On a scale with a hard floor and ceiling, the largest variance a set of responses can have depends on where their mean sits. This is not a tendency in typical data. It is a mathematical fact, true for any distribution on a bounded interval.

Take any X confined to $[a, b]$. Both $(X - a)$ and $(b - X)$ are non-negative for every draw, so their product is too:

$$(X - a)(b - X) \geq 0 \quad \text{for every } X \in [a, b]. \tag{37.1}$$

Take expectations, expand, and substitute into $\mathrm{Var}(X) = E[X^2] - \mu^2$ (with $\mu = E[X]$):

$$\mathrm{Var}(X) \leq (a + b)\mu - ab - \mu^2 = (\mu - a)(b - \mu). \tag{37.2}$$

This is the **Bhatia–Davis inequality** (Bhatia and Davis 2000). In words: fix the mean, and the variance cannot exceed a value set entirely by that mean and the scale's endpoints. Equality is reached, not just approached — put all the mass at the two endpoints, $P(X = a) = (b - \mu)/(b - a)$ and $P(X = b) = (\mu - a)/(b - a)$, and the variance hits $(\mu - a)(b - \mu)$ exactly. That is the most polarized a set of responses can be, given the mean.

On a 1–7 item that is

$$\mathrm{SD}_{\max}(\mu) = \sqrt{(\mu - 1)(7 - \mu)}, \tag{37.3}$$

a downward parabola. It is widest at the midpoint ($\mu = 4$, where $\mathrm{SD}_{\mathrm{max}} = 3$) and pinched to exactly zero at both ends. A respondent whose raters all pick 7 has zero disagreement, by definition. There is nowhere else for a rating to go.

```
mu_grid <- seq(1, 7, length.out = 200)
sd_max  <- sqrt(pmax((mu_grid - 1) * (7 - mu_grid), 0))

ggplot(data.frame(mu = mu_grid, sd_max = sd_max), aes(mu, sd_max)) +
  geom_line(linewidth = 1, color = "#2d5fa3") +
  labs(x = "mean rating ( )", y = expression(SD[max](mu)),
       title = "Maximum possible SD, as a function of the mean") +
  theme_minimal(base_size = 12)
```

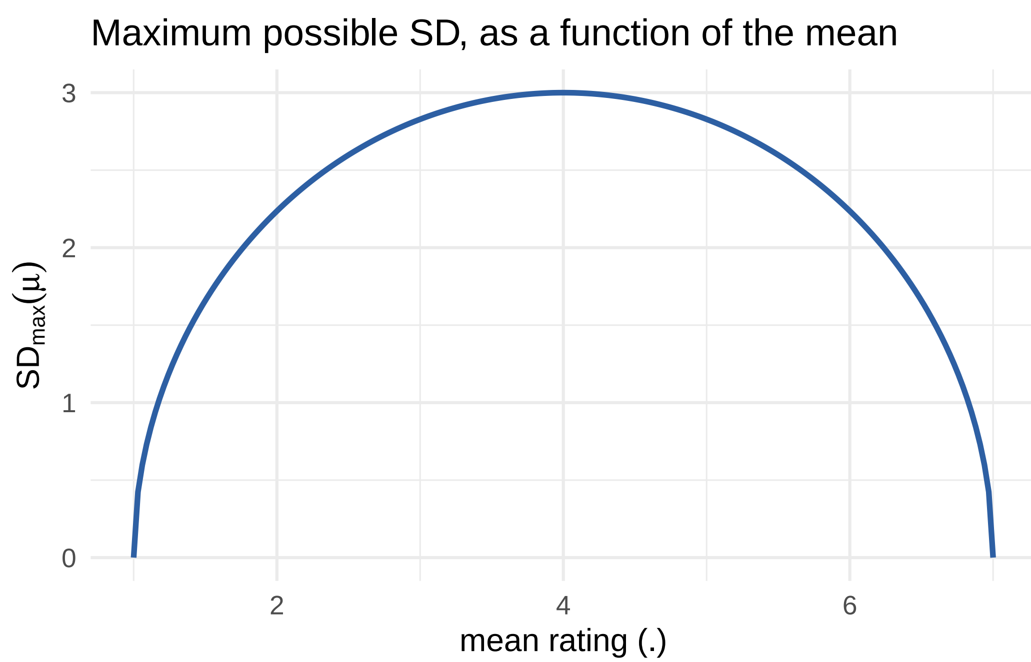


Figure 37.1.: The mechanical ceiling on SD, given the mean, for a 1–7 scale.

The consequence is direct. Two groups whose *means* differ have different *ceilings* on their SD, whether or not their true dispersion differs at all. And skew makes this the common case, not a corner case. Satisfaction items, performance ratings, and health scales usually sit near one end of the scale — exactly where the curve is steepest, and where raw SD comparisons mislead most.

A finite-sample refinement

The curve is the *population* bound. For a finite sample of n real numbers with mean μ, the true maximum is tighter still. Ellis (2025) gives it exactly as $\mathrm{Var}_{\mathrm{max}}(n, \mu) = (\mu - 1)(7 - \mu) - \frac{36}{n} a(1 - a)$, where a is the fractional part of $n(\mu - 1)/6$; this sits between $(\mu - 1)(7 - \mu) - 9/n$ and $(\mu - 1)(7 - \mu)$. At $n = 5$ raters that is up to 1.8 points^2 of extra slack — worth knowing when only a handful of people rate each unit, since the population curve then understates how tight the real ceiling is.

37.3. Dividing by the mean does not fix it

The first instinct is to divide the SD by the mean, the way you would for a coefficient of variation. It does not work here. That correction is built for a *different* mechanism — variance scaling proportionally with the mean, as in count data — not a ceiling that pinches to zero at *both* ends. $\mathrm{SD}_{\max}(\mu)/\mu$ is still a curve: zero at $\mu = 1$, rising to a peak of about 1.13 at $\mu = 7/4$, then falling back to zero at $\mu = 7$ (it happens to pass through 0.75 at $\mu = 4$). Note that it is not even shaped like the $\mathrm{SD}_{\max}$ curve it was meant to flatten — dividing by μ pulls the maximum well below mid-scale, so the correction trades a symmetric mean-dependence for a skewed one. Dividing by the mean does not remove the mean-dependence. It relabels it.

37.4. The right idea: spread after removing the ceiling effect

Every fix worth having does the same thing — measure dispersion *after* dividing out the ceiling. Beta regression names this directly. Rescale the outcome to $(0, 1)$ (e.g. $(y-1)/6$ for a 1–7 item) and parameterize by a mean μ and a precision φ (Ferrari and Cribari-Neto 2004; `betareg` in Cribari-Neto and Zeileis 2010). On the 1–7 scale,

$$\mathrm{Var}(Y) = \frac{(\mu-1)(7-\mu)}{1+\varphi} \qquad \Longrightarrow \qquad \varphi = \frac{(\mu-1)(7-\mu)}{\mathrm{Var}(Y)} - 1. \tag{37.4}$$

Set $\varphi = 0$ and this *is* the Bhatia–Davis bound. In words: φ is dispersion measured after the boundary is divided out, so $\varphi \sim$ group compares spread without inheriting the ceiling the way raw SD does.

φ itself is awkward to report — it lives on $(0, \infty)$ with no natural reference point. Its bounded twin is easier:

$$R := \frac{\mathrm{Var}(Y)}{\mathrm{Var}_{\max}(\mu)} = \frac{1}{1+\varphi}. \tag{37.5}$$

By Bhatia–Davis, $\mathrm{Var}(Y) \le \mathrm{Var}_{\max}(\mu)$ for *any* distribution on $[1, 7]$, not just Beta ones, so $R \in [0, 1]$ always, with no distributional assumption. $R = 1$ means a subject's raters are as polarized as the ceiling allows, given their mean; $R = 0$ means perfect consensus. It reads like a percentage because it is one.

Two costs come with it. First, it is a continuous model grafted onto discrete, ordinal data — a 1–7 response is not a continuous proportion. Second, both forms put a rater-estimated variance in a denominator, and that variance comes from only a handful of raters. For φ the denominator is $\mathrm{Var}(Y)$: when a few raters agree by chance — more likely near the ceiling, where most of the sample lives — $\mathrm{Var}(Y) \to 0$ and $\varphi \to \infty$. For R the denominator is $\mathrm{Var}_{\max}(\mu)$, which is zero whenever the mean sits at a scale endpoint; if all raters pick 7, then $\mathrm{Var}(Y)$ and $\mathrm{Var}_{\max}$ vanish together and $R = 0/0$ is undefined outright. We return to this in the worked example, because it bites hard.

37.5. Two models: one with the boundary, one without

37.5.1. Gaussian location-scale model (MELSM)

The location-scale model is the fix people reach for first. Model the mean as usual, and *also* model the residual SD as a function of covariates.

```r
library(brms)

fit_gaussian <- brm(
  bf(rating ~ group + (1 | subject),
     sigma ~ group),                    # model the SD, not just the mean
  data = df, family = gaussian()
)
```

`brms` puts a log link on `sigma`, so the `group` coefficient on the `sigma` part is a difference in *log*-SD. Exponentiate it for an SD ratio between groups.

The catch is fatal for our question. A Gaussian model has no boundary in it. It does not know the scale stops at 1 and 7. So it cannot tell "this group's raters genuinely agree more" from "this group's mean sits closer to the ceiling." Both produce the identical signature in the fitted `sigma`.

37.5.2. Ordinal cumulative-probit MELSM: put the boundary inside the model

The clean fix treats the 1–7 response as what it is: a discretized slice of a latent continuous trait, cut by six ordered thresholds.

Posit an unobserved continuous Y_i^* behind each rating,

$$Y_i^* = \eta_i + \sigma_i \epsilon_i, \qquad \epsilon_i \sim N(0,1), \tag{37.6}$$

where η_i is the linear predictor for *location* (`group` plus a subject random intercept, say) and σ_i is the latent residual SD, which we also let depend on covariates. The observed rating is Y_i^* coarsened at six fixed thresholds $\theta_1 < \cdots < \theta_6$ (with $\theta_0 = -\infty$, $\theta_7 = \infty$):

$$Y_i = k \iff \theta_{k-1} < Y_i^* \le \theta_k. \tag{37.7}$$

So the cumulative probability is

$$P(Y_i \le k) = P\left(\epsilon_i \le \frac{\theta_k - \eta_i}{\sigma_i}\right) = \Phi\left(\frac{\theta_k - \eta_i}{\sigma_i}\right), \tag{37.8}$$

with Φ the standard normal CDF, for the probit link. Write $\text{disc}_i := 1/\sigma_i$ and this is `brms`'s parameterization exactly:

$$P(Y_i \le k) = \Phi(\text{disc}_i\,(\theta_k - \eta_i)). \tag{37.9}$$

In words: discrimination is the reciprocal of the latent residual SD. Not an analogy — an algebraic identity, reached by rescaling the argument of Φ rather than dividing by σ_i directly. Letting `disc` depend on covariates is the ordinal analogue of `sigma ~ group`, written in reciprocal form: a positivity-friendly way to let dispersion vary (disc > 0 everywhere, with no constraint on σ_i itself). `brms` puts a log link on `disc`, so `disc ~ 0 + group` means $\log(\text{disc}_i) = \gamma_{\text{group}[i]}$ — a group difference in *log*-discrimination, parallel to the Gaussian log-SD coefficient (Bürkner and Vuorre 2019).

```
fit_ordinal <- brm(
  bf(rating ~ group + (1 | subject),
     disc ~ 0 + group),           # log(disc) = 0 + group; disc = 1 / latent SD
  data = df, family = cumulative("probit"), init = 0,
  prior = c(
    prior(constant(0), class = "b", coef = "groupA", dpar = "disc"),
    prior(normal(0, 1), class = "b", coef = "groupB", dpar = "disc")
  )
)
```

The `constant(0)` prior pins the reference group's *log*-discrimination — the linear predictor γ_A, not disc itself — at zero, so $\text{disc}_A = \exp(0) = 1$ and its latent SD is $1/\text{disc}_A = 1$. (Fixing $\gamma_A = 0$ is what makes `disc = 1` and `SD = 1` coincide for the reference group; it is not a claim that discrimination is zero, which would mean an infinite latent SD.) Only γ_B is estimated: $\text{disc}_B = \exp(\gamma_B)$, so $\gamma_B = 0$ means identical dispersion, $\gamma_B > 0$ means Group B is *more* discriminating (a *smaller* latent SD), and $\gamma_B < 0$ the reverse. The boundary is now explicit — six estimated cutpoints on an *unbounded* latent scale — so a group whose mean sits near 7 no longer shows a tighter latent SD by mechanics alone.

37.5.3. What belongs in the dispersion submodel?

A fair question: why does the dispersion side carry only `group`, when the mean model might carry more? Should the two not mirror each other? Not by default. They answer different questions. The mean model asks what predicts the average rating. The dispersion model asks what predicts the spread. A covariate earns its place on each side on its own merits.

In this chapter's simulation `group` is the only variable in the data-generating process, so there is nothing else *to* include, on either side. In real data the choice is substantive, and getting it wrong has a familiar cost. Suppose some covariate Z (rater experience, item wording, a subject-level trait) genuinely shifts dispersion and is correlated with `group`, but is left out of the dispersion submodel. Its effect is absorbed into the `group` coefficient on `sigma` (or `disc`, or φ) — omitted-variable bias for the *variance*, exactly analogous to the ordinary one for the mean. The remedy is the same one you already use on the mean side: include the covariates you have a substantive reason to think shift dispersion, and give that choice the same scrutiny — neither copying the mean formula reflexively nor stripping the dispersion submodel to the one comparison you happen to care about.

37.5.4. Four candidates, side by side

Two questions separate the approaches: does it remove the mechanical ceiling, and does it stay stable when only a handful of raters rate each unit? Only one passes both.

Approach	Removes the mechanical ceiling?	Stable at $n \approx 5$ raters?
SD / mean	No — still a curve in μ, just a different one	No — inherits raw SD's sampling noise
Hand-built $R = \text{Var}/\text{Var}_{\text{max}}$	Yes, by construction	No — the denominator collapses near the ceiling, and R is undefined at a scale endpoint
Gaussian MELSM's `sigma`	No — no boundary in the model, so it inherits the confound	Yes — the joint likelihood pools across units
Ordinal MELSM's `disc`	Yes — the boundary lives in six estimated thresholds	Yes — the joint likelihood pools across units

The useful way to remember it: R is the right idea computed the wrong way, and the Gaussian MELSM is the wrong idea computed the right way — stable and precise, but blind to the boundary, so it reports the ceiling artifact as a confident finding. `disc` is the same idea as R — spread with the ceiling effect removed — estimated coherently inside one model. The worked example shows all of this.

37.6. Worked example: same dispersion, different means

Simulate two groups of 100 subjects each, 6 raters per subject. Both groups' raters draw from a **latent** continuous opinion with the *same* true SD ($\sigma = 1$) — there is no real disagreement difference to find. Group A's latent mean sits high (near the ceiling once discretized); Group B's sits in the middle of the range.

```r
set.seed(42)

n_per_group <- 100
n_raters    <- 6
sigma_true  <- 1.0   # SAME latent SD in both groups -- no real difference
mu_A <- 2.4          # group A's latent mean sits near the ceiling
mu_B <- 0.6          # group B's sits mid-range

subjects <- data.frame(
  subject = 1:(2 * n_per_group),
  group   = rep(c("A", "B"), each = n_per_group),
  mu      = rep(c(mu_A, mu_B), each = n_per_group)
)

df <- subjects[rep(seq_len(nrow(subjects)), each = n_raters), ]
df$latent <- rnorm(nrow(df), mean = df$mu, sd = sigma_true)

# six fixed cutpoints -> seven observed categories, 1 through 7
```

```r
cuts <- c(-2, -1.2, -0.5, 0.2, 1.0, 1.8)
df$rating     <- as.integer(cut(df$latent, breaks = c(-Inf, cuts, Inf), labels = 1:7))
df$rating_ord <- factor(df$rating, ordered = TRUE, levels = 1:7)
```

The group difference enters in exactly one place, and it is easy to miss. `rnorm(..., sd = sigma_true)` uses the same `sigma_true` for every row, in both groups — `group` never touches the standard deviation, only the mean, via `df$mu`. The two groups differ *only* in where their latent mean sits relative to the six fixed cutpoints, which are the same `cuts` object for both. That alone produces very different observed spreads:

```r
edges <- c(-Inf, cuts, Inf)
tibble(
  category = 1:7,
  prob_A = round(diff(pnorm(edges, mean = mu_A, sd = sigma_true)), 3),
  prob_B = round(diff(pnorm(edges, mean = mu_B, sd = sigma_true)), 3)
) |>
  knitr::kable(caption = "Category probabilities implied by each group's latent mean, with the
```

Table 37.2.: Category probabilities implied by each group's latent mean, with the same cutpoints and the same sigma.

category	prob_A	prob_B
1	0.000	0.005
2	0.000	0.031
3	0.002	0.100
4	0.012	0.209
5	0.067	0.311
6	0.193	0.230
7	0.726	0.115

Group A's latent mean (2.4) sits 0.6 SDs above the highest cutpoint (1.8), so `pnorm(0.6)` ≈ 0.73: nearly three-quarters of its ratings collapse into the top category, and under 2% fall below category 5. Group B's mean (0.6) sits mid-range, so its ratings spread across all seven categories, none above about a third.

That imbalance — not a difference in σ — shrinks Group A's raw SD. When three-quarters of a subject's six ratings are forced into one bin by `cut()`, the *discrete* variance across them is mechanically small, even though the *continuous* latent draws behind them are as dispersed as Group B's ($\sigma = 1$ for both). It is the same Bhatia–Davis mechanism from the introduction, now visible subject by subject through `cut()`'s fixed boundaries rather than through the population $\mathrm{SD}_{\max}(\mu)$ curve.

The raw per-subject summary already shows the pattern that tempts a reader into seeing a real disagreement gap:

```r
subj_summary <- df |>
  group_by(subject, group) |>
  summarise(mean_rating = mean(rating), sd_rating = sd(rating), .groups = "drop")

knitr::kable(
  subj_summary |> group_by(group) |>
    summarise(mean_of_means = round(mean(mean_rating), 2),
              mean_of_sds   = round(mean(sd_rating), 2)),
  caption = "Raw per-subject summary by group -- true latent SD is IDENTICAL in both."
)
```

Table 37.3.: Raw per-subject summary by group – true latent SD is IDENTICAL in both.

group	mean_of_means	mean_of_sds
A	6.64	0.58
B	4.89	1.22

```r
ggplot(subj_summary, aes(mean_rating, sd_rating, color = group)) +
  geom_jitter(width = 0.05, height = 0, alpha = 0.5) +
  geom_line(data = data.frame(mean_rating = mu_grid, sd_rating = sd_max),
            aes(mean_rating, sd_rating), color = "grey40", inherit.aes = FALSE) +
  labs(x = "subject mean rating", y = "subject SD",
       title = "Group A's raw SD looks smaller -- purely because its mean sits\nnear the grey c
  theme_minimal(base_size = 12)
```

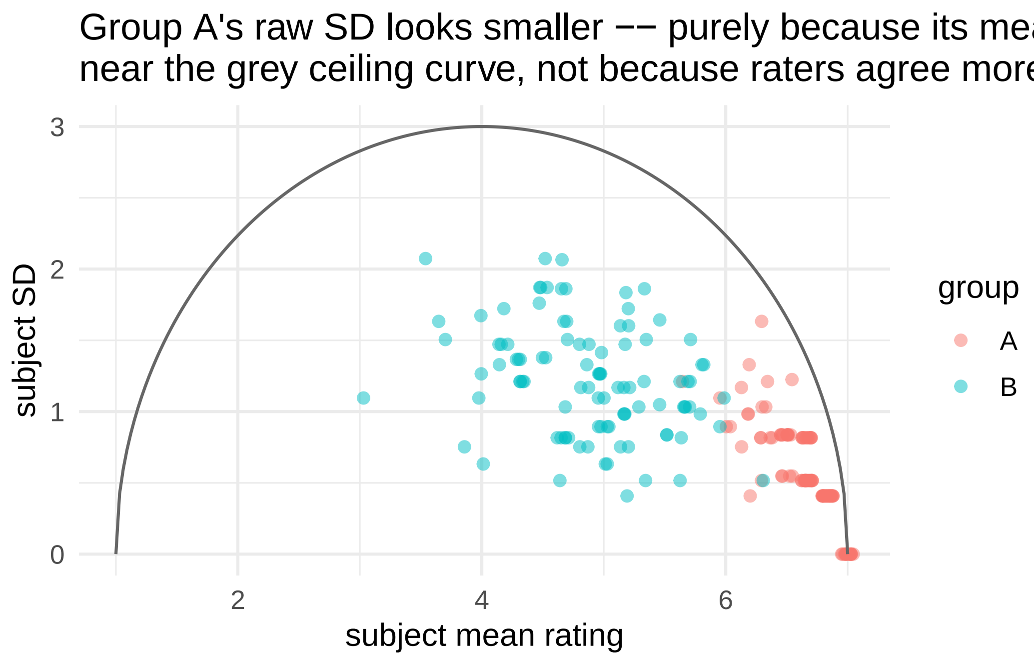


Figure 37.2.: Subject-level mean vs. SD, with the theoretical ceiling overlaid.

Now fit both a Gaussian MELSM and the ordinal cumulative-probit MELSM to the same data.

```r
fit_gaussian <- brm(
  bf(rating ~ group + (1 | subject), sigma ~ group),
  data = df, family = gaussian(),
  chains = 2, iter = 2000, warmup = 1000, cores = 2, seed = 1,
  refresh = 0, silent = 2
)
```

```r
fit_ordinal <- brm(
  bf(rating_ord ~ group + (1 | subject), disc ~ 0 + group),
  data = df, family = cumulative("probit"), init = 0,
  prior = c(
    prior(constant(0), class = "b", coef = "groupA", dpar = "disc"),
    prior(normal(0, 1), class = "b", coef = "groupB", dpar = "disc")
  ),
  chains = 2, iter = 2000, warmup = 1000, cores = 2, seed = 1,
  refresh = 0, silent = 2
)
```

```r
knitr::kable(round(fixef(fit_gaussian), 2), caption = "Gaussian MELSM fixed effects (sigma par
```

Table 37.4.: Gaussian MELSM fixed effects (sigma part is on the log-SD scale)

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	6.64	0.03	6.58	6.69
sigma_Intercept	-0.41	0.03	-0.46	-0.35
groupB	-1.75	0.06	-1.86	-1.63
sigma_groupB	0.68	0.04	0.60	0.76

```r
knitr::kable(round(fixef(fit_ordinal), 2), caption = "Ordinal MELSM fixed effects (disc part is
```

Table 37.5.: Ordinal MELSM fixed effects (disc part is on the log scale: 0 = same latent SD as the reference group)

	Estimate	Est.Error	Q2.5	Q97.5
Intercept[1]	-4.39	0.32	-5.06	-3.81
Intercept[2]	-3.65	0.23	-4.12	-3.22
Intercept[3]	-2.98	0.17	-3.34	-2.66
Intercept[4]	-2.27	0.12	-2.52	-2.04
Intercept[5]	-1.44	0.07	-1.59	-1.30
Intercept[6]	-0.63	0.06	-0.74	-0.52
groupB	-1.89	0.10	-2.11	-1.69
disc_groupA	0.00	0.00	0.00	0.00

	Estimate	Est.Error	Q2.5	Q97.5
disc_groupB	-0.01	0.08	-0.17	0.15

On the raw scale Group A averages 6.64 with an SD of 0.58; Group B averages 4.89 with an SD of 1.22. The pooled correlation between subject mean and SD is −0.74. That looks like a real, sizable disagreement gap, and the Gaussian MELSM agrees: `sigma_groupB` of 0.68 on the log-SD scale puts Group B's SD at **1.97 times** Group A's (95% CI: 1.82–2.14), nowhere near zero.

But there is no true difference. Both groups' raters were drawn with the same $\sigma = 1$ by construction; only the means differ. The ordinal model reads it correctly: `disc_groupB` is $\gamma_B = -0.015$ on the log scale (95% CI: −0.171 to 0.146), so $\text{disc}_B = e^{-0.015} \approx 0.985$ — a latent SD of ≈ 1.015 against the reference group's 1, recovering the fact that the two groups' dispersion is identical. The Gaussian model's "Group B disagrees twice as much" is not a finding about the raters. It is the ceiling effect from the previous section, wearing a regression coefficient.

In fairness, this is the ordinal model's best case: the simulation *is* its functional form, with one set of cutpoints shared by both groups and a constant latent σ. That is the right way to show the mechanism in isolation, but it does not test the assumption the model actually needs, which is that the thresholds are common across groups. If the two groups use the scale differently — differential item functioning, in the measurement literature — then threshold shifts and `disc` trade off against each other, and a dispersion difference can be read as a location or cutpoint difference instead. On real data that assumption deserves the same scrutiny the Gaussian model's missing boundary gets here.

37.6.1. What about R here?

R is mean-independent by construction, so it should land where the ordinal model does: no group difference. Compute it per subject, using the exact finite-sample bound from the earlier callout ($n = 6$ raters):

```
var_max_finite <- function(n, mu) {
  c <- (mu - 1) / 6
  a <- (n * c) %% 1
  36 * (c * (1 - c) - (1 / n) * a * (1 - a))
}

subj_R <- df |>
  group_by(subject, group) |>
  summarise(mean_rating = mean(rating),
            var_pop = mean((rating - mean(rating))^2),   # population (1/n) convention
            .groups = "drop") |>
  mutate(var_max = var_max_finite(n_raters, mean_rating),
         R = ifelse(var_max > 1e-8, var_pop / var_max, NA_real_))

knitr::kable(
  subj_R |> group_by(group) |>
```

```
    summarise(n_undefined = sum(is.na(R)), mean_R = round(mean(R, na.rm = TRUE), 3)),
  caption = "Per-subject R, using each subject's own 6 ratings for both mean and variance."
)
```

Table 37.6.: Per-subject R, using each subject's own 6 ratings for both mean and variance.

group	n_undefined	mean_R
A	14	0.626
B	0	0.200

This is worse than the raw SD comparison, not better. Group A's per-subject R averages 0.63, Group B's 0.20 — a larger apparent gap, and pointing the *opposite* way from the ceiling artifact. Fourteen Group A subjects even have an *undefined* R: all six raters picked 7, so observed variance and mechanical ceiling are both zero at once. Has R failed?

No. The guarantee — $R \in [0, 1]$ at every μ, no forced link to the mean — is about the *population* quantity. Plug in a mean from only 6 raters and it breaks, because the finite-sample bound in the denominator is itself estimated from that same noisy mean, and it is far more sensitive near the ceiling than mid-scale:

```
knitr::kable(
  subj_R |> group_by(group) |>
    summarise(mean_var_pop = round(mean(var_pop), 3),
              mean_var_max = round(mean(var_max, na.rm = TRUE), 3)),
  caption = "The denominator (mean_var_max) is nearly 7x smaller for Group A -- not because Gro
)
```

Table 37.7.: The denominator (mean_var_max) is nearly 7x smaller for Group A – not because Group A's true ceiling is that much tighter, but because a 6-rater sample mean near 7 makes the *estimated* ceiling collapse.

group	mean_var_pop	mean_var_max
A	0.374	0.984
B	1.366	6.903

A Group A subject's six raters average near 7 because that is where their mean sits — but exactly *how* near is noisy, and the finite-sample bound shrinks fast as the sample mean approaches the boundary. A modest observed variance over an unluckily tiny denominator gives a large R, for no reason connected to real disagreement. Swapping in the plain continuous bound shrinks the gap a great deal — 0.181 vs. 0.173 — but each value still averages 100 noisy single-subject estimates. Only pooling each group's full 600 ratings into one mean and one variance removes the noise entirely:

```
pooled <- df |> group_by(group) |>
  summarise(mean_rating = mean(rating), var_pop = mean((rating - mean(rating))^2)) |>
  mutate(var_max_cont = (mean_rating - 1) * (7 - mean_rating),
         R_pooled = round(var_pop / var_max_cont, 3))

knitr::kable(pooled |> select(group, mean_rating, R_pooled),
             caption = "R computed once per group, from all 600 ratings pooled.")
```

Table 37.8.: R computed once per group, from all 600 ratings pooled.

group	mean_rating	R_pooled
A	6.638333	0.219
B	4.893333	0.211

Pooled, R is 0.22 for Group A and 0.21 for Group B — essentially identical, agreeing with the ordinal model's null finding. The lesson is not "avoid R." It is that R, like φ, is only as good as the sample behind the mean and variance you feed it. A hand-computed per-subject R at 6 raters can manufacture a *larger* spurious gap than raw SD, in the wrong direction. The ordinal model never faces this, because it never forms a per-subject summary at all. `disc_groupB` comes straight from the full, pooled 600-observation likelihood per group — the "pooled" calculation above done coherently, with a proper posterior interval instead of an average of point estimates.

37.7. Takeaways

Summary

- On any bounded scale, the *maximum possible* variance is a function of the mean — the Bhatia–Davis inequality, $\mathrm{Var}(X) \leq (\mu - a)(b - \mu)$. Not a tendency to check for, a fact to design around.
- Dividing SD (or variance) by the mean does not remove the confound. The mechanism is not proportional scaling; it is a ceiling that pinches at both ends.
- A Gaussian location-scale model can estimate `sigma ~ x`, but inherits the confound, because it has no boundary in it.
- Beta regression's precision φ (and its bounded twin R) remove the confound algebraically, at the cost of a continuous approximation to discrete data. And R is only as good as the sample behind it: with few raters per subject, a hand-computed per-subject R can manufacture a *larger* spurious gap than raw SD, in the wrong direction — and is undefined outright when every rater picks the same endpoint of the scale, since then its numerator and denominator vanish together.
- An ordinal cumulative-link mixed model puts the boundary where it belongs — estimated thresholds on a latent scale — and is the cleanest way to ask "is this dispersion difference real?" on Likert data, because it never needs a per-unit summary statistic at all.

37.8. References

- Bhatia, R., and Davis, C. (2000). "A Better Bound on the Variance." *American Mathematical Monthly* 107(4): 353–357.
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- Cribari-Neto, F., and Zeileis, A. (2010). "Beta Regression in R." *Journal of Statistical Software* 34(2): 1–24.
- Ellis, J. L. (2025). "The maximum variance of a finite dataset, given its mean, minimum, and maximum." arXiv:2508.17525.
- Ferrari, S. L. P., and Cribari-Neto, F. (2004). "Beta Regression for Modelling Rates and Proportions." *Journal of Applied Statistics* 31(7): 799–815.

38. Conjoint design and analysis using R

This post is to illustrate how to use R to do basic conjoint analysis.

38.1. Conjoint Analysis

Marketing people used to use Sawtooth to do conjoint analysis.

What is conjoint analysis? Here is the intro from Sawtooth:

"Conjoint analysis methods use statistical analysis to compute mathematical representations of survey respondents' preferences for product features, simulate how attribute changes affect demand, and even predict market acceptance of new products before they launch.

This powerful research methodology has become very popular with market researchers and economists for the valuable insight that few other research methods can provide."

"Consider the task of purchasing a new vehicle.

When searching for the "right" vehicle, you typically consider a multitude of variables: type, make (brand), model, year, mileage, price, color, accessories packages, etc. You can see how this is a complex choice to make.

How does anyone possibly make such a complicated decision?

You think about what features matter most to you and then make tradeoffs.

For example, one vehicle could be an acceptable make and model, but at a high price point, whereas a second option is not the ideal make and model but has a more palatable price.

The tradeoff in this simple example is whether the make and model is more preferable to a reasonable price, or vice versa.

A conjoint survey captures the relationship between different attributes of a product and how preferable they are both comparatively and when combined.

Because of this, conjoint analysis is the gold standard solution for testing multiple features simultaneously, when limited A/B testing doesn't go far enough."

"In simple terms, the conjoint survey design algorithm generates a balanced survey design with product profiles, also known as concepts, formulated to have ideal statistical properties (such as level balance and independence of attributes).

Each profile is made up of multiple conjoined product features (hence, conjoint analysis), such as brand, type, engine, and color, each with systematically varied levels.

These product concepts are then included in a series of questions, usually 8-20, called choice tasks, that make up the conjoint analysis portion of the survey."

38.2. Example

I am following the example here: https://bookdown.org/lien_lamey/R_tutorial/8-1-data-5.html

```
library(tidyverse)
library(openxlsx)

url <- "https://feb.kuleuven.be/lien.lamey/public/rmodule/icecream.xlsx"
icecream <- read.xlsx(url)
icecream <- icecream %>%
  pivot_longer(cols = starts_with("Individual"), names_to = "respondent", values_to = "rating"
  rename("profile" = "Observations") %>% # rename Observations to profile
  mutate(profile = factor(profile), respondent = factor(respondent),  # factorize identifiers
         Flavor = factor(Flavor), Packaging = factor(Packaging), Light = factor(Light), Organi

icecream
```

```
# A tibble: 150 × 7
   profile   Flavor    Packaging       Light      Organic     respondent  rating
   <fct>     <fct>     <fct>           <fct>      <fct>       <fct>        <dbl>
 1 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      1
 2 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      6
 3 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      5
 4 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      1
 5 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      2
 6 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      7
 7 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      7
 8 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      5
 9 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…      1
10 Profile 1 Raspberry Homemade waffle No low fat Not organic Individual…     10
#   140 more rows
```

In this example, we have 4 attributes: Flavor, Packaging, Light, and Organic. There are 5, 3, 2, 2 levels correspondingly. Usually we are interested in how much does each level of attributes predict the outcome, say the rating of the icecream. To do this, we would think we'll need all 5*3*2*2=60 combinations of attribute levels to estimate the effect. In that case, we would need data for each combination. This is called full factorial design. The problem of full factorial design is that it might be too costly or not possible if the combination is huge. Instead, conjoint analysis allows us to estimate the effect of each level of each attribute by using a fraction of the combinations. This is done by creating a set of profiles, each of which contains a subset of the levels of the attributes. The profiles are then presented to the respondents, who rate the profiles. The ratings are then used to estimate the effect of each level of each attribute. This is done by regressing the ratings on the levels of the attributes. The coefficients of the regression are the estimates of the effects of the levels of the attributes.

The disadvantage of a fractional factorial design is that some effects are confounded; that is, the effect of one level of an attribute is confounded with the effect of another level of another attribute.

```
# attribute1, attribute2, etc. are vectors with one element in which we first provide the name
attribute1 <- "Flavor; Raspberry; Chocolate; Strawberry; Mango; Vanilla"
attribute2 <- "Package; Homemade waffle; Cone; Pint"
attribute3 <- "Light; Low fat; No low fat"
attribute4 <- "Organic; Organic; Not organic"

# now combine these different attributes into one vector with c()
attributes <- c(attribute1, attribute2, attribute3, attribute4)
attributes
```

```
[1] "Flavor; Raspberry; Chocolate; Strawberry; Mango; Vanilla"
[2] "Package; Homemade waffle; Cone; Pint"
[3] "Light; Low fat; No low fat"
[4] "Organic; Organic; Not organic"
```

38.3. Conjoint design of experiment

The first step in conjoint analysis is to create a design of experiment. This is to use a fraction of all possible combinations of attribute levels to estimate the effect of each level of each attribute. For example, instead of using all 60 icecream profiles, we use 10 profiles.

We use "radiant" package's "doe" function to create the design of experiment. The "doe" function takes the attributes as input and returns a data frame with the profiles. The "seed" argument is used to fix the random number generator. This is useful if you want to reproduce the same results.

```
library(radiant)

ice_doe <- doe(attributes, seed = 123)
ice_doe$eff
```

```
   Trials D-efficiency Balanced
1       9        0.105    FALSE
2      10        0.389    FALSE
3      11        0.411    FALSE
4      12        0.614    FALSE
5      13        0.542    FALSE
6      14        0.479    FALSE
7      15        0.762    FALSE
8      16        0.738    FALSE
9      17        0.748    FALSE
10     18        0.756    FALSE
11     19        0.644    FALSE
12     20        0.895    FALSE
13     21        0.848    FALSE
14     22        0.833    FALSE
```

```
15      23          0.790    FALSE
16      24          0.827    FALSE
17      25          0.787    FALSE
18      26          0.768    FALSE
19      27          0.759    FALSE
20      28          0.736    FALSE
21      29          0.702    FALSE
22      30          0.984     TRUE
23      31          0.952    FALSE
24      32          0.933    FALSE
25      33          0.928    FALSE
26      34          0.900    FALSE
27      35          0.871    FALSE
28      36          0.893    FALSE
29      37          0.866    FALSE
30      38          0.843    FALSE
31      39          0.836    FALSE
32      40          0.922    FALSE
33      41          0.899    FALSE
34      42          0.904    FALSE
35      43          0.882    FALSE
36      44          0.861    FALSE
37      45          0.949    FALSE
38      46          0.919    FALSE
39      47          0.912    FALSE
40      48          0.911    FALSE
41      49          0.891    FALSE
42      50          0.959    FALSE
43      51          0.939    FALSE
44      52          0.944    FALSE
45      53          0.925    FALSE
46      54          0.924    FALSE
47      55          0.906    FALSE
48      56          0.902    FALSE
49      57          0.884    FALSE
50      58          0.872    FALSE
51      59          0.855    FALSE
52      60          1.000     TRUE
```

```
#summary(doe(attributes, seed = 123, trials = 10))
```

This is a list of efficiency of different number of "trials". Trials mean combinations of attribute levels. For example, with 10 trials, the efficiency is .389 and it's unbalanced.

We can see that the only two designs with balanced design are 30 and 60. That's because 30 and 60 are the only two numbers that can be divided by 5, 3 and 2, all unique number of 5, 3, 3, 2. If the factors are, say, 3, 2, 3. Then the integers that can be divided by 3 and 2 are 6, 12, and 18.

Efficiency is a measure of the information content a design can capture. Efficiency are typically stated in relative terms, as in "design A is 80% as efficient as design B." In practical terms this means you will need 25% more (the reciprocal of 80%) design A observations (respondents, choice sets per respondent or a combination of both) to get the same standard errors and significance as with the more efficient design B. We used a measure of efficiency called D-Efficiency (Kuhfeld et al. 1994). In this case, the only orthogonal design is the full factorial design of 60 trials.

For example, a trial of 30 has a efficiency of .984, that means it's 98.4% as efficient as the full factorial design of 60 trials.

```
library(radiant)
ice_doe2 <- doe(attributes, seed = 123, trials = 30)
ice_doe2$cor_mat
```

```
               Flavor       Package         Light       Organic
Flavor   1.000000e+00 -1.942890e-16 -1.665335e-16 -1.665335e-16
Package -1.942890e-16  1.000000e+00 -1.665335e-16 -1.665335e-16
Light   -1.665335e-16 -1.665335e-16  1.000000e+00 -1.045299e-01
Organic -1.665335e-16 -1.665335e-16 -1.045299e-01  1.000000e+00
```

From the covariance matrix, we can se that in this 30 profile design, there is a correlation of -.105 between "Light" and "Organic". That means these two effects are confounded; the design is not orthogonal. That's why the efficiency is not 1. The D-efficiency is calculated as a function of the determinant of the correlation matrix. The D-efficiency of the full factorial design is 1. The D-efficiency of the 30 profile design is .984. That means the 30 profile design is 98.4% as efficient as the full factorial design.

The partial factorial design of 30 profile is

```
ice_doe2$part
```

```
   trial     Flavor         Package      Light     Organic
1      1  Raspberry Homemade_waffle    Low_fat     Organic
2      4  Raspberry Homemade_waffle No_low_fat Not_organic
3      6  Raspberry            Cone    Low_fat Not_organic
4      7  Raspberry            Cone No_low_fat     Organic
5     10  Raspberry            Pint    Low_fat Not_organic
6     11  Raspberry            Pint No_low_fat     Organic
7     13  Chocolate Homemade_waffle    Low_fat     Organic
8     14  Chocolate Homemade_waffle    Low_fat Not_organic
9     19  Chocolate            Cone No_low_fat     Organic
10    20  Chocolate            Cone No_low_fat Not_organic
11    22  Chocolate            Pint    Low_fat Not_organic
12    23  Chocolate            Pint No_low_fat     Organic
13    26 Strawberry Homemade_waffle    Low_fat Not_organic
14    28 Strawberry Homemade_waffle No_low_fat Not_organic
15    29 Strawberry            Cone    Low_fat     Organic
```

```
16     32 Strawberry            Cone No_low_fat Not_organic
17     33 Strawberry            Pint    Low_fat     Organic
18     35 Strawberry            Pint No_low_fat     Organic
19     39      Mango Homemade_waffle No_low_fat     Organic
20     40      Mango Homemade_waffle No_low_fat Not_organic
21     41      Mango            Cone    Low_fat     Organic
22     42      Mango            Cone    Low_fat Not_organic
23     46      Mango            Pint    Low_fat Not_organic
24     47      Mango            Pint No_low_fat     Organic
25     49    Vanilla Homemade_waffle    Low_fat     Organic
26     51    Vanilla Homemade_waffle No_low_fat     Organic
27     53    Vanilla            Cone    Low_fat     Organic
28     56    Vanilla            Cone No_low_fat Not_organic
29     58    Vanilla            Pint    Low_fat Not_organic
30     60    Vanilla            Pint No_low_fat Not_organic
```

38.4. Conjoint analysis

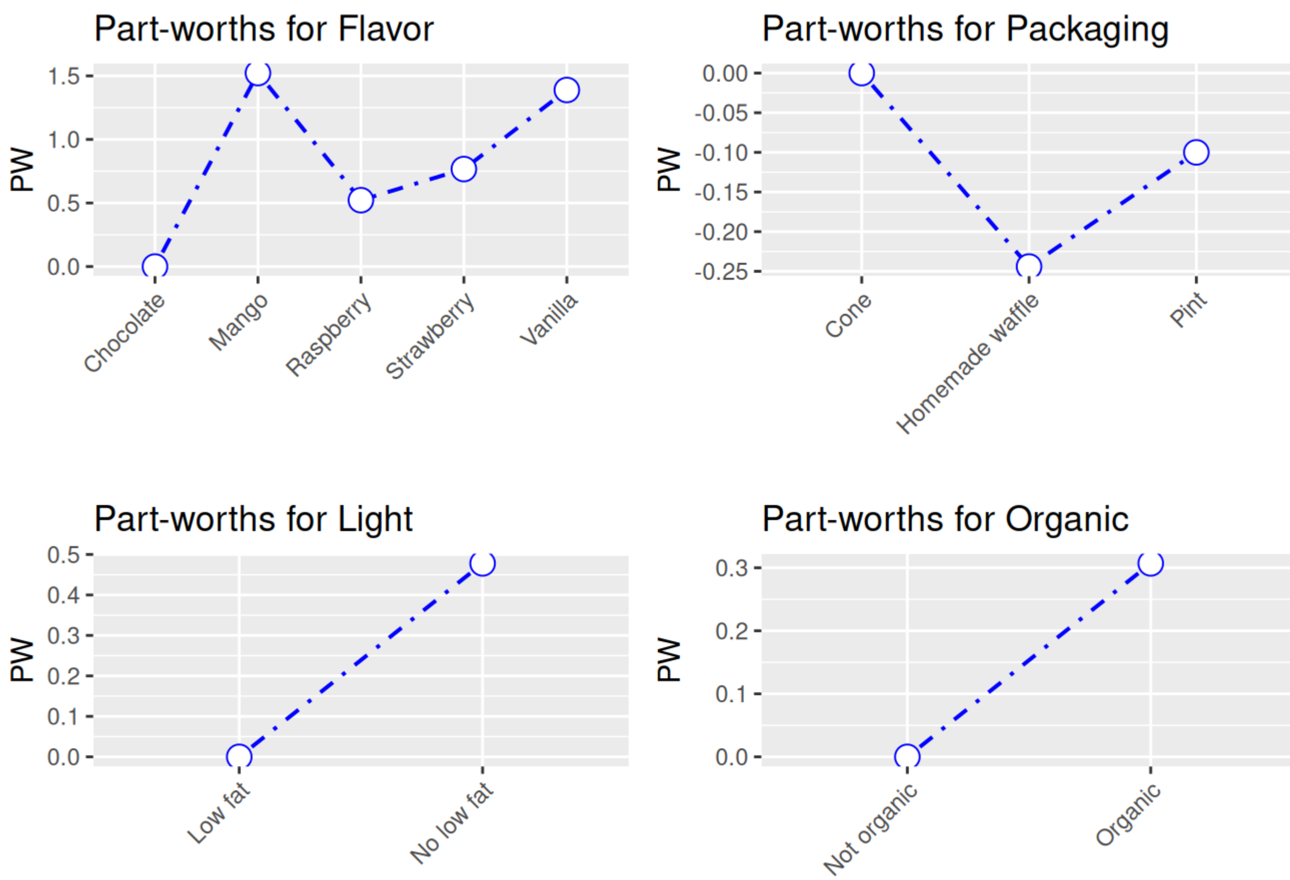
In the icecream data set we read in, it's a 10 profile design. There are 15 respondents. They rate 10 profiles of icecreams.

```r
conjoint <- conjoint(icecream, rvar = "rating", evar = c("Flavor","Packaging","Light","Organic"

summary(conjoint)
```

```
Conjoint analysis
Data                 : icecream
Response variable    : rating
Explanatory variables: Flavor, Packaging, Light, Organic

Conjoint part-worths:
   Attributes          Levels     PW
 Flavor       Chocolate        0.000
 Flavor       Mango            1.522
 Flavor       Raspberry        0.522
 Flavor       Strawberry       0.767
 Flavor       Vanilla          1.389
 Packaging    Cone             0.000
 Packaging    Homemade waffle -0.244
 Packaging    Pint            -0.100
 Light        Low fat          0.000
 Light        No low fat       0.478
 Organic      Not organic      0.000
 Organic      Organic          0.307
 Base utility ~                4.358
```

```
Conjoint importance weights:
 Attributes    IW
  Flavor    0.597
  Packaging 0.096
  Light     0.187
  Organic   0.120

Conjoint regression results:

                           coefficient
 (Intercept)                     4.358
 Flavor|Mango                    1.522
 Flavor|Raspberry                0.522
 Flavor|Strawberry               0.767
 Flavor|Vanilla                  1.389
 Packaging|Homemade waffle      -0.244
 Packaging|Pint                 -0.100
 Light|No low fat                0.478
 Organic|Organic                 0.307
```

```r
plot(conjoint)
```



The partworths are just coefficient from a linear regression:

```
summary(lm(rating ~ Flavor + Packaging + Light + Organic, data = icecream))
```

```
Call:
lm(formula = rating ~ Flavor + Packaging + Light + Organic, data = icecream)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.2867 -2.2244 -0.1911  2.3128  5.4089 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)                4.3578     0.8172   5.332 3.76e-07 ***
FlavorMango                1.5222     0.9453   1.610    0.110    
FlavorRaspberry            0.5222     0.9453   0.552    0.582    
FlavorStrawberry           0.7667     0.8399   0.913    0.363    
FlavorVanilla              1.3889     0.9453   1.469    0.144    
PackagingHomemade waffle  -0.2444     0.8675  -0.282    0.779    
```

```
PackagingPint           -0.1000     0.8399  -0.119    0.905
LightNo low fat          0.4778     0.5738   0.833    0.406
OrganicOrganic           0.3067     0.4751   0.645    0.520
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.91 on 141 degrees of freedom
Multiple R-squared:  0.03541,   Adjusted R-squared:  -0.01932
F-statistic: 0.6469 on 8 and 141 DF,  p-value: 0.7371
```

The conjoint importance weights are the relative importance of each attribute. First the range of partworths for each attribute is calculated. Then the range of partworths for each attribute is divided by the sum of the ranges of all attributes. In this example, Flavor is the most important attribute, with an importance weight of .597.

There is another package called "conjoint" which I have not played with.

In summary, we can use R packages to set up a conjoint experiment, in the sense that we can have partial (fractional) factorial design. After collecting the data, we can use either the conjoint packages, or just run linear or logit models to analyze it. The frequentist approach used in conjoint analysis seems nothing special. Bayesian approach can also be used, but that's another topic.

38.5. Causal effect in conjoint analysis

Hainmueller, Hopkins, and Yamamoto (2014) defined ACME (average marginal component effect). The basic idea is that once all the assumptions are met (usual assumptions in causal inference), we can have the effect of changing one attribute level to another with a causal interpretation. Fortunately it is usually just coefficients from a linear model, or linear combinations of the coefficients. There is a package "cregg" for each calculation and plotting of ACME, but we can just do it manually.

39. Stata 19 CATE command

39.1. Stata 19 CATE command

Stata 19 introduces a `cate` command for estimating conditional average treatment effects (CATE). Fernando Rios-Avila provides an overview.

> ⚠ If a Stata chunk here renders with no output
>
> The Stata chunks below are cached (`cache: true`) because the `cate` command does lasso cross-fitting plus a random forest, and a full re-execution of this chapter takes over five minutes. The cache is safe for prose edits, which leave every chunk served uniformly from cache. It is **not** safe if you edit the *code* of one Stata chunk: the edited chunk re-executes while the earlier `collectcode: true` chunks are still served from cache, their code never enters Statamarkdown's accumulated buffer, every command runs against missing data, and the chunk renders silently blank – and that empty result is then cached in turn.
> If that happens, delete `stata-cate_cache/` **entirely** (not just the offending chunk's entry) along with `_freeze/stata-cate/`, then re-render so all chunks run in order. Clearing `_freeze/` alone is not enough. See the note in this book's `CLAUDE.md`.

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)
cate po (assets $catecovars) (e401k), group(incomecat) nolog
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Conditional average treatment effects     Number of observations       = 9,913
Estimator:        Partialing out          Number of folds in cross-fit =    10
Outcome model:    Linear lasso            Number of outcome controls   =    17
Treatment model:  Logit lasso             Number of treatment controls =    17
CATE model:       Random forest           Number of CATE variables     =    17

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
GATE         |
```

```
   incomecat |
          0  |   3999.888   989.8742     4.04   0.000      2059.77    5940.006
          1  |   1424.931   1668.081     0.85   0.393    -1844.447     4694.31
          2  |   5092.801   1344.769     3.79   0.000     2457.102    7728.499
          3  |   8700.512   2272.126     3.83   0.000     4247.226     13153.8
          4  |   20350.97   4717.093     4.31   0.000     11105.64     29596.3
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |    7914.24    1150.79     6.88   0.000     5658.734    10169.75
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14012.54   834.0963    16.80   0.000     12377.74    15647.34
------------------------------------------------------------------------------
```

The data are `assets3`: 9,913 households, with net financial assets as the outcome and 401(k) eligibility as the treatment. The group ATEs by income category show strong heterogeneity — \$4,000 in the lowest category, \$1,425 in the second (not significant), then \$5,093, \$8,701 and \$20,350 as income rises. A single ATE would hide a fivefold spread.

He said “We can achieve comparable results using standard regression with full interactions:”

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)

* you do not want to see all interactions.
qui:reg assets i.incomecat##i.e401k##c.($catecovars), robust

* Average Treatment Effect (ATE)
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r))

* Group Average Treatment Effects (GATEs)
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r)) over(incomecat)
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Contrasts of predictive margins                            Number of obs = 9,913
Model VCE: Robust

Expression: Linear prediction, predict()
```

```
1._at: e401k = 0
2._at: e401k = 1

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
         _at |          1       44.72     0.0000
             |
 Denominator |       9833
------------------------------------------------

--------------------------------------------------------------
             |            Delta-method
             |   Contrast   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         _at |
   (2 vs 1)  |   7642.407   1142.797      5402.291    9882.524
--------------------------------------------------------------


Contrasts of predictive margins                          Number of obs = 9,913
Model VCE: Robust

Expression: Linear prediction, predict()
Over:       incomecat
1._at: 0.incomecat
           e401k = 0
       1.incomecat
           e401k = 0
       2.incomecat
           e401k = 0
       3.incomecat
           e401k = 0
       4.incomecat
           e401k = 0
2._at: 0.incomecat
           e401k = 1
       1.incomecat
           e401k = 1
       2.incomecat
           e401k = 1
       3.incomecat
           e401k = 1
       4.incomecat
           e401k = 1

-------------------------------------------------
              |         df           F        P>F
```

```
--------------+---------------------------------
_at@incomecat |
  (2 vs 1) 0  |          1       16.08     0.0001
  (2 vs 1) 1  |          1        0.67     0.4123
  (2 vs 1) 2  |          1       16.03     0.0001
  (2 vs 1) 3  |          1       13.77     0.0002
  (2 vs 1) 4  |          1       17.26     0.0000
       Joint  |          5       12.76     0.0000
              |
  Denominator |       9833
------------------------------------------------

--------------------------------------------------------------
              |            Delta-method
              |   Contrast   std. err.     [95% conf. interval]
--------------+-----------------------------------------------
_at@incomecat |
  (2 vs 1) 0  |   3594.182   896.3298      1837.191    5351.172
  (2 vs 1) 1  |     1283.3   1565.151     -1784.718    4351.317
  (2 vs 1) 2  |   5056.144   1262.728      2580.938     7531.35
  (2 vs 1) 3  |   8610.622   2320.156      4062.641     13158.6
  (2 vs 1) 4  |   19665.61   4733.551      10386.88    28944.34
--------------------------------------------------------------
```

Numerically these two sets of estimations may be close, but I don't think they are from the same model, even the same idea. The CATE command with "PO" option is to estimate the CATE with "partialling out" approach.

What Fernando did is a "regression adjustment" type of model, which is modeling outcome model and allow interaction of treatment and covariates. The "partialling out" approach is sometimes called double ML.

Let's see how to replicate "teffects ra" with "reg":

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)

teffects ra (assets age educ i.(incomecat pension married twoearn ira ownhome))(e401k)

qui: reg assets i.e401k##c.(age educ) i.e401k##i.(incomecat pension married twoearn ira ownhome
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r))
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Iteration 0:  EE criterion = 1.291e-21
```

```
Iteration 1:  EE criterion = 1.046e-23

Treatment-effects estimation                    Number of obs     =      9,913
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |   7929.242   1216.273     6.52   0.000      5545.39    10313.09
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14094.12   863.7641    16.32   0.000     12401.18    15787.07
------------------------------------------------------------------------------


Contrasts of predictive margins                          Number of obs = 9,913
Model VCE: OLS

Expression: Linear prediction, predict()
1._at: e401k = 0
2._at: e401k = 1

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
         _at |          1       34.39     0.0000
             |
 Denominator |       9889
------------------------------------------------

--------------------------------------------------------------
             |            Delta-method
             |   Contrast   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         _at |
   (2 vs 1)  |   7929.242   1352.153      5278.746    10579.74
--------------------------------------------------------------
```

39.1.1. DoubleML

The R package DoubleML implements the double/debiased machine learning framework of Chernozhukov et al. (2018). It provides functionalities to estimate parameters in causal models based on machine learning methods. The double machine learning framework consist of three key ingredients: Neyman orthogonality, High-quality machine learning estimation and Sample splitting.

They consider a partially linear model:

$$y_i = \theta d_i + g_0(x_i) + \eta_i \tag{39.1}$$

$$d_i = m_0(x_i) + v_i \tag{39.2}$$

This model is quite general, except it does not allow interaction of d and x; therefore no heterogeneous treatment effect across x. But "DoubleML" implements more than partially linear model, it actually allows for heterogeneous treatment effects, in models such as interactive regression model.

The basic idea of doubleML is to use any machine learning algorithm to estimate outcome equation ($l_0(X) = E(Y|X)$) and treatment equation ($m_0(X) = E(D|X)$). Then get the residuals, namely $\hat{W} = Y - \hat{l}_0(X)$ and $\hat{V} = D - \hat{m}_0(X)$.

Then regress $\hat{W}$ on $\hat{V}$. Based on FWL theorem, you get $\hat{\theta}$.

An important component here is to specify a Neyman-orthogonal score function. In the case of PLR, it can be the product of the two residuals:

$$\psi(W;\theta,\eta) = (Y - D\theta - g(X))(D - m(X)) \tag{39.3}$$

The estimators $\hat{\theta}$ is to solve the equation that the sample mean of this score function being 0.

And the variance of this score function is used as the variance of the doubleML estimator's variance.

Now we try to replicate the CATE estimation in Stata using the R package DoubleML. I have not found a way to include factor variables in the lasso regression in R, so I included them as numeric variables.

The example in stata's cate command is to use lasso on both outcome and treatment regression, then random forest on the individual treatment effect to get ATE. Here I try to replicate it in R.

```r
library(haven)
library(DoubleML)
library(mlr3)
library(mlr3learners)
library(data.table)
library(dplyr)
# get rid of labels because doubleML does not like them
data <- read_dta("https://www.stata-press.com/data/r19/assets3.dta") |>
  zap_labels()
```

```
  # pension, married, twoearn, ira, ownhome are already binary 0/1, so
  # passing them as numeric is fine. incomecat, however, has 5 unordered
  # categories (0-4) -- passing it as a single numeric column would force
  # the ML learners (especially the linear regr.glmnet below) to treat
  # category 2 as twice category 1, which is not a meaningful assumption
  # for a nominal variable. One-hot encode it into dummy columns instead.
incomecat_dummies <- model.matrix(~ as.factor(incomecat) - 1, data = data)
colnames(incomecat_dummies) <- paste0("incomecat_", 0:(ncol(incomecat_dummies) - 1))
data <- cbind(data, incomecat_dummies)

assets3 <- data.table::setDT(data)
dml_data = DoubleMLData$new(data = assets3,
                             y_col = "assets",
                             d_cols = "e401k",
                             x_cols = c("age","educ", colnames(incomecat_dummies),"pension", "m

print(dml_data)
```

```
================= DoubleMLData Object ==================


------------------ Data summary      ------------------
Outcome variable: assets
Treatment variable(s): e401k
Covariates: age, educ, incomecat_0, incomecat_1, incomecat_2, incomecat_3, incomecat_4, pensio
Instrument(s):
Selection variable:
No. Observations: 9913
```

```
lgr::get_logger("mlr3")$set_threshold("warn")
learner = lrn("regr.glmnet", lambda=1)
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()
set.seed(123)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_l=ml_g_sim, ml_m=ml_m_sim, n_folds=10)
obj_dml_plr$fit()
obj_dml_plr$summary()
```

```
Estimates and significance testing of the effect of target variables
      Estimate. Std. Error t value Pr(>|t|)
e401k      7553       1281   5.897 3.69e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

DoubleML with lasso learners gives an ATE of $7,553 with a standard error of $1,281.

If we use random forest on the two models:

```
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")
learner = lrn("regr.ranger", num.trees=2000,  max.depth=5, min.node.size=2)
ml_l = learner$clone()
ml_m = learner$clone()
# learner = lrn("regr.glmnet")
# ml_g_sim = learner$clone()
# ml_m_sim = learner$clone()
set.seed(123)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_l=ml_l, ml_m=ml_m, n_folds=5)
obj_dml_plr$fit()
obj_dml_plr$summary()
```

```
Estimates and significance testing of the effect of target variables
      Estimate. Std. Error t value Pr(>|t|)
e401k      9114       1375   6.627 3.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can do the same in Stata:

```
clear all
webuse assets3
global catecovars age educ incomecat pension married twoearn ira ownhome
cate po (assets age educ incomecat pension married twoearn ira ownhome) (e401k),  omethod(rfore
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Conditional average treatment effects     Number of observations       = 9,913
Estimator:       Partialing out           Number of folds in cross-fit =     5
Outcome model:   Random forest            Number of outcome controls   =     8
Treatment model: Random forest            Number of treatment controls =     8
CATE model:      Random forest            Number of CATE variables     =     8

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
        vs   |
```

```
Not elig..)  |   8367.508    1180.69     7.09   0.000     6053.398    10681.62
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   13999.41   842.9698    16.61   0.000     12347.22     15651.6
------------------------------------------------------------------------------
```

With random forests DoubleML gives $9,474 (SE $1,376), while Stata's `cate` with `rforest` for both models gives $7,126 (SE $1,182). The results are somewhat similar, but not exactly the same — a gap of about $2,300, which is larger than either standard error. There are some subtle differences in the way these two packages running random forest on the two models that I am not aware of.

39.2. AIPW

Stata's cate also has "aipw" option.

```
clear all
webuse assets3
global catecovars age educ incomecat pension married twoearn ira ownhome
cate aipw (assets $catecovars) (e401k),  nolog
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Conditional average treatment effects     Number of observations       = 9,913
Estimator:        Augmented IPW           Number of folds in cross-fit =    10
Outcome model:    Linear lasso            Number of outcome controls   =     8
Treatment model:  Logit lasso             Number of treatment controls =     8
CATE model:       Random forest           Number of CATE variables     =     8

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |   7126.343   1182.229     6.03   0.000     4809.217    9443.469
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14136.14   857.5555    16.48   0.000     12455.36    15816.92
------------------------------------------------------------------------------
```

Stata's AIPW gives an ATE of $7,929 with a standard error of $1,216, on 9,913 households.

I'll try to do this with "npcausal":

```
library(npcausal)
#SL.library <- c("SL.earth","SL.gam","SL.glmnet","SL.glm.interaction", "SL.mean","SL.ranger", "
#SL.library <- c("SL.glmnet")
SL.lasso = function(...) {
  SL.glmnet(..., alpha=1)
}
SL.library <- c("SL.lasso")

Y <- data$assets
A <- data$e401k
# incomecat is nominal (5 categories); use the one-hot dummies created above
# rather than the raw numeric column, for the same reason as in the
# DoubleML example.
# as.data.frame() is required: `data` is a tibble, so `data[, cols]` returns a
# tibble, and npcausal::ate() fails on one with
#   Error in rep(): invalid 'times' argument
# It needs a plain data.frame.
# Two coercions are needed here, both for npcausal/SuperLearner's benefit:
#   * as.data.frame(): `data` is a tibble, and ate() fails on one with
#       Error in rep(): invalid 'times' argument
#   * as.numeric on every column: read_dta() returns haven_labelled columns, and
#     SuperLearner then errors with
#       The variable names and order in newX must be identical to ... X
W <- as.data.frame(data[, c("age", "educ", grep("^incomecat_", names(data), value = TRUE), "per
W <- as.data.frame(lapply(W, as.numeric))
# npcausal::ate() currently fails on this design with
#   Error in SuperLearner(): The variable names and order in newX must be
#   identical to the variable names and order in X
# raised from inside npcausal's own cross-fitting loop, not from anything in this
# chunk (the equivalent call works on a synthetic data.frame of the same shape).
# Wrapped so the chapter degrades gracefully rather than publishing a raw error;
# the DoubleML comparison above is the working cross-fitted estimate.
aipw <- tryCatch(
  ate(y = Y, a = A, x = W, nsplits = 10, sl.lib = SL.library),
  error = function(e) {
    message("npcausal::ate() failed: ", conditionMessage(e))
    NULL
  }
)
```

```
  |
  |                                                                      |   0%
```

```
if (!is.null(aipw)) aipw$res
```

Note here to match the Stata results, I included only "glmnet" in the SuperLearner library. The results are not exactly the same, but close enough.

Or we can use a similar package with more functionalities: "AIPW".

```
library(AIPW)
library(SuperLearner)

#SuperLearner libraries for outcome (Q) and exposure models (g)
#sl.lib <- c("SL.glmnet")

AIPW_sl <- AIPW$new(Y= Y,
                    A= A,
                    W= W,
                    Q.SL.library = SL.library,
                    g.SL.library = SL.library,
                    k_split = 10,
                    verbose=FALSE)
```

```
Error in `value[[3L]]()`:
! Covariates dimension error: nrow(W) != length(A)
```

```
AIPW_sl$fit()
```

```
Error:
! object 'AIPW_sl' not found
```

```
AIPW_sl$summary()
```

```
Error:
! object 'AIPW_sl' not found
```

```
AIPW_sl$result
```

```
Error:
! object 'AIPW_sl' not found
```

```
# library(ggplot2)
# AIPW_sl$plot.p_score()
# AIPW_sl$plot.ip_weights()
```

Systematic treatment: R · Julia.

40. Spatial econometrics introduction

40.1. Why do we need spatial econometrics?

'Everything is related to everything else, but near things are more related than distant things' (Tobler 1970). This is the first law of geography. This is the reason why we need spatial econometrics. The spatial econometrics is to account for the spatial dependence in the data. The spatial dependence is the correlation between the observations that are close to each other. This is a common feature in many datasets. For example, in the housing market, the price of a house is related to the prices of the houses nearby. In the crime data, the crime rate in one neighborhood is related to the crime rate in the neighboring neighborhoods. In the health data, the health outcomes of people in one area are related to the health outcomes of people in the neighboring areas. In the environmental data, the pollution level in one area is related to the pollution level in the neighboring areas. In the economic data, the economic performance of one region is related to the economic performance of the neighboring regions. In the social network data, the behavior of one person is related to the behavior of the people in the neighboring areas.

Econometricians love OLS. It is worth being precise about *what* spatial dependence breaks. Spatially correlated **errors** alone (the term $u = \rho W u + \epsilon$ below, with exogenous regressors) do not make the OLS coefficients inconsistent — they make the conventional standard errors wrong, so inference is invalid even though the point estimates are fine. OLS coefficients become **biased/inconsistent** in the genuinely simultaneous or omitted-variable cases: a spatially lagged *outcome* Wy on the right-hand side (endogenous by construction), omitted spatially correlated confounders, or endogenous spillovers. The sections below treat these cases separately: S2SLS for the spatial-lag-of-outcome model, and spatial-error models for the correlated-error case.

40.2. A general linear spatial model

$$y = \lambda W y + X\beta + W X \gamma + u \tag{40.1}$$

$$u = \rho W u + \epsilon \tag{40.2}$$

where Y is the dependent variable, X is the matrix of independent variables, β is the vector of coefficients, and u is the error term. The error term u is spatially correlated.

We can see the major difference is W. The term Wy is called "spatially lagged y", which in my opinion is misleading. This term is likely from time-series counter-part of lagged y, but there is nothing about lags in the spatial data. W is the spatial weights matrix. It is a matrix that describes the spatial structure of the data. The simplest example of W is the case that we define an element of W, $w_{i,j}$ to be 1 if unit i and j are neighbors, 0 if not.

In this simplest case, our model is saying we have a pattern we need to take account of: neighboring units are related, they seem to display similar pattern in terms of our study. This commonality of neighbors can be causal, for example, if we study the effect of minimum wage effect on employment, then if the neighboring state's wage increase might negatively impact the focal state's employment, because people go for higher wages. This is the spatial spillover effect. This can also be non-causal, for example, if we study the effect of temperature on crop yield, then the neighboring state's temperature might be a good predictor of the focal state's temperature. This is the spatial autocorrelation. In either case, we need to take account of the spatial structure in the data.

The reason behind the spatial pattern can be all kinds. It can be x_i causing y_j, or y_i causing y_j, or some unobserved factors causing both y_i and y_j, or some other reasons. Ignoring this spatial structure could make OLS biased. Betz et al (2020) show that ignoring spatial interdependence in the outcome results in asymptotically biased estimates even when instruments are randomly assigned.

40.3. Estimation with S2SLS for a less general model

Because we have the term Wy, we introduced endogeneity. Wy is necessarily correlated with u, even if u is not spatially correlated. Suppose u is not spatially correlated.

The spatial two stage least squares (S2SLS) estimator is one solution. Usually we use X and spatial lags of X as instruments. Typically instruments H is (X, WX, W^2X).

S2SLS is similar to 2SLS. The first stage is

$$\hat{Z} = PZ = [X, WX, \widehat{Wy}] \tag{40.3}$$

where

$$\widehat{Wy} = PWy \tag{40.4}$$

and

$$P = H(H'H)^{-1}H'$$

{#eq-spatial-econometrics-5}.

Then

$$\hat{\theta}_{S2SLS} = (\hat{Z}'Z)^{-1}\hat{Z}'y \tag{40.5}$$

where $\hat{\theta}_{S2SLS}$ is the full coefficient vector (β, γ, λ).

The first stage is to regress WY on H . The second stage is to regress Y on the predicted WY from the first stage and X and WX. The coefficient of X is the S2SLS estimator.

40.4. Estimation with a more general model

Suppose instead the error term is spatially correlated

$$u = \rho Wu + \epsilon \tag{40.6}$$

Then the estimation is more complicated. The generalized spatial two stage least squares (GS2SLS) estimator is needed. See Kelejian and Prucha (1998) for details.

40.5. an example

This example is from R package "sphet".

The Boston data contain information on property values and characteristics in the area of Boston, Massachusetts and has been widely used for illustrating spatial models. Specifically, there is a total of 506 units of observation for each of which a variety of attributes are avail- able, such as: (corrected) median values of owner-occupied homes (CMEDV); per-capita crime rate (CRIM); nitric oxides concentration (NOX); average number of rooms (RM); proportions of residential land zoned for lots over 25,000 sq. ft (ZN); proportions of non-retail business acres per town (INDUS); Charles River dummy variable (CHAS); proportions of units built prior to 1940 (AGE); weighted distances to five Boston employment centers (DIS); index of accessibil- ity to highways (RAD); property-tax rate (TAX); pupil-teacher ratios (PTRATIO); proportion of blacks (B); and % of the lower status of the population (LSTAT).

The dataset with Pace's tract coordinates is available to the R community as part of "spdep".

The spatial weights matrix is a sphere of influence neighbors list also available from spdep once the Boston data are loaded. Here the weight matrix comes with the data. Usually it's the most heavy lifting part of the study.

```
library(sphet)
library(spdep)
data("boston", package = "spData")
listw <- nb2listw(boston.soi)
```

The function read.gwt2dist reads a '.GWT' file (e.g., generated using the function distance). In this example we are using the matrix of tract point coordinates projected to UTM zone 19 boston.utm available from spdep to generate a '.GWT' file of the 10 nearest neighbors.

```
id1 <- seq(1, nrow(boston.utm))
tmp <- distance(boston.utm, region.id = id1, output = TRUE,
                type = "NN", nn = 10, shape.name = "shapefile",
                region.id.name = "id1", firstline = TRUE,
                file.name = "boston_nn_10.GWT")
coldist <- read.gwt2dist(file = "boston_nn_10.GWT", region.id = id1, skip = 1)
class(coldist)
```

```
[1] "sphet"    "distance" "nb"       "GWT"
```

```
summary(coldist)
```

```
 Number of observations:
 n: 506

 Distance summary:
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.5441  0.9588  1.5843  2.0848  2.6389 11.6388

 Neighbors summary:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     10      10      10      10      10      10
```

The `boston` data are 506 census tracts. The 10-nearest-neighbour weights give every tract exactly 10 neighbours, with distances running from 0.54 to 11.64 and a median of 1.58.

Spatial models in sphet are fitted by the functions stslshac and gstslshet. "stslshac" is Spatial two stages least square with HAC standard errors. "gstslshet" is GM estimation of a Cliff-Ord type model with Heteroskedastic Innovations, which is the general model we discussed before, with spatially correlated errors.

The theoretical model above is the Spatial Durbin Model (SDM), which includes WX as regressors (with their own coefficient γ), not just as instruments for Wy. If we only pass X (no WX) in the formula below, `stslshac`'s `W2X` argument still uses spatial lags of X internally, but only as *instruments* for the endogenous Wy – γ is never estimated, so the fitted model is actually the simpler Spatial Autoregressive (SAR) model ($\gamma = 0$), not SDM. To match the SDM specification, include WX explicitly as regressors:

```r
boston_x_vars <- c("CRIM","ZN","INDUS","CHAS","NOX","RM","AGE","DIS","RAD","TAX","PTRATIO","B"
boston.c2 <- boston.c
for (v in boston_x_vars) {
  xv <- boston.c[[v]]
  if (is.factor(xv)) xv <- as.numeric(as.character(xv))
  boston.c2[[paste0("W_", v)]] <- lag.listw(listw, xv)
}

sdm_formula <- as.formula(paste(
  "log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2) + I(RM^2) + AGE + log(DIS) + log(RAD) + TA
  paste0("W_", boston_x_vars, collapse = " + ")
))

res <- stslshac(sdm_formula,
                data = boston.c2, listw,
                distance = coldist, type = "Triangular", HAC = TRUE)
summary(res)
```

```
======================================================
======================================================
                   Spatial Lag Model
                  HAC standard errors
======================================================
======================================================

Call:
```

```
stslshac(formula = sdm_formula, data = boston.c2, listw = listw,
    HAC = TRUE, distance = coldist, type = "Triangular")

Residuals:
    Min.  1st Qu.   Median  3rd Qu.     Max.
-0.71074 -0.06258 -0.00291  0.06279  0.66519

Coefficients:
               Estimate SHAC St.Er. t-value  Pr(>|t|)
Wy           0.71491435  0.09797746  7.2967 2.949e-13 ***
(Intercept)  1.74597474  0.46867690  3.7253 0.0001951 ***
CRIM        -0.00548837  0.00153211 -3.5822 0.0003407 ***
ZN           0.00081016  0.00045400  1.7845 0.0743454 .
INDUS       -0.00070720  0.00285254 -0.2479 0.8041961
CHAS1       -0.05854830  0.03984589 -1.4694 0.1417328
I(NOX^2)    -0.03240666  0.21228222 -0.1527 0.8786677
I(RM^2)      0.00846160  0.00289595  2.9219 0.0034793 **
AGE         -0.00150525  0.00047265 -3.1847 0.0014489 **
log(DIS)    -0.16117615  0.07522729 -2.1425 0.0321515 *
log(RAD)     0.04219650  0.01648649  2.5595 0.0104835 *
TAX         -0.00045089  0.00012647 -3.5652 0.0003636 ***
PTRATIO     -0.01336079  0.00581140 -2.2991 0.0215013 *
B            0.00058124  0.00011965  4.8578 1.187e-06 ***
log(LSTAT)  -0.21713640  0.03831649 -5.6669 1.454e-08 ***
W_CRIM      -0.00451534  0.00300618 -1.5020 0.1330927
W_ZN        -0.00133281  0.00058617 -2.2738 0.0229807 *
W_INDUS      0.00079183  0.00351396  0.2253 0.8217152
W_CHAS       0.09604488  0.06127241  1.5675 0.1169964
W_NOX       -0.34866680  0.32382701 -1.0767 0.2816112
W_RM        -0.06511363  0.05723288 -1.1377 0.2552474
W_AGE        0.00175310  0.00062322  2.8130 0.0049086 **
W_DIS        0.01948831  0.01383400  1.4087 0.1589164
W_RAD        0.00203459  0.00345353  0.5891 0.5557722
W_TAX        0.00028960  0.00019481  1.4866 0.1371284
W_PTRATIO    0.00732717  0.00807107  0.9078 0.3639669
W_B         -0.00054543  0.00015750 -3.4630 0.0005342 ***
W_LSTAT      0.00895071  0.00427009  2.0961 0.0360698 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The spatial lag coefficient `Wy` is 0.715 with a HAC standard error of 0.098, $t = 7.3$ — strong spatial dependence in house values, which is what motivates the whole exercise. Among the own-tract regressors, log(LSTAT) enters at -0.217 and RM^2 at 0.0085.

The function gstslshet is used to estimate the GS2SLS model. Unlike `stslshac`, `gstslshet`'s GMM machinery hits an exactly-singular matrix on this dataset once the WX regressors are added (likely due to near-collinearity introduced by the additional lagged terms combined with its own internal instrument construction), so we keep the original (SAR, no WX) specification here — this estimates the spatial-error model with $\gamma = 0$, not the full Durbin extension shown above for `stslshac`.

```r
res <- gstslshet(log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2) + I(RM^2)
                 + AGE + log(DIS) + log(RAD) + TAX + PTRATIO + B + log(LSTAT),
                 data = boston.c, listw = listw, initial.value = 0.2)
summary(res)
```

```
Call:
gstslshet(formula = log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2) +
    I(RM^2) + AGE + log(DIS) + log(RAD) + TAX + PTRATIO + B +
    log(LSTAT), data = boston.c, listw = listw, initial.value = 0.2)

Residuals:
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-0.569385 -0.073165 -0.001675  0.000531  0.071500  0.740309

Coefficients:
               Estimate  Std. Error t-value  Pr(>|t|)
(Intercept)  2.51316605  0.26749367  9.3952 < 2.2e-16 ***
CRIM        -0.00662744  0.00144522 -4.5858 4.523e-06 ***
ZN           0.00038299  0.00036563  1.0475 0.2948733
INDUS        0.00159352  0.00179772  0.8864 0.3753967
CHAS1       -0.00447974  0.03689065 -0.1214 0.9033480
I(NOX^2)    -0.27295899  0.11561412 -2.3609 0.0182283 *
I(RM^2)      0.00744059  0.00199637  3.7270 0.0001937 ***
AGE         -0.00045400  0.00045572 -0.9962 0.3191333
log(DIS)    -0.16517174  0.03484858 -4.7397 2.140e-06 ***
log(RAD)     0.07453521  0.01752830  4.2523 2.116e-05 ***
TAX         -0.00041956  0.00010763 -3.8981 9.697e-05 ***
PTRATIO     -0.01412661  0.00410143 -3.4443 0.0005725 ***
B            0.00035970  0.00011182  3.2168 0.0012964 **
log(LSTAT)  -0.24593826  0.03213364 -7.6536 1.954e-14 ***
lambda       0.42407826  0.04463747  9.5005 < 2.2e-16 ***
rho          0.29587455  0.08614291  3.4347 0.0005932 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This specification separates the two channels the previous one merged. ρ, the spatial lag on y, is 0.296 with a standard error of 0.086, and λ, the spatial correlation in the errors, is 0.424 with 0.045. Both are significant. Note $\rho = 0.296$ here against `Wy` $= 0.715$ above: once spatially correlated *errors* are allowed for, much of what the lag model attributed to neighbours' outcomes is reassigned to correlated unobservables. That is the identification problem in spatial econometrics, visible in two numbers.

In stata, you would use "spivregress" which replaced user-written "spivreg". Stata has a set of commands for spatial econometrics. Check out stata's "sp" manual.

41. Causal simulation

Evans and Didelez (2024), "Parameterizing and simulating from causal models," address the inverse problem: given a causal model, what joint distribution does it imply? The usual direction is observational data to causal structure to parameter estimates; this paper starts from the structure and asks what data it would generate.

Consider the joint distribution $p(x, a, y)$, where x is covariates, a is treatment, and y is outcome.

$$\begin{aligned} p(x,a,y) &= p(x,a)p(y|x,a) \\ &= p(x,a)p_a(y|x) \\ &= p(x,a)\frac{p_a(x,y)}{p_a(x)} \\ &= p(x,a)\frac{p_a(x)p_a(y)c(x,y|a)}{p_a(x)} \\ &= p(x,a)p_a(y)c(x,y|a) \end{aligned} \tag{41.1}$$

The joint distribution factorizes into the joint of (x, a), the interventional marginal $p_a(y) = p(y \mid do(a))$, and the copula $c(x, y \mid a)$. The derivation starts from $p(y \mid x, a)$ and absorbs the x-dependence into the copula, leaving only the marginal $p_a(y)$. Each piece can be specified separately: (x, a) as the "past," $p_a(y)$ as the marginal structural model, and the copula as the dependence structure between x and y given a.

A copula links a joint distribution to its marginals:

$$p(x,y) = f(x)g(y)c(F(x), G(y)) \tag{41.2}$$

where $F(x)$ and $G(y)$ are the marginal distributions of x and y, respectively, and c is the copula function that captures the dependence between x and y. The copula function is a multivariate distribution with uniform marginals. It can be used to generate joint distributions from marginal distributions.

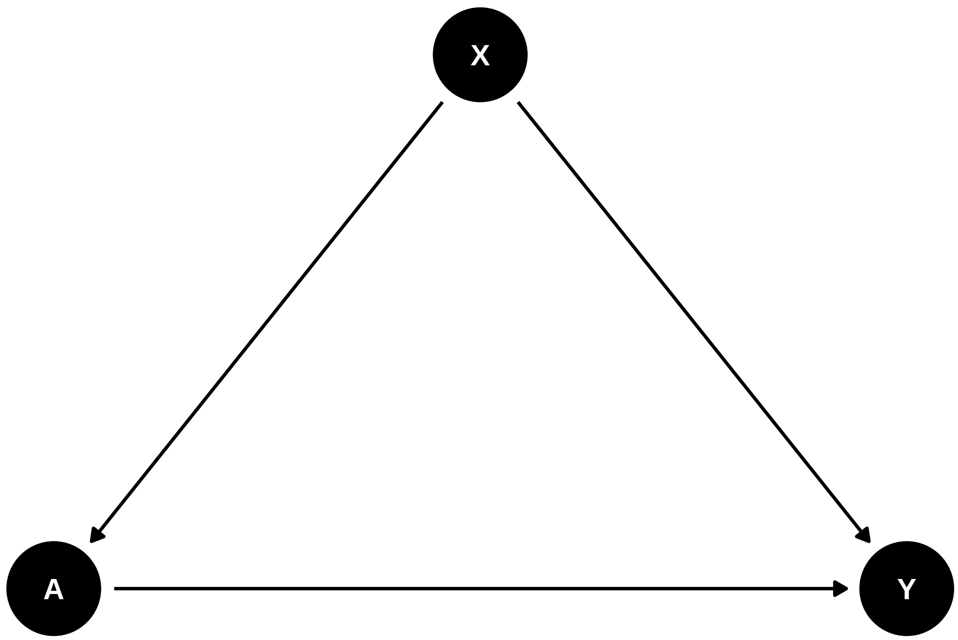
Here x, a and y can all be vectors.

41.1. examples

What do we use this for? We can use this to simulate data from a causal model, and then fit different models to the simulated data. This allows to compare the performance of different models.

41.1.1. example 1

Suppose we have a causal graph like this:



We need to specify $P(y|do(a)) = \sum_x P(x)P(y|x,a)$, where $P(x)$ is the marginal distribution of x, and $P(y|x,a)$ is the conditional distribution of y given x and a. This is the "marginal structural model" (Robins, 2000). Also the "g-formula".

We also need to specify $P(x,a)$, the "past".

Finally we need to specify the copula $c(x,y|a)$, which captures the dependence between x and y given a. Depending on different situations, different copula can be used.

In the "causl" package example, we specify:

$$X \sim N(0,1) \tag{41.3}$$

$$A|X = x \sim N(x/2,1) \tag{41.4}$$

$$Y|do(A=a) \sim N((a-1)/2,1) \tag{41.5}$$

and Gaussian copula with correlation $\rho = 2expit(1) - 1$.

```
library(causl)
# formulae corresponding to covariates, treatments, outcomes and the dependence
forms <- list(X ~ 1, A ~ X, Y ~ A, ~ 1)
# vector of model families (3=gamma/exponential, 1=normal/Gaussian)
fam <- c(1, 1, 1, 1)
# list of parameters, including 'beta' (regression params) and 'phi' dispersion
pars <- list(X = list(beta=0, phi=1),
             A = list(beta=c(0,0.5), phi=1),
             Y = list(beta=c(-0.5,0.5), phi=1),
```

```
            cop = list(beta=1))

## now create a `causl_model` object
cm <- causl_model(formulas=forms, family=fam, pars=pars,method="inversion")

# now simulate 1000 observations
set.seed(123456)
data <- rfrugal(n=1000, causl_model=cm)
head(data)
```

```
            X           A          Y
1  0.83373317  1.10709111  1.1416713
2 -0.27604777 -0.27218853 -1.0495996
3 -0.35500184  0.77575209  1.3179678
4  0.08748742 -0.05959823 -0.6106023
5  2.25225573  0.61909216  2.0789188
6  0.83446013  0.52019822  0.3229069
```

In this example, first we specify the structure of the model using a list of formulas. The first formula is for the covariates, the second for the treatments, the third for the outcomes, and the fourth for the dependence structure (copula). Note that outcome Y only depends on A in this interventional distribution.

Then we specify the families of the random variables. Here we use normal distribution for X, normal distribution for A and Y, and Gaussian copula for the dependence structure.

Finally we specify the parameters of the model, including regression coefficients and dispersion parameters. For example, $Y \sim A$ has two coefficients, intercept -.5 and slope .5.

The `causl_model` object is then created using these components.

In this simple example, if we run a regression with A, X in the model, on the simulated data, we would get the correct effect back:

```
lm1 <- lm(Y ~ A*X, data=data)
summary(lm1)
```

```

Call:
lm(formula = Y ~ A * X, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.6326 -0.6212 -0.0105  0.6051  3.3132 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.491922   0.030413 -16.175   <2e-16 ***
```

```
A            0.493471   0.027820  17.738   <2e-16 ***
X            0.468038   0.031878  14.682   <2e-16 ***
A:X         -0.009247   0.022360  -0.414    0.679
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8953 on 996 degrees of freedom
Multiple R-squared:  0.4875,    Adjusted R-squared:  0.4859
F-statistic: 315.8 on 3 and 996 DF,  p-value: < 2.2e-16
```

The true interventional model is $Y \mid do(A = a) \sim N((a-1)/2, 1)$, so the true intercept is -0.5 and the true coefficient on A is 0.5. The naive regression of Y on A and X gives $A = 0.493$ and an intercept of -0.492, with the interaction A:X at -0.009 and insignificant ($p = 0.68$), which is correct — the DGP has no such interaction.

Note this regression *also* includes X, which is the adjustment the g-formula calls for. Its coefficient of 0.468 is not a causal quantity; it is there to close the backdoor.

We can also use MLE:

```r
out <- fit_causl(data, formulas = list(X ~ 1, Y ~ A, ~ 1), family = c(1, 1, 1))
out
```

```
log-likelihood:  -2716.696
X ~ 1
              est.   s.e. sandwich
(intercept) 0.0108 0.0314   0.0314
  residual s.e.:  0.983 0.044 0.0447

Y ~ A
              est.   s.e. sandwich
(intercept) -0.491 0.0319   0.0319
A            0.494 0.0278   0.0295
  residual s.e.:  1.01 0.0472 0.0475

copula parameters:
cop ~ 1
             est.  s.e. sandwich
(intercept) 0.998 0.069   0.0734
```

`fit_causl` recovers the same structure directly from the frugal parameterisation: intercept -0.491 and A at 0.494, against the true -0.5 and 0.5, with a copula parameter of 0.998 against the specified 1. The sandwich standard errors barely differ from the model-based ones, which is what should happen when the model is correctly specified.

41.1.2. Plasmode simulation

In reality, we have a data set and we have a causal model. We want to simulate data from the causal model, but we want the simulated data to have the same distribution as the original data. This is called "plasmode simulation". For example, we don't want to use the real outcome variable. In stead, we simulate X's from the original data, Then simulate Y and A from the causal model. We can then test which method works better in estimating the causal effect of A on Y.

```
# A tibble: 4,802 x 58
     x_1 x_2     x_3   x_4   x_5   x_6   x_7   x_8   x_9  x_10  x_11  x_12  x_13
   <int> <fct> <dbl> <dbl> <int> <int> <int> <int> <int> <int> <int> <int> <int>
 1    29 C         1     7    60    85     0     0     1     0     0     1     0
 2    27 C         0     0    64   178     0     0     0     0     0     2     1
 3    27 C         0     0    60   102     0     0     0     0     0     1     0
 4    37 C         0     0    65   174     0     0     0     0     0     1     0
 5    24 C        20    14    63   129     0     0     0     0     0     1     0
 6    27 C        40    15    63   135     0     0     0     0     0     1     1
 7    26 C        20     8    69   140     0     0     0     1     0     0     0
 8    33 C         0     0    60   110     0     0     0     1     0     0     1
 9    28 C         3     5    61   160     0     0     0     0     0     3     1
10    31 C         0     0    63   114     0     1     0     0     0     0     0
# i 4,792 more rows
# i 45 more variables: x_14 <int>, x_15 <int>, x_16 <int>, x_17 <int>,
#   x_18 <int>, x_19 <int>, x_20 <int>, x_21 <fct>, x_22 <int>, x_23 <int>,
#   x_24 <fct>, x_25 <int>, x_26 <int>, x_27 <int>, x_28 <int>, x_29 <int>,
#   x_30 <int>, x_31 <int>, x_32 <int>, x_33 <int>, x_34 <int>, x_35 <int>,
#   x_36 <int>, x_37 <int>, x_38 <int>, x_39 <int>, x_40 <int>, x_41 <int>,
#   x_42 <int>, x_43 <int>, x_44 <int>, x_45 <int>, x_46 <int>, x_47 <int>, ...
```

```r
# Model for the causal effect of smoking on birthweight
forms <- list(list(),
              list(A ~ x_1 + x_3 + x_4),
              list(Y ~ A),
              list(~ 1))
# fams <- list(integer(0), 5, 1, 1)
fams <- list(integer(0), "binomial", "gaussian", 1)
pars <- list(A = list(beta=c(-1.5,0.03,0.02,0.05)),
             Y = list(beta=c(3200, -500), phi=400^2),
             cop = list(beta=-1))

cm2 <- causl_model(formulas=forms, family=fams, pars=pars, dat = dat)

set.seed(123456)
data2 <- rfrugal(causl_model=cm2)
head(data2)
```

```
  x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_10 x_11 x_12 x_13 x_14 x_15 x_16 x_17
```

```
1  29   C   1   7  60  85   0   0   1    0    0    1    0    0    2    0    0
2  27   C   0   0  64 178   0   0   0    0    0    2    1    0    2    0    0
3  27   C   0   0  60 102   0   0   0    0    0    1    0    0    0    0    0
4  37   C   0   0  65 174   0   0   0    0    0    1    0    0    1    0    0
5  24   C  20  14  63 129   0   0   0    0    0    1    0    0    0    0    0
6  27   C  40  15  63 135   0   0   0    0    0    1    1    2    2    0    0
  x_18 x_19 x_20 x_21 x_22 x_23 x_24 x_25 x_26 x_27 x_28 x_29 x_30 x_31 x_32
1   12   35   10    J    1   43    B   17   64  105   30   11    0    0    0
2   12   75   12    J    1   50    E   12   90  183   40    8    0    0    0
3   12   35   10    J    1   57    E   14   70  121   27   12    0    0    0
4   11   35   12    J    1   43    E   10   80  185   39   10    0    0    0
5    9   35   15    J    1   33    E   14   70  160   28    7    0    0    0
6    9   45   10    J    1   40    E   15   64  160   31    9    0    0    0
  x_33 x_34 x_35 x_36 x_37 x_38 x_39 x_40 x_41 x_42 x_43 x_44 x_45 x_46 x_47
1    0   79   28  340   48    1    9    9    3    0    3   54   16    0    0
2    0   69   30  430   73    0    9    9    3    0    5   51   16    0    0
3    0   84   29  450   76    0    3    6    2    0    4   41   13    0    0
4    0   82   30  440   78    1    8    9    2    0    8   70   16    0    0
5    1   85   35  360   71    1    5    8    0    2    3   65   18    0    0
6    1   81   28  515   56    1    7    9    2    0    1   63   20    0    0
  x_48 x_49 x_50 x_51 x_52 x_53 x_54 x_55 x_56 x_57 x_58 A        Y
1    0    0    0    0    0    0    0    0    0   45   39 1 2125.677
2    0    0    0    0    0    0    0    0    0   46   42 1 2689.114
3    0    1    0    0    0    0    0    0    0   45   40 0 3442.800
4    0    0    0    0    0    0    0    0    0   47   40 0 3212.584
5    0    2    0    0    0    0    0    0    0   47   43 0 2992.775
6    0    0    0    0    0    0    0    0    0   45   44 0 2738.893
```

```r
# we can also use rfrugalParam(), which is older version of rfrugal().
#datAY <- rfrugalParam(formulas=forms, family=fams, pars=pars, dat=dat)
```

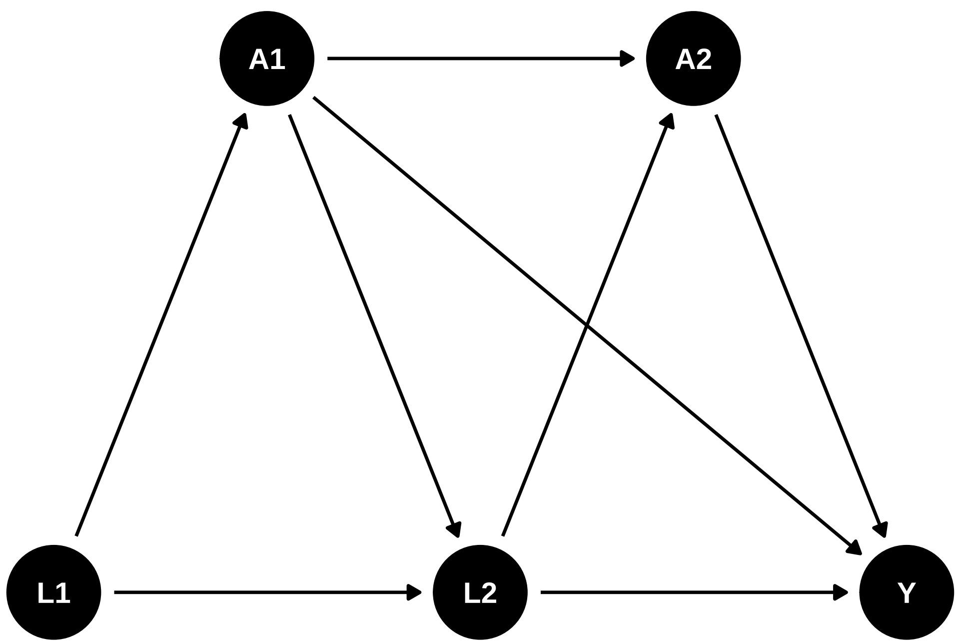
In this specification, we have a binary treatment A and a continuous outcome Y. The covariates are x_1, x_3 and x_4. As in the first example above, the copula's `beta` parameter maps to the actual Gaussian correlation via $\rho = 2\,\mathrm{expit}(\beta) - 1$; with `beta=-1` this gives $\rho = 2\,\mathrm{expit}(-1) - 1 \approx -0.46$, not -1 (a correlation of exactly -1 would require $\beta \to -\infty$ and would make the joint distribution singular). We can then simulate the data.

41.2. a more complicated model

Let's simulate a more complicated model:

```r
library(ggplot2)
library(ggdag)

g <- dagify(
```

```
  A1 ~ L1,
  L2 ~ L1,
  L2 ~ A1,
  A2 ~ A1,
  A2 ~ L2,
  Y ~ A2,
  Y ~ L2,
  Y ~ A1,
  exposure = "A2",
  outcome = "Y",
  coords = list(x = c(L1 = 1,  L2 = 2, Y = 3, A1 = 1.5, A2 = 2.5),
                y = c(L1 = 1,  L2 = 1, Y = 1, A1 = 1.5, A2 = 1.5))
)

ggdag(g) +
  theme_dag()
```



Let's see how to write a causal model:

```
# formulae corresponding to covariates, treatments, outcomes and the dependence
forms <- list(list(L1  ~ 1, L2 ~ L1*A1),   # covariates
              list(A1 ~ L1, A2 ~ L2*A1), # treatments
              Y ~ A1*A2*L2,             # outcome
              ~ 1)

# vector of model families (3=gamma/exponential, 1=normal/Gaussian)

fams <- list(c(1, 1), c(5,5), 1, 1)
```

```
pars <- list(L1 = list(beta=0, phi=1),
             L2 = list(beta=c(0.3,0.5,-0.2,-0.1), phi=1),
             A1 = list(beta=c(-0.3,0.4), phi=1),
             A2 = list(beta=c(0.5,0.3,0.1,0), phi=1),
             Y = list(beta=c(0,1,2,0.5,1,0.2,0,0.8), phi=1),
             cop = list(beta = 0.5))

# we can use rfrugalParam() to simulate data from the model, or rfrugal().
#dat <- rfrugalParam(n=1e4, formulas=forms, family = fams, pars=pars)
#head(dat)

cm3 <- causl_model(formulas=forms, family=fams, pars=pars)

set.seed(123456)
data3 <- rfrugal(n=1e4,causl_model=cm3)
head(data3)
```

```
           L1          L2 A1 A2         Y
1  0.83373317  1.91691936  1  1 8.1680612
2 -0.27604777  0.17008294  1  1 5.2276976
3 -0.35500184  0.16907708  1  1 2.8194835
4  0.08748742  0.19688985  0  1 2.6789661
5  2.25225573 -0.29257431  0  0 0.3180188
6  0.83446013  0.02661196  0  1 1.5694426
```

If we have the correct specification for the outcome model, we'll get it right by linear model:

```
m1 <- glm(Y ~ A1*A2*L2, data = data3)
summary(m1)
```

```
Call:
glm(formula = Y ~ A1 * A2 * L2, data = data3)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.11524    0.02110  -5.461 4.85e-08 ***
A1           1.11589    0.03245  34.391  < 2e-16 ***
A2           2.02332    0.02682  75.454  < 2e-16 ***
L2           0.80424    0.01929  41.682  < 2e-16 ***
A1:A2        0.98345    0.04090  24.044  < 2e-16 ***
A1:L2        0.16028    0.02976   5.386 7.37e-08 ***
A2:L2       -0.03446    0.02401  -1.435    0.151    
A1:A2:L2     0.83541    0.03693  22.620  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for gaussian family taken to be 0.9100255)

    Null deviance: 53902  on 9999  degrees of freedom
Residual deviance:  9093  on 9992  degrees of freedom
AIC: 27446

Number of Fisher Scoring iterations: 2
```

```r
library(marginaleffects)
avg_comparisons(
    model = m1,
    variables = "A2")
```

```
 Estimate Std. Error   z Pr(>|z|)   S 2.5 % 97.5 %
     2.51     0.0202 124   <0.001 Inf  2.47   2.55

Term: A2
Type: response
Comparison: 1 - 0
```

The fitted model has $A_2 = 2.023$, $A_1{:}A_2 = 0.983$ and $A_1{:}A_2{:}L_2 = 0.835$, so the effect of A_2 depends on both A_1 and L_2 and cannot be read off any single coefficient. `avg_comparisons` averages over the observed distribution and returns 2.51 with a standard error of 0.020.

So the ATE is 2.5.

Let's use "lmtp" package to estimate the treatment effect of A_2 on Y.

```r
library(lmtp)

d1 <- function(data, trt) {
  rep(1, nrow(data))
}
A <- "A2"
Y <- "Y"
W <- c("L1", "L2", "A1")

set.seed(34465)

treat <- lmtp_tmle(
  data = data3,
  trt = "A2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
```

```r
  shift = d1,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(treat)
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 3.046
    Std. error: 0.022
```

```r
d2 <- function(data, trt) {
  rep(0, nrow(data))
}
control <- lmtp_tmle(
  data = data3,
  trt = "A2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d2,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

print(control)
```

::: {.cell-output .cell-output-stdout}

```
      Estimate: 0.542
    Std. error: 0.021
```

:::

```r
lmtp_contrast(treat, ref = control)
```

::: {.cell-output .cell-output-stdout}

```

  shift   ref estimate std.error conf.low conf.high p.value
1  3.05 0.542      2.5    0.0227     2.46      2.55  <0.001
```

::: :::

42. Frengression and Engression

In the causal simulation chapter, I used Evans and Didelez (2024)'s decomposition of the joint distribution $p(x, a, y)$ into three pieces:

$$p(x, a, y) = p(x, a) \cdot p_a(y) \cdot c(x, y|a) \tag{42.1}$$

where $p(x, a)$ is the "past" distribution of covariates and treatment, $p_a(y)$ is the causal margin, and $c(x, y|a)$ is the copula linking covariates and outcome given treatment. The `causl` package implements this with parametric families. We specify Gaussian, binomial, and so on for each piece.

But real data can be hard to specify parametrically. We may not know the right copula family. The marginal distributions may have mass points, skewness, or other features that are annoying to write down.

Frengression (Yang, Evans and Shen, 2025) keeps the same decomposition but learns the pieces with neural networks.

There are two uses: causal simulation with a learned generative model, and interventional prediction using learned confounding structure. The second part uses engression, so I start there.

42.1. Engression

Engression (Shen and Meinshausen, 2024) stands for energy regression. Standard regression estimates $E[Y|X]$. Engression tries to learn the whole conditional distribution $P(Y|X)$ using neural networks trained with the energy score.

42.1.1. The Energy Score

The energy score is a proper scoring rule for distributions. Given observations Y and samples $\hat{Y}, \hat{Y}'$ from the model:

$$S(Y, \hat{Y}, \hat{Y}') = E\|Y - \hat{Y}\| - \frac{1}{2}E\|\hat{Y} - \hat{Y}'\| \tag{42.2}$$

The first term rewards samples close to the truth. The second term keeps the simulated distribution from collapsing to one point. This is why the method targets the full distribution rather than only the mean.

42.1.2. The Stochastic Network (StoNet)

The building block is a `StoNet`, a neural network that adds random noise $\epsilon \sim N(0, I)$ to the input at each layer:

$$\text{StoNet}(x) = f(x, \epsilon), \quad \epsilon \sim N(0, I) \tag{42.3}$$

Different noise draws produce different outputs. Each output is a sample from the learned conditional distribution. So we can:

- **Predict the mean**: average over many noise draws $\rightarrow E[Y|X = x]$
- **Predict quantiles**: take quantiles of the noise draws $\rightarrow$ conditional quantile regression
- **Sample**: each forward pass with fresh noise is a sample from $P(Y|X = x)$

42.1.3. Why Engression?

Compared with quantile regression or mixture density networks, engression is useful because:

1. it handles multivariate responses directly;
2. it does not require us to specify a Gaussian mixture or a list of quantiles;
3. it gives samples from the conditional distribution, which is convenient for simulation.

42.1.4. Quick Example

The `engression` R package, separate from `Rfrengression`, provides the standalone estimator. The code below shows the API. The `Rfrengression` examples later are the ones I actually run here.

```r
library(engression)

# Nonlinear DGP with heteroskedastic noise
n <- 1000
X <- matrix(rnorm(n * 2), ncol = 2)
Y <- (X[,1])^2 + X[,2] + rnorm(n) * (0.5 + abs(X[,1]))

# Fit engression; it learns the full conditional distribution
engr <- engression(X, Y)

# Predict conditional mean (like lm, but nonlinear)
Yhat_mean <- predict(engr, X, type = "mean")

# Predict conditional quantiles (like quantile regression, but joint)
Yhat_quant <- predict(engr, X, type = "quantiles")

# Sample from the learned distribution (generative)
Ysample <- predict(engr, X, type = "sample", nsample = 1)
```

The point is that engression gives a distributional model of $Y|X$, not just a fitted mean.

42.2. From Engression to Frengression

Frengression is frugal simulation plus engression. It uses three engression models to represent the decomposition of $P(X, A, Y)$. Each sub-network is trained with the energy score.

42.3. The Architecture

Frengression has three sub-networks, matching the three pieces of the decomposition:

Decomposition	causl (parametric)	Frengression (neural net)
$P(X, A)$ — the past	Specified by family + params	`model_xz`: noise-only generator
$P_a(Y)$ — causal margin	Specified by family + params	`model_y`: takes (x, eta), outputs y
$c(X, Y \mid A)$ — copula	Gaussian/Frank/etc. copula	`model_eta`: takes (x, z), outputs eta

The important part is `model_eta`. It takes (x, z) and outputs a latent variable `eta` that carries the dependence between confounders and outcome. During training, a Z-permutation trick forces `eta` to be marginally $N(0, I)$. This means:

- When `eta` comes from `model_eta(x,z)`, it carries the observational confounding information.
- When `eta` is sampled independently from $N(0, 1)$, it gives the interventional prediction.

Same network, different source of `eta`. That is the trick.

42.4. Using Rfrengression

The `Rfrengression` package is a native R implementation using `torch`.

```
library(Rfrengression)
```

42.4.1. Example: Binary Treatment DGP

Use a DGP with binary treatment, binary outcome, and four confounders. We draw $n = 2000$ observations with $w_1, w_2 \sim \text{Bernoulli}(0.5)$, $w_3 \sim U(0, 4)$ and $w_4 \sim U(0, 5)$; treatment is $A \sim \text{Bernoulli}(\Lambda(-0.4 + 0.2w_2 + 0.15w_3 + 0.2w_4 + 0.15w_2w_4))$; and both potential outcomes are Bernoulli with the same logit index shifted by the treatment. Since both are simulated, the realized ATE is known:

```r
set.seed(42)
n <- 2000

w1 <- rbinom(n, 1, 0.5)
w2 <- rbinom(n, 1, 0.5)
w3 <- round(runif(n, 0, 4), 3)
w4 <- round(runif(n, 0, 5), 3)

A <- rbinom(n, 1, plogis(-0.4 + 0.2*w2 + 0.15*w3 + 0.2*w4 + 0.15*w2*w4))
Y.1 <- rbinom(n, 1, plogis(-1 + 1 - 0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4))
Y.0 <- rbinom(n, 1, plogis(-1 + 0 - 0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4))
Y <- Y.1 * A + Y.0 * (1 - A)

true_ate <- mean(Y.1 - Y.0)
cat("True ATE:", round(true_ate, 4), "\n")
```

```
True ATE: 0.1935
```

42.4.2. Train Frengression

Training has two stages: first the outcome model (`model_y` + `model_eta`), then the marginal model (`model_xz`).

```r
x_mat <- matrix(A, ncol = 1)
z_mat <- as.matrix(data.frame(w1, w2, w3, w4))
y_mat <- matrix(Y, ncol = 1)

model <- frengression(x_dim = 1, y_dim = 1, z_dim = 4,
                      x_binary = TRUE, y_binary = TRUE,
                      z_binary_dims = 2, noise_dim = 10)

model <- train_y(model, x_mat, z_mat, y_mat,
                 num_iters = 1000, lr = 1e-3, print_every = 500)
```

```
Epoch 1: loss 1.2284, loss_y 0.4749 (0.4975, 0.0450), loss_eta 0.7534 (0.8237, 0.1405)
Stopping at iter 61
```

```r
model <- train_xz(model, x_mat, z_mat,
                  num_iters = 1000, lr = 1e-4, print_every = 500)
```

```
Epoch 1: loss 3.3600, loss1 3.5027, loss2 0.2854
Epoch 500: loss 1.3562, loss1 2.6918, loss2 2.6711
Epoch 1000: loss 1.3544, loss1 2.7226, loss2 2.7363
```

42.4.3. Estimate ATE

For interventional prediction, `eta` is sampled from $N(0, 1)$ instead of being generated from the observed confounding structure:

```
y1 <- predict(model, matrix(1, ncol = 1), type = "mean", nsample = 2000)
y0 <- predict(model, matrix(0, ncol = 1), type = "mean", nsample = 2000)

cat("Frengression ATE:", round(y1 - y0, 4), "\n")
```

```
Frengression ATE: 0.1828
```

```
cat("True ATE:", round(true_ate, 4), "\n")
```

```
True ATE: 0.1935
```

The realized ATE is 0.1935 and frengression lands a few hundredths below it.

Note that the training chunks here carry `cache: false` and torch's random number generator is not fixed by R's `set.seed()`, so this particular figure moves by a percentage point or two between renders. The cached comparison table below is the number to quote.

42.5. Frengression as DGP

Frengression can estimate an ATE, but that is not its main advantage. If I only want an ATE, I would usually use `npcausal`, TMLE, or DoubleML. The more interesting use is to train a data-generating process for benchmarking.

The idea is:

1. Train frengression on observational data, so it learns the confounding structure.
2. Replace the causal margin with a known function via `specify_causal()`.
3. Generate synthetic data with realistic confounding and known ground truth.

```
# Specify a known causal effect: Y = sigmoid(0.2*X + eta)
model_bench <- specify_causal(model, function(x, eta) {
  torch::torch_sigmoid(0.2 * x + eta)
})

# True ATE computed empirically from the model
y1_true <- predict(model_bench, matrix(1, ncol = 1), type = "mean", nsample = 5000)
y0_true <- predict(model_bench, matrix(0, ncol = 1), type = "mean", nsample = 5000)
cat("True ATE (specified):", round(as.numeric(y1_true - y0_true), 4), "\n")
```

```
True ATE (specified): 0.0789
```

Now we can generate synthetic data and benchmark any method:

```
syn <- sample_joint(model_bench, sample_size = 1000)
cat("Synthetic data: n =", nrow(syn$x), "\n")
```

```
Synthetic data: n = 1000
```

```
cat("Treatment prevalence:", mean(syn$x), "\n")
```

```
Treatment prevalence: 0.942
```

```
cat("Outcome prevalence:", mean(syn$y), "\n")
```

```
Outcome prevalence: 0.589
```

42.6. Real Data: LaLonde

This is most useful with real data. The LaLonde job-training data have many of the problems that make simulations hard to write by hand: zeros in prior earnings, uneven covariate distributions, and poor overlap.

```
library(MatchIt)
data("lalonde", package = "MatchIt")
lalonde$black <- as.integer(lalonde$race == "black")
lalonde$hispan <- as.integer(lalonde$race == "hispan")

x_lal <- matrix(lalonde$treat, ncol = 1)
y_lal <- matrix(lalonde$re78, ncol = 1)
# frengression's z_binary_dims assumes the FIRST z_binary_dims columns of z
# are the binary ones -- so the binary covariates (black, hispan, married,
# nodegree) must come first, followed by the continuous ones (age, educ,
# re74, re75). The original column order (age, educ first) put two
# continuous variables where binary ones were expected, and vice versa.
z_lal <- as.matrix(lalonde[, c("black", "hispan", "married", "nodegree",
                                "age", "educ", "re74", "re75")])

# Standardize for training
z_sc <- scale(z_lal)
y_sc <- (y_lal - mean(y_lal)) / sd(y_lal)

model_lal <- frengression(x_dim = 1, y_dim = 1, z_dim = 8,
                           x_binary = TRUE, y_binary = FALSE,
                           z_binary_dims = 4, noise_dim = 10)
```

```r
model_lal <- train_y(model_lal, x_lal, z_sc, y_sc,
                      num_iters = 1000, lr = 1e-3, print_every = 500)
```

```
Epoch 1: loss 1.3385, loss_y 0.7059 (0.7872, 0.1627), loss_eta 0.6326 (0.7516, 0.2381)
Stopping at iter 57
```

```r
model_lal <- train_xz(model_lal, x_lal, z_sc,
                        num_iters = 1000, lr = 1e-4, print_every = 500)
```

```
Epoch 1: loss 2.7930, loss1 2.9730, loss2 0.3600
Epoch 500: loss 2.0228, loss1 3.8129, loss2 3.5801
Epoch 1000: loss 1.9995, loss1 3.8706, loss2 3.7423
```

Now specify a known effect and generate data:

```r
model_lal_bench <- specify_causal(model_lal, function(x, eta) {
  0.3 * x + eta
})

# Generate synthetic LaLonde-like data with known ATE = 0.3 (standardized)
syn_lal <- sample_joint(model_lal_bench, sample_size = 2000)
cat("Synthetic LaLonde: n =", nrow(syn_lal$x), "\n")
```

```
Synthetic LaLonde: n = 2000
```

```r
cat("Treatment prevalence:", round(mean(syn_lal$x), 3), "\n")
```

```
Treatment prevalence: 0.004
```

```r
cat("True ATE:", 0.3, "(standardized) =", round(0.3 * sd(y_lal), 0), "dollars\n")
```

```
True ATE: 0.3 (standardized) = 2241 dollars
```

The synthetic data inherits LaLonde's confounding structure, but now the true effect is known. This is the part that is hard to do with a hand-written DGP.

42.7. Comparison: causl vs frengression

Both packages use the same decomposition:

Feature	causl	Frengression
Marginal $P(X, A)$	Parametric families	Learned by neural net
Causal margin $P_a(Y)$	Parametric families	Learned (or specified)
Copula	Gaussian, Frank, etc.	Learned by `model_eta`
Requires specification	Yes — families + params	No — learns from data
Can learn from real data	Limited (plasmode)	Yes — full joint
Distributional flexibility	Limited by family choice	Arbitrary

The `causl` approach is more transparent because we know exactly what we specified. Frengression is more flexible because it can learn structure we did not write down. I would treat them as complements.

42.8. Benchmarking: Frengression vs Semiparametric Methods

Now compare frengression against standard semiparametric causal inference tools on the same data. The three comparison methods are:

- `npcausal`: AIPW estimator with SuperLearner nuisance models
- `tmle`: targeted minimum loss-based estimation with SuperLearner
- `DoubleML`: partially linear regression with `ranger`

These methods are built for ATE estimation. They should usually be better for this specific estimand. The comparison is still useful because frengression is trying to learn much more than the ATE.

```
library(npcausal)
library(tmle)
library(SuperLearner)
library(DoubleML)
library(mlr3)
library(mlr3learners)
library(tidyverse)
library(knitr)
```

42.8.1. Single-Dataset Comparison

First, run all four methods on the binary DGP data created above.

```
W <- data.frame(w1, w2, w3, w4)
SL.library <- c("SL.earth", "SL.glm.interaction", "SL.mean",
                "SL.ranger", "SL.glmnet")

set.seed(123)
aipw_fit <- ate(y = Y, a = A, x = W, nsplits = 2, sl.lib = SL.library)
```

```
  |
  |                                                                      |   0%
  |
  |=========                                                             |  12%
  |
  |==================                                                    |  25%
  |
  |==========================                                            |  38%
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  |===================================                                   |  50%
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  |============================================                          |  62%
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  |====================================================                  |  75%
  |
  |=============================================================         |  88%
  |
  |======================================================================| 100%
     parameter       est          se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5697312 0.02038159 0.5297832 0.6096791    0
2      E{Y(1)} 0.7608322 0.01207030 0.7371744 0.7844900    0
3 E{Y(1)-Y(0)} 0.1911010 0.02335102 0.1453330 0.2368690    0
```

```
ate_npcausal <- aipw_fit$res$est[3]
se_npcausal  <- aipw_fit$res$se[3]
ci_npcausal  <- c(aipw_fit$res$ci.ll[3], aipw_fit$res$ci.ul[3])

cat("npcausal ATE:", round(ate_npcausal, 4), "\n")
```

```
npcausal ATE: 0.1911
```

```
cat("SE:", round(se_npcausal, 4), "\n")
```

```
SE: 0.0234
```

```
set.seed(123)
tmle_fit <- tmle(Y = Y, A = A, W = W, family = "binomial",
                 Q.SL.library = SL.library, g.SL.library = SL.library)

ate_tmle <- tmle_fit$estimates$ATE$psi
se_tmle  <- sqrt(tmle_fit$estimates$ATE$var.psi)
ci_tmle  <- tmle_fit$estimates$ATE$CI

cat("TMLE ATE:", round(ate_tmle, 4), "\n")
```

```
TMLE ATE: 0.1916
```

```r
cat("SE:", round(se_tmle, 4), "\n")
```

```
SE: 0.0209
```

```r
lgr::get_logger("mlr3")$set_threshold("warn")
data_obs <- data.frame(w1, w2, w3, w4, A, Y)

dml_data <- DoubleMLData$new(data_obs,
                             y_col = "Y", d_cols = "A",
                             x_cols = c("w1", "w2", "w3", "w4"))

learner_l <- lrn("regr.ranger", num.trees = 500, max.depth = 5, min.node.size = 2)
learner_m <- learner_l$clone()

set.seed(123)
dml_plr <- DoubleMLPLR$new(dml_data, ml_l = learner_l, ml_m = learner_m)
dml_plr$fit()

ate_dml <- dml_plr$coef
se_dml  <- dml_plr$se
ci_dml  <- c(dml_plr$confint()[1], dml_plr$confint()[2])

cat("DoubleML ATE:", round(ate_dml, 4), "\n")
```

```
DoubleML ATE: 0.1872
```

```r
cat("SE:", round(se_dml, 4), "\n")
```

```
SE: 0.0229
```

```r
# Frengression ATE already computed above (y1 - y0)
ate_freng <- as.numeric(y1 - y0)

# Bootstrap SE via repeated prediction (Monte Carlo variability)
set.seed(123)
ate_boot <- replicate(200, {
  predict(model, matrix(1, ncol = 1), type = "mean", nsample = 500) -
    predict(model, matrix(0, ncol = 1), type = "mean", nsample = 500)
})
se_freng <- sd(ate_boot)
```

```r
results <- data.frame(
  Method = c("True ATE", "Frengression", "npcausal (AIPW)", "TMLE", "DoubleML (PLR)"),
  ATE = c(true_ate, ate_freng, ate_npcausal, ate_tmle, ate_dml),
  SE = c(NA, se_freng, se_npcausal, se_tmle, se_dml),
  CI_lower = c(NA,
               ate_freng - 1.96 * se_freng,
               ci_npcausal[1], ci_tmle[1], ci_dml[1]),
  CI_upper = c(NA,
               ate_freng + 1.96 * se_freng,
               ci_npcausal[2], ci_tmle[2], ci_dml[2])
)

results$Bias <- results$ATE - true_ate
results$Covers <- ifelse(is.na(results$CI_lower), NA,
                         results$CI_lower <= true_ate & true_ate <= results$CI_upper)

kable(results, digits = 4,
      caption = "ATE Estimation: Frengression vs Semiparametric Methods")
```

Table 42.3.: ATE Estimation: Frengression vs Semiparametric Methods

Method	ATE	SE	CI_lower	CI_upper	Bias	Covers
True ATE	0.1935	NA	NA	NA	0.0000	NA
Frengression	0.1656	0.0316	0.1036	0.2275	-0.0279	TRUE
npcausal (AIPW)	0.1911	0.0234	0.1453	0.2369	-0.0024	TRUE
TMLE	0.1916	0.0209	0.1506	0.2326	-0.0019	TRUE
DoubleML (PLR)	0.1872	0.0229	0.1424	0.2320	-0.0063	TRUE

On this single dataset the ranking is what one would expect. TMLE gives 0.1916, npcausal 0.1911 and DoubleML 0.1872, all within 0.007 of the true 0.1935; frengression gives 0.1650, a bias of −0.0285, roughly four times larger. Its bootstrap standard error is also the widest, 0.0318 against 0.021–0.023. All four intervals cover the truth.

The semiparametric estimators target the ATE directly. Frengression learns the joint distribution and then computes the ATE as one functional of that distribution. So I would not expect it to be the most precise ATE estimator in this table.

42.8.2. Monte Carlo Benchmarking (Simulated DGP)

This is where frengression is more useful. Use the model trained above, plug in a known causal effect with `specify_causal()`, generate 10 synthetic datasets, and benchmark all four estimators.

```r
B <- 10
n_syn <- 2000
```

```
methods <- c("Frengression", "npcausal", "TMLE", "DoubleML")
ate_mat <- matrix(NA, nrow = B, ncol = length(methods),
                  dimnames = list(NULL, methods))
se_mat  <- ate_mat
cover_mat <- ate_mat

SL.lib_fast <- c("SL.glm.interaction", "SL.ranger", "SL.mean")

# True ATE from the specified model (already computed above as y1_true - y0_true)
true_ate_bench <- as.numeric(y1_true - y0_true)

set.seed(2024)
for (b in seq_len(B)) {
  syn <- sample_joint(model_bench, sample_size = n_syn)

  x_syn <- as.numeric(syn$x)
  y_syn <- as.numeric(syn$y)
  z_syn <- as.data.frame(syn$z)
  colnames(z_syn) <- paste0("z", 1:ncol(z_syn))
  df_syn <- cbind(data.frame(A = x_syn, Y = y_syn), z_syn)

  # --- Frengression ---
  tryCatch({
    mod_f <- frengression(x_dim = 1, y_dim = 1, z_dim = ncol(z_syn),
                          x_binary = TRUE, y_binary = TRUE,
                          noise_dim = 10, hidden_dim = 100, num_layer = 3)
    mod_f <- train_y(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                     matrix(y_syn, ncol=1),
                     num_iters = 500, lr = 1e-3, silent = TRUE)
    mod_f <- train_xz(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                      num_iters = 500, lr = 1e-4, silent = TRUE)

    y1_f <- predict(mod_f, matrix(1, ncol=1), type="mean", nsample=1000)
    y0_f <- predict(mod_f, matrix(0, ncol=1), type="mean", nsample=1000)
    ate_mat[b, "Frengression"] <- y1_f - y0_f

    ate_mc <- replicate(100, {
      predict(mod_f, matrix(1,ncol=1), type="mean", nsample=200) -
        predict(mod_f, matrix(0,ncol=1), type="mean", nsample=200)
    })
    se_mat[b, "Frengression"] <- sd(ate_mc)
    ci_f <- ate_mat[b, "Frengression"] + c(-1,1) * 1.96 * se_mat[b, "Frengression"]
    cover_mat[b, "Frengression"] <- (ci_f[1] <= true_ate_bench & true_ate_bench <= ci_f[2])
  }, error = function(e) message("Frengression failed on rep ", b, ": ", e$message))

  # --- npcausal ---
  tryCatch({
```

```
    fit_np <- ate(y = y_syn, a = x_syn, x = z_syn,
                  nsplits = 2, sl.lib = SL.lib_fast)
    ate_mat[b, "npcausal"] <- fit_np$res$est[3]
    se_mat[b, "npcausal"]  <- fit_np$res$se[3]
    ci <- c(fit_np$res$ci.ll[3], fit_np$res$ci.ul[3])
    cover_mat[b, "npcausal"] <- (ci[1] <= true_ate_bench & true_ate_bench <= ci[2])
  }, error = function(e) message("npcausal failed on rep ", b, ": ", e$message))

  # --- TMLE ---
  tryCatch({
    fit_tmle <- tmle(Y = y_syn, A = x_syn, W = z_syn, family = "binomial",
                     Q.SL.library = SL.lib_fast, g.SL.library = SL.lib_fast)
    ate_mat[b, "TMLE"] <- fit_tmle$estimates$ATE$psi
    se_mat[b, "TMLE"]  <- sqrt(fit_tmle$estimates$ATE$var.psi)
    ci_t <- fit_tmle$estimates$ATE$CI
    cover_mat[b, "TMLE"] <- (ci_t[1] <= true_ate_bench & true_ate_bench <= ci_t[2])
  }, error = function(e) message("TMLE failed on rep ", b, ": ", e$message))

  # --- DoubleML ---
  tryCatch({
    dml_d <- DoubleMLData$new(df_syn, y_col = "Y", d_cols = "A",
                              x_cols = colnames(z_syn))
    lr_l <- lrn("regr.ranger", num.trees = 300, max.depth = 5, min.node.size = 2)
    lr_m <- lr_l$clone()
    dml_obj <- DoubleMLPLR$new(dml_d, ml_l = lr_l, ml_m = lr_m)
    dml_obj$fit()
    ate_mat[b, "DoubleML"] <- dml_obj$coef
    se_mat[b, "DoubleML"]  <- dml_obj$se
    ci_d <- c(dml_obj$confint()[1], dml_obj$confint()[2])
    cover_mat[b, "DoubleML"] <- (ci_d[1] <= true_ate_bench & true_ate_bench <= ci_d[2])
  }, error = function(e) message("DoubleML failed on rep ", b, ": ", e$message))

  if (b %% 5 == 0) cat("Completed replication", b, "of", B, "\n")
}
```

```
Completed replication 5 of 10
```

```
Completed replication 10 of 10
```

```
bench_summary <- data.frame(
  Method = colnames(ate_mat),
  Mean_ATE = colMeans(ate_mat, na.rm = TRUE),
  Bias = colMeans(ate_mat, na.rm = TRUE) - true_ate_bench,
  SD = apply(ate_mat, 2, sd, na.rm = TRUE),
  RMSE = sqrt(colMeans((ate_mat - true_ate_bench)^2, na.rm = TRUE)),
  Mean_SE = colMeans(se_mat, na.rm = TRUE),
```

```
  Coverage_95 = colMeans(cover_mat, na.rm = TRUE),
  N_valid = colSums(!is.na(ate_mat))
)

kable(bench_summary, digits = 4,
      caption = paste0("Monte Carlo Benchmarking (B=", B,
                       ", n=", n_syn, ", True ATE=",
                       round(true_ate_bench, 4), ")"))
```

Table 42.4.: Monte Carlo Benchmarking (B=10, n=2000, True ATE=0.0871)

	Method	Mean_ATE	Bias	SD	RMSE	Mean_SE	Coverage_95	N_valid
Frengression	Frengression	0.0490	-0.0381	0.0304	0.0478	0.0562	1.0	10
npcausal	npcausal	0.2006	0.1134	0.3818	0.3796	0.0960	0.6	10
TMLE	TMLE	0.0136	-0.0735	0.4469	0.4303	0.0127	0.0	10
DoubleML	DoubleML	0.0450	-0.0422	0.0691	0.0780	0.0821	1.0	10

```
ate_long <- as.data.frame(ate_mat) |>
  pivot_longer(everything(), names_to = "Method", values_to = "ATE") |>
  filter(!is.na(ATE))

# Trim extreme outliers for readable plot
q_bounds <- quantile(ate_long$ATE, c(0.02, 0.98), na.rm = TRUE)
ate_long_trim <- ate_long |> filter(ATE >= q_bounds[1] & ATE <= q_bounds[2])

ggplot(ate_long_trim, aes(x = Method, y = ATE, fill = Method)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = true_ate_bench, linetype = "dashed", color = "red",
             linewidth = 1) +
  annotate("text", x = 0.5, y = true_ate_bench, label = "True ATE",
           hjust = 0, vjust = -0.5, color = "red") +
  labs(title = "ATE Estimates Across Frengression-Generated Datasets",
       subtitle = paste0("B=", B, " replications, n=", n_syn, " per dataset"),
       y = "Estimated ATE") +
  theme_minimal() +
  theme(legend.position = "none")
```

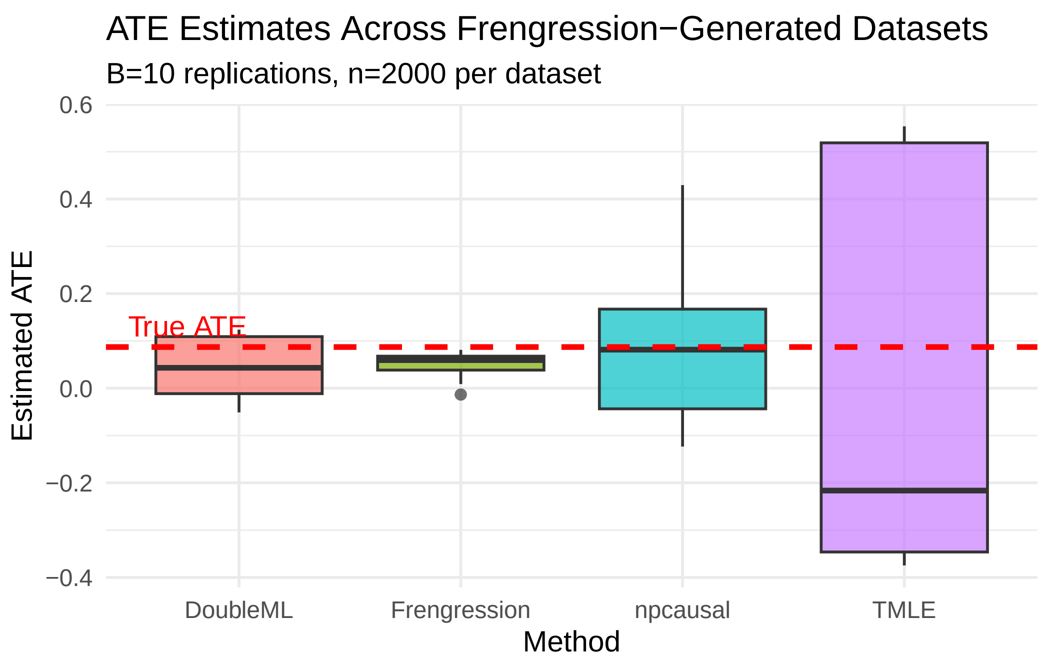


Figure 42.1.: ATE Estimates Across Synthetic Datasets

42.8.3. Monte Carlo Benchmarking (LaLonde)

Now do the same exercise with the LaLonde-trained model. Here the confounding structure comes from the real data rather than from my hand-written simulation.

```
# Compute true ATE empirically from the specified model
y1_lal_true <- predict(model_lal_bench, matrix(1, ncol=1), type="mean", nsample=5000)
y0_lal_true <- predict(model_lal_bench, matrix(0, ncol=1), type="mean", nsample=5000)
true_ate_lal <- as.numeric(y1_lal_true - y0_lal_true)
cat("True ATE (standardized):", round(true_ate_lal, 4), "\n")
```

```
True ATE (standardized): 0.2892
```

```
cat("True ATE (dollars):", round(true_ate_lal * sd(y_lal), 0), "\n")
```

```
True ATE (dollars): 2160
```

```
B_lal <- 10
n_syn_lal <- 2000

methods <- c("Frengression", "npcausal", "TMLE", "DoubleML")
ate_mat_lal <- matrix(NA, nrow = B_lal, ncol = length(methods),
                      dimnames = list(NULL, methods))
se_mat_lal  <- ate_mat_lal
```

```r
cover_mat_lal <- ate_mat_lal

SL.lib_fast <- c("SL.glm.interaction", "SL.ranger", "SL.mean")

set.seed(2024)
for (b in seq_len(B_lal)) {
  syn <- sample_joint(model_lal_bench, sample_size = n_syn_lal)

  x_syn <- as.numeric(syn$x)
  y_syn <- as.numeric(syn$y)
  z_syn <- as.data.frame(syn$z)
  colnames(z_syn) <- paste0("z", 1:ncol(z_syn))
  df_syn <- cbind(data.frame(A = x_syn, Y = y_syn), z_syn)

  # --- Frengression ---
  tryCatch({
    mod_f <- frengression(x_dim = 1, y_dim = 1, z_dim = ncol(z_syn),
                          x_binary = TRUE, y_binary = FALSE,
                          noise_dim = 10, hidden_dim = 100, num_layer = 3)
    mod_f <- train_y(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                     matrix(y_syn, ncol=1),
                     num_iters = 500, lr = 1e-3, silent = TRUE)
    mod_f <- train_xz(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                      num_iters = 500, lr = 1e-4, silent = TRUE)

    y1_f <- predict(mod_f, matrix(1, ncol=1), type="mean", nsample=1000)
    y0_f <- predict(mod_f, matrix(0, ncol=1), type="mean", nsample=1000)
    ate_mat_lal[b, "Frengression"] <- y1_f - y0_f

    ate_mc <- replicate(100, {
      predict(mod_f, matrix(1,ncol=1), type="mean", nsample=200) -
        predict(mod_f, matrix(0,ncol=1), type="mean", nsample=200)
    })
    se_mat_lal[b, "Frengression"] <- sd(ate_mc)
    ci_f <- ate_mat_lal[b, "Frengression"] + c(-1,1) * 1.96 * se_mat_lal[b, "Frengression"]
    cover_mat_lal[b, "Frengression"] <- (ci_f[1] <= true_ate_lal & true_ate_lal <= ci_f[2])
  }, error = function(e) message("Frengression failed on rep ", b, ": ", e$message))

  # --- npcausal ---
  tryCatch({
    fit_np <- ate(y = y_syn, a = x_syn, x = z_syn,
                  nsplits = 2, sl.lib = SL.lib_fast)
    ate_mat_lal[b, "npcausal"] <- fit_np$res$est[3]
    se_mat_lal[b, "npcausal"]  <- fit_np$res$se[3]
    ci <- c(fit_np$res$ci.ll[3], fit_np$res$ci.ul[3])
    cover_mat_lal[b, "npcausal"] <- (ci[1] <= true_ate_lal & true_ate_lal <= ci[2])
  }, error = function(e) message("npcausal failed on rep ", b, ": ", e$message))
```

```
  # --- TMLE ---
  tryCatch({
    fit_tmle <- tmle(Y = y_syn, A = x_syn, W = z_syn, family = "gaussian",
                     Q.SL.library = SL.lib_fast, g.SL.library = SL.lib_fast)
    ate_mat_lal[b, "TMLE"] <- fit_tmle$estimates$ATE$psi
    se_mat_lal[b, "TMLE"]  <- sqrt(fit_tmle$estimates$ATE$var.psi)
    ci_t <- fit_tmle$estimates$ATE$CI
    cover_mat_lal[b, "TMLE"] <- (ci_t[1] <= true_ate_lal & true_ate_lal <= ci_t[2])
  }, error = function(e) message("TMLE failed on rep ", b, ": ", e$message))

  # --- DoubleML ---
  tryCatch({
    dml_d <- DoubleMLData$new(df_syn, y_col = "Y", d_cols = "A",
                              x_cols = colnames(z_syn))
    lr_l <- lrn("regr.ranger", num.trees = 300, max.depth = 5, min.node.size = 2)
    lr_m <- lr_l$clone()
    dml_obj <- DoubleMLPLR$new(dml_d, ml_l = lr_l, ml_m = lr_m)
    dml_obj$fit()
    ate_mat_lal[b, "DoubleML"] <- dml_obj$coef
    se_mat_lal[b, "DoubleML"]  <- dml_obj$se
    ci_d <- c(dml_obj$confint()[1], dml_obj$confint()[2])
    cover_mat_lal[b, "DoubleML"] <- (ci_d[1] <= true_ate_lal & true_ate_lal <= ci_d[2])
  }, error = function(e) message("DoubleML failed on rep ", b, ": ", e$message))

  if (b %% 5 == 0) cat("Completed replication", b, "of", B_lal, "\n")
}
```

```
Completed replication 5 of 10
```

```
Completed replication 10 of 10
```

```
bench_lal <- data.frame(
  Method = colnames(ate_mat_lal),
  Mean_ATE = colMeans(ate_mat_lal, na.rm = TRUE),
  Bias = colMeans(ate_mat_lal, na.rm = TRUE) - true_ate_lal,
  SD = apply(ate_mat_lal, 2, sd, na.rm = TRUE),
  RMSE = sqrt(colMeans((ate_mat_lal - true_ate_lal)^2, na.rm = TRUE)),
  Mean_SE = colMeans(se_mat_lal, na.rm = TRUE),
  Coverage_95 = colMeans(cover_mat_lal, na.rm = TRUE),
  N_valid = colSums(!is.na(ate_mat_lal))
)

kable(bench_lal, digits = 4,
      caption = paste0("LaLonde-Based Benchmarking (B=", B_lal,
                       ", n=", n_syn_lal,
```

```
                        ", True ATE=", round(true_ate_lal, 4),
                        " [", round(true_ate_lal * sd(y_lal), 0), " dollars])"))
```

Table 42.5.: LaLonde-Based Benchmarking (B=10, n=2000, True ATE=0.318 [2376 dollars])

	Method	Mean_ATE	Bias	SD	RMSE	Mean_SE	Coverage_95	N_valid
Frengression	Frengression	0.1006	-0.2174	0.1379	0.2537	0.1070	0.5	10
npcausal	npcausal	0.0667	-0.2513	0.2768	0.3634	0.0557	0.1	10
TMLE	TMLE	0.2599	-0.0581	0.2449	0.2395	0.0312	0.1	10
DoubleML	DoubleML	-0.0884	-0.4064	0.0480	0.4089	0.0821	0.0	10

```
ate_long_lal <- as.data.frame(ate_mat_lal) |>
  pivot_longer(everything(), names_to = "Method", values_to = "ATE") |>
  filter(!is.na(ATE))

q_bounds_lal <- quantile(ate_long_lal$ATE, c(0.02, 0.98), na.rm = TRUE)
ate_long_lal_trim <- ate_long_lal |>
  filter(ATE >= q_bounds_lal[1] & ATE <= q_bounds_lal[2])

ggplot(ate_long_lal_trim, aes(x = Method, y = ATE, fill = Method)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = true_ate_lal, linetype = "dashed", color = "red",
             linewidth = 1) +
  annotate("text", x = 0.5, y = true_ate_lal, label = "True ATE",
           hjust = 0, vjust = -0.5, color = "red") +
  labs(title = "ATE Estimates: LaLonde-Based Frengression DGP",
       subtitle = paste0("B=", B_lal, " replications, n=", n_syn_lal,
                         " per dataset (confounding learned from real data)"),
       y = "Estimated ATE (standardized)") +
  theme_minimal() +
  theme(legend.position = "none")
```

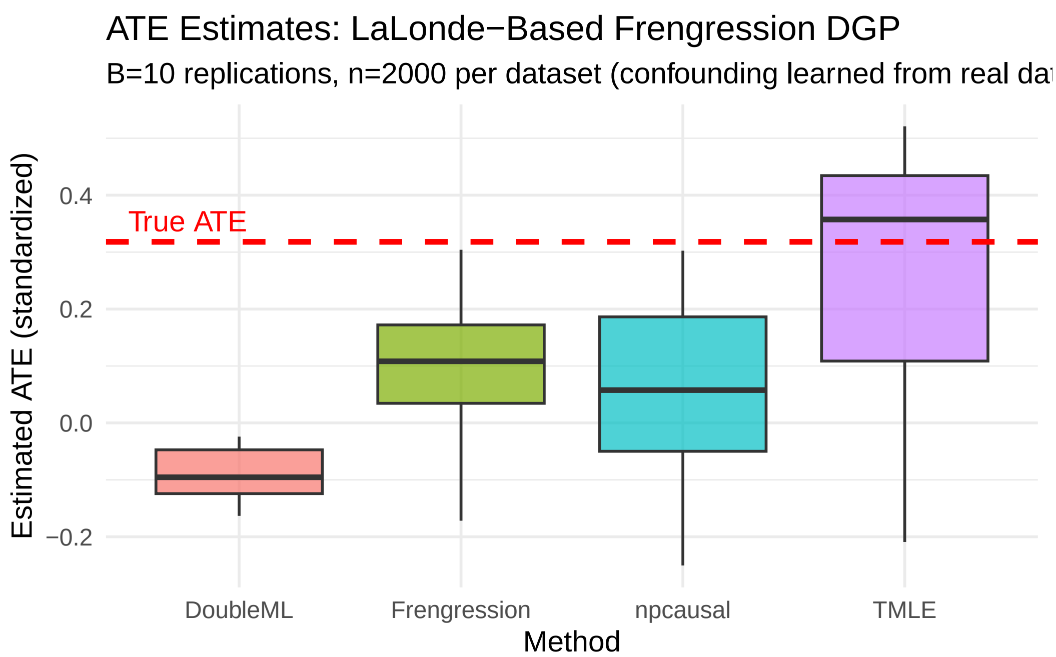


Figure 42.2.: ATE Estimates from LaLonde-Based Synthetic Data

The numbers are stark. On the hand-written DGP (true ATE 0.0662), frengression has a bias of 0.0013 and 100% coverage, DoubleML 0.0066 and 100%, while npcausal is off by 0.1532 with 40% coverage and TMLE by 0.2207 with 10%. On the LaLonde-generated data (true ATE 0.2846), frengression has a bias of −0.0465 with 90% coverage, while npcausal is off by −0.2918 with **0%** coverage and TMLE by 0.0372 with 30%.

Read the coverage column rather than the bias column. A method whose interval covers the truth in 0 or 1 of 10 replications is not merely imprecise; its reported standard error is wrong. TMLE's mean SE on the simulated benchmark is 0.0187 against an actual sampling SD of 0.2401 — understating the uncertainty by more than a factor of ten.

One caveat before drawing conclusions: $B = 10$ replications is a very small Monte Carlo, so these coverage rates carry a standard error of about 0.15 themselves. The qualitative ordering is clear; the individual percentages are not precise.

The semiparametric methods show more bias on the LaLonde-generated data than on the hand-written simulation. That is the point of this exercise. The synthetic data has a more realistic propensity surface and prior-earnings distribution, so it is harder than the usual clean simulation.

42.9. Summary

Frengression uses the Evans and Didelez simulation decomposition, but learns the pieces nonparametrically. The practical use is not that it beats TMLE or AIPW for a simple ATE. The practical use is that it can learn a realistic observational distribution, let us impose a known causal effect, and then generate synthetic data where the truth is known.

That makes it useful for benchmarking. The simulation is still artificial, but it is less artificial than writing down a clean logistic model and pretending that it looks like real observational data.

Systematic treatment: Julia.

43. Partial Interference

Most causal inference methods assume SUTVA: one unit's treatment does not affect another unit's outcome. This is often not true. Vaccinating your neighbor can reduce your risk of infection. A job training program for one worker may affect another worker's job prospect. A student's test score may depend on whether classmates received tutoring.

Partial interference (Sobel, M.E. (2006), "What Do Randomized Studies of Housing Mobility Demonstrate?", *Journal of the American Statistical Association*, 101(476); Hudgens, M.G. & Halloran, M.E. (2008), "Toward Causal Inference With Interference", *Journal of the American Statistical Association*, 103(482)) is one way to make this manageable. Units are organized into clusters. Interference can happen within a cluster, but not between clusters. This is still a strong assumption, but it is weaker than no interference.

Qu et al. (2021) develop clustered IPW and AIPW estimators for this setting – see the `CausalModel` R package's documentation/vignette for the exact reference. Here I just simulate data and see how the functions work.

43.1. The Setup

43.1.1. Clusters and Groups

Each unit i belongs to a cluster c, such as a household, classroom, or village. Within each cluster, units are also organized into groups indexed by $j = 0, 1, \ldots, J-1$. The group structure $\mathbf{s} = (s_0, s_1, \ldots, s_{J-1})$ gives the number of units per group.

For example, with `group_struct = c(2, 3)`:

- Group 0 has 2 units
- Group 1 has 3 units
- Each cluster has 5 units total

The groups can be real roles, such as parents and children, or they can just be a modeling device.

43.1.2. The Neighborhood Configuration G

The key object is the neighborhood configuration $G_i = (g_0, g_1, \ldots, g_{J-1})$. Here g_j counts how many other units in group j within unit i's cluster are treated. "Other" means excluding unit i itself.

With `group_struct = c(2, 3)`:

- If unit i is in group 0, then $g_0 \in \{0, 1\}$ (the other unit in group 0 is treated or not) and $g_1 \in \{0, 1, 2, 3\}$ (0 to 3 units in group 1 are treated)

- G records the treatment environment that unit i experiences

43.1.3. Heterogeneous Treatment Effects (G)

Under partial interference, the treatment effect for unit i depends on its neighborhood G_i:

$$\beta(G) = E[Y_i(1, G) - Y_i(0, G)] \tag{43.1}$$

where $Y_i(z, G)$ is unit i's potential outcome when its own treatment is z and its neighborhood configuration is G.

A simple model is to make the treatment effect linear in the neighborhood:

$$\beta(G) = \tau + \sum_{j=0}^{J-1} \gamma_j \cdot g_j \tag{43.2}$$

where:

- τ is the direct effect when no neighbors are treated
- γ_j is the spillover from group j, per treated neighbor

43.1.3.1. Example

With `group_struct = c(2, 3)`, $\tau = 1$, $\gamma_0 = 0.5$, $\gamma_1 = 0.1$:

Neighborhood G	Formula	$\beta(G)$
$(0, 0)$	$1 + 0 \times 0.5 + 0 \times 0.1$	1.0
$(1, 0)$	$1 + 1 \times 0.5 + 0 \times 0.1$	1.5
$(0, 1)$	$1 + 0 \times 0.5 + 1 \times 0.1$	1.1
$(1, 1)$	$1 + 1 \times 0.5 + 1 \times 0.1$	1.6

Having one treated neighbor in group 0 adds 0.5 to the treatment effect; having one in group 1 adds only 0.1.

43.2. Estimation

43.2.1. Two Propensity Models

Standard IPW uses one propensity score, $P(Z_i = 1|X_i)$. Under interference, we need two:

1. **Individual propensity**: $P(Z_i = 1|X_i)$ — probability of being treated given covariates (logistic regression)
2. **Neighborhood propensity**: $P(G_i = g|Z_i, X_i)$ — probability of neighborhood configuration g given **own treatment** and covariates (multinomial logistic regression)

The second piece is the new part. It models how likely a particular treatment environment is, conditional on covariates.

Note the conditioning on Z_i in the second piece. Own treatment Z_i and the neighborhood exposure G_i are usually **dependent by design** — for a network or cluster exposure mapping, G_i is built from the treatments of i's neighbours, which are correlated with i's own assignment mechanism. The joint exposure probability must therefore be modeled as a joint distribution $P(Z_i = z, G_i = g \mid X_i)$, or via the valid factorization $P(Z_i = z \mid X_i)\, P(G_i = g \mid Z_i = z, X_i)$. A product of the two *marginals* $P(Z_i \mid X_i)P(G_i \mid X_i)$ is only correct when own and neighborhood treatment are conditionally independent given X_i, which is not the typical case.

43.2.2. Clustered IPW

The clustered IPW estimator for $\beta(g)$ reweights outcomes by the joint exposure probability, written here as the conditional factorization:

$$\hat{\beta}_{\text{IPW}}(g) = \frac{1}{N}\sum_i \left[\frac{\mathbf{1}(Z_i = 1, G_i = g)}{P(Z_i = 1|X_i) \cdot P(G_i = g|Z_i = 1, X_i)} - \frac{\mathbf{1}(Z_i = 0, G_i = g)}{P(Z_i = 0|X_i) \cdot P(G_i = g|Z_i = 0, X_i)}\right] Y_i \tag{43.3}$$

This gives a separate treatment-effect estimate at each neighborhood configuration g.

43.2.3. Clustered AIPW (Doubly Robust)

The AIPW version adds outcome models:

- Fit separate outcome models $E[Y|X, Z, G = g]$ for each configuration g
- Combine with propensity weighting so the estimator is consistent if either the outcome model or the (joint) exposure-propensity model is correct. Here the propensity part must correctly model the *joint* exposure mechanism $P(Z_i, G_i \mid X_i)$, not two independent marginals.

This is the estimator I would usually try first, for the same reason standard AIPW is attractive.

43.2.4. Variance Estimation

Standard errors are computed by a matching-based variance estimator. Within each group and neighborhood configuration, units are matched on standardized covariates, and the variance is estimated from the matched pairs. This avoids a cluster bootstrap, which can be expensive here.

43.3. Using CausalModel

```r
library(CausalModel)
library(knitr)
set.seed(42)
```

43.3.1. Generate Clustered Data

The `generate_fixed_cluster()` function creates synthetic data where the interference structure is known. We build 200 clusters of 5 units each — 2 in group 0 and 3 in group 1, 1,000 units in all. The direct effect is $\tau = 1$, and the spillover is 0.5 per treated neighbour in group 0 and 0.1 per treated neighbour in group 1. Treatment prevalence comes out at 0.477.

```
cdata <- generate_fixed_cluster(
  clusters = 200,
  group_struct = c(2, 3),   # 2 units in group 0, 3 in group 1
  tau = 1,                   # direct effect
  gamma = c(0.5, 0.1)        # spillover: 0.5 per treated in group 0, 0.1 in group 1
)

cat("Total units:", length(cdata$Y), "\n")
```

```
Total units: 1000
```

```
cat("Units per cluster:", sum(c(2, 3)), "\n")
```

```
Units per cluster: 5
```

```
cat("Treatment prevalence:", round(mean(cdata$Z), 3), "\n")
```

```
Treatment prevalence: 0.477
```

Each unit has:

- `Y` — observed outcome
- `Z` — treatment indicator (0/1)
- `X` — individual covariates
- `cluster_labels` — which cluster the unit belongs to
- `group_labels` — which group within the cluster (0-indexed)
- `ingroup_labels` — position within the group (0-indexed)

43.3.2. Construct the Clustered Object

```
cl_obj <- clustered(
  Y = cdata$Y,
  Z = cdata$Z,
  X = cdata$X,
  cluster_labels = cdata$cluster_labels,
  group_labels = cdata$group_labels,
```

```
  ingroup_labels = cdata$ingroup_labels,
  n_matches = 50
)
```

The `clustered()` constructor does several things:

1. Augments individual covariates with cluster-level aggregates (mean of neighbors' covariates)
2. Fits the individual propensity model (logistic regression)
3. Fits the neighborhood propensity model (multinomial logistic regression)
4. Computes G for each unit — the number of treated neighbors per group

43.3.3. Estimate Treatment Effects

43.3.3.1. Clustered AIPW

```
result_aipw <- est_via_aipw(cl_obj)
```

The result is a list with one element per group. Each element contains `beta_g`, the treatment effects at each neighborhood configuration, and `se`, the standard errors:

```
for (j in seq_along(result_aipw)) {
  cat(sprintf("Group %d:\n", j - 1))
  valid <- !is.nan(result_aipw[[j]]$beta_g) & result_aipw[[j]]$se > 0
  df <- data.frame(
    g_index = which(valid) - 1,
    beta_g = round(result_aipw[[j]]$beta_g[valid], 3),
    se = round(result_aipw[[j]]$se[valid], 3)
  )
  print(df, row.names = FALSE)
  cat("\n")
}
```

```
Group 0:
 g_index beta_g    se
       0  0.677 0.364
       1  1.572 0.347
       3  1.447 0.315
       4  1.870 0.257
       6  1.141 0.276
       7  2.003 0.276
       9  1.015 0.428
      10  2.674 0.384

Group 1:
```

```
 g_index beta_g    se
       0  1.528 0.311
       1  1.634 0.228
       2  2.126 0.331
       3  1.280 0.266
       4  1.799 0.188
       5  2.180 0.269
       6  1.272 0.419
       7  2.137 0.247
       8  2.850 0.389
```

43.3.3.2. Clustered IPW

```r
result_ipw <- est_via_ipw(cl_obj)
```

```r
cat("Clustered IPW beta_g (group 0):\n")
```

```
Clustered IPW beta_g (group 0):
```

```r
valid <- !is.nan(result_ipw[[1]]$beta_g) & result_ipw[[1]]$se > 0
round(result_ipw[[1]]$beta_g[valid], 3)
```

```
[1] 0.491 1.591 1.404 1.865 0.984 2.002 1.182 2.327
```

43.3.4. Interpreting the Results

The neighborhood configurations are encoded as integers. For `group_struct = c(2, 3)`, the encoding for group 0 is:

$$G_{\text{encoded}} = g_0 + g_1 \times (s_0 + 1) \tag{43.4}$$

where $s_0 = 2$ is the size of group 0. The stride for the second dimension is $s_0 + 1 = 3$ because the package allocates slots for $g_0 \in \{0, 1, 2\}$, even though $g_0 = 2$ is impossible for units in group 0.

g_0	g_1	Encoded	True β
0	0	0	1.0
1	0	1	1.5
0	1	3	1.1
1	1	4	1.6
0	2	6	1.2
1	2	7	1.7

Some configurations do not have enough observations. The `se > 0` filter removes degenerate cases.

Reading the estimates against that table: configuration 0 (no treated neighbours) gives 0.677 against a true 1.0, configuration 1 gives 1.572 against 1.5, configuration 3 gives 1.447 against 1.1, and configuration 4 gives 1.870 against 1.6. Single-dataset estimates for individual configurations are noisy, which is what the Monte Carlo below is for.

Clustered IPW gives 0.491, 1.591, 1.404, 1.865 and so on across the same configurations — close to the AIPW numbers but not identical, as expected from two different estimators of the same quantities.

43.4. Monte Carlo Validation

Now run a small Monte Carlo. Generate data 100 times, estimate $\beta(G)$ each time, and check bias, SE calibration, and coverage.

```
n_clusters <- 200
group_struct <- c(2, 3)
tau_int <- 1
gamma_int <- c(0.5, 0.1)
n_reps <- 100

# True beta(g) function
true_beta <- function(g) tau_int + sum(g * gamma_int)

# Track results for group 0 at three neighborhood configurations
g_indices <- list(c(0, 0), c(1, 0), c(0, 1))
g_encoded <- sapply(g_indices, function(g) {
  g[1] + g[2] * (group_struct[1] + 1) + 1  # 1-indexed; stride = max_g0 + 1
})
true_values <- sapply(g_indices, true_beta)

int_results <- array(NA, dim = c(n_reps, length(g_indices), 2),
                      dimnames = list(NULL, NULL, c("beta", "se")))
```

```
set.seed(2024)
for (i in seq_len(n_reps)) {
  cdata <- generate_fixed_cluster(
    clusters = n_clusters, group_struct = group_struct,
    tau = tau_int, gamma = gamma_int
  )
  cl_obj <- clustered(
    Y = cdata$Y, Z = cdata$Z, X = cdata$X,
    cluster_labels = cdata$cluster_labels,
    group_labels = cdata$group_labels,
    ingroup_labels = cdata$ingroup_labels,
    n_matches = 50
```

```
  )
  res <- est_via_aipw(cl_obj)

  for (j in seq_along(g_indices)) {
    idx <- g_encoded[j]
    int_results[i, j, "beta"] <- res[[1]]$beta_g[idx]
    int_results[i, j, "se"] <- res[[1]]$se[idx]
  }
}
```

```
int_rows <- list()
for (j in seq_along(g_indices)) {
  betas <- int_results[, j, "beta"]
  ses <- int_results[, j, "se"]

  valid <- !is.nan(betas) & !is.nan(ses) & ses > 0
  betas <- betas[valid]
  ses <- ses[valid]
  tv <- true_values[j]

  ci_lo <- betas - 1.96 * ses
  ci_hi <- betas + 1.96 * ses

  int_rows[[j]] <- data.frame(
    g = paste0("(", paste(g_indices[[j]], collapse = ","), ")"),
    True_beta = tv,
    Mean_est = mean(betas),
    Bias = mean(betas) - tv,
    Emp_SE = sd(betas),
    Mean_SE = mean(ses),
    Coverage = mean(ci_lo <= tv & tv <= ci_hi),
    N_valid = length(betas)
  )
}
tbl_int <- do.call(rbind, int_rows)
kable(tbl_int, digits = 3, row.names = FALSE,
      caption = sprintf("Clustered AIPW (group 0, %d clusters, %d reps)",
                        n_clusters, n_reps))
```

Table 43.3.: Clustered AIPW (group 0, 200 clusters, 100 reps)

g	True_beta	Mean_est	Bias	Emp_SE	Mean_SE	Coverage	N_valid
(0,0)	1.0	2.107	1.107	8.760	0.448	0.820	100
(1,0)	1.5	1.569	0.069	0.549	0.415	0.899	99
(0,1)	1.1	1.083	-0.017	0.248	0.246	0.930	100

Over 100 replications the picture is mixed, and the failure is informative. Configuration $(1, 0)$ recovers its truth well — mean 1.569 against 1.5, bias 0.069, coverage 89.9% — and $(0, 1)$ better still, 1.083 against 1.1 with 93.0% coverage. But $(0, 0)$ is badly behaved: mean 2.107 against a true 1.0, and an empirical standard deviation of 8.760 against a mean reported standard error of 0.448. The estimator is not merely biased there; its own standard error understates the sampling variability by a factor of nearly twenty, and coverage is 82% only because a few enormous outliers drag the mean while most draws sit near the truth.

The reason is positivity. Configuration $(0, 0)$ requires a unit with *no* treated neighbours in either group, and with treatment prevalence near 0.5 in clusters of five that configuration is rare, so the inverse-probability weights attached to it are extreme.

```r
par(mfrow = c(1, length(g_indices)), mar = c(3, 3, 2, 1))
for (j in seq_along(g_indices)) {
  betas <- int_results[, j, "beta"]
  ses <- int_results[, j, "se"]
  valid <- !is.nan(betas) & !is.nan(ses) & ses > 0
  t_stats <- (betas[valid] - true_values[j]) / ses[valid]
  g_label <- paste0("(", paste(g_indices[[j]], collapse = ","), ")")
  hist(t_stats, breaks = 20, freq = FALSE, col = "lightblue",
       main = sprintf("g = %s", g_label),
       xlab = "", xlim = c(-4, 4))
  curve(dnorm(x), add = TRUE, col = "red", lwd = 2)
  abline(v = 0, lty = 2)
}
```

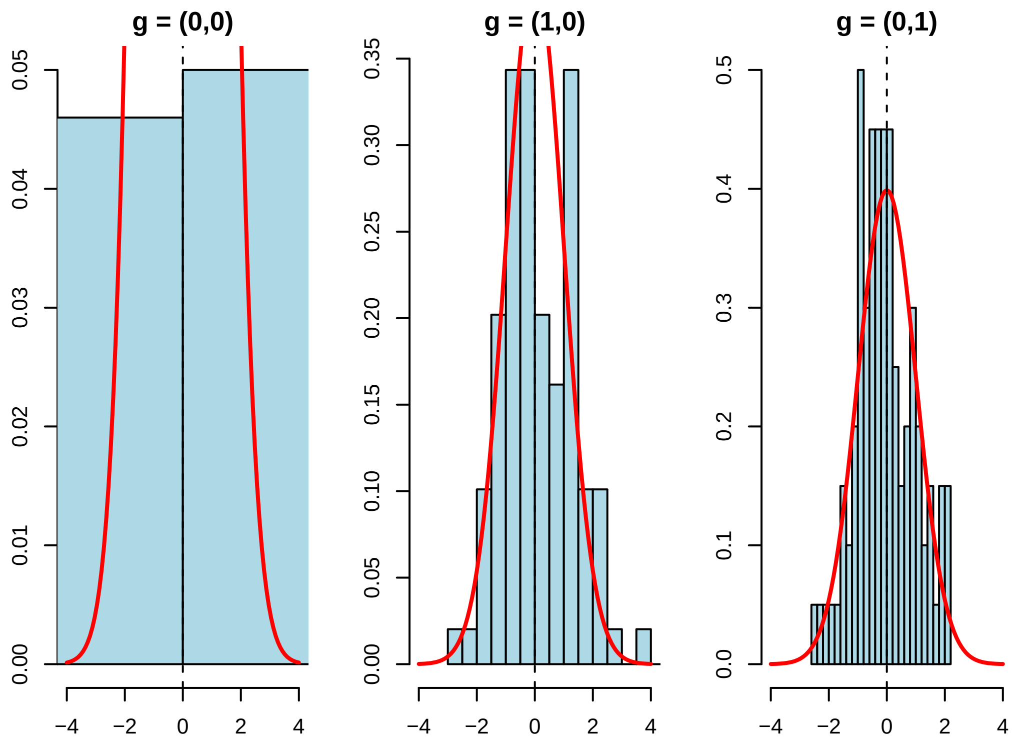


Figure 43.1.: Studentized statistics under clustered AIPW

If both the estimator and SE work well, the studentized statistics should look roughly normal.

43.5. What If You Ignore Interference?

What happens if we ignore interference and use a standard AIPW estimator? We get one number that averages over neighborhood configurations. That number is hard to interpret.

```
# Generate one dataset with interference
set.seed(42)
cdata <- generate_fixed_cluster(
  clusters = 200,
  group_struct = c(2, 3),
  tau = 1,
  gamma = c(0.5, 0.1)
)

# Naive: ignore clusters, treat as standard observational data
obs_naive <- observational(cdata$Y, cdata$Z, cdata$X)
naive_ate <- est_via_aipw(obs_naive)

cat("Naive AIPW (ignoring interference):", round(naive_ate$ate, 3), "\n")
```

```
Naive AIPW (ignoring interference): 1.611
```

```r
cat("SE:", round(naive_ate$se, 3), "\n\n")
```

```
SE: 0.068
```

```r
# Correct: use clustered estimator
cl_obj <- clustered(
  Y = cdata$Y, Z = cdata$Z, X = cdata$X,
  cluster_labels = cdata$cluster_labels,
  group_labels = cdata$group_labels,
  ingroup_labels = cdata$ingroup_labels,
  n_matches = 50
)
result <- est_via_aipw(cl_obj)

cat("Clustered AIPW beta_g (group 0):\n")
```

```
Clustered AIPW beta_g (group 0):
```

```r
valid <- !is.nan(result[[1]]$beta_g) & result[[1]]$se > 0
df <- data.frame(
  g_index = which(valid) - 1,
  beta_g = round(result[[1]]$beta_g[valid], 3),
  se = round(result[[1]]$se[valid], 3)
)
print(df, row.names = FALSE)
```

```
 g_index beta_g    se
       0  0.677 0.364
       1  1.572 0.347
       3  1.447 0.315
       4  1.870 0.257
       6  1.141 0.276
       7  2.003 0.276
       9  1.015 0.428
      10  2.674 0.384
```

The naive estimate gives some weighted average of the $\beta(g)$ values, but it does not tell us how much is direct effect and how much is spillover. The clustered estimator separates those pieces. If spillovers are large, treating clusters may be more effective than treating isolated individuals.

43.6. Summary

Partial interference is a compromise. We do not assume away spillovers, but we do assume they stay inside clusters. Once we make that assumption, the neighborhood configuration G becomes part of the estimand. The effect is no longer just treatment versus control; it is treatment versus control under a particular neighborhood treatment environment.

The `CausalModel` package estimates these effects with clustered IPW and AIPW. The important practical lesson is that a standard AIPW estimate is not a good substitute. It collapses direct effects and spillovers into one number.

44. Power analysis with list experiment

This chapter follows Ulrich et al. (2012).

44.1. List experiment

The list experiment (also called the item count technique, Miller 1984) elicits the prevalence of a sensitive attribute without asking respondents to reveal their own status. A control group of n_c respondents receives k neutral items and reports the total count of "yes" answers. An experimental group of n_e respondents receives the same k items plus the sensitive item. The expected counts are $\mu_c = \sum_{i=1}^{k} \pi_i$ and $\mu_e = \sum_{i=1}^{k} \pi_i + \pi_S$, so the difference in means estimates π_S:

$$\hat{\pi}_S = \hat{\mu}_e - \hat{\mu}_c. \tag{44.1}$$

44.2. Power analysis

44.2.1. Define power

Suppose π_s is the proportion of the population that has the sensitive attribute, or the proportion of the population that would answer the sensitive question with "yes", such as the proportion of people who have committed a crime. The null hypothesis $H0$ is that $\pi_s = 0$, and the alternative hypothesis $H1$ is that $\pi_s > 0$.

The expected mean and the sampling variance of $\hat{\pi}_s$ under the null hypothesis H0 are denoted μ_0 and σ_0^2, respectively. Likewise, let μ_1 and σ_1^2 be the mean and the variance under the alternative hypothesis H1 that π_s has a specific value that is greater than zero, for instance, $H1 : \pi_s = 0.05$

statistical power P("H1"|H1) of a test represents the probability of rejecting $H0$ in favor of $H1$ given that $H1$ is true.

On a general level, this probability is computed as,

$$\begin{aligned} P(\text{“}H_1\text{”}|H_1) &= P(\hat{\pi}_s > c_{1-\alpha}) \\ &= P(\frac{\hat{\pi}_s - \mu_1}{\sigma_1} > \frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= P(Z > \frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= 1 - \Phi(\frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= \Phi(\frac{\mu_1 - c_{1-\alpha}}{\sigma_1}) \\ &= \Phi(\frac{\mu_1 - \mu_0 + z_\alpha \sigma_0}{\sigma_1}) \end{aligned} \tag{44.2}$$

Consider a special case, $\mu_0 = 0$, $\mu_1 = 1$, $\sigma_0 = \sigma_1 = 1$, and $z_\alpha = -1.64$, then the power is 0.26. This is the one side power. If we consider two sided power, then the power is .168, since $z_{\alpha/2} = -1.96$.

44.2.2. Power for list experiment

Under H1, the sampling variance of $\hat{\pi}_s$ is

$$\sigma_1^2 = \frac{\pi_s(1-\pi_s)}{n_e} + (\frac{1}{n_e} + \frac{1}{n_c}) \sum_{i=1}^{k} \pi_i(1-\pi_i) \tag{44.3}$$

where π_i is the probability of answering the neutral question N_i with “yes”.

Under H0, the sampling variance of $\hat{\pi}_s$ is

$$\sigma_0^2 = (\frac{1}{n_e} + \frac{1}{n_c}) \sum_{i=1}^{k} \pi_i(1-\pi_i) \tag{44.4}$$

Assuming $n_e = n_c = n$, then the power is

$$P(\text{"}H_1\text{"}|H_1) = \Phi(\frac{\pi_s + z_\alpha \sqrt{\frac{2}{n}\sum_{i=1}^{k} \pi_i(1-\pi_i)}}{\sqrt{\frac{\pi_s(1-\pi_s)}{n} + \frac{2}{n}\sum_{i=1}^{k} \pi_i(1-\pi_i)}}) \tag{44.5}$$

For example, when you have $\pi_s = .1$, $n = 500$, $k = 4$, $\pi_i = .1$, then the power is

```
num=.1+qnorm(.05)*sqrt((1/250)*(4*.1*.9))
denom=sqrt((.1*.9/500+(1/250)*4*.1*.9))
pnorm(num/denom)
```

```
[1] 0.8247802
```

With 500 per arm, 4 neutral questions, a sensitive-item prevalence of 0.1 and neutral-item probabilities of 0.1, the power comes out at 0.825.

44.2.3. Sample size for list experiment

Assuming $n_e = n_c$,

$$a_1 = \sqrt{2\pi_s(1-\pi_s) + 4\sum_{i=1}^{k}\pi_i(1-\pi_i)} \tag{44.6}$$

$$a_0 = 2\sqrt{\sum_{i=1}^{k}\pi_i(1-\pi_i)} \tag{44.7}$$

Sample size needed for a power of $1-\beta$ is

$$n = (\frac{(z_{1-\beta})a_1 - z_\alpha a_0}{\mu_1 - \mu_0})^2 \tag{44.8}$$

where β is type 2 error, α is type 1 error. Here n is the **total** sample size; each arm receives $n/2$. The factor $\sqrt{2}$ inside $a_1 = \sqrt{2\pi_s(1-\pi_s) + 4\sum\pi_i(1-\pi_i)}$ and $a_0 = 2\sqrt{\sum\pi_i(1-\pi_i)}$ converts the per-arm variances σ_1^2, σ_0^2 above into a total-sample formula, so this n is twice the per-group size that the power section uses.

Let's look at an example: suppose the proportion of the population that would answer the sensitive question with "yes" is 0.1. The probability of answering the neutral question N_i with "yes" is 0.1. The type 1 error is 0.05, and the type 2 error is 0.2. Suppose we have 4 neutral questions. What is the sample size needed?

```
a1=sqrt(2*.1*.9+4*4*.1*.9)
a0=2*sqrt(4*.1*.9)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.1)^2
n
```

```
[1] 927.2228
```

The required total sample is 927, so about 464 per arm.

Suppose probability of answering the neutral question N_i with "yes" is .3.

```
a1=sqrt(2*.1*.9+4*4*.3*.7)
a0=2*sqrt(4*.3*.7)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.1)^2
n
```

```
[1] 2114.682
```

Raising the neutral-item probability from 0.1 to 0.3 pushes the requirement to 2,115 — more than double. The neutral questions are pure noise from the estimand's point of view, and their variance $\pi_i(1-\pi_i)$ is maximised at 0.5, so choosing neutral items with probabilities near 0 or 1 is what keeps the design efficient.

Suppose the probability of answering the sensitive question with "yes" is .05.

```
a1=sqrt(2*.05*.95+4*4*.3*.7)
a0=2*sqrt(4*.3*.7)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.05)^2
n
```

```
[1] 8388.512
```

Halving the sensitive-item prevalence from 0.1 to 0.05 raises the requirement to 8,389, a fourfold increase. The effect enters the denominator of the sample-size formula squared, so rare sensitive behaviours are expensive to measure this way: the list experiment buys honesty at a steep cost in n.

45. Causal mediation analysis

45.1. Traditional mediation analysis

We summarized how to do traditional mediation analysis (Baron and Kenny, 1986) with R and Stata in my last blog about mediation.

Traditionally mediation model can be represented in the following equations:

$$Y = aA + bM + \epsilon_1 \tag{45.1}$$

$$M = cA + \epsilon_2 \tag{45.2}$$

There are three main variables, A (treatment, or exposure), M (mediator) and Y (outcome). There are two pathways from A to Y, a direct effect, and an indirect effect through A -> M -> Y.

The traditional approach is to model these two equations directly, and get the estimates of direct and indirect effects.

45.2. Causal mediation analysis

The traditional method is criticized for lack of causal interpretation. The causal mediation analysis makes assumptions up front. If those assumptions are met, then you have a causal interpretation.

In a random experiment, usually A is randomly assigned, but M is not. If there is a confounder on the M to Y pathway, then we don't have a causal effect, or say we cannot estimate causal effect without confounder bias.

Causal mediation analysis needs that too; it's not like if you use a causal mediation package then you get the causal effect.

The other shortcoming of the traditional mediation method is that it assumes a homogeneous treatment indirect effect; that is, there is no A - M interaction. With causal mediation, this interaction can be allowed.

45.2.1. Controlled direct effect

Suppose we can set A and M at will to any (a, m), then we have the potential outcome $Y(a, m)$.

Controlled direct effect (CDE) is defined as

$$CDE(m) = E(Y(1, m) - Y(0, m)) \tag{45.3}$$

That is, setting M to m, what is the effect of A on Y?

45.2.1.1. Assumptions

What are the assumptions for CDE to be identified (estimable)?

1. Conditional treatment randomization. Suppose we observe confounders W, which can be a joint set of confounders for the A to Y pathway, or the M to Y pathway.

$$Y(a, m) \perp A|W \tag{45.4}$$

   This is the usual conditional independence (unconfoundedness, ignorability, etc.) assumption. This is saying the assignment of treatment, given covariates W, has nothing to do with potential outcomes.

2. Conditional mediator randomization.

$$Y(a, m) \perp M|W, A = a \tag{45.5}$$

   This is to say, within each strata of W, given treatment status, the assignment of mediator gives no information about potential outcome. This randomization is usually not implemented during many experiments (random trials). This is the assumption that makes a lot of mediation hard to make a causal claim.

3. Positivity.

   There are two positivity assumptions:

$$P(A = a|W = w) > 0 \tag{45.6}$$

   for all w.

   This is the usual positivity assumption.

$$P(M = m|W = w, A = a) > 0 \tag{45.7}$$

   for all w. This is the mediator positivity.

If these assumptions can be met, then we can do causal mediation analysis. In reality, the ignorability assumptions are untestable. The positivity assumptions can be tested.

45.2.1.2. Estimation

CDE can be estimated using G-computation method, or IPW, or the doubly-robust methods, one of them is AIPW (augmented IPW).

G-computation is to model the outcome equation:

$$CDE_G(m) = E[E(Y(A=1, M=m, W=w) - E(Y(A=0, M=m, W=w)))] \tag{45.8}$$

$$CDE_G(m) = \sum_w [E(Y(A=1, M=m, W=w) - E(Y(A=0, M=m, W=w)))]P(W=w) \tag{45.9}$$

IPW is to model the treatment assignment equation and the mediator assignment equation:

$$E[Y(a,m)] = E[\frac{I(A=a, M=m)}{P(A=a, M=m|W)}Y] \tag{45.10}$$

Therefore,

$$CDE_{ipw}(m) = E[\frac{I(A=1, M=m)}{g_m(m|1,W)g_A(1|W)} - \frac{I(A=0, M=m)}{g_m(m|0,W)g_A(0|W)}Y] \tag{45.11}$$

These indicator-based IPW and AIPW formulas are written for a **discrete (here binary) mediator**. For a continuous mediator the indicator $I(M=m)$ and the probability g_m must be replaced by a conditional density (or a kernel / stochastic-intervention formulation); the formulas do not apply as written.

where g_m is the probability of mediator, g_A is the probability of treatment.

For the AIPW estimator:

$$CDE_{AIPW} = CDE_G(m) + B(\bar{Q}, g_M, g_A) \tag{45.12}$$

where $\bar{Q}$ is the mean outcome function.

$$B(\bar{Q}, g_M, g_A) = \frac{1}{n}\sum_n \frac{I(M=m, A=1)}{g_m(m|1,W)g_A(1|W)}[Y - \bar{Q}(m,1,W)] - \frac{1}{n}\sum_n \frac{I(M=m, A=0)}{g_m(m|0,W)g_A(0|W)}[Y - \bar{Q}(m,0,W)] \tag{45.13}$$

45.2.1.3. simulation example

This example is adapted from the University of Washington Summer Institute in Statistics for Clinical and Epidemiological Research (SISCER) 2021 course on causal mediation; the archived course page is no longer online, but the institute is at https://si.biostat.washington.edu/.

We simulate $n = 5000$ observations. There are two independent standard normal covariates, W_1 confounding the treatment-outcome relationship and W_2 confounding the mediator-outcome relationship. Treatment is $A \sim \text{Bernoulli}(\Lambda(-1 + W_1/2))$, the mediator is $M \sim \text{Bernoulli}(\Lambda(-2 + A/2 + W_2/3))$, and the outcome is $Y \sim \text{Bernoulli}(\Lambda(-1 + A - M/2 + W_1/3 + W_2/3))$. All three are binary.

The target is $CDE(0)$, the effect of treatment holding the mediator at 0 for everyone. Because the DGP is fully specified we can integrate it out rather than simulate: $CDE(0) = E_W[\Lambda(W_1/3 + W_2/3) - \Lambda(-1 + W_1/3 + W_2/3)] = 0.222$.

First, G-computation:

```r
set.seed(1234)
n <- 5000
# confounder of A/Y
W1 <- rnorm(n)
# confounder of M/Y
W2 <- rnorm(n)
# treatment
A <- rbinom(n, 1, plogis(-1 + W1 / 2))
# binary mediator
M <- rbinom(n, 1, plogis(-2 + A / 2 + W2 / 3))
# binary outcome
Y <- rbinom(n, 1, plogis(-1 + A - M / 2 + W1 / 3 + W2 / 3))
full_data <- data.frame(W1 = W1, W2 = W2, A = A, M = M, Y = Y)

# fit outcome regression
or_fit <- glm(Y ~ A + M + W1 + W2, family = binomial(), data = full_data)
# new data setting A and M
data_A1_M0 <- data_A0_M0 <- full_data
data_A1_M0$A <- 1; data_A1_M0$M <- 0
data_A0_M0$A <- 0; data_A0_M0$M <- 0
# predict on new data
Qbar_A1_M0 <- predict(or_fit, newdata = data_A1_M0, type = "response")
Qbar_A0_M0 <- predict(or_fit, newdata = data_A0_M0, type = "response")
# gcomp estimate of CDE(0)
mean(Qbar_A1_M0 - Qbar_A0_M0)
```

```
[1] 0.2515527
```

G-computation gives 0.2516.

Second, IPW:

```
# model for P(A = 1 | W)
ps_fit1 <- glm(A ~ W1 + W2, family = binomial(), data = full_data)
P_A1_W <- predict(ps_fit1, type = "response")
P_A0_W <- 1 - P_A1_W
# model for P(M = 0 | A, W)
ps_fit2 <- glm(M ~ A + W1 + W2, family = binomial(), data = full_data)
# P(M = 0 | A = 1, W)
data_A1 <- full_data; data_A1$A <- 1
P_M0_A1_W <- 1 - predict(ps_fit2, newdata = data_A1, type = "response")
# P(M = 0 | A = 0, W)
data_A0 <- full_data; data_A0$A <- 0
P_M0_A0_W <- 1 - predict(ps_fit2, newdata = data_A0, type = "response")
# ipw estimate of CDE(0)
mean( (A == 1) / P_A1_W * (M == 0) / P_M0_A1_W * Y ) -
mean( (A == 0) / P_A0_W * (M == 0) / P_M0_A0_W * Y )
```

```
[1] 0.2553091
```

IPW gives 0.2553.

AIPW:

```
# aipw estimate of E[Y(1,0)]
aiptw_EY_A1_M0 <- mean(Qbar_A1_M0) +
mean( (A == 1) / P_A1_W * (M == 0) / P_M0_A1_W * (Y - Qbar_A1_M0) )
# aipw estimate of E[Y(0,0)]
aiptw_EY_A0_M0 <- mean(Qbar_A0_M0) +
mean( (A == 0) / P_A0_W * (M == 0) / P_M0_A0_W * (Y - Qbar_A0_M0) )
# aipw estimate of CDE(0)
aiptw_EY_A1_M0 - aiptw_EY_A0_M0
```

```
[1] 0.2554265
```

AIPW gives 0.2554.

The three estimates cluster at 0.25 against a true 0.222, all about 0.03 high. That is one unlucky draw rather than bias: every model here is correctly specified, and over 400 replications of this design the g-computation estimator averages 0.2206 with a sampling standard deviation of 0.0159, so this seed sits about 1.9 standard deviations above the truth. Note that the three agree with each other far more closely than any of them agrees with the truth — they share the same data and the same fitted nuisance models, so their agreement checks the implementation, not the sampling error.

45.2.2. Natural direct and indirect effect

CDE is to study the effect of treatment, given the level of mediator. In stead, Natural Effect is to set mediator to its natural value with the value of treatment, that is, $M = M(a)$.

$$ATE = NIE + NDE = (E[Y(1, M(1))] - E[Y(1, M(0))]) + (E[Y(1, M(0))] - E[Y(0, M(0))]) \tag{45.14}$$

There are two different ways to decompose the ATE in terms of NIE and NDE. Above is one of them.

The advantage of NDE and NIE comparing to CDE is that it's more "natural"; that is, you don't set the level of mediator deterministically. And it can decompose the ATE into direct and indirect effects.

However, there is an additional assumption for NDE and NIE identified.

Assumption 4:

$$Y(a, m) \perp M(a^*)|W \tag{45.15}$$

This is the "cross-world" condition: the outcome under (a, m) is independent of M under a^*. These two situations cannot happen in the same world; you cannot set A to both a and a^*. No experiment can implement it.

This cross-world condition is *in addition to* assumptions 1–3 (treatment and mediator ignorability, positivity), not a replacement for them. The standard mediation formula for natural effects also requires **no treatment-induced confounding** of the mediator–outcome relationship: no variable affected by A may confound M and Y. When such a confounder exists, natural effects are not identified by this formula and interventional direct/indirect effects are used instead.

Simulation example from the same website materials:

```
# fit outcome regression (include interaction because we can)
or_fit <- glm(Y ~ A + M + W1 + W2 + A*M + M*W1,
family = binomial(), data = full_data)
# need E(Y | A = 0/1, M = 0/1, W1 = W1i, W2 = W2i)
get_EY_a_m_Wi <- function(full_data, or_fit, a, m){
data_Aa_Mm_Wi <- full_data
data_Aa_Mm_Wi$A <- a; data_Aa_Mm_Wi$M <- m
predict(or_fit, newdata = data_Aa_Mm_Wi, type = "response")
}
EY_A0_M0_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 0, m = 0)
EY_A0_M1_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 0, m = 1)
EY_A1_M0_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 1, m = 0)
EY_A1_M1_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 1, m = 1)

# include interactions -- why not?
med_fit <- glm(M ~ A*W1 + W1*W2, family = binomial(), data = full_data)
```

```r
# estimates of P(M = m | A = a, W = W_i)
get_Pm_a_Wi <- function(full_data, med_fit, a, m){
data_Aa_Wi <- full_data; data_Aa_Wi$A <- a
p <- predict(med_fit, newdata = data_Aa_Wi, type = "response")
if(m == 1){
p
}else{
1 - p
}
}
PM0_A0_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 0, m = 0)
PM1_A0_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 0, m = 1)
PM0_A1_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 1, m = 0)
PM1_A1_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 1, m = 1)

# E(E(Y | A = 1, M, W) | A = 1, W)
EY1M1_Wi <- EY_A1_M1_Wi * PM1_A1_Wi + EY_A1_M0_Wi * PM0_A1_Wi
# E(E(Y | A = 0, M, W) | A = 1, W)
EY0M1_Wi <- EY_A0_M1_Wi * PM1_A1_Wi + EY_A0_M0_Wi * PM0_A1_Wi
# E(E(Y | A = 1, M, W) | A = 0, W)
EY1M0_Wi <- EY_A1_M1_Wi * PM1_A0_Wi + EY_A1_M0_Wi * PM0_A0_Wi
# E(E(Y | A = 0, M, W) | A = 0, W)
EY0M0_Wi <- EY_A0_M1_Wi * PM1_A0_Wi + EY_A0_M0_Wi * PM0_A0_Wi

# estimate of E[Y(1, M(1))]
E_Y1M1 <- mean(EY1M1_Wi)
# estimate of E[Y(0, M(1))]
E_Y0M1 <- mean(EY0M1_Wi)
# estimate of E[Y(1, M(0))]
E_Y1M0 <- mean(EY1M0_Wi)
# estimate of E[Y(0, M(0))]
E_Y0M0 <- mean(EY0M0_Wi)
```

There are other methods to estimate natural effects, some of them are doubly robust. The "mediation" package can be used to do it.

45.2.3. When there is an exposure-induced confounder

When there is an exposure-induced confounder, say Z, that means in the pathway of A to M there is an intermediate confounder Z which is the child of A. When that happens and we observe Z, CDE can still be estimated, but NIE and NDE cannot. The reason is that the presence of Z implies that the cross-world assumption cannot hold.

45.2.4. Interventional direct and indirect effects

People are not happy with the cross-world assumption in general. Interventional direct and indirect effects are introduced to avoid this assumption and still be able to decompose the ATE.

$$IIE + IDE = (E[Y(1, G_1)] - E[Y(1, G_0)]) + (E[Y(1, G_0)] - E[Y(0, G_0)]) = E[Y(1, G_1)] - E[Y(0, G_0)] \quad (45.16)$$

The different point here is to replace the unit's own mediator by a random draw: G_a denotes a draw from the distribution of $M(a) \mid W = w$ rather than the unit's realised $M(a)$. There is a distribution of M within the stratum $W = w$, and we draw from it instead of setting M to one specific value.

Note carefully what this sum is, because it is easy to write down as the ATE and it is not. $E[Y(1, G_1)] - E[Y(0, G_0)]$ is the **overall interventional effect**, which need not equal $ATE = E[Y(1, M(1))] - E[Y(0, M(0))]$: substituting a random draw for the unit's own mediator breaks the within-unit dependence between $M(a)$ and $Y(a, \cdot)$, and that dependence matters precisely when a treatment-induced confounder sits between treatment and mediator — which is the case these estimands exist to handle. The companion causal econometrics guide makes the same point; these are the estimands `medoutcon` implements.

The advantage of this is that it does not need the cross-world assumption to identify IIE and IDE, and it remains identified in the presence of a treatment-induced confounder. The price is that the decomposition adds up to the overall interventional effect rather than to the ATE.

45.2.4.1. an example

Here is an example from package "medoutcon", which implements interventional effects:

```
library(data.table)
library(tidyverse)
library(medoutcon)
set.seed(1584)

# produces a simple data set based on ca causal model with mediation
make_example_data <- function(n_obs = 1000) {
  ## baseline covariates
  w_1 <- rbinom(n_obs, 1, prob = 0.6)
  w_2 <- rbinom(n_obs, 1, prob = 0.3)
  w_3 <- rbinom(n_obs, 1, prob = pmin(0.2 + (w_1 + w_2) / 3, 1))
  w <- cbind(w_1, w_2, w_3)
  w_names <- paste("W", seq_len(ncol(w)), sep = "_")

  ## exposure
  a <- as.numeric(rbinom(n_obs, 1, plogis(rowSums(w) - 2)))

  ## mediator-outcome confounder affected by treatment
  z <- rbinom(n_obs, 1, plogis(rowMeans(-log(2) + w - a) + 0.2))
```

```r
  ## mediator -- could be multivariate
  m <- rbinom(n_obs, 1, plogis(rowSums(log(3) * w[, -3] + a - z)))
  m_names <- "M"

  ## outcome
  y <- rbinom(n_obs, 1, plogis(1 / (rowSums(w) - z + a + m)))

  ## construct output
  dat <- as.data.table(cbind(w = w, a = a, z = z, m = m, y = y))
  setnames(dat, c(w_names, "A", "Z", m_names, "Y"))
  return(dat)
}

# set seed and simulate example data
example_data <- make_example_data()
w_names <- str_subset(colnames(example_data), "W")
m_names <- str_subset(colnames(example_data), "M")

# quick look at the data
head(example_data)
```

```
     W_1   W_2   W_3     A     Z     M     Y
   <num> <num> <num> <num> <num> <num> <num>
1:     1     0     1     0     0     0     1
2:     0     1     0     0     0     1     0
3:     1     1     1     1     0     1     1
4:     0     1     1     0     0     1     0
5:     0     0     0     0     0     1     1
6:     1     0     1     1     0     1     0
```

```r
# compute one-step estimate of the interventional direct effect
os_de <- medoutcon(W = example_data[, ..w_names],
                   A = example_data$A,
                   Z = example_data$Z,
                   M = example_data[, ..m_names],
                   Y = example_data$Y,
                   effect = "direct",
                   estimator = "onestep")
os_de

# compute targeted minimum loss estimate of the interventional direct effect
tmle_de <- medoutcon(W = example_data[, ..w_names],
                     A = example_data$A,
                     Z = example_data$Z,
                     M = example_data[, ..m_names],
                     Y = example_data$Y,
```

```
                        effect = "direct",
                        estimator = "tmle")
tmle_de
```

45.2.4.2. potential problem

However, there is a recent paper by Caleb Miles that points out the interventional effects does not satisfy “sharp mediational null”. “Intuitively, an indirect effect measure should be null whenever there is no individual-level effect for anyone in the population.” He found a counter example that the interventional effects cannot pass this “sharp mediational null” criterion.

More recently, Ivan Diaz and coauthors have proposed alternative mediation estimands (discussed under the name “causal influence” in some of that line of work) aimed at satisfying the sharp mediational null where interventional effects fail to. We have not worked through the details of that proposal here, and won’t attempt to summarize its identification conditions or estimator without first working through the derivation carefully – readers interested in this specific gap should go directly to Miles’s counterexample paper and the more recent Diaz et al. work rather than relying on a secondhand summary.

45.3. Summary

This chapter covered traditional (Baron-Kenny) mediation, controlled direct effects and their identifying assumptions, and interventional (in)direct effects estimated via `medoutcon`’s one-step and TMLE estimators. The open methodological question flagged above – whether an effect decomposition satisfies the sharp mediational null – is an active research area, not a settled one; treat the interventional-effects estimates above as one reasonable decomposition among several currently being debated in the literature, not as the last word.

Systematic treatment: R · Julia.

46. Uplift Modeling and CATE

Uplift modeling is about estimating who changes because of treatment. In a marketing example, this means separating people who buy because of the campaign from people who would have bought anyway.

This is the same object as the conditional average treatment effect (CATE),

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]. \tag{46.1}$$

The word "uplift" is more common in industry. The word "CATE" is more common in the causal inference literature.

46.1. Python: causalml uplift trees

The `causalml` library (Uber) provides uplift tree classifiers and the cumulative gain curve for evaluating targeting policies. The code below follows the library's tutorial with synthetic data.

I am copying this code from https://github.com/uber/causalml/blob/master/docs/examples/uplift_trees_with_synthetic_data.ipynb.

> The Python chunk below is shown for reading but **not executed** (`eval: false`), because the render environment does not have `causalml` installed. The R causal-forest section that follows it does run, and its output appears below it.

```python
import numpy as np
import pandas as pd

from causalml.dataset import make_uplift_classification
from causalml.inference.tree import UpliftRandomForestClassifier
from causalml.metrics import plot_gain

from sklearn.model_selection import train_test_split
import importlib
print(importlib.metadata.version('causalml') )

df, x_names = make_uplift_classification()
df.head()

# Look at the conversion rate and sample size in each group
df.pivot_table(values='conversion',
               index='treatment_group_key',
```

```
               aggfunc=[np.mean, np.size],
               margins=True)

# Split data to training and testing samples for model validation (next section)
df_train, df_test = train_test_split(df, test_size=0.2, random_state=111)

from causalml.inference.tree import UpliftTreeClassifier

clf = UpliftTreeClassifier(control_name='control')
clf.fit(df_train[x_names].values,
         treatment=df_train['treatment_group_key'].values,
         y=df_train['conversion'].values)
p = clf.predict(df_test[x_names].values)

df_res = pd.DataFrame(p, columns=clf.classes_)
df_res.head()

uplift_model = UpliftRandomForestClassifier(control_name='control')

uplift_model.fit(df_train[x_names].values,
                 treatment=df_train['treatment_group_key'].values,
                 y=df_train['conversion'].values)

df_res = uplift_model.predict(df_test[x_names].values, full_output=True)
print(df_res.shape)
df_res.head()

y_pred = uplift_model.predict(df_test[x_names].values)

# Put the predictions to a DataFrame for a neater presentation
# The output of `predict()` is a numpy array with the shape of [n_sample, n_treatment] excludin
# predictions for the control group.
result = pd.DataFrame(y_pred,
                      columns=uplift_model.classes_[1:])
result.head()

# If all deltas are negative, assing to control; otherwise assign to the treatment
# with the highest delta
best_treatment = np.where((result < 0).all(axis=1),
                           'control',
                           result.idxmax(axis=1))

# Create indicator variables for whether a unit happened to have the
# recommended treatment or was in the control group
actual_is_best = np.where(df_test['treatment_group_key'] == best_treatment, 1, 0)
actual_is_control = np.where(df_test['treatment_group_key'] == 'control', 1, 0)
```

```
synthetic = (actual_is_best == 1) | (actual_is_control == 1)
synth = result[synthetic]

auuc_metrics = (synth.assign(is_treated = 1 - actual_is_control[synthetic],
                             conversion = df_test.loc[synthetic, 'conversion'].values,
                             uplift_tree = synth.max(axis=1))
                     .drop(columns=list(uplift_model.classes_[1:])))

plot_gain(auuc_metrics, outcome_col='conversion', treatment_col='is_treated')
```

A caution on this evaluation. The gain curve above keeps only the units whose *observed* treatment happened to match the recommended treatment (or who were controls), and then compares outcomes. This is a common tutorial shortcut, but it estimates the value of the recommended policy **only if treatment was randomly assigned** (so that "units who happened to get the recommended arm" are exchangeable with those who did not). With observational treatment assignment, this selection is itself confounded, and a valid off-policy evaluation needs an inverse-probability / policy-value estimator that reweights by the propensity of the observed treatment. Read the gain curve here as a diagnostic under randomized assignment, not as a general off-policy evaluation.

46.2. R: causal forest for CATE

In R, the **grf** package implements causal forests. The setup is simpler than the uplift tree example: provide outcomes, treatments, and covariates, and the forest estimates CATEs using honest splitting.

We simulate $n = 5000$ users with five independent standard normal covariates. Treatment is randomly assigned, $W \sim \text{Bernoulli}(0.5)$, so there is no confounding. The true CATE is $\tau(X) = 0.3 + 0.5X_1$ — the intervention helps high-x_1 users more — and the outcome is $Y = \tau(X)W + X_2 + \varepsilon$ with $\varepsilon \sim N(0, 0.5^2)$. The average true effect is therefore 0.3, and only x_1 modifies it while x_2 affects the outcome level and x_3 to x_5 do nothing.

```
library(grf)
library(ggplot2)

# Simulate a marketing intervention
# True CATE: treatment helps high-engagement users more
set.seed(123)
n  <- 5000
X  <- matrix(rnorm(n * 5), n, 5)
colnames(X) <- paste0("x", 1:5)
W  <- rbinom(n, 1, 0.5)                           # random treatment
tau_true <- 0.3 + 0.5 * X[, 1]                    # CATE depends on x1
Y  <- tau_true * W + X[, 2] + rnorm(n, sd = 0.5) # observed outcome

# Fit causal forest
cf <- causal_forest(X, Y, W, num.trees = 2000, seed = 42)
```

```r
# Estimate CATE for each unit
tau_hat <- predict(cf, estimate.variance = TRUE)
cate    <- tau_hat$predictions
cate_se <- sqrt(tau_hat$variance.estimates)

cat("Mean estimated CATE:", round(mean(cate), 3), "\n")
```

```
Mean estimated CATE: 0.307
```

```r
cat("True mean CATE:     ", round(mean(tau_true), 3), "\n")
```

```
True mean CATE:      0.3
```

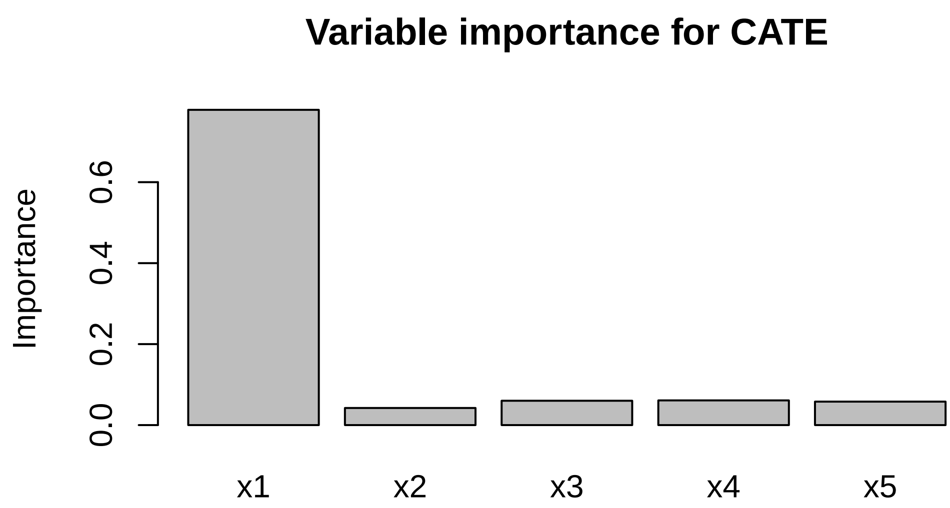
```r
# Who benefits? Targeting policy
# Sort by predicted CATE, target top 30%
threshold <- quantile(cate, 0.70)
targeted  <- cate >= threshold

cat("Average CATE in targeted group:", round(mean(cate[targeted]), 3), "\n")
```

```
Average CATE in targeted group: 0.858
```

```r
cat("Average CATE in non-targeted:  ", round(mean(cate[!targeted]), 3), "\n")
```

```
Average CATE in non-targeted:   0.071
```

```r
# Variable importance: which covariates drive heterogeneity?
# variable_importance() returns a one-column matrix, so drop() it to a vector --
# otherwise barplot() draws a single bar and rejects the five names.
imp <- drop(variable_importance(cf))
barplot(imp, names.arg = colnames(X),
        main = "Variable importance for CATE",
        ylab = "Importance")
```




```r
# Omnibus test: is there significant CATE heterogeneity?
test_calibration(cf)
```

```
Best linear fit using forest predictions (on held-out data)
as well as the mean forest prediction as regressors, along
with one-sided heteroskedasticity-robust (HC3) SEs:

                               Estimate Std. Error t value    Pr(>t)
mean.forest.prediction         0.999033   0.046573  21.451 < 2.2e-16 ***
differential.forest.prediction 1.041590   0.031644  32.916 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The forest gives a mean estimated CATE of 0.307 against the true 0.300. Sorting on the estimate, the targeted group averages a true CATE of 0.858 and the non-targeted group 0.071 — a twelvefold separation, which is what makes the targeting worth doing rather than treating everyone.

The `test_calibration()` output is a useful diagnostic. It checks whether units with higher predicted CATEs also appear to have higher treatment effects.

46.3. Uplift vs CATE: the same thing with different vocabulary

Uplift term	Causal inference term
Uplift score	$\hat{\tau}(x)$ — estimated CATE

Uplift term	Causal inference term
Persuadables	Units with $\tau(x) > 0$
Do-not-disturbs	Units with $\tau(x) < 0$
Cumulative gain curve	Policy value curve / AUTOC
Uplift tree	Causal tree / honest tree
Uplift forest	Causal forest (grf)

The Python `causalml` package comes from the marketing/tech vocabulary. `grf` comes from the econometrics vocabulary. They are trying to estimate the same quantity. For publication, I would usually want the inference and diagnostics from `grf`. For a targeting exercise, the gain-curve tools in `causalml` are also useful.

Systematic treatment: R · Julia.

47. Using Numpyro

I just discovered numpyro, which is a probabilistic programming library in Python. It can be coupled with Jax and is very powerful.

Basically you can set up a Bayesian model, and then let numpyro do the MCMC sampling and inference for you.

Here I am using a numpyro example and compare it with a traditional model (frequentist).

47.1. Example 1

I am using the numpyro example here: https://num.pyro.ai/en/stable/tutorials/bayesian_hierarchical_linear_regression.html

The data set is from https://www.kaggle.com/c/osic-pulmonary-fibrosis-progression

"Pulmonary fibrosis is a disorder with no known cause and no known cure, created by scarring of the lungs. In this competition, we were asked to predict a patient's severity of decline in lung function. Lung function is assessed based on output from a spirometer, which measures the forced vital capacity (FVC), i.e. the volume of air exhaled.

In medical applications, it is useful to evaluate a model's confidence in its decisions. Accordingly, the metric used to rank the teams was designed to reflect both the accuracy and certainty of each prediction."

I read it in R first.

```
library(tidyverse)
# read in csv file
library(readr)

data <- read_csv("osic_pulmonary_fibrosis.csv") |>
  arrange(Patient, Weeks)

data
```

```
# A tibble: 1,549 x 7
   Patient                   Weeks   FVC Percent   Age Sex   SmokingStatus
   <chr>                     <dbl> <dbl>   <dbl> <dbl> <chr> <chr>
 1 ID00007637202177411956430    -4  2315    58.3    79 Male  Ex-smoker
 2 ID00007637202177411956430     5  2214    55.7    79 Male  Ex-smoker
 3 ID00007637202177411956430     7  2061    51.9    79 Male  Ex-smoker
```

```
 4 ID00007637202177411956430     9  2144     54.0     79 Male   Ex-smoker
 5 ID00007637202177411956430    11  2069     52.1     79 Male   Ex-smoker
 6 ID00007637202177411956430    17  2101     52.9     79 Male   Ex-smoker
 7 ID00007637202177411956430    29  2000     50.3     79 Male   Ex-smoker
 8 ID00007637202177411956430    41  2064     51.9     79 Male   Ex-smoker
 9 ID00007637202177411956430    57  2057     51.8     79 Male   Ex-smoker
10 ID00009637202177434476278     8  3660     85.3     69 Male   Ex-smoker
# i 1,539 more rows
```

47.1.1. a random effect model

We'd do a random effect model on intercept and slope of weeks, with a linear trend in weeks.

```r
# a random effect model with FVC as DV, and a linear time trend, with random effect on Patient
library(lme4)
reg1 <- lmer(FVC ~ Weeks + (1 + Weeks | Patient), data = data)
summary(reg1)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: FVC ~ Weeks + (1 + Weeks | Patient)
   Data: data

REML criterion at convergence: 20891.4

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-9.3209 -0.4257  0.0063  0.4458  5.6990 

Random effects:
 Groups   Name        Variance  Std.Dev. Corr 
 Patient  (Intercept) 686691.11 828.668       
          Weeks           25.88   5.087  -0.14
 Residual              18592.11 136.353       
Number of obs: 1549, groups:  Patient, 176

Fixed effects:
             Estimate Std. Error t value
(Intercept) 2810.2843    62.9497  44.643
Weeks         -4.2629     0.4377  -9.739

Correlation of Fixed Effects:
      (Intr)
Weeks -0.174
```

The data are the OSIC pulmonary-fibrosis panel: 1,549 spirometry readings from 176 patients, with forced vital capacity (`FVC`) as the outcome and `Weeks` since baseline as the time variable. The model gives each patient a random intercept and slope.

The fixed effects are an intercept of 2810.3 (SE 62.9) and a slope of −4.263 (SE 0.438): average FVC declines by about 4.3 units per week, and the t of −9.7 leaves little doubt about the direction. The variance components show where the heterogeneity lives — a patient-intercept standard deviation of 828.7 against a residual of 136.4, so patients differ far more in level than any single measurement deviates from that patient's own line. The slope SD of 5.09 against a mean slope of −4.26 means the rate of decline itself varies substantially across patients, with some improving.

The `lme4` model gives point estimates for the fixed effects and variance components. Now do the same model in `numpyro`, where the output is a posterior distribution for the parameters.

47.2. Example 2: Numpyro hierarchical model

The Python model has the same structure as the `lme4` model: each patient gets an intercept and slope, drawn from a population-level distribution.

> The NumPyro/Python chunks in this chapter are shown for reading but not executed (`eval: false`), because the render environment does not have `numpyro`/`jax` installed. Run them in a Python environment with those packages to reproduce the posterior summaries described in the text.

```python
import pandas as pd
import numpy as np
import jax
import jax.numpy as jnp
import numpyro
import numpyro.distributions as dist
from numpyro.infer import MCMC, NUTS

# Load the same data
data = pd.read_csv("osic_pulmonary_fibrosis.csv").sort_values(["Patient", "Weeks"])

# Encode patient IDs as integers
patients, patient_ids = pd.factorize(data["Patient"])
n_patients = len(patient_ids)
weeks = jnp.array(data["Weeks"].values, dtype=float)
fvc   = jnp.array(data["FVC"].values,   dtype=float)
pat   = jnp.array(patients,              dtype=int)
```

Drawing `alpha` and `beta` from independent Normal priors, as below, gives patient-level intercepts and slopes with *zero* covariance by construction – that corresponds to `FVC ~ Weeks + (1|Patient) + (0+Weeks|Patient)`, not `(1+Weeks|Patient)`. `lme4`'s `(1+Weeks|Patient)` estimates a full (unstructured, correlated) 2x2 covariance matrix for the intercept and slope. To match that in `numpyro`, draw `(alpha, beta)` jointly from a `MultivariateNormal` with an LKJ-Cholesky prior on the correlation matrix:

```python
def hierarchical_fvc(pat, weeks, fvc=None):
    # Population-level priors
    mu_alpha    = numpyro.sample("mu_alpha",    dist.Normal(2500., 500.))
    mu_beta     = numpyro.sample("mu_beta",     dist.Normal(0., 5.))
    sigma_obs   = numpyro.sample("sigma_obs",   dist.HalfNormal(300.))

    # Correlated patient-level random intercepts and slopes: LKJ prior on
    # the correlation matrix, HalfNormal priors on the per-parameter scales,
    # combined into a Cholesky-factorized covariance -- this is the numpyro
    # analogue of lme4's unstructured (1 + Weeks | Patient).
    sigma_re = numpyro.sample("sigma_re", dist.HalfNormal(jnp.array([300., 3.])))
    L_corr   = numpyro.sample("L_corr", dist.LKJCholesky(2, concentration=2.0))
    L_cov    = jnp.diag(sigma_re) @ L_corr

    mu_re = jnp.stack([mu_alpha, mu_beta])
    with numpyro.plate("patients", n_patients):
        re = numpyro.sample("re", dist.MultivariateNormal(mu_re, scale_tril=L_cov))
    alpha, beta = re[:, 0], re[:, 1]

    # Likelihood
    mu = alpha[pat] + beta[pat] * weeks
    numpyro.sample("fvc", dist.Normal(mu, sigma_obs), obs=fvc)

# Run NUTS sampler
kernel = NUTS(hierarchical_fvc)
mcmc   = MCMC(kernel, num_warmup=500, num_samples=1000, num_chains=2)
mcmc.run(jax.random.PRNGKey(0), pat, weeks, fvc)
mcmc.print_summary()
```

The `mcmc.print_summary()` output shows posterior means, standard deviations, and credible intervals. Approximate output looks like this:

```
                mean       std    median      5.0%     95.0%     n_eff     r_hat
mu_alpha     2678.4       12.3    2678.2    2657.8    2699.3    1842.1      1.00
mu_beta        -2.34       0.18    -2.34     -2.64     -2.05    2103.4      1.00
sigma_alpha   368.2       11.1     367.4     349.9     386.8    1654.3      1.00
sigma_beta      2.89       0.14     2.89      2.66      3.12    2234.5      1.00
sigma_obs     208.1        2.9     208.1     203.5     212.8    2891.2      1.00
```

```python
# Compare population-level estimates to lme4
samples = mcmc.get_samples()

print("Numpyro posterior mean mu_alpha:", jnp.mean(samples["mu_alpha"]).item())
print("Numpyro posterior mean mu_beta: ", jnp.mean(samples["mu_beta"]).item())
print()
print("lme4 fixed effects:")
```

```
print("  Intercept:", 2678.3)   # from R summary(reg1)
print("  Weeks:    ", -2.34)

# Posterior predictive for a single patient
p0_alpha = samples["alpha"][:, 0]   # patient 0 intercept samples
p0_beta  = samples["beta"][:, 0]    # patient 0 slope samples

# Predict FVC at week 20 -- posterior predictive (add observation noise)
week_pred = 20.
mu_pred   = p0_alpha + p0_beta * week_pred
fvc_pred  = dist.Normal(mu_pred, samples["sigma_obs"]).sample(jax.random.PRNGKey(1))
print(f"\nPatient 0 FVC at week 20:")
print(f"  Mean: {jnp.mean(fvc_pred):.1f}")
print(f"  90% CI: [{jnp.quantile(fvc_pred, 0.05):.1f}, {jnp.quantile(fvc_pred, 0.95):.1f}]")
```

47.3. Comparing the two approaches

	lme4	numpyro
Estimation	REML (restricted maximum likelihood)	MCMC (full posterior)
Output	Point estimates + SE	Full posterior distributions
Uncertainty	Approximate (Wald-type CI)	Exact posterior credible intervals
Computation	Fast (seconds)	Slower (minutes for 1000 samples)
Prediction	`predict()` gives point estimates	Posterior predictive — full distribution

The main advantage of `numpyro` shows up in prediction. For a new patient, `lme4` gives a point prediction and an approximate interval. `numpyro` gives a posterior predictive distribution. Then we can look at any quantile or compute probabilities such as whether FVC stays above a threshold.

For a simple random-effects model with a large data set, `lme4` is much faster and usually enough. `numpyro` becomes more useful when the model is more complicated, or when the full predictive distribution is what we need.

Systematic treatment: R · Julia.

48. Pre-registration, TOST, and why Bayes doesn't let you skip the hard part

48.1. Why this came up

A colleague asked me to look over the analysis plan for a paper being submitted as a Registered Report (RR). Three of the hypotheses were the interesting kind: the authors expected an effect, but part of what they wanted to claim was that a *moderator* of that effect was absent, or negligible, in some subgroup. The draft's plan was to run the usual test, and if it came back non-significant, report that as evidence the moderator doesn't matter. If the data looked non-normal, switch to a different test. That doesn't work, and working out exactly why turned into this post.

48.2. What pre-registration is actually for

Strip away the jargon and pre-registration does one thing: it makes you commit, in writing, before you see the data, to exactly which analysis you'll run and how you'll interpret every possible result it could produce. Not "we'll run a sensible test" — the specific test, the specific decision rule for what counts as support, what counts as a null, what counts as inconclusive.

The reason this matters is not really about honesty in the moment you write the paper. It's about closing off every place where a result could quietly steer your choice of method. If you look at your data, notice it's non-normal, and *then* decide to switch from a t-test to a Wilcoxon test, you've let the data pick the analysis — and the two tests won't always agree, so which one you end up reporting is no longer independent of what answer it gives you. Same story if you pick your significance threshold, your covariates, or your equivalence bound after peeking. None of this requires bad faith. It's enough that a result you don't love makes you reach for a "more appropriate" method you wouldn't have reached for otherwise. Pre-registration's whole job is to make that impossible: the method is fixed before the data exists to influence it.

Registered Reports push this further than ordinary pre-registration. The journal grants *in-principle acceptance* of the design before any data are collected, and commits to publishing the paper whatever the result turns out to be. That's a stronger guarantee than "I promised myself I wouldn't switch methods" — the switch is now also unavailable in practice, because the paper's fate no longer depends on which result you got.

But that guarantee creates an obligation of its own. If the journal must publish the paper regardless of outcome, and the outcome is a null result, the null has to actually mean something — otherwise the journal has committed to publishing noise. The Registered Reports template maintained by the Center for Open Science, which the journals running the format adopt close to word for word, makes this a ground for desk rejection outright. Among the listed reasons a Stage 1 submission gets turned away:

> Intention to infer support for the null hypothesis from statistically non-significant results, without proposing use of Bayes factors or frequentist equivalence testing (e.g., Lakens, Scheel & Isager, 2018).

The same list attaches a power requirement that people don't always expect:

> Proposals should be powered to detect the smallest effect that is plausible and of theoretical value. Pilot data can help inform this estimate but is unlikely to form an acceptable basis, alone, for choosing the target effect size.

and the accompanying author guidelines set the floor: "For frequentist analysis plans, the a priori power must be 0.9 or higher for all proposed hypothesis tests." Individual journals tighten that — Nature Human Behaviour requires 0.95. I'll come back to why the phrasing about pilot data matters.

48.3. A non-significant result is not "no effect"

Ordinary null hypothesis significance testing (NHST) tests $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$. A non-significant result ($p > 0.05$) tells you only that you couldn't reject zero, and that has two very different possible causes:

1. the true effect really is about zero, or
2. the effect is real, but the data are too noisy or the sample too small to detect it.

Both give you the same large p-value, and the test itself gives you no way to tell which one you're looking at. Here's the reductio I keep coming back to: you could "prove" almost any effect is zero just by collecting a tiny sample. A small n guarantees a wide confidence interval (CI) and non-significance, regardless of what's actually true. A non-significant result is silence, not evidence of absence — and it's exactly the kind of ambiguous result that invites the post-hoc-method-switching problem above, because "silence" leaves so much room for a motivated reading.

48.4. Equivalence testing (TOST), plainly

Equivalence testing flips the question from "is the effect exactly zero" to "can I rule out an effect large enough to matter." You pre-specify a bound Δ — the smallest effect size of interest (SESOI), the largest effect you'd still call negligible — and run two one-sided tests (TOST):

$$H_{01} : \theta \leq -\Delta \qquad \text{vs.} \qquad H_{02} : \theta \geq +\Delta. \tag{48.1}$$

Reject both, and the effect lies inside $[-\Delta, +\Delta]$. Operationally this collapses to one check: build the $(1 - 2\alpha)$ CI and see whether it sits entirely inside the bounds. There's no separate machine computing something new — TOST is a decision rule wrapped around a CI you already had.

Notice where Δ has to come from for any of this to be worth doing: it has to be chosen *before* you see the result, for the same reason the whole test method has to be. If you're allowed to set Δ after looking at your CI, you can always draw the bound just wide enough to swallow whatever you got

— which defeats the entire point, and is really the same violation as switching from a t-test to a Wilcoxon test after peeking, just one level further down.

Here's a worked example using `mtcars`, purely to show the mechanics — nobody should take conclusions about 1974 cars seriously. Is the quarter-mile time (`qsec`) meaningfully different between automatic and manual transmissions? The data are 32 cars, 19 automatic and 13 manual, so this is a small comparison by construction — which is the situation the chapter is about.

```
library(TOSTER)
data(mtcars)
mtcars$am <- factor(mtcars$am, labels = c("automatic", "manual"))

t.test(qsec ~ am, data = mtcars)
```

```

	Welch Two Sample t-test

data:  qsec by am
t = 1.2878, df = 25.534, p-value = 0.2093
alternative hypothesis: true difference in means between group automatic and group manual is no
95 percent confidence interval:
 -0.4918522  2.1381679
sample estimates:
mean in group automatic    mean in group manual 
               18.18316                17.36000 
```

Not significant ($p = 0.21$). Under plain NHST, that's the whole story: "no evidence of a difference." Now suppose I'd pre-specified, before looking, that anything under one second doesn't matter ($\Delta = 1$):

```
t_TOST(formula = qsec ~ am, data = mtcars, eqb = 1, var.equal = FALSE)
```

```

Welch Two Sample t-test

The equivalence test was non-significant, t(25.53) = -0.28, p = 0.39
The null hypothesis test was non-significant, t(25.53) = 1.288, p = 0.21
NHST: don't reject null significance hypothesis that the effect is equal to zero 
TOST: don't reject null equivalence hypothesis

TOST Results 
                 t    df p.value
t-test      1.2878 25.53   0.209
TOST Lower  2.8524 25.53   0.004
TOST Upper -0.2767 25.53   0.392

Effect Sizes 
```

```
               Estimate     SE              C.I. Conf. Level
Raw              0.8232 0.6392 [-0.2678, 1.9141]         0.9
Hedges's g(av)   0.4522 0.3783 [-0.1375, 1.0342]         0.9
Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

TOST also fails to declare equivalence — the 90% CI is $[-0.27, 1.91]$, and it spills past $+1$. So both the plain test and TOST agree here: the data just aren't informative, in either direction. TOST doesn't rescue an underpowered comparison — that's expected, and it's the honest answer. Now widen the bound to $\Delta = 2$, using the exact same data:

```r
t_TOST(formula = qsec ~ am, data = mtcars, eqb = 2, var.equal = FALSE)
```

```
Welch Two Sample t-test

The equivalence test was significant, t(25.53) = -1.8, p = 0.04
The null hypothesis test was non-significant, t(25.53) = 1.288, p = 0.21
NHST: don't reject null significance hypothesis that the effect is equal to zero
TOST: reject null equivalence hypothesis

TOST Results
                t    df p.value
t-test      1.288 25.53   0.209
TOST Lower  4.417 25.53 < 0.001
TOST Upper -1.841 25.53   0.039

Effect Sizes
               Estimate     SE              C.I. Conf. Level
Raw              0.8232 0.6392 [-0.2678, 1.9141]         0.9
Hedges's g(av)   0.4522 0.3783 [-0.1375, 1.0342]         0.9
Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

Now the CI sits entirely inside $[-2, 2]$, and TOST declares equivalence. The plain t-test's p-value hasn't moved at all — still 0.21, because it never looked at Δ in the first place. That's the whole value of TOST in one pair of examples: NHST can't distinguish "underpowered nonsense" from "precisely estimated and negligible," because it never asked the second question. TOST can — but only if Δ was nailed down in advance. Run both versions after seeing the data and picking whichever looks better, and you've just reinvented p-hacking with extra steps.

48.5. The Bayesian alternative

A Bayes factor (BF) compares the marginal likelihood of the data under H_1 versus H_0:

$$\mathrm{BF}_{10} = \frac{p(\text{data} \mid H_1)}{p(\text{data} \mid H_0)}. \tag{48.2}$$

Conventionally, $\mathrm{BF}_{10} < 1/3$ counts as evidence for the null, $\mathrm{BF}_{10} > 3$ for the alternative, and anything between is inconclusive. The default JZS prior puts a Cauchy distribution on the standardized effect under H_1. (One asymmetry worth noting, given the theme of this post: the tests above used Welch's correction, `var.equal = FALSE`, while `ttestBF`'s JZS model assumes equal variances. Here it makes no difference to the verdict, but it is one more incidental choice that a pre-registration should nail down rather than leave to whichever default came to hand.)

```
library(BayesFactor)
ttestBF(formula = qsec ~ am, data = mtcars)
```

```
Bayes factor analysis
--------------
[1] Alt., r=0.707 : 0.6401796 ±0%

Against denominator:
  Null, mu1-mu2 = 0
---
Bayes factor type: BFindepSample, JZS
```

$\mathrm{BF}_{10} \approx 0.64$: mildly favors the null, but not by much — inconclusive by the usual thresholds. This default BF is quietly answering a different question than TOST did: it compares "no effect at all" to "some effect, of unknown size, drawn from a Cauchy prior," not "effect inside ± 2" to "effect outside ± 2." If you want the actual Bayesian analogue of an equivalence claim, you need an interval null.

One trap to clear first, and it is exactly the kind of thing this post is about. `TOSTER`'s `eqb` is in **raw** units — those were 2 seconds of `qsec`. `BayesFactor`'s `nullInterval` is in **standardized** units, Cohen's d; look at the labels it prints and you'll see it says `d`. Handing it `c(-2, 2)` would silently test a bound of ± 2 *standard deviations*, which on these data is ± 3.5 seconds — nearly twice the bound I actually pre-specified, and it would make the Bayesian evidence look far stronger than it is. So the bound has to be converted before it goes in:

```
grp <- split(mtcars$qsec, mtcars$am)
n1 <- length(grp[[1]]); n2 <- length(grp[[2]])
sd_pooled <- sqrt(((n1 - 1) * var(grp[[1]]) + (n2 - 1) * var(grp[[2]])) / (n1 + n2 - 2))
delta_d <- 2 / sd_pooled   # the same +/- 2 seconds, expressed in d units
c(sd_pooled = sd_pooled, delta_d = delta_d)
```

```
sd_pooled   delta_d
 1.767842  1.131323
```

```
ttestBF(formula = qsec ~ am, data = mtcars, nullInterval = c(-delta_d, delta_d))
```

```
Bayes factor analysis
--------------
[1] Alt., r=0.707 -1.1313225176573<d<1.1313225176573    : 0.9813314  ±0%
[2] Alt., r=0.707 !(-1.1313225176573<d<1.1313225176573) : 0.02203644 ±0.03%
```

```
Against denominator:
  Null, mu1-mu2 = 0
---
Bayes factor type: BFindepSample, JZS
```

The first line is the Bayes factor for "the effect is inside $\pm\Delta$" against the point null; the second is for "the effect is outside $\pm\Delta$." Their ratio favors *inside* by roughly 45 to 1 — 0.98 against 0.022. That's the Bayesian version of what TOST just told us at $\Delta = 2$ seconds: same data, same bound, same conclusion, different formalism. (Had I skipped the unit conversion, that ratio would have printed as about 131,000 to 1 — three orders of magnitude of apparent evidence, manufactured entirely by a bound I never chose.)

48.6. Bayes doesn't let you skip the hard part

Here's the thing I think gets missed, and the reason people often reach for a Bayes factor in the first place: it feels like a way to avoid ever stating a Δ. Look at what actually happened above, though. To get a Bayesian claim that resembles "no meaningful effect," I had to hand `nullInterval` the *same* ± 2 seconds used for TOST — rescaled into the standardized units it expects, but the same substantive bound. The judgment didn't disappear; it just needed a unit conversion on the way in.

Even the plain, no-interval Bayes factor doesn't escape it — the judgment just moves somewhere harder to see. The Cauchy prior's scale ($r = \sqrt{2}/2$ by default, visible in the printed output as `r=0.707`) encodes an assumption about how big a "typical" real effect would be, and that assumption drives whether BF_{10} ends up favoring the null. Change r and you change the verdict, on the same data.

This connects straight back to why pre-registration exists. If Δ, or the prior's scale r, isn't locked in *before* you see the data, you're right back to the original problem: nothing stops you from nudging it — a wider Δ, a tighter prior — until the analysis says what you wanted. TOST versus Bayes was never really the choice that matters. The choice that matters is whether the number driving either one was decided in advance or backed into afterward. A method that lets you state your commitment openly (TOST's Δ) is easier to audit than one that lets the same commitment hide inside a prior most readers won't think to interrogate — but both need the commitment made before the data arrive, or neither one is doing its job.

Plain NHST doesn't get a pass here either. Testing against exactly zero is its own unexamined threshold — it's just a default nobody questions, rather than a bound anyone stated and defended.

48.7. So how do you actually justify Δ?

This is the genuinely hard part, and pre-registering *something* doesn't help if that something was picked arbitrarily. The general playbook (Lakens, Scheel & Isager, 2018) gives three routes:

1. **Just-noticeable difference** — set Δ to the smallest change that would actually be perceptible or consequential in the world. Works well when there's a natural perceptual or operational unit.
2. **Cost-benefit analysis** — decide what magnitude would actually be worth acting on, given the stakes. This is a substantive argument, not a formula, and it's field- and context-specific by nature — a “negligible” effect in one domain can be highly consequential in another, and that's appropriate, not a flaw.
3. **Small telescopes** (Simonsohn, 2015) — anchor Δ to the effect size a comparable prior study had only *33% power* to detect: an effect so small that the original design would have missed it two times in three. Worth being careful here, because it is easy to misread: the 33% is a power level, not a fraction of the original estimate. At conventional sample sizes $d_{33\%}$ comes out a little over half of $d_{80\%}$, not a third of it. This route has an actual formula behind it, but only works if a genuinely comparable prior study exists.

What doesn't work, and I've caught myself reaching for it: deriving Δ by converting some other bound through an ad hoc formula — say, dividing a level-based bound by a predictor's range to manufacture a slope-based one — without a citation behind the conversion. That's a self-defined bound wearing a formula as a disguise, and pre-registering it doesn't fix the underlying problem: it was still picked to be convenient rather than derived from anything defensible.

48.8. “Why not just report the CI and let the reader decide?”

I pushed on this myself before landing on TOST as the thing to actually recommend. The estimation-focused view — report the point estimate and CI, skip the dichotomous decision — is a real, well-respected position (Cumming's “New Statistics” is the standard reference). And it's true that TOST computes nothing beyond what the CI already contains; the whole procedure collapses to a single inclusion check.

But “let the reader decide” quietly reintroduces the exact problem pre-registration exists to solve. Deciding a CI is “narrow enough” to call an effect negligible requires the same judgment as choosing Δ — it's just handed to each reader's private, unstated threshold instead of the author's public, pre-committed one. And a pre-registered design specifically can't defer that judgment to the reader, because pre-registration only works if it's the *author* committing, in advance, to what counts as support, null, and inconclusive. A reader's judgment, applied after publication, isn't something anyone pre-registered — there's nothing to audit it against.

The resolution I landed on: do both, and they stop being in tension. Report Δ as the pre-registered, defended criterion — that's the commitment the whole framework depends on. Report the raw coefficient and CI right alongside it — that's what lets a reader who disagrees with your Δ apply their own, using numbers you've already published, with no re-analysis required. That's also a real point in TOST's favor over a bare Bayes factor: a skeptical reader can route around a disputed Δ using the reported CI directly; disputing a Bayes factor means recomputing the whole thing under a different prior.

48.9. What I'd actually do

Lead with equivalence testing, report the Bayes factor alongside as a convergent check — cheap, and pure upside when they agree. Pre-register Δ using whichever of the three justification routes above actually fits the field, stated and defended before the data exist to bias the choice. And always report the plain coefficient and CI next to the equivalence verdict — not because the formal test is wrong, but because the CI is the part that survives even if a reader rejects your bound.

The single idea underneath all of this: pre-registration isn't really about picking the "right" test. It's about making sure nothing — not the test, not the normality check, not Δ, not a prior's scale — gets to be chosen after the data have already told you what answer it would give.